Foundations and Pitfalls in Epidemic Modeling:

An Introduction to Formulating, Analyzing, and Fitting Models

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Preface

The purpose of models is not to fit the data, but to sharpen the question.

— Samuel Karlin

Two research teams fit the same compartmental model to the same outbreak data and report basic reproductive numbers that differ by more than a factor of three. Neither team miscalculates, and neither misreads the data; the gap comes entirely from a single modeling assumption each made before the fitting began. Even simple epidemic models can mislead when basic pitfalls like this one are overlooked, and the pitfalls are rarely exotic. They are the recurring choices—about whether a deterministic model applies at all, how to write it down, how to fit it, and how to report what it says—that quietly separate a trustworthy analysis from one that merely looks plausible.

This book is organized around a catalog of these common pitfalls—recurring patterns of mistake that shape modern epidemic modeling. They are not a checklist. Each has cost the field measurable accuracy when it was overlooked, and each a careful modeler can avoid with a little preparation. A reader who internalizes them can recognize the same patterns in unfamiliar problems, which is the operational test of having understood the book.

The book is also written for the moment in which it appears. Its readers increasingly work with powerful AI systems at their elbow—language models, coding assistants, mathematical-reasoning tools—that produce fluent, confident, and sometimes quietly wrong answers on exactly the questions the common pitfalls concern: dimensionally inconsistent equations, conflated forces of infection, mis-specified reproductive-number formulas, and code that runs while implementing the wrong mathematics. A modeler who has internalized these pitfalls is equipped to catch such failures; one who has not can be led confidently astray. The book therefore treats AI-assisted modeling as the working reality it now is, threading a single verification discipline—*verify, verify, verify, then trust*—through its exercises and case studies.

The Common Pitfalls

The common pitfalls fall into four natural groups that follow the workflow of building, fitting, and applying a model, and each group answers one of the modeler's recurring questions. *Pitfalls of regime* ask whether a deterministic compartmental model is the right tool at all, and what it quietly assumes about the setting in which it is used. *Pitfalls of formulation* ask, once that framework is chosen, what must be gotten right in writing the model down: dimensional consistency, the choice between the force *of* and the force *from* infection, a faithful linearization, physically realizable initial conditions, and balanced contact mixing. *Pitfalls of estimation* ask how the model should be fit to data and analyzed so its conclusions are trustworthy: fitting what the data actually report, checking identifiability, respecting the generation-time distribution, and disciplining the sensitivity analysis. *Pitfalls of interpretation* ask how the reproductive number, the most-cited quantity in epidemiology,

should be reported without misleading—distinguishing the basic number that governs invasion from the effective number that governs an epidemic already under way, and invasion analysis from the analysis of an endemic state. Each pitfall is flagged in the text where the error is tempting, and a chapter just before the applications gathers all of them, with a one-page checklist and a worked audit of a constructed model.

Two illustrations make the practical stakes of these common pitfalls concrete. Figure 1 shows the consequence of conflating force-of-infection and force-from-infection formulations Pitfall: "Force of Infection versus Force from Infection" with Pitfall: "Identifiability" when fitting incidence data: the same early-outbreak data, fit from the two formulations, give estimates of $\mathcal{R}_0$ with sharply different sensitivity to uncertainty in the susceptible population. Figure 2 shows the structural distinction between the parameters that control invasion ($\mathcal{R}_0$) and those that control the long-run endemic burden (i^*); the distinction is one of the practical tools the book develops for thinking about Pitfall: "Invasion versus Equilibrium".

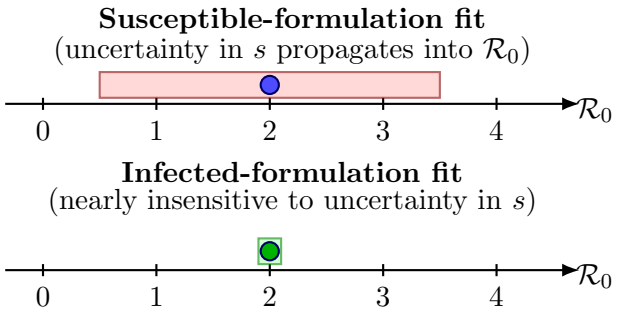


Figure 1: Estimating $\mathcal{R}_0$ from the same incidence data, spanning an outbreak through its peak, using two equivalent formulations of the SIR model. The susceptible-formulation fit (top panel, illustrating Pitfall: "Force of Infection versus Force from Infection") inherits the uncertainty in the susceptible population directly: a three-fold uncertainty in s produces roughly a three-fold uncertainty in $\mathcal{R}_0$. The infected-formulation fit (bottom panel) is nearly insensitive to the same uncertainty, illustrating Pitfall: "Identifiability": which parameters are well-determined by the data depends on the formulation through which the data are interpreted. The two formulations predict identical trajectories; they differ in the sensitivity of fitted parameters to inputs the modeler cannot pin down. The structural reason is a sensitivity asymmetry developed as a result in Chapter 5.

Who the book is for

This book is written for upper-level undergraduates and beginning graduate students in applied mathematics, quantitative public health, and related fields. The mathematical prerequisites are a first course in ordinary differential equations, a first course in linear algebra, and a first course in calculus-based probability and statistics. Prior exposure to epidemic modeling is not assumed. Readers who want to refresh any of these prerequisites should consult Appendix A, which surveys the ODE techniques, matrix-analysis results, and probability-and-statistics material used in the book. Readers who have worked through a first course in mathematical biology at the level of Edelstein-Keshet [2005] will find the material familiar in flavor but different in emphasis.

The book is also written for working modelers who want a practical reference. The case studies of Part V are intended to be directly useful to readers whose question is how to fit an epidemic model

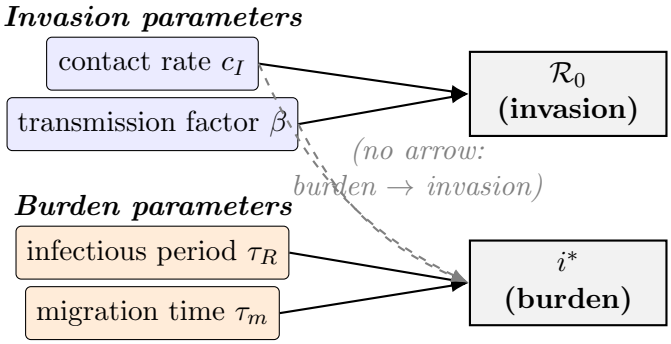


Figure 2: The structural distinction between invasion and equilibrium dynamics. The invasion parameters (c_I, β in the SIR base model) determine whether an outbreak invades, depending on whether $\mathcal{R}_0 > 1$ (invasion) or $\mathcal{R}_0 < 1$ (extinction). The burden parameters (τ_R, τ_m) determine the endemic prevalence i^* once invasion has occurred. Invasion parameters also affect burden (dashed arrows), but burden parameters do not affect invasion (no arrows drawn). This asymmetry is one face of Pitfall: "Invasion versus Equilibrium": the basic reproductive number describes invasion at the disease-free equilibrium, and the analysis of dynamics around the endemic equilibrium requires different tools. Chapter 5 develops the partition rigorously; subsequent chapters generalize it across transmission modes.

to real data and what can be inferred from the fit. The common pitfalls provide the organizing principle that ties the case studies back to the theoretical apparatus of Parts II–III.

What the book does not cover

The book develops deterministic compartmental models in continuous time; several important model classes fall outside that scope by design. Each excluded class has its own book-length treatments. *Stochastic models* capture extinction, takeoff timing and generation-to-generation variability, which matter most early in an outbreak; Allen [2010] gives an accessible treatment and Britton and Pardoux [2019] an advanced one. *Network and individual-based models* relax homogeneous mixing [Keeling and Eames, 2005, Kiss et al., 2017, Willem et al., 2017]. *Spatial and metapopulation models* follow spread across geography [Riley, 2007, Colizza and Vespignani, 2009]. *Optimal control* of interventions is treated by Lenhart and Workman [2007]. Time-dependent parameters appear in the COVID-19 case study (Chapter 17) and in Chapter 6, Section 6.8, but the general theory of fitting them, and ensemble forecasting, lie outside the book. Huang and Morris [2025] surveys these paradigms and is a good next step.

The deterministic regime itself is not the right framework everywhere, and Pitfall: "The Stochastic-to-Deterministic Regime Transition" in particular cautions against using deterministic compartmental models in regimes where stochastic effects dominate. The book treats this not as a limitation to be hidden but as a pitfall to be named.

A note on notation

Lowercase letters are fractions and uppercase letters are counts. Writing $i = 0.01$ says that one percent of the population is infectious; writing $I = 500$ says that five hundred people are. The two meet only through the carrying capacity, as $I = i\,N^{\mathrm{c}}$. The distinction can look fussy on the page.

It is not: a count put where a fraction belongs gives an equation that balances on paper and fails on data, a close relative of Pitfall: "Dimensional Consistency". A book that warns against the mistake should not commit it, so every state variable here carries its kind in its case. Appendix F gives the full conventions and the few places where the rule bends.

A note on AI systems

This book assumes that its readers have an AI assistant at hand and will use it. The boxes headed Verify the AI, AI Learning Prompt and AI Lab, and the tagged exercises described below, are where it teaches that use. Appendix D, *Learning with AI*, develops the verification workflow the opening promised and catalogs the failure modes where AI systems are most confidently wrong, among them hallucinated citations and heterogeneous-setting reproductive-number formulas that look authoritative and are simply incorrect. Three exercise tags—[**Verify the AI**], [**AI Lab**], and [**Open Project**]—mark the exercises built around AI interaction; the rest carry no tag and are meant for unaided work. A companion three-document package treats AI-aware learning more fully: the Student, Instructor, and Institutional Guides [Hyman, 2026c,b,a], all freely available and sharing a common vocabulary.

Acknowledgements

The authors thank colleagues at the Los Alamos National Laboratory, the University of Texas at San Antonio, Tulane University, and Harbin Engineering University for conversations that shaped the material in this book. The book grew out of lectures delivered to students in the Center for Computational Science at Tulane and in the mathematics department at the University of Texas at San Antonio, and we thank those students for their questions and feedback.

The common pitfalls are the spine of the book: each chapter develops one or more in depth, each case study revisits them in context, and each appendix supports their use. The two teams of the opening paragraph reached answers a factor of three apart not through any error in arithmetic but through an unexamined assumption made before the fitting began—and, as Chapter 7 shows, one their shared data could not have settled. Learning to catch such assumptions before they propagate is the practical lesson the book aims to leave with the reader.

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Key Symbols

The full notation, with units and conventions, is in Appendix F.

s, i, r	susceptible, infectious, and recovered compartments as fractions of the carrying capacity
S, I, R, N	the same quantities as counts of individuals
n	current total, as a fraction of the carrying capacity, $n = s + i + r$
N^{c}	carrying capacity, the reference population scale
$j(t)$	model incidence, $j = \alpha i$
c_S, c_I, c_R	contact rates for susceptible, infectious, and recovered individuals
β	per-contact transmission probability
P_S, P_I, P_R	contact-weighted fractions, $P_X = c_X X / C_{\mathrm{total}}$, with $C_{\mathrm{total}} = c_S s + c_I i + c_R r$
λ	force of infection (susceptible viewpoint), $\lambda = c_S \beta P_I$
α	force from infection (infected viewpoint, primary in this book), $\alpha = c_I \beta P_S$
τ_R, $\gamma_R = 1/\tau_R$	infectious period and recovery rate
τ_m, $\nu_m = 1/\tau_m$	mean residence time and migration rate (net demographic turnover)
ν_b	birth rate (per-capita reproduction)
$\mathcal{R}_0$	basic reproductive number, the spectral radius of the next-generation matrix
$\mathcal{R}_e(t)$	effective reproductive number at time t
$\mathcal{R}_c$	controlled reproductive number: $\mathcal{R}_e$ under a specified intervention

How to Read This Book

The book has twenty chapters organized into five parts and seven appendices, and admits four reading paths. The chapter-dependency diagram on the next page shows the prerequisite structure in full; this section describes how to use it.

The Reading Guide that follows the table of contents describes four paths through the book: a one-semester course, a quick introduction to the deterministic compartmental framework, an applied path focused on parameter estimation, and an upper-level undergraduate path that closes with a single case study. The chapter-dependency diagram in the Reading Guide identifies the minimum prerequisite chain for any target chapter. Path 1 is set out as a recommended fifteen-week syllabus, so an instructor considering the book for a one-semester course has a concrete week-by-week starting point to adapt rather than a claim to take on trust.

Each chapter opens with the question it answers and closes with its key takeaways and exercises. Readers who want the whole map first can read the pitfall chapter's checklist early and return to it at the end.

The four reading paths

Path 1: One-semester course. Chapters 1 through 9 in sequence (Parts I, II, and III), with selected chapters from Parts IV and V as time permits. This path develops the full framework (foundations, the SIR model, fitting, and sensitivity) and gives the course instructor a coherent arc from the pitfalls of Part I (regime of validity) through the pitfalls of Part III (estimation). The recommended course-text syllabus is described in the section "Using this book as a course text" below.

Path 2: Quick introduction to the deterministic compartmental framework. The Preface (the common-pitfalls preview), Chapter 1 (foundations), Chapter 3 (force from infection), Chapter 4 (derivation), Chapter 5 (analysis), and Chapter 6 (simulations). This path covers the minimum needed to understand and simulate the SIR model. Chapter 2 (the susceptible-viewpoint companion derivation) can be skipped on a first pass; readers who want to connect the framework to the standard literature should add it back. Fitting (Chapter 7) and sensitivity (Chapter 9) can be added as interest dictates.

Path 3: Applied reader focused on parameter estimation. The Preface (common-pitfalls preview), then Chapters 6 (simulations), 7 (fitting), and 9 (sensitivity), then the case studies of Part V (Chapters 17 through 20). This path skips the derivation and analysis chapters (the framework is taken as given, with the formal results referenced as needed) and goes directly to the practical content. The supplementary chapter on fitting in practice (Chapter 8) is recommended for this path.

Path 4: Upper-level undergraduate path. Chapters 1 through 6 plus selected sections of Chapters 7 and 9, closing with one case study from Part V (typically Chapter 17 for its familiarity). This path gives an upper-level undergraduate the common-pitfalls framework and one worked application without the full analytical apparatus of Part IV. The mathematical prerequisites match a first course in ordinary differential equations, a first course in linear algebra, and some programming comfort; Appendix A supplies the needed reference material for any gaps. An instructor teaching this path should plan to supplement with additional computational exercises drawn from the chapter-end [**L**]-tagged problems and from the companion repository.

All four paths begin with the Preface, which previews the common pitfalls and ties the chapters together via the common pitfalls they develop. After the introduction, the dependency structure branches.

Chapter-dependency diagram

The diagram on the following page shows which chapters are prerequisites for which. Each chapter is a box; an arrow from Chapter A to Chapter B means that B uses material from A. A reader targeting Chapter B should have worked through the chapters reachable by following arrows backward from B. Solid arrows indicate required prerequisites; dashed arrows indicate recommended but not strictly required prior reading.

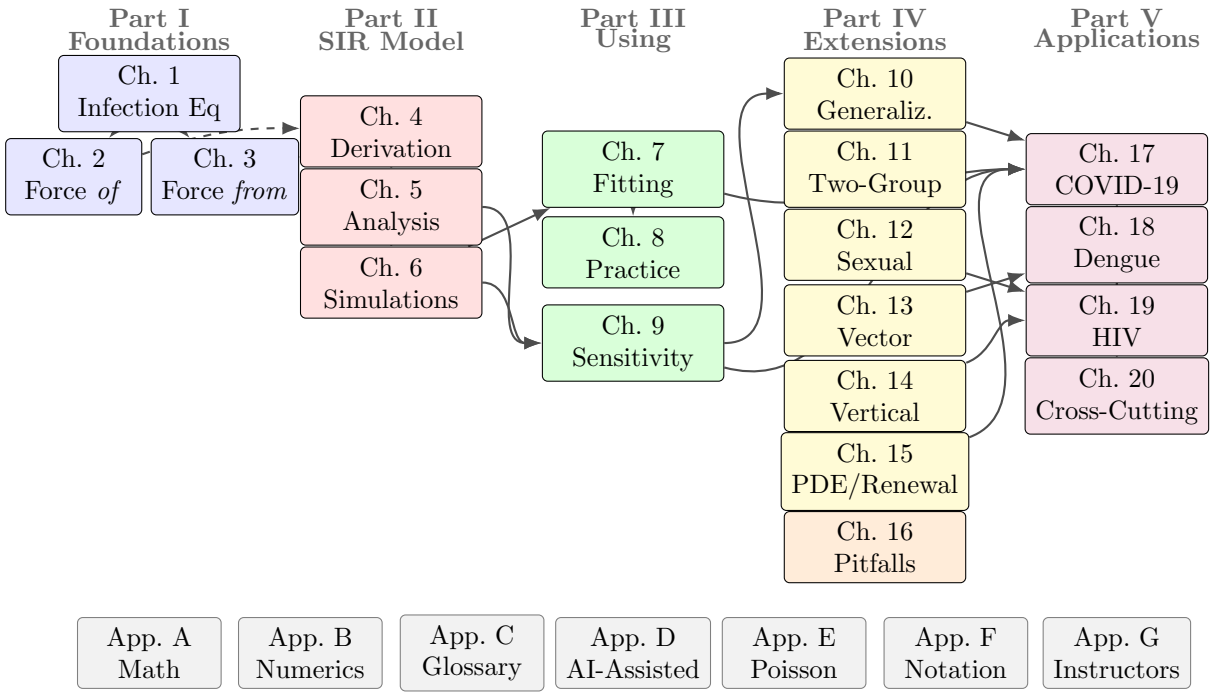


Figure 3: Chapter-dependency structure of the book. Solid arrows indicate required prerequisites; dashed arrows indicate recommended but not strictly required prior reading. The four reading paths described in the preceding section correspond to walks through this diagram. The book's chapters are numbered sequentially from Chapter 1 through Chapter 20; Part I opens at Chapter 1, Part II at Chapter 4, Part III at Chapter 7, Part IV at Chapter 10, and Part V at Chapter 17. Part IV's modeling chapters all extend the Part II framework; this implicit cross-dependence is not drawn as individual arrows. Chapter 16 has no arrows: its checklist can be read early or at the end. Appendices A through C, E and F are reference material cited from many chapters and do not impose dependencies of their own; Appendix D (verification checklists for AI-assisted modeling) is a freestanding guide that can be read at any point, and Appendix G is written for instructors.

List of Common Pitfalls

Where each pitfall is developed

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Part I

Building the Model: The Infection Equation from the Susceptible and Infected Viewpoints

Chapter 1

The Infection Equation

The compartment is not a biological fact. It is the simplification that makes the biology tractable.

Picture a town of ten thousand people in the first weeks of a new respiratory infection. The clinic sees a handful of cases in the first week and more each day after that. The town's health officer must decide what to do, and the decision rests on four things the officer does not yet know: whether the outbreak will grow or fade on its own, when the clinic will be busiest and how busy, how many people will have been infected by the end, and whether closing the schools would change any of that. Each concerns the future or a step the town has not yet taken, so the case counts alone cannot settle it. Each needs a model.

All four turn on one quantity: how fast new infections arise. When new infections outpace recoveries, the outbreak grows; when they fall behind, it turns. The number infected by the end is that rate added up over the whole outbreak. Closing the schools helps only to the extent that it lowers the rate.

The officer's data hold a puzzle that appears only with time. Outbreaks like this one end before everyone has been infected. Many residents never catch the disease, although nothing protected them except the outbreak's own decline. Kermack and McKendrick [1927] explained why. As people are infected and recover, each infectious person meets fewer people who can still catch the disease. Once each case passes the infection to fewer than one other person, the outbreak dies out while susceptible people remain.

So the first question for a model is this one: how fast do new infections arise, and how should we write that rate down? The explanation above already points to an answer. It turns on the number of new infections each infectious person causes, a number that falls as susceptible people are used up. This chapter counts the rate from that side. We assemble it from three things the officer could try to estimate: how often an infectious person meets others, what fraction of those meetings are with people who can still catch the disease, and how likely such a meeting is to pass the infection on. We then ask how each ingredient should be modeled. The same rate can also be counted from the other side, from the people who catch the infection, and that count gives the standard textbook form. The two counts always agree. Section 1.5.4 explains why the book starts from the infectious side, and when the other side is the better place to start.

1.1 Compartments and State Variables

A compartmental epidemic model partitions the population into groups, called compartments, that share the same epidemiological status. The simplest such partition, dating to Kermack and McKendrick [1927], is the three-compartment split into susceptible, infectious, and recovered individuals.

Susceptible. The *susceptible* compartment $S(t)$ contains individuals who can acquire infection on contact with an infectious individual.

Infectious. The *infectious* compartment $I(t)$ contains individuals who currently carry the pathogen and can transmit it through contact.

Recovered. The *recovered* compartment $R(t)$ contains individuals who have cleared the pathogen and are immune to reinfection. This compartment is sometimes called the *removed* compartment.

The basic SIR model and its extensions. The three compartments above define the basic SIR model. The rest of Part I and all of Part II develop the analytical and computational machinery for this baseline structure. Real epidemics often require additional structure: waning immunity (returning recovered individuals to the susceptible compartment), disease-induced mortality (tracking deaths in a separate compartment rather than absorbing them into the recovery rate), exposed-but-not-yet-infectious latency, symptomatic-versus-asymptomatic compartmental splits, age or risk stratification, and others. Chapter 10 develops the systematic extensions; the baseline three-compartment structure is sufficient for the foundational material of Parts I–III.

Total population and closed systems. The total population is

$$N(t) \;=\; S(t) + I(t) + R(t).$$

In a closed population (no births, no deaths, no migration), N is constant and $S + I + R = N$ at all times (Pitfall: "Short-Term Simulation Assumptions"). When migration is present (modeled through a single migration rate ν_m; see Chapter 4), N becomes a function of time rather than a constant.

Remark 1.1 (Counts here, fractions from Chapter 4). In this chapter and the two that follow, $S(t)$, $I(t)$, $R(t)$, and N denote *counts* of individuals. Chapter 4, Section 4.5.3 normalizes them by the carrying capacity and, from that point onward, writes the resulting fractions in lowercase, s, i, r, n, keeping the uppercase letters for counts.

1.2 The Rate of New Infections

The *force from infection* α is the rate at which a typical infectious individual causes new infections: the number of people one infectious person infects per unit time. Its baseline expression is

$$\alpha \;=\; c\,\beta\,\frac{S}{N}. \tag{1.1}$$

This formula assembles three factors: c counts the contacts an infectious person makes per unit time, the fraction S/N selects the contacts with someone who can still be infected, and β is the

probability that such a contact transmits the pathogen. With I infectious people, the rate of new infections in the population is αI. Chapter 3 derives (1.1) from a model of one infectious person's contacts; this chapter takes it as a stated result and explores its two modeling ingredients c and β.

The standard form gives the same rate. Most textbooks count the same infections from the other side. The *force of infection* λ is the rate at which a typical susceptible individual acquires infection; its baseline expression is

$$\lambda \;=\; c\,\beta\,\frac{I}{N}. \tag{1.2}$$

Here c counts a susceptible person's contacts, and I/N selects those with an infectious partner. The rate of new infections is then λS, and it agrees with the count from the infectious side: $\lambda S = c\beta SI/N = \alpha I$. Section 1.5 develops both viewpoints with a separate contact rate for each compartment, and Section 1.5.4 gives the reasons the book treats α as the better default. Chapter 2 derives (1.2) from first principles.

The contact rate. The contact rate c is the average number of epidemiologically relevant contacts a given individual has per unit time. It has units of contacts per person per unit time, which in the book's notation we write as $\mathtt{CP}^{-1}\mathtt{T}^{-1}$. An *epidemiologically relevant contact* is any interaction between two individuals in which pathogen transmission could occur. For respiratory infections a contact is a face-to-face encounter within a few meters for a period of minutes; for sexually-transmitted infections a contact is a sexual encounter; for vector-borne infections the contacts are between vectors and hosts. The definition is specific to the pathogen and its transmission mechanism. In (1.1) the contacts that count are the infectious person's; Section 1.3.2 gives that rate its own symbol, c_I.

The transmission factor. The transmission factor β is the probability that a single contact between a susceptible and an infectious individual results in transmission. The factor β absorbs both the biology of transmission (pathogen dose, environmental conditions, the duration of a typical contact) and the averaging over contacts of variable length. The connection between β and the underlying contact-level transmission rate is developed in Chapter 2; for most practical purposes in this book we take β as a single calibrated parameter.

As a probability it is a pure number between 0 and 1, and it stays dimensionless throughout: it is the fraction of contacts that transmit, not a quantity carrying units of its own. The dimensions of the force from infection come entirely from the contact rate. In (1.1), c counts contacts per infectious person per unit time and contacts are counted events, so c has dimensions $\mathtt{T}^{-1}$; multiplying by the dimensionless β and the dimensionless fraction S/N leaves α with dimensions $\mathtt{T}^{-1}$, a rate per infectious person. The same bookkeeping gives λ the dimensions of a rate per susceptible person. Keeping β dimensionless is what makes that bookkeeping check out, and it is the convention used throughout the book.

Remark 1.2 (Read a parameter's value aloud as a plausibility check). A number that passes every dimensional check can still be biologically absurd, and the fastest way to catch this is to translate the value into plain language before using it. The transmission probability $\beta = 0.0001$, for instance, says that roughly one contact in ten thousand between a susceptible and an infectious individual results in transmission—implausibly low for a directly transmitted pathogen, and a sign that the figure has been confused with a rate or simply mis-transcribed. The same habit works for every rate in the model: a recovery rate $\gamma_R = 0.001$ per day asserts a mean infectious period of about a thousand

days, obviously wrong for an acute infection, and a contact rate $c = 0.5$ per day asserts roughly one transmission-relevant contact every two days. Verbalizing the value is the magnitude-level companion to the dimensional check of Pitfall: "Dimensional Consistency": dimensional analysis rules out the wrong *units*, and reading the number in words rules out the wrong *size*.

Example 1.3 (New infections in the town, counted by hand)**.** Return to the town of the opening a week into its outbreak, and use round numbers. The town has $N = 10{,}000$ residents, of whom $I = 50$ are infectious; ignoring the few who have recovered, $S = 9{,}950$ are susceptible. Suppose each resident makes $c = 10$ epidemiologically relevant contacts per day, and a contact between a susceptible and an infectious resident transmits the infection with probability $\beta = 0.08$. Start with one infectious resident. Of that resident's 10 contacts a day, a fraction $S/N = 0.995$ are with susceptible residents, and each of those transmits with probability 0.08:

$$\alpha \;=\; c\,\beta\,\frac{S}{N} \;=\; 10 \times 0.08 \times 0.995 \;\approx\; 0.80 \text{ new infections per day.}$$

Each infectious resident causes about four new infections every five days. The 50 infectious residents together cause

$$\alpha I \;\approx\; 0.80 \times 50 \;=\; 40 \text{ new infections per day.}$$

Now check the count with the standard form, which works from the other side. Each susceptible resident is infected at the rate

$$\lambda \;=\; c\,\beta\,\frac{I}{N} \;=\; 10 \times 0.08 \times \frac{50}{10{,}000} \;=\; 0.004 \text{ per day,}$$

about one chance in 250 per day, and the 9,950 susceptible residents together acquire $\lambda S = 0.004 \times 9{,}950 \approx 40$ new infections per day. The two counts agree, as they must: every new infection has one person who passed it on and one who caught it. Section 1.5 turns this bookkeeping into the two viewpoints. Chapter 3 derives the first count from a model of contacts, and Chapter 2 derives the second.

The calculation needs only arithmetic.

Verify the AI: The Count of New Infections

Give an AI assistant the numbers of Example 1.3 and ask for the number of new infections per day. Before you accept its answer, apply three checks, none of which needs the AI. *Units:* the answer must be in people per day. *Size:* it cannot exceed the 500 contacts the 50 infectious residents make each day, since every new infection needs a contact. *Two counts:* the infections caused by infectious residents must equal the infections acquired by susceptible ones. An answer that fails any check is wrong, however fluent its explanation. Readers working without AI apply the same three checks to their own hand calculation.

The baseline formula (1.1) uses a single contact rate c and a single transmission factor β applied uniformly across the population. The next two sections develop the modeling decisions that arise when each of these quantities admits additional structure: Section 1.3 examines how c depends on the population size and on the epidemiological compartment of the individual, and Section 1.4 addresses where contact-level variation in transmission should enter the model.

1.3 Modeling the Contact Rate (c)

The baseline contact rate c in the force from infection (1.1) aggregates over two distinct dimensions of variation: the dependence of contacts on total population size, and the variation of contacts

across the susceptible, infectious, and recovered compartments. The next two subsections develop each in turn.

1.3.1 Population-Size Dependence: Density vs. Frequency

The per-capita contact rate c can be treated as either independent of population size or proportional to it (Pitfall: "Density vs. Frequency Dependence"). The two cases correspond to two distinct mathematical formulations of the transmission process, both obtained from the baseline formula (1.1) by a specific assumption about how c depends on N. The density- and frequency-dependent incidence formulations are discussed by Li [2018, Sec. 1.4.2].

Density-dependent transmission. If the contact rate scales proportionally with N (writing $c = c_0 N$ for some constant c_0), the rate of new infections in the population becomes

$$\alpha I \;=\; (c_0 N)\beta \frac{S}{N} I \;=\; (c_0 \beta)\, SI, \tag{1.3}$$

the familiar *density-dependent* form $\beta_d SI$ with $\beta_d = c_0\beta$ a per-pair transmission rate.

Frequency-dependent transmission. If the contact rate is constant, independent of N, the rate of new infections is

$$\alpha I \;=\; c\beta \frac{S}{N} I \;=\; \frac{c\beta}{N}\, SI, \tag{1.4}$$

the familiar *frequency-dependent* form $\beta_f SI/N$ with $\beta_f = c\beta$ aggregating the contact rate and the per-contact probability into a single parameter.

The two formulations agree only when N is held fixed; under interventions or processes that change N (such as population growth or migration), they predict different dynamics. The distinction is the subject of an established literature [Begon et al., 2002, McCallum et al., 2001], in which density-dependent transmission fits livestock diseases, vector-borne diseases where vector populations track host density, and some plant pathogens, while frequency-dependent transmission fits sexually-transmitted infections and most respiratory diseases in modern urban settings. Real systems often sit between the two extremes, with an intermediate scaling $c \propto N^q$ for $0 < q < 1$.

A useful mnemonic: village vs. city. The two regimes have a concrete interpretation. In a village of twenty individuals, an average person has substantial daily contact with most of the others; in a village of forty, they have substantial contact with twice as many: the per-capita contact rate scales with N, leading to the density-dependent transmission formulation (village scenario). In a city of one million, an average person does not have a hundred times more contacts than someone in a city of ten thousand: the per-capita contact rate has saturated at the upper limit imposed by available time and attention, leading to the frequency-dependent transmission formulation (city scenario). The mnemonic should not be over-read: a tightly-clustered sexual-contact network in a city of one million may sit in the density-dependent regime, and a sparse contact pattern in a small rural population may sit in the frequency-dependent regime. The schema captures the typical case, not the universal one.

The transition between the two regimes is gradual in the total population size N, as shown in Figure 1.1. At small N the curves are approximately linear in N (the density-dependent regime); at large N they saturate at constants (the frequency-dependent regime); the crossover happens at the population scale where social structure begins to limit the per-capita number of accessible

contacts. The exact location of the crossover depends on the contact type (close-contact disease, sexually-transmitted disease, and vector-host contact each have different saturation scales) and is an empirical question for the specific disease and population.

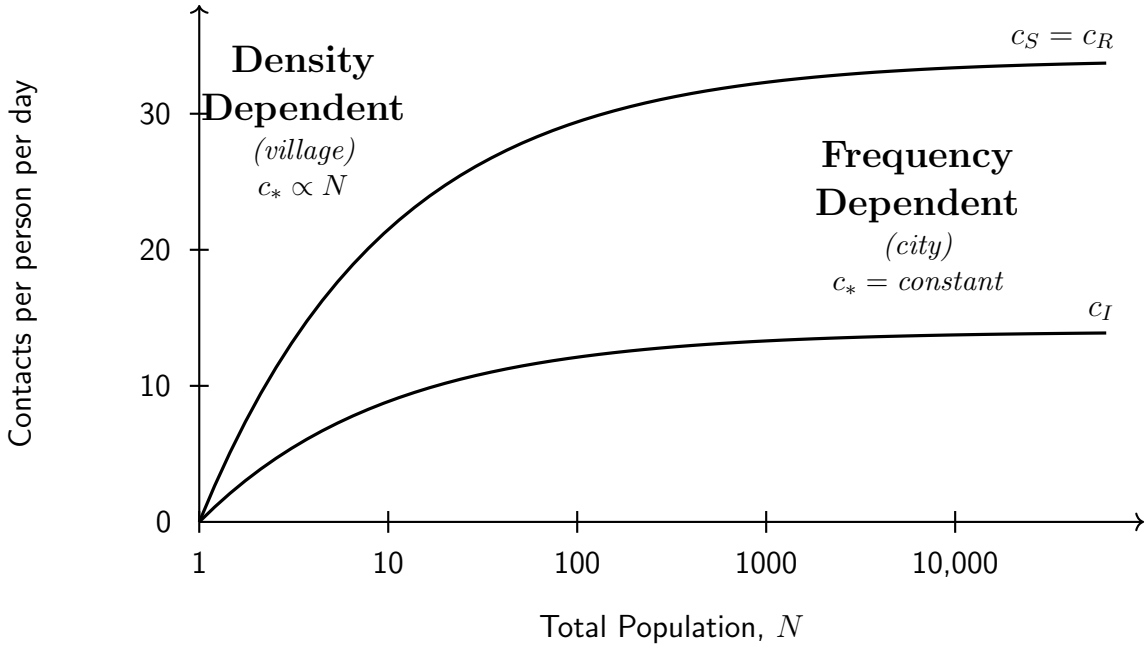


Figure 1.1: Two regimes for the per-capita contact rate as a function of population size N. At small N, per-capita contact rates grow with N (*density-dependent* transmission; the village mnemonic); at large N, they saturate at constants (*frequency-dependent* transmission; the city mnemonic). Here c_S and c_I are the per-capita contact rates (the average number of contacts per individual per unit time) of susceptible and infectious individuals respectively, and c_R is the corresponding rate for recovered individuals; the figure plots them collectively as c_*. The book's analytical and fitting machinery throughout assumes the frequency-dependent regime, in which these rates are constant, independent of N. The two horizontal asymptotes reflect the convention developed in Section 1.3.2: susceptible and recovered individuals have similar contact patterns ($c_S \approx c_R$), while infectious individuals typically reduce their contacts ($c_I < c_S$).

The book's choice. Throughout this book we use frequency-dependent transmission: the contact rate is treated as a constant independent of N (city regime). This choice is consistent with the deterministic ODE framework, which is itself most reliable in the large population regime where stochastic effects are negligible, the same regime where frequency-dependent transmission is empirically supported. A modeler who needs to address a small-population setting (a livestock herd, a school during an outbreak, a refugee camp) should expect to revisit both the frequency-dependent assumption *and* the deterministic regime assumption (see Pitfall: "The Stochastic-to-Deterministic Regime Transition") at the same time, since they are coupled in practice. The two assumptions are logically distinct: a small population may be in the deterministic regime if cumulative incidence is large enough (Pitfall: "The Stochastic-to-Deterministic Regime Transition" gives the threshold), and a large population may need density-dependent transmission if the relevant contact network is small (e.g., a sexually-transmitted infection in a tightly-clustered community). But in the typical application (a respiratory pathogen in a city or region of substantial size) both assumptions hold together.

1.3.2 Distinct Contact Rates for Distinct Compartments

The baseline formula (1.1) uses a single contact rate c that applies uniformly across the population. In reality, the contact rate of an individual depends on which epidemiological compartment they currently occupy. We introduce three compartment-specific contact rates (c_S, c_I, c_R for susceptible, infectious, and recovered individuals respectively) and use them in place of the single c throughout the rest of the book.

The motivation is that interventions and behavioral responses to an epidemic typically act on the contact rates of specific compartments rather than on a uniform population average. Case isolation, symptom-triggered withdrawal, behavioral change following diagnosis, hospitalization, and quarantine all reduce c_I; protective behaviors adopted by uninfected individuals during widespread transmission reduce c_S; recovered individuals typically return to pre-illness contact patterns, so $c_R \approx c_S$ is often a reasonable working assumption. Because these three rates can be impacted independently (and because the dominant interventions act on c_I specifically), treating them as separate parameters is essential for a model that aims to represent the controllable dynamics of an outbreak. The contact rate is among the most important parameters governing model predictions; a model that collapses $c_S = c_I = c_R$ into a single population-average c cannot represent the parameter that the dominant interventions target. That parameter, c_I, is the contact rate in the force from infection.

The book's choice. The book's default throughout is to treat c_S, c_I, c_R as distinct parameters. The simplifying assumption $c_S = c_I = c_R = c$ is used only where classical results simplify enough to justify the loss in realism, chiefly in the endemic-equilibrium derivation of Chapter 5, Section 5.5, where closed-form expressions aid interpretation.

1.4 Modeling the Transmission Factor (β)

A modeler who has chosen distinct contact rates c_S, c_I, c_R might next ask whether the transmission factor β should similarly be allowed to vary: whether contact-level differences in duration, intensity, and protective behavior between, say, an S-I contact and an I-I contact should be represented by distinct transmission probabilities. This is a real modeling decision, but the relevant question is not about state-dependent β per se: only S-I contacts can transmit, so framing the choice as a family of pair-type-specific transmission probabilities would be misleading. The real choice is about *where* contact-duration and behavioral effects should enter the model: absorbed into the transmission factor β, or absorbed into the contact rates c_S, c_I, c_R.

The book absorbs these effects into c. We treat β as a single constant representing the transmission probability per *effective contact*, where the qualifier "effective" already accounts for contact duration and protective behavior. All intervention and behavioral effects are then represented through the compartment-specific contact rates: case isolation, symptom-driven withdrawal, masking, social distancing, and quarantine all modify c_S or c_I; none of them modify β. The biological probability of transmission given an effective contact is a property of the pathogen, the host immune status, and the dose (not of the behavioral overlay), and the book treats it as such. A reader who needs pair-type-specific transmission probabilities for a specific application (a within-host pathogen-dose model, or a setting where personal protective equipment is the central intervention) will find the book's framework extends naturally to that case.

Identifiability note. Whether β is treated as state-dependent or constant, the parameter combination that the rate of new infections identifies from data is the product $c\beta$ (more precisely, $c_I\beta$ from the infected viewpoint, or $c_S\beta$ from the susceptible viewpoint, as developed in Section 1.5 below). The data cannot separate c and β without additional inputs. This identifiability constraint is developed in detail in the parameter-estimation chapter (Chapter 7); we flag it here so the reader understands that the book's choice to absorb behavioral effects into c is well-founded but not data-driven: it is a modeling convention that aligns the parameters with the interventions that act on them.

Exercises for this section

1. **[D]** Verify that the expression $P_S + P_I + P_R = 1$ follows directly from the definitions $P_S = c_S S/C_{\text{total}}$, $P_I = c_I I/C_{\text{total}}$, $P_R = c_R R/C_{\text{total}}$ and the definition of C_{total}.

1.4.1 The Contact Duration Problem

Consider a contact between a susceptible and an infectious individual. Section 1.4 treats the per-contact transmission probability β as a single given constant between 0 and 1. But the biological process of transmission is not instantaneous; it operates during the contact. Suppose the contact lasts τ_c time units, during which transmission opportunities occur continuously at some instantaneous rate ρ (with units of inverse time). What is the probability β that transmission occurs during the contact as a whole?

The tempting linear formula. A tempting answer identifies β with the expected number of transmission events during the contact:

$$\beta \;=\; \rho\,\tau_c \qquad \text{(linear approximation)}. \tag{1.5}$$

This formula has an obvious shortcoming: it can give values greater than 1. If $\rho = 0.1$ per minute and $\tau_c = 15$ minutes, the formula gives $\beta = 1.5$, which is not a valid probability. The linear formula is correct in an important limit (Section 1.4.1) but is not the formula that applies in general.

The Correct Exponential Formula. The correct formula comes from recognizing that transmission during a contact is a counting process. If transmission opportunities occur at rate ρ during the contact, and we assume independence between small time intervals (a standard Poisson-process assumption), then the probability of *no* transmission during a contact of duration τ_c is $e^{-\rho\tau_c}$. The probability that transmission occurs at least once is the complement:

$$\beta \;=\; 1 - e^{-\rho\tau_c}. \tag{1.6}$$

The product $\rho\tau_c$ (the instantaneous transmission rate multiplied by the contact duration) is the *contact-level transmission dose*, the single dimensionless quantity that determines the per-contact transmission probability. The exponential formula automatically satisfies $0 \le \beta \le 1$ for all $\rho \ge 0$ and $\tau_c \ge 0$.

The two extremes of the transmission dose are instructive, and we write them in parallel. As $\rho\tau_c \to 0$ (a short contact with an inefficient pathogen), $\beta \to 0$: transmission becomes vanishingly unlikely. As $\rho\tau_c \to \infty$, which can arise either from a very long contact duration (τ_c large) or from a highly contagious pathogen (ρ large), $\beta \to 1$: a long or efficient contact almost certainly transmits.

Figure 1.2 compares the linear approximation and the exponential formula. The linear approximation tracks the exact exponential closely for small $\rho\tau_c$ but diverges rapidly once $\rho\tau_c$ exceeds a few tenths; beyond $\rho\tau_c = 1$, the linear formula returns values greater than unity and is no longer a probability.

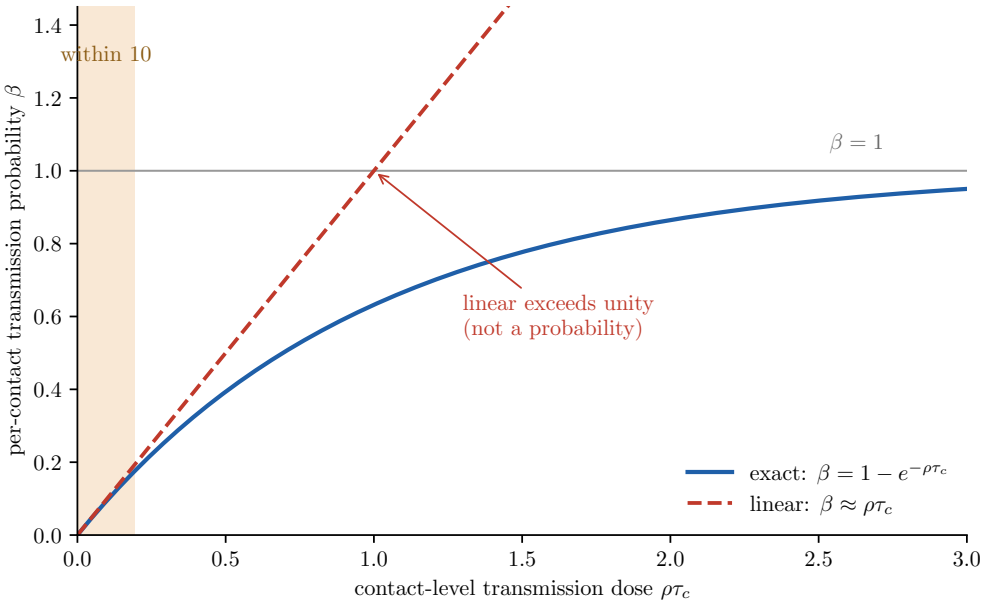


Figure 1.2: The per-contact transmission probability β as a function of the contact-level transmission dose $\rho\tau_c$. The exact exponential formula $\beta = 1 - e^{-\rho\tau_c}$ is bounded by unity; the linear approximation $\beta \approx \rho\tau_c$ tracks it accurately only for $\rho\tau_c \lesssim 0.2$ and fails outside that range. For the parameter values typical of respiratory infections in short casual contacts, the linear formula is an adequate working approximation; for extended contacts or efficient transmission it is not.

When the Linear Approximation is Valid. A Taylor expansion of $e^{-\rho\tau_c}$ gives

$$1 - e^{-\rho\tau_c} \;=\; \rho\tau_c - \frac{(\rho\tau_c)^2}{2} + \frac{(\rho\tau_c)^3}{6} - \cdots .$$

The linear approximation $\beta \approx \rho\tau_c$ is therefore accurate when $\rho\tau_c$ is small compared to unity. The relative error in the linear approximation is

$$\text{relative error} \;=\; \frac{(\rho\tau_c) - (1 - e^{-\rho\tau_c})}{1 - e^{-\rho\tau_c}} \times 100\%.$$

For $\rho\tau_c = 0.1$, the relative error is approximately 5%; for $\rho\tau_c = 0.5$, approximately 27%; for $\rho\tau_c = 1.0$, approximately 58%. The linear approximation is acceptable for short contacts where transmission is unlikely and breaks down when contacts are long enough or transmission is efficient enough that the per-contact probability exceeds a few percent.

Worked example: respiratory infection. Consider a 15-minute face-to-face contact with an instantaneous transmission rate $\rho = 0.01$ per minute (a plausible value for an influenza-like illness). Then $\rho\tau_c = 0.15$, and

$$\beta \;=\; 1 - e^{-0.15} \;\approx\; 0.139.$$

The linear approximation gives $\rho\tau_c = 0.15$, a relative error of about 8%. For this contact, the linear approximation is usable but not exact.

Worked example: extended contact. Consider an hour-long contact ($\tau_c = 60$ minutes) with $\rho = 0.02$ per minute. Then $\rho\tau_c = 1.2$, and

$$\beta \;=\; 1 - e^{-1.2} \;\approx\; 0.699.$$

The linear approximation gives $\rho\tau_c = 1.2$, which is meaninglessly large as a probability. For long contacts or efficient transmission, the exponential formula is required.

Implication for calibration. The choice between the exponential and linear forms affects the value of β, not the structure of either force. In a fitting context, a model that implicitly assumes the linear form absorbs any contact-duration nonlinearity into the fitted β, distorting its interpretation across settings with different contact-duration distributions; this is one reason fitted β values should not be read literally as per-contact transmission probabilities (Chapter 7).

> *Separate the law from the closure. The law tells you what must balance; the closure tells you what you believe.*

Exercises for this section

1. **[D]** Verify by Taylor expansion that $\beta = 1 - e^{-\rho\tau_c}$ reduces to $\beta \approx \rho\tau_c$ when $\rho\tau_c$ is small. Compute the next-order correction explicitly and give the relative error at $\rho\tau_c = 0.1, 0.5$, and 1.0.

AI Learning Prompt: Deriving the Validity Range of the Linear Approximation

The linear form $\beta \approx \rho\tau_c$ is the first-order Taylor approximation of the exact $\beta = 1 - e^{-\rho\tau_c}$. Its error grows from 5% at $\rho\tau_c = 0.1$ to 27% at 0.5 and 58% at 1.0, and the useful request is for the AI to derive the bound, not to recall it. *Prompt:* "*For $\beta = 1 - e^{-\rho\tau_c}$, find the largest $\rho\tau_c$ for which $\beta \approx \rho\tau_c$ has relative error below ϵ. Check your bound at $\epsilon = 0.05$ and $\epsilon = 0.10$.*"
A correct answer uses the next Taylor term to reach $\rho\tau_c \lesssim 2\epsilon$, about 0.10 and 0.20, consistent with the 5% error at 0.1. An answer that repeats the chapter's numbers has answered an easier question. The same move works wherever the book gives a numerical bound: ask for the derivation, then check it against the bound.

1.4.2 Variable Contact Durations

Real populations have contacts of varying duration. A single individual has brief contacts with strangers, longer contacts with coworkers, and extended contacts with household members. This section considers how variable contact duration affects the force-of-infection calculation.

Exponentially distributed contact durations. Suppose contact durations are exponentially distributed with mean $\bar{\tau}_c$. For a contact of duration τ_c, the transmission probability is $\beta(\tau_c) = 1 - e^{-\rho\tau_c}$. Averaging over the exponential distribution of contact durations,

$$\bar{\beta} \;=\; \int_0^\infty \left(1 - e^{-\rho\tau_c}\right) \cdot \frac{1}{\bar{\tau}_c} e^{-\tau_c/\bar{\tau}_c}\, d\tau_c \;=\; 1 - \frac{1}{1+\rho\bar{\tau}_c} \;=\; \frac{\rho\bar{\tau}_c}{1+\rho\bar{\tau}_c}.$$

This is the effective per-contact transmission probability when contact durations are exponentially distributed with mean $\bar{\tau}_c$. Note that $\bar{\beta}$ is a different function of $\rho\bar{\tau}_c$ than the fixed-duration $\beta = 1 - e^{-\rho\bar{\tau}_c}$: for the same mean duration, variability in duration changes the effective transmission probability. This is an instance of *Jensen's inequality* applied to the concave function $f(\tau_c) = 1-e^{-\rho\tau_c}$: averaging f over a distribution of contact durations τ_c yields a smaller value than evaluating f at the mean of τ_c. Quantitatively, $\bar{\beta} = \mathbb{E}[1-e^{-\rho\tau_c}] \leq 1-e^{-\rho\mathbb{E}[\tau_c]}$, with equality only when contact durations are deterministic. The same mechanism underlies the population-averaging warning in Chapter 11, Section 11.8: averaging contact rates before computing $\mathcal{R}_0$ systematically biases the estimate, in a direction set by the curvature of the function being averaged.

For small $\rho\bar{\tau}_c$,

$$\bar{\beta} \;\approx\; \rho\bar{\tau}_c - (\rho\bar{\tau}_c)^2 + O\big((\rho\bar{\tau}_c)^3\big),$$

while the fixed-duration formula gives

$$\beta \;\approx\; \rho\bar{\tau}_c - \frac{(\rho\bar{\tau}_c)^2}{2} + O\big((\rho\bar{\tau}_c)^3\big).$$

The first-order terms agree; the second-order terms differ by a factor of 2. The linear approximation $\beta \approx \rho\bar{\tau}_c$ is insensitive to duration variability; the nonlinear correction is sensitive to it.

Connection to the two forces. Regardless of whether contact durations are fixed or variable, the structure of both forces is preserved with β replaced by the appropriate averaged quantity $\bar{\beta}$. The effect of variable durations is absorbed into the calibrated value of β; the formal structure of the ODE model is unchanged.

Limiting cases. Two limits are worth noting:

- $\rho\bar{\tau}_c \ll 1$: transmission is rare during a typical contact. Then $\bar{\beta} \approx \rho\bar{\tau}_c$, duration variability does not matter at leading order, and β can be safely calibrated from empirical transmission frequencies.
- $\rho\bar{\tau}_c \gg 1$: transmission almost certainly occurs during a typical contact. Then $\bar{\beta} \to 1$, and variability in the duration distribution does not matter: every contact transmits regardless. This regime is unusual for naturally occurring infectious disease contact structures.

Real populations typically sit in the intermediate regime $\rho\bar{\tau}_c \sim 0.1$ to 0.5, where duration variability introduces corrections of order 10% to 50% that are absorbed into calibrated β.

1.4.3 Impact on Epidemic Predictions

The exponential-versus-linear distinction for β has downstream consequences for the basic reproductive number and for the threshold behavior of the model. We illustrate briefly.

Effect on $\mathcal{R}_0$. From Chapter 5, the basic reproductive number of the SIR model is

$$\mathcal{R}_0 \;=\; \frac{c_I\beta}{\gamma_R + \nu_m}.$$

If β is computed from the correct exponential formula but modeled using the linear approximation $\beta \approx \rho\tau_c$, the resulting $\mathcal{R}_0$ is overestimated. For $\rho\tau_c = 0.5$, the linear formula gives $\beta_{\text{lin}} = 0.5$ while the correct value is $\beta = 1-e^{-0.5} \approx 0.393$: a factor-of-1.27 overestimate of $\mathcal{R}_0$ and a correspondingly distorted threshold estimate.

Effect on the pandemic threshold. The condition for pandemic invasion is $\mathcal{R}_0 > 1$. An intervention that reduces the contact duration (for example, public-health messaging that shortens the duration of indoor gatherings) reduces β. If the reduction is computed linearly, the predicted effect of the intervention is overstated; the correct exponential computation gives a smaller but still meaningful reduction.

A practical rule. For most parameter regimes where the book's models apply, $\rho\tau_c < 0.3$ and the linear approximation produces errors under 15%. In such regimes the distinction is pedagogically important but practically unimportant: calibrating β from data absorbs the difference. In regimes where $\rho\tau_c$ exceeds 0.5, the exponential formula is necessary for quantitative accuracy, and the linear approximation should not be used.

Exercises for this section

1. **[L]** *Quantifying the exponential-versus-linear β error on epidemic predictions.* Implement the SIR model in your preferred scientific computing environment, using the baseline parameters of Chapter 6 (in particular $c_S = 20$, $c_I = 4$, $c_R = 20$ contacts per day, $\tau_R = 12$ days, $\tau_m = 7300$ days (≈ 20 years), $S(0)/N^{\mathrm{c}} = 0.999$, $I(0)/N^{\mathrm{c}} = 0.001$). Set $\rho\tau_c$ as the single quantity that the contact-and-conversion model controls and consider three regimes:

   (a) Low: $\rho\tau_c = 0.10$ (the linear approximation should be excellent).

   (b) Moderate: $\rho\tau_c = 0.30$ (the practical-rule boundary).

   (c) High: $\rho\tau_c = 0.70$ (the linear approximation should fail noticeably).

   For each regime, run two simulations: one with $\beta_{\mathrm{exact}} = 1 - e^{-\rho\tau_c}$ and one with $\beta_{\mathrm{lin}} = \rho\tau_c$. From each pair, compute (i) the relative error in $\mathcal{R}_0$, (ii) the relative error in the peak prevalence $\max_t I(t)/N^{\mathrm{c}}$, and (iii) the relative error in the size of the first wave, the fraction of the population infected by the time $I(t)/N^{\mathrm{c}}$ falls back below its starting value of 0.001 (with births and deaths, $S(\infty)$ is the endemic level, not the end of an outbreak). Tabulate the three errors for both comparisons, and produce a single overlaid trajectory plot showing the six $I(t)/N^{\mathrm{c}}$ curves of the first. Then repeat the comparison with $\mathcal{R}_0$ matched to the baseline. In each regime set c_I so that the run with β_{exact} has $\mathcal{R}_0 = c_I\beta_{\mathrm{exact}}/(\gamma_R + \nu_m) \approx 1.92$, keep that c_I for the run with β_{lin}, and report the same three errors. Comment briefly on which output quantity is most sensitive to the linearization error and why: identify whether the error grows multiplicatively through $\mathcal{R}_0$ into the peak and the first-wave size, in each comparison, or whether the nonlinear epidemic dynamics partly compensate. Compare your numerical findings to the practical rule stated in the section text and report whether the $\rho\tau_c < 0.3$ threshold is conservative, accurate, or aggressive in your simulations. (The companion repository provides reference SIR simulation code; only the assignments of β and c_I need modification to perform this study.)

1.5 Two Viewpoints of Infection Transmission

The rate at which new infections appear in a population can be derived from either of two perspectives. The two derivations produce mathematically equivalent expressions, but the parameters they place at the center of the formulation differ. We start, as the book does, from the infectious individual.

1.5.1 The Infected Viewpoint

Fix a single infectious individual and ask: at what rate does this individual generate new infections? The answer involves three factors:

- The rate at which the infectious individual has contacts: c_I contacts per unit time.
- The fraction of those contacts that are with susceptible individuals: $P_S = c_S S / C_{\text{total}}$.
- The probability that a contact with a susceptible individual results in transmission: β.

Multiplying, the per-capita rate at which an infectious individual generates new infections is

$$\alpha \;=\; c_I \beta \, P_S. \tag{1.7}$$

This is the *force from infection*. The total rate of new infections in the population is αI. Chapter 3 develops this derivation rigorously.

1.5.2 The Susceptible Viewpoint

Now fix a single susceptible individual and ask: at what rate does this individual acquire infection? The answer involves three factors:

- The rate at which the susceptible individual has contacts: c_S contacts per unit time.
- The fraction of those contacts that are with infectious individuals: $P_I = c_I I / C_{\text{total}}$, where $C_{\text{total}} = c_S S + c_I I + c_R R$ is the total contact rate in the population.
- The probability that a contact with an infectious individual results in transmission: β.

Multiplying, the per-capita rate at which susceptibles become infected is

$$\lambda \;=\; c_S \beta \, P_I. \tag{1.8}$$

This is the *force of infection*, the traditional quantity in most epidemic-modeling textbooks. The total rate of new infections in the population is λS. Chapter 2 develops this derivation rigorously.

1.5.3 Equivalence of the Two Viewpoints

The total rate of new infections must be the same regardless of the viewpoint used to compute it: the same infections are being counted either way. Using the definitions (1.8) and (1.7),

$$\lambda S \;=\; c_S \beta \cdot \frac{c_I I}{c_S S + c_I I + c_R R} \cdot S \;=\; c_I \beta \cdot \frac{c_S S}{c_S S + c_I I + c_R R} \cdot I \;=\; \alpha I.$$

The algebraic identity $\lambda S = \alpha I$, which the book calls the *detailed-balance identity*, holds as a consequence of the definition of P_S and P_I in terms of the same denominator C_{total}. This identity is the mathematical content of the claim that the two viewpoints produce equivalent dynamics.

1.5.4 Which Viewpoint to Use?

The two viewpoints are two parameterizations of one model, so the choice between them is a matter of convenience rather than correctness (Pitfall: "Force of Infection versus Force from Infection"). Convenience still matters, and the book takes the infected viewpoint as the better default. The reasons are results the book proves. The early-outbreak estimate $\hat{\alpha} = J/I$ does not involve the number of susceptibles, so it is immune to uncertainty in that number, while a relative error in the number of susceptibles passes in full to the matching estimate $\hat{\lambda} = J/S$ (Chapters 7 and 9). The reproduction number $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ displays the policy levers directly (Chapter 5). Table 1.1 summarizes when each viewpoint is the natural choice, with pointers to the chapters that develop the reasons in detail.

Table 1.1: When to use each viewpoint. The infected viewpoint is the primary formulation in this book for the reasons in the left column; the susceptible viewpoint remains the natural choice in the specific contexts in the right column.

Infected viewpoint ($\alpha = c_I\beta P_S$) is preferable when:	**Susceptible viewpoint** ($\lambda = c_S\beta P_I$) is natural when:
Fitting force parameters from early-outbreak incidence data. $\hat{\alpha} = J/I$ (where J is the observed new-infection rate, i.e., the incidence) is structurally immune to uncertainty in S: $\partial\hat{\alpha}/\partial N^{\mathrm{c}} = 0$ exactly (Chapter 7).	The data being fit are seroincidence data, which directly correspond to λ rather than α.
Computing $\mathcal{R}_0$ for policy analysis. The formula $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ features the natural policy levers transparently (Chapter 5).	Studying vaccination policies that alter S directly rather than I; the susceptible viewpoint places the intervention parameter on the natural compartment.
Generalizing to models with multiple infectious compartments. The next-generation matrix method (Chapter 5) is naturally stated from the infected viewpoint.	Comparing with the classical literature (Anderson-May, Keeling-Rohani, Martcheva), where the susceptible-viewpoint formulation is the conventional starting point.
Connecting with stochastic and individual-based models. Branching-process approximations and agent-based simulations natively track infectious individuals as sources of new infections, aligning with α rather than λ (Chapter 3, Section 3.5).	Working from an exposure or seroprevalence dataset where the natural quantity is the per-susceptible hazard λ rather than the per-infected generation rate α.

The book's preference for the infected viewpoint is a pedagogical and practical choice, not a mathematical one; both formulations produce the same dynamics.

A reader comfortable with the orientation provided here can move directly to Chapter 4, where the two forces are assembled into the full SIR model. The subscript I in SIR denotes the infected-viewpoint formulation, not an additional compartment: the model has the same three compartments (S, I, R) as the classical SIR model, only the parameterization changes. The companion susceptible-

viewpoint formulation is denoted SIR_S for the same reason; both formulations describe the same three-compartment system and produce identical trajectories under matched parameters.

Exercises for this section

1. [**C**] State in one sentence each the meaning of the force of infection λ and the force from infection α. Which quantity is measured in units of "probability per susceptible per unit time," and which in "probability per infectious per unit time"?

2. [**C**] Explain in plain English why the identity $\lambda S = \alpha I$ is a conservation law. Your answer should make clear that both sides count the same thing (new infections per unit time), viewed from two different perspectives.

1.6 Preview of the Model Equations

Before this chapter closes, we assemble the ingredients introduced above into the compartmental model that the rest of the book will develop, analyze, and apply. The reader should view this preview as an orientation; every quantity and equation appearing here is treated in depth in the following chapters.

The rest of Part I and all of Part II develop the SIR model

$$\dot{S} = \nu_m N^{\mathrm{c}} - \alpha I - \nu_m S, \tag{1.9a}$$

$$\dot{I} = \alpha I - \gamma_R I - \nu_m I, \tag{1.9b}$$

$$\dot{R} = \gamma_R I - \nu_m R, \tag{1.9c}$$

where $\alpha = c_I \beta P_S$ is the force from infection (the per-infected-individual rate of generating new infections, introduced in Section 1.5), $\gamma_R = 1/\tau_R$ is the recovery rate (the reciprocal of the mean residence time τ_R in the infectious compartment, introduced in Chapter 4), and ν_m is the demographic turnover rate (the inverse of the mean lifetime in the population, introduced in Section 1.1). The first term in the $\dot{S}$ equation, $\nu_m N^{\mathrm{c}}$, represents the inflow of new susceptibles by migration; the remaining terms capture infection and migration outflow.

This chapter has introduced each of the quantities on the right-hand side of (1.9) at a conceptual level. Chapter 3 develops the derivation of α in rigorous detail, and Chapter 2 does the same for λ and the companion contact-level quantities. Chapter 4 then derives the full system (1.9) from first principles.

1.7 Key Takeaways

- A compartmental model partitions the population into disjoint groups (susceptible, infectious, recovered) that share epidemiological status; the state variables $S(t), I(t), R(t)$ count the population in each.

- Each compartment carries its own contact rate c_S, c_I, c_R, reflecting the possibility that infectious individuals behave differently from susceptibles (symptom-driven withdrawal) or that recovered individuals resume baseline activity.

- The rate of new infections can be expressed from two equivalent viewpoints: the infected viewpoint gives αI, where $\alpha = c_I \beta P_S$ is the force from infection; the susceptible viewpoint gives λS, where $\lambda = c_S \beta P_I$ is the force of infection. The two representations agree identically: $\lambda S = \alpha I$. The book takes the infected viewpoint as the better default.
- The equal-contact-rate special case $c_S = c_I = c_R = c$ reduces the two forces to $\alpha = c\beta S/N$ and the classical $\lambda = c\beta I/N$, but the general case in which contact rates differ is biologically necessary and analytically consequential.
- The compartmental structure, contact rates, and forces developed here are the building blocks for the SIR model whose derivation, analysis, and applications occupy the remainder of the book.

Carry-forward. Chapter 3 develops the formal derivation of α from the perspective of a single infectious individual, and sets out five reasons the infected viewpoint is the better default. Chapter 2 develops the companion derivation of λ, starting with a single susceptible individual as the unit of analysis.

Exercises

Five section-level exercises appear above, following Section 1.4 and Section 1.5. The exercises below integrate across the chapter and have been ordered to interleave a midpoint interpretation exercise with the chapter's derivations (this chapter is derivation-heavy, so perfect alternation is not possible). Each is tagged by type: ***C*** *= Concept check (short, conceptual),* ***D*** *= Derivation (analytical),* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab (requires simulation or fitting code). This scheme is used throughout the remainder of the book; the companion repository contains reference solutions and starter code for the L-tagged exercises.*

1. **[D]** Show by direct algebraic manipulation that $\lambda S = \alpha I$ holds identically as a function of (S, I, R) and the parameters (c_S, c_I, c_R, β).
2. **[D]** Consider an infection for which $c_I = c_S/2$ (infectious individuals halve their contact rate), and compare the force from infection α with the force of infection λ at the time when $S = I$. Which is larger, and by what factor?
3. **[I]** Consider two outbreak-response scenarios that differ only in the choice of susceptible population: (a) a compact outbreak affecting a population of 5×10^4 individuals in close contact with the index cases, and (b) the same outbreak modeled at the level of a 1.5×10^6-person metropolitan area. For a fixed observed incidence J and prevalence I, compute how $\hat{\alpha} = J/I$ and $\hat{\lambda} = J/S$ differ between the two scenarios. Which estimator is invariant to the choice of susceptible population?
4. **[D]** In the equal-contact-rate case $c_S = c_I = c_R = c$, show that $P_S = S/N$ and $P_I = I/N$, reducing the force-of-infection expression to the classical $\lambda = c\beta I/N$.
5. **[D]** Verify that at the disease-free equilibrium ($I = 0$, $R = 0$, $S = N^{\mathrm{c}}$), $\alpha = c_I \beta$ and $\lambda = 0$. Explain why the identity $\lambda S = \alpha I$ remains trivially satisfied (both sides equal zero).
6. **[D]** A population has $S = 10^6$ susceptibles and $I = 10^3$ infectious individuals, with contact rates $c_S = c_I = 10$ per day. Compute the ratio α/λ at this time, using the identity from

Exercise 1 above. How does this ratio change as the outbreak progresses and I grows toward $S/100$?

7. **[I]** *Spaced retrieval, the Preface.* Recall the invasion-burden distinction introduced in the Preface (Figure 2): invasion is governed by what happens to a small infectious population in an essentially-fully-susceptible host population, while burden is governed by the equilibrium reached after the outbreak has spread. The distinction was framed there at the level of biological intuition. Now restate it in terms of the two quantities developed in this chapter:

   (a) Which of α and λ is more naturally interpreted as an *invasion* quantity (the rate per infectious individual at which they generate new infections)? Neither is the *aggregate* quantity: the rate of new infections summed across the population, in (people)/(time), is $\lambda S = \alpha I$. State the units of λ and say what it measures instead.

   (b) At the disease-free equilibrium, the force from infection $\alpha = c_I \beta P_S$ involves the susceptible-population proxy P_S, while the force of infection $\lambda = c_S \beta P_I$ involves the infectious-population proxy P_I. State the values of P_S and P_I at the DFE, and explain in one sentence why these values make α the natural starting point for an invasion analysis.

   (c) State, in one sentence each, what specifically goes wrong when one tries to use the susceptible-viewpoint estimator $\hat{\lambda} = J/S$ in a setting where S is poorly known (the invasion problem), and what saves the infected-viewpoint estimator $\hat{\alpha} = J/I$ in the same setting.

   The exercise links the chapter's contact-and-conversion development to the invasion-burden vocabulary that organizes the rest of the book, and is the place where the two viewpoints stop being abstract bookkeeping and start being modeling tools with different operational properties.

Chapter 2

The Force of Infection: Susceptible Viewpoint

Do not add inches to pounds.

This chapter derives the force of infection $\lambda = c_S \beta P_I$ from first principles, starting with a single susceptible individual as the unit of analysis. The derivation resolves a subtle but consequential issue: the relationship between the per-contact transmission factor β and the underlying biology of transmission during individual contact events. Treating β as a purely empirical calibration parameter is adequate for most applications, but understanding its connection to contact duration clarifies when linear approximations are safe and when they fail. The chapter concludes with the derivation of λ from the stochastic contact process and the extension to variable contact durations.

A reader comfortable with the conceptual presentation of Chapter 1 may skip to Chapter 3 and then Chapter 4. This chapter and the next are provided so that the book's treatment of the force terms is self-contained.

2.1 The Susceptible Individual's Perspective

Consider a single susceptible individual in a population that includes some infectious individuals. We wish to compute the rate at which this individual acquires infection. Three quantities enter the calculation:

1. The rate at which the individual has epidemiologically relevant contacts with other people. This is the susceptible contact rate c_S, with units of contacts per unit time.

2. The fraction of those contacts that are with infectious individuals. In a well-mixed population, this fraction is the contact-weighted infectious fraction $P_I = c_I I / C_{\text{total}}$, where $C_{\text{total}} = c_S S + c_I I + c_R R$ is the contact-weighted population size.

3. The probability that a contact with an infectious individual transmits the pathogen. This is the transmission factor β.

Multiplying these three factors yields the rate at which the susceptible individual acquires infection:

$$\text{rate of infection for this susceptible} \;=\; c_S \beta P_I \;=\; \lambda.$$

This expression is the force of infection, the rate quantity that governs the dynamics of the susceptible compartment:

$$\frac{dS}{dt} = \cdots - \lambda S = \cdots - c_S \beta P_I S,$$

where the "$\cdots$" denotes the other terms in the full $\dot{S}$ equation (demographic turnover, for example), which contribute separately; the infection term itself is $-\lambda S$.

Exercises for this section

1. **[C]** The force of infection is $\lambda = c_S \beta P_I$, with units of $\mathtt{T}^{-1}$ (new infections per susceptible individual per unit time). The susceptible contact rate c_S appears explicitly. Explain why λ also depends on the infectious contact rate c_I, even though c_I does not appear explicitly in the formula. (Hint: the contact-weighted infectious fraction $P_I = c_I I / C_{\text{total}}$ is itself a function of c_I.) Distinguish the explicit dependence on c_S from the implicit dependence on c_I, and explain why the explicit factor is the one tied to the susceptible viewpoint.

Check Your Work: The Three Factors of λ

Verifies: equations and model assumptions. Before the next box, answer without looking back.

1. Give the units of c_S, β and P_I, and of their product. Two of the factors are dimensionless and only one carries time, so the product is $\mathtt{T}^{-1}$ (Section 2.1).
2. Write P_I in terms of c_I, I and C_{total}, and say what it becomes when $c_S = c_I = c_R$. Writing $P_I = I/N$ without that condition assumes equal contact rates; it does not derive them.
3. Say which of β and β_{day} is bounded by 1, and which could be reported as "0.3 per day" (Section 2.3).

An uncertain answer marks the gap a fluent but wrong AI derivation slips through, because only the units betray it.

Verify the AI: Dropped Subscripts and the β-vs-βc Conflation

In $\lambda = c_S \beta P_I$, the contact rate c_S is contacts per unit time, while β and P_I are dimensionless, so λ is a rate per susceptible. AI expositions often make two errors here. The first drops the subscript and writes c, which hides the well-mixed assumption $c_S = c_I$; with unequal contact rates the unsubscripted form is wrong. The second blurs β with βc. A value of 0.3 *per contact* is a probability, while 0.3 *per day* is already the effective rate βc; putting the second in place of β in $\lambda = c \beta P_I$ counts the contact rate twice.
Prompt: "What are the units of β in your formula, and of λ? Show that they cancel correctly." A correct answer makes β dimensionless and λ a per-individual rate, and a λ with the wrong units flags the error (Pitfall: "Dimensional Consistency").

2.2 The Force of Infection from the Stochastic Contact Process

We now derive $\lambda = c_S \beta P_I$ from a stochastic model of the contact process. The derivation is not strictly necessary for the deterministic ODE models of this book, but it clarifies which assumptions

enter the force of infection and which are loose. Readers uninterested in the derivation may skip this section without loss of continuity.

The contact process. Fix a single susceptible individual. The individual has contacts with other people that occur as a Poisson process with rate c_S (contacts per unit time). This is a standard assumption in epidemic modeling and is approximately valid for populations in which contacts are numerous and no single contact dominates the dynamics.

Each contact is, with probability P_I, with an infectious individual; with probability P_S, with a susceptible; with probability P_R, with a recovered individual. Only contacts with infectious individuals can result in transmission to our focal susceptible.

The effective infection process. The contacts with infectious individuals form a thinned Poisson process with rate $c_S P_I$ (Appendix E reviews the Poisson process and the thinning property at an elementary level). Each such contact transmits the infection to the focal susceptible with probability β, independently of other contacts. The times at which transmission occurs therefore form a further-thinned Poisson process with rate

$$\lambda \;=\; c_S \beta P_I. \tag{2.1}$$

This is the force of infection. Its interpretation: λ is the rate of an exponential distribution, and the time until the focal susceptible acquires infection is exponentially distributed with mean $1/\lambda$. As the rate of a Poisson process, λ has units of inverse time ($\mathtt{T}^{-1}$), a per-susceptible infection rate.

The deterministic limit. In a deterministic model, we replace the stochastic quantity (time to first infection for an individual) with its expected rate (λ) and treat the susceptible compartment as a continuous quantity. The rate of change of the susceptible compartment due to infection is then

$$\frac{dS}{dt} \;=\; \cdots - \lambda S \;=\; \cdots - c_S \beta P_I S.$$

This is the standard deterministic expression, valid when S is large enough that the law of large numbers applies and fluctuations around the mean can be ignored.

Implicit assumptions. The derivation above relies on several assumptions that should be noted explicitly:

- *Poisson contacts.* Contacts are assumed to occur at rate c_S as a Poisson process. In reality, contacts are correlated in time (repeated contacts with the same person, cluster-structured contacts at home or at school); the Poisson assumption is an approximation that works well when the population is large and no single contact dominates.

- *Independent transmission events.* Each contact is assumed to transmit independently of others. For a homogeneous population this is harmless, but in a structured population, repeated contacts with the same partner can have correlated outcomes that the force of infection does not capture.

- *Well-mixed population.* The fraction P_I seen by the focal susceptible individual equals the population-level fraction of contacts that are with infectious individuals.

These assumptions are standard, and the deterministic force of infection $\lambda = c_S \beta P_I$ is robust to mild violations of them. Chapter 10 discusses situations where more careful modeling of contact structure is warranted, and Chapter 15 treats one such extension (variable time-since-infection distributions).

AI Learning Prompt: Thinning Derived Analytically and Checked by Simulation

Ask the AI for the derivation *and* an independent simulation, then compare; either alone can be wrong and sound right. *Prompt:* "*Take* $S = 0.90\,N^{\mathrm{c}}$, $I = 0.01\,N^{\mathrm{c}}$, $R = 0.09\,N^{\mathrm{c}}$, $c_S = 20$, $c_I = 4$, $c_R = 20$ *per day and* $\beta = 0.040$. *(1) Compute* P_I *and* λ *with units. (2) Simulate* 10^5 *susceptibles for 30 days, drawing each contact's partner in proportion to the members' contact rates. (3) Compare the simulated fraction infected with* $1 - e^{-30\lambda}$."
A correct answer gives $P_I \approx 0.00202$, $\lambda \approx 0.00161$ per day and a 30-day risk of about 4.7%, and the simulation agrees within Monte Carlo error. Using the raw prevalence 0.01 for P_I inflates the risk to about 21%, and units cannot catch it. Step (2) can, because it draws partners from the population, not from a precomputed P_I. A check that reuses the quantity under test verifies nothing.

Treating the two forces as interchangeable is tempting, since detailed balance makes them equal for the population; it is the pitfall Force of Infection versus Force from Infection (Chapter 16).

Exercises for this section

1. **[C]** In one sentence, explain why $\beta = 1 - e^{-\rho \tau_c}$ (with ρ the contact-level transmission rate of Section 1.4.1) takes an exponential rather than linear form. What assumption about the contact-level transmission process produces the exponential?

2.3 Discrete-Time Issues

Some epidemic models are formulated in discrete time rather than continuous time, with the time step typically equal to one day. In these models, the per-time-step transmission probability replaces the continuous-time rate. This section notes the appropriate formula and a common error to avoid.

Daily transmission probability. For a time step of one day, the transmission probability during the day is

$$\beta_{\mathrm{day}} \;=\; 1 - e^{-\lambda} \;=\; 1 - e^{-c_S \beta P_I},$$

where λ is the continuous-time force of infection. The right-hand side uses the continuous-time rate to compute the per-day probability. This formula is the discrete-time analogue of the continuous-time $\dot{S} = -\lambda S$. Note that β_{day} is a per-*day* probability, distinct from the per-*contact* transmission probability β inside $\lambda = c_S \beta P_I$: the former is the chance a susceptible is infected over an entire day, the latter the chance that a single contact transmits.

A common error. A frequently encountered error in discrete-time models is to compute the daily transmission probability as

$$\beta_{\mathrm{day}}^{\mathrm{wrong}} \;=\; \lambda \;=\; c_S \beta P_I.$$

This conflates the rate (with units of inverse time) with the probability (dimensionless). A probability must lie between 0 and 1, but the rate λ carries no such bound and can exceed 1, so setting $\beta_{\mathrm{day}} = \lambda$ can yield a "probability" greater than 1. This happens whenever $\lambda > 1$ per day (that is,

when the force of infection is large, as in a high-prevalence or high-contact-rate setting) rather than at any particular phase of the outbreak. The correct formula $\beta_{\text{day}} = 1 - e^{-\lambda}$ is bounded between 0 and 1, and a Taylor expansion gives $1 - e^{-\lambda} \approx \lambda$ for small λ, so the two agree in the low-force regime.

For continuous-time ODE models, which are the focus of this book, this issue does not arise: the rate λ has the correct dimensions and there is no analogous ambiguity.

2.4 Summary and Exercises

This chapter has developed the force of infection $\lambda = c_S \beta P_I$ from the perspective of a single susceptible individual. The derivation proceeded in three steps:

1. The per-contact transmission probability β is related to the underlying contact-level transmission rate ρ and contact duration τ_c by the exponential formula $\beta = 1 - e^{-\rho\tau_c}$ (derived in Section 1.4.1): a constant transmission hazard ρ acting over a contact of duration τ_c leaves probability $e^{-\rho\tau_c}$ of escaping infection, so the probability of transmission is $1 - e^{-\rho\tau_c}$. The linear approximation $\beta \approx \rho\tau_c$ is valid only when the product $\rho\tau_c$ is small.

2. The force of infection λ arises as the rate of a Poisson process formed by thinning the Poisson contact process with the fraction of contacts that are infectious (P_I) and the per-contact transmission probability (β). The deterministic force of infection $\lambda = c_S \beta P_I$ follows in the large-population limit.

3. Variable contact durations modify the effective β through an averaging over the duration distribution. For exponentially distributed durations, the effective probability is $\bar{\beta} = \rho\bar{\tau}_c/(1 + \rho\bar{\tau}_c)$, which differs from the fixed-duration formula by terms of order $(\rho\bar{\tau}_c)^2$.

Chapter 3 develops the companion derivation of the force from infection $\alpha = c_I \beta P_S$ from the perspective of a single infectious individual. Chapter 4 then assembles the full SIR model from these ingredients.

Pitfall check

The chapter's two named common pitfalls are Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Dimensional Consistency". Before moving on, take a moment to confirm the following points are clear in your mind. *What is the unit of analysis in the susceptible-viewpoint derivation, and how does that choice determine whose contact rate (c_S) appears in the force of infection? What is the dimensional structure of $\lambda = c_S \beta P_I$, and how does each factor's dimensions combine to give a rate (events per unit time)? Why must β be dimensionless, and what would go wrong in the dimensional bookkeeping of λ if it were not?* If any of these are unclear, the pitfall block in Section 2.2 or the corresponding catalog entry (accessible via the List of Common Pitfalls) is the place to revisit.

Exercises

Two section-level exercises appear above, following Section 2.1 and Section 2.2. The exercises below apply the machinery across the chapter. The chapter is entirely derivations, so no interleaving across exercise types is possible at this stage; the C/I/L variants of the chapter's material appear

in Chapter 4 and beyond. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. [**D**] For an infection with instantaneous contact-level transmission rate $\rho = 0.005$ per minute, compute the per-contact transmission probability for contact durations of 5, 15, 30, 60, and 120 minutes. At which of these durations does the linear approximation produce an error exceeding 10%?

2. [**D**] Verify the integral in Section 1.4.2 that yields $\bar{\beta} = \rho\bar{\tau}_c/(1 + \rho\bar{\tau}_c)$ for exponentially distributed contact durations.

3. [**D**] Suppose a population has two classes of contacts: half are 5-minute encounters and half are 60-minute encounters. With $\rho = 0.02$ per minute, compute the effective per-contact transmission probability averaged over these two types.

4. [**D**] In a discrete-time model with time step of one day, suppose the continuous-time force of infection is $\lambda = 0.3$ per day. Compute the correct daily transmission probability using $\beta_{\text{day}} = 1 - e^{-\lambda}$. By what percentage does this differ from the naive $\beta_{\text{day}} = \lambda$?

5. [**D**] The SIR$_S$ model with force of infection $\lambda = c_S\beta P_I$ has $\mathcal{R}_0 = c_S\beta/(\gamma_R + \nu_m)$ in the equal-contact case. Show that this coincides with the infected-viewpoint result $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ when $c_S = c_I$, consistent with the equivalence result of Chapter 4.

6. [**C**] *Dimensional verification of the* SIR$_S$ *infection equation.* The SIR$_S$ infection equation in the force-of-infection formulation is

$$\dot{S} \;=\; -\lambda S \;+\; \nu_m N^{\text{c}} \;-\; \nu_m S, \qquad \lambda \;=\; c_S\,\beta\,P_I,$$

   where S is the susceptible compartment count, N^{c} is the carrying capacity, ν_m is the demographic-turnover rate, and the symbols on the right of the λ definition follow the chapter's conventions.

   (a) State the units of each symbol on the right-hand side: λ, S, ν_m, N^{c}, c_S, β, P_I. Use the conventions from Appendix F (Notation and Conventions).

   (b) Compute the units of λ from your answers in (a), and confirm that λ has units of $\mathtt{T}^{-1}$ as claimed in Section 2.2.

   (c) Compute the units of each of the three terms on the right-hand side of $\dot{S}$. Show that all three terms have identical units, and that those units equal the units of $\dot{S}$ (a population count divided by time).

   (d) State in one sentence the diagnostic question this exercise has just answered, and identify the pitfall from the Preface that the question operationalizes.

7. [**I**] *A dimensional error in the wild.* A student writes the SIR infection equation as

$$\dot{S} \;=\; -c\,\beta\,S\,I \;-\; \nu_m S,$$

   using a single (unsubscripted) contact rate c and the absolute infectious count I rather than a fraction.

   (a) Compute the units of each term on the right-hand side, using the chapter's conventions for c, β, and the compartment counts.

(b) Identify the dimensional inconsistency: which term has units that do not match the units of $\dot{S}$, and by what factor (with units) is it off?

(c) State two corrections that restore dimensional consistency, and explain the modeling difference between them: (i) replacing I by I/N in the bilinear term (so the formula becomes $-c\beta S \cdot I/N$, with $I/N = P_I$ identifiable as the infectious fraction in the well-mixed case), and (ii) replacing β by β/N to absorb the population scale into a redefined transmission probability (so the formula becomes $-c(\beta/N)SI$). Which correction does this book prefer, and why? (Hint: Section 1.4.1 on per-contact probabilities and Section 2.2 on the force-of-infection derivation are both relevant; the answer connects to the AI-audit principle that derived parameters should not absorb population scales.)

(d) Estimate the magnitude of the error introduced by the original (uncorrected) equation if $N \approx 10^7$. What does this imply about the diagnostic value of dimensional analysis as the *first* check applied at every model definition?

8. **[Open Project]** *Force of infection for a structurally richer contact model.* A multi-week capstone project deriving the chapter's force-of-infection expression $\lambda = c_S \beta P_I$ for one of several structurally richer settings (heterogeneous contact rates, age-stratified mixing, network-based contact, or others) from first principles using the chapter's stochastic-Poisson framework as the starting point. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈20–30 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch03-force-of-infection-extended/`.

Chapter 3

The Force from Infection: Infected Viewpoint

The data come from one side of the contact. The model should be written from that side.

Chapter 2 derived the force of infection $\lambda = c_S \beta P_I$ from the perspective of a single susceptible individual. This chapter develops the companion derivation: the *force from infection* $\alpha = c_I \beta P_S$, derived from the perspective of a single infectious individual. The two quantities are mathematically equivalent in the sense that both produce the same population-level rate of new infections. They differ in which parameter is the natural object of fitting and analysis.

The contact-duration and per-contact-transmission-probability material of Section 1.4.1 of Chapter 1 applies here without modification. Rather than repeat that derivation, this chapter cross-references it and focuses on what is structurally different about the infected viewpoint: the direction in which the contact process is unfolded and the connection to the basic reproductive number. Section 3.2 gives the rigorous derivation of α. Section 3.3 previews the connection to $\mathcal{R}_0$, which is developed fully in Chapter 5. Section 3.4 proves the algebraic equivalence of αI and λS in formal detail. Section 3.5 gives the extended case for treating the infected viewpoint as primary in this book.

3.1 The Infectious Individual's Perspective

Consider a single infectious individual in a population that includes susceptible, infectious, and recovered individuals. We wish to compute the rate at which this individual generates new infections in the susceptible population. The ingredients are:

1. The rate at which the infectious individual has contacts with others: c_I contacts per unit time.

2. The fraction of those contacts that are with susceptible individuals: $P_S = c_S S / C_{\text{total}}$, where $C_{\text{total}} = c_S S + c_I I + c_R R$.

3. The probability that a contact with a susceptible individual transmits the pathogen: β.

Multiplying these three factors gives the rate at which the infectious individual generates new infections:

$$\text{rate of infection generation by this infectious individual} \;=\; c_I \beta P_S \;=\; \alpha.$$

This is the force from infection. The total rate of new infections in the population is

$$\frac{dI}{dt} \;=\; \cdots + \alpha I \;=\; \cdots + c_I \beta P_S I,$$

where the "$\cdots$" denotes the other terms in the full $\dot{I}$ equation (recovery and migration), and the infection-generation term itself is $+\alpha I$.

The symmetry with the susceptible viewpoint. Compare the three factors above with the three factors of Chapter 2:

Susceptible viewpoint (λ)	**Infected viewpoint** (α)
c_S (contacts of susceptible)	c_I (contacts of infectious)
P_I (fraction infectious)	P_S (fraction susceptible)
β (per-contact transmission)	β (per-contact transmission)

The first two rows are swapped between the two viewpoints; the third row (transmission probability per contact) is identical. This symmetry is the mathematical content of the equivalence between the two formulations. Section 3.4 makes the equivalence rigorous.

Exercises for this section

1. **[C]** Explain in plain English why the expression $\alpha = c_I \beta P_S$ depends on the *susceptible* fraction P_S rather than on the infectious fraction. Relate your answer to the underlying counting unit (one infectious individual, transmitting to contacts).

3.2 Derivation of the Force from Infection

We now derive $\alpha = c_I \beta P_S$ from a stochastic model of the contact process, paralleling the derivation of λ in Section 2.2.

The infectious individual's contact process. Fix a single infectious individual. Contacts with other individuals occur as a Poisson process with rate c_I. Each contact is, with probability P_S, with a susceptible individual; with probability P_I, with another infectious individual; with probability P_R, with a recovered individual. Only contacts with susceptible individuals can result in new infections generated by our focal infectious person.

The effective generation process. The contacts with susceptible individuals form a thinned Poisson process with rate $c_I P_S$ (Appendix E). Each such contact transmits the infection to the susceptible partner with probability β, independently of other contacts. The times at which our focal infectious individual generates new infections therefore form a further-thinned Poisson process with rate

$$\alpha \;=\; c_I P_S \beta \;=\; c_I \beta P_S. \tag{3.1}$$

This is the force from infection. Its interpretation: α is the rate of an exponential waiting-time distribution for the time between successive new infections generated by a single infectious individual, assuming the population composition (and hence P_S) remains fixed over the interval considered.

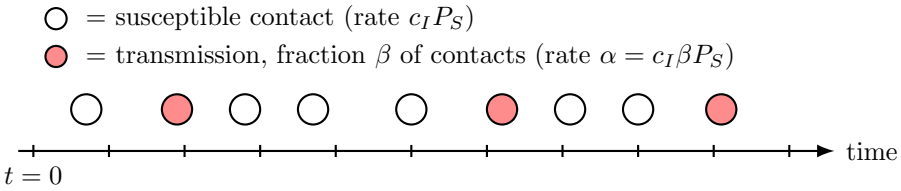


Figure 3.1: Illustration of the two-stage thinning that gives the force from infection α. A single infectious individual encounters susceptible contacts as a Poisson process with rate $c_I P_S$ (all open and shaded circles). Each such contact transmits the infection with probability β independently of the others (shaded circles only). The shaded events are a further-thinned Poisson process with rate $\alpha = c_I \beta P_S$, and form the renewal times at which our focal infectious individual generates new infections.

The population-level rate. If there are I infectious individuals, each generating new infections at rate α, the total rate of new infections in the population is

$$J(t) \;\coloneqq\; \text{population-level new-infection rate} \;=\; \alpha I \;=\; c_I \beta P_S I. \tag{3.2}$$

We use the symbol $J(t)$ throughout the book for this population-level quantity. This is the rate at which the infectious compartment grows due to new infections (before accounting for recovery and migration outflows; see Chapter 4 for the full equation). For the single-infected-compartment SIR model, $J = \alpha I$ recovers the per-source flux directly, and the natural fitting quantity $\hat{\alpha} = J/I$ inverts this relation (throughout the book, a hat denotes a quantity estimated from data, usually by fitting). In multi-compartment, bipartite, and renewal-equation extensions (Chapters 10, 11, 12, 13, 15, and the disease-specific Chapters 17–19), $J(t)$ remains the neutral population-level incidence (the rate at which new infections enter the model, summed across all source-target pairs). Reserving the symbol α for the single-source-and-target-coincident base model and using $J(t)$ as the population-level incidence avoids the source-vs-target ambiguity that arises when force-from-infection notation is applied to multi-compartment systems.

J is a flow, not a compartment. The incidence $J(t)$ has the dimensions of a rate (new infections per unit time) and is therefore dimensionally distinct from the compartment sizes S, I, and R, which count individuals. The visual parallel between the capitals I and J can mislead a reader into treating J as another compartment: I is a *stock* (the number of individuals currently infectious), whereas J is a *flow* (the rate at which new infections are produced). A quick dimensional check catches the error: any equation that adds J to a compartment size, or sets J equal to I, is malformed; the two are linked only through a rate, $J = \alpha I$. The same relation carries a second, equally useful distinction: αI is the *model's* expression for the incidence, while J is the *observable* incidence one measures as a new-case count. The fitting identity $\hat{\alpha} = J/I$ (Chapter 7) is precisely the bridge that recovers the model rate α from the measured flow J and the measured prevalence I, which is why the incidence is given its own symbol rather than written as αI throughout.

Implicit assumptions. The same three assumptions noted in Section 2.2 for the susceptible-viewpoint derivation apply here:

- *Poisson contacts for the infectious individual.* Contacts are assumed to occur at rate c_I as a Poisson process. This is the dual of the susceptible-side assumption.

- *Independent transmissions.* Each contact transmits independently of others.
- *Well-mixed population.* The fraction P_S seen by the focal infectious individual equals the population-level fraction of contacts that are with susceptibles.

These assumptions are the standard ones for homogeneous-mixing deterministic epidemic models. Chapter 15 and the heterogeneous-population chapters (11 through 13) relax the well-mixed assumption in specific ways.

3.3 The Force from Infection and the Basic Reproductive Number

The infected viewpoint gives a clean individual-tracking derivation of the basic reproductive number. Chapter 5 develops $\mathcal{R}_0$ fully; the following preview shows how naturally the formula emerges from the force-from-infection framework.

A single infectious individual introduced into an otherwise fully susceptible population ($P_S \approx 1$) generates new infections at rate $\alpha \approx c_I\beta$ per unit time. The individual remains infectious for an expected time $\tau_R^* = 1/(\gamma_R + \nu_m)$ before exiting by recovery or migration. The star marks the migration-adjusted infectious period: it is shorter than the pure infectious period $\tau_R = 1/\gamma_R$ because an individual may also leave the infectious compartment by migration. The product of the two gives the expected number of secondary infections:

$$\mathcal{R}_0 \;=\; c_I\beta\,\tau_R^* \;=\; \frac{c_I\beta}{\gamma_R + \nu_m}. \tag{3.3}$$

This is the formula that Chapter 5 derives rigorously by the next-generation-matrix construction. The infected-viewpoint framing makes the computation a single line of reasoning: follow a single infectious individual through its infectious period and count offspring. The susceptible-viewpoint derivation of the same formula is less direct because one must first compute the total infection rate and then relate it back to per-capita generation.

The susceptible-viewpoint route, for comparison. To see concretely why the same number is harder to reach from the susceptible side, derive it that way. The susceptible viewpoint supplies the *population-level* new-infection rate $\lambda S = c_S\beta P_I S$, not a per-infective count, so the contribution of a single infectious individual must be isolated first. Introduce infection at trace level into an otherwise susceptible population ($S \approx N^{\mathrm{c}}$, $I \ll N^{\mathrm{c}}$, $R = 0$). The infectious presence fixes the contact-weighted infectious fraction at

$$P_I \;=\; \frac{c_I I}{c_S S + c_I I + c_R R} \;\approx\; \frac{c_I}{c_S}\,I,$$

so the population infection rate becomes

$$\lambda S \;=\; c_S\beta P_I S \;\approx\; c_S\beta\cdot\frac{c_I}{c_S}\,I \;=\; c_I\beta\,I.$$

The susceptible contact rate c_S cancels, and the population rate is linear in I: each infectious individual produces new infections at rate $c_I\beta$, the same quantity the infected viewpoint writes down immediately as α. Multiplying by the mean infectious period $\tau_R^* = 1/(\gamma_R + \nu_m)$ recovers

$\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$. The extra steps (forming the population rate, evaluating P_I, and watching c_S cancel) are exactly what the infected viewpoint bypasses by counting the offspring of one infectious individual directly.

When $P_S < 1$, after some depletion of the susceptible pool, the same calculation applied to a newly-introduced individual gives the effective reproductive number

$$\mathcal{R}_e(t) \;=\; \mathcal{R}_0\, P_S(t),$$

which governs whether the epidemic grows or declines at time t. Chapter 5, Section 5.1 develops $\mathcal{R}_e$ as part of the formal $\mathcal{R}_0$ derivation.

Exercises for this section

1. **[D]** From the force-from-infection derivation, show that the mean waiting time between successive new infections generated by a single infectious individual (at a fixed population composition) is $1/\alpha$. For the baseline parameter set at the moment of introduction, compute this mean waiting time. Interpret.

3.4 Formal Equivalence of the Two Viewpoints

Chapter 1 stated the algebraic identity $\alpha I = \lambda S$ as a consequence of the definitions of P_S and P_I. This section gives the formal proof and explores its implications.

Proposition 3.1 (Equivalence of the two forces)**.** *Let $S, I, R \geq 0$ with $C_{total} = c_S S + c_I I + c_R R > 0$. Define*

$$P_S \;=\; \frac{c_S S}{C_{total}}, \qquad P_I \;=\; \frac{c_I I}{C_{total}},$$

and the two forces

$$\lambda \;=\; c_S \beta P_I, \qquad \alpha \;=\; c_I \beta P_S.$$

Then

$$\lambda S \;=\; \alpha I. \tag{3.4}$$

Proof. From the definitions,

$$\lambda S \;=\; c_S\beta \cdot \frac{c_I I}{C_{\text{total}}} \cdot S \;=\; \frac{c_S c_I \beta\, SI}{C_{\text{total}}},$$
$$\alpha I \;=\; c_I\beta \cdot \frac{c_S S}{C_{\text{total}}} \cdot I \;=\; \frac{c_I c_S \beta\, SI}{C_{\text{total}}}.$$

The two expressions are identical by commutativity of multiplication. □

Interpretation. The identity $\lambda S = \alpha I$ says that the rate of new infections counted from the susceptible side is the same as the rate counted from the infectious side. This is the mathematical expression of a conservation principle: a transmission event is a single event that is counted once, regardless of which side of the contact records it.

A subtle point: which quantity is constrained? The identity $\lambda S = \alpha I$ establishes that the two forces are related, but it does not imply $\lambda = \alpha$. In general, λ and α have different numerical values even at the same time point.

For example, at the disease-free equilibrium, $I = 0$, so $\alpha I = 0$; but α itself is well-defined with $\alpha = c_I\beta$ (since $P_S = 1$). Meanwhile $\lambda = c_S\beta \cdot 0 = 0$ (since $P_I = 0$). The identity $\lambda S = \alpha I$ holds trivially ($0 = 0$) because both sides vanish, but λ and α themselves are unequal: $\alpha = c_I\beta$ while $\lambda = 0$.

More generally, the ratio $\alpha/\lambda = S/I$ is the ratio of susceptible-to-infectious populations. Early in an outbreak when S/I is large, α is much larger than λ. At endemic equilibrium, $\alpha/\lambda = S^*/I^*$ is still positive but finite. This asymmetric behavior is a feature of the two forces, and it underlies the difference in parameter-estimation sensitivity developed in Chapter 7; the sensitivity-analysis paper of Hyman [2026d] gives the full treatment of the asymmetry and its consequences for parameter identifiability.

How the two forces move as an outbreak unfolds. The gap in magnitude is only part of the story. The two forces also behave very differently over time, and that difference—not the difference in size—is what matters for fitting. Because $\alpha = c_I\beta P_S$ depends on the state only through P_S, it holds essentially fixed for as long as the susceptible pool is close to intact, and starts to fall only once susceptible depletion becomes appreciable. Because $\lambda = c_S\beta P_I$ depends on the state only through P_I, it inherits the growth of $I(t)$ directly and rises with it. Figure 3.2 shows both for the baseline outbreak simulated in Chapter 6: across the early-growth window, α moves by about three percent while λ grows roughly twenty-five-fold. The identity $\lambda S = \alpha I$ holds at every instant in that figure; the two curves describe the same infections, counted from opposite sides.

This is where the asymmetry becomes practical. A modeler working in the infected viewpoint is estimating a quantity that early data see as nearly constant, so a plateau in $\hat{\alpha}(t)$ is a feature that can be read directly off the data—and Chapter 7 builds a diagnostic on exactly that plateau. A modeler working in the susceptible viewpoint is estimating a quantity that changes by more than an order of magnitude over the same window, and has no comparable plateau to look for. The infected viewpoint is not merely more convenient here; it puts the near-constant quantity, rather than the fast-varying one, in the place where the estimator needs stability.

Detailed balance gives the two forces the same population rate of new infections but not the same estimation problem, and that difference is the pitfall Force of Infection versus Force from Infection (Chapter 16).

3.5 The Infected Viewpoint as Primary Formulation: Operational Consequences

Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Linearization Fidelity" motivate this book's preference for the infected-viewpoint formulation as the primary one in chapters where it offers an analytical advantage. Five operational consequences make the case.

First (force-of-vs-from and identifiability, fitting under uncertainty in S): Fitted estimates of the force parameter are insensitive to uncertainty in the susceptible population S. In the infected viewpoint, the estimator $\hat{\alpha} = J/I$ is the ratio of incidence to prevalence, and so involves no assumption about S; in the susceptible viewpoint, $\hat{\lambda} = J/S$ scales inversely with the assumed S and inherits its full uncertainty. Early in an outbreak S is the most uncertain quantity in the model, so an estimator that does not depend on it removes a real source of error in outbreak response.

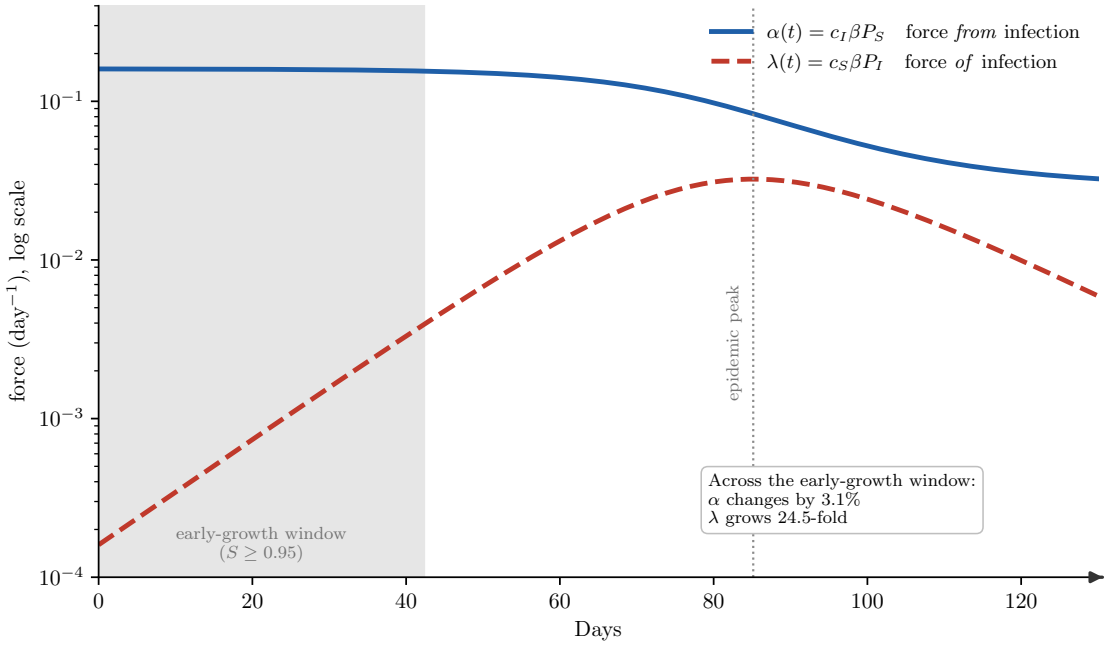


Figure 3.2: The two forces over time in the baseline outbreak of Chapter 6 (Scenario 1: $c_S = 20$, $c_I = 4$, $c_R = 20$ contacts day^{-1}, $\beta = 0.040$, $\tau_R = 12$ d, $\tau_m = 7300$ d, $S(0) = 0.999$, $I(0) = 0.001$, giving $\mathcal{R}_0 \approx 1.92$). The vertical axis is logarithmic. The force from infection $\alpha = c_I\beta P_S$ follows the susceptible fraction and stays nearly flat while that fraction is intact; the force of infection $\lambda = c_S\beta P_I$ follows the infectious fraction and grows with it. The shaded band marks the early-growth window in which S has not yet fallen below 0.95, the same threshold used as a diagnostic in Chapter 7. Both forces generate identical new-infection rates at every instant, since $\alpha I = \lambda S$; what differs is how much each one has to move to do so.

Second (force-of-vs-from, intervention-target interpretation): In the infected viewpoint the basic reproductive number takes the form $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$, with the infectious-contact rate c_I appearing explicitly as a policy lever. This makes the infected-viewpoint formulation the more natural one for intervention analysis, since the parameters a public-health measure acts on appear directly in the expression.

Third (generalizability to richer models): The next-generation matrix method (Chapter 5, Section 5.3) is naturally stated in the infected viewpoint and generalizes without conceptual modification to models with multiple infectious compartments and to heterogeneous mixing (Pitfall: "Heterogeneous-Mixing Balance Conditions"). This makes the infected viewpoint the more extensible formulation as the models grow in structure.

Fourth (cross-cutting, consistency with stochastic and computational extensions): Individual-based simulations and branching-process approximations track infectious agents as sources of new infections, making the infected viewpoint the native framing for stochastic and computational extensions of the deterministic model. This computational advantage is most pronounced for emerging epidemics, when the number of infectious individuals is small and tracking them individually is far cheaper than tracking the much larger susceptible population.

Fifth (Pitfall: "Linearization Fidelity"): a more technical reason worth stating explicitly, because it reveals a slow-fast separation that the infected viewpoint makes transparent and the susceptible viewpoint conceals. In the infected viewpoint the nonlinear incidence term is αI with

$\alpha = c_I \beta P_S$. Early in an outbreak the contact-weighted susceptible fraction P_S is close to unity and changes slowly: it is the slow variable of the system. The prevalence I, by contrast, changes exponentially fast during invasion: it is the fast variable.

Proposition 3.2 (Slow-fast structure of the infected viewpoint)**.** *In the infected-viewpoint formulation of the* SIR *model, freezing the contact-weighted susceptible fraction at its current value, $P_S \approx P_S(0)$, reduces the prevalence equation to the constant-coefficient linear approximation*

$$\dot{I} \;\approx\; (c_I \beta P_S(0) - (\gamma_R + \nu_m))\, I, \tag{3.5}$$

which is the correct linearization of the nonlinear dynamics about the disease-free equilibrium. The corresponding exponential growth rate is $\sigma = c_I \beta P_S(0) - (\gamma_R + \nu_m)$, which recovers the basic reproductive number through $\mathcal{R}_0 = 1 + \sigma\, \tau_R^*$ with $\tau_R^* = 1/(\gamma_R + \nu_m)$.

The susceptible viewpoint does not have this property. Its force of infection $\lambda = c_S \beta P_I$ depends on the fast variable P_I, so freezing λ at its disease-free value (where $P_I = 0$) gives $\dot{I} = -(\gamma_R + \nu_m) I$: pure decay, the wrong linearization. Computing the correct susceptible-viewpoint linearization requires recognizing that λ is itself a function of I and differentiating it accordingly, which reintroduces the coupling that the freeze was meant to eliminate. The infected viewpoint naturally separates fast and slow variables; the susceptible viewpoint entangles them. Chapter 6, Section 6.3 applies the frozen-P_S approximation quantitatively to demonstrate early-outbreak exponential growth, and Chapter 5, Section 5.4 connects it to the formal stability analysis of the disease-free equilibrium.

The case for the infected-viewpoint preference is developed across Chapters 7 and 9: the fitting advantage is demonstrated numerically in Chapter 7, Section 7.6 and its analytical foundation is the sensitivity-index asymmetry in Chapter 9, Section 9.3.4. None of these reasons constitute a mathematical superiority; the two formulations are equivalent. They are reasons of pedagogical clarity and practical utility. The susceptible viewpoint remains the natural choice in specific contexts identified in Chapter 1, Section 1.5.4, particularly when fitting to seroincidence data or when comparing with the classical textbook literature.

Equivalent dynamics, asymmetric estimators. The two viewpoints describe identical dynamics (Proposition 3.1), yet they are not interchangeable in practice: the infected-viewpoint fitting quantity J/I is observable without assuming a value for N^c, while the susceptible-viewpoint quantity J/S inherits any error in N^c. The equivalence and the fitting advantage are consistent: the former holds at the level of the dynamics, the latter at the level of what the data can identify.

Check Your Work: Distinguishing λ from α in Your Own Work

Verifies: equations and interpretations. Before the next box, answer without looking back.

1. Complete "*(probability of ___) per (___ individual) per (unit time)*" for λ and for α. If the two completions match, re-read Sections 2.2 and 3.2.
2. State what λS equals in terms of α. If you cannot, return to Proposition 3.1.
3. Say which of $\hat{\lambda} = J/S$ and $\hat{\alpha} = J/I$ depends on the assumed N^c. If unsure, re-read Section 3.5: the asymmetry, not the equivalence, is the chapter's operational contribution.

The next box shows an AI error that only this distinction makes visible.

Verify the AI: The Single-Symbol Conflation

The susceptible viewpoint's $\lambda = c_S\beta P_I$ is a rate per susceptible. The infected viewpoint's $\alpha = c_I\beta P_S$ is a rate per infectious individual. They meet only at the population level, $\lambda S = \alpha I$. The most consequential AI failure in the book's domain uses one symbol for all three. A typical passage:

> *"The rate of infection is $\lambda = c\beta SI/N$, equivalently the rate at which infectious individuals transmit. Each infectious individual transmits at rate λ, and each susceptible becomes infected at rate λ."*

The formula is a population rate, not a per-individual one, and its bare c hides $c_S = c_I$. The claimed equivalence holds for $\lambda S = \alpha I$ but not for the per-individual rates. *Prompt:* "*Is this rate per individual or per population? Per susceptible or per infectious?*" A correct answer separates λ, α and $\lambda S = \alpha I$. Whenever an AI introduces a transmission rate, ask: per what?

3.6 Summary and Exercises

This chapter has developed the force from infection $\alpha = c_I\beta P_S$ from the perspective of a single infectious individual. The key results are:

1. The force from infection arises naturally from the stochastic contact process of an infectious individual, via Poisson thinning: contacts occur at rate c_I, of which a fraction P_S are with susceptibles, and each susceptible contact transmits with probability β.

2. The basic reproductive number follows from the force from infection in a single step: $\mathcal{R}_0 = \alpha\,\tau_R^* = c_I\beta/(\gamma_R+\nu_m)$. The derivation is cleaner than the corresponding susceptible-viewpoint derivation.

3. The algebraic identity $\lambda S = \alpha I$ (Proposition 3.1) establishes the equivalence of the two viewpoints at the level of the dynamical equations. The two formulations produce identical trajectories.

4. The infected viewpoint is preferable to the susceptible viewpoint in this book for five reasons: fitted-parameter independence from S^*, formula transparency, compatibility with next-generation matrix methods, connection to stochastic and individual-based models, and natural slow-fast separation that yields the correct linearization through the frozen-P_S approximation. This is a pedagogical and practical preference, not a mathematical one.

Chapter 4 assembles the forces from this chapter and Chapter 2 into the full SIR model. Chapters 5 and 6 develop the analysis and numerical simulation of that model. Chapter 7 returns to the practical advantage of the infected viewpoint and quantifies it through the central comparison of Section 6.1's fitting exercises.

Pitfall check

This chapter completes the development of Pitfall: "Force of Infection versus Force from Infection" and exercises Pitfall: "Dimensional Consistency" alongside it. Before moving on, take a moment to confirm the following points are clear in your mind. *What is the formal statement of equivalence*

between the two viewpoints, and how does its single-line proof reflect the conservation principle that a transmission event is a single event regardless of which side counts it? Why are the two formulations equivalent at the population level but not interchangeable for fitting? What is the structural-immunity property of the infected-viewpoint estimator, and how does the asymmetry $\alpha/\lambda = S/I$ explain why it matters most early in an outbreak? If any of these are unclear, the pitfall block in Section 3.4 or the corresponding catalog entry (accessible via the List of Common Pitfalls) is the place to revisit.

Exercises

Two section-level exercises appear above, following Section 3.1 and Section 3.3. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[C]** The five reasons the infected viewpoint is preferable (enumerated in Section 3.6) include "natural slow-fast separation." Explain in two or three sentences what Proposition 3.2 captures about this separation, and why the frozen-P_S linearization matters for parameter fitting in Chapter 7.

2. **[D]** Verify Proposition 3.1 by direct algebraic manipulation of the definitions.

3. **[D]** Using the baseline parameter set from Table 4.4 of Chapter 4, compute α and λ at three times: at $t = 0$ (with $I(0) = 0.001$), at the epidemic peak (where $\mathcal{R}_e = 1$), and at the endemic equilibrium. Report the ratio α/λ at each of these times and interpret.

4. **[I]** Consider a policy intervention that reduces c_I by 50%. Using the infected-viewpoint expression for $\mathcal{R}_0$, compute the resulting change in $\mathcal{R}_0$. Now consider the equivalent computation from the susceptible viewpoint: express the intervention in terms of changes in λ (which changes through $P_I = c_I I/C_{\text{total}}$). Confirm that the effect on $\mathcal{R}_0$ is the same, but note how much more elaborate the susceptible-viewpoint derivation is.

5. **[D]** The branching-process approximation to an early outbreak models each infectious individual as generating offspring at rate α for a time τ_R^* (exponentially distributed), after which the individual recovers. Show that the expected offspring count is $\mathcal{R}_0$. What is the distribution of offspring counts under this model?

6. **[I]** Consider a variant of the model in which contacts are structured by an additional state variable (e.g., age). Argue informally why the force-from-infection expression $\alpha = c_I \beta P_S$ generalizes more naturally than the force-of-infection expression $\lambda = c_S \beta P_I$ in this setting. (A full treatment of structured populations is the subject of Chapter 11.)

7. **[C]** *Dimensional verification of the* SIR *infection equation.* The SIR model in the force-from-infection formulation has the susceptible-compartment equation

$$\dot{S} \;=\; -\alpha I \;+\; \nu_m N^{\mathrm{c}} \;-\; \nu_m S, \qquad \alpha \;=\; c_I \, \beta \, P_S.$$

   Note the structural difference from the SIR_S equation of Chapter 2, Exercise 6: the new-infection term here is $-\alpha I$ (a per-infectious rate times the infectious count) rather than $-\lambda S$ (a per-susceptible rate times the susceptible count), reflecting the change of viewpoint.

   (a) State the units of each symbol on the right-hand side: α, I, ν_m, N^{c}, S, c_I, β, P_S. Use the conventions from Appendix F (Notation and Conventions).

(b) Compute the units of α from your answers in (a), and confirm that α has units of $\mathtt{T}^{-1}$ as claimed in Section 3.2.

(c) Compute the units of each of the three terms on the right-hand side of $\dot{S}$. Show that all three terms have identical units, and that those units equal the units of $\dot{S}$.

(d) Verify dimensionally that the detailed-balance identity $\lambda S = \alpha I$ (Proposition 3.1) is consistent: both sides should have the same units, and those units should match the units of the bilinear infection-rate term that appears in either $\dot{S}$ equation. State in one sentence why this dimensional check, performed on the identity rather than on either viewpoint alone, supports the equivalence claim independently of the algebraic derivation.

8. **[Verify the AI]** *Symbol-overloading audit.* An AI tutor produces the following derivation of $\mathcal{R}_0$: "Each infectious individual transmits at rate β to susceptibles. Over the infectious period $1/\gamma$, the expected number of new infections is $\beta \cdot S/N \cdot 1/\gamma$. Therefore $\mathcal{R}_0 = \beta/\gamma$ when $S = N$." Audit this derivation:

(a) *Symbol identification.* The AI uses β as a rate (with units of probability per unit time) in the first sentence, but β in this book's notation is a dimensionless per-contact transmission probability (Chapter 1, Section 1.4.1). Identify the conflation: what *rate* should appear in the first sentence, and how does it relate to the book's β via the contact rate c_I?

(b) *Restoring the contact rate.* Rewrite the AI's derivation using this book's notation. Show that the corrected first sentence reads "each infectious individual transmits at rate $c_I\beta$ when $P_S = 1$" (the force from infection at the disease-free equilibrium). Confirm that this gives $\mathcal{R}_0 = c_I\beta/\gamma_R$ in the limit of pure recovery without demographic turnover ($\nu_m = 0$).

(c) *The fitting consequence.* The AI's β/γ formula is a "compressed" $\mathcal{R}_0$: it identifies the composite quantity $c_I\beta$ (rather than the factors) but hides this fact by absorbing c_I into a redefined β. Explain in one sentence why this compression makes intervention design ambiguous: an intervention that reduces contact rate (e.g., social distancing) and an intervention that reduces per-contact transmission probability (e.g., masking) act on different factors, but the AI's notation does not distinguish them.

The exercise practices the audit habit on a published-style derivation in which the symbol-overloading is invisible to a reader who has not internalized the book's contact-and-conversion decomposition.

9. **[Open Project]** *Audit an AI's exposition of the* SIR *model.* A multi-day capstone project prompting an AI system to explain the SIR model, the basic reproductive number $\mathcal{R}_0$, and the operational meaning of the force of infection, then auditing the response against the chapter's framework for every conflation, omission and definitional drift, and producing a corrected exposition. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈6–10 hours over 1–2 weeks)—is specified in the companion repository at `capstone-projects/ch04-ai-audit-sir-i/`.

Part II

The SIR Model: The Independent Parameters and Distinguishing Invasion from Burden

Chapter 4

The SIR Compartmental Model: Derivation and Parameters

Big-to-small analysis: forest, trees, branches, bark.

Part I has established the two viewpoints from which the rate of new infections can be derived. This chapter assembles the three-compartment SIR model from the infected viewpoint. The subscript records the viewpoint: SIR denotes the model written with the force from infection α, developed here, while SIR_S denotes the algebraically equivalent form written with the force of infection λ (the susceptible-viewpoint formulation, assembled in Section 4.6). By the end of the chapter, the reader will have a well-posed system of ordinary differential equations, a parameter budget realistic for fitting to data, and a canonical set of baseline values used for numerical illustration in Chapters 5 through 9. The equations assembled here carry the weight of the rest of the book: Chapter 5 analyzes their equilibria, Chapter 6 simulates them, Chapter 7 fits them to data, and every chapter in Part IV builds on their structure.

4.1 Model Assumptions

Before writing equations, we state what the model assumes about the population it describes. Each assumption is strong enough to deserve an explicit statement, and each is relaxed somewhere later in the book. When we introduce richer classes of models in Chapters 10–15, the modification typically amounts to weakening one of the five assumptions below.

1. **Homogeneous random mixing.** Within each compartment, individuals mix at random with the rest of the population according to their compartment-specific contact rate. Every susceptible individual has the same probability per unit time of contacting any given infectious individual, and so on for the other compartment pairs. No individual is more socially connected than any other of the same compartment.

   Homogeneous mixing is the strongest idealization in the model. Real populations are heterogeneous: individuals differ in their number of contacts, in the kinds of contacts they have, and in the persistence of their contact networks. Chapter 11 relaxes this assumption by splitting the population into two groups with distinct contact behaviors; Chapters 12 and 13 go further to bipartite structures. The single-population homogeneous-mixing model of this chapter remains the right starting point because it captures the qualitative behavior of an epidemic with a minimum of structural assumptions.

2. **Differential contact rates.** Susceptible, infectious, and recovered individuals make c_S, c_I, and c_R contacts per person per day, respectively. These three rates are treated as independent parameters. We typically expect $c_I \leq c_S$ because illness-related malaise and self-imposed isolation reduce the contact rate of infectious individuals. We typically expect $c_R \approx c_S$ because recovered individuals have returned to normal daily activity. The model does not require any particular ordering of these three rates, and Chapter 6 explores the consequences of varying their ratios.

   Keeping c_I as an independent parameter is essential to the book's approach. It is the natural lever for modeling behavior change: case isolation, voluntary withdrawal, quarantine, and hospitalization all act primarily by reducing c_I. In the classical susceptible-viewpoint formulation, changes in c_I enter the model only indirectly through the probability P_I that a random contact is with an infectious individual, which obscures their operational meaning. In the infected-viewpoint formulation we develop here, c_I appears directly in the basic reproductive number, making the effect of behavior change transparent.

3. **Migration with an equilibrium population.** Individuals enter the susceptible compartment at a constant rate $\nu_m N^{\mathrm{c}}$, where N^{c} is an carrying-capacity parameter and $\nu_m = 1/\tau_m$ is the per-capita migration rate. Individuals depart from the modeled population at the same per-capita rate ν_m, regardless of which compartment they occupy; this rate governs population exit, not the total rate of leaving a compartment. The parameter τ_m is the mean time an individual spends in the modeled population; it is not the same as the infectious period τ_R.

   The migration term captures several distinct physical processes under one formal device. For a model of disease in a stable geographic region, ν_m corresponds to demographic turnover: births balance deaths, and the population approaches a steady size N^{c}. For a model of disease in a seasonal population, τ_m might represent the duration of a transmission season. For a model of disease in an immigrant-dominated community, ν_m might represent the characteristic rate of arrival and departure. In all of these cases, the modeled population is open in a controlled way: individuals enter and leave, but the total size is stable in the long run.

4. **No disease-induced mortality.** We neglect deaths caused by the disease itself. Equivalently, the per-capita mortality rate of infectious individuals equals that of susceptibles and recovereds. This assumption is appropriate for infections with low case-fatality ratios, for example most respiratory infections in otherwise healthy populations. Chapter 10 extends the model to include a disease-induced mortality rate μ_d; the modification is small in practice but changes the conservation properties of the system.

5. **Permanent immunity upon recovery.** Once an individual recovers, immunity is permanent for the duration of the modeled period. The recovered compartment R is a sink for infection dynamics: individuals in R do not return to S. This assumption is appropriate for many childhood infections (measles, mumps, rubella) over short modeled periods. Chapter 10 relaxes it by introducing a waning-immunity transition $R \to S$ at rate ω, giving the SIRS family of models.

These five assumptions are strong. Nothing in the analysis that follows depends on believing they are quantitatively accurate descriptions of any real disease. The model is a mathematical object we construct to capture the qualitative dynamics of epidemic spread; assessing how closely any particular disease matches these assumptions is a separate empirical question. What we claim

is that the assumptions are self-consistent and that the resulting model is rich enough to illustrate the phenomena the book addresses.

Why ν_b and ν_m are not interchangeable. Both ν_b and ν_m are demographic rates with units of inverse time. Both can replenish the susceptible compartment. They are nevertheless not the same quantity, and conflating them is one of the recurring modeling mistakes the book teaches the reader to avoid.

The migration rate ν_m models the *net* demographic turnover of the population: in a region with stable total population, ν_m absorbs the combined effect of births, natural deaths, in-migration, and out-migration. New susceptibles enter at rate $\nu_m N^c$, and individuals exit each compartment at the per-capita rate ν_m. The single rate captures the dominant demographic flux for human populations on the timescales considered in most of the book (months to a few years).

The birth rate ν_b models a *specific* physical process: per-capita reproduction. It is appropriate when the model needs to track births as a distinct inflow, for example, when newborns can themselves be infected through vertical transmission from an infectious mother. In such a model, ν_b couples the size of the female population to the rate of newborn arrival, with potentially different downstream destinations for newborns of susceptible and infectious mothers.

For human-population models, the order of magnitude is similar but the meaning is different. A typical human birth rate is $\nu_b \approx 12$ births per 1000 people per year, or about 3×10^{-5} per day. A typical demographic-turnover rate over a 75-year mean residence time is $\nu_m \approx 4 \times 10^{-5}$ per day. The numbers can be confused; the modeling roles cannot. **In the simple human-population models that occupy most of this book, the susceptible-replenishment rate is ν_m, not ν_b.** The book uses ν_b only in vertical-transmission contexts (Chapter 14 and the HIV case study in Chapter 19) and in vector-borne models where vector birth and vector mortality are explicitly tracked (Chapter 13, Chapter 18).

Exercises for this section

1. [**I**] *Assumption-sensitivity diagnosis.* For each of the five real-world scenarios below, identify which of the five SIR model assumptions of this section is most prominently violated, briefly justify why, and name the chapter or section of this book where the assumption is relaxed. There may be more than one defensible answer for a given scenario; the exercise asks for the violation that most affects the qualitative dynamics, with reasoning.

   (a) A measles outbreak in a primary school where transmission occurs almost exclusively in classrooms during the academic year and stops when school is out.

   (b) An untreated HIV epidemic in a population whose sexual-behavior survey reveals roughly 20% of individuals reporting partnership-formation rates an order of magnitude higher than the rest of the population.

   (c) Ebola transmission during the 2014–2016 West Africa outbreak, where case-fatality among confirmed infections was approximately 40%.

   (d) Pertussis (whooping cough) in an adolescent and adult population that was vaccinated in childhood, with documented reinfections years to decades later.

   (e) Malaria transmission in a rural region where the mosquito biting rate varies seasonally with rainfall and the human population is approximately stable demographically.

For each answer, state in one sentence what model modification would be needed (without doing the modification): for example, "add a flow $R \to S$ at rate ω" or "introduce two activity classes with distinct contact rates c_H and c_L." Then for two of your five scenarios, briefly describe what bias the homogeneous-mixing SIR model would introduce if applied without modification: would it overstate or understate the early growth rate, the final size, the threshold parameter value for control, or some combination?

4.2 The Force from Infection: Recap

Chapter 3 derived the force from infection from first principles. We recap only the result and definitions here so this chapter is self-contained; readers seeking the derivation should consult Chapter 3. The force from infection is

$$\alpha \;=\; c_I\,\beta\,P_S, \tag{4.1}$$

where $P_S(t)$ is the contact-weighted susceptible fraction

$$P_S(t) \;=\; \frac{c_S\,S(t)}{C_{\text{total}}(t)}, \tag{4.2}$$

and $C_{\text{total}}(t) = c_S S(t) + c_I I(t) + c_R R(t)$. The companion force of infection is $\lambda = c_S \beta P_I$ with $P_I = c_I I / C_{\text{total}}$, and the two forces satisfy the identity

$$\alpha\, I \;=\; \lambda\, S \tag{4.3}$$

established as Proposition 3.1 of Chapter 3. The total rate of new infections in the population is $\alpha I = \lambda S$, identical in both formulations. Here P_S, P_I, C_{total}, α, and λ all vary in time as the state (S, I, R) evolves; following the notation summary, we write them without arguments except where a specific dependence is emphasized, as with the explicit forms $P_S(t)$ and $C_{\text{total}}(t)$ in (4.2). In this chapter and the rest of the book, α is the primary variable; the case for this choice is developed in Chapter 3, Section 3.5 and quantified in Chapters 7 and 9.

Exercises for this section

1. **[C]** Explain in one paragraph why the SIR and SIR$_S$ formulations produce identical trajectories despite writing the new-infection term in two different ways. Which single algebraic identity guarantees this equivalence?

4.3 Independent Parameters

The SIR model is defined by six biologically-grounded parameters plus a carrying capacity N^{c}. We call these *independent* because each represents a distinct physical quantity that can, in principle, be estimated from separate data sources: c_S, c_I, and c_R from contact-tracing surveys, β from household transmission studies, τ_R from clinical observation of recovery times, and τ_m from demographic records.[1]

[1]Throughout this book, "independent parameters" is used in the model-structure sense: each parameter appears individually in at least one model equation rather than only as a composite product, and each can in principle be measured by a distinct experimental procedure. This is not the statistical sense of independence (joint factorizability of a probability distribution); estimated parameters are routinely correlated in their joint posterior, as the fitting analysis of Chapter 7 makes explicit.

Table 4.1 lists the six independent parameters with their dimensions and interpretations. The carrying capacity N^c appears as a structural parameter: it is not estimated from disease data but taken as an input to the model, either from a census or by convention.

Table 4.1: Independent parameters of the SIR model.

Parameter	Description	Dimension
c_S	Mean contacts per susceptible individual per day	$\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1}$
c_I	Mean contacts per infectious individual per day	$\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1}$
c_R	Mean contacts per recovered individual per day	$\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1}$
β	Transmission probability per susceptible-infectious contact	$\mathtt{P}\,\mathtt{C}^{-1}$
τ_R	Mean infectious period	$\mathtt{T}$
τ_m	Mean residence time in model population	$\mathtt{T}$
N^c	Carrying capacity (equilibrium population size, absent disease)	$\mathtt{P}$

The dimensional shorthand is: `C` for contacts, `P` for people, `T` for time (in days). We use these throughout the book.

The durations τ_R and τ_m, not the rates computed from them, are the model's independent parameters; treating the rates as primary is the pitfall The Independence of τ (Chapter 16).

A remark on β is warranted. It is sometimes described loosely as "the probability of transmission," and that is exactly what it is: the probability that a single susceptible–infectious contact transmits infection. It is therefore a pure number, dimensionless, as Chapter 1 and Chapter 2 both state. The dimensions of the incidence come from the contact rate that multiplies it, not from β: c_I counts contacts per unit time, so $c_I \beta P_S$ carries dimensions of inverse time and αI is a flow of individuals per unit time. Keeping β dimensionless is what makes that bookkeeping close, and it is the convention used throughout the book.

Of the six independent parameters, τ_m is the one whose appropriate value is least obvious, because it depends on what one takes the modeled population to be. The model removes individuals from all three compartments at the per-capita rate $\nu_m = 1/\tau_m$ and replaces them with susceptibles, so τ_m is the mean time an individual remains in the population before leaving: by death, by emigration, or by aging out of the group the model describes. What this turnover represents was introduced with the model assumptions in Section 4.1; the more delicate question is what value to assign it.

It is tempting to equate τ_m with the human life expectancy, but that identification holds only for a *closed* population, one in which individuals leave solely by dying. Most modeled populations are open: people move in and out, and many models track a subgroup that individuals enter and leave on a timescale far shorter than a lifetime. A residence time of one year fits only genuinely transient settings, whereas a life expectancy of seventy or eighty years presumes a closed population, an idealization few studies actually intend. A more dependable way to fix τ_m is to ask how long the entire modeled population takes to turn over: for most stable populations that interval falls in a twenty-to-forty-year band, well short of a lifetime because migration and the steady replacement of the group, not mortality alone, set the pace. Table 4.2 gives representative values across the realistic range.

The model's assumption that individuals leave at a constant per-capita rate (a steady daily trickle in and out) can also fail outright, and when it does the reciprocal $\nu_m = 1/\tau_m$ is itself

Table 4.2: Representative values of the mean residence time τ_m by population type, with the endemic-prevalence ceiling each implies. The ceiling $I^*/N^* \approx \tau_R/\tau_m$ is the value approached as $\mathcal{R}_0$ grows large (Chapter 5); the percentages take an infectious period of about one week ($\tau_R \approx 7$ days).

Population type	Mean residence τ_m	Burden ceiling τ_R/τ_m
Transient: refugee camp, seasonal-worker settlement, single student cohort	months to ∼1 yr	∼2%
Mobile community or city with substantial net migration	∼10 yr	∼0.2%
Open national population (mortality and net migration)	∼20–40 yr	∼0.05–0.1%
Closed population (mortality only, ≈ life expectancy)	∼70–80 yr	∼0.03%

misleading. Suppose the modeled population is a single kindergarten class. The class runs for nine months, so the natural impulse is to set τ_m to nine months. But the children do not arrive and depart a few at a time each day: the whole class enters on the first day and leaves on the last. Through the school year almost no one departs, so the per-capita rate of leaving is $\nu_m \approx 0$, not the $1/\tau_m$ the reciprocal would assign: for the daily-rate model the class is effectively a *closed* population, run for nine months and then emptied all at once. The nine-month duration sets the simulation horizon, not a turnover rate. The mismatch is an instance of Pitfall: "The Independence of τ": the relation $\nu_m = 1/\tau_m$ presumes an exponentially-distributed residence time, and a cohort that enters and exits together has its residence time concentrated at a single value, so the reciprocal does not hold.

This uncertainty matters less than it might seem, because τ_m usually has little effect on the solution. The demographic rate $\nu_m = 1/\tau_m$ is far slower than the epidemic rates (for any acute infection $\tau_m \gg \tau_R$), so over the timescales a model is run it contributes negligibly to the dynamics, and the invasion threshold $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ scarcely depends on it. The sensitivity analysis of Chapter 9 makes this quantitative: even a large uncertainty in τ_m typically moves the outputs of interest only slightly, so a rough value usually suffices. The exception is the long-run endemic level, which Chapter 5 shows is approximately $I^*/N^* \approx \tau_R/\tau_m$ and therefore scales inversely with τ_m; a study aimed at endemic prevalence must pin it down more carefully than one aimed at a single outbreak. That τ_m governs the endemic burden while barely touching the invasion threshold is the burden side of the invasion–burden distinction developed in Chapter 5.

Six parameters is a lot for a three-compartment model. Part of the book's argument is that when the full parameter budget is available, the analyst must be explicit about which parameters are being fit from the data at hand (typically one or two dimensionless combinations) and which are being imported from outside data sources (the remainder). Chapter 7 develops this distinction in detail.

4.4 Block Diagram and Model Equations

We now write down the SIR model by reading flows off a block diagram. The block diagram is a compact visual representation of the conservation laws governing the compartmental populations.

4.4.1 Block Diagram

Figure 4.1 shows the flows of individuals among the three compartments. There are four kinds of flows:

- *Inflow to susceptible.* Individuals enter the modeled population at a constant rate $\nu_m N^{\mathrm{c}}$, all entering into the susceptible compartment. (A newborn or new arrival is assumed to be susceptible to the infection under study.)
- *Infection.* Susceptible individuals become infectious at total rate αI, the force from infection times the number of infectious individuals. This flow transfers one individual from S to I per unit time per infection event.
- *Recovery.* Infectious individuals recover at rate $\gamma_R I$, where $\gamma_R = 1/\tau_R$ is the per-capita recovery rate. This flow transfers individuals from I to R.
- *Outflow from each compartment.* Individuals leave each compartment at per-capita rate ν_m, representing departure from the modeled population.

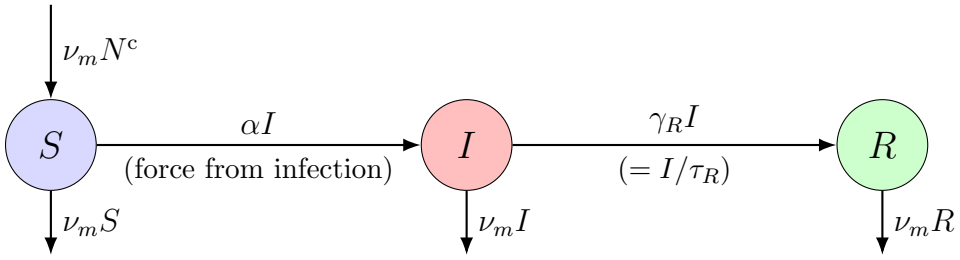


Figure 4.1: Block diagram for the SIR model. Individuals enter the susceptible compartment at constant rate $\nu_m N^{\mathrm{c}}$ and leave all three compartments at per-capita rate $\nu_m = 1/\tau_m$. The force from infection αI governs the rate at which susceptibles become infectious; recovery at rate $\gamma_R I = I/\tau_R$ transfers infectious individuals to the recovered compartment. The model equations (4.4) follow directly by reading the in-rates minus the out-rates for each compartment.

4.4.2 The Model Equations

Reading Figure 4.1, the rate of change of each compartment population equals its inflow minus its outflow. The resulting system is the SIR model:

$$\frac{dS}{dt} = \nu_m N^{\mathrm{c}} - \alpha I - \nu_m S, \tag{4.4a}$$

$$\frac{dI}{dt} = \alpha I - \gamma_R I - \nu_m I, \tag{4.4b}$$

$$\frac{dR}{dt} = \gamma_R I - \nu_m R, \tag{4.4c}$$

with α given by (4.1), P_S given by (4.2), and initial conditions $S(0) \geq 0$, $I(0) \geq 0$, $R(0) \geq 0$ (an epidemic requires $S(0) > 0$, while $S = 0$ lies on an inflowing face of the invariant region Ω:

there $dS/dt = \nu_m N^c > 0$, so a trajectory starting with $S = 0$ immediately enters the interior, as established in Section 4.4.4).

The infectious-compartment equation (4.4b) can be rewritten in a suggestive form:

$$\frac{dI}{dt} \;=\; \big[\alpha - (\gamma_R + \nu_m)\big]\, I. \tag{4.5}$$

The infectious population grows when the force from infection $\alpha(t)$ exceeds the total rate at which individuals leave the infectious compartment, namely $\gamma_R + \nu_m$. This observation motivates the definition of the basic reproductive number in Chapter 5: intuitively, $\mathcal{R}_0$ compares α at the disease-free state to the combined exit rate $\gamma_R + \nu_m$.

4.4.3 Conservation of Total Population

Adding the three equations (4.4) together, the αI, $\gamma_R I$, and migration-in-out pieces cancel except for the imbalance in the migration flow:

$$\frac{dN}{dt} \;=\; \nu_m N^c - \nu_m N \;=\; \nu_m\big(N^c - N\big), \tag{4.6}$$

where $N(t) = S(t) + I(t) + R(t)$ is the total population. The solution of (4.6) is

$$N(t) \;=\; N^c + \big[N(0) - N^c\big]\, e^{-\nu_m t},$$

so the total population approaches the carrying capacity N^c exponentially at rate ν_m, regardless of the initial distribution. In particular, if we set $N(0) = N^c$, then $N(t) = N^c$ for all time.

This is the case we adopt in the analysis below unless otherwise stated. Setting $N(0) = N^c$ removes one parameter from the model (the initial deviation from the equilibrium population) and makes a two-dimensional reduction available, via the constraint $S + I + R = N^c$, should one wish to eliminate a variable.

4.4.4 Non-Negativity and Boundedness

Before solving a model, ask what it is forbidden to do.

The state variables represent populations and must remain non-negative for the model to be biologically meaningful. It is a standard exercise in compartmental-model theory that system (4.4) has the closed region

$$\Omega = \big\{(S, I, R) \in \mathbb{R}^3 : S \geq 0,\; I \geq 0,\; R \geq 0,\; S + I + R = N^c\big\}$$

as a positively invariant set. That is, if $(S(0), I(0), R(0)) \in \Omega$, then $(S(t), I(t), R(t)) \in \Omega$ for all $t \geq 0$. The proof uses the observation that on each face of Ω where one of the variables is zero, the corresponding time derivative points back into Ω.

Remark 4.1. It is worth separating two distinct questions: invariance of the continuous model and positivity of its numerical solutions. Analytically, both the frequency-dependent term $\beta c_I P_S I$ used here and the mass-action alternative $\beta' SI$ (the two are contrasted in Section 1.3.1) vanish when $I = 0$ or $S = 0$, so both vector fields leave the non-negative orthant invariant: neither continuous model can drive a population negative. Negative values in practice are artifacts of discretization, not of the incidence law—an explicit solver taking too large a step can overshoot a boundary the exact flow never crosses. System (4.4) is well-behaved analytically because the infection term αI is automatically zero when $I = 0$ or $S = 0$ (through P_S); preserving that property under numerical integration is a separate matter of step-size control and solver choice.

4.4.5 Generalization: In-Migration into All Compartments

The model equations (4.4) make an implicit assumption about the migration term $\nu_m N^c$: that all new arrivals enter the susceptible compartment. In symbols, the term $\nu_m N^c$ appears only in the $\dot{S}$ equation, not in $\dot{I}$ or $\dot{R}$. The assumption is reasonable for many applications (births contribute to S for childhood-onset diseases; immigration into a population from a disease-free region likewise contributes to S), but it is not always the right choice. In a refugee-camp model, new arrivals may include actively infectious individuals. In a vaccinated-tourist model, new arrivals may include immune (recovered) individuals. In a cross-border surveillance model during a regional outbreak, both possibilities can apply.

A general formulation partitions the arrival rate $\nu_m N^c$ across the three compartments by fractions π_S, π_I, π_R summing to one:

$$\dot{S} = \pi_S \nu_m N^c - \alpha I - \nu_m S, \tag{4.7a}$$

$$\dot{I} = \pi_I \nu_m N^c + \alpha I - \gamma_R I - \nu_m I, \tag{4.7b}$$

$$\dot{R} = \pi_R \nu_m N^c + \gamma_R I - \nu_m R, \tag{4.7c}$$

with $\pi_S + \pi_I + \pi_R = 1$ and all $\pi_X \geq 0$. The fractions π_X are model-defining parameters: an applied modeler must commit to specific values informed by the demographic and epidemiological context. The closed model (4.4) is the special case $\pi_S = 1, \pi_I = \pi_R = 0$.

The book's convention. Throughout the book's main development, we set $\pi_S = 1$. This is the operating convention that the analytical machinery of Chapters 5–7 assumes. The choice serves two purposes. The first is mathematical tractability: with $\pi_S = 1$, the disease-free equilibrium $(N^c, 0, 0)$ is exactly an equilibrium of the closed system, and the linearization machinery introduced in the next chapter applies cleanly. The second is more subtle and is the subject of a common pitfall: with $\pi_I > 0$, the disease-free state $(N^c, 0, 0)$ is no longer an equilibrium of (4.7), and the standard analytical tools for invasion analysis cease to apply directly. This issue is named in Pitfall: "NGM Applicability" and developed in Chapter 5, Section 5.3.1, where a worked counterexample shows what goes wrong when the NGM construction is applied to a model with $\pi_I > 0$ without acknowledging the assumption violation.

Chapter 10 returns to the general case (4.7) as a setting in which the standard machinery must be extended; for the duration of the foundational development in Chapters 5, 6, and 7, we adopt $\pi_S = 1$ silently. Readers are alerted to the assumption every time a new chapter's setup states "a closed population with $\pi_S = 1$."

Exercises for this section

1. **[C]** The SIR model includes migration at rate ν_m in and out of each compartment. Briefly explain why the total population $N(t)$ approaches a constant N^c, and why this makes N^c a natural scale for nondimensionalization.

4.5 Dependent Quantities

The six independent parameters of Section 4.3 combine to produce several *dependent quantities* that recur throughout the analysis. Table 4.3 lists them.

Table 4.3: Dependent quantities of the SIR model. Each is a function of the independent parameters in Table 4.1.

Parameter	Definition	Interpretation
γ_R	$1/\tau_R$	Recovery rate (per day)
ν_m	$1/\tau_m$	Migration rate (per day)
τ_R^*	$1/(\gamma_R + \nu_m)$	Mean time an individual spends in the I compartment
C_{total}	$c_S S + c_I I + c_R R$	Total contacts per day in the population
P_S	$c_S S / C_{\text{total}}$	Probability a random contact is with a susceptible
P_I	$c_I I / C_{\text{total}}$	Probability a random contact is with an infectious individual
P_R	$c_R R / C_{\text{total}}$	Probability a random contact is with a recovered individual
α	$c_I \beta P_S$	Force from infection (per day per infectious individual)

Of these, C_{total}, P_S, P_I, P_R, and α are time-dependent: they vary with the state of the system as S, I, and R evolve. The rates γ_R and ν_m, together with the mean time τ_R^*, are constants determined by the independent parameters.

4.5.1 The Adjusted Infectious Time

The quantity $\tau_R^* = 1/(\gamma_R + \nu_m)$ deserves a brief discussion. It is the mean time an individual actually spends in the I compartment, given that they can leave by two independent routes.

Consider an individual who is in I at time zero. Two independent processes compete for their exit: recovery (at rate γ_R) and migration out of the population (at rate ν_m). Each process is exponentially distributed. The combined exit process is also exponential, with rate equal to the sum of the individual rates:

$$\text{exit rate} \;=\; \gamma_R + \nu_m.$$

The mean time until exit is the reciprocal of the combined exit rate:

$$\tau_R^* \;=\; \frac{1}{\gamma_R + \nu_m}.$$

This is the quantity, rather than τ_R alone, that enters the basic reproductive number in Chapter 5. Whenever an infectious individual could leave the population (through ν_m) before recovering (through γ_R), we must use τ_R^* rather than τ_R when computing average quantities over the infectious career.

In practice, for most modeled populations, $\nu_m \ll \gamma_R$ (migration happens on the timescale of a year, recovery on the timescale of days), so $\tau_R^* \approx \tau_R$. But the distinction is real, and it becomes non-negligible only when τ_m is comparable to τ_R, when migration removes a meaningful fraction of infectives before they recover. The absolute size of τ_m is not the criterion: a seasonal population with $\tau_m \sim 90$ days matters here only if the infectious period is itself on the order of weeks to months.

4.5.2 Dimensional Analysis

Nondimensionalize first. The equation often tells you what matters after you strip off the units.

Every term in the model equations (4.4) must have the same dimension. We verify this now, both as a check on the model's consistency and as an illustration of a technique that we recommend as the first step in any modeling exercise.

The state variables S, I, R, and N all have dimension $\mathtt{P}$. Their time derivatives have dimension $\mathtt{P}\,\mathtt{T}^{-1}$. The force from infection has dimension

$$[\alpha] \;=\; [c_I]\,[\beta]\,[P_S] \;=\; (\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1})\,(\mathtt{P}\,\mathtt{C}^{-1})\,(1) \;=\; \mathtt{T}^{-1},$$

so $[\alpha I] = \mathtt{T}^{-1} \cdot \mathtt{P} = \mathtt{P}\,\mathtt{T}^{-1}$, matching $[dS/dt]$ and $[dI/dt]$. The recovery rate γ_R has dimension $\mathtt{T}^{-1}$, so $[\gamma_R I] = \mathtt{P}\,\mathtt{T}^{-1}$, matching $[dI/dt]$ and $[dR/dt]$. The migration rate ν_m likewise has dimension $\mathtt{T}^{-1}$, so $[\nu_m N^{\mathrm{c}}] = [\nu_m S] = [\nu_m I] = [\nu_m R] = \mathtt{P}\,\mathtt{T}^{-1}$.

All terms in (4.4) have dimension $\mathtt{P}\,\mathtt{T}^{-1}$. The system is dimensionally consistent.

Dimensional analysis is cheap and catches many common modeling errors. A term that fails to balance dimensionally signals either a missing factor or an incorrect interpretation of a parameter. We recommend performing this check every time a new model is formulated, before any further analysis.

Exercises for this section

1. **[D]** For the baseline parameters in Table 4.4, compute $\tau_R^* = 1/(\gamma_R + \nu_m)$ and compare with the recovery-only timescale $\tau_R = 1/\gamma_R$. By what percentage does migration shorten the effective infectious time in this model? Discuss whether this adjustment matters quantitatively.

4.5.3 Normalizing by the Equilibrium Population

The conservation result (4.6) showed that the total population $N(t) = S + I + R$ approaches a fixed value and, under the standing choice, stays there for all time. That value is the *carrying capacity* N^{c}: the population size at which births balance deaths in the absence of disease mortality, a structural demographic parameter rather than a fitted quantity, and the natural scale for the problem. Measuring each compartment in units of N^{c} gives the fractions $s = S/N^{\mathrm{c}}$, $i = I/N^{\mathrm{c}}$, and $r = R/N^{\mathrm{c}}$, with total $n = N/N^{\mathrm{c}}$, and dividing the equations (4.4) by the constant N^{c} gives the same system in these variables: the force from infection $\alpha = c_I \beta\, P_S$ is unchanged, because $P_S = c_S s / C_{\text{total}}$ is a ratio of populations unaffected by the common scale, and the demographic inflow becomes $\nu_m N^{\mathrm{c}}/N^{\mathrm{c}} = \nu_m$. From here on the book works in these fractions and writes them in lowercase: s, i, r, and n are fractions of the carrying capacity, and the uppercase letters S, I, R, and N are counts of individuals, each recovered from its fraction through the carrying capacity, as in $I = i\,N^{\mathrm{c}}$. The distinction is first put to use in the simulations of Chapter 6.

The *equilibrium* total population is a distinct quantity. At the endemic equilibrium the compartments settle at (s^*, i^*, r^*), and the equilibrium total is $n^* = s^* + i^* + r^*$ by definition, consistent with the starred compartment notation. In the basic model the population equilibrates exactly at its carrying capacity, so $n^* = 1$ and the two coincide, and the disease-free equilibrium is $(1, 0, 0)$. They separate only when the total genuinely varies, as under the disease-induced mortality of Chapter 17, where the endemic total is drawn below the carrying capacity and $n^* < 1$.

The normalization buys three things. The state variables are dimensionless and of order one, which improves the conditioning of the numerical integration and lets a single simulation stand for

any population that shares the same rates, so that outbreaks in populations of very different size become directly comparable. The carrying capacity drops out as an explicit parameter, leaving the disease-free equilibrium at the clean point $(1, 0, 0)$, with the infectious fraction read directly as the prevalence and the susceptible depletion that pushes $\mathcal{R}_e = \mathcal{R}_0 P_S$ below one read directly off the state. The resulting unit-scale system is also the one most of the classical literature uses, which keeps the connection to standard results transparent.

What the normalization costs is the direct reading of counts. Public-health quantities are reported as numbers of cases, hospitalizations, and deaths, and the incidence $J = \alpha I$ that drives the fitting chapters carries units of individuals per unit time; in the normalized variables these are recovered only by multiplying back through the carrying capacity N^{c}. The scaling also sets aside the dimensional bookkeeping developed in Section 4.5.2, which is part of the safety net for catching specification errors, and it can obscure the demographic meaning of the inflow term once $\nu_m N^{\mathrm{c}}$ has collapsed to ν_m. The carrying capacity is therefore deferred rather than eliminated: it reappears the moment a result is converted back to counts, as in the initial susceptible count $S(0) = F_0\, N^{\mathrm{c}}$ of Chapter 7.

One consequence of the scaling is worth stating now because it runs through the rest of the book. Because the dynamics are scale-free, any quantity formed as a ratio in which the population size cancels is independent of the value of N^{c}. The early-outbreak estimator $\hat{\alpha} = j/i$ is of exactly this kind, and its independence from the assumed susceptible population is the structural advantage of the infected viewpoint developed in Chapter 7. The normalized formulation makes that scale-invariance visible rather than incidental.

Remark 4.2 (Normalize by N^{c}, not by the current total $n(t)$). The scale to divide by is the *constant* carrying capacity N^{c}, not the *current* total $n(t)$. Under the standing choice $n(0) = 1$ the two coincide, since $n(t) \equiv 1$, and the distinction is moot. It is not moot when the total genuinely varies, as in a model with disease-induced mortality (the COVID-19 model of Chapter 17, where symptomatic individuals exit to a death compartment) or with unbalanced migration. Dividing by $n(t)$ there introduces a chain-rule term proportional to $\dot{n}/n$ and an extra equation for the scaled total $n = N/N^{\mathrm{c}}$, which in the migration model obeys $\dot{n} = \nu_m(1 - n)$; the fraction variables then no longer close on their own and the system becomes non-autonomous. Normalizing by the fixed N^{c} keeps the reduced system three-dimensional and autonomous. The trade-off between density-dependent and frequency-dependent transmission that underlies this scale-freedom is catalogued under Pitfall: "Density vs. Frequency Dependence".

Remark 4.3 (When the carrying capacity itself varies in time). The same quotient-rule term returns, and is easy to miss, when the disease-free population itself is prescribed to change with calendar time. Writing $s(t) = S(t)/N^{\mathrm{c}}(t)$ with a time-dependent carrying capacity,

$$\dot{s} = \frac{\dot{S}}{N^{\mathrm{c}}(t)} - \frac{\dot{N}^{\mathrm{c}}(t)}{N^{\mathrm{c}}(t)}\, s(t), \tag{4.8}$$

so $N^{\mathrm{c}}(t)$ cannot be substituted into the normalized equations as though it were a constant scale. It has to be carried as a disease-free demographic trajectory, for example $\dot{N}^{\mathrm{c}}(t) = \Lambda(t) - \nu_m(t)\, N^{\mathrm{c}}(t)$ for a recruitment rate $\Lambda(t)$, rather than as an instantaneous equilibrium satisfying only $\Lambda(t) = \nu_m(t)\, N^{\mathrm{c}}(t)$. Equivalently, a modeler who prescribes a target disease-free trajectory $N^{\mathrm{c}}(t)$ must use the recruitment that produces it, $\Lambda(t) = \dot{N}^{\mathrm{c}}(t) + \nu_m(t)\, N^{\mathrm{c}}(t)$, and not the equilibrium recruitment. The second term in (4.8) is exactly the quotient-rule term flagged just above; a time-dependent demography only makes it unavoidable even at the disease-free state. The consequences of parameters that vary in calendar time—the transmission parameters as well as the demography—are taken up in Chapter 6, Section 6.8.

4.6 Connection to the Susceptible-Viewpoint Formulation

The standard textbook formulation of the SIR model (Anderson-May, Keeling-Rohani, Martcheva) uses the force of infection λ rather than the force from infection α. We state this formulation here for readers arriving at this book from the standard literature, and we show that the two formulations describe the same dynamics.

4.6.1 The SIR$_S$ Model

Replacing αi in (4.4) by λs using the relation (4.3), we obtain the susceptible-viewpoint formulation:

$$\frac{ds}{dt} = \nu_m - \lambda s - \nu_m s, \tag{4.9a}$$

$$\frac{di}{dt} = \lambda s - \gamma_R i - \nu_m i, \tag{4.9b}$$

$$\frac{dr}{dt} = \gamma_R i - \nu_m r, \tag{4.9c}$$

with $\lambda = c_S \beta P_I$ and $P_I = c_I i / C_{\text{total}}$.

Systems (4.4) and (4.9) are algebraically identical: they differ only in how the same infection rate $\alpha i = \lambda s$ is written. Any trajectory $(s(t), i(t), r(t))$ that solves one system also solves the other. Chapter 6 confirms this equivalence numerically by integrating both systems with the same initial conditions and verifying that the trajectories coincide to machine precision.

4.6.2 When Each Viewpoint Is Natural

The book's reasons for adopting the infected-viewpoint formulation as primary are given in Chapter 3, Section 3.5, with the fitting demonstration in Chapter 7 and the sensitivity-based analytical foundation in Chapter 9. The SIR$_S$ formulation remains the natural choice in three specific situations: when the observables are susceptible-viewpoint quantities (seroincidence cohort studies, vaccine trials where λs is measured directly), when the intervention acts on the susceptible compartment (vaccination reduces s by moving individuals to r without requiring infection), and when comparing with the classical literature where many results are stated in the SIR$_S$ formulation and translating them through the identity (4.3) can introduce notational complications. In this book, where parameter estimation from incidence data is the primary practical concern, the infected-viewpoint formulation is the default throughout.

Exercises for this section

1. **[D]** Re-do the dimensional analysis of Section 4.5.2 for the SIR$_S$ formulation (4.9) and verify dimensional consistency.

4.7 Worked Example: A Respiratory Infection

We close the chapter with a concrete parameter set that will serve as the book's canonical running example throughout Chapters 5 through 9. The parameters are chosen to represent a generic respiratory infection in a small geographic region, with mild illness-induced withdrawal.

4.7.1 Parameter Values

Table 4.4 lists the baseline values. The units are consistent with Tables 4.1 and 4.3.

Table 4.4: Baseline parameter values for the book's canonical running example.

Parameter	Value	Unit	Interpretation
c_S	20	$\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1}$	Susceptible contact rate
c_I	4	$\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1}$	Infectious contact rate (mild withdrawal)
c_R	20	$\mathtt{C}\,\mathtt{P}^{-1}\,\mathtt{T}^{-1}$	Recovered contact rate
β	0.040	$\mathtt{P}\,\mathtt{C}^{-1}$	Transmission factor per contact
τ_R	12	days	Mean infectious period
τ_m	7300	days	Mean residence time ($\approx$ 20 years)
$\gamma_R = 1/\tau_R$	0.0833	day^{-1}	Recovery rate
$\nu_m = 1/\tau_m$	0.000137	day^{-1}	Migration rate
$\tau_R^* = 1/(\gamma_R + \nu_m)$	12.0	days	Adjusted infectious time
$\mathcal{R}_0$ (derived in Chapter 5)	≈ 1.92	—	Basic reproductive number

The values in Table 4.4 are representative of many respiratory infections (seasonal influenza, common cold viruses, early-stage outbreaks of novel respiratory pathogens). We have deliberately chosen parameters that place the basic reproductive number modestly above unity, so that the example system admits both a disease-free and an endemic equilibrium. At this realistic residence time the endemic prevalence is small (below a tenth of a percent, as is typical of endemic respiratory infections), but it is positive and stable, and the baseline remains useful for illustrating every qualitative phenomenon developed in Chapters 5 through 9.

4.7.2 Discussion of the Parameter Choices

A brief defense of each value:

Contact rates. Twenty contacts per person per day sits at the upper range of measured adult contact rates in moderately social environments. Measured rates depend strongly on what a survey counts as a contact: POLYMOD-style population means run near 10–13 contacts per day [Mossong et al., 2008, Prem et al., 2017], and transmission depends only on the product $c\beta$, so the split between contact rate and transmission probability is a modeling convention. We take a round upper-range value for the teaching baseline so that the worked ratios stay clean; the applied case studies of Chapters 7 and 17 use survey-anchored values, with β calibrated to the same products. The reduction from $c_S = 20$ to $c_I = 4$ corresponds to an eighty-percent reduction in contacts for individuals who are symptomatic but not bedridden; larger reductions (for example $c_I = 2$, a ninety-percent reduction) correspond to symptomatic individuals who significantly self-isolate, and are explored in Chapter 6. Setting $c_R = c_S = 20$ assumes that recovered individuals return to normal contact activity.

Transmission factor. A per-contact transmission factor of $\beta = 0.040$ is appropriate for respiratory infections in typical settings. Household or close-contact transmission factors are higher (often $\beta \sim 0.1$–0.3); community transmission in casual contacts is lower. The value 0.040 represents a weighted average appropriate for the homogeneous-mixing model.

Infectious period. A mean infectious period of $\tau_R = 12$ days is typical for seasonal influenza (symptomatic period roughly one week, infectiousness extending a few days beyond) and for many respiratory viruses. Shorter infectious periods ($\tau_R \sim 5$ days) represent the most acute infections; longer ones ($\tau_R \sim 20$ days) represent prolonged carriage.

Residence time. A mean residence time of $\tau_m = 20$ years (7300 days) corresponds to the modeled population turning over at about five percent per year (through death, out-migration, and aging out of the group), a realistic value for a stable but open population. Section 4.3 discusses the range, typically twenty to forty years, and the pitfalls in choosing it.

4.7.3 Preview of Derived Quantities

With the baseline parameters, several derived quantities that we will compute in Chapter 5 take the following values:

$$\begin{aligned}
\mathcal{R}_0 &= \frac{c_I\,\beta}{\gamma_R + \nu_m} \approx 1.92 \\
s^* &\approx 0.521 \quad \text{(endemic susceptible fraction)} \\
i^* &\approx 0.0008 \quad \text{(endemic infectious fraction} \approx 0.08\%) \\
r^* &\approx 0.478 \quad \text{(endemic recovered fraction)} \\
\sigma_2 &\approx 0.077\ \text{day}^{-1} \quad \text{(early-growth rate)} \\
t_{\text{double}} &\approx 9.1\ \text{days} \quad \text{(early doubling time)}
\end{aligned}$$

Here $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ is the early exponential-growth rate (the positive eigenvalue of the linearization about the disease-free equilibrium when $\mathcal{R}_0 > 1$), and $t_{\text{double}} = (\ln 2)/\sigma_2$ is the associated doubling time; both are derived in Chapter 5, Section 5.4. A reader writing their own simulation of (4.4) with these parameters should reproduce all of the above.

4.8 Summary

We have now assembled the SIR model as a system of three ordinary differential equations, described its six independent parameters and the derived quantities that recur throughout the analysis, verified its dimensional consistency, and fixed a canonical baseline parameter set.

The key objects of the chapter are:

- The model equations (4.4), driven by the force from infection $\alpha = c_I \beta P_S$ with $P_S = c_S s / C_{\text{total}}$.
- The six independent parameters $(c_S, c_I, c_R, \beta, \tau_R, \tau_m)$ listed in Table 4.1.
- The dependent quantities γ_R, ν_m, τ_R^*, C_{total}, P_S, P_I, P_R, and α, each a function of the state and the independent parameters.
- The conservation law $dN/dt = \nu_m(N^c - N)$, which implies $n(t) = 1$ for all t when $n(0) = 1$.
- The equivalence with the susceptible-viewpoint formulation (4.9), mediated by the relation (4.3).
- The canonical baseline parameters in Table 4.4.

Chapter 5 derives the basic and effective reproductive numbers, establishes the existence and stability of the disease-free and endemic equilibria, and introduces the invasion–burden distinction in its rigorous form. Chapter 6 presents the numerical trajectories.

Pitfall check

The new structural concept of this chapter is Pitfall: "The Independence of τ": the time spent in a compartment is the independent parameter, the rate is derived. Before moving on, take a moment to confirm the following points are clear in your mind. *Why is τ_R the parameter being estimated rather than γ_R? What assumption makes the relationship $\gamma_R = 1/\tau_R$ exact, and where in the SIR derivation is that assumption being used? What changes when the assumption is judged inadequate, and which chapter develops the alternative?* If any of these are unclear, the pitfall block in Section 4.3 (or the corresponding catalog entry, accessible via the List of Common Pitfalls after the Table of Contents) is the place to revisit.

The chapter has also exercised Pitfall: "Short-Term Simulation Assumptions" (in the model-assumptions section, which states the simulation-horizon assumptions explicitly), Pitfall: "Dimensional Consistency" (in the parameter-table dimensions and the final dimensional check of the model equations), and Pitfall: "Force of Infection versus Force from Infection" (in the connection-to-susceptible-viewpoint section). Pitfall: "Natural Initial Conditions" is previewed here and developed in detail in Chapter 6.

Exercises

*Five section-level exercises appear above, following Section 4.1, Section 4.2, Section 4.4, Section 4.5, and Section 4.6. The exercises below integrate across the chapter and have been ordered so that derivations, interpretation, and computational-lab work alternate. Exercises are tagged by type: **C** = Concept check, **D** = Derivation, **I** = Interpretation / modeling judgment, **L** = Computational lab.*

1. [**D**] Verify by direct computation that (4.6) follows from summing the three equations of (4.4). Integrate to obtain the explicit solution $n(t) = 1 + [n(0) - 1]e^{-\nu_m t}$.

2. [**I**] Consider a variant of the model with disease-induced mortality rate $\mu_d > 0$: the equation for i becomes $di/dt = \alpha i - (\gamma_R + \nu_m + \mu_d)i$. Derive the corresponding equation for dn/dt and discuss whether $N(t) \to N^c$ still holds. For which classes of diseases (e.g., common respiratory infections vs. rabies) is this variant essential, and for which can the baseline model suffice?

3. [**D**] Starting from (4.3) ($\alpha i = \lambda s$), verify that the model equations (4.4) and (4.9) give the same time derivatives for s, i, and r.

4. [**L**] Using the baseline parameters, verify the non-negativity argument of Section 4.4.4 by integrating (4.4) numerically from several near-boundary initial conditions (for example, $s(0) = 0.001, i(0) = 0, r(0) = 0$). Confirm that the trajectory enters the physically admissible region and remains there.

5. [**D**] Using the baseline parameters in Table 4.4, compute C_{total} and P_S at the disease-free equilibrium $(s, i, r) = (1, 0, 0)$. Confirm that $P_S = 1$ at the DFE and that $\alpha = c_I\beta = 0.16$ day^{-1}.

6. [**D**] Derive the reproductive number $\mathcal{R}_0$ for the SIR model in the equal-contact-rate case $c_S = c_I = c_R = c$ by direct substitution into $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$. Compare with the classical susceptible-viewpoint formula $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$ and confirm they coincide.

7. [**D**] An infectious individual can leave the i compartment by recovery or by migration. Show that the probability that the individual exits by recovery (rather than by migration) is $\gamma_R/(\gamma_R+$

ν_m), independent of the time of exit. With baseline parameters, compute this probability numerically.

8. **[I]** *Spaced retrieval, Chapter 3.* Recall the identity $\alpha i = \lambda s$ established as Proposition 3.1 of Chapter 3. Without re-deriving it, explain in two short paragraphs (i) what specifically the proposition guarantees about the equivalence between this chapter's SIR formulation and the susceptible-viewpoint SIR_S formulation of Section 4.6, and (ii) what *single* structural feature of a hypothetical extended model would have to differ for the equivalence to fail. (Hint: examine the C_{total} term in the definition of P_S and P_I and ask what change of normalization would break the symmetry.) The exercise tests whether you can connect a proposition stated in Chapter 3 to a modeling decision that is taken without comment in this chapter.

9. **[Open Project]** *Parameter elicitation and dimensional audit for a chosen pathogen.* A multi-week capstone project assembling the six independent parameters from published sources, verifying the conservation law and the susceptible-viewpoint equivalence numerically, and reporting which of the six the resulting $\mathcal{R}_0$ is least defensible about.

Chapter 5

The SIR Compartmental Model: Analysis

$\mathcal{R}_0 > 1$ tells you the epidemic can begin. It says nothing about how it ends.

This chapter develops the analytical apparatus for the SIR system from the infected viewpoint. Four results carry the weight of the chapter. First, we derive the basic reproductive number $\mathcal{R}_0$ by two independent methods: a heuristic individual-tracking argument (Section 5.1) and the next-generation matrix method (Section 5.3). Second, we establish the existence and local stability of the disease-free equilibrium (Section 5.4). Third, we derive the endemic equilibrium in closed form for both the equal-contact-rate special case and the general case, and establish its existence and local stability for $\mathcal{R}_0 > 1$ (Section 5.5, Section 5.6). Fourth, we give the rigorous form of the invasion–burden distinction introduced informally in the Preface (Figure 2), showing through closed-form expressions for $\mathcal{R}_0$ and i^* that the invasion parameters and the burden parameters are nearly disjoint (Section 5.7). These four results give the analytical infrastructure that Chapters 6 through 9 build on, beginning with the numerical illustrations of Chapter 6.

The analysis assumes the SIR system developed in Chapter 4, written in the fractions of Section 4.5.3:

$$\frac{ds}{dt} = \nu_m - \alpha(t)\, i - \nu_m s, \tag{5.1a}$$

$$\frac{di}{dt} = \alpha(t)\, i - \gamma_R i - \nu_m i, \tag{5.1b}$$

$$\frac{dr}{dt} = \gamma_R i - \nu_m r, \tag{5.1c}$$

with the force from infection $\alpha(t) = c_I \beta P_S(t)$ and the contact-weighted susceptible fraction $P_S(t) = c_S s(t)/C_{\text{total}}(t)$, where $C_{\text{total}} = c_S s + c_I i + c_R r$. The force from infection α is therefore not a constant but a state-dependent rate that varies with (s, i, r) through P_S; we write α for brevity where this dependence is understood. Total population $n(t) = s + i + r$ approaches the carrying capacity N^{c} exponentially at rate ν_m, and we take $n(0) = 1$ throughout the analysis so that $n(t) = 1$ for all $t \geq 0$.

5.1 The Basic Reproductive Number: Heuristic Derivation

The *basic reproductive number* $\mathcal{R}_0$ is defined as the average number of secondary infections produced by a single infectious individual introduced into an otherwise fully susceptible population [Diekmann et al., 1990]. The quantity is dimensionless, and it carries the entire threshold information of the model: an outbreak invades if and only if $\mathcal{R}_0 > 1$.

We derive $\mathcal{R}_0$ for the SIR system by tracking a single newly introduced infectious individual through its time in the i compartment.

At the moment of introduction, the population is entirely susceptible: $s \approx 1$, $i \approx 0$, and $r = 0$. The contact-weighted susceptible fraction is therefore

$$P_S \;=\; \frac{c_S s}{c_S s + c_I i + c_R r} \;\approx\; \frac{c_S s}{c_S s} \;=\; 1.$$

The force from infection for this individual is

$$\alpha \;=\; c_I \beta P_S \;\approx\; c_I \beta.$$

On average, the individual generates $c_I\beta$ new infections per day for as long as it remains in the i compartment.

The individual leaves i by one of two independent exponential processes: recovery at rate γ_R or migration out of the population at rate ν_m. The total departure rate is $\gamma_R + \nu_m$, and the expected time spent in the i compartment is the reciprocal,

$$\tau_R^* \;=\; \frac{1}{\gamma_R + \nu_m}. \tag{5.2}$$

This quantity, which we call the *adjusted infectious time*, is shorter than the mean infectious period $\tau_R = 1/\gamma_R$ when migration is non-negligible, because some individuals leave the compartment via migration before they would have recovered.

Combining, the expected number of secondary infections is the product of the infection rate and the expected residence time:

$$\mathcal{R}_0 \;=\; c_I \beta \cdot \tau_R^* \;=\; \frac{c_I \beta}{\gamma_R + \nu_m}. \tag{5.3}$$

This is the fundamental threshold quantity of the SIR model.

Remark 5.1 (The role of c_I). A distinctive feature of the infected-viewpoint formulation is that $\mathcal{R}_0$ depends on the infectious contact rate c_I rather than the susceptible contact rate c_S. When modeling an intervention that reduces the contact rate of infectious individuals (case isolation, quarantine, symptom-driven self-withdrawal), the effect on $\mathcal{R}_0$ is direct and proportional: a 50% reduction in c_I reduces $\mathcal{R}_0$ by 50%. In the susceptible-viewpoint formulation (Section 4.2 of Chapter 4), the same intervention enters through the probability P_I, which changes in a less transparent way.

Freezing P_S at its initial value gives a valid linearization here, while freezing λ would not, and the difference is the pitfall Linearization Fidelity (Chapter 16).

Exercises for this section

1. **[D]** Using the baseline parameter set from Table 4.4, compute $\mathcal{R}_0$, σ_2, and t_{double} from (5.3), (5.15), and (5.16). Confirm that $\mathcal{R}_0 \approx 1.92$, $\sigma_2 \approx 0.077$ day^{-1}, and $t_{\text{double}} \approx 9.1$ days.

5.2 From the Basic to the Effective Reproductive Number

The basic reproductive number $\mathcal{R}_0$ derived in Section 5.1 is, by construction, the average number of secondary infections produced by one infectious individual introduced into a *fully susceptible* population. The qualifier "fully susceptible" is essential: $\mathcal{R}_0$ characterizes the threshold for invasion at the disease-free equilibrium, where every contact a new infectious individual makes is with a susceptible. Once the epidemic has begun to spread, the susceptible pool shrinks, and the average number of secondary infections produced by a typical infectious individual at time t is no longer $\mathcal{R}_0$.

To capture the operationally-relevant transmission state at any time during an epidemic, we repeat the heuristic count of secondary infections for an individual introduced at time $t > 0$, now without assuming a fully susceptible population. The contact-weighted susceptible fraction $P_S(t)$ is below unity, so the force from infection is $\alpha(t) = c_I \beta P_S(t)$, and a newly-infectious individual is expected to generate $\alpha(t)\,\tau_R^* = \mathcal{R}_0\,P_S(t)$ secondary infections. This defines the *effective reproductive number*

$$\mathcal{R}_e(t) \;=\; \mathcal{R}_0\,P_S(t), \qquad P_S(t) = \frac{c_S s(t)}{c_S s(t) + c_I i(t) + c_R r(t)}. \tag{5.4}$$

Because $P_S(t) \le 1$ at all times, $\mathcal{R}_e(t) \le \mathcal{R}_0$: as the susceptible pool is depleted, transmission progressively decelerates. When the contact rates are equal the weighting cancels and $P_S(t) = s(t)/n(t)$, recovering the familiar form $\mathcal{R}_e(t) = \mathcal{R}_0\,s(t)/n(t)$ found in standard texts; the contact-weighted P_S is the model-correct generalization once c_S, c_I, and c_R differ.

Three observations clarify the relationship between the two quantities.

$\mathcal{R}_0$ is the special case of $\mathcal{R}_e$ at the disease-free equilibrium. At $t = 0$ in an outbreak introduced into a fully naive population, $P_S(0) = 1$ and (5.4) gives $\mathcal{R}_e(0) = \mathcal{R}_0$. The basic reproductive number is the value of the effective reproductive number at the moment of introduction into a population in which no one has prior immunity. From this perspective, $\mathcal{R}_0$ is a particular reading of $\mathcal{R}_e$ at a particular initial state, not a separate quantity.

The threshold $\mathcal{R}_e(t) = 1$ governs whether transmission is currently growing. The growth or decline of $i(t)$ is determined by whether each infectious individual is currently producing more or fewer than one secondary infection on average. When $\mathcal{R}_e(t) > 1$, $i(t)$ is growing; when $\mathcal{R}_e(t) < 1$, $i(t)$ is declining. This is true throughout the epidemic, not just at $t = 0$. The basic reproductive number $\mathcal{R}_0 > 1$ guarantees only that the epidemic can take off from a fully naive population; whether transmission is currently accelerating or receding at any later time is determined by $\mathcal{R}_e(t)$.

Public-health decisions are driven by $\mathcal{R}_e$, not $\mathcal{R}_0$. Every operational question that arises during an outbreak (when interventions can be relaxed, whether a wave has crested, whether the threshold has been crossed in either direction) depends on the current value of $\mathcal{R}_e(t)$. The basic reproductive number $\mathcal{R}_0$ is reported in the literature primarily as a pathogen-and-population characteristic at the moment of emergence; it does not, on its own, tell decision-makers what is happening now.

Herd immunity. *Herd immunity* is the condition in which enough of the population has been removed from effective susceptibility (through immunity or through reduced contact) that $\mathcal{R}_e(t)$ falls below unity and the infection can no longer sustain transmission, even while susceptible individuals

remain. For the SIR model this occurs exactly when the contact-weighted susceptible fraction drops to

$$P_S^{\text{herd}} = \frac{1}{\mathcal{R}_0}, \tag{5.5}$$

at which point $\mathcal{R}_e = 1$ and the infectious population stops growing. The quantity $1 - P_S^{\text{herd}}$ is then the fraction of the population that must be *effectively removed from the susceptible compartment* (through immunity or through reduced contact with infectious individuals) for new outbreaks not to grow. In a homogeneously mixing population this coincides with the fraction that must be immunized; in heterogeneous populations the two diverge, a distinction Chapter 11 examines through two-group mixing models.

Convention for the rest of this book

Building on these observations, the book adopts the following convention for the analyses that follow:

- Where the operationally-relevant question concerns transmission *during* an epidemic (whether $i(t)$ is growing, when interventions become effective, when an endemic equilibrium is approached, how the susceptible pool is depleted), the analysis is conducted in terms of $\mathcal{R}_e(t)$, treating $\mathcal{R}_0$ as the special case $\mathcal{R}_e(0)$ in a naive population.

- Where the relevant question concerns invasion at the disease-free equilibrium (whether an emerging pathogen can establish itself in a never-infected population, whether a vaccination campaign has driven the herd-immunity threshold above the invasion threshold), the analysis is conducted in terms of $\mathcal{R}_0$, since this is the quantity invasion theory characterizes.

- Where heterogeneous mixing is involved (Chapters 11, 13, 14, 15), the next-generation matrix construction of the following section gives $\mathcal{R}_0$ as the spectral radius of the next-generation matrix evaluated at the disease-free equilibrium. The corresponding effective quantity $\mathcal{R}_e(t)$ is the spectral radius of the same next-generation matrix evaluated at the *current* compartment values rather than at the disease-free equilibrium; this is the construction that carries $\mathcal{R}_e(t) = \mathcal{R}_0\, P_S(t)$ over to models with several infected compartments. The heuristic form $\mathcal{R}_e(t) = \mathcal{R}_0\, P_S(t)$ in (5.4) is the explicit instantiation of this general definition for the scalar SIR model: it is what the spectral radius reduces to when the next-generation matrix is evaluated at the current state $(s(t), i(t), r(t))$ rather than at the disease-free equilibrium, so the two are the general framework and its model-specific form, not competing definitions. Optional follow-up subsections in the affected chapters develop this generalization for readers who want it; the main thread of each chapter conducts its analysis in $\mathcal{R}_0$ at the disease-free equilibrium, since this is where the analytical tools are simplest and where the threshold theorem applies in its standard form.

A related quantity that arises naturally in intervention analysis is the *controlled reproductive number* $\mathcal{R}_c$, which describes the expected number of secondary infections produced by one infectious individual under a specific intervention regime (vaccination, contact reduction, treatment) but with the population otherwise still susceptible. The controlled reproductive number $\mathcal{R}_c$ is the special case of $\mathcal{R}_e(t)$ in which the post-intervention state is reached but the susceptible pool has not yet been depleted by the epidemic itself. Because $\mathcal{R}_c$ is mathematically a special case of $\mathcal{R}_e$, this book treats $\mathcal{R}_e(t)$ as the unifying notation and develops $\mathcal{R}_c$ as a particular reading of $\mathcal{R}_e$ when the

relevant context arises (Chapters 7 and 9 discuss how $\mathcal{R}_c$ values are estimated from data and how interventions are ranked by their impact on $\mathcal{R}_e$). A textbook survey of disease-control strategies is given by Martcheva [2015, Ch. 9].

$\mathcal{R}_e(t)$ is the primary quantity during an epidemic, $\mathcal{R}_0$ is a special case at $t = 0$ in a naive population, and $\mathcal{R}_c$ is a special case of $\mathcal{R}_e$ under a specified intervention regime. This aligns the book with current epidemiological practice and clarifies what each quantity does and does not say. The next section returns to the technical machinery: the next-generation matrix construction of $\mathcal{R}_0$ that extends to multi-compartment and heterogeneous-mixing models.

An estimate fitted partway through an outbreak estimates $\mathcal{R}_e$ at the current susceptible fraction, not $\mathcal{R}_0$, and reading one as the other is the pitfall $\mathcal{R}_0$ versus $\mathcal{R}_e(t)$ (Chapter 16).

5.3 The Basic Reproductive Number: Next-Generation Matrix

The heuristic argument of Section 5.1 is transparent but requires case-by-case reasoning for each new model. The next-generation matrix (NGM) method of van den Driessche and Watmough [2002] provides a systematic derivation that extends without change to models with multiple infected compartments (Chapters 10, 11, 13). Textbook developments of the next-generation-matrix construction appear in Brauer et al. [2019, Sec. 5.2], Martcheva [2015, Ch. 5], and Li [2018, Sec. 1.4.9]. In this construction the *infected compartments* are all states in which an individual carries the infection (including any latent or exposed stage that is not yet infectious), not only the infectious compartment(s); in an SEIR model, for example, both E and i are infected compartments and both must be included, since omitting a non-infectious infected stage yields the wrong reproductive number. We present it here first for the SIR system, which has a single infected compartment, where it reduces to a scalar calculation that confirms (5.3).

Decomposition of the infected dynamics. Write the dynamics of the infected compartment (for the SIR model, the single stage i) as

$$\frac{di}{dt} \;=\; \mathcal{F}(s,i,r) \;-\; \mathcal{V}(s,i,r), \tag{5.6}$$

where $\mathcal{F}$ captures new infections entering i and $\mathcal{V}$ captures all other departures from i. The calligraphic forms emphasize that these are *vector fields*, functions of the state (s,i,r) that produce a rate. For the SIR system this decomposition is

$$\mathcal{F}(s,i,r) = \alpha(s,i,r)\, i = c_I \beta\, P_S(s,i,r)\, i = c_I \beta \cdot \frac{c_S s}{c_S s + c_I i + c_R r} \cdot i,$$
$$\mathcal{V}(i) = (\gamma_R + \nu_m)\, i.$$

Linearization at the disease-free equilibrium. The NGM is built from the linearizations of the vector fields $\mathcal{F}$ and $\mathcal{V}$ at the DFE, where $(s,i,r) = (1,0,0)$. The linearizations are matrices (in the scalar single-infected-compartment case here, 1×1 matrices) and we use bold roman $\boldsymbol{F}, \boldsymbol{V}$ to mark them as matrices, distinct from the calligraphic vector fields they are derived from:

$$\boldsymbol{F} \;=\; \left.\frac{\partial \mathcal{F}}{\partial i}\right|_{\text{DFE}}, \qquad \boldsymbol{V} \;=\; \left.\frac{\partial \mathcal{V}}{\partial i}\right|_{\text{DFE}}. \tag{5.7}$$

Differentiating with respect to i and applying the quotient rule,

$$\frac{\partial \mathcal{F}}{\partial i} \;=\; c_I\beta \cdot \frac{-c_S s \cdot c_I \cdot i}{(c_S s + c_I i + c_R r)^2} \;+\; c_I\beta \cdot \frac{c_S s}{c_S s + c_I i + c_R r}.$$

At the disease-free equilibrium $(1, 0, 0)$, the first term vanishes (the factor i is zero) and $P_S = 1$, so the second term reduces to $c_I\beta$. Hence

$$\boldsymbol{F} \;=\; c_I\beta. \tag{5.8}$$

Similarly,

$$\boldsymbol{V} \;=\; \gamma_R + \nu_m. \tag{5.9}$$

The next-generation matrix. The NGM is defined as $\boldsymbol{K} = \boldsymbol{F}\,\boldsymbol{V}^{-1}$. In the scalar case,

$$\boldsymbol{K} \;=\; \frac{c_I\beta}{\gamma_R + \nu_m}.$$

The basic reproductive number is the spectral radius of $\boldsymbol{K}$, which for a scalar is simply $|\boldsymbol{K}|$:

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}) \;=\; \frac{c_I\beta}{\gamma_R + \nu_m}. \tag{5.10}$$

This matches the heuristic result (5.3).

A worked example with two infected compartments. The scalar case confirms the heuristic result, but it hides what the method is for. The SEIR model of Chapter 10 (Section 10.3) adds an exposed compartment E between s and i. Exposed individuals carry the infection but cannot yet transmit it, and they become infectious at rate γ_E. Both E and i are infected compartments. New infections enter E only, so the decomposition of the infected dynamics is

$$\mathcal{F} \;=\; \begin{pmatrix} \alpha\, i \\ 0 \end{pmatrix}, \qquad \mathcal{V} \;=\; \begin{pmatrix} (\gamma_E + \nu_m)\, E \\ -\gamma_E E + (\gamma_R + \nu_m)\, i \end{pmatrix}.$$

Progression from E to i sits in $\mathcal{V}$, with a minus sign in the i row. It moves a person who is already infected, so it is a transition, not a new infection. At the DFE the force from infection is $\alpha = c_I\beta$, and the linearizations are

$$\boldsymbol{F} \;=\; \begin{pmatrix} 0 & c_I\beta \\ 0 & 0 \end{pmatrix}, \qquad \boldsymbol{V} \;=\; \begin{pmatrix} \gamma_E + \nu_m & 0 \\ -\gamma_E & \gamma_R + \nu_m \end{pmatrix},$$

so that

$$\boldsymbol{V}^{-1} \;=\; \begin{pmatrix} \dfrac{1}{\gamma_E + \nu_m} & 0 \\[2ex] \dfrac{\gamma_E}{(\gamma_E + \nu_m)(\gamma_R + \nu_m)} & \dfrac{1}{\gamma_R + \nu_m} \end{pmatrix}.$$

Each matrix can be read before it is multiplied. Entry (j, k) of $\boldsymbol{V}^{-1}$ is the expected time that an individual introduced into compartment k spends in compartment j [van den Driessche and Watmough, 2002]. A newly exposed person spends on average $1/(\gamma_E + \nu_m)$ in E. With probability $\gamma_E/(\gamma_E+\nu_m)$ that person goes on to spend $1/(\gamma_R+\nu_m)$ in i, which is the lower-left entry. Multiplying by the infection rates gives

$$\boldsymbol{K} \;=\; \boldsymbol{F}\,\boldsymbol{V}^{-1} \;=\; \begin{pmatrix} \dfrac{c_I\beta}{\gamma_R + \nu_m} \cdot \dfrac{\gamma_E}{\gamma_E + \nu_m} & \dfrac{c_I\beta}{\gamma_R + \nu_m} \\[2ex] 0 & 0 \end{pmatrix}.$$

Entry (i,k) of $\boldsymbol{K}$ is the expected number of new infections in compartment i caused by one infected individual introduced into compartment k. The bottom row is zero because no infection starts in i. The eigenvalues are the top-left entry and zero, so

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}) \;=\; \frac{c_I\beta}{\gamma_R+\nu_m}\cdot\frac{\gamma_E}{\gamma_E+\nu_m}.$$

This is the SIR value times the probability that an exposed person survives the latent period, the same answer the heuristic argument gives in Chapter 10, equation (10.4).

The same model, split another way. Nothing in the algebra forces this split. Suppose we book progression from E to i as a "new infection" of i and keep only departures in $\mathcal{V}$. Then

$$\tilde{\boldsymbol{F}} \;=\; \begin{pmatrix} 0 & c_I\beta \\ \gamma_E & 0 \end{pmatrix}, \qquad \tilde{\boldsymbol{V}} \;=\; \begin{pmatrix} \gamma_E+\nu_m & 0 \\ 0 & \gamma_R+\nu_m \end{pmatrix}, \qquad \tilde{\boldsymbol{F}}\,\tilde{\boldsymbol{V}}^{-1} \;=\; \begin{pmatrix} 0 & \dfrac{c_I\beta}{\gamma_R+\nu_m} \\ \dfrac{\gamma_E}{\gamma_E+\nu_m} & 0 \end{pmatrix}.$$

The spectral radius of this matrix is the square root of the product of its two off-diagonal entries:

$$\rho(\tilde{\boldsymbol{F}}\,\tilde{\boldsymbol{V}}^{-1}) \;=\; \sqrt{\frac{c_I\beta}{\gamma_R+\nu_m}\cdot\frac{\gamma_E}{\gamma_E+\nu_m}} \;=\; \sqrt{\mathcal{R}_0}.$$

The two numbers differ, yet $\sqrt{\mathcal{R}_0}$ exceeds 1 exactly when $\mathcal{R}_0$ does, so both mark the same threshold. van den Driessche and Watmough [2002, Section 4.1] make the same point with a treatment model: different choices of $\mathcal{F}$ give different spectral radii, and each is a threshold parameter. Only the first split counts secondary infections. The second treats the step from E to i as a generation of its own, so its spectral radius is the geometric mean of two stage numbers. Both splits satisfy the five assumptions stated below, so the assumptions cannot choose between them. The modeler must, and the rule follows from the definition: $\mathcal{F}$ holds new infections and nothing else. The remark on vector-host models below meets the same square root from the other side.

Why the method matters. The heuristic argument also handles the SEIR model, as Chapter 10 shows, but each new model needs its own argument. The NGM method does not. In a two-group model (Chapter 11) the NGM is a 2×2 matrix even though each group has one infected compartment. In a vector-host model (Chapter 13) it is 2×2 because infection passes from host to vector and from vector to host. In each case the same decomposition $\mathcal{F}-\mathcal{V}$ produces $\mathcal{R}_0$ mechanically. We use the method repeatedly in Part IV.

Remark 5.2 (Alternative $\mathcal{R}_0$ conventions for vector-host models). For vector-host models, some authors define $\mathcal{R}_0$ as the spectral radius of $\boldsymbol{F}\boldsymbol{V}^{-1}$ (the convention here), while others define it as the square of that quantity, interpreting $\mathcal{R}_0^2$ as the number of secondary infections accumulated around one full vector-host-vector cycle. The two conventions agree on the threshold (both cross 1 together) but give different numerical values. Chapter 11, Section 11.6 develops a precise stage/cyclic/geometric-mean framework that names the two quantities exactly: the book's $\mathcal{R}_0$ is the geometric mean of the stage reproductive numbers, equivalently the square root of the cyclic reproductive number $\mathcal{C}_0^{(h)}$.

5.3.1 Applicability of the NGM Method

The spectral-radius formula $\mathcal{R}_0 = \rho(\boldsymbol{F}\,\boldsymbol{V}^{-1})$ developed above is not unconditionally valid. It rests on five assumptions stated by van den Driessche and Watmough [2002], which we give in their form and order. Order the n compartments so that the first m are the infected ones, and write each equation as $\dot{x}_k = \mathcal{F}_k(\mathbf{x}) - \mathcal{V}_k(\mathbf{x})$ with $\mathcal{V}_k = \mathcal{V}_k^- - \mathcal{V}_k^+$. Here $\mathcal{F}_k$ is the rate at which new infections appear in compartment k, $\mathcal{V}_k^+$ is the rate of transfer into compartment k by all other means, and $\mathcal{V}_k^-$ is the rate of transfer out of it. Let X_s be the set of disease-free states, those with $\mathbf{x} \geq 0$ and $x_1 = \cdots = x_m = 0$, and let $\mathbf{x}_0$ be the disease-free equilibrium. Each assumption is followed by what it says and what breaks when it fails.

(A1) If $\mathbf{x} \geq 0$, then $\mathcal{F}_k, \mathcal{V}_k^+, \mathcal{V}_k^- \geq 0$ for every k. Each term is a rate at which people move, so none can be negative. A split that breaks this was usually made by algebra rather than by mechanism, for instance by moving a negative term into $\mathcal{F}$ to tidy an expression. The threshold result rests on $\boldsymbol{F}$ having no negative entries and $\boldsymbol{V}$ being a nonsingular M-matrix, so that $\boldsymbol{V}^{-1}$ has none either [van den Driessche and Watmough, 2002, Lemma 1]; once a term changes sign, the conclusion no longer follows.

(A2) If $x_k = 0$, then $\mathcal{V}_k^- = 0$; in particular $\mathcal{V}_k^- = 0$ on X_s for every infected compartment. Nobody can leave an empty compartment. A removal that does not shrink with the compartment, such as treatment at a fixed number of people per day, breaks it: the model then drives the compartment negative, and its linearization describes no real population.

(A3) $\mathcal{F}_k = 0$ for $k > m$. New infections appear only in infected compartments. This is bookkeeping, since an infection by definition creates an infected person. When a model's equations put an infection term in a compartment labeled uninfected, the labels are wrong, and an NGM built on them counts the wrong events.

(A4) If $\mathbf{x} \in X_s$, then $\mathcal{F}_k(\mathbf{x}) = 0$ and $\mathcal{V}_k^+(\mathbf{x}) = 0$ for $k = 1, \ldots, m$. With no infection present, nothing enters the infected compartments, neither by infection nor by any other route. Infected arrivals break it: the disease-free states are then not equilibria, and there is no disease-free population to invade. This is the assumption applied work violates most often, and the worked counterexample below follows it through.

(A5) If $\mathcal{F}$ is set to zero, every eigenvalue of the Jacobian of $\mathbf{f} = \mathcal{F} - \mathcal{V}$ at $\mathbf{x}_0$ has negative real part. With new infections switched off, the disease-free equilibrium is locally asymptotically stable. The assumption asks two things: infected people leave the infected compartments, and the uninfected population settles to $\mathbf{x}_0$. If the uninfected population grows without bound or oscillates, there is no fixed state to invade, and $\rho(\boldsymbol{F}\,\boldsymbol{V}^{-1}) < 1$ no longer guarantees that the infection dies out.

When these assumptions hold, $\mathcal{R}_0 = \rho(\boldsymbol{F}\,\boldsymbol{V}^{-1})$, and the disease-free equilibrium is locally asymptotically stable if $\mathcal{R}_0 < 1$ and unstable if $\mathcal{R}_0 > 1$ [van den Driessche and Watmough, 2002, Theorem 2]. When an assumption fails, the construction can still be carried out mechanically, but the resulting spectral radius need not equal $\mathcal{R}_0$, and using it as such is a category error.

The spectral radius $\rho(\boldsymbol{F}\,\boldsymbol{V}^{-1})$ equals $\mathcal{R}_0$ only when (A1) to (A5) hold, and in-migration of infected individuals breaks (A4); using the formula outside that domain is the pitfall NGM Applicability (Chapter 16).

Worked counterexample: in-migration into the infected compartment. To make the failure mode concrete, consider the SIR model with $\pi_I > 0$: new arrivals partitioned so that a fraction π_I enters the infected compartment. From Chapter 4, Section 4.4.5, the model is

$$\begin{aligned} \dot{s} &= \pi_S\,\nu_m - \alpha i - \nu_m s, \\ \dot{i} &= \pi_I\,\nu_m + \alpha i - \gamma_R i - \nu_m i, \\ \dot{r} &= \pi_R\,\nu_m + \gamma_R i - \nu_m r, \end{aligned} \tag{5.11}$$

with $\pi_S + \pi_I + \pi_R = 1$.

Step 1: check assumption (A4). Infected arrivals are not new infections: they enter i already infected, so they belong to the transfers in, $\mathcal{V}^+ = \pi_I\,\nu_m$, while the new-infection term stays $\mathcal{F} = \alpha(s,i)\,i$. On the disease-free set, where $i = 0$,

$$\mathcal{V}^+ \;=\; \pi_I\,\nu_m.$$

For $\pi_I > 0$, this is strictly positive. Assumption (A4) requires $\mathcal{V}_k^+ = 0$ on X_s; it fails.

Step 2: attempt the NGM construction anyway. Linearize at the candidate disease-free state $(1,0,0)$. The constant inflow $\pi_I\,\nu_m$ has zero derivative, so

$$\boldsymbol{F}_{\text{naive}} \;=\; \left.\frac{\partial \mathcal{F}}{\partial i}\right|_{(1,0,0)} \;=\; c_I\beta, \qquad \boldsymbol{V}_{\text{naive}} \;=\; \gamma_R + \nu_m.$$

The spectral radius of $\boldsymbol{F}_{\text{naive}}\boldsymbol{V}_{\text{naive}}^{-1}$ is

$$\rho(\boldsymbol{F}_{\text{naive}}\boldsymbol{V}_{\text{naive}}^{-1}) \;=\; \frac{c_I\beta}{\gamma_R + \nu_m},$$

formally identical to the $\pi_I = 0$ case.

Step 3: check what this number actually represents. The state $(1,0,0)$ is *not an equilibrium* of system (5.11) when $\pi_I > 0$: substituting $i = 0$ into the $\dot{i}$ equation gives $\dot{i} = \pi_I\,\nu_m > 0$, so the system flows away from $(1,0,0)$ even with $i(0) = 0$. The spectral radius $\rho(\boldsymbol{F}_{\text{naive}}\boldsymbol{V}_{\text{naive}}^{-1})$ was computed at a point that is not the disease-free equilibrium (no such equilibrium exists for the open system with $\pi_I > 0$), and the textbook interpretation "threshold for invasion of a disease-free population" does not apply: there *is* no disease-free population for this system to invade.

Step 4: what is the correct concept? With $\pi_I > 0$, the model has a globally attracting endemic state at some (s^*, i^*, r^*) with $i^* > 0$ for any $\pi_I > 0$, and the question of "invasion threshold" becomes ill-posed: any external perturbation can be absorbed by the system without invasion dynamics in the usual sense. The relevant quantities are the endemic prevalence i^* (a function of π_I and the other parameters) and the local stability of the endemic state via the full-system Jacobian $\boldsymbol{J}$ at (s^*, i^*, r^*), not via the NGM construction. An analyst who reports $\rho(\boldsymbol{F}_{\text{naive}}\boldsymbol{V}_{\text{naive}}^{-1}) = c_I\beta/(\gamma_R + \nu_m)$ as "the $\mathcal{R}_0$ of system (5.11)" is reporting a number that has no operational interpretation as a reproductive number, because the system has no disease-free equilibrium to threshold against. Lenhart [2020] illustrates the same failure mode with a *C. difficile* hospital-shelter model in which arrivals carry the pathogen.

Pedagogical summary. The applicability of the NGM method is the modeler's responsibility, not the algorithm's. The spectral radius $\rho(\boldsymbol{F}\,\boldsymbol{V}^{-1})$ can be computed for any model; it equals $\mathcal{R}_0$ only when the underlying construction is valid. For the book's main development with $\pi_S = 1$, the assumptions hold and the formula carries its standard interpretation. For applications where

$\pi_I > 0$ matters operationally (the worked examples of Chapter 10 and parts of the case studies in Chapters 17–19), the analyst must re-derive the appropriate invasion quantity, not reuse the formula uncritically. This is Pitfall: "NGM Applicability", a pitfall of interpretation: the formula computed and the quantity it is named after must refer to the same underlying construction.

Invasion, not burden. The NGM is built at the disease-free equilibrium, so the number it returns answers one question: can the infection invade a population in which it is absent? It says nothing about how large a share of the population is infected once the infection is established. That level, the endemic prevalence i^*, depends on parameters that $\mathcal{R}_0$ barely sees. In the SIR model, the mean residence time τ_m hardly moves $\mathcal{R}_0$ when recovery is fast compared with turnover, yet it moves i^* almost in proportion. Section 5.7 makes the partition exact. An $\mathcal{R}_0$ from the NGM is a statement about invasion; reading it as a statement about burden is the pitfall Invasion versus Equilibrium (Chapter 16).

Verify the AI: Common Conflations in AI-Generated $\mathcal{R}_0$ Derivations

An AI derivation of $\mathcal{R}_0$ can sit close to the textbook formula and still commit one of three errors, and one question catches each.

Demography dropped. Writing $\mathcal{R}_0 = c_I\beta/\gamma_R$ omits the ν_m of (5.3), $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$, and is off by the factor $1 + \nu_m/\gamma_R$, which exceeds 5% for HIV, with τ_R of years. Ask: "*Where does demographic turnover appear in your formula?*"

Methods mixed. A derivation that blends the heuristic argument with next-generation matrices commits to neither. Ask: "*Which method are you following, and where would the other give a different intermediate quantity?*"

Symbols drifting. $\mathcal{R}_0$ is defined at the disease-free equilibrium, with $P_S = 1$ fixed; $\mathcal{R}_e(t) = \mathcal{R}_0 P_S(t)$ lets it vary. Ask: "*Is P_S fixed at 1 in your formula, or varying?*"

A correct derivation passes all three.

5.4 The Disease-Free Equilibrium

An *equilibrium* of the SIR system (5.1) is a time-independent solution (s^*, i^*, r^*) where all three time derivatives vanish. The *disease-free equilibrium* (DFE) is the equilibrium in which the disease is absent from the population.

5.4.1 Existence of the DFE

Setting $i = 0$ in (5.1), the three equations reduce to

$$\nu_m - \nu_m s \;=\; 0, \qquad 0 \;=\; 0, \qquad -\nu_m r \;=\; 0,$$

which give $s = 1$ and $r = 0$. The DFE is therefore

$$(s^0, i^0, r^0) \;=\; (1, 0, 0). \tag{5.12}$$

The DFE is consistent with $s + i + r = 1$, as required by the conservation identity of Chapter 4.

5.4.2 Local Stability of the DFE

To assess the local stability of the DFE, we linearize the system (5.1) and examine the eigenvalues of the Jacobian matrix at $(1, 0, 0)$.

The nonlinear term $\alpha(s,i,r)i = c_I\beta P_S i$ is the only source of nonlinearity. Its partial derivatives at the DFE are, as computed in Section 5.3,

$$\left.\frac{\partial(\alpha i)}{\partial s}\right|_{\text{DFE}} = 0, \qquad \left.\frac{\partial(\alpha i)}{\partial i}\right|_{\text{DFE}} = c_I\beta, \qquad \left.\frac{\partial(\alpha i)}{\partial r}\right|_{\text{DFE}} = 0.$$

The Jacobian of the full system at the DFE is therefore

$$\mathbf{J}_{\text{DFE}} = \begin{pmatrix} -\nu_m & -c_I\beta & 0 \\ 0 & c_I\beta - (\gamma_R + \nu_m) & 0 \\ 0 & \gamma_R & -\nu_m \end{pmatrix}. \tag{5.13}$$

In the displayed ordering (s,i,r) this matrix is not triangular, but reordering the variables as (i,s,r) makes it lower-triangular: the s and r columns have no off-diagonal entries, and only the i column couples to the other variables. Its eigenvalues are therefore the diagonal entries:

$$\sigma_1 = -\nu_m, \qquad \sigma_2 = c_I\beta - (\gamma_R + \nu_m), \qquad \sigma_3 = -\nu_m. \tag{5.14}$$

Using the definition of $\mathcal{R}_0$ from (5.3), we can rewrite σ_2 as

$$\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1). \tag{5.15}$$

The following theorem summarizes the stability analysis.

Theorem 5.3 (Stability of the disease-free equilibrium)**.** *The disease-free equilibrium* $(1,0,0)$ *of the* SIR *system* (5.1) *is locally asymptotically stable if* $\mathcal{R}_0 < 1$ *and unstable if* $\mathcal{R}_0 > 1$.

Proof. Since $\nu_m > 0$ and $\gamma_R > 0$, the eigenvalues $\sigma_1 = \sigma_3 = -\nu_m$ are strictly negative regardless of $\mathcal{R}_0$. The eigenvalue $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ is strictly negative if and only if $\mathcal{R}_0 < 1$. When $\mathcal{R}_0 < 1$, all three eigenvalues have negative real part and the DFE is locally asymptotically stable. When $\mathcal{R}_0 > 1$, $\sigma_2 > 0$ and the DFE has a one-dimensional unstable manifold; any perturbation with $i(0) > 0$ grows initially at rate σ_2. □

Early exponential growth. When $\mathcal{R}_0 > 1$, a small introduction of infectious individuals grows initially as

$$i(t) \approx i(0)\, e^{\sigma_2 t},$$

where $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$. This approximation holds while $s \approx 1$, which for most outbreaks is the first few weeks. Rearranging gives a useful connection between the epidemic doubling time t_{double} and $\mathcal{R}_0$:

$$t_{\text{double}} = \frac{\ln 2}{\sigma_2} = \frac{\ln 2}{(\gamma_R + \nu_m)(\mathcal{R}_0 - 1)}. \tag{5.16}$$

If one measures t_{double} from early incidence data and has independent estimates of γ_R and ν_m, then (5.16) gives $\mathcal{R}_0$ directly, without requiring knowledge of the susceptible population size. Chapter 7 develops this approach in detail.

Boundary case $\mathcal{R}_0 = 1$. The stability analysis is inconclusive when $\mathcal{R}_0 = 1$, because $\sigma_2 = 0$: the linearization has a zero eigenvalue and higher-order terms determine the dynamics. A center manifold analysis (see Diekmann and Heesterbeek [2000]) shows that the DFE is a degenerate equilibrium that transitions from stable (for $\mathcal{R}_0 < 1$) to unstable (for $\mathcal{R}_0 > 1$) as $\mathcal{R}_0$ passes through unity. This transition is called a *transcritical bifurcation*: at the threshold the disease-free and endemic branches exchange stability, and the endemic equilibrium becomes the biologically relevant, stable attractor (Section 5.5). In practice, the case $\mathcal{R}_0 = 1$ is a measure-zero event in parameter space, and its mathematical subtlety does not affect the qualitative behavior of the model away from the boundary.

Exercises for this section

1. **[C]** Explain in one sentence the physical meaning of the second eigenvalue σ_2 of the DFE Jacobian. How does its sign determine whether a small introduction of infectious individuals grows or decays?

2. **[D]** *Explicit next-generation-matrix construction.* The next-generation-matrix (NGM) method introduced by Diekmann et al. [1990] and van den Driessche and Watmough [2002] computes $\mathcal{R}_0$ as the spectral radius of $\boldsymbol{K} = \boldsymbol{F}\,\boldsymbol{V}^{-1}$, where $\boldsymbol{F}$ is the Jacobian of the new-infection vector field $\mathcal{F}$ at the DFE and $\boldsymbol{V}$ is the Jacobian of the transition vector field $\mathcal{V}$ at the DFE. For the SIR model with the single infected compartment i, identify $\boldsymbol{F}$ and $\boldsymbol{V}$ as 1×1 matrices by linearizing $\dot{i} = \alpha(s,i,r)\,i - (\gamma_R + \nu_m)\,i$ where $\alpha(s,i,r) = c_I \beta P_S(s,i,r)$ at the DFE. Differentiate using the product rule: $\partial(\alpha i)/\partial i|_{\mathrm{DFE}} = \alpha|_{\mathrm{DFE}} + i \cdot \partial\alpha/\partial i|_{\mathrm{DFE}} = c_I\beta$ (since $P_S(\mathrm{DFE}) = 1$ and the second term vanishes at $i = 0$), giving $\boldsymbol{F} = c_I\beta$ and $\boldsymbol{V} = \gamma_R + \nu_m$. Compute $\boldsymbol{F}\,\boldsymbol{V}^{-1}$ explicitly and verify that its spectral radius matches the heuristic $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ obtained in Section 5.1 from the survival-probability argument. State the reason the SIR NGM reduces to a 1×1 matrix and identify one extension from Chapter 10 for which the NGM becomes higher-dimensional.

5.5 The Endemic Equilibrium

When $\mathcal{R}_0 > 1$, the DFE is unstable and the system admits a second equilibrium with $i^* > 0$, called the *endemic equilibrium* (EE). Endemic disease models are developed at length by Brauer et al. [2019, Ch. 3]. This section derives the EE in closed form and establishes its existence. Stability of the EE is addressed in Section 5.6.

5.5.1 Derivation of the Endemic Equilibrium

At an equilibrium of (5.1) with $i^* > 0$, the three equations $\dot{s} = 0$, $\dot{i} = 0$, $\dot{r} = 0$ give three conditions. From $\dot{i} = 0$ and $i^* \neq 0$,

$$\alpha^* \;=\; \gamma_R + \nu_m. \tag{5.17}$$

From $\dot{r} = 0$,

$$r^* \;=\; \frac{\gamma_R}{\nu_m}\, i^*. \tag{5.18}$$

Population conservation $s^* + i^* + r^* = n^*$, together with (5.18), gives

$$s^* \;=\; n^* - i^* - \frac{\gamma_R}{\nu_m}\, i^* \;=\; n^* - \frac{\gamma_R + \nu_m}{\nu_m}\, i^*. \tag{5.19}$$

To close the system for i^*, we substitute into the equilibrium condition $\alpha^* = \gamma_R + \nu_m$ using the explicit form of α:

$$\alpha^* \;=\; c_I\beta \cdot P_S^* \;=\; c_I\beta \cdot \frac{c_S s^*}{c_S s^* + c_I i^* + c_R r^*} \;=\; \gamma_R + \nu_m.$$

The full derivation is algebraic; we state the result and sketch the key steps. Using (5.18), (5.19), and $c_I\beta = \mathcal{R}_0(\gamma_R + \nu_m)$, the equilibrium condition becomes

$$c_S n^*(\mathcal{R}_0 - 1) \;=\; \frac{i^*}{\nu_m}\big[c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I\nu_m + c_R\gamma_R\big]. \tag{5.20}$$

Solving for i^* gives the closed-form expression for the endemic prevalence.

$$\boxed{i^* \;=\; \frac{n^* c_S \nu_m(\mathcal{R}_0 - 1)}{c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I\nu_m + c_R\gamma_R}.} \tag{5.21}$$

The other two endemic values follow:

$$r^* \;=\; \frac{\gamma_R}{\nu_m} i^*, \tag{5.22a}$$

$$s^* \;=\; \frac{n^*(c_I\nu_m + c_R\gamma_R)}{c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I\nu_m + c_R\gamma_R}. \tag{5.22b}$$

5.5.2 Existence of the Endemic Equilibrium

Proposition 5.4 (Existence of the endemic equilibrium)**.** *The* SIR *system* (5.1) *admits a unique endemic equilibrium* (s^*, i^*, r^*) *with* $i^* > 0$*,* $s^* > 0$*, and* $r^* > 0$ *if and only if* $\mathcal{R}_0 > 1$*.*

Proof. Consider the expression (5.21). The denominator is

$$c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I\nu_m + c_R\gamma_R.$$

The second term $c_I\nu_m$ is strictly positive because $c_I, \nu_m > 0$, and the third term $c_R\gamma_R$ is strictly positive because $c_R, \gamma_R > 0$. When $\mathcal{R}_0 \geq 1$ the first term is non-negative, so the denominator is strictly positive. The numerator $n^* c_S \nu_m(\mathcal{R}_0 - 1)$ is positive for $\mathcal{R}_0 > 1$, zero at $\mathcal{R}_0 = 1$, and negative for $\mathcal{R}_0 < 1$. Hence $i^* > 0$ iff $\mathcal{R}_0 > 1$. The converse also follows directly from the equilibrium condition: at any endemic equilibrium with $i^* > 0$, dividing $\dot{i} = 0$ by i^* gives $\alpha^* = \gamma_R + \nu_m$, and since $P_S^* < 1$ whenever $i^*, r^* > 0$, the identity $\alpha^* = c_I\beta P_S^*$ forces $c_I\beta = (\gamma_R + \nu_m)/P_S^* > \gamma_R + \nu_m$, i.e. $\mathcal{R}_0 > 1$.

Given $i^* > 0$, the expression (5.22) gives $r^* > 0$ directly. Positivity of s^* follows from the positivity of the numerator and denominator of the expression in (5.22). Uniqueness follows from the fact that (5.20) is linear in i^* and has a unique positive solution when the right-hand side and the coefficient of i^*/ν_m have the same sign, which occurs exactly when $\mathcal{R}_0 > 1$. □

5.5.3 The Equal-Contact-Rate Special Case

When the three contact rates coincide, $c_S = c_I = c_R = c$, the formulas (5.21) and (5.22) simplify substantially. With $c_S = c_R = c$ and $c_I = c$, the denominator of (5.21) becomes

$$c(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c\nu_m + c\gamma_R \;=\; c\big[(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + (\gamma_R + \nu_m)\big] \;=\; c\mathcal{R}_0(\gamma_R + \nu_m),$$

so

$$i^* = \frac{n^*\nu_m(\mathcal{R}_0 - 1)}{\mathcal{R}_0(\gamma_R + \nu_m)} = \frac{n^*\nu_m}{\gamma_R + \nu_m}\left(1 - \frac{1}{\mathcal{R}_0}\right). \tag{5.23}$$

Substituting into (5.22) and simplifying,

$$s^* = \frac{n^*}{\mathcal{R}_0}, \qquad r^* = \frac{n^*\gamma_R}{\gamma_R + \nu_m}\left(1 - \frac{1}{\mathcal{R}_0}\right). \tag{5.24}$$

The relation $s^* = n^*/\mathcal{R}_0$ is the classical result of SIR theory with demographic turnover [Hethcote, 2000]: at endemic equilibrium, the susceptible fraction equals the reciprocal of the basic reproductive number. The result has an intuitive interpretation: s^*/n^* is the susceptible fraction required to make the effective reproductive number equal to 1, so that the infection persists at steady state without growing or decaying. Substituting $s^* = n^*/\mathcal{R}_0$ into $\mathcal{R}_e = \mathcal{R}_0\, s^*/n^*$ confirms this: $\mathcal{R}_e = \mathcal{R}_0\,(n^*/\mathcal{R}_0)/n^* = 1$.

Recovery of the classical expressions. Most textbook treatments of SIR dynamics with demographic turnover [Anderson and May, 1991, Keeling and Rohani, 2008, Martcheva, 2015] present (5.23) and (5.24) as the canonical endemic equilibrium. The infected and susceptible viewpoints differ not in this equilibrium but in the sensitivity of parameter estimates to uncertainty in s, the subject of Chapters 7 and 9.

Figure 5.1 displays the resulting transcritical bifurcation: the disease-free equilibrium $i^* = 0$ is stable for $\mathcal{R}_0 < 1$ and loses stability at $\mathcal{R}_0 = 1$, where it exchanges stability with the endemic branch (5.23). For $\mathcal{R}_0 > 1$ the endemic branch carries a positive prevalence $i^* > 0$ and is the stable attractor; the exchange of stability between the two branches at the threshold is the signature of a transcritical bifurcation. The endemic relation (5.23) solves the equilibrium equations for every $\mathcal{R}_0$, but for $\mathcal{R}_0 < 1$ it returns $i^* < 0$, which has no biological meaning and is not plotted.

5.5.4 The Final Size of a Closed-Population Outbreak

The endemic equilibrium analysis above assumes the population is open: births and deaths keep the demography turning over so that new susceptibles continually replenish the susceptible pool. In the complementary limit of a closed population ($\nu_m = 0$, so no births and no disease-unrelated deaths), the dynamics reach a different steady state. The infection eventually runs its course, the infectious compartment empties ($i(\infty) = 0$), and the susceptible and recovered compartments settle at asymptotic values $s(\infty)$ and $r(\infty) = n - s(\infty)$. The question of how large the outbreak grows before it ends (how many people ultimately get infected) is answered by the *final size relation*.

The derivation, carried out here for equal contact rates ($c_S = c_I = c_R$), is a one-page calculation that exploits the separation of fast and slow dynamics already used in Chapter 3. For the closed-population SIR system, conservation gives $s(t)+i(t)+r(t) = n$ at all times. Dividing the $\dot{s}$ equation by the $\dot{r}$ equation eliminates time. In the closed population, infection is the only outflow from s ($\dot{s} = -\alpha i$) and recovery the only inflow to r ($\dot{r} = \gamma_R i$), so the infectious count cancels:

$$\frac{ds}{dr} = \frac{\dot{s}}{\dot{r}} = \frac{-\alpha i}{\gamma_R i} = -\frac{\alpha}{\gamma_R} = -\frac{c_I\beta}{\gamma_R}\frac{s}{n} = -\mathcal{R}_0\frac{s}{n}.$$

The fourth equality uses the force from infection $\alpha = c_I\beta P_S$ with the equal-contact reduction $P_S = s/n$; the last collects $\mathcal{R}_0 = c_I\beta/\gamma_R$, the basic reproductive number of the closed population

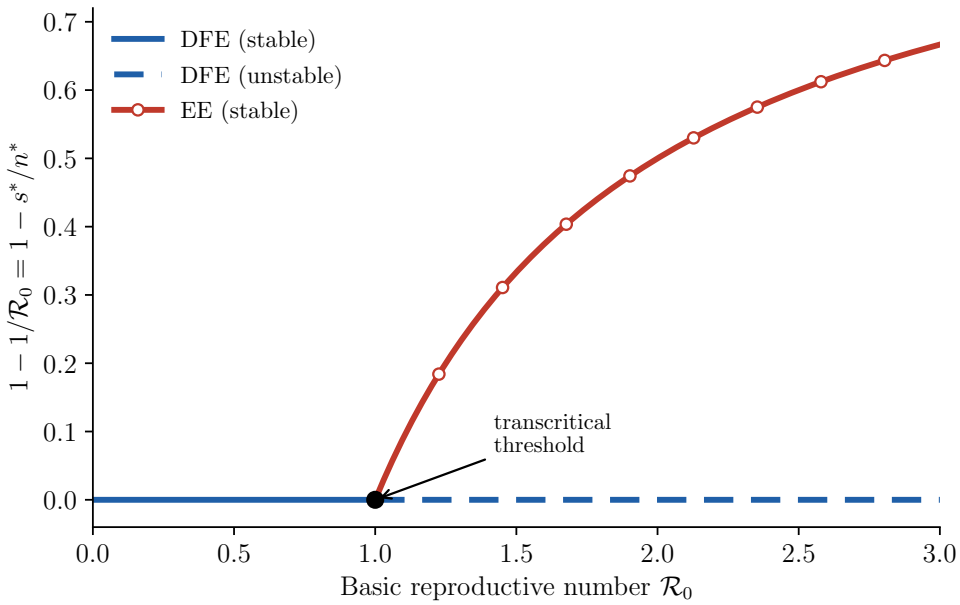


Figure 5.1: Bifurcation diagram for the equal-contact-rate SIR model: the scaled quantity $1 - 1/\mathcal{R}_0$ as a function of $\mathcal{R}_0$. The disease-free equilibrium (DFE) is stable (solid blue) for $\mathcal{R}_0 < 1$ and unstable (dashed blue) for $\mathcal{R}_0 > 1$. The endemic equilibrium (EE, solid red) is positive and stable for $\mathcal{R}_0 > 1$, exchanging stability with the DFE at the transcritical threshold $\mathcal{R}_0 = 1$. The actual prevalence is $i^*/n^* = (1 - 1/\mathcal{R}_0)\,\nu_m/(\gamma_R + \nu_m)$; plotting the scaled form $(1 - 1/\mathcal{R}_0) = 1 - s^*/n^*$ makes the bifurcation structure independent of the residence time τ_m.

($\nu_m = 0$). Separating variables and integrating from $t = 0$ (where $s(0) = s_0$, $r(0) = 0$) to $t = \infty$:

$$\ln \frac{s(\infty)}{s_0} \;=\; -\mathcal{R}_0 \cdot \frac{r(\infty)}{n} \;=\; -\mathcal{R}_0 \cdot \frac{n - s(\infty)}{n}. \tag{5.25}$$

This is the classical final size relation of Kermack and McKendrick [1927]. Modern textbook derivations of the final-size relation appear in Brauer et al. [2019, Sec. 2.4] and Li [2018, Sec. 2.1.3]. It expresses the asymptotic susceptible fraction implicitly through a transcendental equation solvable by fixed-point iteration or root-finding. Two features deserve emphasis. First, $s(\infty) > 0$ strictly: some fraction of the population always escapes infection, even when $\mathcal{R}_0 > 1$. The outbreak stops when the effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0 s(t)/n$ falls below 1, which happens before the susceptible pool is fully depleted. Second, $s(\infty)/s_0$ depends only on $\mathcal{R}_0$, not on the separate values of c_I, β, or γ_R. The total epidemic size is an invasion-facing quantity in the sense of Section 5.7: it is determined by the same combination of parameters that sets the threshold for whether invasion happens at all.

For $\mathcal{R}_0 = 2$ and $s_0/n = 1$, solving (5.25) gives $s(\infty)/n \approx 0.203$, so roughly 80% of the population has been infected (cumulatively) by the end of the outbreak. For $\mathcal{R}_0 = 3$, the asymptotic susceptible fraction drops to about 6%, and for $\mathcal{R}_0 = 5$, to less than 1%. The transcendental nonlinearity of (5.25) means that modest changes in $\mathcal{R}_0$ near the threshold value 1 produce large changes in the attack rate, while changes well above threshold produce diminishing changes, a pattern visible in any classical SIR final-size curve.

The final size relation is the closed-population analog of the endemic equilibrium. Both are answers to "where does the system end up" for their respective demographic settings. The open-population SIR settles at an endemic equilibrium $i^* > 0$; the closed-population SIR settles at a disease-free asymptotic state with $s(\infty) > 0$. The relation (5.25) quantifies how much of the susceptible pool is consumed before the outbreak self-terminates.

Exercises for this section

1. **[D]** Using the baseline parameter set, compute s^*, i^*, r^* from (5.21) and (5.22). Verify that $s^* + i^* + r^* = n^* = 1$ (the normalization convention of Chapter 4, Section 4.5.3).

5.6 Stability of the Endemic Equilibrium

The stability analysis at the endemic equilibrium is more involved than at the DFE, because the partial derivatives of $\alpha(s, i, r)i$ are all nonzero when $i^* > 0$. We first handle the equal-contact-rate case, where the analysis reduces to a tractable Routh–Hurwitz calculation, and then extend to distinct contact rates by a continuity argument.

5.6.1 Equal-Contact-Rate Case

When $c_S = c_I = c_R = c$, the contact-weighted susceptible fraction reduces to $P_S = s/n$, and the system (5.1) becomes

$$\dot{s} = \nu_m - c\beta\frac{si}{n} - \nu_m s, \quad \dot{i} = c\beta\frac{si}{n} - (\gamma_R + \nu_m)i, \quad \dot{r} = \gamma_R i - \nu_m r,$$

where $n = s + i + r = 1$ (constant). The Jacobian at the EE given by (5.24) and (5.23) is

$$\mathbf{J}_{\mathrm{EE}} \;=\; \begin{pmatrix} -\nu_m \mathcal{R}_0 & -(\gamma_R + \nu_m) & 0 \\ \nu_m(\mathcal{R}_0 - 1) & 0 & 0 \\ 0 & \gamma_R & -\nu_m \end{pmatrix}.$$

(The derivation of the matrix entries is straightforward but algebraically tedious; we omit the details.) The block structure shows that $\sigma_3 = -\nu_m$ is always an eigenvalue, and the remaining two eigenvalues are those of the upper-left 2×2 block.

The characteristic polynomial of the 2×2 block is

$$\sigma^2 + \nu_m \mathcal{R}_0 \sigma + \nu_m(\gamma_R + \nu_m)(\mathcal{R}_0 - 1) \;=\; 0. \tag{5.26}$$

Both coefficients are positive when $\mathcal{R}_0 > 1$: the coefficient of σ is $\nu_m \mathcal{R}_0 > 0$, and the constant term is $\nu_m(\gamma_R + \nu_m)(\mathcal{R}_0 - 1) > 0$. By the Routh–Hurwitz criterion for quadratic polynomials, both roots have negative real part, and the equal-contact-rate EE is locally asymptotically stable when $\mathcal{R}_0 > 1$.

5.6.2 General Case

For distinct contact rates, the algebraic computation is heavier than the equal-contact case but tractable, and we present it in full because Theorem 5.5 below depends on more than continuity from the equal-contact baseline. We linearize the SIR system at the endemic equilibrium (s^*, i^*, r^*) and analyze the characteristic polynomial of the Jacobian.

The general-case Jacobian. At the EE, the Jacobian of the right-hand side of (5.1) with respect to (s, i, r) is

$$\mathbf{J}^* \;=\; \begin{pmatrix} -A^* - \nu_m & -B^* & -C^* \\ A^* & B^* - \gamma_R - \nu_m & C^* \\ 0 & \gamma_R & -\nu_m \end{pmatrix}, \tag{5.27}$$

where $A^* = \partial(\alpha i)/\partial s|_*$, $B^* = \partial(\alpha i)/\partial i|_*$, and $C^* = \partial(\alpha i)/\partial r|_*$ are the three partial derivatives of the saturating incidence $\alpha(s,i,r)i = c_I \beta P_S(s,i,r) i$ evaluated at the EE. These coefficients are the linearization of the infection term about the endemic equilibrium: writing $(s, i, r) = (s^*, i^*, r^*) + (\tilde{s}, \tilde{i}, \tilde{r})$, the first-order change in $\alpha(s,i,r)\, i$ is $A^* \tilde{s} + B^* \tilde{i} + C^* \tilde{r}$. Explicit calculation using $P_S = c_S s/(c_S s + c_I i + c_R r)$ gives, after collecting terms,

$$A^* \;=\; \frac{c_I \beta\, c_S\, i^* (c_I i^* + c_R r^*)}{(c_S s^* + c_I i^* + c_R r^*)^2},$$

$$B^* \;=\; \alpha^* - \frac{c_I^2 \beta\, c_S s^*\, i^*}{(c_S s^* + c_I i^* + c_R r^*)^2}, \qquad C^* \;=\; -\frac{c_I \beta\, c_R\, c_S s^*\, i^*}{(c_S s^* + c_I i^* + c_R r^*)^2},$$

with $\alpha^* = c_I \beta P_S^* = \gamma_R + \nu_m$ at the EE. The exact algebraic forms of A^*, B^*, C^* matter less than their signs and orders of magnitude: $A^* > 0$ (susceptibles entering the infected compartment at a positive rate), $B^* < \alpha^*$ (the per-infectious indirect effect is dominated by the direct α term but reduced by self-saturation), and $C^* < 0$ (recovered individuals dilute the infectious-contact-weighted denominator).

The characteristic polynomial. Computing $\det(\mathbf{J}^* - \sigma\mathbf{I}) = 0$ gives a cubic

$$\sigma^3 + a_1\sigma^2 + a_2\sigma + a_3 \;=\; 0, \tag{5.28}$$

with coefficients

$$\begin{aligned} a_1 &= A^* + (\gamma_R + \nu_m - B^*) + 2\nu_m, \\ a_2 &= A^*(\gamma_R + \nu_m) + \nu_m(\gamma_R + \nu_m - B^*) - \gamma_R C^* + \nu_m\big[(A^* + \nu_m) + (\gamma_R + \nu_m - B^*)\big], \\ a_3 &= \nu_m\big[A^*(\gamma_R + \nu_m) + \nu_m(\gamma_R + \nu_m - B^*) - \gamma_R C^*\big]. \end{aligned}$$

Sign analysis under $\mathcal{R}_0 > 1$. At the EE with $\mathcal{R}_0 > 1$, three structural inequalities hold:

(i) $A^* > 0$ (susceptibles are flowing into infectives at a positive rate at the EE).

(ii) $\gamma_R + \nu_m - B^* > 0$. To see this, note that $B^* = \alpha^* -$ (positive correction); since $\alpha^* = \gamma_R + \nu_m$ at the EE and the correction $c_I^2\beta\, c_S s^* i^*/(c_S s^* + c_I i^* + c_R r^*)^2$ is strictly positive when $i^* > 0$, we have $B^* < \alpha^* = \gamma_R + \nu_m$, giving $\gamma_R + \nu_m - B^* > 0$ strictly.

(iii) $C^* < 0$, as computed above (recovered individuals dilute the contact-weighted denominator), so $-\gamma_R C^* > 0$. No sign condition on the combination $A^*B^* + \gamma_R C^*$ is needed; it can be negative at the endemic equilibrium (Exercise 6).

From (i)–(iii), all three coefficients a_1, a_2, a_3 are strictly positive.

Routh–Hurwitz check. For a cubic $\sigma^3 + a_1\sigma^2 + a_2\sigma + a_3$ with $a_1, a_2, a_3 > 0$, the Routh–Hurwitz criterion requires the additional inequality $a_1 a_2 > a_3$. The verification is cleanest after a change of notation that collects the three quantities established as positive above:

$$P \;=\; A^* + \nu_m, \qquad Q \;=\; \gamma_R + \nu_m - B^*, \qquad E \;=\; A^*(\gamma_R + \nu_m) + \nu_m Q - \gamma_R C^*,$$

so that $P > 0$ by (i), $Q > 0$ by (ii), and $E > 0$ by (i)–(iii). In these variables the coefficients take the compact form

$$a_1 \;=\; P + Q + \nu_m, \qquad a_2 \;=\; E + \nu_m(P + Q), \qquad a_3 \;=\; \nu_m E.$$

Expanding the product and cancelling gives the factorization

$$a_1 a_2 - a_3 \;=\; (P + Q)\,\big[\,E \;+\; \nu_m(P + Q) \;+\; \nu_m^2\,\big]. \tag{5.29}$$

The cancellation is worth noting: the term $\nu_m E$ produced by the ν_m in a_1 is exactly a_3, so a_3 is removed in full and no negative contribution survives. Every factor on the right of (5.29) is then a sum of strictly positive quantities, so $a_1 a_2 - a_3 > 0$ whenever (i)–(iii) hold. The Routh–Hurwitz criterion is satisfied; all three roots of (5.28) have negative real part; the EE is locally asymptotically stable.

Theorem 5.5 (Stability of the endemic equilibrium)**.** *Assume $\mathcal{R}_0 > 1$ and the contact rates satisfy $c_S, c_I, c_R > 0$. Then the endemic equilibrium (s^*, i^*, r^*) given by (5.21) and (5.22) is locally asymptotically stable.*

Proof. The equal-contact-rate case is established by the Routh–Hurwitz analysis of (5.26). For the general case, we have shown above that the coefficients of the characteristic cubic (5.28) are all strictly positive (consequences of inequalities (i)–(iii)) and that $a_1 a_2 > a_3$ strictly. By the Routh–Hurwitz criterion for cubics, all three roots of (5.28) have negative real part, so the EE is locally asymptotically stable. □

Global behavior. Theorem 5.5 establishes local stability only: perturbations sufficiently close to the EE decay back to it. The corresponding global result would establish that every initial condition in the biologically relevant region $\{(s,i,r) : s,r \geq 0,\ i > 0,\ s+i+r = 1\}$ converges to the EE when $\mathcal{R}_0 > 1$. Global stability can be proven for the equal-contact-rate case using a Lyapunov function of the form $V(s,i) = s - s^* - s^* \ln(s/s^*) + i - i^* - i^* \ln(i/i^*)$; see Korobeinikov [2004] for the general theory. For distinct contact rates, global stability follows from the same construction under mild additional assumptions on the parameters; we do not pursue the full argument here.

Figure 5.2 shows the global picture numerically for a supercritical regime. Trajectories from a range of initial conditions spiral into the endemic equilibrium (red star); the disease-free equilibrium at $(s,i) = (1,0)$ is unstable, and no trajectory with $i(0) > 0$ approaches it. The structure is consistent with global asymptotic stability of the EE in the interior of the positive orthant.

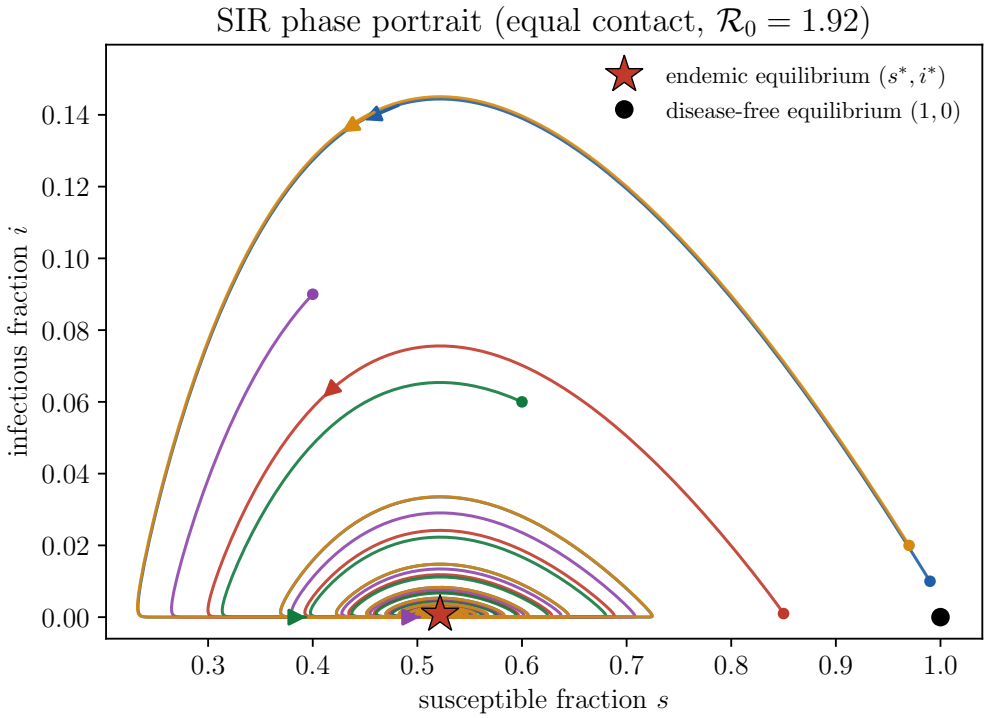


Figure 5.2: Phase portrait of the equal-contact-rate SIR model in the (s,i) plane at $\mathcal{R}_0 = 1.92$. Trajectories from five initial conditions (color-coded) converge to the endemic equilibrium (red star) via damped oscillations in the i-component. Colors distinguish the five initial conditions, not time; the arrows along each trajectory indicate increasing time. The disease-free equilibrium at $(1,0)$ (black dot) is a saddle: small perturbations with $i > 0$ grow away from it. Parameters: $c_I\beta = 0.16$ day^{-1}, $\gamma_R + \nu_m \approx 0.0835$ day^{-1}.

5.7 The Invasion–Burden Distinction

The Preface (Figure 2) introduced the invasion–burden distinction informally. With the reproductive numbers and the endemic equilibrium now in hand, we can state the distinction formally and draw out its consequences.

$\mathcal{R}_0$ governs invasion and i^* governs burden, and the two depend on structurally different parameter sets; reading one for the other is the pitfall Invasion versus Equilibrium (Chapter 16).

The six independent parameters of the SIR model are $(c_S, c_I, c_R, \beta, \tau_R, \tau_m)$. The basic reproductive number (5.3) depends on four of them:

$$\mathcal{R}_0 \;=\; \frac{c_I \beta}{\gamma_R + \nu_m} \;=\; \frac{c_I \beta\, \tau_R \tau_m}{\tau_R + \tau_m}.$$

It depends on c_I, β, τ_R, and τ_m, but not on c_S or c_R. Of these four parameters, c_I and β enter multiplicatively (Pitfall: "Independence of Parameters of Interest"): each produces a proportional change in $\mathcal{R}_0$. The time constants τ_R and τ_m enter through their harmonic mean $\tau_R^* = 1/(\gamma_R + \nu_m)$, whose structure we examine shortly.

The endemic prevalence i^* in the equal-contact-rate case is (5.23):

$$i^* \;=\; \frac{n^* \nu_m}{\gamma_R + \nu_m} \left(1 - \frac{1}{\mathcal{R}_0}\right).$$

This expression depends on three parameters: ν_m, γ_R, and $\mathcal{R}_0$ (which in turn absorbs c_I and β).

5.7.1 The Parameter Partition

Consider the question: which parameters, changed in isolation, change $\mathcal{R}_0$ at the leading order, and which change i^* at the leading order? We compute the logarithmic derivatives of both quantities with respect to each of the six parameters, which gives the relative sensitivity indices of Section 9.2 (Chapter 9 develops this machinery in detail).

These indices are collected once, exactly, in the table of Theorem 5.6 below (equation (5.30)); we read the partition off that table here. The reproductive number $\mathcal{R}_0$ has unit sensitivity to c_I and β, a smaller sensitivity to the infectious period τ_R, a sensitivity to the demographic constant τ_m that is negligible in the typical regime $\gamma_R \gg \nu_m$ (recovery on the scale of days, migration on the scale of a year or more), and identically zero sensitivity to c_S and c_R. At leading order, then, $\mathcal{R}_0$ is dominated by c_I, β, and τ_R.

The burden sensitivities, the $\Upsilon_p^{i^*}$ column of the same table, show the opposite pattern. When $\mathcal{R}_0$ is modestly above unity (say $\mathcal{R}_0 = 2$), the sensitivity of i^* to c_I is $1/(\mathcal{R}_0 - 1) = 1$, comparable to that of $\mathcal{R}_0$. But the sensitivity of i^* to τ_m has magnitude close to unity (the -1 term dominates when $\nu_m \ll \gamma_R$), while the sensitivity of $\mathcal{R}_0$ to τ_m is close to zero. Conversely, when $\mathcal{R}_0$ is large (say $\mathcal{R}_0 = 10$), the sensitivity of i^* to c_I is only $1/9$, while the sensitivity of $\mathcal{R}_0$ to c_I remains exactly 1.

The negative sign in $\partial \log i^* / \partial \log \tau_m$ reflects a real mechanism: longer demographic residence (larger τ_m, smaller ν_m) means fewer new susceptibles flowing into the population per unit time, which lowers the endemic prevalence at the steady state. The magnitude of this sensitivity approaches one regardless of $\mathcal{R}_0$ in the typical regime $\gamma_R \gg \nu_m$, a fact we formalize below.

These observations formalize the partition. Define:

Invasion parameters. The invasion parameters are c_I and β. They appear in $\mathcal{R}_0$ with full unit sensitivity and in i^* with sensitivity that decays as $1/(\mathcal{R}_0 - 1)$.

Burden parameters. The burden parameter is primarily τ_m (equivalently ν_m). It appears in i^* with sensitivity near unity for $\mathcal{R}_0$ modestly above 1 but in $\mathcal{R}_0$ with sensitivity near zero when $\gamma_R \gg \nu_m$.

Mixed parameter. The infectious period τ_R (equivalently γ_R) is a mixed parameter: it appears substantially in both $\mathcal{R}_0$ and i^*, because it controls the rate of transition from i to r, which affects both the generation of new infections (via τ_R^*) and the equilibrium balance.

The contact rates c_S and c_R enter neither $\mathcal{R}_0$ nor the equal-contact-rate i^* directly. Their influence on the endemic equilibrium in the distinct-contact-rate case is through the denominator structure of (5.21); the resulting effect is small for typical parameter regimes.

The partition is shown graphically in Figure 5.3, which plots each parameter as a point in the plane of its two sensitivity magnitudes: invasion sensitivity $|\Upsilon_p^{\mathcal{R}_0}|$ along the horizontal axis and burden sensitivity $|\Upsilon_p^{i^*}|$ along the vertical. The figure makes visible what the algebra describes: c_I and β cluster in the invasion-facing corner, τ_m sits in the burden-facing corner, τ_R occupies the mixed region, and the passive contact rates c_S and c_R sit near the origin.

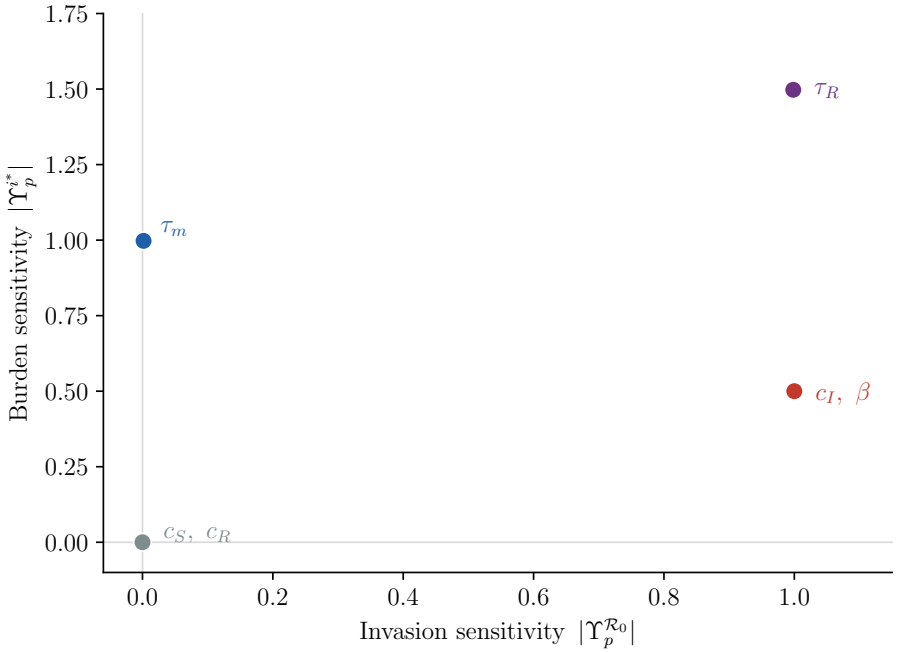


Figure 5.3: The six SIR parameters plotted in the plane of their invasion sensitivity ($|\Upsilon_p^{\mathcal{R}_0}|$) and burden sensitivity ($|\Upsilon_p^{i^*}|$) at $\mathcal{R}_0 = 3$ (equal-contact-rate case). Invasion-facing parameters (c_I, β) sit in the lower-right quadrant; the burden-facing parameter (τ_m) sits in the upper-left; the mixed parameter (τ_R) sits in the upper-right; and the passive contact rates (c_S, c_R) cluster near the origin. The transmission parameters' burden sensitivity scales as $1/(\mathcal{R}_0 - 1)$, so the separation between the invasion-facing and burden-facing groups sharpens as $\mathcal{R}_0$ increases above threshold. The partition clarifies which interventions affect the outbreak threshold versus the long-run endemic burden.

5.7.2 The Sensitivity Asymmetry Theorem

The informal partition above becomes a sharp statement in the separation-of-timescales regime $\gamma_R \gg \nu_m$ that governs all acute infections. We record the result as a theorem because it is the central structural claim behind the book's organizing framework.

Theorem 5.6 (Invasion–burden sensitivity asymmetry)**.** *Consider the equal-contact-rate* SIR *model at endemic equilibrium with* $\mathcal{R}_0 > 1$. *The six relative sensitivity indices* $\Upsilon_p^{\mathcal{R}_0} = \partial \log \mathcal{R}_0 / \partial \log p$ *and*

$\Upsilon_p^{i^*} = \partial \log i^* / \partial \log p$ *for* $p \in \{c_I, \beta, \tau_R, \tau_m, c_S, c_R\}$ *are given exactly by*

p	$\Upsilon_p^{\mathcal{R}_0}$	$\Upsilon_p^{i^*}$
c_I	1	$\dfrac{1}{\mathcal{R}_0 - 1}$
β	1	$\dfrac{1}{\mathcal{R}_0 - 1}$
τ_R	$\dfrac{\gamma_R}{\gamma_R + \nu_m}$	$\dfrac{\gamma_R}{\gamma_R + \nu_m} \cdot \dfrac{\mathcal{R}_0}{\mathcal{R}_0 - 1}$
τ_m	$\dfrac{\nu_m}{\gamma_R + \nu_m}$	$-1 + \dfrac{\nu_m}{\gamma_R + \nu_m} \cdot \dfrac{\mathcal{R}_0}{\mathcal{R}_0 - 1}$
c_S	0	0
c_R	0	0

(5.30)

In the separation-of-timescales regime $\gamma_R \gg \nu_m$*, the table contracts at leading order to*

p	$\Upsilon_p^{\mathcal{R}_0}$	$\Upsilon_p^{i^*}$
$c_I,\ \beta$	1	$1/(\mathcal{R}_0 - 1)$
τ_R	1	$\mathcal{R}_0/(\mathcal{R}_0 - 1)$
τ_m	ν_m/γ_R	-1
$c_S,\ c_R$	0	0

(5.31)

Consequently the invasion-facing and burden-facing parameters exhibit opposite asymmetries:

$$\frac{|\Upsilon_{c_I}^{i^*}|}{|\Upsilon_{c_I}^{\mathcal{R}_0}|} = \frac{1}{\mathcal{R}_0 - 1}, \qquad \frac{|\Upsilon_{\tau_m}^{i^*}|}{|\Upsilon_{\tau_m}^{\mathcal{R}_0}|} \approx \frac{\gamma_R}{\nu_m}. \tag{5.32}$$

For typical acute-infection parameters ($\gamma_R \sim 1/\text{week}$, $\nu_m \sim 1/\text{lifetime}$)*, the right-hand ratio in* (5.32) *exceeds* 10^3.

Proof. Starting from the explicit equal-contact-rate formulas $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ and $i^* = n^*\nu_m(\mathcal{R}_0 - 1)/[\mathcal{R}_0(\gamma_R + \nu_m)]$, each entry in (5.30) follows by direct logarithmic differentiation. For illustration, the τ_m row: writing $\nu_m = 1/\tau_m$, we compute

$$\frac{\partial \log \mathcal{R}_0}{\partial \log \tau_m} = \frac{\partial \log \mathcal{R}_0}{\partial \log \nu_m} \cdot \frac{\partial \log \nu_m}{\partial \log \tau_m} = \left(-\frac{\nu_m}{\gamma_R + \nu_m}\right)(-1) = \frac{\nu_m}{\gamma_R + \nu_m}.$$

For the burden sensitivity, differentiating $\log i^* = \log n^* + \log \nu_m + \log(\mathcal{R}_0 - 1) - \log \mathcal{R}_0 - \log(\gamma_R + \nu_m)$ with respect to $\log \tau_m$ and using the chain rule gives

$$\frac{\partial \log i^*}{\partial \log \tau_m} = -1 + \frac{\mathcal{R}_0}{\mathcal{R}_0 - 1} \cdot \frac{\nu_m}{\gamma_R + \nu_m} - \frac{\nu_m}{\gamma_R + \nu_m} + \frac{\nu_m}{\gamma_R + \nu_m} = -1 + \frac{\nu_m}{\gamma_R + \nu_m} \cdot \frac{\mathcal{R}_0}{\mathcal{R}_0 - 1},$$

where the -1 comes from $\partial \log \nu_m / \partial \log \tau_m = -1$ and the remaining terms collect the indirect effects through $\mathcal{R}_0$ and $\gamma_R + \nu_m$. The other rows are analogous. That c_S and c_R produce zero sensitivity follows from the equal-contact-rate assumption $c_S = c_I = c_R$ reducing them to the single quantity c, which we identify with c_I in the table.

The asymptotic regime $\gamma_R \gg \nu_m$ yields (5.31) by Taylor expansion in ν_m/γ_R. The asymmetry ratios (5.32) are a direct reading of the asymptotic table. □

Corollary 5.7 (Policy asymmetry)**.** *Let* $\varepsilon > 0$ *be small. Consider three classes of intervention applied to the equal-contact-rate* SIR *model at endemic equilibrium with* $\mathcal{R}_0 > 1$*:*

(a) *An intervention reducing a transmission parameter (c_I or β) by fraction ε reduces $\mathcal{R}_0$ by fraction ε and reduces i^* by fraction $\varepsilon/(\mathcal{R}_0 - 1)$.*

(b) *An intervention reducing the infectious period τ_R by fraction ε reduces $\mathcal{R}_0$ by fraction $\varepsilon\gamma_R/(\gamma_R + \nu_m)$ (near ε for acute infections) and reduces i^* by fraction approximately $\varepsilon\mathcal{R}_0/(\mathcal{R}_0 - 1)$.*

(c) *An intervention that* increases *the demographic timescale τ_m by fraction ε raises $\mathcal{R}_0$ by only $\varepsilon\nu_m/(\gamma_R + \nu_m) \approx 0$ while reducing i^* by fraction approximately ε (for acute infections). The direction matters: because $\Upsilon^{i^*}_{\tau_m} \approx -1$ while $\Upsilon^{\mathcal{R}_0}_{\tau_m} \approx 0$,* decreasing *$\tau_m$ lowers $\mathcal{R}_0$ negligibly but* raises *the endemic burden. No choice of direction reduces both quantities; the demographic parameter trades invasion against burden rather than improving both.*

In particular, transmission-focused interventions are strongly invasion-directed, demographic interventions are strongly burden-directed, and infectious-period interventions affect both. A transmission intervention strong enough to drive $\mathcal{R}_0 < 1$ eliminates the disease, an outcome inaccessible to burden-only interventions.

Remark 5.8 (On the qualifier "approximately"). The approximations in parts (b) and (c) of Corollary 5.7 hold at leading order in ν_m/γ_R; for acute infections this ratio is of order 10^{-3} or smaller, so the approximations are accurate to three significant figures. For slow infections (tuberculosis latency, HIV progression) where the infectious or latent stages occupy years, ν_m/γ_R can be closer to unity and the exact formulas (5.30) must be used. Chapter 9 develops the finite ν_m/γ_R corrections in numerical detail.

Remark 5.9 (Extension to generalizations). Theorem 5.6 is stated for the basic SIR model, but the structural message persists across the generalizations of Chapter 10. Proposition 10.2 in Section 10.7 makes the extension precise: for any compartmental model with a well-defined next-generation-matrix reproductive number and disease-free and endemic equilibria, parameters separate into invasion-facing (entering $\mathcal{R}_0$), burden-facing (entering only the endemic distribution), and mixed (entering both), with sensitivity asymmetries of the same qualitative form as in (5.32).

5.7.3 Policy Implications

The parameter partition has concrete implications for the design of epidemic interventions.

Interventions targeting invasion parameters. An intervention that reduces c_I (case isolation, quarantine, voluntary withdrawal by symptomatic individuals) acts directly on $\mathcal{R}_0$ with unit sensitivity: a 30% reduction in c_I reduces $\mathcal{R}_0$ by 30%. If the intervention is strong enough to bring $\mathcal{R}_0$ below 1, the outbreak cannot sustain itself at all, and the DFE becomes the globally attracting equilibrium (Theorem 5.3).

Similarly, an intervention that reduces β (masking, improved ventilation, hand hygiene for respiratory infections; condom use for sexually-transmitted infections) acts on $\mathcal{R}_0$ with unit sensitivity.

Such interventions are most consequential when applied early, where their primary effect is to prevent invasion. A post-invasion reduction in c_I still lowers the endemic burden, but the mechanism differs: in the equal-contact-rate case i^* depends on c_I only through $\mathcal{R}_0$, so reducing c_I moves i^* only by moving $\mathcal{R}_0$; in the general model c_I also enters the endemic prevalence (5.21) directly, so it changes i^* through both channels. The invasion–burden distinction concerns these differing mechanisms and sensitivities; it does not mean that c_I is absent from i^*.

Interventions targeting burden parameters. An intervention that affects the endemic burden i^* without substantially changing $\mathcal{R}_0$ operates on a different lever. Demographic turnover (ν_m) is typically not an intervention parameter (we cannot in practice change how long people live or reside in a region). The infectious period τ_R is, in principle, an intervention parameter: treatment that shortens the infectious period reduces both $\mathcal{R}_0$ and the endemic value.

The picture for τ_R is more complicated than for c_I: reducing τ_R is an invasion intervention and a burden intervention at once.

Designing combined interventions. The practical implication of the partition is that combined invasion-and-burden interventions can be designed with each intervention targeting its natural parameter group. A case-isolation policy (reducing c_I) addresses invasion; a treatment program (reducing τ_R) addresses both invasion and burden; demographic and environmental factors govern burden through τ_m. Understanding which parameter an intervention targets clarifies what the intervention can and cannot accomplish and prevents the common error of expecting an invasion intervention to reduce burden when in fact the threshold has already been crossed and the system is approaching its endemic equilibrium.

Observability of the partition. The partition also shapes what different data can reveal. Early-outbreak data constrain the invasion parameters via the initial growth rate σ_2 and the doubling time t_{double} (see (5.16)), while the burden parameters can only be constrained by longer-term data covering the epidemic peak or the approach to endemic equilibrium.

Constant parameters as a standing assumption. The partition just drawn takes the model parameters to be constant in time (Pitfall: "Constant-Transmission Assumption Ignoring Behavioral Feedback"); all of the time variation lives in the state, through the susceptible depletion that carries $\mathcal{R}_e(t) = \mathcal{R}_0\, P_S(t)$ below one. When the parameters themselves vary with calendar time—under seasonal forcing, behavioral change, or staged interventions—the invasion question changes character, and a reproduction number formed from time-averaged parameters need not retain its threshold meaning. Chapter 6, Section 6.8, develops this once the constant-parameter $\mathcal{R}_e(t)$ has been seen evolve numerically. The constant-parameter reading should be revisited where behavioral feedback is empirically strong: Gozzi et al. [2025] document that models incorporating behavioral feedback outperform constant-β models in retrospective fits of the first COVID-19 wave across multiple geographies.

Exercises for this section

1. **[I]** Consider a reduction in c_I from its baseline value (4 contacts/day) to 2 contacts/day, with all other parameters unchanged. Compute the new $\mathcal{R}_0$ and verify that it is below 1. What does this say about the DFE's stability after the intervention?

5.8 Summary and Forward Pointers

This chapter has developed the analytical core of the SIR model. The principal results are:

1. The basic reproductive number, derived by a heuristic argument and confirmed by the next-generation matrix method, is
$$\mathcal{R}_0 = \frac{c_I \beta}{\gamma_R + \nu_m}.$$
The effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0 P_S(t)$ governs whether the infectious population grows or declines at time t.

2. The disease-free equilibrium $(1, 0, 0)$ is locally asymptotically stable if and only if $\mathcal{R}_0 < 1$ (Theorem 5.3). When $\mathcal{R}_0 > 1$, small introductions grow exponentially at rate $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$, giving the doubling time (5.16).

3. The endemic equilibrium exists uniquely with $i^* > 0$ if and only if $\mathcal{R}_0 > 1$ (Proposition 5.4). Its closed-form expression (5.21) reduces in the equal-contact-rate case to the classical $s^* = n^*/\mathcal{R}_0$ result.

4. The endemic equilibrium is locally asymptotically stable (Theorem 5.5). The equal-contact-rate case follows from a Routh–Hurwitz analysis of a quadratic polynomial; the general case follows from a Routh–Hurwitz analysis of the cubic characteristic polynomial, whose coefficients and factorization Exercise 6 asks the reader to verify.

5. The parameters of the model partition into invasion parameters (c_I, β) which dominate $\mathcal{R}_0$, burden parameters (primarily τ_m) which dominate i^* when $\mathcal{R}_0$ is modestly above 1, and a mixed parameter τ_R which contributes to both. This partition has concrete implications for intervention design.

Chapter 6 illustrates these analytical results numerically for the baseline parameter set introduced in Chapter 4. Scenario 1 of Chapter 6 shows the epidemic trajectory and its approach to endemic equilibrium; Scenario 2 shows how a reduction in c_I implements the invasion-control intervention discussed in Section 5.7. Chapter 7 turns to the practical question of estimating the model parameters from incidence data, and Chapter 9 makes the invasion–burden partition precise through formal sensitivity analysis.

Pitfall check

This chapter has developed three of the book's common pitfalls: Pitfall: "Linearization Fidelity" (in the heuristic $\mathcal{R}_0$ derivation, where the frozen-P_S approximation is the structurally valid linearization), Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$" (in Section 5.2, where the relationship between the basic and effective reproductive numbers is made formal and the convention for the rest of the book is fixed), and Pitfall: "Invasion versus Equilibrium" (in Section 5.7, where the structural difference between the parameter sets that govern $\mathcal{R}_0$ and i^* is made formal). Before moving to Chapter 6, take a moment to confirm the following points are clear in your mind:

1. Why is the frozen-P_S approximation a valid linearization while a frozen-λ approximation is not?

2. What is the operational difference between $\mathcal{R}_0$ and $\mathcal{R}_e(t)$, and why does most of the literature actually estimate $\mathcal{R}_e$ even when calling it $\mathcal{R}_0$?

3. How does the invasion–burden distinction explain why an intervention that reduces $\mathcal{R}_0$ may have a different effect on endemic burden than the reduction in $\mathcal{R}_0$ alone would suggest?

If any of these are unclear, the corresponding pitfall block (or the catalog entry, accessible via the List of Common Pitfalls) is the place to revisit.

Exercises

Five section-level exercises appear above, following Section 5.1, Section 5.4, Section 5.5, and Section 5.7. The exercises below integrate across the chapter and have been ordered so that concept checks, derivations, and interpretation exercises alternate. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[C]** At the endemic equilibrium, $\mathcal{R}_e^* = 1$ exactly. Explain in plain English why this is necessarily true: what balance must hold in a stationary state of a deterministic transmission model?

2. **[D]** Verify by direct substitution that the equal-contact-rate expressions (5.23) and (5.24) follow from the general expressions (5.21) and (5.22) when $c_S = c_I = c_R = c$.

3. **[C]** Using only the sign of $\mathcal{R}_0 - 1$, state which of the two equilibria—the disease-free equilibrium (DFE) or the endemic equilibrium (EE)—is biologically meaningful and locally stable when (a) $\mathcal{R}_0 < 1$, (b) $\mathcal{R}_0 = 1$, and (c) $\mathcal{R}_0 > 1$. Your answer should not require any computation.

4. **[D]** Compute $\mathcal{R}_e(t)$ at the endemic equilibrium using (5.4) and verify that $\mathcal{R}_e^* = 1$, confirming that the endemic equilibrium is the state where the effective reproductive number sits exactly at threshold.

5. **[I]** Consider an intervention that reduces τ_R from 12 days to 6 days (treatment halving the infectious period) while holding all other parameters fixed. Compute the new $\mathcal{R}_0$ and the new equal-contact-rate i^*. What is the relative reduction in each? Compare with the relative reductions produced by halving c_I.

6. **[D]** (Extended exercise.) Expand $\det(\sigma\mathbf{I} - \mathbf{J}^*)$ for the Jacobian (5.27) and verify the coefficients a_1, a_2, a_3 of Section 5.6 and the factorization (5.29). Then evaluate $A^*B^* + \gamma_R C^*$ at the baseline endemic equilibrium. An earlier draft of this proof required that quantity to be positive; is it? What does the example teach about checking a lemma numerically before relying on it?

7. **[I]** Suppose baseline conditions give $\mathcal{R}_0 = 2$ and one wishes to reduce $\mathcal{R}_0$ to 1 by reducing c_I alone. What fraction of the baseline c_I value must be retained? Suppose instead one wishes to reduce $\mathcal{R}_0$ to 0.8. What fraction of the baseline c_I value is then required? Comment on the diminishing returns of case-isolation interventions as the target $\mathcal{R}_0$ approaches zero.

8. **[L]** *Spaced retrieval, Chapter 4.* Reach back to the model equations (4.4) of Chapter 4, Section 4.4. Linearize the system at the disease-free equilibrium $(1, 0, 0)$ symbolically, write out the 3×3 Jacobian $\mathbf{J}_{\text{DFE}}$, and identify its three eigenvalues by inspection (the system is block-triangular at the DFE). Then numerically: with the baseline parameters of this chapter, compute the dominant eigenvalue of $\mathbf{J}_{\text{DFE}}$ (the one with the largest real part; for this DFE all three eigenvalues are real) and verify that it equals $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$, where $\mathcal{R}_0$ is the heuristic value derived in Section 5.1. The exercise reinforces two ideas at once: (a) the model equations of Chapter 4 carry directly into the analytic machinery of this chapter, and (b) $\mathcal{R}_0 = 1$ is the threshold at which the dominant DFE eigenvalue changes sign: the spectral meaning of "threshold" is that the dominant DFE eigenvalue crosses zero.

9. **[I]** *Reasoning visibility.* A colleague shares a manuscript estimating $\mathcal{R}_0 = 4.2$ for a respiratory pathogen using parameters that look superficially similar to your own analysis, which gives $\mathcal{R}_0 = 2.1$. The discrepancy is a factor of two. List, in priority order, the first five items you would check to localize the source of the disagreement. For each item, state *why* you placed it where you did in the priority order: what is the relative likelihood that this is the source of the discrepancy, and how cheap is it to check? Distinguish parameter-value sources of disagreement (different choices of c_I, β, or τ_R) from formula-level sources (e.g., a per-cycle vs per-generation $\mathcal{R}_0$ convention) and from data-level sources (e.g., different fitted incidence series). For your top item, sketch the diagnostic test (one or two sentences) that would either confirm the source or rule it out. The exercise asks you to make explicit the reasoning that an experienced modeler would otherwise execute in their head; doing so once builds a reusable diagnostic template for parameter-disagreement questions.

10. **[Verify the AI]** *Auditing an AI-generated $\mathcal{R}_0$ derivation.* Prompt an AI tool: *"Derive the basic reproductive number $\mathcal{R}_0$ for the SIR model with vital dynamics, including the demographic turnover term. Show the derivation step by step."* Save the response. Audit it against the three structural-error categories from this chapter's Verify the AI box:

    (a) *Demographic-term audit.* Does the AI's final formula include ν_m in the denominator (correct: $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$) or omit it (wrong: $\mathcal{R}_0 = c_I\beta/\gamma_R$)? If the AI omits the term, the derivation has lost the demographic competition between recovery and migration during the infectious period. State the resulting error magnitude in terms of the dimensionless ratio $D = \nu_m/\gamma_R$: for an acute infection ($D \ll 1$) and for a chronic infection ($D \sim 0.1$), how much is the AI's $\mathcal{R}_0$ off?

    (b) *Method-consistency audit.* Identify whether the AI is following the heuristic derivation (count expected secondary infections per primary case during the infectious period) or the next-generation-matrix derivation (compute spectral radius of $\boldsymbol{F}\boldsymbol{V}^{-1}$, where $\boldsymbol{F}$ and $\boldsymbol{V}$ are the Jacobians at the DFE of the new-infection and transition vector fields). The two are mathematically equivalent for the SIR model, which has a single infected compartment, but use different intermediate quantities. Does the AI commit to one method consistently throughout the derivation, or does it mix the two? For each step where the AI uses a quantity, state whether that quantity belongs naturally to the heuristic or the NGM picture.

    (c) *Equilibrium-vs-invasion audit.* The AI's derivation produces a number that is implicitly evaluated at the disease-free equilibrium ($P_S = 1$). Check whether the AI's notation makes this evaluation explicit or leaves it implicit. If the AI writes $\mathcal{R}_0$ with P_S floating (suggesting the formula applies post-invasion as well), correct the notation: the post-invasion analog is $\mathcal{R}_e(t) = \mathcal{R}_0 \cdot P_S(t)$. Why is this distinction operationally important for intervention design (e.g., when computing the herd-immunity threshold)?

    The exercise practices the audit habit on the most-cited derivation in epidemic modeling. Disclosure: per Appendix D, Section D.4, document the AI system, prompt, and audit findings.

11. **[Open Project]** *Endemic-equilibrium analysis for a chosen pathogen.* A multi-week capstone project applying the chapter's analytical, numerical, and AI-collaborative tools to a pathogen with substantial endemic prevalence (suggested options include hepatitis B, tuberculosis, measles, or HIV in defined subpopulations). The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit,

disclosure, and final memo (≈13–20 hours over 2–3 weeks)—is specified in the companion repository at `capstone-projects/ch06-endemic-equilibrium/`.

Chapter 6

The SIR Compartmental Model: Simulations

The purpose of computing is insight, not numbers.

— Richard Hamming

This chapter complements the analytical results of Chapter 5 with numerical simulations. Two scenarios illustrate the two qualitative regimes of the SIR model (derived in Chapter 4): a supercritical outbreak with $\mathcal{R}_0 > 1$ that approaches a positive endemic equilibrium (Scenario 1, Section 6.3), and a subcritical case with $\mathcal{R}_0 < 1$ in which the infection dies out (Scenario 2, Section 6.4). A third numerical experiment (Section 6.5) verifies to machine precision that the infected-viewpoint and susceptible-viewpoint formulations produce identical trajectories when they are given identical parameters and initial conditions. Sections 6.6 and 6.7 compare the numerical results against the analytical predictions from Chapter 5, verifying the doubling time from linearization at the disease-free equilibrium, the endemic values from Proposition 5.4, and the identity $\mathcal{R}_e^* = 1$ at the endemic equilibrium.

The numerical experiments in this chapter use the baseline parameter set introduced in Table 4.4 of Chapter 4. Every simulation in Chapters 6–9 uses the same parameter set as its starting point, so that the results across chapters are directly comparable.

6.1 Introduction and Baseline Parameters

This section establishes the numerical setting for the rest of the chapter. We first recall the canonical baseline parameter set from Chapter 4 Table 4.4, which gives a supercritical regime ($\mathcal{R}_0 \approx 1.92$) in which an outbreak invades and approaches a positive endemic equilibrium. We then compute the basic reproductive number from the baseline parameters to confirm the supercritical regime, and identify the single parameter modification that crosses the invasion threshold for Scenario 2.

We recall the baseline parameter set for convenience:

$$c_S = 20, \qquad c_I = 4, \qquad c_R = 20 \quad \text{(contacts/person/day)},$$
$$\beta = 0.040 \quad \text{(infections/contact)},$$
$$\tau_R = 12 \quad \text{(days)}, \qquad \tau_m = 7300 \quad \text{(days} \approx 20 \text{ years)},$$

with every population a fraction of the carrying capacity (Section 4.5.3). The derived dependent parameters are $\gamma_R = 1/\tau_R \approx 0.0833$ day^{-1} and $\nu_m = 1/\tau_m \approx 0.000137$ day^{-1}. A residence time of

$\tau_m = 7300$ days corresponds to $\nu_m \approx 5\%$ of the population turning over each year, a reasonable value for many human populations.

From the heuristic derivation in Section 5.1,

$$\mathcal{R}_0 = \frac{c_I \beta}{\gamma_R + \nu_m} = \frac{4 \times 0.040}{0.0833 + 0.000137} \approx \frac{0.16}{0.0835} \approx 1.92.$$

The baseline places the model in the supercritical regime. Scenario 1 (Section 6.3) uses this parameter set unchanged. Scenario 2 (Section 6.4) modifies only one parameter, reducing c_I from 4 to 2 contacts per person per day, to cross the invasion threshold.

Exercises for this section

1. **[L]** Reproduce Figure 6.2 by integrating the SIR system with the baseline parameters of Section 6.1, the initial conditions (6.1), and your preferred RK45 solver over 1000 days. Compare your computed values of i_{peak} and t_{peak} with those reported in Section 6.3.

6.2 Numerical Implementation

The SIR system (5.1) is a three-dimensional system of autonomous ODEs with smooth, bounded right-hand sides over the physically relevant region $\{(s, i, r) : s, i, r \geq 0,\ s + i + r \leq 1\}$. Any standard adaptive ODE solver is adequate for its numerical integration.

All simulations in this book use a Runge–Kutta method of order 4(5) with adaptive step-size control. In MATLAB this corresponds to the `ode45` integrator; in Python with SciPy, `solve_ivp` with the `RK45` method; in R with `deSolve`, the `lsoda` integrator. Absolute and relative tolerances are set to 10^{-9} and 10^{-8} respectively. The integration window covers the epidemic trajectory from the time of introduction ($t = 0$) to a time horizon long enough for the trajectory to approach its equilibrium within a tolerance of 10^{-5} in each compartment. For the supercritical baseline the recurrent epidemics of Figure 6.3 damp slowly, and the trajectory comes within that tolerance only after more than 60,000 days; the closed form of Section 5.5 is the efficient check on the endemic values.

Implementation note: $\alpha(t)$ is state-dependent. The force from infection $\alpha(t) = c_I \beta P_S(s, i, r)$ depends on the current state through the contact-weighted susceptible fraction P_S. It must be evaluated *inside* the ODE right-hand-side function at each solver step, not precomputed once before the integrator is called. Pre-computing α as a constant from initial conditions and passing it as a fixed parameter to the right-hand side is the most common implementation error in the SIR system; the result is a misspecified linear ODE that never reproduces the peak-and-decline of the correct nonlinear system, so the two trajectories remain qualitatively different for the entire simulation, not merely at early times.

Implementation note: initial conditions. Begin the integration from a physically realizable state, never from the disease-free equilibrium $(1, 0, 0)$: that point is an equilibrium the trajectory cannot leave, so it yields a flat, unfittable simulation. Pitfall: "Natural Initial Conditions" below explains why, and how to choose the fit-start state instead.

The initial conditions for the scenarios in this chapter are

$$s(0) = 0.999, \qquad i(0) = 0.001, \qquad r(0) = 0. \tag{6.1}$$

This choice seeds the outbreak with a small initial infectious fraction (0.1% of the population), and no prior recovered individuals. The initial condition satisfies $s(0)+i(0)+r(0)=1$, so that $n(t)=1$ for all $t \geq 0$ by the conservation result of Chapter 4. As established in Chapter 4, Section 4.5.3, the state variables $s(t), i(t), r(t)$ here are fractions of the carrying capacity N^c. Multiplying any of these fractions by the modeled population size N^c recovers a count in individuals; for the long-term recurrent dynamics and inter-epidemic fade-out (Figure 6.3) we take a representative $N^c = 10^5$, a small community at the scale where sustained endemic transmission and fade-out are the operative regime, so that the endemic prevalence $i^* \approx 8 \times 10^{-4}$ is about eighty infectious individuals and a single infectious individual corresponds to $i = 10^{-5}$. Peak prevalence, total burden, and intervention effect sizes may be reported in either form without affecting the dynamics.

Starting at the disease-free equilibrium would give the flat line $i(t) \equiv 0$, since the trajectory can never leave it; choosing a start the data-generating setting could produce is the pitfall Natural Initial Conditions (Chapter 16).

First infection, epidemic start, and first reported case. Three moments are easy to conflate, and they can be days to weeks apart. The *first infection* is the arrival of the pathogen in a single host. The *first reported case* is later, for two independent reasons: a reporting delay between infection and report, and the near-certainty that the first *reported* case is not the first *actual* case, since early infections go undetected until surveillance catches one. Both push the true onset earlier than the first report suggests. Figure 6.1 illustrates the consequence with a single early-growth curve run three ways: one that takes the first reported case to be the start of the epidemic; one that allows a five-day delay between the first case's infection and its report; and one in which the first reported case is really the fifth case of the epidemic, treated as if it were the first. The infection curve $i(t)$ has the same shape in all three; only the time origin moves. Anchoring a model at the first reported case, as though the epidemic began then, therefore slides the entire timeline: the epidemic is already further along at the moment of first report than the naive interpretation assumes, and its peak arrives earlier. This is Pitfall: "Natural Initial Conditions" read on the time axis: the fit-start is a point at which the state is observable, not the moment the epidemic began, and the two must be kept distinct when a simulation is compared with the earliest surveillance data.

The figures in this chapter are produced by reproducible scripts in the book's companion repository; each caption records the parameters and integration settings needed to regenerate the figure independently.

AI Lab: Debugging an SIR Simulation

A simulation of the SIR model must reproduce the model's analytical features, and those features are the checks. The workflow is a *three-step audit-and-refine*: the AI does the bookkeeping, and you keep the checks.

Step 1: steady state. Run to at least $10\,\tau_m$, and the solution should approach the disease-free equilibrium if $\mathcal{R}_0 < 1$, or the endemic equilibrium of Section 5.5 if $\mathcal{R}_0 > 1$. If it does not, give the AI your parameters and both values and ask for the likely causes in order. A good list puts first a run too short for the slow timescale $1/\nu_m$, then a sign error in the ν_m term, then a bad initial condition.

Step 2: conservation. Plot $|N(t)/N^c - 1|$ on a log scale; it should stay near the integrator's tolerance. Linear growth means a flux imbalance: ask the AI which term is unbalanced, and check that every out-flux has a matching in-flux.

Step 3: scenario features. For Scenario 1, compare the simulation with three analytical results: the early growth rate $\sigma \approx (\gamma_R+\nu_m)(\mathcal{R}_0-1)$, the condition $\mathcal{R}_e = \mathcal{R}_0 P_S = 1$ at the simulated peak,

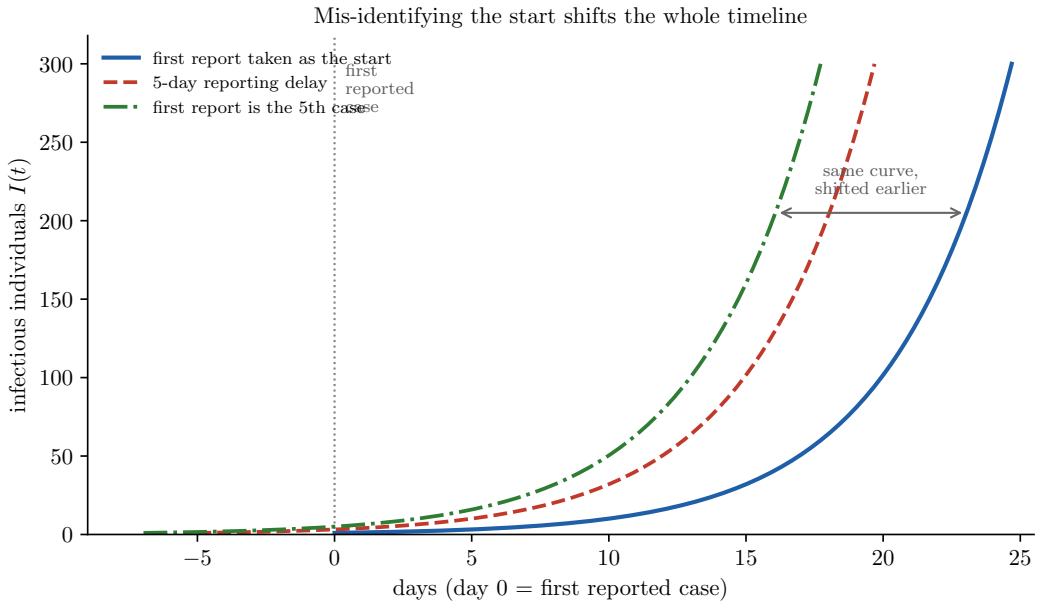


Figure 6.1: The same early epidemic run under three assumptions about when it began, relative to the first reported case (dotted line, day 0). *First report taken as the start:* the first reported case is treated as the first infection. *Reporting delay:* the first case was infected five days before it was reported. *First report is the 5th case:* earlier infections went undetected, so the epidemic began about a week earlier (the time to reach the fifth case, $\ln 5/\sigma$). The infectious curve $i(t)$ has the same shape in each; only its position on the time axis moves. Illustrative early exponential growth at a doubling time of three days.

and the endemic equilibrium of Section 5.5. (The conservation law $i + s - (1/\mathcal{R}_0)\ln s$ and the final size of Section 5.5.4 hold only for equal contact rates; at the baseline a correct simulation misses them by about 20% and 5%.) Ask the AI to compute them from your parameters; a gap above 1% points to a parameter mismatch or a loose tolerance.
Disclose AI assistance in reported work (Appendix D, Section D.4).

Exercises for this section

1. **[L]** *Convergence study.* Re-run the Scenario 1 integration with three different relative tolerances $(10^{-6}, 10^{-9}, 10^{-12})$ and three different time-span resolutions. Report the resulting variation in i_{peak}, t_{peak}, and the endemic value i^*. Identify a tolerance setting at which further tightening produces no change at the four-digit level. This exercise establishes a default solver setting for the rest of the book's numerical work.

6.3 Scenario 1: Baseline Supercritical Outbreak

Scenario 1 uses the baseline parameters of Section 6.1 without modification. With $\mathcal{R}_0 \approx 1.92 > 1$, Theorem 5.3 predicts that the disease-free equilibrium is unstable and that the introduction $i(0) = 0.001$ triggers a self-sustaining outbreak.

6.3.1 The Epidemic Trajectory

Figure 6.2 shows the numerically computed trajectories of $s(t)$, $i(t)$, and $r(t)$ over 1000 days.

Several qualitative features of the trajectory are worth noting.

Initial exponential growth. In the first few weeks, while $s(t) \approx 1$ and $P_S(t) \approx 1$, the infectious population grows as

$$i(t) \;\approx\; i(0)\, e^{\sigma_2 t},$$

where $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1) \approx 0.0835 \times 0.917 \approx 0.077$ day^{-1} from (5.15). The corresponding doubling time (5.16) is $t_{\text{double}} = \ln(2)/\sigma_2 \approx 9.1$ days. Section 6.7 verifies this prediction against the numerical trajectory.

Epidemic peak. The infectious population reaches its peak value at the time when $di/dt = 0$, which from the second equation of (5.1) is when $\alpha = \gamma_R + \nu_m$. Equivalently, this is the time when $\mathcal{R}_e(t) = 1$ (Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$"): the effective reproductive number has fallen from its initial value $\mathcal{R}_0$ to the critical value 1 through depletion of the susceptible pool. From the closed-form analysis of Section 5.1, this occurs when $P_S(t) = 1/\mathcal{R}_0 \approx 0.522$. The peak time in Figure 6.2 is approximately day 85, at which point $i_{\text{peak}} \approx 0.174$: roughly 17% of the population is simultaneously infectious at the peak of the outbreak.

Post-peak decline and endemic approach. After the peak, $P_S(t) < 1/\mathcal{R}_0$ drives $\mathcal{R}_e < 1$ and $i(t)$ declines steeply. At a realistic residence time the decline overshoots dramatically: the epidemic burns through the susceptible pool far faster than demographic turnover ($\nu_m N^{\text{c}}$ new susceptibles per unit time) can replenish it, so $i(t)$ falls not merely below its endemic value i^* but to a deep trough orders of magnitude lower. Figure 6.2 shows it returning to near-zero levels within a few

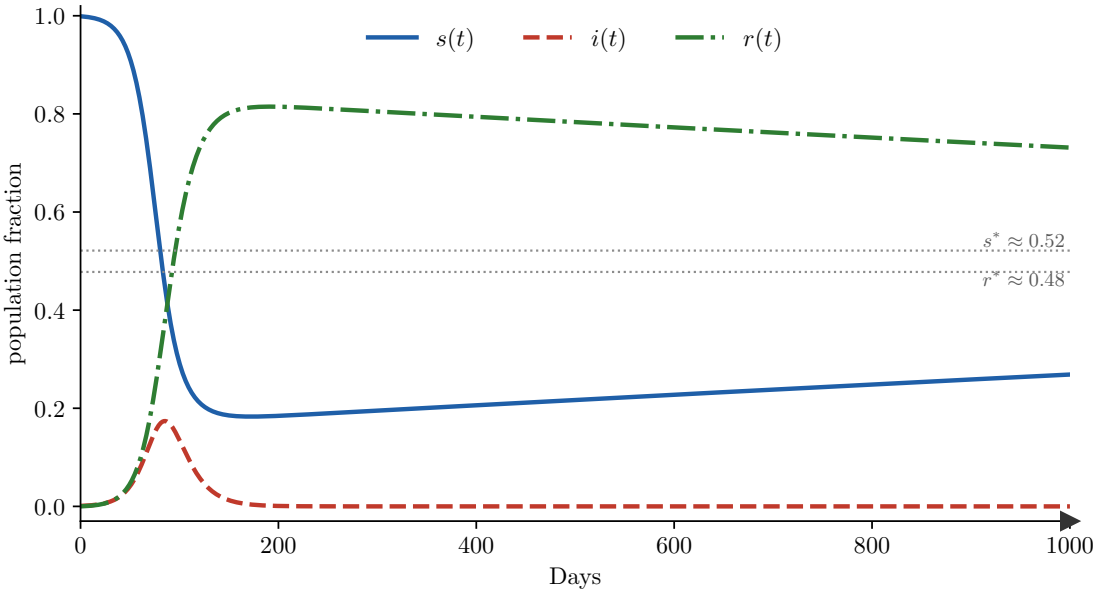


Figure 6.2: Trajectories of $s(t)$ (blue, solid), $i(t)$ (red, dashed), $r(t)$ (green, dash-dotted) for Scenario 1 over 1000 days. Initial conditions $s(0) = 0.999$, $i(0) = 0.001$, $r(0) = 0$. Parameters: $c_S = 20$, $c_I = 4$, $c_R = 20$, $\beta = 0.040$, $\tau_R = 12$, $\tau_m = 7300$, giving $\mathcal{R}_0 \approx 1.92$. Integrated with RK45, tolerances 10^{-9} absolute, 10^{-8} relative. The infectious curve $i(t)$ exhibits the characteristic outbreak shape: exponential rise, peak near day 85 at $i_{\text{peak}} \approx 0.17$, and steep decline. At this realistic residence time, $i(t)$ then falls to a near-zero trough rather than settling directly: the epidemic burns through the susceptibles faster than demographic turnover replenishes them. The susceptible curve $s(t)$ falls steeply during the outbreak and then begins a slow rebuild toward $s^* \approx 0.52$ (dotted); the recovered curve $r(t)$ peaks and slowly relaxes toward $r^* \approx 0.48$ (dotted). The approach to these endemic values takes several decades (Figure 6.3).

hundred days. Only as births and in-migration slowly rebuild the susceptible pool past the invasion threshold does the infection return. The approach to the endemic equilibrium is therefore slow and non-monotonic: the trajectory executes large, slowly damped oscillations (a sequence of recurrent epidemics) around (s^*, i^*, r^*), because the eigenvalues of the Jacobian at the endemic equilibrium (Section 5.6) have small negative real parts with non-zero imaginary parts. The next subsection examines this slow approach.

6.3.2 Recurrent Epidemics and the Slow Approach to Endemicity

Over a horizon of roughly a century, Figure 6.3 shows how the trajectory approaches the endemic equilibrium through a sequence of recurrent epidemics separated by deep inter-epidemic troughs, with the oscillation amplitude damping only slowly toward the low endemic level.

The endemic values from the closed-form expression (5.21) are

$$\begin{aligned} i^* &= \frac{n^* c_S \nu_m (\mathcal{R}_0 - 1)}{c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I \nu_m + c_R \gamma_R} \approx 0.000786, \\ s^* &= \frac{n^* (c_I \nu_m + c_R \gamma_R)}{c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I \nu_m + c_R \gamma_R} \approx 0.521, \\ r^* &= \frac{\gamma_R}{\nu_m} i^* \approx 0.478. \end{aligned}$$

where the numerical values use the normalization $n^* = 1$. These match the numerical equilibrium values to the solver's tolerance, providing a first check on the analytical derivation.

Period of the damped oscillation. The oscillation visible in Figure 6.3 corresponds to the imaginary part of the complex-conjugate eigenvalue pair of the Jacobian at the endemic equilibrium. For the equal-contact-rate case, the characteristic polynomial is (5.26):

$$\sigma^2 + \nu_m \mathcal{R}_0 \sigma + \nu_m(\gamma_R + \nu_m)(\mathcal{R}_0 - 1) = 0,$$

with roots

$$\sigma = \frac{-\nu_m \mathcal{R}_0 \pm \sqrt{\Delta}}{2},$$

where the discriminant is $\Delta = \nu_m^2 \mathcal{R}_0^2 - 4\nu_m(\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$. For the baseline parameters, the discriminant is negative, so the eigenvalues are a complex-conjugate pair with real part $-\nu_m \mathcal{R}_0/2 \approx -0.000131$ day^{-1} and imaginary part $\sqrt{\nu_m(\gamma_R + \nu_m)(\mathcal{R}_0 - 1) - (\nu_m \mathcal{R}_0/2)^2} \approx 0.00324$ rad/day. The period of oscillation is $2\pi/0.00324 \approx 1940$ days, about 5.3 years, consistent with the spacing of the later, smaller epidemics in Figure 6.3. The oscillation decays with timescale $1/|-0.000131| \approx 7600$ days, about 21 years, so the full approach to the endemic equilibrium spans several such timescales (a matter of decades, not years). The early epidemics in Figure 6.3 are larger and more widely spaced than this linear estimate predicts, because the first few excursions are far from the equilibrium about which the linearization is taken; the period and decay quoted here describe the small-amplitude oscillations near the endemic point. Distinct-contact-rate corrections to these values are small for the baseline parameter set.

Inter-epidemic troughs and the separation of timescales. Two features of this slow approach matter for modeling practice. First, the dynamics show a stark *separation of timescales*:

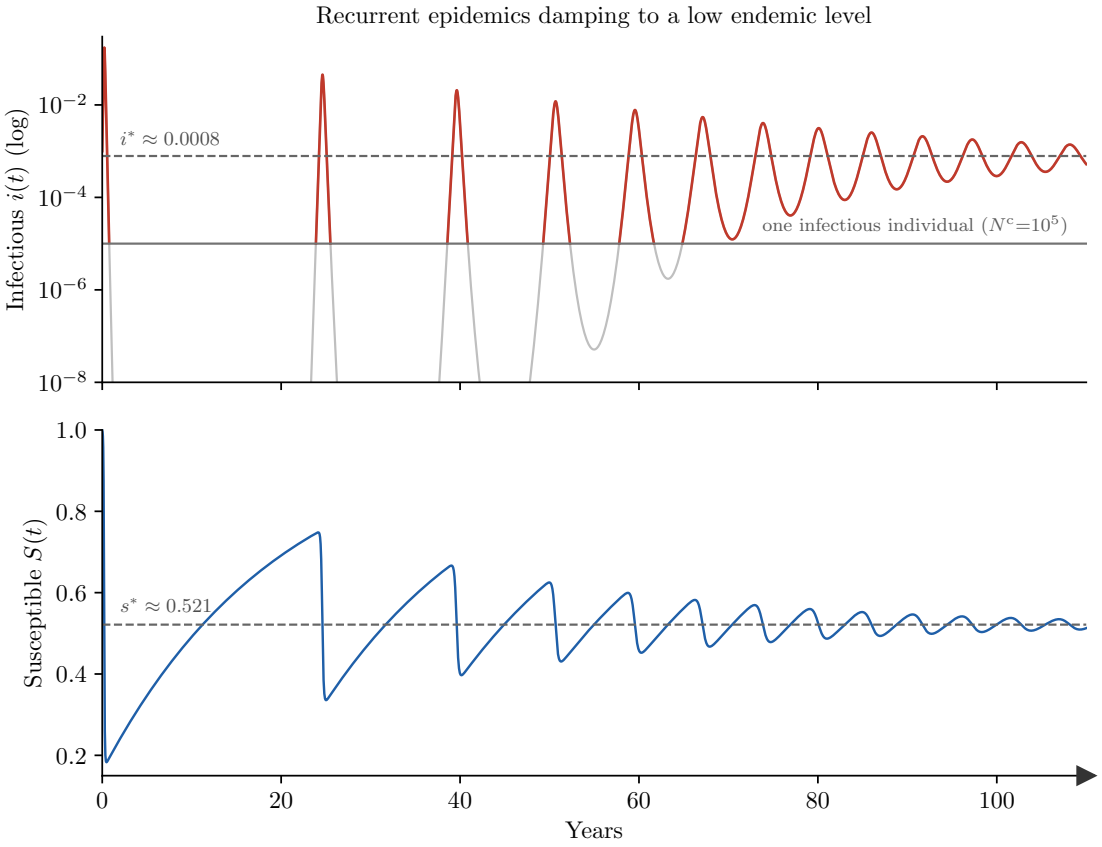


Figure 6.3: Long-term trajectories for Scenario 1 over roughly 110 years. *Upper panel:* the infectious fraction $i(t)$ on a logarithmic scale, showing recurrent epidemics that damp slowly toward the endemic value $i^* \approx 0.0008$ (dashed). The solid horizontal line marks the level of a single infectious individual for a representative modeled population of $N^c = 10^5$ ($i = 10^{-5}$); between the early epidemics the deterministic trajectory plunges many orders of magnitude below it (the faded segments), into a regime of far less than one infectious individual where a finite population would instead fade out to local extinction and require re-introduction. *Lower panel:* the susceptible fraction $s(t)$, which rebuilds slowly between epidemics and is sharply depleted by each one, converging to $s^* \approx 0.521$ (dashed). The small-amplitude oscillation period (about 5 years near the equilibrium) and the slow amplitude decay (timescale about 21 years) follow from the eigenvalues of the endemic Jacobian; the full approach to equilibrium spans several decades.

the outbreak itself plays out in weeks, while the approach to endemicity unfolds over decades, governed by the demographic rate ν_m rather than the epidemic rate γ_R. A simulation run only to the end of the first outbreak (or to the few-year horizon that suffices when τ_m is taken unrealistically short) misses the recurrent-epidemic structure entirely. Second, in the inter-epidemic troughs the infectious fraction falls far below the level of a single infectious individual: for the representative population $N^c = 10^5$ this is below $i = 10^{-5}$, and in the early troughs many further orders of magnitude below (Figure 6.3, upper panel). In a finite population this means fewer than one infectious individual, so a stochastic model would assign substantial probability to *fade-out*: local extinction of the pathogen between epidemics, with each subsequent epidemic requiring re-introduction. The deterministic model, which permits arbitrarily small positive fractions, cannot represent fade-out: it rebuilds every epidemic from an exponentially small seed. This is a genuine limitation of the deterministic SIR model at realistic residence times, and a reason its recurrent epidemics should be read as the deterministic skeleton of what is, between epidemics, a fundamentally stochastic process; capturing fade-out requires a stochastic treatment beyond the scope of this chapter.

6.3.3 From Simulation to Inference

The Scenario 1 trajectory is generated by a deterministic ODE, but in any realistic surveillance setting the observed $j(t)$ and $i(t)$ are noisy samples of the true trajectory rather than the trajectory itself. A natural worry is that observation noise might swamp the empirical force from infection $\hat{\alpha}(t) = j(t)/i(t)$ on which the fitting workflow depends. It does not: because $\hat{\alpha}$ is a ratio of the two observed streams, it is structurally robust to multiplicative observation noise, and reporting errors common to both streams cancel in the ratio. This robustness, and the observation-noise model behind it, are developed with the rest of the inference machinery in Chapter 7, Section 7.4.3.

Exercises for this section

1. **[C]** Looking at the Scenario 1 trajectory (Figure 6.2), explain in plain English why the infectious population $i(t)$ peaks well before the susceptible population $s(t)$ reaches its endemic value. What sets the timescale separation between these two events?

6.4 Scenario 2: Subcritical Case with Reduced Infectious Contact

Scenario 2 modifies a single parameter from the baseline: the infectious contact rate is reduced from $c_I = 4$ to $c_I = 2$ contacts per person per day. All other parameters are unchanged. The new basic reproductive number is

$$\mathcal{R}_0 \;=\; \frac{2 \times 0.040}{0.0835} \;\approx\; 0.958 \;<\; 1.$$

By Theorem 5.3, the disease-free equilibrium is now locally asymptotically stable, and the introduction $i(0) = 0.001$ should decay to zero.

Figure 6.4 compares the two scenarios on a linear scale, one per panel with a shared time axis, so the qualitative difference between an outbreak and a die-out is immediate.

The two trajectories look qualitatively different even though the two scenarios differ in a single parameter. Scenario 1 exhibits all the phases of an outbreak; Scenario 2 exhibits monotone decay

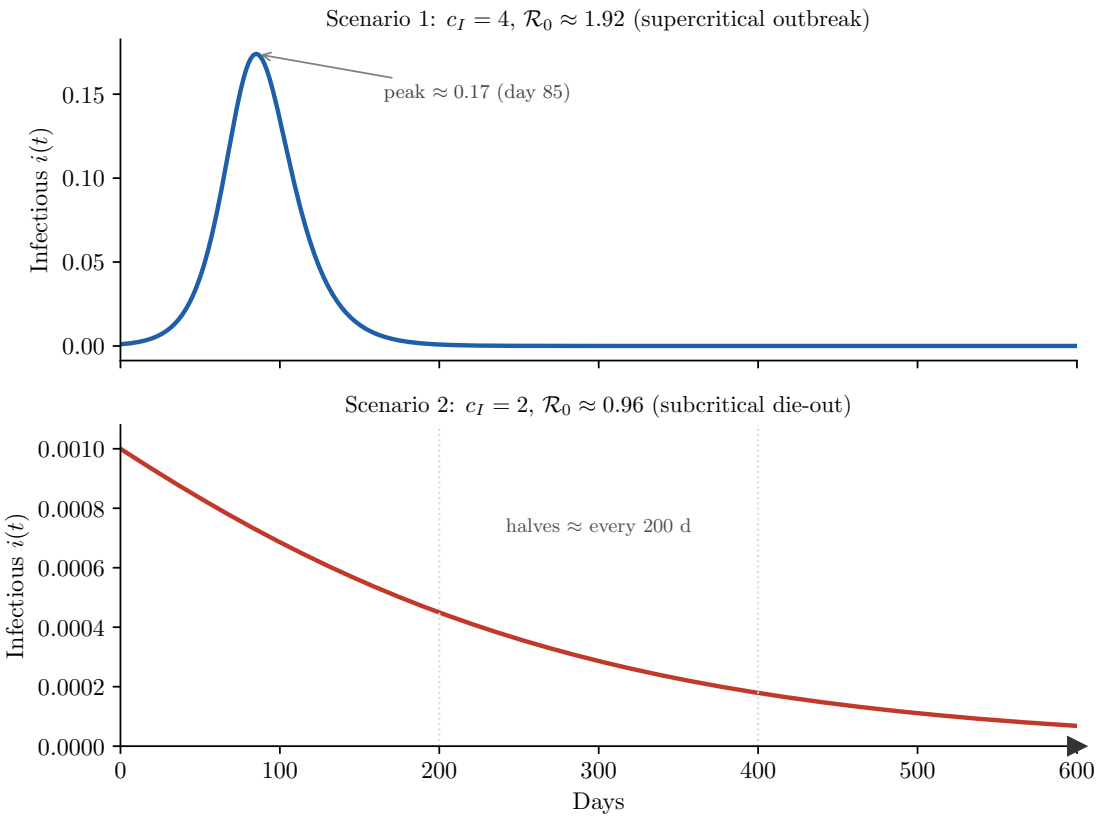


Figure 6.4: Scenario 1 and Scenario 2 compared on a linear vertical scale over 600 days, one per panel with a shared time axis (note the different vertical scales). Parameters are identical except for the infectious contact rate, and both start from $i(0) = 0.001$. *Upper panel:* Scenario 1 ($c_I = 4$, $\mathcal{R}_0 \approx 1.92$) is supercritical: $i(t)$ rises from the small seed, passes through the epidemic peak near day 85 at $i_{\text{peak}} \approx 0.17$, and then crashes. *Lower panel:* Scenario 2 ($c_I = 2$, $\mathcal{R}_0 \approx 0.96$) is subcritical and never invades: $i(t)$ decays monotonically, roughly halving every 200 days, consistent with the linearized decay rate $|\sigma| = (\gamma_R + \nu_m)(1 - \mathcal{R}_0) \approx 0.0035$ day^{-1} (halving time $\ln 2/|\sigma| \approx 200$ days). Its peak prevalence stays below 0.001, more than two orders of magnitude under the Scenario 1 outbreak. The linear scale makes the qualitative contrast immediate: crossing the invasion threshold at $\mathcal{R}_0 = 1$ turns an outbreak into a die-out.

of $i(t)$. The transition occurs at the boundary $\mathcal{R}_0 = 1$, which for fixed β, τ_R, and τ_m corresponds to a critical contact rate

$$c_I^{\text{crit}} = \frac{\gamma_R + \nu_m}{\beta} = \frac{0.0835}{0.040} \approx 2.09.$$

Reducing c_I from 4 to 2 brings the system across this threshold, from supercritical to subcritical. This is the prototype of an invasion-targeting intervention (Section 5.7): by reducing c_I, the intervention lowers $\mathcal{R}_0$ below 1, so the infection cannot invade.

Decay rate in the subcritical case. The DFE Jacobian (5.13) gives the decay rate $|\sigma_2| = (\gamma_R + \nu_m)(1 - \mathcal{R}_0)$ for the subcritical case. For Scenario 2, $|\sigma_2| \approx 0.0835 \times 0.042 \approx 0.0035$ day^{-1}, giving a halving time $\ln(2)/|\sigma_2| \approx 200$ days. The fit of this prediction to the numerical trajectory in Figure 6.4 verifies the analysis. Note that $|\sigma_2|$ becomes small as $\mathcal{R}_0$ approaches 1 from below: the closer $\mathcal{R}_0$ is to the invasion threshold, the slower the decay. An intervention that reduces $\mathcal{R}_0$ from 1.92 to 0.96 (roughly halving) produces quite slow decay; a more aggressive intervention reducing $\mathcal{R}_0$ to 0.5 would produce much faster decay.

A practical observation. In real outbreak response, the distinction between the controlled reproductive number $\mathcal{R}_c$ being slightly below 1 and substantially below 1 matters operationally. A marginally subcritical outbreak can persist at very low prevalence for many months before definitively dying out, during which time a single lapse in intervention (e.g., a relaxation of contact restrictions) can push $\mathcal{R}_c$ back above 1 and restart the outbreak. More aggressive interventions, bringing $\mathcal{R}_c$ decisively below 1, produce extinction times that are shorter than typical intervention durations.

Exercises for this section

1. **[L]** Consider the subcritical Scenario 2 and vary c_I in the range $1 \le c_I \le 2$. For each value, compute $\mathcal{R}_0$ and the time $t_{1/10}$ at which $i(t)$ has decayed to one-tenth of its initial value. Plot $t_{1/10}$ as a function of $\mathcal{R}_0$. As $\mathcal{R}_0 \to 1^-$, does $t_{1/10}$ diverge? Interpret in terms of the eigenvalue σ_2.

6.5 Numerical Equivalence and Behavior Under Uncertainty in s

Chapter 4 established algebraically that the SIR formulation (5.1) and the SIR$_S$ formulation produce the same dynamics when given the same parameters and initial conditions. This section first verifies the equivalence numerically to machine precision (Section 6.5.1), then previews how the two formulations diverge when they are fit to the same outbreak data under different assumptions about the susceptible population size N^{c} (Section 6.5.2), a practical difference between the two viewpoints that Chapter 7 develops in full.

6.5.1 Exact Equivalence Under Identical Parameters

When both formulations are given identical parameters and initial conditions, their trajectories must coincide. We verify this numerically. Both systems are integrated with the baseline parameters of Section 6.1 and the initial conditions (6.1). The SIR right-hand side uses $\alpha = c_I \beta\, c_S s/(c_S s + c_I i +$

$c_R r$); the SIR_S right-hand side uses $\lambda = c_S \beta\, c_I i/(c_S s + c_I i + c_R r)$. The two force expressions are algebraically distinct but satisfy $\alpha i = \lambda s$ (equation (4.3) of Chapter 4). Both systems use the same RK45 solver with the same tolerances and the same 1000-day integration horizon.

The maximum absolute difference $|i_{\text{SIR}}(t) - i_{\text{SIR}_S}(t)|$ over the 1000-day horizon remains at the level of floating-point round-off ($\sim 10^{-16}$), with no order-one mismatch. The two formulations are mathematically equivalent, and their numerical implementations agree to the precision of floating-point arithmetic.

6.5.2 Dynamics Under Uncertainty in s

The equivalence in Section 6.5.1 assumes the two formulations are given the same value of the susceptible population size N^{c}. In practice, at the onset of an outbreak, N^{c} is rarely known to within better than a factor of several: the metropolitan area's full population, the population in close contact with the index cases, and the population within a transit network can differ by factors of three to ten. This uncertainty propagates into the fitted force parameters, though not symmetrically across the two viewpoints, and into every forward-simulated trajectory.

The asymmetry is clearest through the two estimators of the force. A modeler taking the infected viewpoint estimates $\hat{\alpha} = j(t)/i(t)$ from observed incidence j and prevalence i; this ratio never involves the susceptible population, so it is unchanged by any error in the assumed N^{c}. A modeler taking the susceptible viewpoint estimates $\hat{\lambda} = J(t)/S(t) \approx J(t)/N^{\text{c}}_{\text{assumed}}$, which scales inversely with the assumed population: a factor-of-three overestimate of N^{c} yields a fitted force three times too small. The infected-viewpoint estimate is structurally immune to uncertainty in N^{c}; the susceptible-viewpoint estimate inherits that uncertainty factor for factor. A forward simulation needs N^{c} in either viewpoint, because depletion is measured against it. A model fitted to the early growth with the seed counted in individuals (Chapter 7) and run forward at three times the population puts the peak about two weeks later and three times higher in counts. Chapter 3, Figure 3.2 shows the companion effect in time rather than in scale: over the early-growth window of this same baseline outbreak, α is nearly constant while λ grows by more than an order of magnitude.

This distinction is the practical payoff of the SIR formulation, and Chapter 7 develops it in full. Section 7.6 gives the central numerical comparison of $\hat{\alpha}$ and $\hat{\lambda}$ under uncertainty in the susceptible population, and Chapter 9, Section 9.3.4 gives the closed-form sensitivity statement ($\Upsilon_s^{\hat{\alpha}} = 0$ versus $\Upsilon_s^{\hat{\lambda}} = -1$) that makes the asymmetry exact. The algebraic root of the effect was established in Chapter 3, Section 3.5.

6.6 The Effective Reproductive Number Over Time

The effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0 P_S(t)$ from (5.4) is a useful diagnostic quantity. At any time t, $\mathcal{R}_e(t)$ gives the expected number of secondary infections that would be generated by a single infectious individual introduced at time t. The epidemic grows when $\mathcal{R}_e(t) > 1$ and declines when $\mathcal{R}_e(t) < 1$.

Figure 6.5 shows $\mathcal{R}_e(t)$ for Scenario 1.

The crossing at the epidemic peak. The time at which $\mathcal{R}_e(t) = 1$ corresponds to the epidemic peak ($di/dt = 0$). This follows from the i equation of (5.1): $di/dt = (\alpha - \gamma_R - \nu_m)i = (\mathcal{R}_e - 1)(\gamma_R + \nu_m)i$. The infectious population grows when $\mathcal{R}_e > 1$ and declines when $\mathcal{R}_e < 1$, so i is stationary

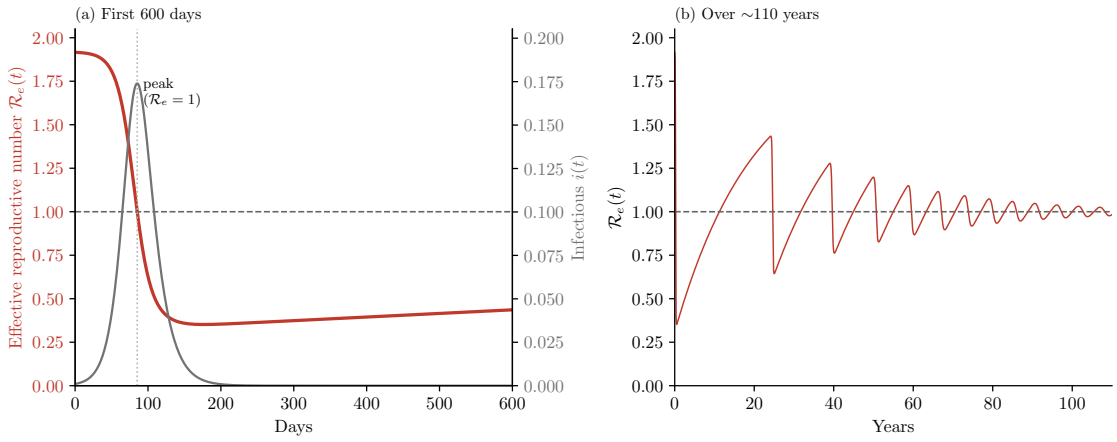


Figure 6.5: Effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0\, P_S(t)$ for Scenario 1. **(a)** Over 600 days, with the infectious fraction $i(t)$ overlaid on the right axis: $\mathcal{R}_e$ starts at $\mathcal{R}_0 \approx 1.92$ when $P_S(0) \approx 1$, decreases as susceptibles are depleted, crosses the invasion threshold $\mathcal{R}_e = 1$ near day 85—exactly at the peak of $i(t)$—and undershoots to approximately 0.35 at the bottom of the post-epidemic trough. **(b)** Over roughly 110 years: $\mathcal{R}_e(t)$ climbs back toward 1 only slowly as demographic turnover replenishes the susceptible pool, and across the recurrent epidemics it oscillates around 1, settling to the endemic value $\mathcal{R}_e^* = 1$ exactly. The dashed horizontal line marks $\mathcal{R}_e = 1$ in both panels.

exactly when $\mathcal{R}_e = 1$. Figure 6.5(a) shows this crossing near day 85 for Scenario 1, coinciding with the peak of $i(t)$ now plotted on the right axis of the same panel.

Approach to unity at the endemic equilibrium. At the endemic equilibrium, $\mathcal{R}_e^* = \mathcal{R}_0 P_S^*$ must equal 1 by the same stationarity condition. Using $P_S^* = c_S s^*/C_{\text{total}}^*$, one can verify analytically that $\mathcal{R}_e^* = 1$ (Chapter 5, Section 5.8, end-of-chapter Exercise 4). Figure 6.5(b) shows the numerical trajectory approaching $\mathcal{R}_e = 1$ from below as $t \to \infty$, consistent with the analytical prediction.

6.7 Verification Against the Analysis

This section compares the numerical trajectory of Scenario 1 against the quantitative predictions of Chapter 5. The comparison provides a first-principles check on both the analysis and the numerical implementation.

6.7.1 Doubling Time From the Linearization

The DFE linearization of Section 5.4 predicts early exponential growth

$$i(t) \;\approx\; i(0)\, e^{\sigma_2 t}, \qquad \sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1) \approx 0.077 \text{ day}^{-1},$$

with doubling time $t_{\text{double}} = \ln(2)/\sigma_2 \approx 9.1$ days. To verify this against the numerical trajectory, we compute the effective growth rate

$$\sigma_{\text{eff}}(t) \;=\; \frac{d \log i(t)}{dt} \;=\; \frac{\dot{i}(t)}{i(t)}$$

and compare it to the prediction σ_2. For the first few weeks, while $P_S(t) \approx 1$, the trajectory should have $\sigma_{\text{eff}}(t) \approx \sigma_2$. Numerically, we observe

$$\sigma_{\text{eff}}(5) \;=\; 0.0764, \qquad \sigma_{\text{eff}}(10) \;=\; 0.0763, \qquad \sigma_{\text{eff}}(20) \;=\; 0.0758.$$

The prediction $\sigma_2 = 0.0765$ agrees with the initial value to better than 1%. The slow drift in σ_{eff} is due to the departure of $P_S(t)$ from unity as susceptibles begin to be depleted; this drift grows until the peak of the outbreak.

6.7.2 Endemic Values

The closed-form endemic values from (5.21) and (5.22) are

$$s^* \approx 0.521, \qquad i^* \approx 0.000786, \qquad r^* \approx 0.478.$$

Integrating the system past the recurrent-epidemic transient (Figure 6.3), until the oscillations have damped, reproduces these equilibrium values to the solver's tolerance (10^{-5} or better in each compartment), confirming both the analytical derivation and the numerical integration. At the realistic residence time this requires a long integration horizon (several decades) in contrast to the few-year horizon that would suffice if τ_m were taken to be a single year.

6.7.3 The Identity $\mathcal{R}_e^* = 1$

From $\mathcal{R}_e^* = \mathcal{R}_0 P_S^*$, using the endemic values and the definition of P_S,

$$P_S^* \;=\; \frac{c_S s^*}{c_S s^* + c_I i^* + c_R r^*} \;\approx\; \frac{20 \times 0.5214}{20 \times 0.5214 + 4 \times 0.000786 + 20 \times 0.4779} \;\approx\; \frac{10.428}{19.989} \;\approx\; 0.5217.$$

This gives $\mathcal{R}_e^* = \mathcal{R}_0 \times 0.5217 \approx 1.92 \times 0.5217 \approx 1.000$, confirming the analytical identity. Four decimals are carried in this verification because the identity closes only at that precision; elsewhere the endemic values are quoted to three.

Note that when all three contact rates coincide ($c_S = c_I = c_R$), this identity simplifies to $P_S^* = s^*/n^*$ and $s^*/n^* = 1/\mathcal{R}_0$, reproducing the classical formula $s^* = n^*/\mathcal{R}_0$ from (5.24). In the distinct-contact-rate case of Scenario 1, $s^*/n^* \approx 0.521$ remains close to $1/\mathcal{R}_0 \approx 0.522$ without equalling it: the two differ by about 0.06%, visible in the third decimal. At the realistic residence time the endemic infectious fraction is tiny, so the infectious contribution to the contact-weighted denominator, $c_I i^* \approx 0.003$, is negligible beside the recovered contribution $c_R r^* \approx 9.56$ and the susceptible contribution $c_S s^* \approx 10.43$. The distinct-contact correction that separates s^*/n^* from $1/\mathcal{R}_0$ is therefore small, well under a tenth of a percent, but unlike the equal-contact case it does not vanish.

6.8 When the Parameters Themselves Change in Time

The average of a function is not the function of the averages.

The reproduction numbers of Chapter 5 and this one are built for constant model parameters. There, all of the time variation in transmission is carried by the epidemic state, principally by susceptible depletion, and is summarized by

$$\mathcal{R}_e(t) = \mathcal{R}_0 \, P_S(t). \tag{6.2}$$

It is easy to leave those chapters believing that (6.2) captures all of the relevant time dependence. It does not, once the transmission, recovery, migration, recruitment, or intervention parameters themselves vary with calendar time. This section makes the correction, and targets one specific and consequential error: reading a reproduction number computed from *averaged* parameters as though it were the invasion threshold of the time-varying system. We keep to the single-infectious-compartment SIR model, for which a clean scalar result is available; the closing remark says why that cleanliness does not survive the move to several infected compartments.

6.8.1 The instantaneous ratio, and why averaging fails

Near the disease-free state, where $P_S \approx 1$, the time-dependent SIR model has an infected equation of the form

$$\dot{i} = \left[\, c_I(t)\beta(t) - \gamma_R(t) - \nu_m(t) \,\right] i. \tag{6.3}$$

It is natural to define an instantaneous ratio,

$$\mathcal{R}_{\text{inst}}(t) = \frac{c_I(t)\beta(t)}{\gamma_R(t) + \nu_m(t)}, \tag{6.4}$$

and then to reach for a single scalar summary. Two such summaries suggest themselves—the time average of the instantaneous ratio and the reproduction number of the averaged parameters—and they are in general different from each other and from the true invasion threshold:

$$\overline{\mathcal{R}_{\text{inst}}} = \frac{1}{T}\int_0^T \frac{c_I(t)\beta(t)}{\gamma_R(t) + \nu_m(t)}\, dt, \qquad \mathcal{R}_0^{\text{avg}} = \frac{\overline{c_I}\,\overline{\beta}}{\overline{\gamma_R} + \overline{\nu_m}}. \tag{6.5}$$

Two mismatches separate these quantities. The first is elementary but easy to forget: the average of a product is not the product of the averages,

$$\overline{c_I(t)\beta(t)} \neq \overline{c_I(t)}\;\overline{\beta(t)}, \tag{6.6}$$

unless the covariance between contact rate and per-contact transmission happens to vanish. The second is more serious, because it survives even when each factor is treated correctly: by the usual Jensen-type inequality for a ratio,

$$\overline{\left(\frac{c_I\beta}{\gamma_R + \nu_m}\right)} \;\neq\; \frac{\overline{c_I\beta}}{\overline{\gamma_R + \nu_m}} \tag{6.7}$$

in general. An "average $\mathcal{R}_0$" is therefore not the average of $\mathcal{R}_{\text{inst}}(t)$, and neither quantity should be read off as the invasion threshold without justification.

6.8.2 The correct threshold in the scalar periodic case

Because (6.3) is a scalar linear equation, its solution over one period is explicit. The one-cycle multiplier is

$$\frac{i(T)}{i(0)} = \exp\!\left(\int_0^T \left[\, c_I(t)\beta(t) - \gamma_R(t) - \nu_m(t) \,\right] dt\right), \tag{6.8}$$

so invasion is decided by the integrated net growth rate: the infection grows over a cycle exactly when the integral in (6.8) is positive. Equivalently, writing the criterion as a ratio of integrated

transmission to integrated removal,

$$\mathcal{T} := \frac{\displaystyle\int_0^T c_I(t)\beta(t)\,dt}{\displaystyle\int_0^T \big[\gamma_R(t) + \nu_m(t)\big]\,dt} > 1 \tag{6.9}$$

signals invasion. The threshold is thus a ratio of integrals—not an integral of ratios, and not a ratio of averaged parameters. It gives a correct scalar threshold in the simplest forced setting and shows directly why naive averaging can fail.

Remark 6.1 ($\mathcal{T}$ is a threshold criterion, not the periodic $\mathcal{R}_0$). The quantity $\mathcal{T}$ in (6.9) agrees in *sign* with the rigorous periodic reproduction number—it is above or below one exactly when the periodic $\mathcal{R}_0$ is—but it is not equal to it in value. We write $\mathcal{T}$ rather than calling it "the periodic $\mathcal{R}_0$" to keep that distinction honest.

6.8.3 Why the scalar result does not generalize

The clean form of (6.9) depends on SIR having a single infected compartment, so that (6.3) is a genuine scalar equation with an elementary integrating factor. For a model with two or more infected compartments the invasion condition is a Floquet problem for the monodromy matrix of the linearized periodic system (Floquet theory is reviewed by Li 2018, Sec. 3.4), and the periodic reproduction number is defined through a next-generation operator on a function space rather than by any ratio of integrals [Bacaër and Guernaoui, 2006, Wang and Zhao, 2008]. The integrated-ratio rule (6.9) does not carry over, and applying it to the integrated new-infection and transition matrices of a multi-compartment periodic model would be its own averaging error. We flag the boundary here and do not develop the periodic-operator theory; the cited references give the rigorous treatment.

6.8.4 A two-season example

Consider a year split into two equal seasons. Winter has intense contact and transmission and a longer infectious period; summer has weaker transmission and a seasonal treatment campaign that shortens the infectious period. Recovery and demography are otherwise as in the running baseline. Table 6.1 gives the parameters.

Table 6.1: A deliberately simple two-season forcing (each season lasts half the year; rates per day).

	Winter (6 mo)	Summer (6 mo)
Contact rate c_I	12	5
Per-contact transmission β	0.020	0.008
Transmission rate $c_I\beta$	0.240	0.040
Infectious period τ_R (days)	9	7
Removal rate $\gamma_R + \nu_m$	0.111	0.143

The three scalar summaries then take materially different values,

$$\mathcal{R}_0^{\text{avg}} \approx 0.94, \qquad \overline{\mathcal{R}_{\text{inst}}} \approx 1.22, \qquad \mathcal{T} \approx 1.10. \tag{6.10}$$

The naive quantity, the reproduction number of the averaged parameters, is $\mathcal{R}_0^{\text{avg}} \approx 0.94 < 1$ and predicts that the infection dies out. The correct threshold criterion is $\mathcal{T} \approx 1.10 > 1$: the infection actually invades. The average of the instantaneous ratio, $\overline{\mathcal{R}_{\text{inst}}} \approx 1.22$, agrees with neither and overstates growth. The disagreement is not a rounding effect. Evaluating the one-cycle multiplier (6.8) for these parameters, with each season half of a 360-day year, gives

$$\frac{i(T)}{i(0)} = \exp\left(\int_0^T \left[c_I\beta - \gamma_R - \nu_m\right] dt\right) \approx e^{4.64} \approx 103, \tag{6.11}$$

so the infectious population grows by roughly a hundredfold over the year that the averaged-parameter model predicts to be subcritical. The invasion verdict itself flips, not merely its magnitude. This is a linear statement about growth from the disease-free state; the nonlinear saturation that follows—susceptible depletion, and the possibility of a large excursion that overshoots and then fades—is a separate question the threshold criterion does not settle. A further consequence is a pitfall in its own right: through the sub-threshold season the deterministic model does not extinguish the infection but carries an exponentially small—often sub-individual—infectious fraction and reignites it once $\mathcal{R}_e$ climbs back above one. A finite population would instead fade out during the unfavorable season and require re-introduction before the next, the deterministic-continuum limitation catalogued under Pitfall: "The Stochastic-to-Deterministic Regime Transition" and visible in the inter-epidemic troughs of Figure 6.3.

6.8.5 The lesson

A static fit to a multi-month epidemic with seasonal forcing, behavioral change, or staged interventions estimates some form of time-averaged parameter. That averaged parameter does not automatically inherit the threshold interpretation of the constant-model $\mathcal{R}_0$. For the single-compartment case the correct scalar threshold is the ratio of integrated transmission to integrated removal, (6.9), which can sit on the opposite side of one from the averaged-parameter value. The safe habit is to integrate the net growth rate over the forcing period rather than to average the parameters and then form a reproduction number as though the model were autonomous: the net growth rate is linear in the parameters and averages correctly, whereas the reproduction number is a ratio and does not. *Average rates, not ratios.* This pitfall deepens the constant-transmission assumption catalogued under Pitfall: "Constant-Transmission Assumption Ignoring Behavioral Feedback" and sharpens the invasion-versus-equilibrium distinction of Pitfall: "Invasion versus Equilibrium"; it returns in the piecewise-constant interventions of the COVID-19 case study (Chapter 17) and in the seasonal forcing of the influenza fitting example (Chapter 7).

6.9 Summary and Exercises

This chapter has illustrated the SIR model numerically through two scenarios and three verification exercises. The main results are:

1. Scenario 1 (baseline, $\mathcal{R}_0 \approx 1.92$) produces the characteristic outbreak trajectory: exponential growth from a small introduction and a peak near day 85, followed (at the realistic residence time) by recurrent epidemics with deep inter-epidemic troughs that damp to the endemic equilibrium $(s^*, i^*, r^*) \approx (0.521, 0.0008, 0.478)$ over several decades.

2. Scenario 2 (reduced c_I, $\mathcal{R}_0 \approx 0.96$) produces monotone decay. The decay rate is slow when $\mathcal{R}_0$ is marginally below 1. Bringing $\mathcal{R}_0$ decisively below 1 produces faster extinction.

3. The SIR and SIR_S formulations produce identical numerical trajectories to machine precision when given identical parameters and initial conditions, confirming their algebraic equivalence.

4. The effective reproductive number $\mathcal{R}_e(t)$ crosses 1 at the epidemic peak and approaches 1 from below at the endemic equilibrium, as predicted by the analysis.

5. Quantitative verification against Chapter 5: doubling time (≈ 9.1 days), endemic values, and the identity $\mathcal{R}_e^* = 1$ all agree with the numerical trajectory to solver tolerance.

Chapter 7 turns to the inverse problem: given incidence data from a real outbreak, how do we estimate the parameters of the SIR model, and how sensitive is the estimate to uncertainty in the susceptible population size? Chapter 9 extends this with a systematic sensitivity analysis of the output $\mathcal{R}_0$ and i^* with respect to the six independent parameters, making the invasion–burden partition of Section 5.7 quantitative.

Exercises

Four section-level exercises appear above, following Section 6.1, Section 6.2, Section 6.3, and Section 6.4. The exercises below integrate across the chapter and have been ordered so that concept checks, derivations, and computational labs alternate rather than clump (this chapter is computational-lab heavy, so perfect alternation is not possible). Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[L]** Modify the baseline by increasing c_I from 4 to 6 (a 50% increase) while holding all other parameters fixed. Compute the new $\mathcal{R}_0$, integrate over 1000 days, and compare the peak time and peak height with Scenario 1. Is the peak earlier or later? Higher or lower? Explain why.

2. **[C]** The chapter shows that the SIR and SIR_S formulations agree to round-off ($\sim 10^{-16}$, Section 6.5.1). Explain in one sentence why this difference is not zero, and what would need to change in the implementation to make it exactly zero.

3. **[L]** Modify the baseline by decreasing τ_m from 7300 to 100 days (simulating a highly mobile population) while holding all other parameters fixed. Compute the new $\mathcal{R}_0$, integrate over 10,000 days, and compute the new endemic equilibrium. How does the endemic prevalence i^* change? Interpret the change in terms of the invasion–burden partition of Section 5.7.

4. **[D]** Compute $\mathcal{R}_e(t)$ for Scenario 1 from the numerical trajectory by the definition $\mathcal{R}_e(t) = \mathcal{R}_0 P_S(t)$ and verify that it crosses 1 at the time of the epidemic peak. Report the peak time to two decimal places.

5. **[L]** Compute the critical contact rate c_I^{crit} such that $\mathcal{R}_0 = 1$ at the baseline values of β, τ_R, and τ_m. Integrate the SIR system with $c_I = c_I^{\text{crit}}$ and verify that the infectious population neither grows nor decays at the initial time, but drifts slowly due to the nonlinear terms.

6. **[L]** Implement the SIR_S formulation in your preferred scientific computing environment, using the force of infection $\lambda = c_S \beta P_I$ rather than the force from infection α. Integrate with the baseline parameters and compare with the SIR result. Is your numerical difference consistent with the machine-precision agreement (a maximum difference at the level of round-off, $\sim 10^{-16}$) established in Section 6.5.1?

7. **[I]** *Spaced retrieval, Chapter 4.* Look back at Chapter 4, Section 4.1, where five model assumptions are stated explicitly. Pick *one* assumption, propose a numerical diagnostic that can be applied to a Scenario 1 simulation trajectory of this chapter, and explain what observable signature would warn that the assumption was being violated by, say, a 10% deviation in the underlying real population. Then rank the five assumptions of Section 4.1 from most to least diagnosable from simulation output alone (without auxiliary external data). Justify your ranking in two or three sentences. (Note: "diagnosable from simulation output" is not the same as "important"; the assumption of homogeneous mixing is among the most consequential modeling choices but may be among the hardest to diagnose without independent contact-network data.) The exercise links the abstract assumptions of Chapter 4 to the concrete numerical-experiment workflow of this chapter, and asks you to think about which modeling assumptions can be checked from internal evidence and which require external comparison.

8. **[AI Lab]** *Simulation-debugging workflow.* Implement the SIR system in your preferred scientific computing environment using the baseline parameters of Section 6.1. Then deliberately introduce *one* of the following bugs (chosen randomly or assigned by an instructor) and execute the chapter's three-step audit-and-refine debugging workflow:

   (1) *Sign error.* Flip the sign of the ν_m term in the $\dot{r}$ equation (so $\dot{r} = \gamma_R i + \nu_m r$ instead of $\gamma_R i - \nu_m r$).

   (2) *Missing demographic term.* Drop the $\nu_m r$ term entirely from $\dot{r}$ (so $\dot{r} = \gamma_R i$).

   (3) *Step-size mismatch.* Use a fixed-step explicit Euler integrator with step size $\Delta t = 15$ days (longer than τ_R).

   (4) *Initial-condition typo.* Set $i(0) = -0.001$ (negative, by typo on a sign).

   Then execute the workflow:

   (a) *Step 1: Steady-state check.* Run the buggy simulation for $10\tau_m$ and report whether the trajectory approaches the predicted endemic equilibrium. If not, prompt an AI tool to diagnose the failure based on the symptoms (you describe the discrepancy without revealing the bug). Audit the AI's response against the chapter's priority list.

   (b) *Step 2: Conservation check.* Plot $|N(t)/N^{\mathrm{c}} - 1|$ on a log scale. Is the deviation at the level of integrator round-off, or does it grow systematically? If systematic, prompt the AI tool to identify the unbalanced term. Audit the response.

   (c) *Step 3: Scenario-feature cross-check.* Compute the analytical early-time growth rate σ_2, and check that $\mathcal{R}_e = 1$ at the simulated peak. Compare with the buggy simulation. The discrepancy localizes the bug.

   (d) *Bug identification.* From the workflow, identify which of the four bugs you introduced. Document the audit chain that led to the identification.

   The exercise practices the audit-fix-verify loop on a controlled bug. Disclosure: per Appendix D, Section D.4, document the AI system, prompts, and audit findings.

9. **[Verify the AI]** *Auditing AI-generated* SIR *implementation code.* Prompt an AI tool: *"Write Python code using* `scipy.solve_ivp` *to simulate the* SIR *model with three compartments* (s, i, r)*, contact rates* c_S*,* c_I*,* c_R*, transmission factor* β*, recovery rate* γ_R*, and demographic*

turnover ν_m. The force from infection is $\alpha = c_I \beta P_S$, where P_S is the contact-weighted susceptible fraction." Save the response. Audit the AI-generated code against this chapter's three-error structure for ODE implementation:

(a) *State-dependence audit.* Does the AI's right-hand-side function recompute $\alpha = c_I \beta P_S(s, i, r)$ at each solver step, or does it precompute α once from the initial state and pass it as a fixed parameter? The Implementation Note at Section 6.3 of this chapter is explicit on this point: α depends on the state (s, i, r) through P_S, and pre-computing it is the most common SIR implementation error. The diagnostic question: *"If I change the function signature to* `rhs(t, y, alpha_const)` *with alpha precomputed, what happens to the trajectory?"* A correct implementation produces a misspecified linear ODE whose trajectory diverges substantially from the correct nonlinear one within a few infectious-period timescales; an AI that has misimplemented this will not flag the divergence.

(b) *Conservation audit.* Does the AI's $\dot{s}$, $\dot{i}$, $\dot{r}$ equations sum to 0 at the disease-free equilibrium ($\dot{s} + \dot{i} + \dot{r} = 0$ when $s = 1$, $i = r = 0$)? The total population should be conserved up to demographic-turnover balance: $\sum_k \dot{X}_k = \nu_m(1 - n)$, which equals zero at $n = 1$. The diagnostic question: *"Verify by direct algebraic substitution that $\dot{s} + \dot{i} + \dot{r} = 0$ at $(1, 0, 0)$."* An AI that has dropped a ν_m term, mis-signed a flux, or omitted the births-into-s term will produce a non-conservative system that drifts away from the equilibrium population.

(c) *Tolerance audit.* Does the AI's `solve_ivp` call set absolute and relative tolerances appropriate for the dynamic range of the trajectory ($i(t)$ varies from 10^{-3} at outbreak start to about 0.17 at the peak, and far lower in the troughs that follow)? Default tolerances of 10^{-3} are inadequate for trajectories with $i(t)$ values of that magnitude; the chapter's standard is 10^{-9} absolute and 10^{-8} relative. The diagnostic question: *"At default tolerances of 10^{-3}, what is the relative error at the peak prevalence? Compute and compare to the chapter's standard tolerances."* An AI defaulting to standard tolerances may produce visibly correct trajectories whose peak time and height differ from the chapter's standard in the third significant figure, enough to fail a four-digit comparison.

The exercise practices the audit habit on the three highest-impact failure modes in AI-generated SIR implementation code: state-dependence of the force from infection (the chapter's central implementation pitfall), conservation of total population (the standard sanity check), and integrator-tolerance adequacy (a quantitative-accuracy issue invisible to visual inspection). Disclosure: per Appendix D, Section D.4, document the AI system, prompts, and audit findings.

10. **[Open Project]** *Full* SIR *implementation, verification, and debugging project.* A multi-week capstone project building a production-quality SIR simulator from scratch with a complete verification suite, applying the chapter's audit-fix-verify loop to catch the structural errors the chapter identifies. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈15–25 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch07-debugging-project/`.

Part III

Fitting the Model: The Right Estimator, Identifiable Parameters, and Reporting Uncertainty

Chapter 7

Parameter Estimation for the SIR Model

A model that fits the data has not been confirmed. It has failed to be refuted.

This chapter addresses the inverse problem of epidemic modeling: given incidence data from a real outbreak, how do we estimate the parameters of the SIR model? The answer has two parts. The first is methodological: what parameters can be estimated from what data, and by what procedure. The second is the practical contribution for which the book develops the infected viewpoint as its primary formulation.

Central result of this chapter.

> *The infected-viewpoint estimator $\hat{\alpha} = j/i$ of the force from infection, applied to observed incidence j and observed prevalence i, is exactly independent of the assumed susceptible population size s: its normalized sensitivity satisfies $\Upsilon_s^{\hat{\alpha}} = 0$. The susceptible-viewpoint estimator $\hat{\lambda} = j/s$ inherits the uncertainty in s directly: $\Upsilon_s^{\hat{\lambda}} = -1$. When s is poorly known, as it typically is during an emerging outbreak, $\hat{\lambda}$ inherits the error in the assumed s with unit elasticity while $\hat{\alpha}$ is untouched by it. Carrying either force estimate forward to $\mathcal{R}_0$ is a separate question, and a more delicate one: Section 7.6 shows that when prevalence is observed the assumed population cancels and the two viewpoints agree exactly, while a fit to incidence alone identifies $\mathcal{R}_0$ only through a compound of the assumed population and the reporting fraction that the data cannot resolve. The two-research-groups vignette of the Preface is the observable consequence of that second result.*

The rest of the chapter develops the formal statement, proves the asymmetry, and applies it on real outbreak data. The susceptible-viewpoint asymmetry is not a minor technical point: it is the primary reason to prefer the infected-viewpoint formulation whenever prevalence can be observed, and it resolves a practical puzzle that has generated sizeable disagreements in published $\mathcal{R}_0$ estimates for the same disease outbreak. Textbook treatments of fitting epidemic models to data are given by Martcheva [2015, Ch. 6] and Li [2018, Sec. 4.3].

Section 7.1 introduces the inverse problem and its structural challenges. Section 7.2 explains how surveillance data should be preprocessed before fitting, including the distinction between report date and event date, day-of-week cycles, holiday gaps, reporting delays, and incomplete recent observations. Section 7.3 discusses what can and cannot be inferred from early-outbreak data alone. Section 7.4 develops the least-squares fitting of α from incidence data. Section 7.5 treats the susceptible-fraction question. Section 7.6 gives the central numerical comparison of $\hat{\alpha}$ and $\hat{\lambda}$ under

uncertainty in s. Section 7.7 works through a case study using seasonal influenza data. Section 7.8 closes with a summary and exercises.

7.1 The Inverse Problem

In the forward problem addressed in Chapters 5 and 6, the parameters of the SIR model are given and the dynamics are determined. The inverse problem reverses this relationship: a time series of observed incidence (or some other epidemiological indicator) is given, and the parameters that produced it are sought. In practice, the inverse problem is solved by posing a loss function that measures the discrepancy between the model's output at a given parameter set and the observed data, and minimizing this loss over the parameter space.

The data. The most commonly reported epidemic data are counts of new cases per day or per week. Let $J_{\text{true}}(t)$ denote the true incidence: the rate of new infections in the population at time t. The reported data $\{J_{\text{obs}}(t_k)\}$ are noisy, possibly undercounted observations of $J_{\text{true}}(t)$ at the discrete times t_k.

For the SIR model, the incidence is

$$j(t) \;=\; \alpha(t)\, i(t) \;=\; c_I \beta P_S(t)\, i(t), \tag{7.1}$$

which is the rate at which individuals leave the susceptible compartment through infection. This expression connects the observed incidence to the model parameters through the force from infection α and the infectious-compartment size i.

In practice, incidence is often reported as a cumulative count rather than a rate: the cumulative number of reported cases through day t_k is $T(t_k) = \int_0^{t_k} J_{\text{obs}}(t')\, dt'$, and the single-day incidence is reconstructed by differencing, $J_{\text{obs}}(t_k) \approx T(t_k) - T(t_{k-1})$.

Four kinds of incidence. Three of these quantities are easy to conflate, and the fitting procedure adds a fourth; most fitting mistakes trace back to treating one as another. It helps to separate them once, explicitly (Figure 7.1).

- The *true incidence* $J_{\text{true}}(t)$ is the actual rate of new infections in the population. It is latent: no surveillance system observes it directly.
- The *observed incidence* $J_{\text{obs}}(t_k)$ is what surveillance reports—a sampled, noisy, typically undercounted view of J_{true}. Only a fraction ρ of infections is ascertained, reporting is delayed, and the counts carry day-of-week and backfill artifacts.
- The *estimated incidence* $J_{\text{est}}(t_k)$ is what remains once those reporting artifacts are removed by the preprocessing of Section 7.2. It is the series the model is actually fit to, and the best available proxy for the truth, $J_{\text{est}} \approx J_{\text{true}}$.
- The *model incidence* $j(t) = \alpha(t) i(t)$ is what the model predicts for a given parameter set, written $j(t;\theta)$ when the parameter dependence matters.

Fitting chooses θ so that the model prediction in counts, $j(t_k;\theta)\, N^{\text{c}}$, is fit to match the estimated incidence $J_{\text{est}}(t_k)$. The subtlety Figure 7.1 makes visible is that the model *represents* J_{true}—the quantity we actually care about—but is *fit* to J_{est}, both through the count bridge $j\, N^{\text{c}}$ since the

model works in normalized units, because J_{true} is never observed. Every gap between J_{est} and J_{true} (imperfect ascertainment, residual reporting artifacts) therefore propagates directly into the fitted parameters. The same obs/true/est vocabulary applies to any observed quantity—for prevalence, $I_{\text{obs}}, I_{\text{true}}, I_{\text{est}}$—whenever that quantity is what surveillance reports.

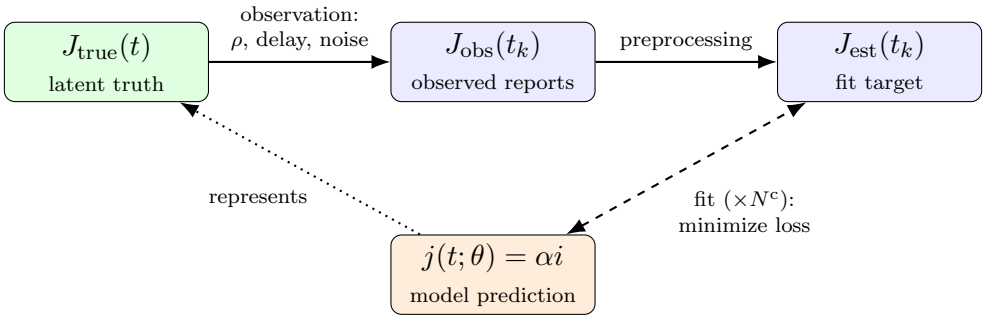


Figure 7.1: The four kinds of incidence and the fitting pipeline. The true incidence J_{true} is latent; observation degrades it to the reported J_{obs} (ascertainment fraction ρ, reporting delay, and noise), and the preprocessing of Section 7.2 recovers the estimate $J_{\text{est}} \approx J_{\text{true}}$. The model incidence $j(t;\theta) = \alpha i$ is normalized, so it *represents* the true incidence and is *fit* to the estimate through the count bridge $j\,N^{\text{c}}$, since J_{true} is never observed directly; fitting chooses θ to minimize the discrepancy between $j(t_k;\theta)\,N^{\text{c}}$ and $J_{\text{est}}(t_k)$.

What else might be observed. Beyond incidence, epidemic surveillance sometimes provides:

- *Prevalence*: the current number of infectious individuals, the count $I(t)$ (normalized fraction $i(t)$), estimated from active case surveillance or mass testing.
- *Serology*: the fraction of the population that has recovered (carries antibodies) from past infection, which gives $r(t)/n(t)$.
- *Mortality*: the rate of disease-induced deaths, which at appropriate time lag is proportional to $j(t)$ under the assumption of a fixed case-fatality ratio.

Each data type constrains the model in a different way. This chapter focuses on fitting to incidence data, which is the most commonly available and the most commonly used data type. Fitting to multiple data types simultaneously is more informative but also more complex; we touch on the joint-fitting problem briefly in Section 7.7.

The challenge. Even with perfect incidence data, the inverse problem for the SIR model has structural challenges. A recurring feature of outbreak inference is the distinction between *observed* and *latent* quantities. The reported incidence $J_{\text{obs}}(t_k)$ is observed, though imperfectly so. The compartmental populations $s(t)$, $i(t)$, and $r(t)$ are latent: none is directly measured, and each must be inferred from the observed incidence through the model's dynamics. The force from infection $\alpha(t) = j(t)/i(t)$, the book's central quantity, is a ratio of a partially-observed numerator to a latent denominator; its estimation depends on the inference of $i(t)$ from the incidence time series. The basic reproductive number $\mathcal{R}_0$ is further downstream still, derived from α and the estimated infectious period. This layering (observation at the top, model-mediated inference in the middle, derived parameters at the bottom) is characteristic of epidemic fitting and governs where the main sources of uncertainty enter.

Returning to the parameter-level challenge: the six independent parameters $(c_S, c_I, c_R, \beta, \tau_R, \tau_m)$ cannot all be estimated from incidence data alone. The parameters τ_R and τ_m have natural values from independent sources (clinical infectious-period data; demographic turnover data) and are typically treated as known inputs rather than fitted quantities. The remaining four parameters (c_S, c_I, c_R, β) enter $\mathcal{R}_0$ through the single combination $c_I\beta$ at the early stages of the outbreak, and through the combinations $c_S\nu_m + c_R\gamma_R$ and $c_I\nu_m + c_R\gamma_R$ at the endemic equilibrium. The data constrain these combinations but not the individual parameters.

This observation motivates a particular form of the fitting problem: rather than trying to estimate six parameters from incidence data, one estimates $\mathcal{R}_0$ (or equivalently, the force parameter α during the early phase) and $F_0 = S(0)/N^{\mathrm{c}}$, with the individual contact rates and the transmission factor taken from independent sources.

Exercises for this section

1. **[I]** In the worked numerical example of Section 7.6.7, the two research groups arrive at identical $\hat{\alpha}$ but different $\hat{\lambda}$ values, despite using the same incidence data. Explain in your own words why this happens. Then give one concrete example (real or hypothetical) of an outbreak where the two groups' geographic scope assumptions might differ by a factor of 30. Finally, use Corollary 7.4 to describe a single change in the reporting system—with no change at all to either group's geographic assumption—that would produce exactly the same disagreement in $\hat{\mathcal{R}}_0$.

7.2 Preprocessing Surveillance Data

Data preparation is most of the work. Preprocessing is one stage of a larger data-preparation effort (Pitfall: "Data Preprocessing"). Before an incidence series can be corrected for reporting artifacts it must be *found* (which surveillance system, which case definition, which geography), *assembled* (merging line lists, laboratory feeds, and death records that rarely share keys or timestamps), *cleaned* (removing duplicates, reconciling revised counts, resolving impossible values), and *restructured* (aggregated to a common time step and geography, and aligned to event date). Each stage is a modeling commitment that shapes the series the model ultimately sees, and each should be documented alongside the estimates. Practitioner folklore holds that data preparation consumes eighty percent of a modeling project; the honest figure for epidemic work is lower—often closer to half—but this remains the stage where the most consequential and least visible decisions are made, and the one that textbook treatments most often skip. This section concentrates on the preprocessing decisions that most directly bias a fit.

The inverse problem of the previous section assumed that an observed incidence time series $\{J_{\mathrm{obs}}(t_k)\}$ is given, and asked which parameter set produced it. Preprocessing is the step that turns those raw reports into the estimated series $J_{\mathrm{est}}(t_k)$ the fit actually uses—the $J_{\mathrm{obs}} \to J_{\mathrm{est}}$ stage of Figure 7.1. Real surveillance data complicate this premise. Reported case counts are not a clean record of the epidemic; they are the output of a reporting pipeline whose internal structure is systematically present in the data. Testing capacity follows business hours, lab schedules generate a day-of-week reporting cycle, public holidays produce gaps that are filled in days later, and the most recent observations in any dataset are systematically incomplete because the reporting delay has not yet elapsed for the most recent events. Each of these features is a structural artifact of the reporting process, not a feature of the underlying transmission dynamics.

A least-squares fit to uncorrected data treats this reporting structure as signal. The Friday-spike/Sunday-trough pattern is absorbed into the transmission parameters β and $\hat{\alpha}$, biasing them toward whatever values produce a 7-day periodicity in the model's output. The holiday gap is interpreted as a transient drop in transmission and gets fit accordingly. The artificial dip at the most recent days pulls the fitted $\mathcal{R}_e(t)$ downward at exactly the moment the modeler is most likely to be asked for a forecast or a current-state assessment. Each of these failures is observable, common, and preventable.

Surveillance data carry reporting artifacts that a fit absorbs as signal unless they are removed first, which is the pitfall Data Preprocessing (Chapter 16).

The remainder of this section develops the principle in five parts. We first distinguish the timestamps that surveillance data carry (Section 7.2.1). We then examine the day-of-week reporting cycle and a simple correction (Section 7.2.2), the related artifacts of holiday gaps and reporting delays (Section 7.2.3), the recent-edge problem of backfilling (Section 7.2.4), and the missing-data problem that cuts across them all (Section 7.2.5). Beyond the incidence series itself, fitting also needs a susceptible denominator, whose preparation—and the good news that the infected-viewpoint estimator largely sidesteps it—is taken up in Section 7.2.6. A worked example showing the magnitude of the effect on parameter estimates appears in Chapter 8. The section closes (Section 7.2.7) by relating preprocessing to the chapter's other named pitfalls.

7.2.1 Report Date versus Event Date

Surveillance data carry at least two timestamps that the modeler must distinguish. The *event date* is the date on which the case actually occurred: the date of symptom onset, the date of hospitalization, the date of death, or some other epidemiologically meaningful date. The *report date* is the date on which the case was entered into the surveillance system. The distribution of the difference (report date minus event date) is the *reporting-delay distribution*.

The reporting-delay distribution is jurisdiction-specific, time-varying, and typically heavy-tailed. Medians of three to seven days are common in well-functioning systems; tails extending several weeks are not unusual. The distinction between report date and event date matters for fitting because the model's incidence $j(t)$ is naturally indexed by event date (it is the rate of new infections at time t), while raw surveillance data are typically released indexed by report date. Fitting a model to data indexed by report date while treating the model's $j(t)$ as event-date incidence is a structural mismatch that the fit will absorb as biased parameter estimates.

The pragmatic recommendation is to fit to event-date data wherever it is available. Most public-health surveillance systems make both timestamps available; the modeler's task is to identify which one the dataset is indexed by. Where only report-date data are available, the next subsections develop simpler corrections; full deconvolution against a reporting-delay distribution is treated in the literature [for a representative review, see Goldstein et al., 2009].

7.2.2 The Day-of-Week Reporting Cycle

The day-of-week reporting cycle is the most visible artifact in most surveillance datasets. Its mechanism is straightforward: testing capacity is concentrated on weekdays, lab processing is concentrated on weekdays, and reporting forms are filed on weekdays. A test administered on Saturday is processed Monday and reported Tuesday; a test administered on Wednesday is processed Wednesday afternoon and reported Thursday. The result is a 7-day cycle in reported counts, although the precise days on which counts peak depend on the local testing, processing, and reporting schedule.

The diagnostic is the periodogram. A periodogram decomposes a time series into its frequency components, showing how much of the signal's variance sits at each frequency; it is the standard tool for detecting periodic structure, so a regular weekly cycle stands out as a peak. To compute it, take the discrete Fourier transform of the daily case-count series and plot the power as a function of frequency (Figure 7.3 works through an example). A peak at frequency $1/7$ day^{-1} is the signature of the day-of-week effect. The presence of this peak is a near-universal feature of raw daily-resolution surveillance data; its magnitude varies by jurisdiction but is rarely small.

The simplest correction is a 7-day centered moving average. Replace each day's reported count $J_{\text{obs}}(t_k)$ with the average of the seven counts centered at t_k:

$$\widetilde{J}(t_k) \;=\; \frac{1}{7}\sum_{j=-3}^{3} J_{\text{obs}}(t_{k+j}). \tag{7.2}$$

This correction has three properties worth naming explicitly. First, it is lossy: it smooths real epidemic signal at scales shorter than seven days, including any genuine super-exponential acceleration during early outbreak phases. Second, it is edge-degraded: at the start and end of the dataset, the centered average uses fewer than seven points, increasing the variance of $\widetilde{J}_t$ at the recent edge (where reliability matters most for forecasting). Third, it is incomplete: the 7-day moving average reduces the day-of-week peak in the periodogram but does not always eliminate it, particularly when the day-of-week pattern is asymmetric across the week.

A more careful correction is day-of-week deconvolution: estimate a multiplicative weight w_d for each day-of-week $d \in \{\text{Mon}, \ldots, \text{Sun}\}$ from the data—the day's relative reporting level, above one on days that systematically over-report and below one on days that under-report, with mean one across the week—then divide each count by its day-of-week weight, removing that day-specific inflation or deflation to produce a corrected series. The weights are estimated jointly with the underlying smooth incidence, often by fitting a parametric or smoothing model that explicitly separates the slow trend from the periodic component. This approach is less lossy than the moving average but introduces estimation error in the weights themselves. Pang et al. [2025] extend this approach by fusing a compartmental model with a generalized additive model that estimates the day-of-week weights jointly with the time-varying effective reproductive number, treating the day-of-week correction as part of the inference problem rather than as a separate preprocessing step. Their construction is the contemporary state of the art for one-step day-of-week-aware $\mathcal{R}_e(t)$ estimation.

Validation. Whichever correction is applied, the modeler should recompute the periodogram of the preprocessed series. The peak at $1/7$ day^{-1} should be substantially reduced. If it is not, the preprocessing did not catch the day-of-week effect, and the fit will inherit the artifact regardless of the correction's nominal intent.

7.2.3 Holiday Gaps and Reporting Delays

Public holidays produce reporting gaps that are filled in over the next several days. In the United States, the conspicuous gaps are around Memorial Day, July 4, Thanksgiving, and the Christmas/New Year week; in other jurisdictions the calendar is different but the structure is the same. A 1-to-4 day reporting gap is typically followed by elevated counts as the backlog clears, producing a downward spike followed by an upward spike in the time series.

Treating these gaps as low-transmission periods is a common error. The day-of-week correction of Equation (7.2) partially smooths holiday gaps but does not eliminate them, because the gap is

a one-time event rather than a periodic feature. The pragmatic correction is to identify holiday-affected days explicitly and either (i) drop them from the fit, discarding those observations, or (ii) retain them but treat them as missing under a likelihood-based fitting framework that handles missingness honestly (Section 7.2.5)—the two coincide for a plain least-squares fit but diverge once the framework models *why* a value is missing.

Reporting delays produce a related but distinct artifact. The reporting-delay distribution interacts with the day-of-week cycle: a case occurring on a Saturday with a 2-day median reporting delay produces a report on Monday, contributing to Monday's reported count rather than to Saturday's. When the reporting delay is short and stable, fitting to event-date data sidesteps the issue entirely. When the reporting delay is long or varies in time, even event-date data are subject to revision; the next subsection takes up this case.

7.2.4 The Recent-Edge Problem

The most recent observations in any surveillance dataset are systematically incomplete. A case occurring today will be reported, on average, in several days; a case occurring a week ago may still be in the pipeline. The result is that the most recent days appear artificially low in any dataset accessed before the reporting delay has fully elapsed.

The implication for fitting is that the recent edge is the part of the data least suited to the simple fitting machinery developed elsewhere in this chapter. A least-squares fit applied to the full time series, including the recent edge, will pull fitted $\mathcal{R}_e(t)$ downward at exactly the moment forecasts are most often requested. The connection to Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$" is direct: the recent-edge artifact is one of the principal practical reasons that $\mathcal{R}_e(t)$ from raw data should not be reported uncritically.

Two responses are practical. The pragmatic response is to drop the most recent n days from the fit, where n is chosen as a high quantile (e.g., the 90th percentile) of the reporting-delay distribution. This is crude but defensible: the dropped days are the ones the dataset is most actively still reporting, so excluding them from the fit avoids the bias they would introduce. The principled response is *nowcasting*: explicitly model the reporting-delay distribution, treat the recent days' observed counts as truncated samples from the distribution, and project what the eventual backfilled counts will be.

Both problems are, at bottom, a *deconvolution*. The reporting pipeline convolves the true incidence with a delay kernel: writing $d(\tau)$ for the probability that a case is reported τ days after it occurs, the expected reported count is

$$\mathbb{E}[J_{\text{obs}}(t)] \;=\; \rho \sum_{\tau \geq 0} d(\tau)\, J_{\text{true}}(t-\tau), \tag{7.3}$$

the same convolution drawn in Figure 7.1. The ascertainment fraction ρ scales the delayed true incidence to the reported level, so that only the reported fraction survives; its role in the observation model is developed further in Chapter 8. The sum runs over whole-day delays τ because surveillance is reported at daily resolution, with the continuous-time analogue an integral over τ. Recovering J_{true}—and in particular its most recent values, which (7.3) has only partially populated because the later-arriving terms of the sum have not yet been reported—means inverting this relation. Richardson–Lucy deconvolution, introduced to epidemic surveillance for reconstructing influenza incidence from mortality data [Goldstein et al., 2009], is the classical tool; modern nowcasting frameworks such as EpiNow2 [Abbott et al., 2020] embed the same deconvolution inside a probabilistic model that also propagates its uncertainty. The inversion is ill-posed at the recent

edge—the most recent true incidence is the least constrained by what has been reported so far—which is why nowcast estimates there carry wide, often asymmetric uncertainty bands. This chapter names the method rather than developing it, but the modeler should take away two things: the recent-edge problem has a principled solution, and that solution necessarily returns large error bars for the most recent days.

7.2.5 Missing Data

The holiday gaps of Section 7.2.3 and the incomplete recent edge of Section 7.2.4 are specific instances of a problem that runs through every stage of data preparation: some of the observations the fit would like to use are missing. Missingness deserves treatment in its own right, because the correct response depends on *why* the data are missing, and the wrong response introduces bias that is invisible in the fit diagnostics.

Kinds of missingness. Three patterns are common in epidemic surveillance.

- *Structural gaps*: no report is expected—weekends and holidays for jurisdictions that do not report daily, or days before a surveillance system came online. The value was never generated on that schedule rather than merely unrecorded.

- *Unascertained cases*: infections that occurred but were never captured, the aggregate of which is the ascertainment shortfall $1 - \rho$. This is missingness in J_{true} rather than in the reported series; no amount of data cleaning recovers it, and it is handled by the observation model (Chapter 8), not by preprocessing.

- *Suppressed counts*: values withheld for privacy, typically small counts (often those below five) redacted in publicly released data. Here the value exists but is censored, and—critically—the censoring is correlated with the value, since it is precisely the small counts that are removed.

Handling, and the bias each risks. Given a gap in the reported series, three responses are available.

- *Drop* the missing observations from the fit. This is safe when the data are missing completely at random—missingness independent of the underlying value—so that the remaining observations are an unbiased sample. It is *not* safe for suppressed small counts: dropping the censored days removes the low-incidence periods preferentially and biases the fitted trajectory upward.

- *Impute* the missing values, most simply by interpolation. Imputation fills the gap but fabricates data: it injects the interpolation model's assumptions (usually smoothness) as though they were observations, understating uncertainty and smoothing over any real feature that occurred during the gap. Interpolation is acceptable across short structural gaps and dangerous across long or informative ones.

- *Mask* the missing observations in a likelihood-based fit: let the model predict the missing days but exclude them from the loss, so they contribute nothing rather than a fabricated value. This is the honest default when the fitting framework supports it, because it neither discards information from the observed days nor invents information for the missing ones.

The distinction that governs the choice is the standard missing-data taxonomy. Data missing completely at random can be dropped without bias. Data missing at random—missingness that depends on observed covariates, such as the day of the week—can be handled by conditioning on those covariates, which is exactly what the day-of-week correction of Section 7.2.2 does. Data missing *not* at random—suppressed small counts, or unascertained cases whose rate depends on the very incidence being estimated—bias any naive approach and require the missingness mechanism to be modeled explicitly. As with the other preprocessing choices, the missingness assumption is part of the modeling commitment and should be reported with the estimates.

7.2.6 The Susceptible Denominator

The incidence series is not the only quantity a fit consumes. The susceptible-viewpoint estimator $\hat{\lambda} = J_{\text{obs}}/s$, and any force written as a fraction j/n of the population, require a *denominator* (Pitfall: "Force of Infection versus Force from Infection"): the number of people at risk in the population the outbreak is actually drawing from. Preparing this denominator is its own data problem, and a particularly stubborn one: even careful work leaves the susceptible count poorly constrained.

The difficulty is that the epidemiologically relevant population is not an administrative object. A census counts residents by address; it does not count the population that mixes with the index cases, which may be a neighborhood, a transit corridor, a workplace network, or a whole metropolitan area, and which no single table reports. Section 7.5.1 returns to this as a modeling choice and shows that plausible catchment definitions span three orders of magnitude. Prior immunity compounds the problem: the denominator is the *susceptible* count, so infections and vaccinations from earlier waves must be subtracted, and the only direct measurement of accumulated immunity—serological surveys—is itself sparse, lagged, and assay-dependent. No preprocessing step removes this uncertainty; it is structural.

The good news is the reason the book develops the infected viewpoint. The infected-viewpoint estimator of the force from infection, $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$, never refers to the susceptible count: its denominator is the observed *prevalence* i, not the susceptible population s. It therefore sidesteps the denominator-preparation problem entirely, while the susceptible-viewpoint estimator inherits the denominator's uncertainty factor for factor—a discrepancy that can reach three orders of magnitude during the early-outbreak phase, when $S \approx N^{\text{c}}$ is large and poorly known while i is small and comparatively well estimated. This is a data-preparation argument for the infected viewpoint, made quantitative in the estimator comparison of Sections 7.4.3 and 7.6.

7.2.7 Connections to Other Pitfalls

The data-preprocessing pitfall is woven into the chapter's other named pitfalls in several ways.

The connection to Pitfall: "Identifiability" is operational: preprocessing changes which parameters are identifiable from the resulting time series. A heavily smoothed dataset has less information about high-frequency dynamics and tightens identifiability of slow-varying parameters at the cost of fast-varying ones. The Buckingham-Pi count of identifiable dimensionless groups developed in Section 7.3 does not change under preprocessing, but the empirical informativeness of the data about each group does.

The connection to Pitfall: "Estimator Honesty" is methodological: fitting to uncorrected data and reporting the parameter estimates without disclosing the preprocessing choices is a form of estimator dishonesty. The Scenario 4 fix developed later in this chapter (Section 7.6) and the preprocessing principle here are both aspects of a single commitment: the modeling output is

honest only if the structural choices that produced it are reported alongside the numerical results. The reader of a fitted estimate should be able to recover what the modeler did to the data before fitting.

The next subsection, Section 7.2.8, makes these connections concrete with a worked example demonstrating the magnitude of the bias that uncorrected data introduce into fitted parameter estimates.

7.2.8 Worked Example: How Much Does Preprocessing Matter?

To make the preprocessing principle concrete, we generate a synthetic incidence trajectory from the book's baseline SIR parameter set (Table 4.4), inject a multiplicative day-of-week reporting cycle calibrated to published patterns from JHU CSSE COVID-19 surveillance data [Abbott et al., 2020, Goldstein et al., 2009], and ask how the fitted parameters change as the preprocessing improves.

The synthetic dataset. The ground-truth incidence J_{true} is obtained by integrating the SIR model with $c_S = 20$, $c_I = 4$, $c_R = 20$, $\beta = 0.040$, $\tau_R = 12$ days, and $\tau_m = 7300$ days (yielding $\mathcal{R}_0 \approx 1.92$) and scaling the normalized incidence j to absolute daily case counts in a population of $n_{\text{pop}} = 10^6$, modulated by day-of-week weights $w_{\text{Mon}} = 1.10$, $w_{\text{Tue}} = 1.10$, $w_{\text{Wed}} = 1.10$, $w_{\text{Thu}} = 1.15$, $w_{\text{Fri}} = 1.20$, $w_{\text{Sat}} = 0.65$, $w_{\text{Sun}} = 0.70$, whose mean is one. The pattern (Friday peak, weekend trough, and a flat Monday–Wednesday weekday baseline that climbs into the Thursday–Friday peak) matches the structure reported across multiple JHU-based studies [Abbott et al., 2020]. Poisson-scale observation noise is added on top, producing the raw daily count series.

Figure 7.2 shows the raw data alongside the same series after each of the three preprocessing levels developed in the previous subsections. The day-of-week artifact is conspicuous in the raw series; both the 7-day moving average and the day-of-week deconvolution suppress it almost entirely, with the moving average producing a slightly smoother trajectory at the cost of attenuating the trend's curvature.

The periodogram diagnostic introduced in Section 7.2.2 confirms what the eye sees. Figure 7.3 shows the power spectral density of each series. The raw data have a sharp peak at $1/7$ day^{-1}, several times the background; both preprocessing methods reduce this peak to baseline.

The parameter-recovery comparison. We now fit the SIR model to each preprocessed series. To represent an applied modeler estimating parameters from the rising limb of an outbreak, we fit all 60 days of the series, which end before the peak. The optimizer minimizes the squared residual between the model's incidence in counts, $j(t) \cdot n_{\text{pop}}$, and the observed counts J_{obs} on each day, each residual divided by the square root of the model count (the Poisson weighting of Section 7.4.2), treating c_S, c_I, c_R, and τ_m as known and fitting only β and τ_R.

Figure 7.4 shows the fitted trajectories under each preprocessing level. Table 7.1 summarizes the recovered parameters and their relative errors against the known truth.

The bias in $\hat{\mathcal{R}}_0$ from raw data is +3.2%, while either preprocessing method brings the recovery to within 1.8%. Repeating the experiment across twenty random seeds of the observation noise confirms the direction: the raw-data estimate is biased upward in every run, by about three percent on average with a run-to-run standard deviation of about one percent, while the moving average sits about two percent low in every run and the deconvolution about one percent high on average.

What the magnitude means operationally. A three-percent bias in $\hat{\mathcal{R}}_0$ is small in absolute terms but systematic—always in the same (upward) direction, so it does not average away across

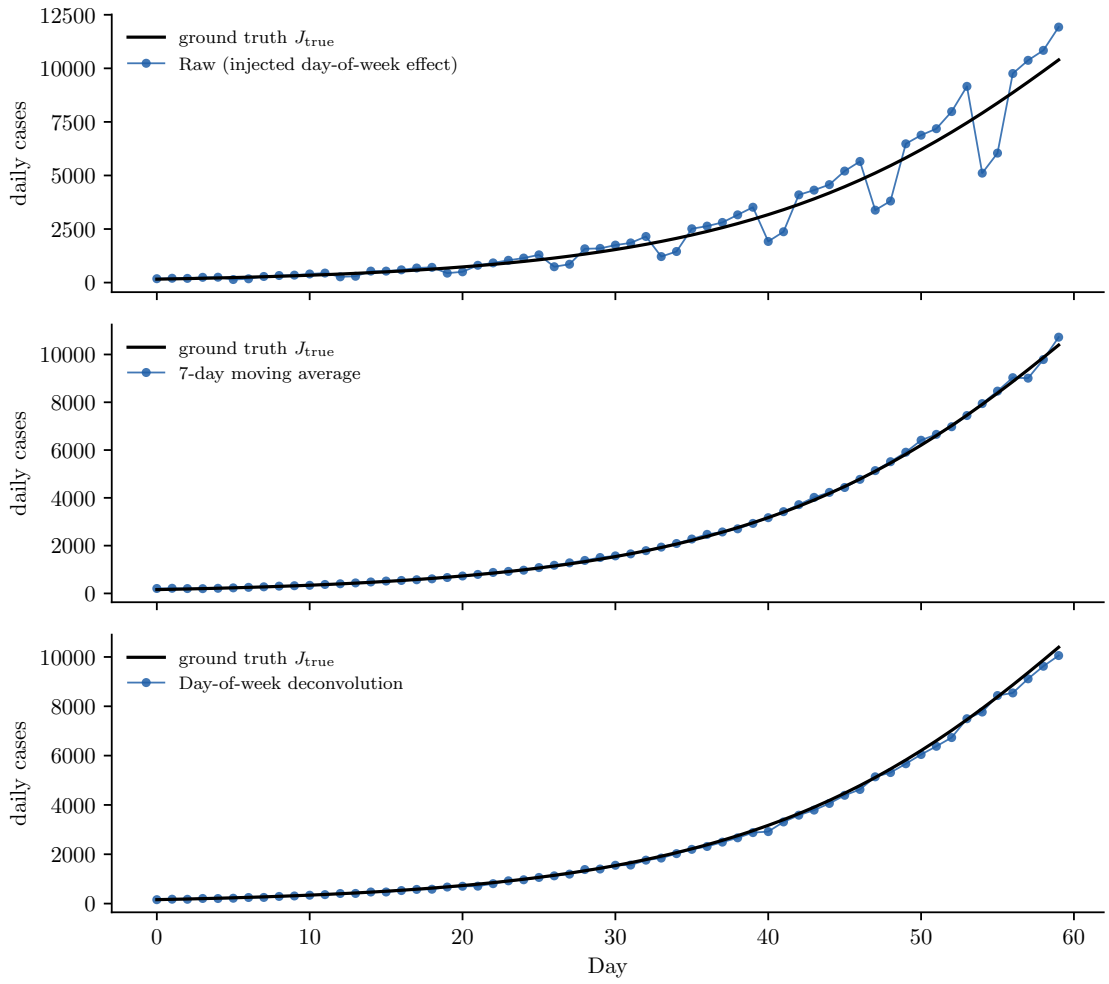


Figure 7.2: Synthetic incidence data over a 60-day window centered on the outbreak's exponential-growth phase. *Top:* the raw series with an injected day-of-week effect calibrated to JHU CSSE COVID-19 surveillance patterns. *Middle:* the same series after a 7-day centered moving average. *Bottom:* the same series after day-of-week multiplicative deconvolution. The ground-truth trajectory $j(t)$ (solid black) is shown for reference in all three panels.

repetitions. In coarse-grained analyses (comparing whether $\mathcal{R}_0$ is above or below one, classifying an outbreak by order of magnitude), it is ignorable. In applied work where small differences matter (vaccine efficacy estimation, intervention attribution, sub-doubling-time forecasting, or any analysis where the difference between $\mathcal{R}_0 = 1.92$ and $\mathcal{R}_0 = 1.98$ has policy consequences), a three-percent systematic bias is operationally important. The preprocessing pitfall is not always a dramatic failure; it is more often a small, systematic error too subtle to be caught by the modeler's usual quality checks.

The bias attenuates when the fit window extends through the peak: the curvature of the trajectory near the peak provides additional constraints that pull the fit toward the truth, but a fit over a 180-day window narrows the raw-data bias only to about two percent. This is the second pedagogical lesson: *preprocessing matters most where the data matter most*, which is the period of an outbreak when the trajectory's structure is least informative and the modeler is most reliant on getting each observation right. Reproducing this attenuation on the raw series is left as an exercise.

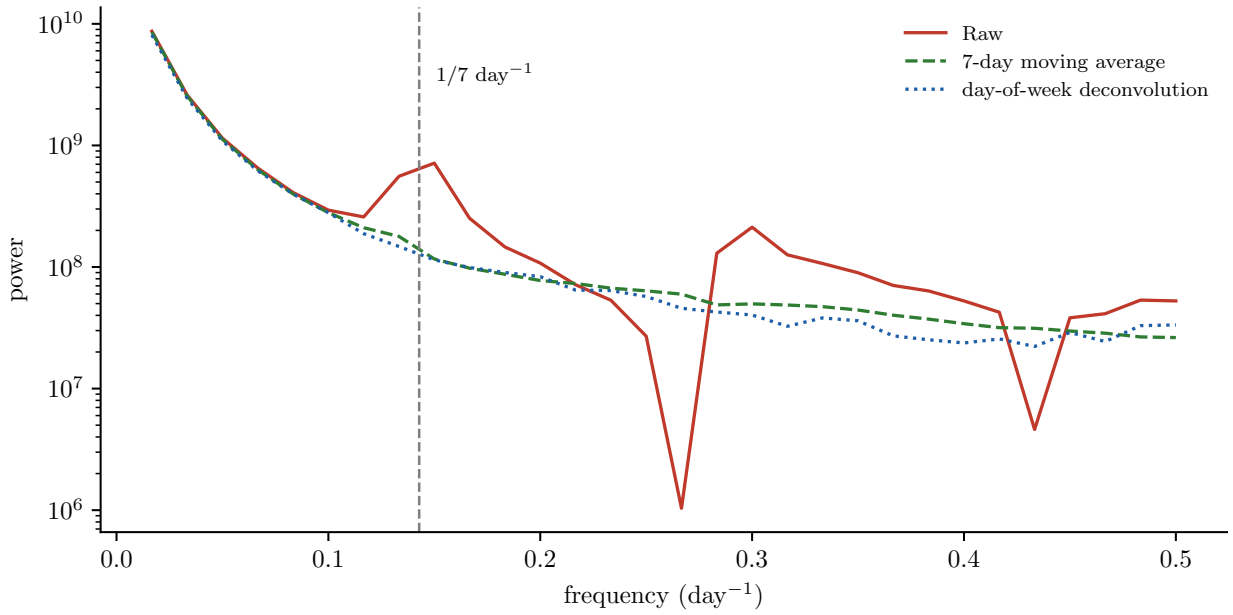


Figure 7.3: Periodograms of the raw and preprocessed series. The periodogram is the unnormalized squared magnitude of the discrete Fourier transform of the mean-subtracted daily counts, so power has units of cases^2. The vertical dashed line marks the 1/7 day^{-1} frequency where the day-of-week artifact concentrates. Both preprocessing methods eliminate the peak; the raw series retains it at substantial amplitude.

Table 7.1: Recovered SIR parameters under three preprocessing levels (60-day fit, Poisson-weighted residuals; ground truth $\beta = 0.040$, $\tau_R = 12$ days, $\mathcal{R}_0 = 1.9169$).

Preprocessing	$\hat{\beta}$	$\hat{\tau}_R$ (days)	$\hat{\mathcal{R}}_0$	Rel. error in $\hat{\mathcal{R}}_0$
Raw (no correction)	0.0393	12.61	1.979	+3.24%
7-day moving average	0.0407	11.59	1.884	−1.72%
Day-of-week deconvolution	0.0393	12.37	1.941	+1.27%

Honesty about the preprocessing choice. The three preprocessing levels examined here are not the only options, and none is unambiguously correct. The 7-day moving average attenuates real epidemic signal at scales shorter than seven days, including any genuine super-exponential acceleration during early outbreak phases. The day-of-week deconvolution assumes that the weekly cycle is multiplicative and stationary, which is true on average but fails during periods of changing test capacity or new variant detection. The raw data require no assumption but introduce the bias documented above. Pitfall: "Estimator Honesty" requires that whichever choice is made, it is reported alongside the parameter estimates, so that the reader of a fitted result can audit the preprocessing assumptions and not just the model.

Exercises for this section

1. [**C**] A surveillance system administers tests only on weekdays, with each test reported the next business day. Sketch the expected pattern of reported counts across a typical week,

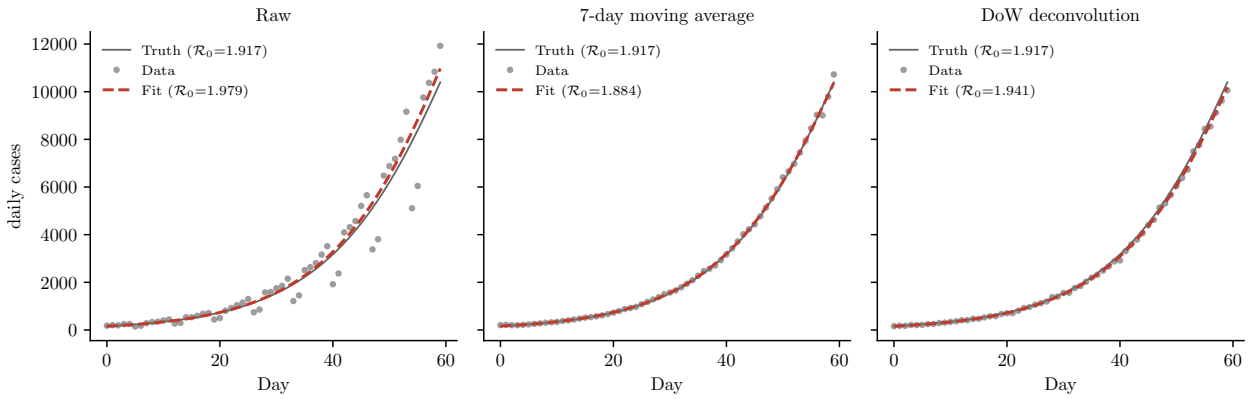


Figure 7.4: Fitted SIR trajectories under each of the three preprocessing levels, fitted to all 60 days. Recovered $\mathcal{R}_0$ values shown in each panel's legend.

identifying which day of the week will have the highest and lowest reported counts. Explain why this pattern is an artifact of the reporting pipeline rather than a feature of the underlying transmission dynamics.

2. [**I**] Use the synthetic surveillance series of Figures 7.2 to 7.4, whose generating script in the companion repository builds in a known day-of-week effect, or a public daily case series of your choice. (i) Plot the raw daily counts and compute the periodogram. Identify the peak at $1/7$ day^{-1} (trim the series to a whole number of weeks, or the peak splits between frequency bins). (ii) Apply the 7-day centered moving average of Equation (7.2). Replot and recompute the periodogram. Discuss what the moving average preserves and what it loses.

3. [**L**] On the same dataset as the previous exercise, implement a day-of-week multiplicative deconvolution: estimate seven weights $w_{\text{Mon}}, \ldots, w_{\text{Sun}}$ such that the corrected series $\widetilde{J}(t_k) = J_{\text{obs}}(t_k)/w_{d(t_k)}$ has minimal day-of-week structure. Fit a SIR model to (a) the raw data, (b) the 7-day moving average, and (c) the deconvolved data. Report the three sets of fitted parameters and discuss which preprocessing approach is most appropriate for the dataset and why.

7.3 What Can Be Fit: An Identifiability Discussion

Before discussing fitting procedures, we clarify what the data actually support. Informally, the number of distinct pieces of information in an incidence time series during the exponential-growth phase of an outbreak is approximately two: the growth rate and the starting time. No amount of fitting sophistication can extract more than two parameters from this information. Comparative work on SEIR-type models documents this constraint quantitatively: Saucedo et al. [2024] compare structural and practical identifiability across observable data types and show that incidence-indexed data yields the most identifiable parameters, prevalence-indexed data is intermediate, and cumulative-incidence data is the least identifiable. The chapter's preference for incidence as the fitting target (rather than cumulative case counts) is consistent with this empirical hierarchy. Chapter 8 develops the diagnostic for distinguishing practical-identifiability failures from genuine non-identifiability via the profile likelihood.

Table 7.2 summarizes the status of each quantity in the SIR fitting workflow. Three categories matter for the practical discussion of identifiability and asymmetry: the model parameters (most

externally calibrated, though the transmission probability β is identifiable only in the combination $c_I\beta$), the modeling choice of susceptible population N^c (which has no canonical value and is the source of the fitting asymmetry developed in this chapter), and the observed data with its derived estimators.

7.3.1 The Two Degrees of Freedom in Exponential Growth

In the exponential-growth regime at the start of an outbreak, $P_S(t) \approx 1$ (Pitfall: "Linearization Fidelity") and the infectious population evolves as

$$i(t) \;\approx\; i(0)\, e^{\sigma_2 t},$$

with $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ from Chapter 5. The incidence is

$$j(t) \;\approx\; \alpha i(t) \;\approx\; c_I\beta \cdot i(0)\, e^{\sigma_2 t}.$$

Taking logarithms,

$$\ln j(t) \;\approx\; \ln(c_I\beta i(0)) + \sigma_2\, t.$$

This is a linear function of t with two parameters: a slope (σ_2) and an intercept ($\ln(c_I\beta i(0))$). Any fitting procedure applied to exponential-growth data can recover at most these two combinations. A procedure that tries to recover c_I and β separately, with $i(0)$ also unknown, is underdetermined: the data do not distinguish between a model with $(c_I, \beta, i(0))$ and a model with $(2c_I, \beta/2, i(0))$ that has the same slope and intercept. Formally, the Fisher information matrix for the exponential-growth model has rank two, confirming that early-outbreak data identifies at most two scalar functionals of the parameters; this rank condition is the formal content of the "two degrees of freedom" statement above. The structural-identifiability literature treats this rank-deficiency analysis systematically [Raue et al., 2009, Villaverde et al., 2016].

A consequence. The early-growth data constrain σ_2, and hence $\mathcal{R}_0$ once γ_R and ν_m—the recovery and demographic-turnover rates, taken throughout this chapter as externally calibrated (Table 7.2)—are substituted in. The data also constrain the product $c_I\beta i(0)$, a three-way ambiguity that is not resolved by more data from the same growth regime. Resolving the ambiguity requires additional information: a known $i(0)$ (from genomic surveillance of an outbreak's first cases, for example), or additional data from a later regime of the epidemic.

Caution: the conversion from σ_2 to $\mathcal{R}_0$ assumes an exponential infectious period. The relationship $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ rests on the SIR model's assumption that the infectious period is exponentially distributed with mean $\tau_R^* = 1/(\gamma_R + \nu_m)$, equivalently that the mean generation time T_g equals τ_R^*, an aspect of the time-in-compartment treatment that the catalog handles under Pitfall: "The Independence of τ" (with the data-resolution implications under Pitfall: "Correct Data-Fitting Practice"). For pathogens with strongly pre-peak infectiousness profiles (influenza and SARS-CoV-2 are the canonical examples, where viral shedding peaks before or at symptom onset rather than uniformly across the infectious period), the true mean generation time is shorter than τ_R^*, typically with $T_g/\tau_R^* \approx 0.5$–0.7 (Figure 7.5). In that case the exponential-period conversion overestimates $\mathcal{R}_0$: the same observed growth rate σ_2 implies a smaller $\mathcal{R}_0$ when the generation time is shorter (for an exponential interval, $1 + \sigma_2 T_g$ in place of $1 + \sigma_2\tau_R^*$). Chapter 15, Section 15.5 develops the renewal-equation correction that handles arbitrary generation-time distributions; for the case studies in this book where the exponential approximation is materially wrong (specifically

Quantity	Role in model	Source	Identifiability status
Model parameters			
c_S, c_R	Contact rates of susceptibles, recovered	Contact-tracing surveys, behavioral data	Externally calibrated; often $c_S \approx c_R$
c_I	Contact rate of infectious individuals	Contact-tracing of cases; symptomatic-isolation studies	Externally calibrated; enters $\mathcal{R}_0$ only through $c_I\beta$
$\tau_R = 1/\gamma_R$	Mean infectious period	Clinical observation of symptom duration; viral-shedding studies	Externally calibrated, typically known to ±10–20%
$\tau_m = 1/\nu_m$	Demographic turnover time	Census birth/death rates	Externally calibrated, essentially known
Modeling choice (source of fitting asymmetry)			
$N^{\mathrm{c}}, S(0)$	Susceptible population size	Geographic / contact scope of outbreak; modeling judgment	*No canonical value*; can vary by factor of 5–10; root cause of the susceptible-viewpoint asymmetry
Observed data			
$J_{\mathrm{obs}}(t)$	Daily incidence (new cases per day)	Public-health surveillance reports	Observed (with reporting lag and undercount; see Section 7.7)
$I_{\mathrm{obs}}(t)$	Current prevalence	Active case surveillance, contact tracing, mass testing	Observed (when available; otherwise reconstructed)
Fitted and derived quantities			
$\hat{\alpha} = J_{\mathrm{obs}}/I_{\mathrm{obs}}$	Infected-viewpoint estimator of the force from infection	Direct ratio of observed quantities	Fitted; $\Upsilon_s^{\hat{\alpha}} = 0$ (*s-independent*)
$\hat{\lambda} = J_{\mathrm{obs}}/s$	Susceptible-viewpoint estimator	Ratio using assumed s	Fitted; $\Upsilon_s^{\hat{\lambda}} = -1$ (*inherits s uncertainty*)
$\hat{\alpha}/(\gamma_R + \nu_m)$	Effective reproductive number $\mathcal{R}_e$; equals $\mathcal{R}_0$ only when $P_S \approx 1$	Derived from $\hat{\alpha}, \gamma_R, \nu_m$	Derived; s-independent as an estimate of $\mathcal{R}_e$, but reading it as $\mathcal{R}_0$ requires $P_S \approx 1$
β	Per-contact transmission probability	Enters only via the product $c_I\beta$	Not separately identified; recoverable only in the combination $c_I\beta$ (through $\mathcal{R}_0$), never alone

Table 7.2: Status of each quantity in the SIR fitting workflow from incidence and prevalence data. The susceptible-population quantities N^{c} and $S(0)$ are the only quantities without a canonical external source; they are modeling choices that can vary by a factor of 5–10 across reasonable scope definitions, which is the practical source of the asymmetry between $\hat{\alpha}$ and $\hat{\lambda}$ developed in this chapter.

the COVID-19 case study of Chapter 17), the renewal-equation result is used. For pathogens like seasonal influenza where the exponential approximation is adequate to within published uncertainty in τ_R, the simpler conversion is used.

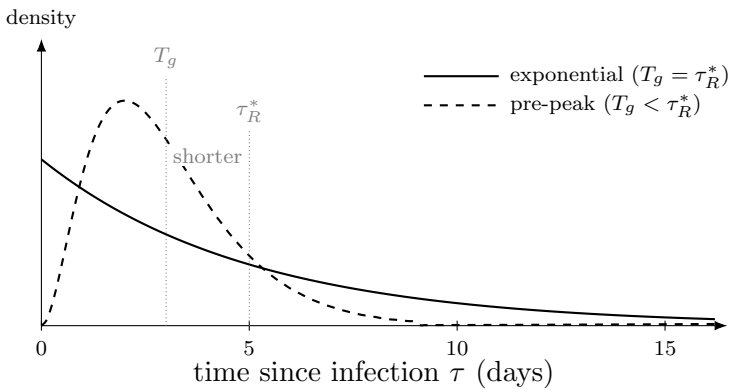


Figure 7.5: Generation-time distributions and the exponential-period assumption. The SIR model treats the infectious period as exponentially distributed (solid), for which the mean generation time equals the mean infectious period, $T_g = \tau_R^*$. Pathogens whose infectiousness peaks before the end of the infectious period have a more concentrated generation-time distribution (dashed) with a shorter mean, $T_g < \tau_R^*$ (drawn here at $T_g \approx 0.6\,\tau_R^*$). Because the observed growth rate σ_2 constrains $\mathcal{R}_0$ only through the generation time, the exponential assumption overestimates $\mathcal{R}_0$ when the true generation time is shorter.

Verify the AI: Joint MLE Convergence vs. Identifiability

Ask an AI to "fit the SIR model" to early-outbreak data and it may fit all four of $(c_I, \beta, \gamma_R, i(0))$, reporting a converged optimum, small residuals and a confidence interval for each. The fit looks successful and is not. Early-growth data carry about two pieces of information, σ_2 and a time shift, so the optimizer's starting values and bounds pick the answer, not the data.
Ask two questions. "*What is the rank of the Fisher information matrix at the optimum, and which parameter combinations are unidentifiable?*" A correct answer names $c_I\beta$ as one combination and ties $i(0)$ to it. "*Re-run from three starting points and report the spread.*" An under-identified fit gives different optima with the same residuals. A converged optimizer is not evidence of identifiability (Pitfall: "Identifiability"): find the identifiable combinations first, then fit them.

7.3.2 Worked Example: Three Parameter Sets, One Trajectory

The under-determination just described is easier to believe once it is seen. Consider three parameter sets, each paired with the same seed $i(0) = 0.001$:

$$(c_I,\ \beta) \ = \ (8,\ 0.02), \qquad (4,\ 0.040), \qquad (16,\ 0.01). \tag{7.4}$$

The middle pair is the book's baseline. All three give the same product,

$$c_I\beta \ = \ 0.16 \text{ day}^{-1},$$

and therefore the same α, the same $\mathcal{R}_0 = 0.16/(\gamma_R + \nu_m) \approx 1.92$, and the same early-growth rate σ_2. Simulated forward from the same seed, the three trajectories are indistinguishable during the

exponential-growth phase, because while $c_I i$ is negligible in C_{total} the model's behavior depends on c_I and β only through their product. They separate once prevalence grows (by about 2% in incidence after 30 days and 12% after 60), which is where the depletion phase carries information about c_I itself.

An optimizer handed all three as starting points will report three different "best fits" with three different values of c_I and three different values of β, each with a confidence interval, and each fitting the data equally well. Nothing in the residuals distinguishes them. The contact rate has been doubled and halved across these sets, and the data are silent about it.

This is what Pitfall: "Identifiability" means operationally. The identifiable quantity is the product $c_I\beta$; the factors are not identifiable from early-outbreak incidence alone. Reporting a fitted c_I from such data is reporting the optimizer's starting point, not a measurement. Exercise 4 asks you to verify the equality numerically and to find what additional data would separate the factors.

7.3.3 What Endemic-Phase Data Adds

If incidence data extend through the epidemic peak and into the approach to endemic equilibrium, additional parameters become identifiable. The endemic values s^*, i^*, r^* are functions of the six parameters and N^c; data extending through the peak and into this approach constrain additional parameter combinations.

From Chapter 5, equation (5.21), the endemic infectious population is

$$i^* \;=\; \frac{n^* c_S \nu_m (\mathcal{R}_0 - 1)}{c_S(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) + c_I \nu_m + c_R \gamma_R}.$$

This expression depends on $\mathcal{R}_0$, already constrained by early growth, and on the three contact rates. It is worth carrying out the reduction explicitly, because the count of new constraints is not evident from the equation as written, and because the reduction separates two mechanisms that are easily conflated: what the endemic *level* adds, and what the through-peak *transient* adds.

What the endemic level adds: exactly one constraint. Divide numerator and denominator by c_S:

$$\frac{i^*}{n^*} \;=\; \frac{\nu_m(\mathcal{R}_0 - 1)}{(\mathcal{R}_0 - 1)(\gamma_R + \nu_m) \;+\; \dfrac{c_I}{c_S}\,\nu_m \;+\; \dfrac{c_R}{c_S}\,\gamma_R}\,. \tag{7.5}$$

Two features of this form matter. First, the three contact rates enter only through the two ratios c_I/c_S and c_R/c_S; the overall scale of the contact rates cancels, so the endemic level carries no information about it. Second, those two ratios enter only through the single weighted sum

$$K \;\equiv\; \frac{c_I}{c_S}\,\nu_m \;+\; \frac{c_R}{c_S}\,\gamma_R\,. \tag{7.6}$$

Now treat as known everything the earlier analysis already supplies: $\mathcal{R}_0$ from the growth rate, and γ_R and ν_m from the externally calibrated residence times. With those fixed, (7.5) inverts to

$$K \;=\; (\mathcal{R}_0 - 1)\left[\frac{\nu_m n^*}{i^*} \;-\; (\gamma_R + \nu_m)\right], \tag{7.7}$$

a single scalar equation in the two unknown ratios. This is the precise sense in which endemic data add *one* constraint. A measured endemic prevalence i^*/n^* determines K and nothing further: any pair $(c_I/c_S,\, c_R/c_S)$ on the line (7.6) with K held fixed reproduces the same endemic level exactly. The two ratios are not separately identifiable from the equilibrium, and neither are the contact rates themselves.

What the transient adds: the initial susceptible fraction. The same reduction shows what the endemic level *cannot* do. The initial susceptible fraction F_0 does not appear in (7.5) at all. The endemic state is a property of the parameters, not of where the trajectory started; the model forgets its initial condition on the way there. Information about F_0 therefore comes not from the level but from the approach to it. With $\mathcal{R}_0$, the residence rates, and now K all fixed, the one remaining degree of freedom in matching the trajectory from the exponential phase through the peak is F_0: it governs how far the susceptible pool is drawn down on the way to the endemic level, and the height and timing of the peak record that drawdown. This is why F_0 becomes identifiable once the data extend through the peak, whereas the early-growth phase alone—where the susceptible fraction is approximately one—carries no information about it. The practical limits of this identifiability, and the profile-likelihood diagnostics that expose them, are examined in Chapter 8.

For many purposes, the two parameters of primary interest are $\mathcal{R}_0$ (invasion threshold) and $F_0 = S(0)/N^{\mathrm{c}}$ (initial susceptible fraction), both identifiable from combined early-growth and through-peak data.

A useful sanity check at the endemic equilibrium: by construction, $\mathcal{R}_e^* = \mathcal{R}_0 \cdot s^*/n^* = 1$ identically (Chapter 5, Section 5.2). The endemic equilibrium is the state where each infectious individual produces, on average, exactly one secondary infection. A fitted endemic equilibrium that violates this identity ($\mathcal{R}_e^* \neq 1$ to within numerical tolerance) indicates either a fitting failure or a model-data mismatch. The check is mechanical and catches a class of subtle errors that visual inspection of the trajectory would miss.

7.3.4 Dimensionless Parameters

In general, the SIR model's behavior depends on the parameters only through certain dimensionless combinations. The dimensionless quantities that matter here are:

- $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$: the basic reproductive number.
- $F_0 = S(0)/N^{\mathrm{c}}$: the initial susceptible fraction.
- $D = \nu_m/\gamma_R = \tau_R/\tau_m$: the ratio of demographic-turnover rate to recovery rate, bounded in practice between 0.001 and 0.1 for most respiratory infections. Because τ_R and τ_m are externally calibrated, D is fixed by them rather than estimated from the data; it sets the separation of timescales between fast epidemic dynamics and slow demographic turnover.

Of these, $\mathcal{R}_0$ and F_0 are estimated from the data while D is fixed externally by the residence times. Fitting $\mathcal{R}_0$ and F_0 this way is more robust than fitting the six dimensional parameters directly, because these dimensionless combinations are the quantities the data actually constrain. Chapter 9 develops this point quantitatively through formal sensitivity indices.

An aside on dimensional analysis. It is worth separating a question of *dimensional analysis* from the identifiability question that occupies the rest of this section, because the two are easily conflated. Dimensional analysis, through the Buckingham Pi Theorem [Barenblatt, 2003], counts how many independent dimensionless groups a model *possesses*: the number of dimensional parameters minus the number of independent dimensions they span. This is a property of the model's structure, not of any dataset, and it is not the same as the number of combinations a given set of *data* can determine—the property the rest of this section is concerned with. The theorem answers "how many groups are there," not "how many can be estimated."

Applied to the SIR model, the count depends on which quantities are admitted to the reduction, and this book uses two of them: a parameters-only reduction that yields five groups, and a fuller reduction including the population-dimensioned quantities that yields six. Both are correct. They answer slightly different questions, and the difference between them is exactly where F_0 enters.

Reducing over the six *parameters* alone—c_S, c_I, c_R (contact rates, of dimension inverse time under the book's convention), β (a dimensionless per-contact transmission probability), and τ_R, τ_m (residence times, of dimension time)—these span a single independent dimension, time, since none of them carries a population dimension, so the reduction gives $6 - 1 = 5$ dimensionless combinations, among them $\mathcal{R}_0$, D, and ratios of contact rates such as c_I/c_S. The initial-condition ratio $F_0 = S(0)/N^{\mathrm{c}}$ is *not* among them, and cannot be, since neither $S(0)$ nor N^{c} is one of those parameters. Recovering F_0 requires enlarging the reduction to include the two population-dimensioned quantities, the count Appendix A.3 carries out: eight quantities spanning two dimensions, time and population, giving $8-2 = 6$ groups. The appendix's is the reduction that yields the full set of groups the fitting discussion uses.

How many of these groups the *data* can determine is a separate matter, settled not by dimensional analysis but by the observation model. For early-growth incidence data, the exponential-growth analysis earlier in this section shows that the data fix at most the growth rate σ_2 and the amplitude combination $c_I\beta i(0)$—fewer than the groups the reduction yields. That gap between what the model contains and what the data can resolve is the subject of the identifiability consideration below. It is worth being explicit about what this does and does not settle: nothing in the dimensional argument establishes that $\mathcal{R}_0$, F_0, and D are the right quantities to *fit*. Dimensional analysis supplies the candidate groups; the case for these three rests on identifiability, developed in the Fisher-information analysis later in this section and in the correlation and practical-identifiability diagnostics of Chapter 9. A reader who wants the justification rather than the inventory should read those.

Early data carry about two pieces of information, a growth rate and a start time, so a fit to them can identify little more; that limit is the pitfall Identifiability (Chapter 16).

Exercises for this section

1. **[C]** Using Table 7.2, classify each of the following quantities as *observed*, *externally calibrated*, *fitted from data*, or *modeling choice*: (a) the recovery rate γ_R, (b) the susceptible contact rate c_S, (c) the daily incidence $j(t)$, (d) the assumed geographic scope N^{c} of the outbreak, (e) the force-from-infection estimator $\hat{\alpha}$, (f) the basic reproductive number $\mathcal{R}_0$.

7.4 Least-Squares Fitting of α

The standard approach to fitting compartmental epidemic models is to minimize a weighted sum of squared residuals between model predictions and observed data, treating the model's parameters as decision variables. This section develops the procedure for the SIR model and then specializes to a simplified form that exposes the central advantage of the infected viewpoint.

7.4.1 The Loss Function

Let $\theta = (\theta_1, \ldots, \theta_p)$ denote the parameters to be fit. For the SIR model, a natural choice is $\theta = (\mathcal{R}_0, F_0)$. The residence times τ_R, τ_m and the contact rates c_S, c_I, c_R are supplied from independent sources and used to integrate the model; the transmission probability β is not fit separately, since

it enters only through the force coefficient $c_I\beta$, which $\mathcal{R}_0$ captures. Fitting $\mathcal{R}_0$ rather than c_I and β individually is exactly what the identifiability analysis of Section 7.3 requires, since early-growth data recover only the product $c_I\beta$. The model, integrated with parameters θ and initial conditions consistent with F_0, produces predicted incidence $j(t_k;\theta)$ at each observation time.

The observation model. Before specifying any loss function, we make explicit the statistical model that connects the deterministic SIR model to the observed data. Case-count data $J_{\text{obs}}(t_k)$ are non-negative integers reflecting counts of new infections during a reporting interval $[t_k-\Delta t, t_k]$. The standard observation model is the inhomogeneous Poisson process: conditional on the deterministic-model intensity $\alpha(t)i(t)$,

$$J_{\text{obs}}(t_k) \;\sim\; \text{Poisson}\Big(N^{\text{c}}\int_{t_k-\Delta t}^{t_k}\alpha(t')\,i(t')\,dt'\Big) \;\approx\; \text{Poisson}\big(\alpha(t_k)i(t_k)\,N^{\text{c}}\,\Delta t\big), \tag{7.8}$$

where the second form takes the standard right-endpoint approximation, valid when $\alpha(t)i(t)$ varies slowly within the reporting interval. This Poisson structure follows from the same thinned-Poisson-process argument that motivates the force-from-infection in Chapter 3: each infectious individual generates new infections according to a Poisson process at rate α, and the count of new infections in a fixed reporting window is the resulting count from i independent rate-α Poisson processes, which is itself Poisson with rate $\alpha i N^{\text{c}}\Delta t$. The factor N^{c} is the count bridge required by the book's dimensional convention: αi is a *normalized* flow (a fraction of the population per unit time), while J_{obs} is a *count*, so the Poisson mean must be the count-valued incidence $j\,N^{\text{c}}$ integrated over the reporting interval.

The Poisson log-likelihood is, up to a constant,

$$\ell(\theta) \;=\; \sum_{k=1}^{K}\Big[J_{\text{obs}}(t_k)\log\big(N^{\text{c}}j(t_k;\theta)\big) - N^{\text{c}}j(t_k;\theta)\Big], \tag{7.9}$$

where $j(t_k;\theta) = \alpha(t_k;\theta)i(t_k;\theta)\Delta t$ is the model-predicted *normalized* incidence over the reporting interval and $N^{\text{c}}j(t_k;\theta)$ is the corresponding count-valued Poisson rate, in agreement with the observation model (7.8). The count bridge N^{c} must be carried here for the same reason it appears in (7.8): the observed J_{obs} is a count, so the quantity it is compared against must also be a count. Maximum-likelihood estimation maximizes (7.9); the fitted estimator $\hat{\theta}$ has standard asymptotic properties (consistency, asymptotic normality, Fisher-information variance).

> **Caution: time units must match between data and model.** The reporting interval Δt in equation (7.8) carries operational consequences that are easy to mishandle. Most surveillance data are reported as weekly counts (CDC FluView, ECDC seasonal surveillance, ProMED weekly digests), while most SIR model parameters in this book are quoted in per-day units (this chapter's case study uses $\gamma_R \approx 0.143$ day^{-1} and $c_I \approx 6$ contacts day^{-1}). The count-valued Poisson rate $\alpha(t_k)i(t_k)N^{\text{c}}\Delta t$ in (7.8) must be computed with consistent units: if α is in day^{-1} and Δt in days, then $\Delta t = 7$ for weekly data, and $N^{\text{c}}j$ is the predicted weekly count. The most common error in undergraduate fitting projects is to compute $\alpha(t_k)i(t_k)$ from per-day parameters and forget the factor of 7, producing $\hat{\alpha}$ estimates that are systematically too small by a factor of 7 and corresponding reproductive-number estimates that are too small by the same factor. The reverse error (using α in week^{-1} but quoting per-day parameters) produces estimates too large by a factor of 7. Always state the time unit of each fitted quantity in the methods section, and verify by dimensional analysis that the predicted-incidence and observed-incidence have the same units before computing the residual.

Connection to weighted least squares. Expanding the Poisson log-likelihood to second order in the predicted mean, about the value $N^{\mathrm{c}}j = J_{\mathrm{obs}}(t_k)$ at which each term is maximized, gives the standard quadratic approximation

$$\ell(\theta) \;\approx\; \ell(\hat{\theta}) - \tfrac{1}{2}\sum_{k=1}^{K} \frac{\big[J_{\mathrm{obs}}(t_k) - N^{\mathrm{c}}j(t_k;\theta)\big]^2}{J_{\mathrm{obs}}(t_k)},$$

which is, up to additive constants, the negative of a weighted sum of squares. Weighted least squares is therefore an *approximation* to Poisson maximum likelihood, adequate for moderately large counts ($J_{\mathrm{obs}}(t_k) \gtrsim 10$); for small counts—early-outbreak windows, small populations, the regime of Pitfall: "The Stochastic-to-Deterministic Regime Transition"—the likelihood (7.9) should be used directly, as it should whenever the dispersion is large enough that the quadratic approximation is doing real work.

The step just taken generalizes, and stating the general form is worth more than the Poisson instance alone. For *any* observation model in which the kth measurement has mean $\mu_k(\theta)$ and variance σ_k^2, the same quadratic approximation gives $-\frac{1}{2}\sum_k (y_k - \mu_k)^2/\sigma_k^2$. The rule is therefore

$$w_k \;=\; \frac{1}{\sigma_k^2}\,, \tag{7.10}$$

weight each squared residual by the inverse variance of its observation, and everything else in this section is an instance of it (Pitfall: "Weighting Residuals by Their Uncertainty").

Equation (7.10) is also the only dimensionally coherent choice, which is a useful check on any weighting a reader meets in the literature. With counts as the unit, a squared residual carries units of counts^2, so w_k must carry counts^{-2} for the loss to be a dimensionless χ^2. The Poisson relation $\sigma^2 = \mu$ appears to violate this, and does not: it carries a suppressed constant of one count, and restoring it returns $w_k = 1/(\mu_k u)$ with the expected units. A weighting that leaves the loss with residual dimensions is a sign that some σ has been replaced by something that is not a standard deviation. Two practical consequences follow. Loss values computed under different weightings are not comparable with one another, so "the weighted fit achieved a lower loss" is a statement about units rather than about fit; and because the loss transforms differently under a change of reporting unit for each weighting, model comparison must run through the likelihood, an information criterion, or held-out prediction, never through the raw value of $\mathcal{L}$.

The weighted sum-of-squares loss is

$$\mathcal{L}(\theta) \;=\; \sum_{k=1}^{K} w_k \big[J_{\mathrm{obs}}(t_k) - N^{\mathrm{c}}j(t_k;\theta)\big]^2, \tag{7.11}$$

where the weights come from (7.10). The familiar cases are the familiar variance relations: Poisson counts have $\sigma_k^2 = \mu_k$, giving $w_k = 1/\mu_k$; data with a constant coefficient of variation have $\sigma_k^2 \propto \mu_k^2$, giving $w_k \propto 1/\mu_k^2$; homoscedastic Gaussian noise has σ_k^2 constant, giving $w_k = 1$. Negative-binomial counts, with $\sigma_k^2 = \mu_k + \mu_k^2/\kappa$, interpolate between the first two, approaching Poisson weighting as $\kappa \to \infty$ and constant-coefficient-of-variation weighting as the dispersion grows.

Two choices inside the weights that are easy to get wrong. The first is whether μ_k in the weight means the model's prediction $N^{\mathrm{c}}j(t_k;\theta)$ or the observation $J_{\mathrm{obs}}(t_k)$. Use the model's prediction. Weighting by the observation gives extra influence to points that happened to fluctuate low, which pulls the fit downward, and it is undefined at $J_{\mathrm{obs}}(t_k) = 0$—a common value in exactly

the early-outbreak window this chapter cares most about. The cost is that the weights are then updated as the fit proceeds, which every nonlinear least-squares routine already accommodates.

The second is which variance relation to assume. It is tempting to treat Poisson as the default for case counts and reach for the negative binomial only on evidence of overdispersion, but that puts the burden of proof the wrong way round: surveillance counts are overdispersed as a matter of course, through reporting artifacts, batching, and clustering, and Poisson weights are poorly calibrated on them. A workable order of operations is to fit with Poisson weights, then read the scale factor $\hat{\phi} = \mathcal{L}(\hat{\theta})/(K-p)$ at the optimum. If the observation model is right, $\hat{\phi} \approx 1$. If $\hat{\phi}$ is much larger, the data are overdispersed, and the choice is either to estimate κ and reweight or to fall back on constant-coefficient-of-variation weights, which behave nearly as well as correctly specified negative-binomial weights once the dispersion is substantial and require nothing to be estimated.

The fitted parameters are

$$\hat{\theta} \;=\; \arg\min_{\theta} \mathcal{L}(\theta),$$

computed by any standard nonlinear optimization routine (Levenberg–Marquardt, quasi-Newton, or the simpler Nelder–Mead simplex method).

7.4.2 What Unweighted Least Squares Does to the Fit

Most modelers meet least squares in its unweighted form, and most epidemic fits in the literature are unweighted. It is worth being explicit about what that choice does, because the damage is not where intuition puts it.

Unequal uncertainties are the ordinary condition, not a special case. The weighting question only bites when observations differ appreciably in their uncertainty. In epidemic data they always do. Across a single outbreak of the baseline model, expected weekly counts run from a few to several thousand—a range of nearly three orders of magnitude—so the Poisson standard deviations attached to them range over a factor of roughly twenty-five. A modeler fitting an outbreak curve is never in the homoscedastic case that unweighted least squares assumes.

The fit concentrates where the counts are largest. Each observation contributes to the estimate in proportion to $w_k(\partial\mu_k/\partial\theta)^2$, its share of the Fisher information. Under unweighted least squares that share is dominated by the peak: for the baseline outbreak, the eleven observations within twenty days of the peak—barely more than a quarter of the data—carry about 79% of the information about $\mathcal{R}_0$, against 58% under Poisson weighting and 24% under constant-coefficient-of-variation weighting. Put the other way, an unweighted fit draws roughly four-fifths of what it knows about $\mathcal{R}_0$ from a quarter of the observations, and constant-coefficient-of-variation weighting cuts that concentration by more than a factor of three. The exponential-growth phase, which is where the identifiability arguments of Section 7.3 locate the cleanest signal, contributes very little to an unweighted fit.

Figure 7.6 shows both halves of the problem: which observations an unweighted fit is actually listening to, and what that does to the interval it reports.

The effect is starkest when two data streams of different magnitude are fitted together. Suppose weekly cases and weekly deaths are fitted jointly at an infection fatality ratio of 0.5%. Under unweighted least squares the death series carries about 0.0025% of the total information; under inverse-variance weighting it carries about 0.50%, its proper share—a factor of two hundred differ-

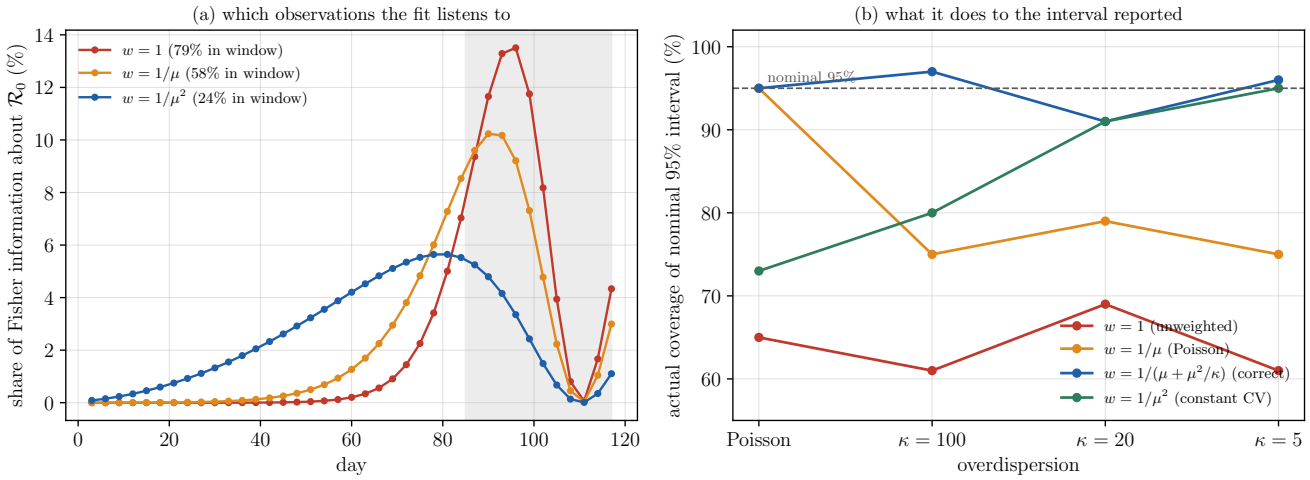


Figure 7.6: What the choice of residual weights does to a fit of the baseline outbreak. **(a)** Each observation's share of the Fisher information about $\mathcal{R}_0$, $w_k(\partial\mu_k/\partial\mathcal{R}_0)^2$ normalized, under three weightings. The outbreak peaks near day 106 and the fitting window ends at day 117, so the shaded twenty-day peak window runs from day 87 and is truncated on the right by the data rather than by the plot; it holds eleven of the thirty-nine observations. **(b)** Actual coverage of nominal 95% intervals against increasing overdispersion, from Table 7.3. An unweighted fit concentrates its attention near the peak and never reaches its nominal coverage; Poisson weighting is well calibrated only when the data really are Poisson; constant-coefficient-of-variation weighting improves as the dispersion grows, which is the direction real surveillance data lie in.

ence in how much the deaths are allowed to say. The modeler has not down-weighted the deaths. The deaths have not been fitted at all.

The estimate survives; the uncertainty statement does not. The consequences separate cleanly. When the observation model is genuinely Poisson and the model is correctly specified, unweighted least squares is nearly unbiased and only modestly less efficient than the correctly weighted fit. Under the overdispersion that real surveillance data carry, the picture changes: correct weighting cuts the root-mean-square error by more than a factor of two at ordinary surveillance noise, and by a factor of three at the heaviest dispersion shown. In both regimes the reported interval is the more serious casualty. Table 7.3 gives interval coverage and error for the baseline outbreak at several dispersion levels.

Four things in the table are worth reading carefully. Only the correctly specified weight holds its nominal coverage across the whole range, staying between 91% and 97% at every dispersion level; no other row does. An unweighted fit reaches nominal coverage nowhere: the interval it reports is about half the width it should be, so it misses the truth roughly seven times as often as advertised, and that shortfall is flat across dispersion rather than worsening with it. Poisson weighting is not a safe default either—it is well calibrated only when the data really are Poisson, and settles near three-quarters coverage once they are not, better than leaving the fit unweighted but still far short of what it claims, which is the practical reason for the order of operations recommended in Section 7.1. And constant-coefficient-of-variation weighting, which requires no dispersion parameter to be estimated, matches the correctly specified fit once the overdispersion is substantial, while paying for that robustness at the Poisson end.

Table 7.3: Coverage of nominal 95% confidence intervals for $\mathcal{R}_0$, and root-mean-square error, under four weighting choices, for the baseline outbreak fitted at 39 observations with $N^{\mathrm{c}} = 100{,}000$. Counts are negative-binomial with $\sigma^2 = \mu + \mu^2/\kappa$; $\kappa = \infty$ is Poisson, and $\kappa = 20$ corresponds to a coefficient of variation near 22% at the peak, ordinary week-to-week surveillance noise. Intervals are scale-corrected, as returned by standard nonlinear least-squares routines. All four schemes are fitted to the same simulated datasets (common random numbers), so the rows may be compared directly; with 150 replicates the Monte Carlo standard error on any coverage entry is about four percentage points, and differences smaller than that should not be read as real. At $\kappa = \infty$ the correctly specified weight reduces to the Poisson weight, so those two rows necessarily agree—a check on the simulation rather than a finding. The table assumes the fitted model is structurally correct; under misspecification the weighting additionally decides which part of the curve the fit honours, and coverage degrades further.

	coverage of nominal 95% interval			
weighting	Poisson	$\kappa = 100$	$\kappa = 20$	$\kappa = 5$
$w = 1$ (unweighted)	65%	61%	69%	61%
$w = 1/\mu$ (Poisson)	95%	75%	79%	75%
$w = 1/(\mu + \mu^2/\kappa)$ (correct)	95%	97%	91%	96%
$w = 1/\mu^2$ (constant CV)	73%	80%	91%	95%
	RMSE of $\hat{\mathcal{R}}_0$			
$w = 1$ (unweighted)	0.0037	0.0248	0.0551	0.1097
$w = 1/\mu$ (Poisson)	0.0032	0.0188	0.0440	0.0860
$w = 1/(\mu + \mu^2/\kappa)$ (correct)	0.0032	0.0112	0.0239	0.0348
$w = 1/\mu^2$ (constant CV)	0.0177	0.0193	0.0250	0.0340

The summary is that unweighted least squares quietly decides that the largest observations are the only observations, and then reports an interval computed as though it had used them all (Pitfall: "Weighting Residuals by Their Uncertainty"). Section 7.6 shows the same mechanism deciding the answer outright rather than only its error bar, and Chapter 8 turns the diagnosis into a routine check.

7.4.3 A Simplified Estimator: $\hat{\alpha} = j/i$

A particularly simple and informative estimator of α arises directly from the definition of the force from infection. From (7.1),

$$j(t) \;=\; \alpha(t)\, i(t),$$

so if incidence j and prevalence i are both observed at time t, the moment-estimator of α follows immediately. This is an estimator of the force from infection α, not of the classical per-contact transmission coefficient β; the relation between β and incidence depends on the model's incidence convention (Chapter 3).

Definition 7.1 (Infected-viewpoint force estimator)**.** The *infected-viewpoint force estimator* of the force from infection at time t is

$$\hat{\alpha}(t) \;=\; \frac{J_{\mathrm{obs}}(t)}{I_{\mathrm{obs}}(t)}, \tag{7.12}$$

where $J_{\text{obs}}(t)$ is the observed daily incidence and $I_{\text{obs}}(t)$ is the observed current prevalence. The corresponding estimator of the basic reproductive number is

$$\hat{\mathcal{R}}_0 \;=\; \frac{\hat{\alpha}}{\gamma_R + \nu_m}, \tag{7.13}$$

using externally calibrated values of γ_R and ν_m, and valid as an estimate of $\mathcal{R}_0$ when the population is approximately naive, $P_S \approx 1$. Away from that limit the same ratio returns the effective reproductive number: since $\alpha(t) = c_I \beta P_S(t)$, the conversion gives $\mathcal{R}_0 P_S(t) = \mathcal{R}_e(t)$ (Section 7.6). The force estimator (7.12) itself depends only on observed quantities and never on the assumed susceptible population size; the conversion to $\mathcal{R}_0$ depends on it through P_S, and that is the one place the susceptible fraction re-enters.

Each observation of (j, i) gives a direct estimate of α at that time, requiring no fitting procedure and no model integration. The full time course of $\alpha(t)$ can be read off the data. For readability the chapter refers to this estimator by the shorthand $\hat{\alpha} = j/i$, with the observed-count subscripts of (7.12) suppressed; wherever j and i appear bare in that ratio they denote the observed series J_{obs} and I_{obs} of Definition 7.1.

Interpretation. The estimator (7.12) is the natural way to read α from the data: it asks, per infectious individual, how many new infections are being generated per unit time, which is exactly the definition of α.

Connection to $\mathcal{R}_0$. If $\hat{\alpha}$ is constant over some time window, then the data are consistent with an exponential-growth regime of the outbreak, and equation (7.13) gives the corresponding estimate of $\mathcal{R}_0$, using the known values of γ_R and ν_m.

When $\hat{\alpha}$ is reliable: $i(t)$ as observed vs. reconstructed. The estimator (7.12) requires *some* estimate of the infectious prevalence $i(t)$. The book uses the word "observed" for $i(t)$ throughout most of this chapter, but this is a deliberate simplification of three operationally distinct cases:

- *Directly observed prevalence* (best case): For pathogens with low-cost rapid diagnostic tests deployed at population scale (chronic infections like HIV via cross-sectional serosurveys; diseases with mandatory case registers like tuberculosis), $i(t)$ is reported directly from prevalence surveillance. The estimator $\hat{\alpha} = j/i$ uses two independently-collected data streams, and the assumption $i(t) = I_{\text{obs}}(t)$ is well-supported.

- *Reconstructed prevalence from incidence* (typical case): For acute infections in routine surveillance, only incidence $j(t)$ (new-case reports) is collected operationally. The prevalence $i(t)$ is then reconstructed from the incidence time series under an assumed infectious-period distribution: assuming exponential infectious period with rate $\gamma_R + \nu_m$,

$$i(t) \;\approx\; \int_0^\infty j(t-\tau)\, e^{-(\gamma_R+\nu_m)\tau}\, d\tau \;\approx\; \sum_{k:t_k \le t} j(t_k)\, e^{-(\gamma_R+\nu_m)(t-t_k)},$$

  which convolves the incidence with the infectious-period kernel. With this reconstruction, $\hat{\alpha}(t) = J_{\text{obs}}(t)/I_{\text{est}}(t)$ reduces to a quantity computable from incidence alone, up to the approximation error in the infectious-period distribution.

- *Hybrid* (intermediate case): A few cross-sectional prevalence anchor points (from a serosurvey at a known date, or from a known-prevalence cohort) calibrate the reconstruction in the previous case, reducing reliance on the assumed $\gamma_R + \nu_m$ value. Operationally most realistic for emerging-pathogen contexts where one or two seroprevalence studies appear weeks into an outbreak.

Implication for fitting. The structural-immunity property of $\hat{\alpha} = j/i$ to uncertainty in the susceptible population s is preserved in all three cases above: s does not appear in any reconstruction. What differs across cases is the noise structure of the recovered $i(t)$, which propagates into the noise structure of $\hat{\alpha}$. Best-case observation gives the cleanest estimator; reconstruction adds approximation error from the infectious-period assumption. The Caution box in Section 10.6 of Chapter 10 addresses a related issue specific to pathogens with asymptomatic transmission: when the observable prevalence excludes asymptomatic carriers (as for most acute infections), $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$ recovers the symptomatic-track force only, providing a lower bound on $\mathcal{R}_0$ rather than a full estimate.

Observation noise on the estimator. The reconstruction error above is systematic; there is also random observation noise on the two surveillance streams themselves. A standard model for surveillance counts is multiplicative log-normal, $J_{\text{obs}}(t) = J_{\text{true}}(t)\, e^{\epsilon_J(t)}$ and $I_{\text{obs}}(t) = I_{\text{true}}(t)\, e^{\epsilon_I(t)}$ with $\epsilon_J, \epsilon_I \sim \mathcal{N}(0, \sigma^2_{\text{obs}})$ independent, multiplicative rather than additive because incidence counts are nonnegative and their relative error is roughly scale-invariant (additive Gaussian noise is the wrong model here, since it admits negative counts). Taking $\sigma_{\text{obs}} = 0.10$, about $\pm 20\%$ noise at two standard deviations, the estimator becomes $\hat{\alpha}_{\text{obs}}(t) = J_{\text{obs}}/I_{\text{obs}} = \alpha(t)\, e^{\epsilon_J - \epsilon_I}$, log-normally distributed about the true $\alpha(t)$ with log-scale standard deviation $\sigma_{\text{obs}}\sqrt{2}$. Two features carry into the fitting. The early-time plateau in $\hat{\alpha}(t)$ remains detectable, because the systematic plateau structure averages down the multiplicative noise over a window of time points. And because $\hat{\alpha}$ is a ratio of the two streams, reporting errors common to both (a systematic under-ascertainment that scales j and i together) cancel in the ratio, whereas independent errors add in quadrature. This cancellation is a structural advantage of the infected-viewpoint estimator that $\hat{\lambda} = j/s$ does not share, since s is not measured by the same surveillance pipeline as j.

Data sources for compartment-specific contact rates. When the framework is applied with distinct c_S and c_I (the compartment-specific refinement carried through Chapters 4, 11, 12, and 19), the two rates typically come from different data streams. Susceptible-host contact rates c_S are estimated from general-population contact surveys (the POLYMOD study [Mossong et al., 2008] and its successors) which record self-reported contacts across representative cross-sectional samples. Infected-host contact rates c_I require contact-tracing of index cases or self-reported post-symptom-onset contact reductions, both of which are collected in outbreak investigations rather than routine surveillance. The ratio c_I/c_S is not directly measurable in a single dataset; it is inferred by combining POLYMOD-style baseline contact data with outbreak-specific contact-tracing data for the same pathogen and population. The sensitivity-analysis paper of Hyman [2026d] develops estimators that remain well-posed even when c_I itself is weakly identified, by exploiting the structure in which c_I enters $\mathcal{R}_0$ linearly while $c_I\beta$ appears as a single product in $\hat{\alpha}$.

AI Lab: Early-Time $\hat{\alpha} = j/i$ Fitting Workflow

The estimator $\hat{\alpha} = j/i$ is simple to compute but needs three judgment calls: which points form the early plateau, how to summarize them, and how to attach uncertainty. Use the AI for each in a *three-step audit-and-refine*, and check its proposal against the chapter.

Step 1: the plateau. Compute and plot $\hat{\alpha}(t_k)$, and ask the AI to find the early window where it is roughly constant. A generic change-point rule is not enough: the plateau ends when $S(t)/N^{c}$ falls below about 0.95, after which $\hat{\alpha}$ no longer estimates α at the disease-free equilibrium.

Step 2: estimate and interval. Take the mean over the plateau and ask the AI for a bootstrap. A simple resample ignores autocorrelation; a block bootstrap with blocks about τ_R long suits the series, and the report should state the block size and the number of replicates.

Step 3: the reproductive number. By (7.13), $\hat{\alpha}/(\gamma_R + \nu_m)$ is $\hat{\mathcal{R}}_e$, and it is $\hat{\mathcal{R}}_0$ only when $P_S \approx 1$. Ask the AI for the published range for your pathogen and setting, and whether those values are $\mathcal{R}_0$ or $\mathcal{R}_e$. Check that the cited papers exist and report what the AI says they do. An estimate outside the range points to the plateau, to γ_R or ν_m, or to comparing $\hat{\mathcal{R}}_e$ with $\mathcal{R}_0$.

Disclose the AI system, the prompts and your audit findings (Appendix D).

Checkpoint before going further. The next subsection introduces a second estimator and then compares the two. That comparison is only meaningful if the first estimator is working, so verify it now on a case where the answer is known. Simulate the baseline SIR model of Chapter 4 (Table 4.4), take j and i from the trajectory at an early time, and form $\hat{\alpha} = j/i$. You should recover

$$\hat{\alpha} \;=\; c_I\beta \;=\; 4 \times 0.040 \;=\; 0.16 \text{ day}^{-1}, \qquad \hat{\mathcal{R}}_0 \;=\; \frac{\hat{\alpha}}{\gamma_R + \nu_m} \;\approx\; \frac{0.16}{0.0835} \;\approx\; 1.92,$$

to within the tolerance of the integrator. If your implementation returns something else, the problem is in the estimator or in how i was extracted from the trajectory, and no amount of comparison against a second method will reveal which. Fix it here. This section's Exercise 1 extends the check by asking when $\hat{\alpha}$ begins to drift away from this value, and why.

7.4.4 The Alternative: $\hat{\lambda} = j/s$

This estimator is developed for comparison, not as a recommendation. It is included because it is in wide use and because the contrast with $\hat{\alpha} = j/i$ is instructive: the two solve different inverse problems, and Section 7.5.1 shows that the difference is decisive whenever the susceptible denominator is uncertain, which early in an outbreak it almost always is. The workflow this book recommends is the infected-viewpoint estimator of Section 7.4.3. Read what follows as the alternative being characterized, not as a second option of equal standing.

The corresponding estimator in the susceptible viewpoint is derived analogously. Since $j(t) = \lambda(t)s(t)$, if both j and s are known,

$$\hat{\lambda}(t) \;=\; \frac{j(t)}{s(t)}. \tag{7.14}$$

This is the moment-estimator of λ at time t. To convert it to $\mathcal{R}_0$, use the early-outbreak form of λ. In counts, frequency-dependent transmission gives $\lambda \approx c_S\beta\, I/N$, where N is the total mixing population that sets the transmission denominator. Inverting, $c_S\beta = \hat{\lambda}\, N/I$, and combining with $\mathcal{R}_0 = c_S\beta/(\gamma_R + \nu_m) = c_I\beta/(\gamma_R + \nu_m)$ in the equal-contact-rate case (Chapter 5, eq. (5.10)) gives

$$\hat{\mathcal{R}}_0^{(\lambda)} \;=\; \frac{\hat{\lambda}(t)\, N}{I(t)\,(\gamma_R + \nu_m)} \;=\; \frac{J(t)\, N}{S(t)\, I(t)\,(\gamma_R + \nu_m)}. \tag{7.15}$$

What the two conversions actually return. Equation (7.15) is worth reading carefully, because the two populations appearing in it are not the same quantity. The N in the numerator is the total mixing population; the S in the denominator is the susceptible subset. Writing $F = S/N$ for the susceptible fraction and using $\hat{\lambda} = J/S$,

$$\hat{\mathcal{R}}_0^{(\lambda)} \;=\; \frac{J}{I\,(\gamma_R + \nu_m)} \cdot \frac{1}{F} \;=\; \frac{\hat{\mathcal{R}}_0^{(\alpha)}}{F}\,. \tag{7.16}$$

Two consequences follow, and neither is the one a reader might expect.

First, when the population is approximately naive ($F \approx 1$, the early-emergence setting in which this section's approximation is stated), the assumed population *cancels*: $\hat{\mathcal{R}}_0^{(\lambda)}$ reduces to $J/[I(\gamma_R + \nu_m)] = \hat{\mathcal{R}}_0^{(\alpha)}$. Doubling the assumed N^{c} halves $\hat{\lambda}$ and doubles the conversion factor, and the two effects cancel exactly. Under a consistent conversion with prevalence observed, the two viewpoints return the *same* $\mathcal{R}_0$. This is the point made from the sensitivity side in Chapter 9: the exact estimator-level asymmetry $\Upsilon_s^{\hat{\lambda}} = -1$ is a statement about the fitted *force*, and it does not propagate to a fitted $\mathcal{R}_0$.

Second, when the population is *not* naive, the factor $1/F$ survives, and it is $\hat{\mathcal{R}}_0^{(\alpha)}$ that needs the correction rather than $\hat{\mathcal{R}}_0^{(\lambda)}$. Since $\alpha(t) = c_I \beta P_S(t)$, the infected-viewpoint conversion $\hat{\alpha}/(\gamma_R + \nu_m)$ returns $\mathcal{R}_0 P_S$, which is the *effective* reproductive number $\mathcal{R}_e$, not $\mathcal{R}_0$ (Section 7.6). The susceptible-viewpoint conversion divides by S rather than N, which restores the missing P_S factor. The honest summary is that the infected viewpoint buys an assumption-free estimate of the quantity governing *current* transmission, and the price of insisting on $\mathcal{R}_0$ instead is that the susceptible fraction must be reintroduced—which is precisely the trade-off catalogued under Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$".

The catch. The denominator in (7.14) is $s(t)$, the current susceptible population. Early in an outbreak, $s(t)$ is well-approximated by N^{c}, the carrying capacity of the modeled population. The value of N^{c} is, in turn, a modeling choice: is the relevant population the full metropolitan area, the subset at risk, or the contact-traced cohort? Different choices give values of N^{c} that differ by factors of 5 or 10, producing correspondingly different values of $\hat{\lambda}$.

That spread in $\hat{\lambda}$ is real, but by (7.16) it is not by itself what makes the two research teams of the Preface disagree about $\mathcal{R}_0$; a ratio estimate with prevalence in hand would leave them in agreement. Their disagreement arises because they fit $\mathcal{R}_0$ to an incidence series covering the approach to the peak, without an independent prevalence series. By Proposition 7.2, such a fit determines $\mathcal{R}_0$ only through the compound ratio k that combines the assumed population with the ascertainment fraction, so two teams with different scopings are fitting at different values of k and their shared data cannot adjudicate between them. That is an identifiability failure rather than a scaling of the estimator, and Section 7.6 develops it quantitatively.

Implementation in practice. The nonlinear least-squares fit of α to incidence data is, at the code level, a few dozen lines in any modern scientific-computing environment. In Python, `scipy.optimize.least_squares` solves the problem directly once the observational model (the mapping from parameters to predicted incidence) is written as a callable. In MATLAB, `lsqnonlin` plays the same role; in R, the `nls()` function in base R or `nlsLM()` from the `minpack.lm` package (Levenberg–Marquardt) provide the equivalent capability. What matters pedagogically is not the specific tool but the workflow: read the data, define the observational model that maps parameters to predicted observations, define the residual vector as observed minus predicted, and call the solver. The book's companion repository provides a reference Colab notebook for this fit, which

generates the case-study incidence series reproducibly, so that readers can step through a complete fit end-to-end and then modify it for their own applications.

Check Your Work: Before Trusting Your $\hat{\alpha}$ Estimate

Verifies: numerical values, model assumptions and interpretations. Before turning $\hat{\alpha} = j/i$ into $\hat{\mathcal{R}}_0$ or a recommendation, run three checks.

Plateau. Plot $\hat{\alpha}(t)$ and look for an early window where it is roughly constant. If $S(t)/N^{\mathrm{c}}$ is already below 0.95 by the second observation, there is no plateau and the estimator does not apply.

Uncertainty. The bootstrap interval should be wider than the scatter of $\hat{\alpha}$ on the plateau. If it is narrower, the series is autocorrelated: use a block bootstrap.

Range. Convert to $\hat{\mathcal{R}}_0$ and compare with published values for the pathogen and setting. A gap larger than a factor of 2 usually means a wrong γ_R or ν_m, or published values that report $\mathcal{R}_e$.

If a check fails, fix the method or report the caveat, and do not base an intervention recommendation on the fit alone.

Exercises for this section

1. **[L]** *A guided walkthrough of the checkpoint of Section 7.4.3.* For the baseline parameter set from Chapter 4 (Table 4.4), compute $\hat{\alpha}$ exactly from j/i at $t = 0, 5, 10, 20, 50$ days using the true simulated trajectory. At which time does $\hat{\alpha}$ first deviate from the initial value by more than 10%?

2. **[C]** Explain in one paragraph why $\hat{\alpha} = j/i$ is independent of the assumed susceptible population s, while $\hat{\lambda} = j/s$ inherits the uncertainty in s directly. Your explanation should not rely on equations; write as if addressing a colleague who understands the SIR model but has not read this chapter.

7.5 Initial Conditions and the Susceptible-Fraction Question

The initial conditions of the SIR system are $(S(0), I(0), R(0))$ with the constraint $S(0) + I(0) + R(0) = N^{\mathrm{c}}$ when the population is at its carrying capacity. The number of initial conditions that need to be specified is three minus the constraint, or two free parameters.

A natural parameterization in terms of dimensionless quantities is

$$F_0 = \frac{S(0)}{N^{\mathrm{c}}} \qquad \text{initial susceptible fraction,} \tag{7.17a}$$

$$i_0 = \frac{I(0)}{N^{\mathrm{c}}} \qquad \text{initial infectious fraction.} \tag{7.17b}$$

The initial recovered fraction is $1 - F_0 - i_0$. When the population is mostly naive (no prior immunity) at the outbreak's start, $r(0) = 0$ and $F_0 = 1 - i_0$; in practice, some partial pre-existing immunity (from past outbreaks, cross-protection from related pathogens, or partial vaccination) gives $F_0 < 1 - i_0$.

7.5.1 The Susceptible Population Size as a Modeling Choice

The product $S(0) = F_0 N^{\mathrm{c}}$ expresses the initial susceptible number as the fraction F_0 of the carrying capacity N^{c}. Neither F_0 nor N^{c} is directly measurable. Their product is.

N^{c} is the population size of the region assumed to be epidemiologically relevant for the outbreak. Plausible choices for an outbreak starting in a single city include:

- The full metropolitan population (perhaps 10^7 people for a major metropolitan area).
- The urban core population (perhaps 10^6).
- The population in a specific neighborhood or transit network (perhaps 10^5).
- The population in close contact with the index cases (perhaps 10^4 for early contact tracing).

These plausible choices span three orders of magnitude. The choice made by the modeler determines the interpretation of F_0 and the downstream numerical estimates of contact rates c_S, c_I, c_R (which depend on what "contact" means in a population of a given density), but in a well-designed model the dimensionless quantities $\mathcal{R}_0$ and F_0 are reasonably stable across choices.

7.5.2 Why F_0 is Weakly Identifiable From Early-Outbreak Data

Early in the outbreak, when $i(t) \ll 1$, the trajectory of $j(t)$ is essentially determined by $\mathcal{R}_0$ and the product $c_I \beta i(0)$. The parameter F_0 enters only through the slight suppression of P_S below unity, which during exponential growth is negligible. This makes F_0 weakly identifiable from early-outbreak data; its value affects the trajectory only at the leading order of $(1 - F_0)$, which is small when pre-existing immunity is modest.

For later-time fitting (past the epidemic peak), F_0 becomes more strongly identifiable because the trajectory's peak height and decline rate depend on F_0. The staged fitting procedure is therefore:

1. From early-outbreak data, estimate $\mathcal{R}_0$ and the combination $c_I \beta i(0)$ through the exponential-growth slope and intercept.
2. From later-time data (through the peak and into the approach to equilibrium), refine the estimate of F_0 and the endemic combinations.
3. Use prior information (clinical infectious-period data, demographic migration rates) to pin down τ_R and τ_m.

This staged approach minimizes the number of parameters being fit simultaneously at any one step and is more robust than a monolithic joint fit over all parameters at once. The staged procedure is approximately efficient (in the statistical sense of approaching the Fisher-information lower bound on parameter variance) when $\mathcal{R}_0$ and F_0 are nearly orthogonal in the Fisher information matrix, i.e., when early-phase data constrains essentially only $\mathcal{R}_0$ and late-phase data constrains essentially only F_0. For the SIR model this orthogonality holds approximately because the early-time growth rate σ depends on $\mathcal{R}_0$ but not on F_0, while the endemic prevalence depends on both; staged fitting therefore loses little efficiency relative to a properly-conditioned joint MLE while gaining substantial robustness against the conditioning failures that joint MLE incurs in practice.

Exercises for this section

1. **[L]** A modeler fits weekly cases and weekly deaths jointly, with deaths running at an infection fatality ratio of 0.5%, using unweighted least squares. Taking the Poisson variance relation, compute the ratio of the two series' contributions to the total $\sum_k w_k \mu_k^2$ under $w_k = 1$ and under $w_k = 1/\mu_k$. Show that the death series contributes a factor of 200 less under unweighted fitting than under inverse-variance weighting, and state in one sentence what the modeler has actually fitted. No simulation is required.

2. **[L]** For the 2017–2018 case study, suppose the assumed F_0 is varied from 0.3 to 1.0 in steps of 0.1. Compute the resulting $\hat{\mathcal{R}}_0^{(\lambda)}$ estimates from the susceptible viewpoint and plot them against F_0. Verify that the curve is $\hat{\mathcal{R}}_e/F_0$ with $\hat{\mathcal{R}}_e \approx 1.40$, the infected-viewpoint estimate, which is independent of F_0. Explain in two sentences why the two viewpoints coincide at $F_0 = 1$ and diverge below it, and state which of the two quantities the incidence data alone can determine.

AI Learning Prompt: Diagnosing Identifiability Before Fitting

The Verify the AI box earlier in this chapter showed the cost of fitting before checking identifiability; here the AI does the check first. *Prompt:* "*For the SIR model with parameters $(c_I, \beta, \gamma_R, \nu_m, I(0), N^{\mathrm{c}})$ and Poisson daily incidence $j(t)$ over the first 30 days of an outbreak: (1) derive the Fisher information matrix; (2) give its rank and null space; (3) rewrite the model in identifiable combinations; (4) say what further data would identify the rest.*"
A correct answer finds rank two, so four null directions: $c_I\beta$ against $i(0)$, c_I against β, γ_R against ν_m (only their sum enters the growth rate), and N^{c}, which does not enter at all. The identifiable pair is $(\sigma_2, c_I\beta i(0))$, and later-time data or a prior on $i(0)$ breaks the remaining ambiguity. An answer that skips the rank calculation has not done the diagnosis (Pitfall: "Identifiability").

7.6 Comparison of Two Estimation Methods Under Uncertainty in the Susceptible Population

This section gives the quantitative version of the book's central practical claim. Stated carefully, the claim is an identifiability result rather than a numerical one: an estimate of $\mathcal{R}_0$ built from incidence alone cannot separate the assumed susceptible population from the fraction of infections actually reported, so it rests on a compound assumption that the data cannot check. The infected-viewpoint estimator $\hat{\alpha} = J_{\mathrm{obs}}/I_{\mathrm{obs}}$ requires neither quantity, and is invariant to both by construction. What it requires instead is observed prevalence, and the rest of this section makes both the benefit and its price precise. Chapter 6, Section 6.5.2 demonstrates the dynamics-level version: the two estimators, fit from the same observed data under different assumed N^{c}, produce forward trajectories that coincide (infected viewpoint) or diverge (susceptible viewpoint). This section gives the corresponding fitting-level result.

Reconciling equivalence with asymmetry. Two facts coexist without contradiction. First, the full force from infection $\alpha(t) = c_I\beta P_S(t)$ depends on s through the contact-weighted susceptible fraction P_S, just as the susceptible-viewpoint force $\lambda(t) = c_S\beta P_I(t)$ depends on I_{obs}; the two formulations are structurally symmetric (Chapter 3, Section 3.4). Second, the early-outbreak estimator $\hat{\alpha} = J_{\mathrm{obs}}/I_{\mathrm{obs}}$ does not depend on the assumed N^{c}, while $\hat{\lambda} = J_{\mathrm{obs}}/s$ inherits any error in N^{c} with unit elasticity. The advantage of the infected viewpoint is not that α avoids s: it is that the natural

fitting quantity derived from α is the ratio $J_{\text{obs}}/I_{\text{obs}}$, which is observable without assuming a value of N^{c}. The remainder of this section is the quantitative development of this asymmetry.

7.6.1 Setup of the Comparison

Let $J_{\text{obs}}(t)$ and $I_{\text{obs}}(t)$ be the observed incidence and prevalence, respectively, at time t. These are directly measurable quantities that do not depend on modeling assumptions about the susceptible population size. Let N^{c} denote the modeler's assumed susceptible population; we consider how estimates depend on N^{c} over a range of plausible values.

The infected-viewpoint estimator is

$$\hat{\alpha} \;=\; \frac{J_{\text{obs}}}{I_{\text{obs}}} \qquad \text{(depends on } J_{\text{obs}} \text{ and } I_{\text{obs}} \text{ only)}.$$

The susceptible-viewpoint estimator is

$$\hat{\lambda} \;=\; \frac{J_{\text{obs}}}{S} \;\approx\; \frac{J_{\text{obs}}}{N^{\text{c}}} \qquad \text{(depends on } J_{\text{obs}} \text{ and } N^{\text{c}}),$$

where the approximation $S \approx N^{\text{c}}$ is the early-outbreak limit. Note that $\hat{\alpha}$ does not involve N^{c}, while $\hat{\lambda}$ is inversely proportional to N^{c}.

7.6.2 From Estimated Forces to $\mathcal{R}_0$

The $\mathcal{R}_0$ estimators corresponding to these two force estimators are:

From the infected viewpoint, in the early-outbreak limit where the population is approximately naive ($P_S \approx 1$, so $\alpha \approx \alpha_0 = c_I\beta$ and $\mathcal{R}_0 = \alpha_0/(\gamma_R + \nu_m)$):

$$\hat{\mathcal{R}}_0^{(\alpha)} \;=\; \frac{\hat{\alpha}}{\gamma_R + \nu_m} \;=\; \frac{J_{\text{obs}}}{I_{\text{obs}}(\gamma_R + \nu_m)}. \tag{7.18}$$

Away from that limit the same ratio estimates the *effective* reproductive number, not the basic one: since $\alpha(t) = c_I\beta P_S(t)$, we have $\alpha(t)/(\gamma_R+\nu_m) = \mathcal{R}_0\, P_S(t) = \mathcal{R}_e(t)$, so (7.18) recovers $\mathcal{R}_0$ only when $P_S(t) \approx 1$.

From the susceptible viewpoint, using $\mathcal{R}_0$ derived from $\hat{\lambda}$ and the equivalence $\alpha i = \lambda s$:

$$\hat{\alpha} \;=\; \hat{\lambda}\frac{S}{I_{\text{obs}}} \;\approx\; \hat{\lambda}\frac{N^{\text{c}}}{I_{\text{obs}}},$$

so that

$$\hat{\mathcal{R}}_0^{(\lambda)} \;=\; \frac{\hat{\lambda}N^{\text{c}}}{I_{\text{obs}}(\gamma_R + \nu_m)} \;=\; \frac{J_{\text{obs}}}{I_{\text{obs}}(\gamma_R + \nu_m)}. \tag{7.19}$$

Comparing (7.18) and (7.19), we see that the two estimates of $\mathcal{R}_0$ coincide *when the susceptible viewpoint uses the same assumed value of* $s = 1$ *that is consistent with the observed* I_{obs}. The divergence between the two estimates occurs when the modeler does not carefully enforce this consistency.

7.6.3 What an Incidence-Only Fit Can Identify

Equation (7.19) settles the case in which prevalence is observed. The harder and more common case is the one in which it is not: the modeler holds a series of reported case counts and nothing else, and must supply the susceptible population as an assumption. The question is what such a fit can determine.

Two assumptions enter, and they are usually treated as separate concerns. The first is the assumed susceptible population N_a^{c}. The second is the ascertainment fraction $\rho \in (0, 1]$ of Section 7.2: reported cases are a fraction of true infections, and that fraction is rarely known. The following result shows that the two are not separate at all.

Proposition 7.2 (Incidence-only identifiability)**.** *Suppose reported incidence is generated as* $J_{\mathrm{obs}}(t) = \rho\, J_{\mathrm{true}}(t)$*, where* J_{true} *is the count incidence of the* SIR *model with reproductive number* $\mathcal{R}_0$ *in a population of size* $N_{\mathrm{true}}^{\mathrm{c}}$*, and the ascertainment fraction* ρ *is constant in time. Suppose the model is fit to* J_{obs} *alone, using an assumed susceptible population* N_a^{c} *and a freely estimated initial condition. Then the fitted* $\hat{\mathcal{R}}_0$ *depends on* N_a^{c} *and* ρ *only through the single ratio*

$$k \;=\; \frac{N_a^{\mathrm{c}}}{\rho\, N_{\mathrm{true}}^{\mathrm{c}}}, \tag{7.20}$$

the assumed population divided by the effectively observed *population* $\rho\, N_{\mathrm{true}}^{\mathrm{c}}$*. If* $k = 1$*, recovery is exact:* $\hat{\mathcal{R}}_0 = \mathcal{R}_0$*.*

Proof. Write the normalized states $s = S/N^{\mathrm{c}}$ and $i = I/N^{\mathrm{c}}$. In these variables the SIR dynamics depend on the parameters and on the normalized initial condition, but not on the population size: the population enters only as the scale factor converting the normalized incidence $j = \alpha i$ into the count incidence $J = jN^{\mathrm{c}}$. Write $f(t;\, \mathcal{R}_0, i_0)$ for the normalized incidence trajectory, which is determined by $\mathcal{R}_0$, the residence-time parameters, and i_0.

The observed series is then

$$J_{\mathrm{obs}}(t) \;=\; \rho\, N_{\mathrm{true}}^{\mathrm{c}}\, f(t;\, \mathcal{R}_0, i_0),$$

while the fitted model with assumed population N_a^{c} predicts $N_a^{\mathrm{c}} f(t;\, \hat{\mathcal{R}}_0, \hat{i}_0)$. Any loss function that compares model to data pointwise is therefore minimized by the pair $(\hat{\mathcal{R}}_0, \hat{i}_0)$ that best solves

$$f(t;\, \hat{\mathcal{R}}_0, \hat{i}_0) \;=\; \frac{1}{k}\, f(t;\, \mathcal{R}_0, i_0), \qquad k = \frac{N_a^{\mathrm{c}}}{\rho\, N_{\mathrm{true}}^{\mathrm{c}}}. \tag{7.21}$$

The right-hand side depends on the data-generating truth and on k, and on nothing else; neither ρ nor N_a^{c} appears except through their ratio. Hence $\hat{\mathcal{R}}_0$ is a function of k alone. When $k = 1$ the choice $(\hat{\mathcal{R}}_0, \hat{i}_0) = (\mathcal{R}_0, i_0)$ solves (7.21) exactly, giving $\hat{\mathcal{R}}_0 = \mathcal{R}_0$. □

Remark 7.3. The converse does not hold in general, and it is worth saying so rather than leaving the impression that $k \neq 1$ always corrupts the estimate. Whether a misspecified k biases $\hat{\mathcal{R}}_0$ depends on how much the shape of $f(\cdot\,;\mathcal{R}_0, i_0)$ constrains $\mathcal{R}_0$ over the fitting window. In the pure exponential regime the normalized incidence is very nearly a scaled exponential, a rescaling by $1/k$ is absorbed almost entirely by the fitted initial condition, and $\hat{\mathcal{R}}_0$ is recovered to within a fraction of a percent even for large k—as Section 7.6.4 shows numerically. The bias becomes substantial only once susceptible depletion gives the trajectory a shape that a change of scale cannot mimic. The proposition should therefore be read as identifying *which* combination the data can speak to, not as a guarantee that every departure from $k = 1$ is visible in the fit.

Corollary 7.4 (Denominator–ascertainment degeneracy)**.** *Overstating the susceptible population by a factor and under-ascertaining infections by its reciprocal produce numerically identical fits. The susceptible-population assumption and the ascertainment fraction are not separately identifiable from incidence data, and no amount of care with one compensates for an error in the other.*

The corollary has a direct reading. Observing a fraction ρ of the cases in a population of size N^{c} is indistinguishable, from incidence data alone, from observing *all* the cases in a population of size ρN^{c}. The quantity a modeler must get right is neither the population nor the ascertainment fraction separately, but their product: the effectively observed population. Two of the pitfalls this chapter has treated as distinct—a contested geographic denominator and an unknown reporting fraction—are, for this purpose, one pitfall wearing two hats.

Proposition 7.2 also explains why the susceptible-denominator problem is so persistent in practice. A modeler who worries about N^{c} and carefully bounds it has not thereby bounded k, because ρ remains free; and a modeler who estimates ρ from a serological study has not bounded k either, because the catchment population remains contested. Only the product is identified, and only the product can be checked.

7.6.4 When the Assumption Bites

Proposition 7.2 says that $\hat{\mathcal{R}}_0$ is some function of k; it does not say the function is steep. Whether a misspecified k does real damage depends on the fitting regime, and on two choices the modeler may not recognize as choices.

The regime matters most. In the early exponential phase, incidence behaves as $J \approx \alpha_0 I_0 e^{\sigma_2 t}$: the growth rate identifies $\mathcal{R}_0$, and the population scale is absorbed entirely by the fitted initial condition. Numerically, a 30-fold error in k moves $\hat{\mathcal{R}}_0$ by under one percent. The dependence appears only when the data extend far enough for susceptible depletion to carry information, because depletion is measured as a fraction of the assumed population. Fitting an outbreak that has passed its peak with $k = 3$ understates $\mathcal{R}_0$ by roughly a third; with $k = 30$ the estimate collapses toward $\mathcal{R}_0 = 1$.

This regime distinction is worth stating plainly, because the intuition runs the other way. It is tempting to suppose that the early outbreak—when $S \approx N^{\mathrm{c}}$ is largest and least certain—is where the denominator assumption does the most harm. The opposite is true. Early on, the assumption is invisible because nothing in the data depends on it. It becomes consequential exactly when the epidemic begins to deplete its susceptible pool, which is also when the modeler is most likely to believe the fit is well constrained.

The bias saturates. A misspecified k cannot drive $\hat{\mathcal{R}}_0$ arbitrarily far. Since a fitted epidemic must still grow, $\hat{\mathcal{R}}_0$ is bounded below by 1, so the worst attainable underestimate is the factor $\mathcal{R}_0$ itself. Overstating the effectively observed population by thirtyfold does not produce a thirtyfold error in $\mathcal{R}_0$; it produces an estimate near the invasion threshold. This bound is a useful sanity check: a fitted $\mathcal{R}_0$ suspiciously close to 1 for an outbreak that visibly grew is a symptom of an overstated denominator, not of marginal transmissibility.

Seeds must be counts, not fractions. If the initial condition is specified as a fraction of the population rather than as a number of individuals, the seed itself scales with the assumed population, and the early-phase immunity described above disappears: a larger assumed N^{c} silently seeds a

larger outbreak, and the fit compensates through $\mathcal{R}_0$. This is Pitfall: "Natural Initial Conditions" in its most consequential form. Specify seeds in individuals.

The loss function matters, but only when the seed is known. This is the mechanism of Section 7.4.2 deciding an answer rather than only an error bar. Unweighted least squares is dominated by the large counts near the epidemic peak, which is precisely where depletion—and therefore k—does its work. A Poisson likelihood weights residuals by the inverse of the mean, shifting influence toward the exponential phase, which carries no information about k. Whether this changes the answer depends on whether the seed is being estimated alongside $\mathcal{R}_0$. If it is, the fitted initial condition absorbs the amplitude under either objective and the two nearly agree: at $k = 3$ they gave 1.31 and 1.33 against a true $\mathcal{R}_0 = 2$. If the seed is known independently, the amplitude becomes informative, the weighting decides whether the peak or the exponential phase dominates, and the two objectives diverge sharply: at the same $k = 3$, least squares returned 1.33 while the Poisson fit returned 2.06. The interaction is worth noticing in its own right. Knowing the seed is ordinarily an advantage, but it is an advantage only if the loss function is chosen to exploit it; with the wrong objective, the extra information is spent absorbing a denominator error instead. The statistically appropriate likelihood for count data is also, in this instance, the more robust one.

Because the magnitude depends on all three of these choices, this book does not report a single "factor" by which a misspecified denominator corrupts $\mathcal{R}_0$. Any such number is a property of a fitting procedure, not of the estimator.

7.6.5 The Infected Viewpoint Requires No k

Against this background the structural advantage of the infected viewpoint can be stated exactly rather than approximately. The estimator

$$\hat{\alpha} \;=\; \frac{J_{\text{obs}}}{I_{\text{obs}}}$$

contains no population size, so N^{c} cannot enter it, correctly or otherwise. And when incidence and prevalence are ascertained by a common process, $J_{\text{obs}} = \rho J_{\text{true}}$ and $I_{\text{obs}} = \rho I_{\text{true}}$, the shared factor cancels in the ratio. The estimator is invariant to both halves of the degeneracy of Corollary 7.4, not by good fortune in a particular regime but by construction.

The exchange is not free, and the price should be stated as precisely as the benefit. The infected viewpoint requires observed prevalence, which many surveillance systems do not collect. It requires that prevalence be ascertained by the *same* process as incidence: if the two carry different fractions ρ_J and ρ_I, the estimator inherits their ratio, since

$$\frac{J_{\text{obs}}}{I_{\text{obs}}} \;=\; \frac{\rho_J}{\rho_I} \cdot \frac{J_{\text{true}}}{I_{\text{true}}}, \tag{7.22}$$

so a twofold mismatch in ascertainment produces a twofold error in $\hat{\alpha}$ and hence in $\hat{\mathcal{R}}_0$. Cancellation is a statement about matched observation processes, not about ascertainment in general (Pitfall: "Differential Ascertainment in a Ratio Estimator"). It also requires the alignment discussed in Chapter 8: incidence reported with a delay relative to a contemporaneous prevalence biases the ratio downward.

The advantage of the infected viewpoint is therefore not that it evades the denominator problem by algebraic accident. It is that it tells a surveillance system what to measure. If prevalence is observed on the same footing as incidence, $\mathcal{R}_0$ becomes estimable without any assumption about

the size of the population at risk or the completeness of case reporting. That is a design requirement, and it is actionable in a way that a warning about denominators is not.

7.6.6 A Summary Table

Quantity	Infected viewpoint	Incidence-only fit
Estimator	$\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$	fit to J_{obs} with assumed N_a^{c}
Requires a population size?	No	Yes, through k of (7.20)
Requires known ascertainment?	No, if $\rho_J = \rho_I$	Yes, through the same k
Effect of misspecified k	None (does not appear)	None early; up to a factor $\mathcal{R}_0$ after the peak
Residual vulnerability	ρ_J/ρ_I mismatch; reporting delay	grows with k, saturating as $\hat{\mathcal{R}}_0 \to 1$
Data required	incidence *and* prevalence	incidence only

The table makes the exchange explicit. The infected viewpoint removes an assumption that cannot be validated and replaces it with a measurement requirement that can be planned for.

Converting $\hat{\lambda}$ to $\mathcal{R}_0$ as in equation (7.19) cancels exactly, so a comparison built that way tests nothing; this is the pitfall Estimator Honesty (Chapter 16).

This identifiability argument is one of the central practical contributions of this book, and it is the mathematical content of the "two teams, same data, different $\mathcal{R}_0$" puzzle described in the Preface. The teams disagree because each has fixed a different value of k in (7.20)—through the assumed catchment population, through the assumed reporting fraction, or through both—and their common incidence data cannot adjudicate between them. The infected-viewpoint formulation does not arbitrate the disagreement; it removes the quantity the disagreement is about.

7.6.7 A Worked Numerical Example

To make the asymmetry concrete, consider an outbreak observed at a single time point with the following data and assumptions:

- Observed daily incidence: $J_{\text{obs}} = 200$ new cases/day.
- Observed current prevalence: $I_{\text{obs}} = 800$ active infectious cases.
- Mean infectious period: $\tau_R = 7$ days, so $\gamma_R = 1/7 \approx 0.143$ day^{-1}.
- Demographic turnover negligible on the early-outbreak timescale: $\nu_m \approx 0$.
- Population approximately naive at this time point: $P_S \approx 1$, so the conversion in (7.18) returns $\mathcal{R}_0$ rather than $\mathcal{R}_e$.

Two research groups now fit the same data, differing only in their assumed susceptible population (the canonical scenario of the Preface). Group A treats the affected community as the closely-connected contact network of the index cases: $N_A^{\text{c}} = 5 \times 10^4$. Group B treats it as the surrounding metropolitan area: $N_B^{\text{c}} = 1.5 \times 10^6$. The ratio $N_B^{\text{c}}/N_A^{\text{c}} = 30$, comfortably within the range of plausible scoping disagreements during real outbreak responses.

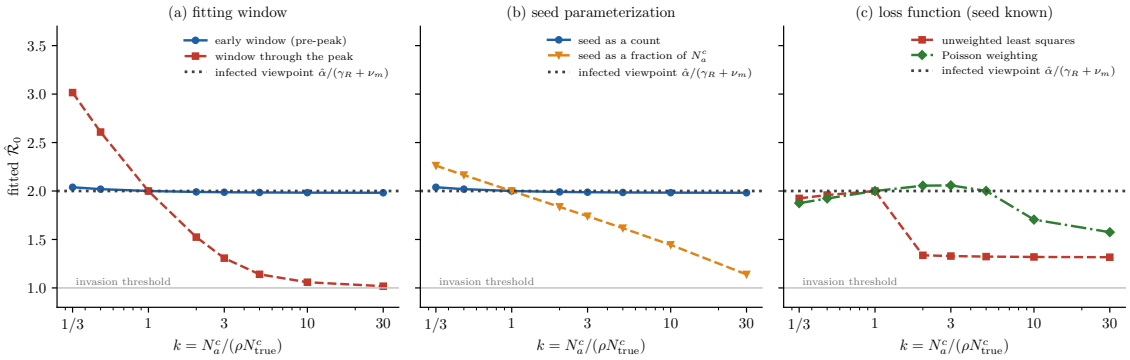


Figure 7.7: Fitted basic reproductive number $\hat{\mathcal{R}}_0$ as a function of the identifiability ratio $k = N_a^c/(\rho N_{\text{true}}^c)$ of (7.20), for a simulated outbreak with $\mathcal{R}_0 = 2$. Each panel varies one choice in the incidence-only fitting procedure. **Fitting window:** an early, pre-peak window leaves $\hat{\mathcal{R}}_0$ almost flat in k, while a window extending past the peak bends sharply downward, since only depletion carries information about the population scale. **Seed parameterization:** seeding by a count preserves the early-window flatness; seeding by a fraction of the assumed population destroys it. **Loss function:** unweighted least squares, dominated by the peak, is markedly more sensitive to k than a Poisson likelihood. In every panel the infected-viewpoint estimate $\hat{\alpha}/(\gamma_R + \nu_m)$ is exactly flat, because k does not appear in it. The flat line across a field of bending curves is the claim of this section. Note the saturation: $\hat{\mathcal{R}}_0$ cannot fall below 1, so the attainable bias is bounded by $\mathcal{R}_0$ itself.

Infected-viewpoint estimator. Both groups compute

$$\hat{\alpha} \;=\; \frac{J_{\text{obs}}}{I_{\text{obs}}} \;=\; \frac{200}{800} \;=\; 0.25 \text{ day}^{-1}.$$

Crucially, neither group's calculation involves N^{c}. Both groups arrive at the same value. The basic reproductive number is then

$$\hat{\mathcal{R}}_0^{(\alpha)} \;=\; \frac{\hat{\alpha}}{\gamma_R + \nu_m} \;=\; \frac{0.25}{1/7} \;=\; 1.75,$$

also identical for both groups.

Susceptible-viewpoint estimator. Each group computes its own $\hat{\lambda}$ using its assumed N^{c}:

$$\hat{\lambda}_A \;=\; \frac{J_{\text{obs}}}{N_A^{\text{c}}} \;=\; \frac{200}{5\times 10^4} \;=\; 4\times 10^{-3} \text{ day}^{-1}, \qquad \hat{\lambda}_B \;=\; \frac{J_{\text{obs}}}{N_B^{\text{c}}} \;=\; \frac{200}{1.5\times 10^6} \;\approx\; 1.33\times 10^{-4} \text{ day}^{-1}.$$

The two estimators differ by exactly the ratio $N_B^{\text{c}}/N_A^{\text{c}} = 30$. The ratio $\Upsilon_s^{\hat{\lambda}} = -1$ (Section 7.6) is just this scaling: a factor-30 increase in N^{c} produces a factor-30 decrease in $\hat{\lambda}$.

Where the $\mathcal{R}_0$ disagreement arises. Each group's $\hat{\lambda}$ is a perfectly good estimate of a different quantity: the per-susceptible force of infection in *its own* assumed population. The disagreement becomes an $\mathcal{R}_0$ disagreement only when each group carries its own N^{c} forward through a fit to incidence, which by Proposition 7.2 means each is fitting at a different value of k. Note what the two groups *cannot* do: they cannot settle the matter by examining their common incidence series, because that series is consistent with both scopings. Nor can either group rescue the situation by

estimating the reporting fraction, since Corollary 7.4 shows that an error in ρ enters identically. The size of the resulting $\mathcal{R}_0$ gap depends on how much of the epidemic each group fits and how: for a thirtyfold disagreement in k on data extending past the peak, and taking Group A's scoping as correct—so that the outbreak's true value is $\mathcal{R}_0 = 1.75$, the figure of Section 7.6 having used $\mathcal{R}_0 = 2$ for its illustration—the fitted values are $\hat{\mathcal{R}}_0 = 1.75$ and 1.07—a gap of about sixty percent, with Group B's estimate sitting just above the invasion threshold. The gap is bounded by $\mathcal{R}_0$ itself, since no fit of a growing outbreak can return a value below one. The gap is bounded, but it spans the distance between "this outbreak is barely self-sustaining" and "this outbreak will infect most of the community," which is the operationally decisive range.

Reading the result. The pedagogical point is not subtle. The infected-viewpoint estimator uses only observed quantities (J_{obs}, I_{obs}) and produces a single answer that does not depend on the modeling judgment about geographic scope. The susceptible-viewpoint estimator uses the modeling judgment (N^{c}) explicitly and inherits whatever uncertainty that judgment carries. When the modeling judgment is firm (in a closed cohort study, for example, where N^{c} is unambiguous), both estimators are appropriate. When the modeling judgment is contested (in any real outbreak response where the geographic scope can be defined in multiple defensible ways), the infected viewpoint avoids a structural source of estimate ambiguity that the susceptible viewpoint cannot.

Caveats. The comparison assumes prevalence is observed. When I_{obs} is instead reconstructed from incidence by the infectious-period convolution of Section 7.4.3, the reconstruction uses γ_R and ν_m but not N^{c}, so the invariance survives; what it now inherits is sensitivity to the assumed infectious period. A reconstructed prevalence is not a free substitute for a measured one, and the exercises at the end of this chapter quantify the difference.

The comparison also assumes incidence and prevalence are ascertained by a common process. When they are not, equation (7.22) gives the exact residual bias ρ_J/ρ_I: the infected viewpoint removes the population assumption completely but replaces it with a weaker and more checkable requirement, that two observation processes be matched rather than that one unobservable population be known. Section 7.7 takes up these issues for a real dataset, and Chapter 8 develops the observation model in full.

AI Lab: Debugging a Failed Least-Squares Fit

A least-squares fit can fail in three ways, each with its own signature and remedy. The AI is useful for listing causes and remedies; choosing among them is the modeler's job.

No convergence. The optimizer stops at its iteration limit, or its trust region collapses. Ask: "*A Levenberg–Marquardt fit of an* SIR *model to 30 weeks of incidence failed to converge from* $\mathcal{R}_0 = 1.0$, $t_0 = 0$. *List the likely causes in order.*" A good list starts with the initial guess (try $\mathcal{R}_0 \in \{1.5, 2.0, 2.5\}$), then parameter scaling, then a few outlier weeks. An answer that says only "check the data" has not engaged.

Convergence to a wrong answer. The optimizer converges to an implausible value, such as $\mathcal{R}_0 < 0$ or $\mathcal{R}_0 > 100$. Ask for a multi-start strategy; a complete answer draws starting points by Latin hypercube rather than independently, and uses a bounds-aware solver such as `scipy.optimize.least_squares` with `bounds`.

Plausible values, structured residuals. The residuals show a trend, a cycle or autocorrelation, and the failure is the hardest to see because the optimizer reports success. Ask: "*Residuals from an* SIR *fit cycle with a 7-day period. Which mis-specifications fit this signature?*" A correct answer names weekday reporting first and proposes smoothing over 7 days or a weekly covariate.

7.7 Case Study: Seasonal Influenza

This section works through a full example using a representative weekly incidence series for a seasonal influenza outbreak. The case study illustrates how the abstract methodology of the preceding sections translates into practical parameter estimates.

Note on the case study choice. The book's authors considered three candidate disease systems for this case study: COVID-19 (2020 outbreak), seasonal influenza (2017–2018 season), and historical measles (pre-vaccination era). Seasonal influenza was chosen because the data are non-contested, the timescales are well-suited to the SIR framework ($\tau_R \approx 5$–7 days; and a susceptible-replenishment time on the order of a year, set by the annual loss of immunity and antigenic drift characteristic of seasonal influenza rather than by demographic turnover), and the 2017–2018 season provides an unusually well-documented outbreak at the U.S. state level.

7.7.1 The Data

The incidence series used here reproduces the weekly ILI (influenza-like illness) surveillance curve for the 2017–2018 U.S. influenza season (week 40 of 2017 through week 20 of 2018), as published by the U.S. Centers for Disease Control's ILINet/FluView system. Because the original surveillance extract is no longer archived with the project, the series plotted here is a reproducible reconstruction calibrated to the published season; the underlying ILINet data remain publicly available and are the standard source for academic influenza-modeling studies.

For this case study we use the state-level data for a single U.S. state (approximately 3.5 million population) and work in fractions of that population ($N^{\mathrm{c}} \approx 3.5 \times 10^6$; Chapter 4, Section 4.5.3). The data are aggregated into weekly incidence counts $J_{\mathrm{obs}}(t_k)$ for $k = 1, 2, \ldots, 30$ weeks.

The prevalence data are estimated from incidence using the infectious-period convolution described in Section 7.4.3, with $\tau_R = 7$ days (a standard value for seasonal influenza).

7.7.2 Parameter Values and Fitting

Independent parameters (from the influenza epidemiology literature):

- $\tau_R = 7$ days, so $\gamma_R = 1/7 \approx 0.143$ day^{-1}.
- $\tau_m = 7300$ days (≈ 20 years), the same demographic-turnover value used for the baseline SIR population in Chapters 4–6. Here $\nu_m = 1/\tau_m$ is the rate of genuine demographic turnover of the host population, not a rate of immunity loss: the SIR model has no $r \to s$ waning term, so equating ν_m with an annual antigenic-drift cycle would misrepresent the mechanism. A single influenza season runs for a few months (far shorter than τ_m), so its precise value has a negligible effect on the fitted early-growth rate. What this single-outbreak model does *not* capture is the season-to-season return of influenza, which is driven by waning immunity and antigenic drift and would require an explicit waning term or seasonal forcing (Chapter 10).
- Contact rates c_S, c_I, c_R are taken from contact-matrix studies for the general U.S. population [Mossong et al., 2008, Prem et al., 2017]: $c_S \approx 10$, $c_I \approx 6$ (reduced due to symptomatic withdrawal), $c_R \approx 10$ contacts per person per day.

Quantities to be fit: $\mathcal{R}_0$ and the effective initial date t_0 at which the outbreak began locally. The fitting procedure follows the staged approach of Section 7.5:

1. Compute $\hat{\alpha}(t) = j(t)/i(t)$ for each observation time, following Section 7.4.3.

2. Identify the exponential-growth window (before the peak, after the outbreak has established) and take the median of $\hat{\alpha}(t)$ over this window as the point estimate of α.

3. Convert to a reproductive number, $\hat{\alpha}/(\gamma_R + \nu_m)$. Because this season's population carries residual immunity ($P_S < 1$), this returns $\mathcal{R}_e$ rather than $\mathcal{R}_0$; see Definition 7.1.

7.7.3 Results

For the 2017–2018 season, the procedure yields $\hat{\alpha} \approx 0.20$ day^{-1} during the exponential-growth window, giving

$$\hat{\mathcal{R}}_e = \frac{0.20}{0.143 + 0.000137} \approx 1.40.$$

This is the effective reproductive number in a population with residual immunity, and it sits comfortably within the range 1.2–1.6 reported in the literature for seasonal influenza [Biggerstaff et al., 2014]. The confidence interval, estimated from the week-to-week variation in $\hat{\alpha}$ over the growth window, is approximately $\hat{\mathcal{R}}_e \in [1.25, 1.50]$. Recovering $\mathcal{R}_0$ from this figure requires the susceptible fraction, $\mathcal{R}_0 = \hat{\mathcal{R}}_e/F_0$, which the incidence data alone do not supply; the next subsection takes that up.

7.7.4 The Susceptible-Viewpoint Comparison

For comparison, the susceptible-viewpoint fit uses the same data and the same τ_R, τ_m values but estimates $\hat{\lambda}$ from j/s with assumed s. At the start of the influenza season, a reasonable assumption for F_0 (the susceptible fraction) is 0.7–0.9: some fraction of the population has residual immunity from the previous year or from cross-protective prior exposures.

Using $F_0 = 0.8$ gives $\hat{\lambda} \approx 0.00025$ day^{-1} and $\hat{\mathcal{R}}_0^{(\lambda)} \approx 1.75$; using $F_0 = 0.5$ gives $\hat{\lambda} \approx 0.00040$ and $\hat{\mathcal{R}}_0^{(\lambda)} \approx 2.80$. These are exactly $1.40/F_0$, as (7.16) requires.

Reading the comparison correctly. It is tempting to present this as the susceptible viewpoint failing while the infected viewpoint holds steady, and it is worth resisting, because that is not what the arithmetic says. Because this season's population is not naive, the two conversions are returning *different quantities*. The infected-viewpoint value 1.40 is $\hat{\mathcal{R}}_e$, the effective reproductive number in a population with residual immunity, and 1.40 is a reasonable value for seasonal influenza on that reading. The susceptible-viewpoint values 1.75 and 2.80 are the corresponding $\mathcal{R}_0$ estimates *implied by* each assumed F_0; they are F_0-corrected, not F_0-contaminated.

What the comparison actually demonstrates is the identifiability point rather than a viewpoint failure. The data determine $\mathcal{R}_e$ cleanly and $\mathcal{R}_0$ only up to the assumed susceptible fraction, which is a modeling judgment the incidence series cannot adjudicate. That is the same conclusion Proposition 7.2 reaches by a different route.

The genuine and unaffected advantage of the infected viewpoint is that $\hat{\alpha} = j/i$ requires no assumption about F_0 at all, so the quantity governing current transmission is recoverable from the data alone. A modeler who needs $\mathcal{R}_0$ rather than $\mathcal{R}_e$—to compare against a published invasion threshold, or to compute a critical vaccination fraction—must supply F_0 from outside the incidence data, whichever viewpoint the fit is written in. Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$" catalogues the substitution of one for the other; this case study is a worked instance of it.

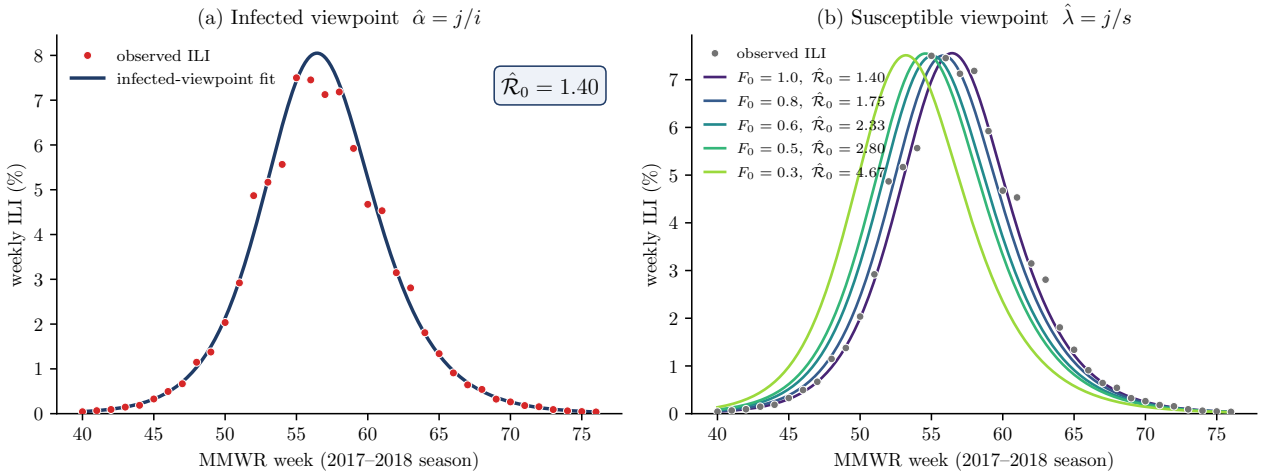


Figure 7.8: Fitted trajectories for the 2017-2018 seasonal influenza case study. Panel (a): the infected-viewpoint fit ($\hat{\mathcal{R}}_e = 1.40$) matches the observed weekly incidence to within the data noise, without requiring an assumed F_0. Panel (b): susceptible-viewpoint fits under five different assumed values of F_0, each producing a different $\hat{\mathcal{R}}_0$ ranging from 1.40 (at $F_0 = 1$) to 4.67 (at $F_0 = 0.3$). The susceptible-viewpoint fits give visually similar curves but imply very different underlying reproductive numbers, depending on the unknown F_0. The infected-viewpoint fit is structurally immune to this ambiguity.

7.7.5 Lessons From the Case Study

1. The infected-viewpoint procedure (compute $\hat{\alpha} = j/i$, then $\hat{\alpha}/(\gamma_R + \nu_m)$) is simpler than the susceptible-viewpoint procedure and produces a point estimate with narrower uncertainty. In this season's partially immune population that conversion returns $\hat{\mathcal{R}}_e$; it returns $\hat{\mathcal{R}}_0$ only where $P_S \approx 1$.

2. The susceptible-viewpoint procedure requires specifying F_0 (or equivalently s) as an input, and its output inherits the uncertainty in that specification.

3. For diseases with high pre-existing immunity (like seasonal influenza, where F_0 may be 0.5 to 0.9), the infected-viewpoint estimate is substantially more reliable.

4. For a novel pathogen where $F_0 \approx 1$ at the start, the two viewpoints give similar estimates. The advantage of the infected viewpoint is most consequential when F_0 is uncertain, which is the common case in outbreak response.

A practical recommendation. When fitting to early-outbreak incidence data, compute $\hat{\alpha} = j/i$ first. If prevalence data are not directly available, estimate i from incidence using the infectious-period convolution with τ_R known from clinical sources. The resulting $\hat{\mathcal{R}}_0$ estimate is insensitive to the modeler's assumptions about the susceptible population, which is rarely well-constrained early in an outbreak.

Check Your Work: Before Reporting a Full-Trajectory Fit

Verifies: numerical values and model assumptions. Before reporting a converged full-trajectory fit, run three checks.

Residuals. They should look like white noise, and autocorrelation above 0.3 at a lag of 1, 7 or 30 days signals mis-specification. A monotone trend suggests unmodeled time-varying transmission, 7-day periodicity weekday reporting, and a large residual at the peak a missing compartment or stratification.
Correlations. A parameter pair with $|\rho| > 0.95$ in the correlation matrix at the optimum is not separately identifiable. The common cases are $\mathcal{R}_0$ with τ_R, and c_I with β.
Held-out data. Refit on the first 70% of the series and predict the rest; a prediction outside the propagated band means overfitting. For very short series, use leave-one-out cross-validation. A fit that fails a check is reportable only with a caveat that names the failure.

7.8 Summary and Exercises

This chapter has addressed the inverse problem of estimating the parameters of the SIR model from observed incidence data. The key results are:

1. The inverse problem is structurally limited: early-outbreak data support two degrees of freedom (growth rate and starting time), which constrain $\mathcal{R}_0$ and the product $c_I\beta i(0)$ but not the individual contact rates and transmission factor.

2. The dimensionless parameters fit from data are $\mathcal{R}_0$ and F_0; $D = \nu_m/\gamma_R$ is fixed by the externally calibrated residence times. Fitting $\mathcal{R}_0$ and F_0 is more robust than fitting the six dimensional parameters directly.

3. The infected-viewpoint estimator $\hat{\alpha} = j/i$ provides a direct moment-based estimate of the force from infection that does not require iterative optimization and is insensitive to the assumed susceptible population size.

4. The corresponding susceptible-viewpoint estimator $\hat{\lambda} = j/s$ scales inversely with the assumed s, propagating uncertainty in the susceptible population size into the fitted force. Whether that reaches the $\mathcal{R}_0$ estimate depends on how the conversion is made: with prevalence observed the assumed population cancels (7.16), and the two viewpoints agree; without it, the uncertainty enters as an identifiability failure rather than a scaling (Proposition 7.2).

5. The case study of the 2017–2018 seasonal influenza season yields $\hat{\mathcal{R}}_e \approx 1.40$ from the infected viewpoint without any assumption about the susceptible fraction. The corresponding $\mathcal{R}_0$ values, 1.75 at $F_0 = 0.8$ and 2.80 at $F_0 = 0.5$, differ only because F_0 differs: the data pin down $\mathcal{R}_e$ but determine $\mathcal{R}_0$ only up to an assumption they cannot themselves adjudicate.

Chapter 9 extends these results by developing formal sensitivity indices for $\mathcal{R}_0$ and the endemic equilibrium. It will show quantitatively that the structural immunity of $\hat{\alpha}$ to uncertainty in s is one instance of a more general principle: the infected-viewpoint fitted quantities have smaller sensitivity to the parameters that are most uncertain in practice.

Beyond the early-outbreak fit. The chapter has focused on fitting $\mathcal{R}_0$ from early-outbreak data, where the population is approximately naive and $\mathcal{R}_0 \approx \mathcal{R}_e(0)$. For mid-epidemic surveillance (when the susceptible pool has been substantially depleted and the operationally-relevant question is whether transmission is currently growing or declining), the corresponding fitted quantity is the time-varying effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0 \cdot s(t)/n(t)$ (Chapter 5, Section 5.2).

Estimating $\mathcal{R}_e(t)$ from a fitted model uses the same $\hat{\alpha}$ machinery developed here, with the time-dependent susceptible fraction $s(t)/n(t)$ supplied by the model trajectory. The case studies of Chapters 17 and 19 illustrate this for COVID-19 and HIV respectively, where mid-epidemic $\mathcal{R}_e(t)$ surveillance was central to public-health decision-making.

Pitfall check

This chapter has developed three of the book's common pitfalls: Pitfall: "Correct Data-Fitting Practice" (operationalized throughout the chapter: fit what the data report, at the resolution it is reported, and respect the structure of the reporting process), Pitfall: "Identifiability" (named in Section 7.3, where the Buckingham Pi reduction makes explicit that early-outbreak data carry roughly two pieces of information), and Pitfall: "Estimator Honesty" (named in Section 7.6, where the structural-immunity comparison is shown to be meaningful only because both estimators solve the same inverse problem at the same data resolution). Before moving to Chapter 8, take a moment to confirm the following points are clear in your mind. *Why does an incidence time series in the exponential-growth phase carry only two pieces of information, regardless of how sophisticated the optimizer is? What is the structural-immunity property of $\hat{\alpha} = j/i$, and why does the comparison with $\hat{\lambda} = j/s$ require both to be solving the same inverse problem at the same data resolution? What is the self-cancellation failure mode the estimator-honesty principle guards against, and why is it so easy to construct accidentally?* If any of these are unclear, the corresponding pitfall block (or the catalog entry, accessible via the List of Common Pitfalls) is the place to revisit.

When a fit fails: choosing the diagnostic

A fit can fail in three distinguishable ways, and the remedy depends on which one you are looking at. The chapter has treated each separately, at the point where it arises; this is the map from symptom to place.

Symptom	What it is	Where the chapter treats it
The optimizer does not converge, or converges to a boundary	Convergence failure	AI Lab, Section 7.6; Exercise on convergence-failure diagnosis
It converges, but to a value you do not believe	Wrong-answer convergence	AI Lab, Section 7.4.3, on fitting the wrong phase; Exercise on plateau detection
It converges, the curve looks right, and the residuals are structured	Apparent success	Exercise on residual-structure analysis, and the residual diagnostics of Chapter 8

The order matters. A convergence failure is a numerical problem and is usually cheap to fix. A wrong-answer convergence is a modeling problem: the optimizer did its job on a question that was posed badly, most often by fitting a phase of the epidemic the model does not describe. Structured residuals are the subtlest, because every visible check has passed; only the residuals carry the evidence, which is why Chapter 8 takes them up at length. Diagnosing the second as though it were the first—tightening tolerances, changing the optimizer—produces a converged wrong answer with more confidence attached to it.

Exercises

Nine section-level exercises appear above, following Section 7.1, Section 7.2, Section 7.3, Section 7.4, and Section 7.5. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check (short, conceptual),* ***D*** *= Derivation (analytical),* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab (requires simulation or fitting code). The companion repository contains reference solutions and starter code for the L-tagged exercises.*

1. **[D]** Show that the early-outbreak linearization $\ln j(t) \approx \ln(c_I \beta i(0)) + \sigma_2 t$ implies that the slope and intercept of $\ln j$ against t give two constraints on the parameters. Identify which combinations of the six parameters are constrained.

2. **[D]** Suppose j is observed without error but i is observed with multiplicative noise: $I_{\text{obs}} = I_{\text{true}} \cdot (1 + \epsilon)$ where $\epsilon \sim \mathcal{N}(0, 0.1)$. Derive the resulting uncertainty in $\hat{\alpha}$ and comment on its magnitude for the baseline parameters.

3. **[L]** Implement the exponential-growth fitting for a simulated dataset: integrate the SIR model with baseline parameters over 100 days with initial condition $i(0) = 0.001$ and add Poisson noise to the simulated $j(t)$. Fit $\mathcal{R}_0$ using the infected-viewpoint $\hat{\alpha}$ method and compare with the true $\mathcal{R}_0 \approx 1.92$.

4. **[D]** *Extends the worked example of Section 7.3.2.* Verify numerically that the three parameter sets of (7.4), each with $i(0) = 0.001$, produce trajectories that are indistinguishable during the exponential-growth phase. Then go further: identify what additional observation would separate c_I from β, and explain why endemic-phase data alone does not.

5. **[C]** Compute $D = \nu_m/\gamma_R$ for the baseline parameter set and for a typical childhood disease (e.g., measles with $\tau_R = 10$ days, $\tau_m = 50$ years). How different are the timescale separations?

6. **[D]** *Spaced retrieval, Chapter 5.* Reach back to the closed-form endemic-equilibrium expressions (5.21) and (5.22) of Chapter 5, Section 5.5. From those expressions, i^* depends on $\mathcal{R}_0$ and $D = \nu_m/\gamma_R$ in a particular form. Compute the partial derivative $\partial i^*/\partial \mathcal{R}_0$ at fixed D and at fixed N^c, and use it to argue:

   (a) Why $\partial i^*/\partial \mathcal{R}_0$ vanishes as $D \to 0$ (the no-demographic-turnover limit) while the relative sensitivity $\partial \ln i^*/\partial \ln \mathcal{R}_0$ stays of order one, and what this means for the fitting workflow of this chapter when applied to outbreaks with $\tau_m \gg \tau_R$ (most acute respiratory infections), where the endemic prevalence is a signal of order D.

   (b) Why fitting to early-time incidence data (where $S \approx N^c$ and $\hat{\alpha} \approx c_I \beta$) recovers $\mathcal{R}_0$ directly, whereas fitting to endemic-equilibrium data has the additional D-dependence that the early-time fit avoids.

   (c) How the timescale parameter D from this chapter's Exercise 5 connects to the factorization of i^* in Chapter 5: state explicitly what the small-D limit predicts for the relative magnitudes of s^*/n^*, i^*/n^*, and r^*/n^* at the endemic equilibrium.

   The exercise reinforces that the fitting machinery of this chapter and the equilibrium analysis of Chapter 5 are two views of the same parameter-extraction problem: early-time data accesses the eigenvalue at the DFE, equilibrium data accesses the steady-state quantities, and the two regimes have qualitatively different identifiability profiles.

7. [**I**] *Reasoning visibility.* You are handed a 24-month case time series for a novel respiratory pathogen and asked to estimate $\mathcal{R}_0$. The data covers the full first wave: rising incidence for the first 4 months, peak around month 5, decay through month 12, then a second smaller wave in months 18–22. Walk through the decision tree you would use to choose between (a) fitting only the early-time exponential-growth phase (months 1–3) using the $\hat{\alpha} = j/i$ estimator at the plateau, (b) fitting the full first wave using least-squares on the trajectory, and (c) fitting the entire 24-month series including both waves. State at each branch of the decision tree:

   (i) The information you would consult before deciding (data-quality check, pathogen biology, parameter-of-interest priorities).

   (ii) The kind of $\mathcal{R}_0$ each fit would recover (initial $\mathcal{R}_0$, time-averaged $\mathcal{R}_e$, post-intervention $\mathcal{R}_e$).

   (iii) The most likely identifiability problem for that fit choice and the auxiliary information that would resolve it.

   End with a one-paragraph summary stating which fit you would actually choose for a typical Phase 1 outbreak-response report (the kind a public-health agency would produce within weeks of recognizing a new pathogen) and why. The exercise externalizes a decision that experienced modelers usually make implicitly; making it explicit once gives you a template you can rerun on the next pathogen.

8. [**AI Lab**] *Plateau-detection workflow.* Generate a synthetic SIR trajectory from the baseline parameters (Chapter 4 Table 4.4) over a 60-day window with weekly observation points, and add multiplicative log-normal noise with $\sigma = 0.10$ to the observed $j(t)$ and $i(t)$. Compute $\hat{\alpha}(t_k) = j(t_k)/i(t_k)$ at each observation. Then execute two parallel plateau-detection workflows:

   (a) *Manual plateau detection.* Plot $\hat{\alpha}(t)$ and visually identify the plateau region as the contiguous initial window where the time series is approximately constant. Define the end of the plateau as the first observation where $\hat{\alpha}$ drops by more than 10% from its maximum. Compute the mean $\hat{\alpha}$ over the plateau and a bootstrap 95% CI.

   (b) *AI-assisted plateau detection.* Prompt an AI tool to identify the plateau region in the same time series. Provide it with the data and the chapter-specific criterion (the plateau ends when $S(t)/N^{\mathrm{c}}$ drops below 0.95). Compare the AI's identified plateau to your manual identification: are the boundaries the same? Does the AI's recommended bootstrap procedure (simple resample vs block bootstrap) match the autocorrelation structure of the data? Audit the AI's response against the three-step pattern in this chapter's first AI Lab box.

   Report the resulting $\hat{\mathcal{R}}_0$ point estimates and CIs from both workflows, and compute the relative difference. Comment on which discrepancies (if any) reflect substantive disagreement vs noise. Disclosure (per Appendix D, Section D.4): state the AI system, the prompt, the date, and the audit findings.

9. [**AI Lab**] *Convergence-failure diagnostic.* Set up a least-squares fit of an SIR model to the synthetic data from the previous exercise, but *deliberately* initialize the optimizer poorly: $\mathcal{R}_0 = 0.5$ (well below the true value), $t_0 = -50$ days (far from the truth), and bounds set so the optimizer can wander into unphysical regions ($\mathcal{R}_0 > 100$ allowed). Run the fit. The optimizer should fail to converge to a meaningful answer. Now execute the diagnostic workflow:

(a) *Document the failure mode.* Record the optimizer's final state: did it fail to converge (max iterations reached, gradient norm not decreasing), converge to an unphysical value, or converge to a physically plausible but wrong answer? This is the failure-mode classification from the chapter's second AI Lab box.

(b) *Prompt for fixes.* Ask an AI tool: "A Levenberg–Marquardt fit of an SIR model failed in the following way: [describe failure]. What are the most likely causes, in priority order?" Save the response.

(c) *Audit and rank.* Audit the AI's response against the chapter's priority list (initial guess too far from optimum; parameter scaling problems; data outliers). Rank the AI's suggestions: which match the chapter's priority list, which were missed, and which (if any) are spurious for this specific failure?

(d) *Implement the fix.* Implement the highest-ranked AI suggestion and verify it produces a converged, physically plausible fit. Compare the recovered $\mathcal{R}_0$ to the true value used to generate the data.

The exercise practices the audit-fix-verify loop on a real fitting failure; experienced modelers run this loop dozens of times in a typical project.

10. **[AI Lab]** *Residual-structure analysis.* For the converged fit from the previous exercise, plot the residuals (observed j minus predicted j) as a time series. Compute the residual autocorrelation at lags of one, two and four observations (one, two and four weeks). Then:

(a) *Initial interpretation.* Visually inspect the residual plot. Do you see any of the canonical signatures from the chapter's second Check Your Work box: monotone trend, 7-day periodicity, or large residual at the peak? Note your interpretation.

(b) *AI interpretation.* Show the residual plot and autocorrelation values to an AI tool. Ask: "What model mis-specifications are consistent with this residual structure?" Save the response.

(c) *Audit.* For each candidate mis-specification the AI identifies, state whether it is plausible given how the synthetic data were generated (you generated them from the SIR baseline with simple multiplicative noise, so the only structural mis-specifications that should appear are noise-related). The AI should not be inventing structural causes for noise that you generated yourself; if it does, this is a hallucination failure mode worth recording.

The exercise tests both the workflow (AI tools as residual-interpretation collaborators) and a specific failure mode (AI tools confidently inventing structural causes for purely-random residuals when no structure exists).

11. **[AI Lab]** *Cross-validation workflow.* For a real (or synthetic) outbreak time series of at least 20 weekly observation points:

(a) *Hold-out split decision.* A naive approach is to use the first 70% for fitting and the last 30% for validation. For an outbreak time series with autocorrelation, this is the wrong cross-validation design. Ask an AI tool: "What cross-validation design is appropriate for fitting an epidemic model to time-series data?" Save the response.

(b) *Audit the AI's design.* A chapter-aware response will identify rolling-window cross-validation (or expanding-window cross-validation) as the appropriate design, with the

held-out segment immediately following the training segment to preserve causality. A hand-waving response will recommend k-fold cross-validation with random splits, which is wrong for time series because it allows the validation set to be earlier than the training set.

(c) *Implement and compare.* Implement both the AI's recommended design and the chapter-correct rolling-window design. Refit the model under each design and compare the held-out prediction error. Which design gives a more honest assessment of fit quality?

The exercise highlights a specific failure mode of AI tools on time-series workflows: applying generic cross-validation patterns without adjusting for the time-series structure that distinguishes epidemic-modeling fits from typical regression problems.

12. **[Verify the AI]** *Auditing an AI-generated $\mathcal{R}_0$-fitting workflow.* Prompt an AI tool: *"Given weekly incidence data $j(t_k)$ from the first 8 weeks of an outbreak in a U.S. city of 1 million people, write a complete workflow to estimate $\mathcal{R}_0$. Show all derivations, formulas, and code."* Save the response. Audit the AI's workflow against this chapter's three-error structure for fitting:

(a) *Susceptible-population dependence audit.* Does the AI's workflow use the susceptible-viewpoint estimator $\hat{\lambda} = j/s$ (which depends on the assumed N^{c}) or the infected-viewpoint estimator $\hat{\alpha} = j/i$ (which does not)? If $\hat{\lambda}$, ask how far the assumed population reaches: a factor-of-three uncertainty in N^{c} produces a factor-of-three uncertainty in $\hat{\lambda}$, but an incidence fit's $\hat{\mathcal{R}}_0$ depends on the assumed population only through the identifiability ratio of Proposition 7.2, and only once depletion enters the window. The diagnostic question: *"If the city's population were estimated as 300,000 instead of 1,000,000, would your $\hat{\mathcal{R}}_0$ change? By how much?"* A chapter-aware AI will recognize that $\hat{\alpha}$ does not change, that $\hat{\lambda}$ changes by a factor of about 3.3, and that an incidence fit's $\hat{\mathcal{R}}_0$ moves only if the eight weeks reach depletion (Section 7.6).

(b) *Growth-rate-to-$\mathcal{R}_0$ conversion audit.* Does the AI's workflow assume an exponentially-distributed infectious period when converting growth rate σ to $\mathcal{R}_0 = 1 + \sigma\tau_R^*$? For pathogens with strongly pre-peak shedding (influenza, SARS-CoV-2), this conversion overestimates $\mathcal{R}_0$, because the true mean generation time is only about 0.5–0.7 of τ_R^* (the caution paragraph following the early-growth log-linearization above). Identify whether the AI's response acknowledges this assumption and whether the response cites the renewal-equation correction of Chapter 15, Section 15.5. The diagnostic question: *"Is the infectious-period distribution exponential in your model? If not, how should the conversion change?"*

(c) *Identifiability audit.* Does the AI's workflow attempt to fit c_I, β, and $i(0)$ as separately identifiable parameters from early-growth data? The chapter's identifiability discussion (Section 7.3) shows the early-growth data identifies only two scalar combinations (the slope σ and the intercept $\ln(c_I\beta i(0))$), not the three individual parameters. If the AI presents three-parameter point estimates from early-growth data alone, the workflow is misspecified. The diagnostic question: *"How many degrees of freedom does the early-growth data support? Justify by counting independent functionals."*

The exercise practices the audit habit on the three highest-impact failure modes in AI-generated fitting workflows for epidemic models: N^{c}-dependence (the chapter's central distinction), generation-time-distribution assumption (the renewal-equation correction), and identi-

fiability rank (the Fisher-information rank-2 condition). Disclosure: per Appendix D, Section D.4, document the AI system, prompts, and audit findings.

13. [**Open Project**] *Full AI-collaborative fitting project.* A multi-week capstone project fitting an SIR-type model to a real or synthetic outbreak dataset (suggested options include the chapter's seasonal influenza data, COVID-19 weekly incidence, or a synthetic dataset with realistic noise overlay), with full uncertainty quantification and AI-collaborative validation. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (15–25 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch08-fitting-uncertainty/` .

Chapter 8

Practical Issues in Fitting Epidemic Models to Real Data

A precise answer to the wrong likelihood is the most dangerous mistake in fitting.

Before You Read

Before you read: a forecasting attempt.
It is September 4, 2009. You are a campus epidemiologist at Washington State University (WSU). The student health center has reported the following new influenza cases for the first 14 days of the outbreak:

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Cases	11	14	18	22	29	34	41	50	61	74	89	107	128	154

These counts are an idealized series on the scale of the real 2009 WSU outbreak, not its recorded case counts; the data note in the chapter introduction says where the real data and its literature can be found. Section 8.4 gives the diagnostic that establishes this, equation (8.7); it is worth returning here and applying it to the table above once you have read it.

Using only the methods you know from Chapters 1–6, fit a SIR model to these 14 days of data and answer:

(a) What is your best estimate of $\mathcal{R}_0$ for this outbreak?

(b) When do you predict the daily case count will peak?

(c) What is your confidence level in prediction (b)? Express it as a number (e.g., "I am 80% confident the peak falls between days X and Y").

Record your answers before reading further. You will return to them in Section 8.6.

One further question, which most readers will not have thought to ask: what did you minimize? If the answer is a plain sum of squared residuals, note that too. Chapter 7, Section 7.4.2 shows that an unweighted fit to a curve spanning three orders of magnitude concentrates almost all of its information near the peak and returns an interval about half as wide as it should be. The estimate you wrote down may be fine; the uncertainty you attached to it probably is not.

A first diagnosis, before the chapter begins. You have three answers written down and the chapter will not settle them for some time, so it is worth saying now which of the three the data can actually support.

Fit a straight line to ln(cases) against day. The fourteen points lie on it almost exactly, $R^2 \approx 0.997$, with slope

$$\sigma \approx 0.20 \text{ day}^{-1}, \tag{8.1}$$

a doubling time near 3.5 days. That number is solid. It is the one thing in this exercise the fourteen days determine on their own.

Question (a) is where the trouble starts, and not for the reason most readers expect. Converting σ into $\mathcal{R}_0$ requires the relation $\mathcal{R}_0 = 1 + \sigma/(\gamma_R + \nu_m)$, and $\gamma_R = 1/\tau_R$ is not in the data. It comes from outside, from clinical estimates of the infectious period. Take the $\tau_R = 7$ days used for influenza in Chapter 7 and $\sigma = 0.20$ gives $\mathcal{R}_0 \approx 2.4$. Take the book's baseline $\tau_R = 12$ days and the same data give $\mathcal{R}_0 \approx 3.4$. The data did not change. The assumption did. If you reported a single $\mathcal{R}_0$ without recording which τ_R produced it, you reported an assumption and called it a measurement — the same failure the identifiability discussion of Chapter 7, Section 7.3 was written to make visible.

Questions (b) and (c) are not deferred out of tidiness. Peak timing depends on the susceptible denominator and on whether behavior stays constant, neither of which fourteen days of case counts observe; that is Section 8.6. And the confidence you attached to (b) is almost certainly too tight for a reason that has nothing to do with arithmetic: early counts this small are dominated by chance, and a deterministic model absorbs that chance as though it were signal. That is Section 8.7.

So: keep (a) provisionally, with its τ_R written beside it. Hold (b) and (c). The chapter returns to all three.

Chapter 7 established the theoretical foundation for fitting the SIR model: the infected-viewpoint estimator $\hat{\alpha} = j/i$, the staged fitting procedure, and the structural immunity to uncertainty in N^c. This chapter puts those tools to work on real surveillance data, using the 2009 pandemic influenza A(H1N1) outbreak on the Washington State University (WSU) campus as a running example throughout.

Real data are imperfect in ways that synthetic examples cannot prepare you for. Case counts are incomplete. Population sizes are uncertain. Reporting systems close on weekends. Students go home for Thanksgiving and return with new contacts. Seven failure modes appear repeatedly in the epidemic modeling literature, and each one can silently corrupt an otherwise correct fitting procedure (Pitfall: "Correct Data-Fitting Practice"). This chapter presents each failure mode alongside its diagnostic and its fix.

The WSU campus outbreak (a geographically contained epidemic tracked daily from August through December 2009) is ideal for our purposes: the population is approximately closed, the data span a complete epidemic wave, and multiple independent groups have analyzed the same dataset, providing a rich literature for comparison Vaidya et al. [2015], Schwartz et al. [2015a,b].

Data source. The series used throughout this chapter is illustrative: it is generated by the chapter's own code in the SIR regime of the 2009 WSU outbreak, not retrieved from an archive. The real outbreak's data were collected by WSU Health and Wellness Services and are described in the studies cited above. The MIDAS Network data catalog (`catalog.midasnetwork.us`) indexes datasets and software for infectious-disease modeling more broadly. All Python code for this chapter is available in the companion repository at `github.com/machyman/hyman2026avoiding`.

Exercise types. The exercises use the book's tags (Appendix G). Exercises marked **[D]** are done without AI assistance; they build the foundational skills needed to audit AI output reliably. Exercises marked **[Verify the AI]** present an AI-generated derivation or claim; your task is to find

the error, provide the correct argument, and explain why that type of error is plausible. Exercises marked [**L**] use Python and AI collaboration for computation; the deliverable is your interpretation of the result, not the code. Exercises marked [**Open Project**] are open projects where AI may be used freely, but your intellectual contribution (domain judgment, biological interpretation, policy reasoning) must be evident and non-delegatable.

8.1 Accounting for Uncertainty in the Assumed Susceptible Population

The failure mode. A modeler downloads the WSU H1N1 incidence data, assumes the susceptible population is $N^c = 20{,}000$ (the approximate student enrollment), and fits the susceptible-viewpoint model to the daily incidence, so that its force of infection $\hat{\lambda} = J/S \approx J/N^c$ rests on that assumption (Pitfall: "Force of Infection versus Force from Infection"). Her fit, run through the epidemic peak, gives $\hat{\mathcal{R}}_0 = 1.47$.

A second modeler uses $N^c = 10{,}000$, reasoning that many students live off-campus and mix primarily with non-students. The same fit gives him $\hat{\mathcal{R}}_0 = 1.59$.

Both modelers used the same data, the same model, and the same method. The discrepancy traces entirely to the assumed N^c. Published H1N1 analyses report a wide range of values; the assumed susceptible population is one of the choices that differ between them.

The fix: use the infected-viewpoint estimator. Recall from Chapter 7 that the infected-viewpoint estimator

$$\hat{\alpha} = \frac{j(t)}{i(t)\,\Delta t} \tag{8.2}$$

does not involve N^c. The normalized sensitivity index satisfies $\Upsilon^{\hat{\alpha}}_{N^c} = 0$ exactly: no matter how wrong the assumed N^c is, $\hat{\alpha}$ returns the same value. The susceptible-viewpoint estimator $\hat{\lambda} = j/s$ satisfies $\Upsilon^{\hat{\lambda}}_{N^c} = -1$: a factor-$k$ error in N^c produces a factor-k error in $\hat{\lambda}$. It does not produce a factor-k error in the susceptible-viewpoint $\hat{\mathcal{R}}_0$. A fit absorbs most of the error into its other parameters: $\hat{\mathcal{R}}_0$ depends on the assumed population only through the identifiability ratio of Proposition 7.2, and only as far as the fitted window carries depletion.

Figure 8.1 measures this contrast on the chapter's illustrative series. As N^c varies from 10,000 to 40,000 (a plausible range for the effective susceptible population in a rural university community), the infected-viewpoint $\hat{\mathcal{R}}_0^{(\alpha)}$ stays at 1.35, because nothing in it depends on N^c. Fitted to days 0 to 66, before ten percent of the population is infected, the susceptible-viewpoint $\hat{\mathcal{R}}_0^{(\lambda)}$ moves only from 1.44 to 1.35, and the fits barely differ in quality. Fitted through the peak, it moves from 1.59 to 1.29. Even at the true 20,000 it reads 1.47, because the single-phase model misses the rise in transmission after day 70 (Section 8.4). The data also begin to object: at $N^c = 10{,}000$ the model produces only about half the reported cases, and its deviance is nearly seven times that of the fit at 20,000. A wrong N^c leaves early data almost untouched; once depletion enters the window it moves $\hat{\mathcal{R}}_0$ and degrades the fit. The infected-viewpoint estimate needs no N^c at either stage.

Verify the AI

The estimator-force confusion. A common AI error in epidemic fitting is conflating the infected-viewpoint estimator $\hat{\alpha} = j/i$ with the infected-viewpoint force of infection $\alpha(t) = c_I \beta P_S(t)$. These are not the same quantity. The force $\alpha(t)$ depends on $P_S(t)$, which involves s

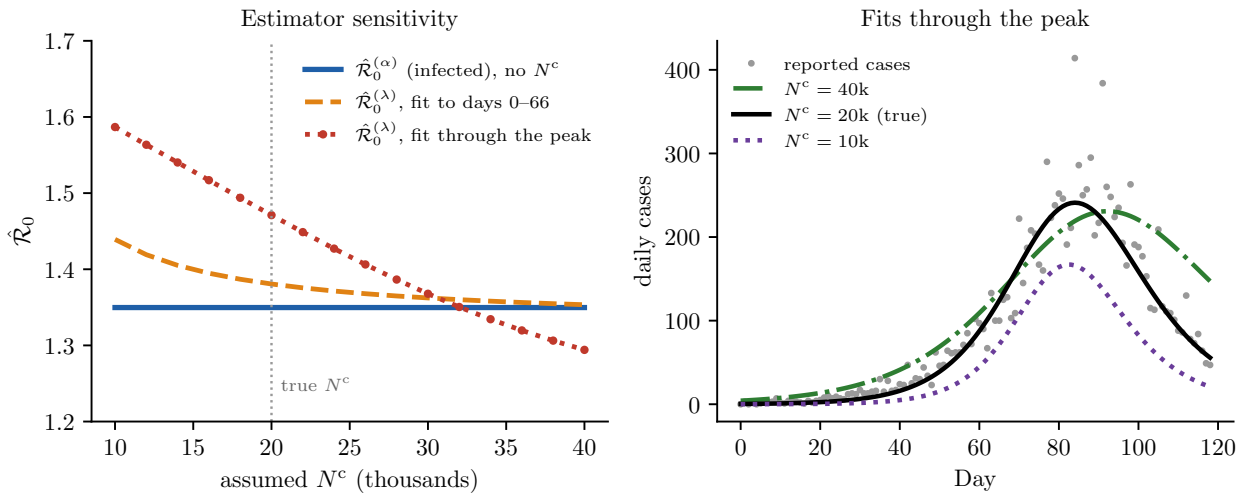


Figure 8.1: Sensitivity of $\hat{\mathcal{R}}_0$ to the assumed susceptible population N^c in the chapter's illustrative series. **Left:** The infected-viewpoint estimate $\hat{\mathcal{R}}_0^{(\alpha)}$ (solid blue) involves no N^c and stays at 1.35. The susceptible-viewpoint estimate $\hat{\mathcal{R}}_0^{(\lambda)}$ from an incidence-only fit moves from 1.44 to 1.35 when the fit sees days 0 to 66 (dashed orange) and from 1.59 to 1.29 when it runs through the peak (dotted red). **Right:** The through-peak fits at $N^c = 10{,}000$, 20,000 and 40,000 against the reported daily cases; the smallest assumed population cannot produce the outbreak. Illustrative data generated in the WSU H1N1 regime, not the recorded WSU counts.

and hence N^c. The estimator $\hat{\alpha} = j/i$ does not involve s at all; N^c cancels because both j and i are observed without knowing N^c.
Ask an AI assistant: *"Does the infected-viewpoint force of infection depend on the susceptible population?"* A correct answer: *yes, the full force $\alpha(t) = c_I \beta P_S$ depends on s through P_S, but the estimator $\hat{\alpha} = j/i$ does not.* An incorrect answer: *no, the infected viewpoint never involves S.*
Structural verification: compute $\partial\alpha(t)/\partial s$ from $\alpha = c_I \beta P_S(t)$. Is it zero or nonzero? Now compute $\partial\hat{\alpha}/\partial N^c$ from $\hat{\alpha} = j/i$. The two computations take under two minutes and settle the question definitively.

Exercise 1. ***[D]***

Given daily incidence $J = 47$ cases, current prevalence $I = 312$ active cases, and $\Delta t = 1$ day, compute $\hat{\alpha}$ from equation (8.2). If the mean infectious period is $\tau_R = 5$ days and demographic turnover is negligible ($\nu_m \approx 0$), compute $\hat{\mathcal{R}}_0 = \hat{\alpha}\,\tau_R$. State the units of each quantity. Is this outbreak growing? What does your answer say about when $\hat{\alpha}\,\tau_R$ estimates $\mathcal{R}_0$?

Exercise 2. ***[Verify the AI]*** *Auditing the estimator-force distinction*

An AI assistant produces the following response to the question *"Does the infected-viewpoint estimator avoid the N^c problem?"*

> *Yes. In the infected-viewpoint formulation, the force of infection is $\alpha = c_I \beta P_S$, where $P_S = s/(s + i + r)$. This expression does not involve N^c because s, i, and r are the observed state variables, not the assumed susceptible population. Therefore, $\hat{\alpha}$ estimated from this formula is immune to uncertainty in N^c.*

Identify the error, provide the correct argument, and explain in 2–3 sentences why this confusion is plausible for an AI system to make on this question. *Verification:* compute $\partial\alpha(t)/\partial s$ and $\partial\hat{\alpha}/\partial N^{c}$. Report both with signs.

Exercise 3. ***[L]***

Using the chapter's illustrative WSU-regime series from the companion repository, compute $\hat{\mathcal{R}}_0^{(\alpha)}$ and $\hat{\mathcal{R}}_0^{(\lambda)}$ for $N^{c} \in \{10{,}000, 12{,}000, \ldots, 40{,}000\}$ (the range of Figure 8.1). Plot both estimates vs. N^{c}. At what value of N^{c} do the two estimates agree numerically? Explain why agreement at that particular N^{c} does not validate either estimate.

8.2 Reporting Confidence Intervals, Not Just Point Estimates

The failure mode. A modeler reports $\hat{\mathcal{R}}_0 = 1.37$ and concludes that an epidemic is inevitable. No uncertainty is reported. A reader has no way to assess whether this conclusion is based on tight or loose constraints from the data.

The fix: the delta-method confidence interval. Recall the sensitivity-index uncertainty formula from Chapter 9:

$$\frac{\mathrm{SE}(\hat{\mathcal{R}}_0)}{\hat{\mathcal{R}}_0} \approx \sqrt{\sum_{p} \left(\Upsilon_p^{\mathcal{R}_0}\right)^2 \mathrm{CV}(\hat{p})^2} \tag{8.3}$$

For the WSU H1N1 fit with $\mathcal{R}_0 = \alpha\,\tau_R^{*}$, using $\Upsilon_{\alpha}^{\mathcal{R}_0} = 1$, $\Upsilon_{\tau_R}^{\mathcal{R}_0} = 1$, $\mathrm{CV}(\hat{\alpha}) \approx 0.048$ (the standard error of the plateau mean: the 19 daily values from days 48 to 66 scatter with a coefficient of variation of 0.21, and the mean's is smaller by $\sqrt{19}$), and $\mathrm{CV}(\hat{\tau}_R) \approx 0.14$ (literature range [4, 6] days De Serres et al. [2010]):

$$\frac{\mathrm{SE}(\hat{\mathcal{R}}_0)}{\hat{\mathcal{R}}_0} \approx \sqrt{1^2 \times 0.048^2 + 1^2 \times 0.14^2} \approx 0.148, \tag{8.4}$$

giving, for the plateau estimate $\hat{\mathcal{R}}_0 = 1.35$ from the chapter's illustrative series, the approximate 95% confidence interval [0.96, 1.74]. Almost all of that width, 89 percent of the variance, comes from τ_R, which the incidence data cannot inform. The tempting shortcut uses the day-to-day spread, 0.21, as if it were the estimator's error. That gives [0.68, 2.02], an interval 1.7 times as wide as it should be.

Two conditions make that label honest, and both should be stated whenever this calculation is reported. First, (8.3) keeps only the diagonal of the parameter covariance matrix: it assumes the input uncertainties are *independent*. The general form in Chapter 9 carries the off-diagonal terms, and correlated inputs—which is the usual case when transmission parameters and ascertainment are estimated together—can widen or narrow the interval substantially. Second, the interval inherits whatever the input coefficients of variation actually represent. Here $\mathrm{CV}(\hat{\alpha})$ is a sampling quantity, estimated from the observed variability of the plateau, while $\mathrm{CV}(\hat{\tau}_R)$ encodes a literature range rather than a sampling distribution; the result is a confidence interval only to the extent that those inputs behave like standard errors. When the inputs are plausible ranges chosen by judgement rather than estimated from data, the output is an uncertainty interval and should be called one—it quantifies the consequences of stated assumptions, not the sampling variability of an estimator. The cross-check below is the practical guard: a delta-method interval that disagrees materially with a bootstrap is reporting the assumptions, not the data.

Remark 8.1. The interval straddles $\mathcal{R}_0 = 1$: its lower limit is 0.96, and the bootstrap below puts 4% of its resamples under one. That is exactly the information a decision-maker needs. The point estimate says "outbreak inevitable"; the interval says the threshold is probably crossed, but not by a margin the data can guarantee, because the margin rests on the infectious period taken from the literature.

Check Your Work

Cross-check: delta method vs. bootstrap. *Verifies: numerical values.* Before accepting the delta-method interval, verify it numerically. Using the plateau region of the chapter's illustrative WSU-regime series (companion repository): (1) resample the plateau-region $\hat{\alpha}(t_k)$ values with replacement, 1,000 times; (2) for each resample draw τ_R from its literature distribution (mean 5 days, CV = 0.14) and compute $\hat{\mathcal{R}}_0 = \bar{\hat{\alpha}} \cdot \tau_R$; (3) take the 2.5th and 97.5th percentiles. **Expected result:** [0.96, 1.74], within 0.01 of the delta-method interval at each end. Resampling $\hat{\alpha}$ alone, with τ_R fixed at 5 days, gives [1.21, 1.46]: a bootstrap sees only the uncertainty it resamples. A larger discrepancy means the linear approximation behind the formula is failing; prefer the bootstrap in that case.

Bootstrap comparison. Figure 8.2 shows the bootstrap distribution of $\hat{\mathcal{R}}_0$ from 1,000 resamples of the plateau values of the chapter's illustrative series, with τ_R drawn independently, and the delta-method interval overlaid. The two agree to within 0.01 at each end, confirming that the linear approximation is adequate here. They cannot test independence, because the bootstrap imposes it by drawing τ_R separately. Independence holds because $\hat{\alpha}$ comes from the data and $\hat{\tau}_R$ from the literature, by separate procedures. The outline histogram resamples $\hat{\alpha}$ alone: the data pin $\hat{\alpha}$ down, and the width belongs almost entirely to τ_R.

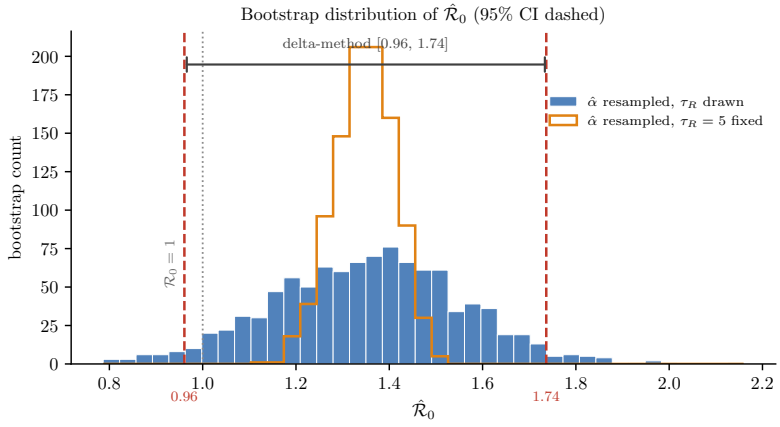


Figure 8.2: Bootstrap distribution of $\hat{\mathcal{R}}_0$ from 1,000 resamples of the plateau-region $\hat{\alpha}(t_k)$ values (days 48 to 66) of the chapter's illustrative WSU-regime series. Filled: τ_R drawn independently (mean 5 days, CV = 0.14); outline: $\tau_R = 5$ days fixed. Vertical dashed lines mark the bootstrap 95% interval [0.96, 1.74] of the filled histogram; the bracket marks the delta-method interval [0.96, 1.74]. Both put the lower limit just below $\mathcal{R}_0 = 1$.

Verify the AI

The independence assumption in the delta method. The formula $\mathrm{SE}(\hat{\mathcal{R}}_0)/\hat{\mathcal{R}}_0 \approx \sqrt{\sum_p (\Upsilon_p^{\mathcal{R}_0})^2 \,\mathrm{CV}(\hat{p})^2}$ assumes the parameters are estimated *independently*. A common AI error:

applying this formula to a joint MLE fit of all parameters without noting that the covariance term $\mathrm{Cov}(\hat{\alpha}, \hat{\tau}_R)$ is omitted. When parameters are estimated jointly, the omitted covariance can cause the formula to understate $\mathrm{SE}(\hat{\mathcal{R}}_0)$.
The formula is valid, to first order, when parameters are estimated by separate procedures (as in the staged fitting procedure of Chapter 7). For joint MLE, replace the sum-of-squares formula with the full delta-method version involving the parameter-covariance matrix.
Verification: when using equation (8.3), confirm that each parameter in the sum was estimated by a separate procedure. If parameters were estimated jointly, compare the delta-method interval to the bootstrap; a significant discrepancy signals omitted covariance.

Exercise 1. ***[D]***

Using equation (8.3) with sensitivity indices from Chapter 9, Section 9.3, compute $\mathrm{SE}(\hat{\mathcal{R}}_0)$ under two scenarios: (a) $\mathrm{CV}(\hat{\alpha}) = 0.10$ (tight plateau); (b) $\mathrm{CV}(\hat{\alpha}) = 0.30$ (noisy plateau). Which parameter (α or τ_R) dominates the uncertainty in each case?

Exercise 2. ***[Verify the AI]*** *Auditing the delta-method application*

An AI assistant applies equation (8.3) to a joint MLE fit of all four SIR parameters and concludes:

> *Using the formula with* $\Upsilon_{\alpha}^{\mathcal{R}_0} = 1$, $\mathrm{CV}(\hat{\alpha}) = 0.18$, $\Upsilon_{\tau_R}^{\mathcal{R}_0} = 0.97$, *and* $\mathrm{CV}(\hat{\tau}_R) = 0.14$, *I obtain the* 95% *CI* [0.78, 1.96]. *This CI is valid regardless of how the parameters were estimated.*

Identify the error in the final sentence. State the condition under which equation (8.3) is exact, and describe the correction needed when that condition fails. Provide a one-sentence rule a practitioner can apply before using the formula.

Exercise 3. ***[D]***

Show that under the consistent conversion of Equation (7.15) the elasticity $\Upsilon_{N^{\mathrm{c}}}^{\hat{\mathcal{R}}_0^{(\lambda)}}$ is zero, so the delta-method term $(\Upsilon_{N^{\mathrm{c}}}^{\mathcal{R}_0^{(\lambda)}})^2\,\mathrm{CV}(N^{\mathrm{c}})^2$ vanishes, as the infected-viewpoint term does. Then estimate the elasticity for an incidence-only fit from the figure in Section 8.1, for each window, and explain why the extra term grows with the depletion the window contains.

8.3 Diagnosing and Resolving Practical Identifiability Failures

The failure mode. A modeler fits all four parameters (α, τ_R, τ_m, i_0) simultaneously to the WSU early-outbreak data. The optimizer reports convergence, but different starting points give i_0 values spanning three orders of magnitude at essentially the same likelihood. The data cannot distinguish between these initial conditions.

This is *practical non-identifiability* (Pitfall: "Identifiability"): the likelihood function is flat along the i_0 direction, and no amount of data from the observed window can resolve it.

The profile-likelihood diagnostic. The diagnostic fixes i_0 at each value on a grid and minimizes the fitting objective over the remaining parameters. In deviance form $D = -2\ell$, the profile deviance for parameter i_0 is

$$D_{\mathrm{prof}}(i_0) = \min_{\alpha,\, \tau_R,\, \tau_m} D(\alpha,\, \tau_R,\, \tau_m,\, i_0), \tag{8.5}$$

plotted relative to its minimum, so that the χ^2_1 threshold of 3.84 delimits the approximate 95% confidence region. A well-identified parameter produces a parabolic profile with a clear minimum; a non-identifiable parameter produces a flat or weakly curved profile. For i_0 in the WSU early-phase data, $D_{\text{prof}}(i_0)$ is approximately flat over three orders of magnitude (Figure 8.3). The choice of which observable to fit also affects identifiability quantitatively: Saucedo et al. [2024] compare practical identifiability across incidence, prevalence, and cumulative-incidence data for an SEIR model and show that incidence yields the most identifiable parameter estimates while cumulative incidence yields the least. This corroborates the chapter's running practice of fitting incidence rather than cumulative case counts and supplies recent empirical justification for the convention.

The generation-time sensitivity problem. The data fix the growth rate, not $\mathcal{R}_0$. The chapter's series grows at about 0.076 per day; read through an exponential generation interval, $\mathcal{R}_0 = 1 + \sigma T$, that is 1.15 for $T = 1.91$ days Fraser et al. [2009], 1.27 for $T = 3.6$ days Hsieh et al. [2010], and 1.38 for the model's own $\tau_R = 5$ days. The data cannot distinguish these biologically plausible values; the generation time must come from elsewhere.

The fix: penalized fitting with literature-based bounds. Minimize the penalized loss function

$$\mathcal{L}_{\text{pen}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{data}}(\boldsymbol{\theta}) + \sum_j \frac{\lambda_j}{\sigma_j^2}\left(p_j - \bar{p}_j\right)^2 \tag{8.6}$$

where $\bar{p}_j$ is the literature-based central estimate and σ_j its uncertainty.

Remark 8.2 (Connection to Bayesian inference). Each penalty term $\lambda_j(p_j - \bar{p}_j)^2/\sigma_j^2$ is the negative log of a Gaussian prior with mean $\bar{p}_j$ and standard deviation $\sigma_j/\sqrt{\lambda_j}$. The penalized estimator equals the maximum a posteriori (MAP) estimate under this prior. We do not need the full machinery of Bayesian computation to use this idea; the penalty is simply a regularizer that keeps the optimizer in biologically plausible regions.

Table 8.1: Literature-based parameter bounds for the penalized SIR fit to the 2009 H1N1 outbreak at Washington State University.

Parameter	Symbol	Central estimate	Range	Source
Infectious period (days)	τ_R	5	[4, 6]	De Serres et al. [2010]
Generation time (days)	T	1.91	[1.30, 2.71]	Fraser et al. [2009]
Demographic time (days)	τ_m	4×365	$[2, 6] \times 365$	university context
Initial fraction infectious	i_0	$1/N^{\text{c}}$	$[0.5, 5]/N^{\text{c}}$	plausible index cases (in i compartment at $t = 0$)

In the SIR fit the mean generation time equals τ_R^*; the T row applies only to the renewal-equation conversion of Chapter 15 and is not a second prior.

Verify the AI

Reporting convergence without checking identifiability. A standard optimizer reports "convergence" when it finds a local minimum. This does not mean the minimum is unique. A common AI error: reporting fitted parameters from a single optimizer run without verifying that the objective has a clear minimum in every parameter direction.

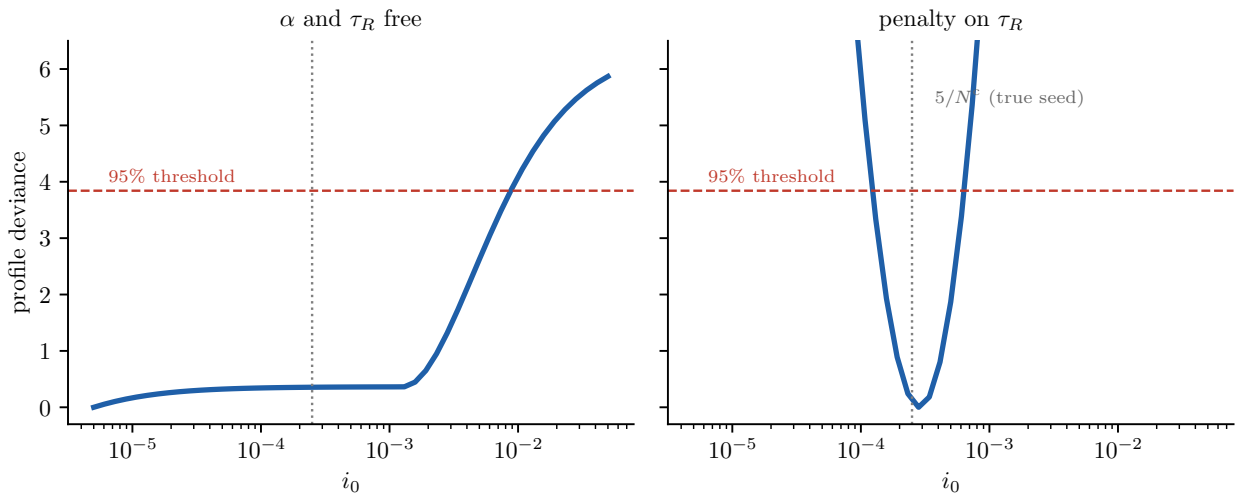


Figure 8.3: Profile deviance $D_{\text{prof}}(i_0)$ for the initial infected fraction, computed on the first 21 days of the chapter's illustrative WSU-regime series. **Left:** α and τ_R free: flat over three orders of magnitude, because τ_R trades against the seed. **Right:** with the literature penalty on τ_R (Table 8.1): a minimum consistent with about 2.5 to 13 index cases.

Verification protocol: run the optimizer 20 times with randomly selected starting points log-uniformly distributed over the parameter's plausible range. If the 20 fitted values cluster tightly, away from every parameter bound, the minimum is well-defined; a cluster on a bound is a boundary solution, not an estimate. If they scatter across orders of magnitude, the parameter is practically non-identifiable and requires a penalty (or prior) to resolve. Never report a fitted parameter without this multi-start check.

Exercise 1. ***[L]***

Implement the penalized loss function (8.6) in Python using `scipy.optimize.minimize`. Fit the WSU H1N1 data with the bounds from Table 8.1. Report the penalized $\hat{\mathcal{R}}_0$ and compare with the unpenalized estimate.

Exercise 2. ***[Verify the AI]*** *Auditing a convergence report*

An AI assistant returns:

> *I fitted the SIR model using L-BFGS-B. The optimizer converged. Fitted parameters:* $\hat{\alpha} = 0.271$ *day*$^{-1}$, $\hat{\tau}_R = 4.8$ *days,* $\hat{F}_0 = 8.3 \times 10^{-5}$, *giving* $\hat{\mathcal{R}}_0 = 1.30$.

(a) Identify the missing diagnostic step. Write the Python code (2–4 lines) that performs it for i_0.
(b) Run the diagnostic using the WSU data. Is i_0 well-identified from the first 21 days of data?
(c) Explain in 2 sentences why the optimizer's convergence report does not guarantee that i_0 is uniquely determined.

Exercise 3. ***[L]***

Vary the penalty strength for τ_R from $\lambda = 0$ to $\lambda = 100$. Plot $\hat{\mathcal{R}}_0$ vs. λ. At what λ does the estimate stabilize? Interpret: what does a large λ assume about the reliability of the literature estimate relative to the data?

AI Learning Prompt

Finding literature-based parameter bounds. An AI assistant can accelerate the literature search for penalty bounds.
Prompt pattern:

> *I am fitting an SIR model to 2009 H1N1 influenza data. I need a literature-based estimate for [parameter name] with a plausible range. Provide: (1) the central estimate with units, (2) the range reported across studies, (3) the two or three most-cited sources for this estimate. If you are uncertain, say so; do not invent citations.*

How to verify: Look up the cited sources and confirm the reported values match. For H1N1, compare the AI's estimates to Table 8.1. If they differ substantially, the AI may be citing a different H1N1 strain (seasonal vs. pandemic) or a different study population. Always verify the citation exists and reports the stated value before using it as a penalty bound.

8.4 Residual Diagnostics: Detecting Model Misspecification

The failure mode. A modeler produces a fitted trajectory that looks visually consistent with the observed data and stops there. The residuals $\varepsilon_t = J_{\text{obs}}(t) - j(t;\,\hat{\boldsymbol{\theta}})\,N^{\text{c}}$ are never examined.

Why residuals matter. Under the observation model $J_{\text{obs}}(t_k) \sim \text{Poisson}(\alpha i(t_k)\,N^{\text{c}}\Delta t)$, standardized residuals should be approximately independently distributed with mean zero and variance one. Each residual decomposes as observation noise plus the model's approximation error, $\varepsilon_t = (\text{noise}) + \big(J_{\text{true}}(t) - j(t;\hat{\boldsymbol{\theta}})\,N^{\text{c}}\big)$, so a good fit leaves only the noise term and the diagnostic question becomes whether the residuals look like the noise the loss function assumed. Any systematic departure signals either an observation artifact or a missing model mechanism, what statisticians call *model discrepancy*.

Three complementary tests. No single statistic captures every way residuals can depart from noise, so it is standard to apply three tests that probe different aspects; each can pass while another fails.

- *Sign.* Under the null, each residual is positive with probability one half, so the number of positive residuals is Binomial$(n, \frac{1}{2})$; a binomial test flags gross asymmetry, such as a model that sits systematically below the data. The Wald–Wolfowitz runs test complements it by counting sign runs, catching the serial dependence the binomial test misses—too few runs indicate positive autocorrelation.

- *Correlation.* The autocorrelation function and the Ljung–Box test probe pairwise dependence between residuals at each lag; these are the diagnostics used for Patterns A and B below.

- *Frequency.* The periodogram probes the whole correlation structure at once. For white noise it is flat in expectation, so the *normalized cumulative periodogram* tracks the diagonal and a residual cycle appears as a jump at its frequency. Bartlett's Kolmogorov–Smirnov test rejects whiteness when the maximum deviation from the diagonal exceeds $1.36/\sqrt{\lfloor n/2 \rfloor}$ [Bartlett, 1955]. Applied to the day-of-week artifact of Section 7.2.2, it registers the excess power at $1/7$ day^{-1} directly.

Before the diagnostics themselves, a connection worth making explicit. Every test in this section is applied to *standardized* residuals—each residual divided by the standard deviation the observation model assigns to its observation. That division is the same one that defines the inverse-variance weights of Chapter 7, Section 7.4.2, and it is not a coincidence: a residual diagnostic asks whether the noise looks like what the loss function assumed, which presupposes that the loss function assumed something. A fit performed with unweighted least squares has, in effect, assumed homoscedastic noise, so its standardized residuals will fail these tests on any epidemic curve—not because the model is misspecified but because the loss function was. Checking the weighting before reading the residual plots saves diagnosing the wrong problem (Pitfall: "Weighting Residuals by Their Uncertainty").

Check the variance model first: the scale factor. There is one diagnostic that should be read before any of the others, because it tests the assumption the others rest on. After a weighted fit, compute

$$\hat{\phi} \;=\; \frac{1}{K-p}\sum_{k=1}^{K} w_k\big[y_k - \mu_k(\hat{\theta})\big]^2, \tag{8.7}$$

the weighted residual sum of squares per degree of freedom, for K observations and p fitted parameters. If the assumed variance model is correct, each weighted squared residual has expectation one, and $\hat{\phi} \approx 1$.

Departures in either direction are informative, and both are common.

If $\hat{\phi}$ is much larger than one, the data are more variable than the weights assumed. Fitting the baseline outbreak of Chapter 7 with Poisson weights when the counts are in fact negative-binomial illustrates how loud this signal is: at $\kappa = 100$ the scale factor comes back near 13, at $\kappa = 20$ near 62, and at $\kappa = 5$ near 220. These are not marginal readings requiring a formal test. The approximation behind them is $\mathbb{E}[\hat{\phi}] \approx 1 + \bar{\mu}/\kappa$, where $\bar{\mu}$ is the mean fitted count, which inverts to a usable estimate of the dispersion, $\hat{\kappa} \approx \bar{\mu}/(\hat{\phi} - 1)$. Refitting with negative-binomial weights, or falling back on constant-coefficient-of-variation weights, then returns the interval to something near its nominal coverage (Chapter 7, Table 7.3).

If $\hat{\phi}$ is much smaller than one, the data are *smoother* than counting noise permits. This is the less familiar failure and the more troubling one, because nothing looks wrong: the fit appears excellent and the residuals appear small. A series of case counts cannot be smoother than its own Poisson variability unless something has removed that variability, and the usual causes are that the series has been smoothed or interpolated before the modeler received it, that it has been reconstructed from a model rather than observed, or that it is a nowcast still being revised. A scale factor of, say, 0.05 on a daily case series is not a sign of a good fit but a sign that the series is not raw counts, and the provenance should be established before anything is inferred from it.

The reason to run this check before the residual tests below is that a misspecified variance model makes standardized residuals fail for reasons that have nothing to do with the mean model. A modeler who reaches for the autocorrelation function first will find structure, conclude the epidemic model is wrong, and start adding compartments to fix a problem that lives in the loss function (Pitfall: "Weighting Residuals by Their Uncertainty").

Two failure patterns in the WSU H1N1 data:

Pattern A: Weekly reporting cycle. The student health center operates Monday through Friday. Case counts drop on weekends and spike on Mondays as the backlog is cleared. The SIR model produces a smooth trajectory; the residuals show a 7-day sawtooth. The autocorrelation function (ACF) has a strong peak at lag 7. *Fix:* use weekly-aggregated case counts and verify the weekly residuals pass the Ljung-Box test.

Pattern B: A change in transmission. Weekly aggregation removes the cycle but not everything. The weekly residuals of the single-phase fit still fail the Ljung–Box test ($p = 0.02$ at four lags), and a runs test on the daily residuals rejects randomness ($p = 0.02$). The single-phase fit has absorbed a rise in transmission at day 70, about two weeks before the incidence peak near day 86, by raising $\hat{\alpha}$ for the whole epidemic, which leaves its weekly residuals negative around the change (weeks 10 and 11) and positive elsewhere. *Interpretation:* a 15 percent rise in transmission at day 70. *Fix:* fit a two-phase model with a behavioral change point at day 70. Its weekly residuals pass ($p = 0.63$), and its two rates, 0.274 and 0.315 per day, recover the 15 percent rise built into the series.

Table 8.2 summarizes the residual diagnostic workflow.

Table 8.2: Residual diagnostic workflow for epidemic model fitting. The *Response* column names which of the three responses discussed below each fix belongs to—*noise* for changing the noise model, *extend* for extending the model, *restrict* for restricting the data—so an observed pattern leads to the class of remedy as well as to the particular one.

Pattern	Diagnostic	Diagnosis	Fix	Response
Random scatter	Ljung-Box $p > 0.05$	Observation noise	None needed	—
7-day cycle	ACF peak lag 7	Reporting artifact	Aggregate to weekly	noise
Trend after a change	Runs test $p < 0.05$	Missing mechanism	Two-phase model	extend
Early underfit	Stage 1 data	Stochastic boundary	Use Stage 2 only (Section 8.7)	restrict

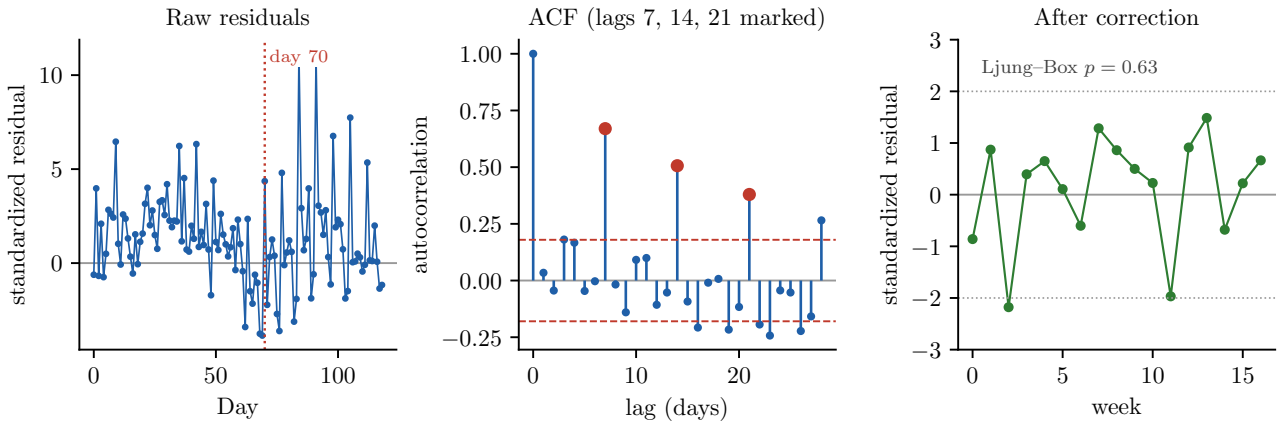


Figure 8.4: Residual analysis for the single-phase SIR fit to the chapter's illustrative WSU-regime series. **Left:** Standardized daily residuals; the 7-day reporting cycle dominates (scale factor $\hat{\phi} = 8.4$). **Center:** Autocorrelation function, with peaks at lags 7, 14 and 21 (0.67, 0.51, 0.38; band ±0.18). **Right:** Weekly residuals after weekly aggregation and a two-phase model with a change point at day 70, consistent with white noise (Ljung–Box $p = 0.63$ at four lags).

When residuals are correlated: three responses. A failed residual test means the model has not captured all the signal, and there are three standard responses, differing in where they locate the misfit.

- *Change the noise model.* Keep the trajectory model but fit it with a loss that accounts for the correlation—generalized least squares with a correlated-error covariance, or a coarser time resolution. Aggregating the WSU counts to weekly (Pattern A) is the simplest instance: it removes the day-of-week correlation by changing the resolution at which residuals are defined.

- *Extend the model.* Add the missing mechanism so that the structure now in the residuals moves into the trajectory. The two-phase model with a change point at day 70 (Pattern B) is this response: the rise in transmission that appeared as a residual trend becomes an explicit model feature. Whether the added complexity is warranted—rather than fitting noise—is a model-selection question: penalized criteria such as the Akaike and Bayesian information criteria (AIC and BIC) weigh the improved fit against the extra parameters, favoring the richer model only when the improvement outpaces the penalty for its added degrees of freedom.

- *Model the discrepancy.* When the missing structure is real but its mechanism is unknown, add an explicit discrepancy term—a correlated error process or a Gaussian-process residual—that absorbs the structure and widens the parameter uncertainty to reflect it, rather than letting the misfit bias the estimates. This is the honest option when neither a better noise model nor a known mechanism is available.

The responses run from cheapest (change the noise model) to most flexible (model the discrepancy); the choice turns on whether the misfit lives in the noise, in a known mechanism, or in structure one can model but not yet explain.

What a failed test establishes, and what a passed one does not. The three responses above all begin from a failed test, and that asymmetry is not accidental. A failed independence test is strong evidence that the fitted structure is incomplete. A passed test is much weaker evidence that it is complete, and the reason is worth stating, because it is the point at which residual analysis for a mechanistic model parts company with the version taught for linear regression.

In both settings the residual carries only the part of the misfit that the fitted model could not imitate. For a linear model that part is well defined: the residual is orthogonal to a fixed linear span. A compartmental model has no such span. Its parameters trace a curved family of trajectories that bends to follow the data, so it absorbs far more of a misspecification than a design matrix can, and what survives into the residuals is correspondingly less. The same flexibility that makes the remedy structural rather than statistical also makes the diagnostic weaker.

What survives is not the largest part of a misspecification but the part that *varies* across the fitting window. A discrepancy that is nearly constant over the window is absorbed exactly, by rescaling a transmission parameter; only variation the parameters cannot reproduce is left behind. The practical form of the question is therefore not whether a suspected mechanism matters, but whether the window spans a substantial change in the quantity that mechanism acts through.

The SIR contact structure makes this concrete. Writing $c_S = c_R = c$ and $c_I = \xi\, c$, the contact-weighted population size of Chapter 2 gives a force of infection proportional to $1/\big(1-(1-\xi)i\big)$, so assuming equal contact rates distorts the fit through prevalence. Fitting across the rising limb, where prevalence changes by an order of magnitude, leaves structure a runs or Ljung–Box test can find. Fitting a flat window near the peak leaves almost none, even though the distortion is largest there, because a nearly constant factor is what the fitted transmission parameter absorbs best. A

misspecification that varied with calendar time instead, an intervention or a behavioral change, would reverse the two windows.

The same ratio governs an analytical question in Chapter 9, Section 9.3.6, where the equal-contact approximation is shown to give sensitivity indices accurate to a fraction of a percent at the book's baseline, $c_I/c_S = 0.2$, because those indices are evaluated at the endemic equilibrium, where prevalence is of order 10^{-3} and the correction $(1-\xi)i$ is negligible. The analytical and the empirical statements are governed by the same product and differ only in where the prevalence is taken: at equilibrium it is vanishing, while across the rising limb fitted here it reaches percent levels, which is why the same contact ratio that is invisible there is detectable in the residuals below. It is also a reminder that they are not the same statement: the first is a property of the parameters and the operating point, the second depends on how much data one has and how noisy it is.

The same reasoning in other settings. The contact-rate example is convenient because the misspecification enters through a single factor, but the reasoning transfers. Unmodeled *heterogeneity* in susceptibility or contact rates depletes the high-risk part of the population first, so its effect grows with cumulative incidence and is read best across a window over which that quantity moves. *Behavioral change* and *intervention* act through calendar time rather than through prevalence, so they are visible in a window containing the change and invisible in one that sits entirely on either side of it, whatever the prevalence there. A shift in *reporting*, whether a definition change or a ramp in ascertainment, differs from the other three in kind rather than in timing: it rescales the predicted counts without altering the trajectory that produced them. Because the fitted family can reshape the trajectory but cannot rescale one part of a curve relative to another, a reporting shift leaves a signature the other three do not, and in the one case we have examined closely a mistaken ascertainment ramp was rejected decisively while a rival *structural* explanation was not. How far that separation extends is not something a single example settles. In every case the question to ask before reading the residual plot is the same: over this window, how much does the suspected mechanism's effect actually change, and could the fitted parameters have imitated it?

The scale factor also sets the power of everything below it. Whether any of this structure is visible depends on the noise it competes with, and here the scale factor (8.7) does a second job. Under Poisson counting noise the relative noise on an observation falls as $1/\sqrt{\mu}$, so a larger outbreak will eventually expose any fixed relative bias. Under overdispersion it falls instead toward $1/\sqrt{\kappa}$ and then stops: a floor that no amount of additional data lowers, because the noise grows with the mean as fast as the signal does. Since $\hat{\phi}$ already estimates κ through $\hat{\kappa} \approx \bar{\mu}/(\hat{\phi}-1)$, reading it tells you not only whether the variance model was right but how much the tests below it can be expected to see.

Refitting with the correct negative-binomial weights does not recover what is lost. Correct weighting restores calibration, and with it the nominal coverage of Table 7.3; it cannot manufacture information the data do not contain. On strongly overdispersed counts a residual independence test can be properly calibrated and still have essentially no power against a structural error. The consequence for practice is uncomfortable but simple: on such data a clean residual diagnostic is close to no evidence at all, and a modeler who treats it as confirmation has learned nothing (Pitfall: "Reading Residuals in a Mechanistic Model").

Table 8.3 puts numbers on this. A synthetic outbreak is generated at the book's baseline contact ratio, $c_I/c_S = 0.2$, fitted over the rising limb with the equal-contact model, and the three tests of this section are applied at the five percent level; the table reports how often at least one of them rejects. Two columns are needed to read it. *Size* is the rejection rate when the model is in fact correctly specified, so it is the floor against which power must be judged; because three tests are

applied together it sits nearer 0.15 than 0.05: if the three were independent the union would reject a correct model $1-(0.95)^3 \approx 0.14$ of the time. *Relative noise* is the standard deviation of an observation as a fraction of its mean, which under a Poisson model falls as $1/\sqrt{\mu}$ and under a negative binomial approaches $1/\sqrt{\kappa}$.

Setting	Relative noise	Size	Power
(a) Effect of dispersion, at $N^c = 10^6$ (peak count $\approx 1.6 \times 10^4$)			
Poisson	0.008	0.16	**1.00**
$\kappa = 100$	0.100	0.25	0.20
$\kappa = 30$	0.183	0.19	0.18
$\kappa = 10$	0.316	0.19	0.12
$\kappa = 3$	0.577	0.20	0.13
(b) Effect of more data, at $\kappa = 30$			
$N^c = 10^5$ (peak $\approx 1.6 \times 10^3$)	0.184	0.16	0.17
$N^c = 10^6$ (peak $\approx 1.6 \times 10^4$)	0.183	0.19	0.18
$N^c = 10^7$ (peak $\approx 1.6 \times 10^5$)	0.183	0.14	0.24
(c) Effect of correcting the weights, at $N^c = 10^6$			
$\kappa = 30$, Poisson MLE	0.183	0.19	0.19
$\kappa = 30$, NB MLE	0.183	0.13	0.09
$\kappa = 10$, Poisson MLE	0.316	0.19	0.12
$\kappa = 10$, NB MLE	0.316	0.11	0.08

Table 8.3: How often the residual independence tests of this section detect an equal-contact misspecification. Data are generated from the SIR model at the baseline contact ratio $c_I/c_S = 0.2$ and fitted, over a forty-day window on the rising limb, with the equal-contact model; rejection means at least one of the sign, Ljung–Box, and Bartlett tests rejects at the five percent level. *Size* is the same quantity computed when the fitted model is correct. Because three tests are applied together the size sits near 0.15 rather than 0.05, and power must be read against it: a power that merely equals the size is no detection at all. Panel (a): under Poisson noise the misspecification is found every time, and under every dispersion tested the power falls to the size. Panel (b): raising the population a hundredfold does not recover it, because the relative noise has reached the floor $1/\sqrt{\kappa}$ and stops falling. Panel (c): fitting overdispersed counts with Poisson weights inflates the size to 0.19; refitting with the correct negative-binomial weights removes that inflation and returns it to the level a correctly specified fit shows. The power falls with it, so the corrected test is properly calibrated and still unable to distinguish the misspecified fit from the correct one. All entries are Monte Carlo estimates over 100 replicates and carry a standard error near 0.04; the gap between 1.00 and the rest is far larger than that, while differences among the dispersed rows are not.

When the residuals cannot settle the question, the discriminating evidence has to come from elsewhere: prediction on a held-out window, an independent data stream such as hospitalizations or serology, or a designed contrast in which the candidate mechanisms make different predictions. Model selection among mechanisms is a different exercise from detecting that some mechanism is missing, and the residuals are far better at the second than the first.

Verify the AI

Accepting visual fit without formal residual diagnostics. A common AI error: declaring that a model "fits well" based on a plot where the trajectory passes through the data, without computing or examining residuals. Visual fit is necessary but not sufficient.

Ask an AI assistant to evaluate your model fit and check whether its assessment is based on a residual diagnostic (ACF, Ljung-Box, runs test) or only on the appearance of the fitted trajectory. An AI that says "the model fits well, as shown by the close agreement between the fitted curve and the data" without mentioning residuals has not completed the diagnostic.
Verification: after any model fit, plot and examine the residuals first, even if the trajectory looks good. Random scatter (observation noise) and a systematic pattern (model discrepancy) look alike at the trajectory level and differ at the residual level.

Exercise 1. ***[L]***

Compute and plot the residuals from the single-phase SIR fit to the WSU data. Run the Ljung-Box test for white noise at lags 1–14. Is there evidence of a 7-day cycle? Quantify the strength of the cycle by reporting the spectral power at frequency $1/7$ day^{-1}.

Exercise 2. ***[L]***

Generate an 80-day synthetic epidemic with Poisson observation noise, fit the single-phase model by *unweighted* least squares, and compute residuals standardized under the homoscedastic assumption that the unweighted loss implies. Report the Ljung-Box statistic and the residual-versus-fitted funnel. Now recheck the variance model with the scale-factor statistic (8.7), refit with inverse-variance weights, and rerun both diagnostics. Explain, in the language of this section, why the first fit's residual structure was a loss-function artifact rather than model misspecification, and why the scale-factor check therefore precedes the autocorrelation function in the workflow (Pitfall: "Weighting Residuals by Their Uncertainty").

Exercise 3. ***[Open Project]***

Two-phase model with behavioral change point. Implement a two-phase SIR model with a contact-rate change point at day 70 (Thanksgiving break return). Estimate α_1 (pre-Thanksgiving) and α_2 (post-Thanksgiving) jointly using penalized fitting. Compare the residual ACF of the single-phase and two-phase models.

Creative-transformation component: Find 2–3 papers from the 2009 H1N1 literature that document post-holiday epidemic resurgence in university or school settings. Use those findings to justify your choice of change point at day 70 rather than another day, and to assess whether $\alpha_2 > \alpha_1$ is consistent with what those papers report. An AI assistant may help you locate papers; you must evaluate whether their settings are comparable to WSU and whether the behavioral mechanism matches.

AI Learning Prompt

Socratic questioning about residual patterns. Once you have diagnosed a residual pattern, an AI assistant can help you deepen your understanding of the mechanism, but only after you have made your own diagnosis first.
Prompt pattern (after your own diagnosis):

> *I have fitted an SIR model to weekly H1N1 incidence data. The residuals show [describe the pattern you found]. I believe this is caused by [your diagnosis]. Instead of telling me whether I am right, ask me a series of questions that will help me verify or refute my diagnosis. Start with the most basic question and work toward the mechanistic explanation.*

What to verify: The AI's questions should become progressively more specific as your answers improve. If the AI reveals the answer in its second question, the Socratic constraint was not strong enough: refine the prompt. The goal is to arrive at the mechanistic explanation through your own reasoning.

8.5 Correcting for Reporting Delays and Under-ascertainment

The failure mode. A modeler fits $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$ as if every true case were reported.

The scale of the problem. For the 2009 H1N1 pandemic nationally, only 43,677 laboratory-confirmed cases were reported through July 2009. Correcting for under-ascertainment, CDC estimated 1.8 to 5.7 million symptomatic cases, a median multiplier of 79 Reed et al. [2009]. For the WSU campus, seroprevalence suggests a final attack rate of approximately 44% Furushima et al. [2017] while campus health records captured approximately 10%, implying $\hat{\rho} \approx 0.23$.

The observation model.

$$J_{\text{obs}}(t) \sim \text{Poisson}\big(\rho \cdot \alpha \cdot i(t) \cdot N^{c} \cdot \Delta t\big) \tag{8.8}$$

where $\rho \in (0, 1]$ is the ascertainment fraction.

The infected-viewpoint advantage is preserved, when ρ is constant. When ρ is constant in time, it cancels from $\hat{\alpha} = J_{\text{obs}}/(I_{\text{obs}}\,\Delta t)$: the estimator recovers the true α regardless of how small ρ is (the structural reason—that $\hat{\alpha}$ never invokes the susceptible denominator, only the observed prevalence—is developed in Section 7.2.6). When $\rho(t)$ changes (improved testing, changed case definition), the ratio $J_{\text{obs}}/I_{\text{obs}}$ absorbs the change as an apparent change in transmission (Pitfall: "Differential Ascertainment in a Ratio Estimator").

Reporting delay. A mean reporting delay d causes $\hat{\mathcal{R}}_0$ to be underestimated (Pitfall: "Data Preprocessing"), because the reported incidence reflects transmission that occurred d days earlier while the prevalence denominator is current. During exponential growth at rate σ, the ratio estimator is biased downward by the factor $e^{-\sigma d}$, so the relative underestimate is $1-e^{-\sigma d}$. For the chapter's series ($\sigma \approx 0.076$ day^{-1}), a 3-day delay gives $1 - e^{-\sigma d} \approx 20\%$. With the influenza generation time $T = 1.91$ days and $\mathcal{R}_0 = 1.37$ the growth rate would be $(\mathcal{R}_0 - 1)/T \approx 0.19$ and the bias 44%, where the linearization $\sigma d \approx 58\%$ overstates it; the linearization should not be used beyond $\sigma d \ll 1$; note in particular that the delay enters through the growth rate, not as the ratio d/T alone.

Verify the AI

The constant-ascertainment condition. A common AI error: stating that "the infected-viewpoint estimator is robust to under-ascertainment" without noting that this holds only when ρ is constant.

Ask an AI: *"Is $\hat{\alpha} = J/I$ unaffected by under-ascertainment?"* A correct answer specifies the constant-ρ condition. An incorrect answer omits it.

Structural verification: Surveillance builds prevalence from past reports, $I_{\text{obs}}(t) = \sum_{s \le t} \rho(s)\, J_{\text{true}}(s)\, e^{-\gamma_R (t-s)}$, so $\hat{\alpha}_t = \rho(t) J_{\text{true}}(t)/[I_{\text{obs}}(t)\,\Delta t]$. Does ρ cancel when it changes? Only if incidence and prevalence are ascertained by the same factor at the same time; after a step in ρ, $\hat{\alpha}$ moves with it for about two infectious periods.

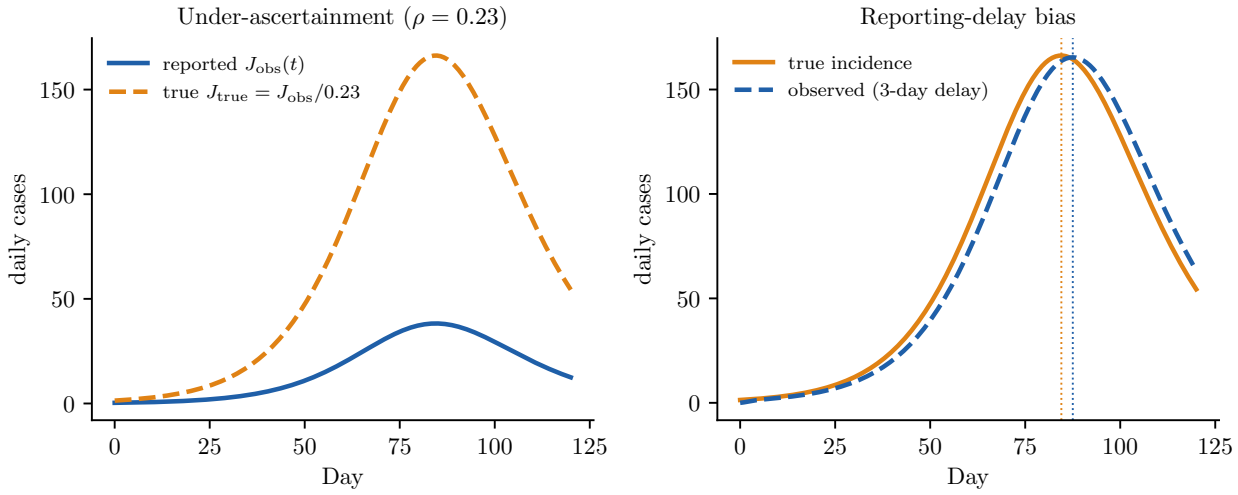


Figure 8.5: Under-ascertainment and reporting delay in the chapter's illustrative WSU-regime series. **Left:** Reported incidence $J_{\text{obs}}(t)$ (blue) and estimated true incidence $J_{\text{true}}(t) = J_{\text{obs}}(t)/0.23$ (orange), illustrating the roughly 4-fold gap on campus. **Right:** Effect of a 3-day reporting delay: the observed peak is shifted about three days later; dotted lines mark the two peaks.

Exercise 1. ***[D]***

If $\rho = 0.23$ is constant and the reporting delay is $d = 3$ days, compute the corrected $\hat{\mathcal{R}}_0$ for the WSU data. Show algebraically that the infected-viewpoint structural immunity to N^{c} is preserved under this correction.

Exercise 2. ***[Verify the AI]*** *Auditing a robustness claim*

An AI assistant states:

> *The infected-viewpoint estimator $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$ is robust to under-ascertainment. Even if the true case count is 79 times larger than reported, the ratio recovers the correct α because the ascertainment fraction cancels from numerator and denominator.*

(a) Identify the hidden assumption. State it as a mathematical condition on $\rho(t)$.
(b) Generate a synthetic time series where $\rho(t)$ doubles at day 20. Compute $\hat{\alpha}(t)$ before and after. Does the AI's claim hold?
(c) Describe in 2–3 sentences how the residual plot would reveal this violation.

Exercise 3. ***[L]***

Simulate a time-varying $\rho(t)$ that doubles in week 5 (improved campus testing). Generate synthetic $J_{\text{obs}}(t)$ and fit the SIR model. Plot the residuals. Does the time-varying ρ produce detectable structure? Describe the signature and how you would distinguish it from a true change in transmission.

8.6 Distinguishing Model Uncertainty from Real-World Uncertainty

Return to the productive-failure opener. At the start of this chapter you estimated $\mathcal{R}_0$, the peak timing, and your confidence level from 14 days of WSU H1N1 data. Retrieve your answers now.

The failure this section addresses is subtle: you may have expressed correct *parameter* uncertainty (a range of $\mathcal{R}_0$ values given the model) while implicitly treating it as *forecast* uncertainty (a range of future outcomes given reality). These are not the same thing.

Scope of Uncertainty Principle

A confidence interval computed from fitting a deterministic model to past data quantifies *parameter uncertainty*: uncertainty about α, τ_R, and related quantities, assuming the model's structural assumptions remain valid.
It does not quantify *model-scope uncertainty*: uncertainty about whether contact rates, behavioral patterns, and mixing structure will remain stable through the forecast horizon.
Parameter uncertainty is reducible by collecting more data from the current outbreak. Model-scope uncertainty is not reducible by any amount of outbreak data, because it concerns future conditions that the current data cannot observe.
When presenting a fitted trajectory, state explicitly which kind of uncertainty is being quantified and which is not.

Illustration with WSU data. Three scenarios fitted to the Stage 2 plateau, days 48 to 66, of the chapter's illustrative WSU-regime series (Table 8.4).

Table 8.4: Three scenarios illustrating the Scope of Uncertainty Principle. All are calibrated to the same Stage 2 plateau, days 48 to 66, of the chapter's illustrative WSU-regime series.

Scenario	Assumption	Predicted final size
Baseline	Constant α, plateau $\hat{\mathcal{R}}_0 = 1.35$	9,385 students
Behavioral	α drops 30% at day 70	3,812 students
95% parameter CI	Constant α, τ_R fixed; $\hat{\mathcal{R}}_0 \in [1.21, 1.46]$	[6,602, 11,182] students

The parameter CI does not contain the behavioral scenario outcome. The interval correctly captures uncertainty within the model's assumptions; it cannot capture uncertainty about whether those assumptions hold.

The correct forecast statement is:

> *Under our model's assumptions (constant contact rates and homogeneous mixing), the fitted SIR model predicts a final size of 9,385 students (95% parameter CI, τ_R fixed: 6,602–11,182). If contact rates change by more than 20% due to campus closure, behavioral response, or reseeding from returning students, actual outcomes may fall substantially outside this interval.*

Verify the AI

Presenting a confidence band as a prediction interval. A widespread AI error: presenting a 95% confidence band from a fitted deterministic model as a "95% prediction interval for the

epidemic's future course" without noting the model-scope limitation. This error appeared in numerous automated COVID-19 forecasting reports in 2020–2021.
When reviewing a modeling report that presents uncertainty bands, ask: does the report state whether these bands capture only parameter uncertainty, or also model-scope uncertainty? A band that captures only parameter uncertainty should be labeled explicitly; a band that claims to capture forecast uncertainty requires a different method (scenario ensemble, ensemble Kalman filter, or Bayesian model averaging).
Test: ask an AI to generate a "prediction interval" for the WSU epidemic peak using the fitted SIR model. Does its response note the model-scope limitation? If not, the AI has committed the error this section describes.

Exercise 1. ***[D]***

Compute the 95% parameter CI for the WSU H1N1 trajectory using residual bootstrap (1,000 resamples of the plateau-region $\hat{\alpha}$). Write a two-sentence figure caption that correctly communicates what the CI does and does not represent, following Box 8.6.

Exercise 2. ***[Verify the AI]*** *Auditing a prediction interval claim*

An AI assistant produces the following paragraph for a campus health briefing:

> *We fitted the SIR model to four weeks of WSU H1N1 data. The model predicts a peak prevalence of 890 students per day. The 95% prediction interval, obtained by propagating parameter uncertainty through the fitted model, is [750, 1050] students per day. Campus health services should plan for a peak in the upper range of this interval.*

(a) Identify the error in the phrase "95% prediction interval."
(b) Rewrite the paragraph to communicate the uncertainty honestly, in no more than the original word count, accessible to a campus administrator.
(c) Name one event that would cause the actual peak to fall outside [750, 1050] even if the parameter estimates are exactly correct.

Exercise 3. ***[Open Project]*** *Scenario ensemble with behavioral grounding*

Design and implement a scenario-ensemble approach for the WSU H1N1 outbreak that quantifies both parameter uncertainty and model-scope uncertainty.

Part A (behavioral literature): Using the H1N1 pandemic behavioral literature, identify and justify three plausible post-week-4 contact-rate scenarios (no-change, moderate reduction, strong reduction). For each, state the assumed percentage change in α and cite empirical evidence. *Your scenario bounds must be grounded in cited findings, not chosen for convenience.*

Part B (implementation): Run the SIR model under each scenario. Add the 95% parameter CI band around each trajectory. Produce a single figure showing: (a) the observed data, (b) the three scenario trajectories with CI bands, and (c) a horizontal line at campus health center capacity (200 cases/day).

Part C (communication): Write a 150-word briefing paragraph for campus administration that (a) distinguishes parameter from model-scope uncertainty, (b) states which scenario is most consistent with observed behavioral patterns in comparable campus outbreaks, and (c) makes a specific resource-planning recommendation. Apply Box 8.6 explicitly.

Discriminating test: a student who understood nothing about epidemic dynamics could not assess which scenario bounds are empirically grounded, could not evaluate comparability of cited

campus outbreaks, and could not write a briefing that correctly distinguishes the two uncertainty types.

8.7 Recognizing When Stochastic Effects Dominate

The failure mode. A modeler fits a deterministic SIR model to the first two weeks of WSU data ($i < 50$ cumulative cases) and reports $\hat{\mathcal{R}}_0 = 2.1$ with tight confidence intervals. The estimate is unreliable: early-outbreak data are dominated by demographic stochasticity (Pitfall: "The Stochastic-to-Deterministic Regime Transition"), and the deterministic model absorbs random early-growth fluctuations as if they were signal.

The three-stage framework. Geneus et al. [2025] identify three stages at which the relationship between deterministic models and stochastic reality changes qualitatively.

Stage 1: Stochastic early stage (cumulative incidence below 3–6%). Individual contact patterns dominate. Trajectory is unpredictable per realization. What IS predictable: the probability distribution of arrival times at each infection level. **Rule:** do not fit a deterministic *trajectory* to Stage 1 data alone. The distinction matters, because it is not the same as ignoring early data. What is unreliable in Stage 1 is the realized path: a single stochastic trajectory need not track the mean-field solution, so matching an ODE curve to it fits noise. The exponential *growth rate*, by contrast, is estimable from the same window and is precisely what identifies $\mathcal{R}_0$ when susceptible depletion has not yet begun—the regime Chapter 7, Section 7.6.4 shows to be insensitive to the assumed population. Estimate the rate from Stage 1 if the data allow; do not ask a deterministic trajectory to reproduce a stochastic one.

Stage 2: Mean-field middle stage (cumulative incidence from 3–6% to about 75%). Stochastic trajectories synchronize. Deterministic ODE models are accurate. **Rule:** fit $\hat{\alpha} = j/i$ to Stage 2 data.

Stage 3: Stochastic late stage (cumulative incidence above about 75%). Individual random contacts dominate as susceptibles become scarce. **Rule:** report a final-size distribution, not a point prediction.

For WSU ($N^{\mathrm{c}} \approx 20{,}000$), Stage 2 begins at approximately 600 cumulative cases, reached around day 48 in the chapter's illustrative series. Fitting $\hat{\alpha}$ from days 1–15 (before Stage 2) gives unstable daily estimates, anywhere from 0 to 1 day^{-1}. Fitting from days 48–66 (from Stage 2 onset to 10 percent infected) gives the plateau mean $\hat{\alpha} \approx 0.270$ day^{-1}. End the window by the time 10 percent have been infected (2,000 cumulative cases here, day 66): $\hat{\alpha} = \alpha s(t)$, so depletion biases it low by the fraction already infected. A rolling slope of $\hat{\alpha}(t_k)$ that turns down shows the same bend, but only on smooth data; on this series the bend is a third of the slope's standard error and cannot be seen (the last exercise of Section 8.8).

The early extinction probability. When i is small, the probability that the epidemic goes extinct before becoming established is approximately

$$P(\text{extinction}) \approx \frac{1}{\mathcal{R}_0} \tag{8.9}$$

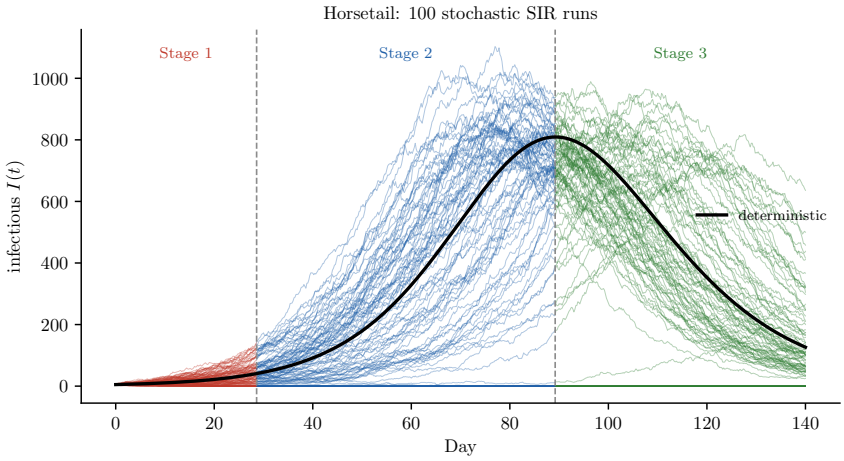


Figure 8.6: Horsetail plot of 100 stochastic SIR simulations with WSU H1N1 parameters ($N^c = 20{,}000$, $\hat{\mathcal{R}}_0 = 1.37$, $\tau_R = 5$ days). Stage 1 (red) shows high variability in early growth; Stage 2 (blue) shows synchronized trajectories; Stage 3 (green) shows divergence as susceptibles become scarce. The deterministic trajectory (black) runs through the Stage 2 ensemble. Twenty-two of 100 simulations went extinct before the epidemic became established, eighteen of them in Stage 1, consistent with $(1/\mathcal{R}_0)^5 \approx 0.21$ for the five initial infectives. Thresholds from Geneus et al. [2025].

for a single introduction, and $(1/\mathcal{R}_0)^k$ for k independent introductions. For the WSU outbreak with $\mathcal{R}_0 \approx 1.37$: $P(\text{extinction}) \approx 1/1.37 \approx 0.73$. The deterministic model says "epidemic inevitable"; the stochastic model says "epidemic occurs 27% of the time."

Verify the AI

Deterministic certainty vs. stochastic probability. A widespread AI error: when $\mathcal{R}_0 > 1$, stating that "an epidemic will occur" without qualification. This is the deterministic model's answer. The stochastic answer is that an epidemic occurs with probability approximately $1 - 1/\mathcal{R}_0$. For $\mathcal{R}_0 = 1.37$, that probability is 27%, not 100%.
Ask an AI: *"If $\mathcal{R}_0 = 1.5$ for a respiratory infection introduced into a fully susceptible campus of 20,000 students, will an epidemic occur?"* A correct answer: *"The deterministic model predicts an epidemic; the stochastic model gives a probability of approximately 1/1.5 ≈ 67% that the outbreak will go extinct before becoming established."*
Structural verification: equation (8.9) gives the formula. Verify numerically using the horsetail simulation code in the companion repository: count the fraction of simulations that went extinct in Stage 1. Does it match $(1/\mathcal{R}_0)^5$, for the simulation's five initial infectives?

Exercise 1. ***[D]***

With $N^c = 20{,}000$ and a Stage 2 threshold of 3% cumulative incidence: at what cumulative case count does Stage 2 begin? From the chapter's illustrative WSU-regime series in the companion repository, on approximately what day does Stage 2 begin?

Exercise 2. ***[D]***

Use equation (8.9) to compute the early extinction probability for $\mathcal{R}_0 \in \{1.1, 1.37, 2.0, 5.0\}$. For which value is the deterministic statement "an epidemic is certain" most misleading? At what value of $\mathcal{R}_0$ is the stochastic correction negligible?

Exercise 3. ***[L]***

Fit $\hat{\alpha}$ using only Stage 2 data (days 48–66) and compare with the fit using all data (days 1–80). Compute bootstrap SE($\hat{\mathcal{R}}_0$) in each case. Which window gives a more stable estimate? Explain why Stage 2 data produce narrower confidence intervals.

AI Learning Prompt

Verifying the early-extinction probability formula.
First derive equation (8.9) yourself. With an exponentially distributed infectious period, as in the chapter's simulation, the number of secondary cases is geometric with mean $\mathcal{R}_0$ and generating function $G(z) = 1/(1 + \mathcal{R}_0(1 - z))$; the extinction probability is the smallest non-negative root of $z = G(z)$.
Then, using the AI:

> *My derivation of $P(\text{extinction}) = 1/\mathcal{R}_0$ from the branching-process approximation is [describe your attempt]. Check it step by step. If a step is wrong, say which, and ask me a question that leads me to the correct step without giving it.*

Numerical verification: Run the horsetail simulation with $\mathcal{R}_0 = 1.37$ and count Stage 1 extinctions. Compare with $(1/1.37)^5 \approx 0.21$ (the simulation starts with five infectives). If they differ by more than 5 percentage points, check the simulation's $\mathcal{R}_0$ formula (ν_m in the denominator) and its Stage 1 threshold.

8.8 The Complete Fitting Workflow

The seven pitfalls arise in sequence as an epidemic unfolds. The following algorithm integrates all seven fixes into a single reproducible procedure.

Algorithm 1 Infected-Viewpoint Fitting Workflow for Real Surveillance Data

1. **Acquire data.** Download incidence $J_{\text{obs}}(t)$ and (if available) prevalence $I_{\text{obs}}(t)$ from MIDAS or the national surveillance system. Record the data source, reporting frequency, case definition, and known gaps.

2. **Apply reporting correction.** Estimate $\hat{\rho}$ from seroprevalence surveys or test-positivity rates. Apply the delay correction $J_{\text{est}}(t) = J_{\text{obs}}(t + d)$.

3. **Identify Stage 2.** Compute cumulative incidence and find $t^*_{1\to 2}$ (the day the 3–6% threshold is crossed). Restrict fitting to Stage 2 data only. End the window by the time 10% have been infected: $\hat{\alpha} = \alpha s(t)$, so depletion biases it low by the fraction already infected.

4. **Compute $\hat{\alpha}$ from the plateau.** Identify the plateau region where $\hat{\alpha}(t_k)$ is approximately constant. Compute $\hat{\alpha}$ as the mean of the plateau values.

5. **Apply penalized fitting for remaining parameters.** Minimize $\mathcal{L}_{\text{pen}}(\boldsymbol{\theta})$ from equation (8.6) with bounds from Table 8.1.

6. **Compute $\hat{\mathcal{R}}_0$ and its CI.** Compute $\hat{\mathcal{R}}_0 = \hat{\alpha}\,\tau^*_R$. Apply equation (8.3) for $\text{SE}(\hat{\mathcal{R}}_0)$. Cross-check with a residual bootstrap.

7. **Diagnose the residuals.** Compute $\varepsilon_t = J_{\text{est}}(t) - j(t; \hat{\boldsymbol{\theta}})\, N^{\text{c}}$. Plot vs. time; compute the ACF; apply the Ljung-Box test. If systematic patterns are found, identify and model the underlying mechanism.

8. **State the scope of uncertainty explicitly.** Distinguish parameter uncertainty (Step 6 CI) from model-scope uncertainty (contact-rate change, behavioral response). Report both separately in any public communication.

Check Your Work

Workflow verification on synthetic data. *Verifies: numerical values and model assumptions.* Before applying Algorithm 1 to a new dataset, verify that it recovers the true parameters from synthetic data: (1) choose a "true" parameter set: $\alpha_{\text{true}} = 0.28$ day^{-1}, $\tau_{R,\text{true}} = 5$ days, $\rho_{\text{true}} = 0.25$, $N^{\text{c}} = 20{,}000$; (2) simulate $J_{\text{obs}}(t) \sim \text{Poisson}(\rho_{\text{true}} \cdot \alpha_{\text{true}} \cdot i(t)\, N^{\text{c}}\, \Delta t)$ for 80 days; (3) apply the workflow and check that $\hat{\mathcal{R}}_0$ falls within the 95% CI of the true $\mathcal{R}_0 = 1.40$.

The figure code in the companion repository carries out the workflow's computations for the chapter's illustrative WSU-regime series and produces all figures in this chapter.

Exercise 1. ***[D]***

Walk Steps 4 and 6 of Algorithm 1 by hand for the synthetic-recovery setting of the verification box: the plateau mean gives $\hat{\alpha} = 0.276$ day^{-1} with $\text{SE}(\hat{\alpha}) = 0.012$ day^{-1}, and the infectious period is taken as $\tau^*_R = 5$ days with $\text{SE}(\tau_R) = 0.4$ days, independent of $\hat{\alpha}$. Compute $\hat{\mathcal{R}}_0 = \hat{\alpha}\,\tau^*_R$, apply equation (8.3) to obtain $\text{SE}(\hat{\mathcal{R}}_0)$, and state the 95% confidence interval. Does the true $\mathcal{R}_0 = 1.40$ of the synthetic generator fall inside?

Exercise 2. ***[D]***

Repeat the synthetic-recovery run of the verification box with Step 2 of Algorithm 1 deliberately skipped, so the constant ascertainment $\rho_{\text{true}} = 0.25$ is never corrected. Working from the observation

model (8.8), derive which quantities in Steps 4–6 are biased and which are exactly immune. In particular, show that the plateau estimator $\hat{\alpha} = j/(i\,\Delta t)$ is immune precisely when incidence and prevalence are under-ascertained by the same factor, and state what happens to $\hat{\mathcal{R}}_0$ when prevalence is instead reconstructed from corrected incidence. Connect your conclusion to Pitfall: "Estimator Honesty".

Exercise 3. ***[Open Project]***

Step 3 of Algorithm 1 restricts fitting to Stage 2 via the 3–6% cumulative-incidence threshold. Formalize the bias of the plateau mean $\hat{\alpha}$ when the fitting window erroneously includes Stage 1 days (stochastic regime) or Stage 3 days (susceptible depletion): determine the sign of the bias in each case and justify it from the stage definitions. Propose a data-driven contamination diagnostic based on the rolling slope of $\hat{\alpha}(t_k)$ across the window, then test both the bias signs and your diagnostic on the synthetic generator with the window boundaries shifted ±10 days.

Seven Practical Fitting Pitfalls at a Glance

Table 8.5: Summary of the seven practical fitting pitfalls developed in this chapter, their diagnostic signals, and their fixes. These are the chapter's local enumeration of failure modes that arise when applying the book's fitting framework to real data; see the chapter opener for how these connect to the book-wide Common Pitfalls catalog.

§	Pitfall	Diagnostic signal	Fix
8.1	Ignoring N^{c} uncertainty	$\hat{\mathcal{R}}_0$ changes with assumed N^{c}	Use $\hat{\alpha} = j/i$
8.2	Point estimates only	No CI reported	Delta method or bootstrap
8.3	Identifiability failure	Optimizer wanders; flat profile likelihood	Penalized fitting
8.4	No residual analysis	Model looks good visually	ACF, Ljung-Box, runs test
8.5	Ascertainment ignored	Seroprevalence ≫ clinical reports	Multiplier model; ρ prior
8.6	Model CI ≠ forecast CI	CI labeled as prediction interval	Scope of Uncertainty Principle
8.7	Deterministic in stochastic stages	Fitting early or late-stage data	Stage thresholds from Geneus et al. [2025]

Pitfall check

This chapter has carried Pitfall: "Correct Data-Fitting Practice" from synthetic data to surveillance data. It estimated the force from infection, not the force of infection, so that $\hat{\mathcal{R}}_0$ does not move with an uncertain N^{c} (Pitfall: "Force of Infection versus Force from Infection"; Section 8.1). It met Pitfall: "Identifiability" as a flat likelihood and answered it with penalized fitting (Section 8.3). It checked the weights before reading the residuals, then read their structure for mechanism (Pitfall: "Weighting Residuals by Their Uncertainty", Pitfall: "Reading Residuals in a Mechanistic Model";

Section 8.4). It corrected for ascertainment and reporting delay (Pitfall: "Differential Ascertainment in a Ratio Estimator", Pitfall: "Data Preprocessing"; Section 8.5), and it fitted only Stage 2 data (Pitfall: "The Stochastic-to-Deterministic Regime Transition"; Section 8.7). Six rows of the table above are these entries met in practice, with Pitfall: "Estimator Honesty" behind "Point estimates only." The remaining row, a model interval read as a forecast interval, is Pitfall: "Constant-Transmission Assumption Ignoring Behavioral Feedback" seen from the data side: the interval holds only while transmission stays constant (Section 8.6).

Chapter Summary

Chapter Summary

- The infected-viewpoint estimator $\hat{\alpha} = j/i$ is structurally immune to uncertainty in N^{c}; the susceptible-viewpoint estimator is not (Section 8.1).
- A point estimate of $\mathcal{R}_0$ without a CI is incomplete. The delta-method formula quantifies how parameter uncertainty propagates to $\hat{\mathcal{R}}_0$ (Section 8.2).
- When parameters cannot be identified from a single incidence curve, penalized fitting with literature-based bounds resolves the non-identifiability without requiring full Bayesian computation (Section 8.3).
- Residual analysis distinguishes observation noise from model discrepancy. A weekly pattern signals a reporting artifact; a post-peak trend signals a missing behavioral mechanism (Section 8.4).
- Under-ascertainment cancels in $\hat{\alpha} = j/i$ when ρ is constant, but time-varying ascertainment produces detectable residual structure (Section 8.5).
- A CI from a fitted model quantifies parameter uncertainty, not model-scope uncertainty. These require different methods and communicate different things (Section 8.6).
- Deterministic fitting is unreliable in Stages 1 and 3. Restrict $\hat{\alpha}$ estimation to Stage 2 data, identified by a cumulative incidence of about 3–6% (Section 8.7).

Chapter 9 turns from fitting to analysis: given the estimated parameters and their uncertainties, which parameters most strongly drive $\hat{\mathcal{R}}_0$ and i^*? The sensitivity indices introduced there complete the link from the confidence interval in equation (8.3) to the intervention-ranking visualization of the tornado plot, the analytical foundation of what this chapter applied numerically.

Chapter 9

Sensitivity Analysis and Uncertainty Quantification

Sensitivity tells you which lever moves the machine. It does not tell you which lever you can reach.

This chapter develops the tools of sensitivity analysis for the SIR model, and uses them to make rigorous the invasion–burden partition introduced informally in the Preface and derived algebraically in Chapter 5. The central question is: to which model parameters are the outputs (the basic reproductive number $\mathcal{R}_0$, the endemic prevalence i^*, the epidemic peak height, the peak time) most sensitive, and which parameters can the modeler afford to leave poorly characterized?

The sensitivity-analysis framework developed in this chapter follows the philosophy laid out in the companion volume [Hyman and Arriola, 2026]: sensitivity analysis is a mathematical discipline organized around three unifying ideas. First, the sensitivity index is a normalized derivative, dimensionless and unit-free, that measures the percentage response of an output to a percentage change in a parameter. Second, the adjoint method (when sensitivities of one output with respect to many parameters are needed) is the transpose of the forward problem, instantiated as a matrix transpose for algebraic systems, as a computation-graph reversal for finite-state computations, and as a backward ODE for dynamical systems. Third, local sensitivity analysis is a linearization, with all the limitations that linearizations imply: it is valid in a neighborhood of the chosen baseline and fails near bifurcation points or for highly nonlinear models. The present chapter develops the first idea quantitatively for the SIR model, applies it to the invasion–burden partition, and indicates where the third idea limits the validity of the resulting indices. The adjoint method (the second idea) is left to the companion volume, where it is developed end-to-end with the SIR model as a running example.

Two complementary approaches appear throughout the chapter. *Local sensitivity analysis* computes partial derivatives of the model output with respect to each parameter at a chosen baseline. It is fast to evaluate, gives closed-form expressions for the SIR model, and produces rankings of parameter importance at the baseline. *Global sensitivity analysis* samples the parameter space broadly (for example, by Latin hypercube sampling) and measures how much of the variation in the model output is explained by variation in each parameter. It is slower, requires a large number of simulations, but gives rankings that are not tied to a single baseline and can capture nonlinear interactions among parameters.

Three related questions travel together in this part of the book, and it is worth separating them at the outset. *Sensitivity analysis* asks the forward question of response: how much does an output

change when an input parameter changes? *Uncertainty quantification* asks the forward question of propagation: given the uncertainty in the inputs, how uncertain is the output? *Identifiability* asks the inverse question: can the inputs be recovered from observations of the output? The three share machinery but answer different questions, and each can fail independently of the others. This chapter develops the first two: the indices of Sections 9.2 and 9.3 answer the response question, and the delta method of Section 9.3.5 converts those indices into output uncertainty, answering the propagation question. The inverse question belongs to Chapters 7 and 8, where the profile-deviance diagnostic detects its failure in practice; it re-enters this chapter in Section 9.4.2, because identifiability is a property of the quantity of interest as much as of the model.

Section 9.1 motivates the analysis with examples drawn from the SIR model. Section 9.2 develops the local sensitivity-index framework. Section 9.3 derives closed-form sensitivity indices for $\mathcal{R}_0$ and i^* in the SIR model, which is one of the book's distinctive technical contributions. Section 9.4 addresses what is sometimes called the most common mistake in sensitivity analysis (the assumption that all parameters can be varied independently) and develops the diagnostic and total-derivative responses. Section 9.5 introduces tornado plots as the canonical visualization of local sensitivity results. Section 9.6 develops the Latin hypercube sampling method paired with partial rank correlation coefficients (PRCC) for global sensitivity analysis. Section 9.7 gives the quantitative version of the invasion–burden partition, drawing on both local and global analysis. Section 9.8 works through a case study. Section 9.9 closes with a summary and exercises.

Central result of this chapter.

> *The infected-viewpoint estimator $\hat{\alpha} = j/i$ has* exactly zero *normalized sensitivity to the assumed susceptible population: $\Upsilon_s^{\hat{\alpha}} = 0$. The susceptible-viewpoint estimator $\hat{\lambda} = j/s$ has unit normalized sensitivity: $\Upsilon_s^{\hat{\lambda}} = -1$. Both indices are exact and follow from the algebraic forms alone. They are the sensitivity-theoretic half of the fitting-advantage argument made in Chapter 7; the other half is an identifiability result rather than a sensitivity one, and Section 9.3.4 is careful to keep the two apart. Taken together they are the chapter's central distinctive contribution: a local index can show that one estimator ignores a quantity the other inherits, but only an identifiability argument can say what a fit to a given body of data is able to determine at all.*

The remainder of the chapter develops the local and global sensitivity machinery, derives closed-form sensitivity indices for $\mathcal{R}_0$ and i^* that quantitatively confirm the invasion-burden partition introduced in the Preface and Chapter 5, and applies the framework to a worked case study.

9.1 Motivation: Why Sensitivity Analysis Matters

Epidemic models have parameters that are known with widely varying degrees of precision. The recovery rate γ_R is constrained by clinical data on the duration of symptomatic infection and is typically known to within ten or twenty percent. The transmission factor β is constrained only indirectly, through fitting against outbreak data, and is typically known to within a factor of two. The contact rate c_I of infectious individuals is constrained by contact-tracing studies of specific diseases and is typically known to within a factor of two or three. The effective susceptible population size N^{c} is a modeling choice that can vary by factors of five or ten depending on how the population is defined.

A natural question arises: given these disparate uncertainties, which of the parameters most strongly affect the model's predictions? If $\mathcal{R}_0$ is much more sensitive to c_I (uncertain by a factor

of 3) than to γ_R (uncertain by a factor of 1.2), then the modeler's uncertainty budget is dominated by c_I, and effort to refine the estimate of c_I is well-invested. If instead $\mathcal{R}_0$ is equally sensitive to c_I and to τ_m (the migration time, essentially known for demographic reasons), then effort to refine τ_m yields diminishing returns.

Sensitivity analysis answers this question quantitatively. It assigns each parameter a number (or a small number of numbers) measuring how strongly the model output responds to variation in that parameter. The numbers can then be used to:

- Rank the parameters by importance, identifying those to which the model is most and least sensitive.
- Guide the choice of parameters to estimate from data (Chapter 7): parameters with high sensitivity must be pinned down; parameters with low sensitivity can be left at default values with little cost.
- Design intervention policies: if $\mathcal{R}_0$ is much more sensitive to c_I than to β, then policies that reduce c_I (case isolation) are preferable to policies that reduce β (behavioral risk reduction), all else equal.
- Propagate parameter uncertainty to output uncertainty: given a distribution over the parameters, what is the resulting distribution over $\mathcal{R}_0$ and i^*?

Two common confusions. Before developing the methodology, two common confusions about sensitivity analysis are worth clarifying.

First, a parameter with a high sensitivity index is not necessarily the parameter most in need of intervention. Sensitivity measures how the output changes with the parameter; whether the parameter can be changed by an intervention is a separate question. A parameter like ν_m (migration rate) might have a high sensitivity for i^* but be essentially immutable from a policy perspective.

Second, the sensitivity of $\mathcal{R}_0$ to a parameter is not the same as the sensitivity of the endemic prevalence i^* to that parameter. A parameter may strongly affect invasion (through $\mathcal{R}_0$) and weakly affect burden (through i^*), or vice versa. The Preface (Figure 2) and Chapter 5, Section 5.7 noted this distinction informally and algebraically; Section 9.7 of this chapter makes it fully quantitative.

Roadmap for the rest of the chapter. Local sensitivity analysis (Sections 9.2 through 9.5) gives the most tractable set of tools, and the SIR model is unusually well-suited to local analysis because closed-form expressions for $\mathcal{R}_0$ and i^* yield closed-form sensitivity indices. Global sensitivity analysis (Section 9.6) complements the local picture and reveals nonlinear behavior that local analysis misses. The invasion–burden discussion (Section 9.7) uses both to demonstrate the structural partition of parameters into two groups.

A note on sensitivity-index notation. The chapter uses a compact index notation for sensitivities that the reader should keep in mind. The *normalized sensitivity index* of an output Y to a parameter X is written

$$\Upsilon_X^Y = \frac{X}{Y}\frac{\partial Y}{\partial X},$$

read as "the sensitivity of Y with respect to X." The superscript names the output, and the subscript names the parameter. When $|\Upsilon_X^Y| = 1$, a one percent change in X produces a one percent change in Y (unit elasticity). When $|\Upsilon_X^Y| \gg 1$, small changes in X amplify into large changes in

Y; when $|\Upsilon_X^Y| \ll 1$, even large changes in X barely move Y. The sign of Υ_X^Y indicates whether Y increases $(+)$ or decreases $(-)$ with X. When a partial rank correlation coefficient appears, it is written η_X^Y, by analogy. Readers meeting this notation for the first time will find the sensitivity indices of Section 9.3 the most useful point of reference.

9.2 Local Sensitivity: The Normalized Sensitivity Index

The central quantity of local sensitivity analysis is the *normalized sensitivity index*,[1] defined for a scalar output q (say, $\mathcal{R}_0$ or i^*) and a scalar parameter p (say, c_I or τ_R) as

$$\Upsilon_p^q \;=\; \frac{p}{q}\frac{\partial q}{\partial p} \;=\; \frac{\partial \log q}{\partial \log p}. \tag{9.1}$$

The first equality is the standard definition; the second shows that Υ_p^q is equivalently a logarithmic derivative, measuring percentage change in q per percentage change in p. The normalized form was developed in this notation by Arriola and Hyman [2007] and applied to mathematical-biology models in Chitnis et al. [2008]; Hyman and Arriola [2026] develops the full unified framework, including the connection to global variance-based methods through the sigma-normalized index.

Interpretation. If $\Upsilon_p^q = 1$, a 1% increase in p produces a 1% increase in q at the baseline. The output responds linearly and proportionally to the parameter.

If $\Upsilon_p^q = 0$, small variations in p do not affect q at the baseline. The output is insensitive to p.

If $\Upsilon_p^q = -0.5$, a 1% increase in p produces a 0.5% decrease in q. The output responds in the opposite direction, with half the magnitude.

The normalized index is dimensionless, comparable across parameters with different units, and its magnitude is directly interpretable as a local elasticity.

Why the normalization matters. The un-normalized partial derivative $\partial q/\partial p$ is dimensional and depends on the scales of q and p. Two parameters with the same dimensional derivative but different units or magnitudes contribute differently to the output. Dividing by q/p normalizes out these effects: Υ_p^q is a relative measure, making comparisons across parameters meaningful.

For example, consider $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$ with baseline $c_I = 4$, $\beta = 0.040$, $\gamma_R = 0.083$, $\nu_m = 0.0001$. The partial derivative $\partial\mathcal{R}_0/\partial c_I = \beta/(\gamma_R + \nu_m) \approx 0.48$, while $\partial\mathcal{R}_0/\partial\beta = c_I/(\gamma_R + \nu_m) \approx 48$. Comparing 0.48 and 48 gives the false impression that β is 100 times more important than c_I; the true comparison is via the normalized indices, which (as we shall see) are each equal to 1.

9.2.1 Computing Indices: Analytic Versus Numerical

For a model output q that is known in closed form (as $\mathcal{R}_0$ is for the SIR model, from Chapter 5), the sensitivity index is computed by direct differentiation:

$$\Upsilon_p^q \;=\; \frac{p}{q}\frac{\partial q}{\partial p}.$$

[1] We use "sensitivity index" in the mathematical sense of normalized partial elasticity: the percent change in an output produced by a one-percent change in a parameter. This is unrelated to the biostatistical use of "sensitivity" for the true-positive rate of a diagnostic test (and the related "specificity" for the true-negative rate). The two senses are unfortunately homonymous in the same broad subject area; we use the qualifier "sensitivity index" consistently to signal the elasticity meaning, and reserve unqualified "sensitivity" for the diagnostic-testing sense in chapters where both meanings would otherwise collide.

This is the case for $\mathcal{R}_0$ and for the endemic-equilibrium values s^*, i^*, r^*, all of which admit closed-form expressions in the SIR model.

For outputs that are not known in closed form (such as the epidemic peak height i_{peak} or the time to peak t_{peak}, which are defined by the trajectory of the ODE system), the index is computed numerically. The standard approach is the forward-difference approximation:

$$\Upsilon_p^q \approx \frac{p}{q}\frac{q(p+\epsilon p)-q(p)}{\epsilon p} = \frac{q(p+\epsilon p)-q(p)}{\epsilon q(p)},$$

with a small relative perturbation ϵ (typically 10^{-3}). Each evaluation of $q(p+\epsilon p)$ requires a full trajectory integration; for a model with k parameters, this costs $k+1$ full integrations per sensitivity analysis.

Symmetric-difference refinement. A more accurate numerical approximation uses the symmetric difference:

$$\Upsilon_p^q \approx \frac{p}{q}\frac{q(p+\epsilon p)-q(p-\epsilon p)}{2\epsilon p}.$$

This has $O(\epsilon^2)$ error rather than $O(\epsilon)$ error, at the cost of one additional integration per parameter.

Choice of ϵ. The perturbation ϵ must be small enough that the derivative approximation is accurate, but large enough that the finite difference is not dominated by round-off noise in the integrator. For double-precision arithmetic and tolerances of 10^{-8}, values of ϵ in the range 10^{-3} to 10^{-4} are typical. Values much smaller than 10^{-4} begin to suffer from round-off; values much larger than 10^{-2} begin to capture nonlinear effects that the local index does not represent.

Implementation notes. The closed-form sensitivity indices of Section 9.3 require no numerical machinery; they are direct algebraic evaluations once the baseline parameters are chosen. The global sensitivity analyses of Section 9.6 require Latin hypercube sampling and partial rank correlation routines, both standard in modern scientific-computing environments: the `SALib` library for Python; the `lhsdesign` and `partialcorr` functions in MATLAB; and the `lhs` package (Latin hypercube sampling) together with the `sensitivity` package (`pcc` with rank-transformation for PRCC) in R each provide production-quality implementations. The book's companion repository contains worked implementations in all three environments, applied to the SIR baseline of Table 4.4; the scripts reproduce every sensitivity figure and tornado plot in this chapter end-to-end.

9.2.2 What Local Indices Do Not Capture

The local sensitivity index (9.1) is a local (pointwise) quantity: it measures the response of q to an infinitesimal perturbation of p at the baseline. Two important effects are not captured by local analysis.

Nonlinear effects at large perturbations. If $\Upsilon_p^q = 1$ at the baseline, the linear prediction is that a 50% increase in p produces a 50% increase in q. The actual response may be larger or smaller depending on the higher-order structure of $q(p)$. For the SIR model, $\mathcal{R}_0$ is exactly linear in c_I and β (so the linear prediction is exact for these parameters), but nonlinear in γ_R and ν_m. Large perturbations in the latter parameters will deviate from the linear prediction; quantifying this deviation requires evaluating q at multiple perturbation sizes.

Interactions among parameters. Local analysis treats each parameter in isolation. Joint variation of two parameters may produce effects that the individual indices do not reveal. For the SIR model, $\mathcal{R}_0$ depends on the ratio $(c_I\beta)/(\gamma_R + \nu_m)$; a simultaneous halving of c_I and doubling of β leaves $\mathcal{R}_0$ unchanged, a coordinated variation that the individual local indices do not predict. Global sensitivity analysis (Section 9.6) captures some of these interaction effects through rank-correlation measures.

Despite these limitations, the local index framework is the starting point of all sensitivity analysis, and the closed-form indices developed in the next section are the cleanest quantitative statement available for the SIR model.

A sensitivity index means little until its output and its parametrization are chosen, and this book reports sensitivities with respect to the durations throughout; the two choices are the pitfall Sensitivity Analysis Discipline (Chapter 16).

Exercises for this section

1. **[C]** A parameter has $|\Upsilon_p^{\mathcal{R}_0}| = 0.8$ at the baseline. A colleague interprets this as "$\mathcal{R}_0$ is 80% dependent on p." Is this interpretation correct? If not, state the correct interpretation in terms of elasticity.

9.3 Closed-Form Sensitivity Indices for the SIR Model

The SIR model admits closed-form expressions for both the basic reproductive number $\mathcal{R}_0$ and the endemic equilibrium (s^*, i^*, r^*), and consequently admits closed-form sensitivity indices. This is unusual among epidemic models: the closed-form indices make the model an especially informative example for teaching sensitivity analysis, because every index can be computed without numerical differentiation and the resulting expressions are directly interpretable. This section derives the indices and comments on their structure.

Choosing interpretable parameters. Before any sensitivity index is computed, the modeler must choose the parameterization in which results will be reported. Model equations are written in the language that is mathematically convenient, which is not always the language that is operationally natural. The recovery process can be parameterized either by the recovery rate γ_R (units: day^{-1}) or by the mean infectious period $\tau_R = 1/\gamma_R$ (units: days). These two parameterizations are mathematically equivalent. Their sensitivity indices are not equal: they have the same magnitude and opposite signs, related by

$$\Upsilon_{\tau_R}^{q} = -\Upsilon_{\gamma_R}^{q}, \qquad \text{for any quantity of interest } q. \tag{9.2}$$

This identity follows directly from the chain rule: for any rate κ with reciprocal $\tau = 1/\kappa$, $\partial q/\partial \tau = -\kappa^2(\partial q/\partial \kappa)$, so $(\tau/q)(\partial q/\partial \tau) = -(\kappa/q)(\partial q/\partial \kappa)$, which is exactly $\Upsilon_{\tau}^{q} = -\Upsilon_{\kappa}^{q}$ when both are well-defined. The discipline that follows from Pitfall: "The Independence of τ" is to choose POIs that can be discussed in ordinary language without confusion: durations rather than rates, probabilities rather than their complements, counts rather than densities where the distinction matters. A clinician reading $\Upsilon_{\gamma_R}^{\mathcal{R}_0} = -1$ must mentally invert γ_R before the public health interpretation is clear; reading $\Upsilon_{\tau_R}^{\mathcal{R}_0} = +1$ is immediately interpretable as a longer infectious period producing higher transmission. Sensitivity indices that require mental inversion to interpret are sensitivity indices that will be misread.

Throughout the rest of this chapter we report indices using the interpretable parameters $\{c_I, c_S, c_R, \beta, \tau_R, \tau_m\}$, where $\tau_R = 1/\gamma_R$ is the mean infectious period and $\tau_m = 1/\nu_m$ is the mean residence time in the population. Where a derivation is more transparent with the rate parameters, we use them and report the result in the duration form via equation (9.2).

9.3.1 Indices for $\mathcal{R}_0$

From Chapter 5, Section 5.1, the basic reproductive number is

$$\mathcal{R}_0 \;=\; \frac{c_I \beta}{\gamma_R + \nu_m}.$$

Taking logarithmic derivatives,

$$\Upsilon^{\mathcal{R}_0}_{c_I} \;=\; \frac{\partial \log \mathcal{R}_0}{\partial \log c_I} \;=\; 1, \tag{9.3}$$

$$\Upsilon^{\mathcal{R}_0}_{\beta} \;=\; \frac{\partial \log \mathcal{R}_0}{\partial \log \beta} \;=\; 1. \tag{9.4}$$

The reproductive number depends linearly on both c_I and β through direct multiplication, so the normalized indices are each unity.

For the rate parameters in the denominator, the indices are less trivial:

$$\Upsilon^{\mathcal{R}_0}_{\gamma_R} \;=\; -\frac{\gamma_R}{\gamma_R + \nu_m}, \tag{9.5}$$

$$\Upsilon^{\mathcal{R}_0}_{\nu_m} \;=\; -\frac{\nu_m}{\gamma_R + \nu_m}. \tag{9.6}$$

These two indices are negative because $\mathcal{R}_0$ decreases when either rate increases; their sum is -1, reflecting that $\mathcal{R}_0$ is inversely proportional to $\gamma_R + \nu_m$. Equivalently, letting $\tau_R = 1/\gamma_R$ and $\tau_m = 1/\nu_m$,

$$\Upsilon^{\mathcal{R}_0}_{\tau_R} \;=\; \frac{\gamma_R}{\gamma_R + \nu_m}, \qquad \Upsilon^{\mathcal{R}_0}_{\tau_m} \;=\; \frac{\nu_m}{\gamma_R + \nu_m}. \tag{9.7}$$

The sensitivities of $\mathcal{R}_0$ to the three contact rates are:

$$\Upsilon^{\mathcal{R}_0}_{c_S} \;=\; 0, \qquad \Upsilon^{\mathcal{R}_0}_{c_R} \;=\; 0. \tag{9.8}$$

The reproductive number is insensitive to the susceptible and recovered contact rates: a change in c_S or c_R does not change $\mathcal{R}_0$ at all. This is the mathematical content of the claim in Chapter 5 that $\mathcal{R}_0$ depends on only four of the six independent parameters.

Numerical values at the baseline. Using the baseline parameters from Table 4.4 ($c_S = 20$, $c_I = 4$, $c_R = 20$, $\beta = 0.040$, $\tau_R = 12$, $\tau_m = 7300$, giving $\gamma_R \approx 0.083$, $\nu_m \approx 0.0001$, $\mathcal{R}_0 \approx 1.92$):

$$\Upsilon^{\mathcal{R}_0}_{c_I} = 1, \quad \Upsilon^{\mathcal{R}_0}_{\beta} = 1, \quad \Upsilon^{\mathcal{R}_0}_{\tau_R} \approx 1.00, \quad \Upsilon^{\mathcal{R}_0}_{\tau_m} \approx 0.002,$$

$$\Upsilon^{\mathcal{R}_0}_{c_S} = 0, \quad \Upsilon^{\mathcal{R}_0}_{c_R} = 0.$$

Interpretation: the dominant contributors to $\mathcal{R}_0$ are c_I and β (each with sensitivity 1), followed closely by τ_R (sensitivity 1.00). The migration time τ_m has negligible effect on $\mathcal{R}_0$ for the baseline (sensitivity 0.002). The susceptible and recovered contact rates contribute nothing.

9.3.2 Indices for the Endemic Equilibrium i^*

The sensitivity indices for i^* are more involved. We work in the equal-contact-rate special case, where i^* takes the particularly simple form (5.23):

$$i^* \;=\; \frac{n^*\nu_m}{\gamma_R+\nu_m}\left(1-\frac{1}{\mathcal{R}_0}\right).$$

For the general case, the sensitivity expressions are algebraically more complex; we treat the equal-contact-rate case and note the qualitative features that extend to the general case.

Differentiating with respect to c_I:

$$\frac{\partial i^*}{\partial c_I} \;=\; \frac{n^*\nu_m}{\gamma_R+\nu_m}\cdot\frac{1}{\mathcal{R}_0^2}\cdot\frac{\partial \mathcal{R}_0}{\partial c_I} \;=\; \frac{n^*\nu_m}{(\gamma_R+\nu_m)\mathcal{R}_0^2}\cdot\frac{\mathcal{R}_0}{c_I}.$$

Multiplying by c_I/i^*,

$$\Upsilon^{i^*}_{c_I} \;=\; \frac{1}{\mathcal{R}_0-1}. \tag{9.9}$$

By the same calculation applied to β,

$$\Upsilon^{i^*}_{\beta} \;=\; \frac{1}{\mathcal{R}_0-1}. \tag{9.10}$$

These two indices are small when $\mathcal{R}_0$ is large (above 3 or 4) and large when $\mathcal{R}_0$ is just above 1. As $\mathcal{R}_0 \to 1^+$, $\Upsilon^{i^*}_{c_I}$ diverges: at the invasion threshold, the endemic prevalence is very sensitive to small changes in c_I. Figure 9.1 shows the divergence, together with the contrasting flatness of the τ_m index, across the range $\mathcal{R}_0 \in [1.05, 5]$.

For γ_R (and correspondingly τ_R),

$$\Upsilon^{i^*}_{\gamma_R} \;=\; -\frac{\gamma_R}{\gamma_R+\nu_m} - \frac{1}{\mathcal{R}_0-1}\cdot\frac{\gamma_R}{\gamma_R+\nu_m} \;=\; -\frac{\gamma_R}{\gamma_R+\nu_m}\cdot\frac{\mathcal{R}_0}{\mathcal{R}_0-1}. \tag{9.11}$$

For ν_m,

$$\Upsilon^{i^*}_{\nu_m} \;=\; 1-\frac{\nu_m}{\gamma_R+\nu_m}\cdot\frac{\mathcal{R}_0}{\mathcal{R}_0-1}. \tag{9.12}$$

The reciprocal-time sensitivity follows by the chain rule. Since $\tau_m = 1/\nu_m$, differentiation of $\log\tau_m = -\log\nu_m$ gives $\partial\log\nu_m/\partial\log\tau_m = -1$, hence

$$\Upsilon^{i^*}_{\tau_m} \;=\; -\,\Upsilon^{i^*}_{\nu_m}. \tag{9.13}$$

The structure of (9.12) is informative: for typical parameter values where $\nu_m \ll \gamma_R$ and $\mathcal{R}_0$ is modestly above 1, the second term is small and $\Upsilon^{i^*}_{\nu_m} \approx 1$, so $\Upsilon^{i^*}_{\tau_m} \approx -1$. The endemic prevalence responds linearly to ν_m at leading order: a 1% increase in ν_m produces approximately a 1% increase in i^*. Equivalently, a 1% increase in τ_m (longer demographic timescale) produces approximately a 1% *decrease* in i^*, because slower susceptible turnover reduces the available pool of new infections needed to sustain the endemic equilibrium.

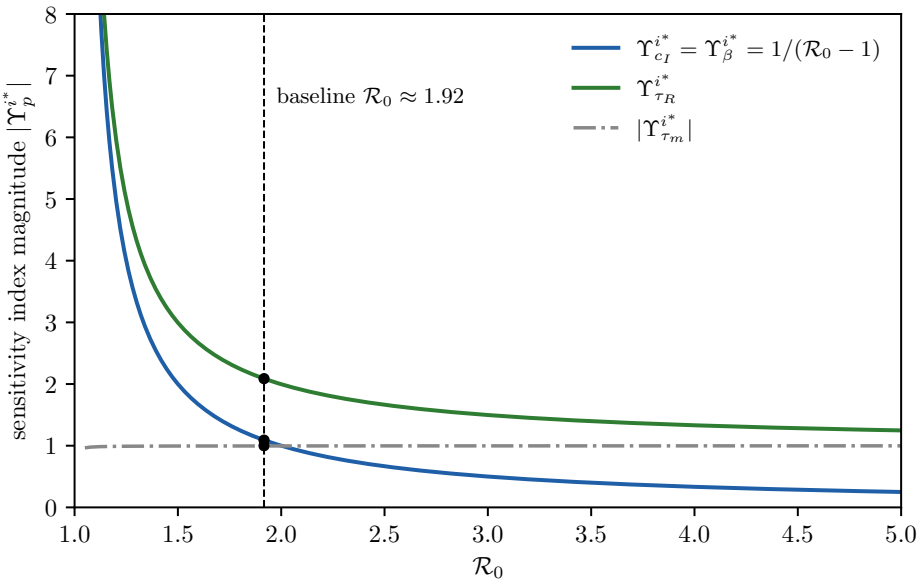


Figure 9.1: Magnitudes of the closed-form i^* sensitivity indices as functions of $\mathcal{R}_0$ (equal-contact-rate case, with the recovery and demographic rates fixed at their baseline values and $\mathcal{R}_0$ swept through the product $c_I\beta$). The invasion-parameter index $\Upsilon_{c_I}^{i^*} = \Upsilon_\beta^{i^*} = 1/(\mathcal{R}_0 - 1)$ and the mixed-parameter index $\Upsilon_{\tau_R}^{i^*}$ both diverge as $\mathcal{R}_0 \to 1^+$, while $|\Upsilon_{\tau_m}^{i^*}|$ stays near 1 across the range. The dashed line marks the baseline $\mathcal{R}_0 \approx 1.92$, where the curves take the values +1.09, +2.1, and 1.00 derived in this section. Near the invasion threshold, every percentage prediction read off a linear index is suspect (Exercise 6); a perturbation that crosses the threshold invalidates the prediction outright, the failure mode the Policy Implications of Section 9.7 are careful to avoid (Exercise 7).

Numerical values at the baseline. Using the baseline parameters ($\mathcal{R}_0 \approx 1.92$, $\tau_R = 12$, $\tau_m = 7300$):

$$\Upsilon^{i^*}_{c_I} \approx \Upsilon^{i^*}_{\beta} \approx \frac{1}{0.92} \approx 1.09,$$
$$\Upsilon^{i^*}_{\tau_R} \approx 1.00 \cdot 1.92/0.92 \approx 2.1,$$
$$\Upsilon^{i^*}_{\tau_m} = -\Upsilon^{i^*}_{\nu_m} \approx -\big(1 - 0.002 \cdot 1.92/0.92\big) \approx -1.00.$$

The dominant contributors to i^* in magnitude are τ_R (sensitivity $+2.1$, the largest), c_I and β (sensitivity $+1.09$ each), and τ_m (sensitivity -1.00, negative). All four contribute substantially in magnitude; the pattern is qualitatively different from that of $\mathcal{R}_0$. The negative sign of $\Upsilon^{i^*}_{\tau_m}$ is biologically meaningful: longer demographic timescales (slower susceptible turnover) reduce endemic prevalence, while shorter timescales increase it. At the realistic baseline residence time the elasticity is essentially exactly -1, so i^* scales inversely with τ_m to leading order.

9.3.3 The Structural Difference Between the Two Indices

The sensitivity patterns for $\mathcal{R}_0$ and i^* differ in important ways:

- c_I and β dominate $\mathcal{R}_0$ (each sensitivity $+1$) but contribute modestly to i^* (each sensitivity $1/(\mathcal{R}_0 - 1) \approx +1.09$ for the baseline).
- τ_m has negligible sensitivity for $\mathcal{R}_0$ ($\approx +0.002$ at baseline) but substantial sensitivity for i^* (≈ -1.00 at baseline; longer demographic timescales reduce endemic prevalence).
- τ_R contributes to both, with sensitivity $\approx +1.00$ for $\mathcal{R}_0$ and $\approx +2.1$ for i^*.

This quantitative pattern is the formal statement of the invasion–burden partition. Section 9.7 returns to this after developing the global sensitivity tools.

9.3.4 The Sensitivity Argument for the Infected-Viewpoint Fit

The structural difference between the two viewpoints has a sensitivity-theoretic consequence for parameter fitting. Recall the two force estimators developed in Chapter 7, Section 7.6: the infected-viewpoint estimator $\hat{\alpha} = j/i$ and the susceptible-viewpoint estimator $\hat{\lambda} = j/s$. We compute the normalized sensitivity of each estimator with respect to the susceptible population s, treating j and i as directly observable quantities and s as the source of modeling uncertainty.

Estimator-level sensitivity asymmetry. For the infected-viewpoint estimator, $\hat{\alpha} = j/i$ does not depend on s at all:

$$\Upsilon^{\hat{\alpha}}_{s} = \frac{s}{\hat{\alpha}} \cdot \frac{\partial \hat{\alpha}}{\partial s} = 0. \tag{9.14}$$

For the susceptible-viewpoint estimator, $\hat{\lambda} = j/s$ is inversely proportional to s:

$$\Upsilon^{\hat{\lambda}}_{s} = \frac{s}{\hat{\lambda}} \cdot \frac{\partial \hat{\lambda}}{\partial s} = \frac{s}{j/s} \cdot \left(-\frac{j}{s^2}\right) = -1. \tag{9.15}$$

These two relations are the *estimator-level* asymmetry: the infected-viewpoint estimator has zero s-sensitivity exactly, while the susceptible-viewpoint estimator has unit s-sensitivity exactly. They are properties of the algebraic forms j/i and j/s alone, holding for all j, i, $s > 0$. They do not depend on outbreak phase, on $\mathcal{R}_0$, or on the specific model.

What the asymmetry does *not* say. It is tempting to carry these two indices forward and conclude that the fitted $\mathcal{R}_0$ inherits the same contrast—that a susceptible-viewpoint $\hat{\mathcal{R}}_0$ scales with the assumed N^c while an infected-viewpoint $\hat{\mathcal{R}}_0$ does not. That step does not survive scrutiny, and Chapter 7 shows why. Converting $\hat{\lambda}$ to $\mathcal{R}_0$ using the observed prevalence cancels the assumed population exactly, so under a consistent conversion the two viewpoints return the *same* $\mathcal{R}_0$. The real difference appears only when prevalence is not observed at all, and then it is not a sensitivity but an identifiability failure: by Proposition 7.2, a fit to incidence alone determines $\mathcal{R}_0$ only through the ratio k combining the assumed population with the ascertainment fraction, and no local index measures that.

The distinction is worth keeping straight because the two statements have different logical types. The estimator-level asymmetry (9.14)–(9.15) is a derivative: exact, local, and true of the algebraic forms j/i and j/s whatever the data. The identifiability result is not a derivative at all; it is a statement about which parameter combinations the data can distinguish. A sensitivity index answers "how much does the answer move when this input moves," and cannot answer "is this input recoverable from these data." Conflating the two produces exactly the kind of overstated propagation claim that Pitfall: "Estimator Honesty" warns against.

The ratio s/i does appear in this setting, but it is not an amplification factor. It is the ratio of the two forces themselves, $\alpha/\lambda = s/i$, which follows immediately from $\alpha i = \lambda s$: the two viewpoints describe the same flow of new infections normalized by different populations, so their numerical values differ by exactly that factor. That is a statement about units, not about error propagation, and it carries no sensitivity content.

This sensitivity argument, together with the identifiability result of Chapter 7, is the analytical foundation for preferring the infected viewpoint whenever prevalence can be observed. The sensitivity-analysis paper of Hyman [2026d] develops the local theory in full generality.

9.3.5 The Delta Method: From Sensitivity Indices to Output Uncertainty

Sensitivity indices give the local responsiveness of an output Y to each parameter p_k at the baseline. They become operationally useful when combined with parameter uncertainties to produce a calibrated uncertainty on Y. The delta method is the standard first-order tool for this propagation, used throughout statistics, econometrics, and physics.

Theorem 9.1 (Delta method for output uncertainty)**.** *Let $Y(p_1, \ldots, p_d)$ be a smooth function of d parameters $p_1, \ldots, p_d$, with $Y(\vec{p}^*) = Y^*$ and gradient ∇Y evaluated at $\vec{p}^*$. Let $\hat{\vec{p}}$ be an unbiased estimator of $\vec{p}^*$ with covariance matrix $\mathbf{\Sigma}_{\hat{p}}$. Then the linearized estimator $\hat{Y} = Y(\hat{\vec{p}})$ has, to first order in the parameter uncertainties,*

$$\mathbb{E}[\hat{Y}] \;\approx\; Y^*, \qquad \mathrm{Var}(\hat{Y}) \;\approx\; \nabla Y^{\top}\, \mathbf{\Sigma}_{\hat{p}}\, \nabla Y \;=\; \sum_{j,k=1}^{d} \frac{\partial Y}{\partial p_j} \frac{\partial Y}{\partial p_k}\, \Sigma_{jk}. \tag{9.16}$$

In normalized-sensitivity-index form, defining the relative variance $V^Y = \mathrm{Var}(\hat{Y})/(Y^*)^2$ and the parameter relative-variance matrix $\Lambda_{jk} = \Sigma_{jk}/(p_j^* p_k^*)$,

$$V^Y \;\approx\; \sum_{j,k=1}^{d} \Upsilon_{p_j}^{Y}\, \Upsilon_{p_k}^{Y}\, \Lambda_{jk}, \tag{9.17}$$

where $\Upsilon^Y_{p_k} = (p_k/Y)\cdot \partial Y/\partial p_k$ *is the normalized sensitivity index defined in* (9.1).

When the parameters are uncorrelated, $\Lambda_{jk} = \delta_{jk} V^{p_k}$*, and the formula reduces to a sum of independent contributions:*

$$V^Y \;\approx\; \sum_{k=1}^{d} \left(\Upsilon^Y_{p_k}\right)^2 V^{p_k}. \tag{9.18}$$

Proof. A first-order Taylor expansion of Y around $\vec{p}^*$ gives

$$\hat{Y} \;=\; Y(\hat{\vec{p}}) \;\approx\; Y^* + \nabla Y^\top (\hat{\vec{p}} - \vec{p}^*).$$

Taking expectations, $\mathbb{E}[\hat{Y}] \approx Y^* + \nabla Y^\top \mathbb{E}[\hat{\vec{p}} - \vec{p}^*] = Y^*$ since $\hat{\vec{p}}$ is unbiased. Taking variances, $\mathrm{Var}(\hat{Y}) \approx \mathrm{Var}(\nabla Y^\top \hat{\vec{p}}) = \nabla Y^\top \boldsymbol{\Sigma}_{\hat{p}} \nabla Y$. The normalized form (9.17) follows by dividing through by $(Y^*)^2$ and using $\partial Y/\partial p_k = \Upsilon^Y_{p_k} \cdot Y/p_k$ at the baseline. The uncorrelated-parameters reduction (9.18) follows from the diagonal structure of Λ. □

Interpretation and use. Equation (9.18) is the chapter's most useful single formula. It states that the squared relative uncertainty in Y is a sum of contributions, one per parameter, each weighted by the squared sensitivity index. Three immediate consequences for practice:

- *Squared weighting.* Parameters enter through their squared sensitivity indices: a parameter with $|\Upsilon^Y_p| = 2$ contributes four times as much to the output variance as a parameter with $|\Upsilon^Y_p| = 1$, even though both are described as "sensitive." Variance attribution differs from ranking.
- *Sign cancellation.* Equation (9.18) sums squared terms, so positive and negative sensitivity contributions both add to output variance. The sign of Υ^Y_p matters for interpreting the direction of error (does $\hat{Y}$ go up or down with parameter increase?), but not for the magnitude of $\mathrm{Var}(\hat{Y})$.
- *Correlated parameters.* When parameters are correlated (a common case in joint MLE fitting where $\hat{\vec{p}}$ has a non-diagonal $\boldsymbol{\Sigma}$), the cross-terms $\Upsilon^Y_{p_j}\Upsilon^Y_{p_k}\Lambda_{jk}$ in (9.17) can either increase or reduce V^Y. Sensitivity indices alone are not sufficient; the covariance structure of the parameter estimates is essential. This is the formal reason why a fitting workflow that reports point estimates without their joint covariance is incomplete.

Application to the SIR baseline. For the equal-contact-rate SIR baseline with uncorrelated parameter uncertainties of 20% on τ_R, 5% on β, and 1% on ν_m, the delta-method approximation (9.18) gives

$$V^{\mathcal{R}_0} \;\approx\; (\Upsilon^{\mathcal{R}_0}_{\tau_R})^2 \cdot (0.20)^2 + (\Upsilon^{\mathcal{R}_0}_{\beta})^2 \cdot (0.05)^2 + (\Upsilon^{\mathcal{R}_0}_{\nu_m})^2 \cdot (0.01)^2.$$

Substituting the baseline values $\Upsilon^{\mathcal{R}_0}_{\tau_R} \approx +1.00$, $\Upsilon^{\mathcal{R}_0}_{\beta} = +1$, $\Upsilon^{\mathcal{R}_0}_{\nu_m} \approx -0.002$:

$$V^{\mathcal{R}_0} \;\approx\; (1.00)^2(0.20)^2 + (1.00)^2(0.05)^2 + (0.002)^2(0.01)^2 \;\approx\; 0.0400 + 0.0025 + 4\times 10^{-10} \;\approx\; 0.0425.$$

The relative standard deviation is $\sqrt{0.0425} \approx 0.21$, so $\hat{\mathcal{R}}_0$ inherits a 21% relative uncertainty from these three parameter uncertainties. The dominant contribution comes from τ_R (94% of the variance); β contributes 6%; ν_m contributes negligibly. The delta-method propagation identifies τ_R as the priority parameter for uncertainty reduction *at the baseline parameter set*, a quantitative version of the invasion-controlling parameter ranking.

A more elaborate application of the same machinery to $\hat{\mathcal{R}}_e$ in the case study of Section 9.8 appears below; the calculation of Section 9.8 subsection "Propagating the Uncertainty" is a one-parameter special case of equation (9.18) with τ_R as the single uncertain input.

AI Learning Prompt: Delta-Method Propagation with Monte-Carlo Verification

The delta method (Theorem 9.1) is the standard first-order tool for propagating parameter uncertainty, and a common AI slip is to drop its cross-term when parameters are correlated. Ask for the propagation *and* an independent Monte Carlo check. *Prompt:* "*Fitted* $\hat{\alpha} = 0.30$ *per day and* $\hat{\tau}_R = 5$ *days have relative variances* 0.01 *and* 0.04 *and correlation* $+0.4$. *(1) Use the delta method for the relative variance of* $\hat{\mathcal{R}}_0 = \hat{\alpha} \cdot \hat{\tau}_R$*, with the cross-term. (2) Draw 10,000 correlated log-normal pairs and report the empirical relative variance. (3) Explain any gap.*"
A correct answer gives $0.01 + 0.04 + 2 \cdot 0.4 \cdot \sqrt{0.0004} \approx 0.066$, a relative standard deviation near 26%, and a Monte Carlo value within a few percent of it. Dropping the cross-term gives 0.050, a 24% underestimate. A Monte Carlo run with independent draws agrees with that wrong value and seems to confirm it, which is why step (2) must keep the correlation (Pitfall: "Sensitivity Analysis Discipline").

9.3.6 Sensitivity Indices for the Equal-Contact Formula: A Cautionary Note

The expressions (9.9) through (9.12) are derived in the equal-contact-rate case, where $c_S = c_I = c_R = c$. For the general case with distinct contact rates, the expression for i^* is (5.21), and the indices acquire terms in c_S and c_R separately. Writing $\xi = c_I/c_S$, the contact-weighted population $c_S s + c_I i + c_R r$ equals $c_S\big(1-(1-\xi)i\big)$ when $c_R = c_S$, so the correction to the equal-contact formula is then set not by the ratio alone but by the *product* $(1-\xi)i$: by how far the ratio sits from unity and by the prevalence at which the indices are evaluated. Strong case isolation makes $(1-\xi)$ large, but a small prevalence can still leave the correction negligible.

For the book's baseline ($c_S/c_I = c_R/c_I = 20/4 = 5$), the equal-contact formula reproduces i^* to 0.07%, and the indices for c_I, β, τ_R and τ_m to 0.001. The reason is not that the ratios are near unity, which they are not. It is that $c_R = c_S$ and that the endemic prevalence $i^* \approx 7.9 \times 10^{-4}$ makes $(1-\xi)i^*$ of order 10^{-4}. The agreement does not extend to c_S and c_R separately. At equilibrium the recovered are about half the population ($r^* \approx 0.48$), and their contacts dilute the pool the susceptibles draw from, so $\Upsilon^{i^*}_{c_S} \approx +0.52$ and $\Upsilon^{i^*}_{c_R} \approx -0.52$ at the baseline; the two cancel only when c_S and c_R change together. With $c_R = 4$ in place of 20, the general i^* is 72% above the equal-contact value. Evaluated at the percent-level prevalence of an epidemic transient, the $(1-\xi)i$ correction is also roughly two orders of magnitude larger, and the general formula (5.21) is then required. We derive the general indices in Exercise 3.

The threshold this subsection sets analytically has an empirical counterpart. Chapter 8, Section 8.4, asks when the same simplification becomes visible *in data*, through the structure it leaves in fitting residuals, and finds the same product at work. The two routes evaluate it at different prevalences, which is why they can appear to disagree: at the endemic equilibrium the correction is negligible, while across the rising limb, where prevalence climbs by an order of magnitude, it is exactly the structure the residual tests detect. The routes differ in one further respect worth keeping in mind. The condition derived here depends only on the parameters and the prevalence; whether the discrepancy can actually be *seen* depends also on how much data there is and how noisy it is, so a regime that is safe by this calculation may still register in residuals given a large, clean series, and an unsafe one may leave no trace at all in a small or overdispersed one.

The closed-form derivations of this section parallel those of Hyman and Arriola [2026, §2.5.4], where the same indices are derived in the foundations-of-SA pedagogical context; the technical predecessor of both treatments is Arriola and Hyman [2009]. The two book-length developments are complementary: the present treatment emphasizes the modeling-practice consequences of each index for parameter interpretation and intervention design, while the foundations-of-SA treatment emphasizes the indices' role as the canonical example of normalized-derivative SA on a nonlinear compartmental system.

Table 9.1 collects the closed forms and their baseline values in one place for reference; the chapter returns to these six rows repeatedly, and the invasion–burden partition of Section 9.7 is read directly off the table's columns.

Table 9.1: Closed-form normalized sensitivity indices for the SIR model, with values at the baseline parameter set ($\mathcal{R}_0 \approx 1.92$). The $\mathcal{R}_0$ column is exact for the general model; the i^* column is the equal-contact-rate case (5.23), whose accuracy at the baseline is quantified in Section 9.3.6. The final column gives the dominant classification from the invasion–burden partition of Section 9.7.

Parameter	$\Upsilon_p^{\mathcal{R}_0}$	baseline	$\Upsilon_p^{i^*}$	baseline	**Dominant class**
c_I	1	$+1$	$\dfrac{1}{\mathcal{R}_0 - 1}$	$+1.09$	Invasion
β	1	$+1$	$\dfrac{1}{\mathcal{R}_0 - 1}$	$+1.09$	Invasion
τ_R	$\dfrac{\gamma_R}{\gamma_R + \nu_m}$	$+1.00$	$\dfrac{\gamma_R}{\gamma_R + \nu_m}\dfrac{\mathcal{R}_0}{\mathcal{R}_0 - 1}$	$+2.1$	Mixed (burden-leaning)
τ_m	$\dfrac{\nu_m}{\gamma_R + \nu_m}$	$+0.002$	$-\left(1 - \dfrac{\nu_m}{\gamma_R + \nu_m}\dfrac{\mathcal{R}_0}{\mathcal{R}_0 - 1}\right)$	-1.00	Burden
c_S	0	0	general only†	$+0.52$	Burden†
c_R	0	0	general only†	-0.52	Burden†

†General-contact-rate case (5.21): the equal-contact form does not separate c_S and c_R. The two indices cancel when c_S and c_R change together (Section 9.3.6; Exercise 3).

Exercises for this section

1. **[L]** Using the baseline parameters of Table 4.4, compute the sensitivity indices $\Upsilon_p^{\mathcal{R}_0}$ for all six parameters and verify the numerical values given in Section 9.3.

9.4 Correlated Parameters: The Most Common Mistake in Sensitivity Analysis

The sensitivity indices derived in Section 9.3 were computed by varying each parameter at a time while holding all others fixed. This is the standard procedure: it underlies every numerical sensitivity calculation, every finite-difference scheme, and every PRCC sample. Before computing any sensitivity index, there is a prior question that must be asked: can the parameters actually be varied independently in the system being modeled? In many models, the answer is no.

When parameters are correlated, the one-at-a-time sensitivity calculation produces indices that misrepresent the true response of the output to a joint change in the correlated parameters. Hyman

and Arriola [2026, Chapter 3] frames this as the most common mistake in sensitivity analysis. The disciplined response is the topic of this section.

9.4.1 Three Sources of Correlation

Correlation among parameters of interest arises from three structurally distinct sources.

Structural correlation. Parameters appear in the model equations only as a product or ratio. In the disease-free expression for the basic reproductive number of the SIR model,

$$\mathcal{R}_0 = \frac{c_I \beta}{\gamma_R + \nu_m},$$

the infectious contact rate c_I and the transmission probability β appear only as the product $c_I\beta$. Static-$\mathcal{R}_0$ data alone (such as estimates obtained from early-outbreak doubling times) can identify $c_I\beta$ as a unit but cannot separate the two factors. Computing $\Upsilon_{c_I}^{\mathcal{R}_0}$ and $\Upsilon_{\beta}^{\mathcal{R}_0}$ as independent indices is mathematically valid but operationally misleading when $\mathcal{R}_0$ is the only quantity of interest: any intervention that targets $\mathcal{R}_0$ must target the product $c_I\beta$, either by reducing infectious-individual contacts, by reducing the per-contact transmission probability, or by both. Reparameterizing to use $c_I\beta$ as a single POI is more natural in this regime.

Empirical correlation. Parameter values are correlated across data sources. Two parameters that are not structurally linked in the model equations may nevertheless covary across field studies because they share an unmeasured driver, are constrained by a common observation process, or are estimated jointly from the same data. The mosquito biting rate σ_v and the human-side biting saturation σ_h in the Chitnis–Hyman–Cushing malaria model [Chitnis et al., 2008] are a published example: both appear in the force of infection and tend to be positively correlated across malaria-prevalence settings, even though the model equations treat them as independent.

Constraint correlation. A physical or biological requirement links the parameters. Probabilities that must sum to one, stage durations that must sum to a total lifecycle, fractions allocated across compartments that must respect a conservation law. In all of these cases, no parameter can change independently of the others. The constraint is part of the model's structure, not an artifact of how the parameters are estimated.

9.4.2 QOI-Dependent Identifiability

Whether two parameters are correlated depends not only on the model but also on which output is being studied. Returning to the SIR static $\mathcal{R}_0$: c_I and β are structurally inseparable for $\mathcal{R}_0$ because the only way they enter is the product $c_I\beta$. But c_I also appears outside that product: it enters the contact-weighted denominator $C_{\text{total}} = c_S s + c_I i + c_R r$ that defines P_S in the force from infection $\alpha = c_I \beta P_S$. For dynamic quantities of interest such as $i(t)$ at a fixed time or the integrated burden $j = \int_0^T i\,dt$, the model output responds differently to a perturbation of c_I than to a perturbation of β. In the early outbreak, where $i \ll 1$ and the $c_I i$ term in C_{total} is negligible, the dynamic sensitivities $\Upsilon_{c_I}^{i(t)}$ and $\Upsilon_{\beta}^{i(t)}$ are very close. As the outbreak matures and i becomes a non-negligible fraction of the population, the two indices diverge.

A quantitative anchor for the divergence: at the standard SIR baseline (Table 4.4), the integrated burden $j = \int_0^T i\,dt$ over a one-year horizon ($T = 365$ days) has sensitivity indices $\Upsilon_{c_I}^{j} \approx +0.72$ and

$\Upsilon^j_\beta \approx +0.74$, differing by approximately 3%; the construction follows Hyman and Arriola [2026, Chapter 3]. The static $\mathcal{R}_0$ has the two indices both exactly equal to +1. The product structure that makes c_I and β inseparable for static $\mathcal{R}_0$ becomes separable for the integrated burden, because the contact-weighted denominator places c_I outside the product. The implication is operationally important: structural identifiability is a property of the QOI-parameter pair, not of the model alone. Two parameters that are inseparable for one QOI may be separable for another, and a sensitivity-analysis report should state which QOIs were used.

9.4.3 The Impact of Ignoring Correlation: A Worked Example

To see concretely what goes wrong when correlation is ignored, consider a simple QOI $q = p_1 + p_2$ where the two POIs satisfy the linear constraint $p_2 = 2p_1$. The standard one-at-a-time sensitivity indices are

$$\Upsilon^q_{p_1} = \frac{p_1}{q}\cdot\frac{\partial q}{\partial p_1} = \frac{p_1}{3p_1} = \frac{1}{3}, \qquad \Upsilon^q_{p_2} = \frac{p_2}{q}\cdot\frac{\partial q}{\partial p_2} = \frac{2p_1}{3p_1} = \frac{2}{3}.$$

These look reasonable: the second parameter appears twice as important as the first, consistent with its larger magnitude. But p_1 and p_2 cannot be varied independently: any change to p_1 forces the corresponding change $p_2 = 2p_1$. The true effect of a 1% increase in p_1 is

$$\Delta q = \frac{\partial q}{\partial p_1}\,\delta p_1 + \frac{\partial q}{\partial p_2}\,\delta p_2 = 1\cdot(0.01\,p_1) + 1\cdot(0.02\,p_1) = 0.03\,p_1.$$

The normalized total-derivative sensitivity index for p_1 is then $(p_1/q)\cdot(0.03\,p_1/0.01\,p_1) = (1/3)\cdot 3 = 1$, not 1/3. The standard one-at-a-time index understates the true sensitivity by a factor of three. A modeler relying on the standard index would conclude that p_1 is the less important parameter; the truth is that they are equally important: under the constraint each total-derivative index is 1.

The same effect, in muted form, arises in any compartmental model with correlated parameters. The factor-of-three understatement is the cleanest case; real models exhibit underestimates in the 20–50% range when correlation is empirically modest and the structural-correlation case (parameters appearing only as a product) effectively makes the separate sensitivity indices uninterpretable as a ranking.

9.4.4 Four Practical Responses

The discipline that follows from Pitfall: "Independence of Parameters of Interest" has four operational components. The first three are alternatives to the standard one-at-a-time procedure; the fourth applies whenever the others cannot be carried through.

Response 1: Diagnose correlation before computing sensitivity indices. Examine the correlation structure of the parameters before any sensitivity calculation. When parameter values are available across studies or populations, compute the Pearson correlation matrix. A correlation magnitude $|\rho_{\ell j}| > 0.5$ warrants attention. For the static $\mathcal{R}_0$ analysis of the SIR model, the structural correlation between c_I and β is exact (correlation 1.0 because they appear only as a product), and the diagnostic is satisfied by simple inspection of the expression for $\mathcal{R}_0$. For more complex models or for empirical correlation, the diagnostic step is non-trivial and is the most commonly skipped part of the discipline.

Response 2: Reparameterize structurally correlated pairs. When two parameters appear only as a product or ratio in the QOI of interest, define that product or ratio as the natural POI. For the static $\mathcal{R}_0$ of the SIR model, define the composite parameter $\psi = c_I\beta$ and compute the single sensitivity index

$$\Upsilon^{\mathcal{R}_0}_{\psi} \;=\; \frac{\partial \log \mathcal{R}_0}{\partial \log \psi} \;=\; 1,$$

since $\mathcal{R}_0 = \psi/(\gamma_R + \nu_m)$ is linear in ψ. This single index is the operationally meaningful sensitivity, because only the product $\psi = c_I\beta$ is identifiable from the static $\mathcal{R}_0$: a 1% increase in ψ—achievable by a 1% increase in either factor alone, or a 0.5% increase in each—raises $\mathcal{R}_0$ by approximately 1%. The one-at-a-time sum $\Upsilon^{\mathcal{R}_0}_{c_I} + \Upsilon^{\mathcal{R}_0}_{\beta} = 2$ answers a different question, the response to a simultaneous 1% increase in *both* factors (a 2% increase in the product), and so overstates the sensitivity to the identifiable combination. The reparameterization is not always appropriate: for dynamic QOIs, where c_I appears outside the product through the contact-weighted denominator, the parameters are genuinely separable and the composite would discard information.

Response 3: Total-derivative sensitivity for empirically correlated pairs. When empirical data establish that a proportional change in p_ℓ induces a proportional change in p_j with coefficient $c_{\ell j}$ (so that $\Delta p_j/p_j \approx c_{\ell j}\Delta p_\ell/p_\ell$), the total-derivative sensitivity index is

$$\Upsilon^{q}_{p_\ell}\big|_{\text{tot}} \;=\; \Upsilon^{q}_{p_\ell}\big|_{\text{std}} \;+\; \sum_{j\neq\ell} \Upsilon^{q}_{p_j}\, c_{\ell j}. \tag{9.19}$$

The standard one-at-a-time index is the special case $c_{\ell j} = 0$ for all $j \neq \ell$. The coefficient $c_{\ell j}$ represents the causally induced proportional change in p_j when p_ℓ is actively perturbed, not merely the observed statistical correlation. If a 1% increase in c_I tends to be accompanied by a 0.6% increase in β from field data, then $c_{c_I,\beta} = 0.6$ and the total-derivative SI for c_I at static $\mathcal{R}_0$ is $\Upsilon^{\mathcal{R}_0}_{c_I}|_{\text{tot}} = 1 + 1 \cdot 0.6 = 1.6$, 60% larger than the standard index. Estimate $c_{\ell j}$ from regression of observed parameter pairs across studies or from domain knowledge about shared drivers. A caution on causal interpretation: if the empirical correlation between c_I and β across field sites arises from a shared lurking variable (for example, both are higher in densely-populated urban settings) rather than from a direct biological mechanism, then a policy intervention that reduces c_I through social distancing may not induce the correlated change in β, and the total-derivative SI will overstate the intervention's effect. When the causal structure is uncertain, report both the standard and total-derivative SIs and state the assumption underlying the latter.

Response 4: Report transparently. When correlations exist but cannot be fully quantified (either because parameter-correlation data are not available or because the causal interpretation is uncertain), state explicitly in the sensitivity-analysis report that the independence assumption was made, identify the correlated pairs, and flag which SI values are most likely affected by the correlation. The discipline of transparent reporting protects readers who would otherwise misinterpret an underestimated SI as a ranking that they can act on.

9.4.5 The Connection to Multicollinearity in Regression

Readers with a regression background will recognize the structure of correlated POIs as the SA analog of multicollinearity. In regression with nearly collinear predictors, the design matrix $X^\top X$ is near-singular, coefficient estimates become unstable, and variance inflation factors grow large.

In sensitivity analysis with correlated POIs, varying one parameter while holding others fixed is physically impossible, and the resulting sensitivity indices misrepresent the true response for the same underlying reason: two predictors that move together cannot both be estimated or ranked in isolation. The tools differ (variance inflation factors in regression versus reparameterization or total-derivative SI in sensitivity analysis), but the underlying statistical problem is identical.

The Fisher-spectrum view: stiff and sloppy directions. Chapter 7's identifiability discussion, whose diagnostics are developed in Chapter 8 (Section 8.3), gives this analogy a spectral form. There, the Fisher (equivalently Hessian) eigenvalues of a fit typically span many orders of magnitude: a few large eigenvalues mark the *stiff* directions the data constrain tightly, while a long tail of small eigenvalues marks the *sloppy* directions along which the loss barely changes—the spread sometimes called model *sloppiness*. Correlated POIs are the sensitivity-analysis face of the same geometry. A structurally correlated pair such as (c_I, β) has one stiff combination, the product $c_I\beta$ that static-$\mathcal{R}_0$ information pins down, and one sloppy combination, the ratio c_I/β, along which $\mathcal{R}_0$ is exactly flat. The one-at-a-time indices $\Upsilon_{c_I}^{\mathcal{R}_0} = \Upsilon_{\beta}^{\mathcal{R}_0} = 1$ report the response along neither eigendirection, which is precisely why they misstate what the data can constrain. The disciplined responses of this section are spectral operations in this language: reparameterizing to the product (Response 2) aligns the parameter of interest with the stiff direction, and the total-derivative index (Response 3) measures the response along the empirically realized direction rather than along an axis the system cannot traverse alone. When a sensitivity ranking and a sloppiness analysis are both available, they should agree about which parameter combinations are constrained; disagreement means one of them is being read in the wrong parametrization.

9.5 Tornado Plots

A *tornado plot* is the standard visualization of local sensitivity results. It displays the normalized sensitivity indices as horizontal bars, ordered by magnitude, with the largest at the top. The shape of the resulting plot resembles a tornado, hence the name. A tornado plot for $\mathcal{R}_0$ shows which parameters dominate the invasion threshold; a tornado plot for i^* shows which parameters dominate the endemic burden. Side-by-side tornado plots make the invasion–burden partition visually immediate.

9.5.1 Constructing a Tornado Plot

For each parameter p in the model, compute the normalized sensitivity index Υ_p^q at the baseline. Arrange the indices by decreasing magnitude: the parameter with the largest $|\Upsilon_p^q|$ at the top, the smallest at the bottom. Draw a horizontal bar for each parameter, with length proportional to $|\Upsilon_p^q|$ and colored (by convention) red if $\Upsilon_p^q > 0$ (a positive relationship) and blue if $\Upsilon_p^q < 0$ (a negative relationship).

Figure 9.2 shows the tornado plots for $\mathcal{R}_0$ and for i^* in the SIR model at the baseline parameter set.

The left panel of Figure 9.2 highlights three of the six parameters as dominant (c_I, β, τ_R); the other three (c_S, c_R, τ_m) are negligible. The right panel shows a different pattern: four of the six parameters contribute substantially (τ_R most strongly with sensitivity $+2.1$, then c_I, β at $+1.09$ each, then τ_m at -1.00), and c_S and c_R at about ± 0.52, through the recovered people's share of the contact pool (Section 9.3.6).

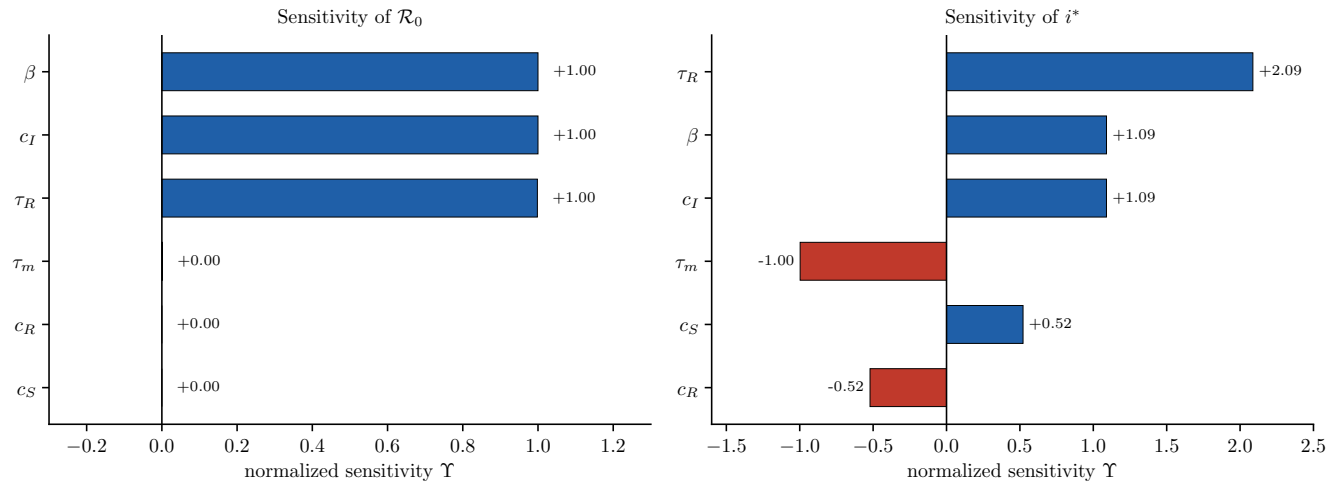


Figure 9.2: Tornado plots at the baseline parameter set. Left: sensitivity of $\mathcal{R}_0$. The dominant contributors are c_I and β (each $s = +1$), followed by τ_R ($s \approx +1.00$). Migration time τ_m contributes negligibly ($s \approx +0.002$); c_S and c_R contribute zero. Right: sensitivity of i^*. The dominant contributor is τ_R ($s \approx +2.1$); c_I and β contribute substantially ($s \approx +1.09$ each); τ_m contributes substantially *negatively* ($s \approx -1.00$, by the chain rule of equation (9.13)); c_S and c_R, from the general formula (5.21), contribute $+0.52$ and -0.52, which cancel when they change together. The two lists share c_I, β and τ_R; τ_m, c_S and c_R act on i^* alone. At this baseline the invasion–burden partition of Section 5.7 is therefore partial; it sharpens as $\mathcal{R}_0$ grows, because the burden index of c_I and β is $1/(\mathcal{R}_0 - 1)$.

The pattern visible in the figure is what the algebra of Section 9.3 predicted, and at this baseline it is only partly separated. Three parameters are common to both top lists: c_I and β, whose burden index $1/(\mathcal{R}_0 - 1)$ is $+1.09$ here, and τ_R, which appears with sensitivity $+1.00$ in the $\mathcal{R}_0$ tornado and $+2.1$ in the i^* tornado (in fact τ_R is the single largest contributor to i^*). Only τ_m, c_S and c_R act on one output alone. The lists separate as $\mathcal{R}_0$ grows: at $\mathcal{R}_0 = 5$ the burden index of c_I and β falls to 0.25.

9.5.2 Reading a Tornado Plot: Three Uses

The tornado plot condenses the sensitivity information into a form that supports three distinct inferences.

Identifying the key parameters. At a glance, the top-three contributors to each output are obvious. The modeler can focus attention on these parameters; for parameters at the bottom of the tornado, precision of the estimate is of secondary importance.

Comparing two outputs' sensitivities side by side. When two tornado plots are drawn side by side, the visual juxtaposition reveals which parameters contribute to each output and which do not. Figure 9.2's two panels illustrate that τ_m is negligible for invasion (left) but substantial for burden (right); this is the invasion–burden partition made visible.

Prioritizing data collection. If a tornado identifies c_I as the most-sensitive parameter for $\mathcal{R}_0$ and τ_m as the most-sensitive parameter for i^*, the modeler knows what data will most reduce the uncertainty of each output. Funds for additional contact-tracing studies (which pin down c_I) should be prioritized if the modeler cares about $\mathcal{R}_0$; funds for residence-time studies (pinning down τ_m) should be prioritized if the modeler cares about i^*.

Decomposing the output uncertainty. A tornado plot drawn at $\pm 1\sigma$ parameter ranges is also a graphical decomposition of the output's standard deviation. By Theorem 9.1, the relative standard deviation of $\hat{Y}$ satisfies $\sqrt{V^Y} \approx \sqrt{\sum_p (\Upsilon_p^Y)^2 (V^p)}$ when parameters are uncorrelated, so each parameter's bar in the $\pm 1\sigma$ tornado has length proportional to $|\Upsilon_p^Y| \cdot \sqrt{V^p}$, which is exactly that parameter's contribution to the output standard deviation. The squared bar length is the parameter's contribution to the output variance, and the sum of squared bar lengths is V^Y. The same plot supports two distinct readings: an *intervention-ranking* reading (which parameter, if changed, would most shift Y?) and a *variance-decomposition* reading (which parameter, given its current uncertainty, contributes most to the uncertainty in $\hat{Y}$?). The two rankings can differ: a parameter with $|\Upsilon_p^Y| = 2$ but well-known (V^p small) contributes less variance than a parameter with $|\Upsilon_p^Y| = 0.5$ but poorly known (V^p large). When drawing tornado plots intended for uncertainty quantification, label the axis explicitly as $\pm 1\sigma$ contributions to clarify which reading is intended; the case study of Section 9.8 below applies this convention.

9.5.3 Limitations of the Tornado Plot

The tornado plot has two important limitations.

First, it is built from local indices at a single baseline. If the parameter ranges of interest extend far from the baseline (for example, if $\mathcal{R}_0$ varies from 1.2 to 5.0 across the range of parameter values being considered), the local indices at $\mathcal{R}_0 = 1.92$ may not be representative of the sensitivities at $\mathcal{R}_0 = 1.2$ or $\mathcal{R}_0 = 5.0$. Global sensitivity analysis (Section 9.6) produces rankings that integrate over the full parameter range and are not tied to a single baseline.

Second, the tornado displays only first-order sensitivities. Interactions between parameters (where the effect of two parameters together differs from the sum of their individual effects) are not shown. For the SIR model, first-order effects dominate at parameter values where the formulas for $\mathcal{R}_0$ and i^* are smooth, and tornado plots are a reliable guide. For more complex models where nonlinearities or interactions are prominent, global analysis is the appropriate tool.

9.5.4 Two Readings of a Tornado Plot

The tornado plot supports two distinct readings, and the difference between them matters operationally [Hyman and Arriola, 2026, Chapter 9].

The first reading is intervention-ranking. The plot answers the question, "which parameter, if changed by an intervention, would most strongly shift the output?" This is the reading implicit in the construction so far: bars are sorted by $|\Upsilon_p^Y|$, the absolute normalized sensitivity index, and the parameter at the top is the most influential lever the modeler can pull. Two parameters with the same $|s|$ are equally good intervention targets; a parameter with $|s| = 2$ is twice as good a target as one with $|s| = 1$.

The second reading is variance-decomposition. The plot answers the question, "which parameter, given its current uncertainty, contributes most to the uncertainty in the output?" The relevant

quantity is no longer the raw sensitivity but the sigma-normalized sensitivity index

$$s_\ell^\sigma \;=\; \frac{\sigma_\ell}{\sigma_Y}\,\frac{\partial Y}{\partial Q_\ell}, \tag{9.20}$$

which reweights each sensitivity by the relative uncertainty of the parameter. Two parameters with identical $|\Upsilon_p^Y|$ but different uncertainties contribute differently to the output uncertainty: the well-known parameter contributes little even if it has high influence, and the poorly-known parameter contributes substantially even if it has modest influence.

The two readings can give different rankings. In the SIR model, the static $\mathcal{R}_0$ has $|\Upsilon_{c_I}^{\mathcal{R}_0}| = |\Upsilon_\beta^{\mathcal{R}_0}| = 1$: by the intervention-ranking reading, c_I and β are equally good targets. But β is typically known to within a factor of two from outbreak fitting, while c_I is constrained more tightly by contact-tracing data; in a setting where $\sigma_\beta/\beta \approx 0.3$ and $\sigma_{c_I}/c_I \approx 0.1$, the sigma-normalized indices are $s_\beta^\sigma \approx 0.3 \cdot \Upsilon_\beta^{\mathcal{R}_0}/(\sigma_{\mathcal{R}_0}/\mathcal{R}_0)$ and $s_{c_I}^\sigma \approx 0.1 \cdot \Upsilon_{c_I}^{\mathcal{R}_0}/(\sigma_{\mathcal{R}_0}/\mathcal{R}_0)$: the variance-decomposition reading ranks β three times more important than c_I for explaining the uncertainty in $\mathcal{R}_0$. A modeler answering "which parameter should I refine?" should look at the variance-decomposition reading; a policy-maker answering "which intervention should I prioritize?" should look at the intervention-ranking reading. The two are not always aligned.

The variance-decomposition reading is also the conceptual bridge from local sensitivity analysis to global sensitivity analysis. The sigma-normalized index is what variance-based global methods (Sobol indices, the PRCC framework of Section 9.6) generalize from a single baseline to the full parameter range. Reading a tornado plot in the variance-decomposition mode is the first step toward variance-based global SA.

AI Lab: Generating a Tornado Plot

A tornado plot is the most readable one-page summary of a local sensitivity analysis, and its value depends on ranges that represent real uncertainty. The workflow is a *three-step audit-and-refine.*
Step 1: parameter ranges. Ask the AI: "*For the* SIR *parameters* $\{c_I, c_S, c_R, \beta, \tau_R, \tau_m\}$ *in [pathogen X, setting Y], what is the published* 1-σ *range for each?*" A good answer gives pathogen- and setting-specific ranges with sources; a default of uniform $\pm 50\%$ is the failure. Spot-check two or three ranges against their sources.
Step 2: computation. Compute the output at each parameter's upper and lower bound, holding the others at baseline. When the index is analytic (Section 9.3), keep perturbations small, $\Delta p/p \sim 0.01$; when it is simulated, larger steps are fine, but check that the response is monotone over the range.
Step 3: plot and check. Sort the bars by size and compare the ranking with the closed-form indices. If the plot ranks τ_R above $c_I\beta$ and the closed form says the opposite, the plot is wrong, most likely through the ranges of Step 1.
Disclose AI assistance in published work (Appendix D).

9.6 Global Sensitivity: Latin Hypercube Sampling and PRCC

Local sensitivity analysis at a single baseline gives partial information. If the modeler's uncertainty about the parameters is small (each known to within a few percent), the local indices are reliable guides. If the uncertainty is large (parameters known within factors of 2 or 3), the local indices at the baseline may mislead: the true behavior across the parameter range may differ from the local approximation.

Global sensitivity analysis samples the parameter space broadly, integrating over the uncertainty distribution of each parameter. The standard procedure combines Latin hypercube sampling (LHS) for the sample design with partial rank correlation coefficients (PRCC) for the sensitivity measure [Blower and Dowlatabadi, 1994, Helton and Davis, 2003, Marino et al., 2008]. Appendix B, Section B.2 develops the methodology in full, and the corresponding Python implementation is in the `code/python/ch10_sensitivity/` module of the companion GitHub repository. This section applies the methodology to the SIR model and discusses when the local and global analyses disagree.

9.6.1 What PRCC Computes Statistically

Partial rank correlation is a two-stage construction with a specific statistical meaning that is essential for interpreting the rankings produced by the LHS-PRCC procedure. We state both stages explicitly because reviewers regularly conflate PRCC with simpler quantities (Pearson correlation, sensitivity index) that are mathematically distinct.

Stage 1: Rank transformation. Given samples $(p_{\ell,k}, Y_\ell)$ for parameter p_k and output Y across $\ell = 1, \ldots, M$ LHS-generated parameter vectors, replace each value with its rank within the sample. The rank-transformed pair $(\rho_{\ell,k}, \rho_\ell^Y)$ takes integer values $1, 2, \ldots, M$. The rank transformation linearizes any monotonic relationship between p_k and Y regardless of the original functional form: a quadratic, exponential, or sigmoidal monotonic relationship becomes a linear relationship in the ranks. PRCC rankings therefore depend only on the *monotonicity* of the parameter–output relationship, not on its specific functional form.

Stage 2: Partial correlation on ranks. Given the rank-transformed sample, the partial correlation between ρ^Y and ρ_k controlling for the other parameters $\rho_{j\neq k}$ is the standard partial correlation coefficient: regress ρ^Y on $\{\rho_j : j \neq k\}$, regress ρ_k on the same set, and compute the Pearson correlation between the two residuals. By the Frisch-Waugh-Lovell theorem of econometrics, this is equivalent to the coefficient on ρ_k in a multivariate linear regression of ρ^Y on all $\{\rho_j\}$, suitably normalized.

In compact notation, denoting by $\mathrm{pcor}(X, Y \mid Z)$ the partial correlation of X and Y given the set Z,

$$\mathrm{PRCC}(p_k, Y) \;=\; \mathrm{pcor}\big(\rho_k,\, \rho^Y \mid \{\rho_j : j \neq k\}\big) \;=\; \frac{\mathrm{Cov}(r_k, r^Y)}{\sqrt{\mathrm{Var}(r_k)\,\mathrm{Var}(r^Y)}}, \tag{9.21}$$

where r_k and r^Y are the residuals of ρ_k and ρ^Y after regressing on $\{\rho_j : j \neq k\}$ respectively.

Statistical interpretation. PRCC values lie in $[-1, +1]$, with $+1$ indicating perfect monotonic increase, -1 perfect monotonic decrease, and 0 no monotonic relationship after controlling for the other parameters. The "partial" qualifier matters: PRCC isolates the contribution of p_k to Y after accounting for the linear (in ranks) effects of the other parameters. Without the partial qualifier, simple rank correlation $\mathrm{corr}(\rho_k, \rho^Y)$ would conflate the direct effect of p_k with effects mediated through correlations among parameters, a serious issue when LHS-generated parameter samples have small but non-negligible spurious correlations.

The PRCC value is itself a sample statistic with sampling variability. The standard error scales as $1/\sqrt{M}$; for $M = 5000$ samples and a PRCC of moderate magnitude (0.3 to 0.9), the 95% confidence interval has width approximately 0.04. Confidence intervals on PRCC values can be computed by bootstrap resampling or by the closed-form Fisher z-transformation; both methods are implemented in the `sensitivity` package in R and the `SALib` library in Python.

The monotonicity assumption and how to check it. PRCC's rank construction linearizes any monotonic relationship between a parameter and the output, but it does not detect non-monotonic relationships well. A parameter whose effect on the output rises and then falls across the sampled range (a unimodal response) can produce a PRCC value near zero even when the parameter is in fact highly influential. Pitfall: "Hidden Monotonicity Assumption" names this assumption explicitly. The practical safeguard is to plot the output against each parameter using the LHS sample and visually verify that the relationship is monotonic, or at least close enough that the rank-correlation summary is meaningful, before reporting the PRCC ranking [Marino et al., 2008]. When a parameter's response is genuinely non-monotonic, PRCC is the wrong tool, and a variance-based method such as eFAST (Section 9.6.3) is preferred.

Significance testing via a dummy parameter. A PRCC magnitude near zero is ambiguous: it could mean the parameter has no monotonic effect on the output, or it could mean that the sampling produced a non-zero value purely by chance. The standard test is to include a *dummy parameter* in the LHS design: a parameter that the model does not actually use, so any non-zero PRCC value computed for it reflects only sampling noise [Marino et al., 2008]. Report only PRCC magnitudes that are significantly larger than the dummy parameter's PRCC magnitude at a stated confidence level (typically $p < 0.005$). Parameters whose PRCC magnitudes are indistinguishable from the dummy's should be reported as having no detectable monotonic effect on the output, regardless of the numerical value of their PRCC.

What PRCC does *not* compute. PRCC is not a measure of variance attributable to a parameter; that is the domain of variance-based global sensitivity analysis (Sobol indices). PRCC is also not a sensitivity index in the local sense of Section 9.2: it does not equal $(\partial Y/\partial p_k) \cdot (p_k/Y)$ at the baseline. PRCC and local sensitivity indices typically agree on rankings (which parameters dominate) for models with smooth, nearly-monotonic parameter–output relationships in the tested range, but they can disagree numerically and can disagree on rankings when the model has strongly nonlinear or non-monotonic responses.

9.6.2 LHS-PRCC Applied to the SIR Model

Applying the full LHS-PRCC procedure to the SIR model, with uniform distributions around the baseline (each parameter varied within a factor of 2 above or below its baseline value) and $M = 5000$ samples, produces PRCC estimates for $\mathcal{R}_0$ and for i^*.

Figure 9.3 shows the results.

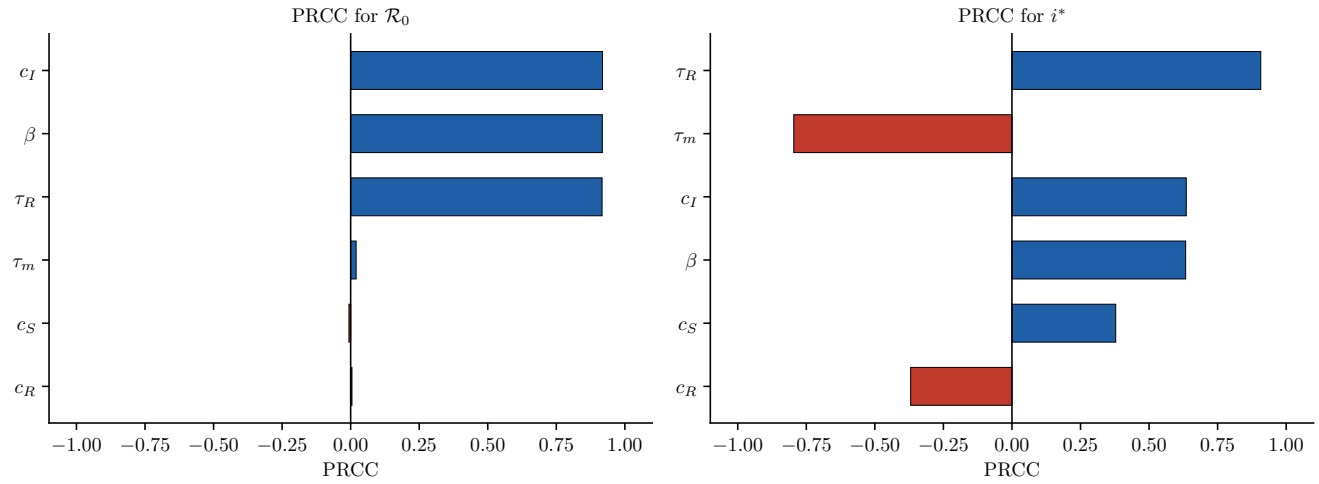


Figure 9.3: PRCC estimates for (left) $\mathcal{R}_0$ and (right) i^*, computed from $M = 5000$ LHS samples with each parameter varied within a factor of 2 above or below its baseline. Parameters: $c_S, c_I, c_R, \beta, \tau_R, \tau_m$. For $\mathcal{R}_0$, the dominant parameters are c_I, β and τ_R, each with PRCC near 0.9; τ_m, c_S and c_R have PRCC consistent with zero. For i^*, computed from the general formula (5.21), the dominant parameter is τ_R with PRCC above 0.9, followed by τ_m with PRCC near -0.8, then c_I and β with PRCC near 0.65, and c_S and c_R near ± 0.37. The same parameters dominate as in the local indices of Figure 9.2: τ_R, c_I and β matter for both outputs, and τ_m, c_S and c_R only for i^*.

The PRCC rankings in Figure 9.3 closely match the local-index rankings from Section 9.3. This agreement is expected: $\mathcal{R}_0$ and i^* are smooth, nearly linear functions of most of their parameters over the baseline $\times 2$ range, so local and global methods return similar answers. For a strongly nonlinear or interaction-heavy output (not typical for the SIR model but common for more complex models), the two methods can diverge meaningfully.

9.6.3 eFAST: When PRCC's Monotonicity Assumption Fails

PRCC's reliance on monotonicity (Section 9.6.1) leaves a gap. When the output response to a parameter is genuinely non-monotonic, when parameters interact strongly so that a parameter's

effect on the output depends on the values of the others, or when correlations among parameters (Section 9.4) confound the partial-correlation construction, PRCC may misrank the truly influential parameters. The variance-based methods address these limitations directly.

The most commonly used variance-based method in mathematical epidemiology is the extended Fourier amplitude sensitivity test (eFAST) of Marino et al. [2008]. The method assigns each parameter a distinct sinusoidal frequency, samples the parameter space along curves parameterized by these frequencies, and uses Fourier analysis of the resulting output signal to decompose the output variance into contributions from each parameter and from their interactions.

The two quantities eFAST reports are:

- The *first-order sensitivity index* $\mathcal{S}_\ell$, defined as the fraction of the output variance attributable to parameter p_ℓ alone, averaged over all values of the other parameters. This is the variance-based analog of a main-effect sensitivity.

- The *total sensitivity index* $\mathcal{S}_{T_\ell}$, defined as the fraction of the output variance attributable to parameter p_ℓ together with all of its interactions with other parameters. The difference $\mathcal{S}_{T_\ell} - \mathcal{S}_\ell$ quantifies the interaction effects involving p_ℓ.

The two indices coincide ($\mathcal{S}_{T_\ell} = \mathcal{S}_\ell$) for models without parameter interactions; when they differ, the gap is a quantitative statement that interactions matter. A parameter with small $\mathcal{S}_\ell$ but large $\mathcal{S}_{T_\ell}$ is one whose main effect is modest but whose interactions with other parameters meaningfully drive output variance. Such a parameter would likely be missed by a PRCC analysis, which has no analog of $\mathcal{S}_{T_\ell}$.

The eFAST procedure requires no monotonicity assumption: the Fourier decomposition captures the contribution of p_ℓ to output variance regardless of whether the parameter–output relationship is monotonic, unimodal, or oscillatory. eFAST also handles correlated parameters more gracefully than PRCC, because the variance decomposition does not depend on the residualization step that PRCC's partial-correlation construction requires. These advantages come at computational cost: eFAST requires more samples than PRCC for comparable precision, and the Fourier-frequency design is more involved than LHS to set up correctly. The reference MATLAB implementation is the Kirschner-laboratory toolbox accompanying Marino et al. [2008]; a Python implementation is in the `SALib` library.

For the SIR model analyzed in Section 9.6 above, PRCC suffices: the model's outputs $\mathcal{R}_0$ and i^* are smooth and nearly monotonic in each parameter over the tested range, and the agreement of PRCC with the closed-form local indices confirms that the rankings are not artifacts of PRCC's assumptions. For more complex compartmental models (multi-strain dynamics, age-structured mixing, models with feedback between population dynamics and behavior), eFAST is often the right choice, both because monotonicity can fail and because interaction effects are typically more pronounced.

9.6.4 A Compact Sobol Panel from the Closed Forms

The verdict that PRCC suffices can be checked rather than assumed, and the check is nearly free here: both quantities of interest have closed forms, so the $M(k+2)$ model evaluations of a Saltelli design [Sobol, 1993, Saltelli et al., 2008] cost nothing, and none of eFAST's frequency machinery is needed. The panel of Figure 9.4 uses the same design as the PRCC analysis above—six parameters, independent and uniform within a factor of 2 of the baseline—with $M = 65{,}536$ base samples, a fixed seed, and the standard Monte Carlo estimators for $\mathcal{S}_\ell$ and $\mathcal{S}_{T_\ell}$. The quantities are $\mathcal{R}_0$ and

the equal-contact endemic prevalence (5.23), the latter set to zero wherever a sampled parameter vector gives $\mathcal{R}_0 \leq 1$, since the endemic equilibrium then does not exist and the long-run prevalence is zero. Because c_S and c_R enter neither closed form, their indices are structural zeros, and the computation returns them as exact zeros—the variance-based analog of the dummy-parameter check of Section 9.6.1.

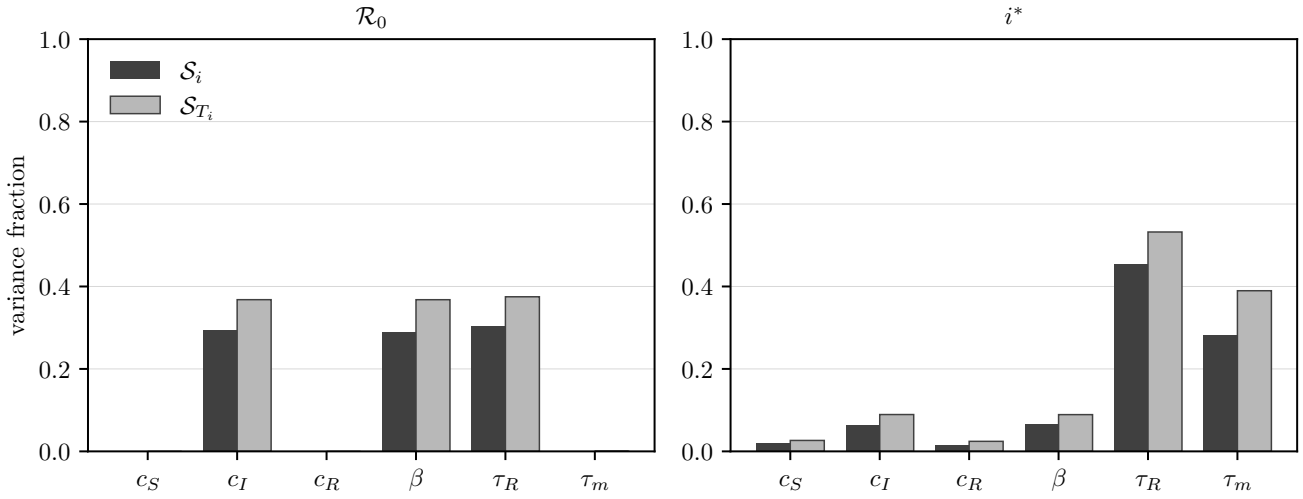


Figure 9.4: Sobol first-order ($\mathcal{S}_\ell$) and total-order ($\mathcal{S}_{T_\ell}$) indices for (left) $\mathcal{R}_0$ and (right) i^* from the general formula (5.21), from a Saltelli design with $M = 65{,}536$ base samples and each parameter uniform within a factor of 2 of its baseline (the design of Figure 9.3). c_S and c_R do not enter $\mathcal{R}_0$ and return exact zeros there; for i^* they carry small shares ($\mathcal{S}_\ell \approx 0.02$), because over this range the variance is dominated by τ_R and τ_m. For $\mathcal{R}_0$, the near-symmetric gap $\mathcal{S}_{T_\ell} - \mathcal{S}_\ell \approx 0.07$ on each active parameter is the multiplicative interaction share, carried by the pairwise products and led in interpretive weight by the transmission product $c_I\beta$; for i^*, τ_R dominates, and the invasion threshold contributes additional interaction variance.

For $\mathcal{R}_0$ the three active parameters land almost symmetrically: $\mathcal{S}_\ell \approx 0.29$–$0.30$ and $\mathcal{S}_{T_\ell} \approx 0.37$ for each of c_I, β, and τ_R, with the first-order effects summing to 0.89. The missing tenth of the variance is exactly what a multiplicative form must produce: for a product, main effects cannot exhaust the variance, and the remainder sits in the pairwise products of the active factors—roughly 0.04 per pair here, led in interpretive weight by the transmission product $c_I\beta$, the stiff direction of Section 9.4. An analytic variance decomposition of the product form reproduces every Monte Carlo entry in the left panel to within 0.01, a check that the sampled panel is measuring structure rather than noise. For i^* the panel sharpens the PRCC picture rather than changing it: τ_R dominates ($\mathcal{S}_\ell = 0.45$, $\mathcal{S}_{T_\ell} = 0.53$), τ_m follows (0.28 and 0.39), c_I and β contribute modestly (0.06 to 0.07, and 0.09), and c_S and c_R least (0.02 each, and 0.02 to 0.03)—the same ranking PRCC reported, now expressed as fractions of variance. The interaction share is again about a tenth of the total, fed both by the multiplicative prefactor and by the invasion threshold itself: whether a perturbation of one parameter changes i^* at all can depend on whether the others hold $\mathcal{R}_0$ above 1, the same threshold effect that breaks linear extrapolation in the Policy Implications of Section 9.7 and that Figure 9.1 displays near $\mathcal{R}_0 = 1$.

The panel's message is calibration rather than revolution. The rankings match the closed-form indices and the PRCC analysis, which is the agreement the preceding subsections predicted; the new information is quantitative—how much of each output's variance is attributable to single

parameters and how much is irreducibly joint. When closed forms make evaluations free, the check costs essentially nothing; for simulation-priced models the same panel costs $M(k+2)$ runs, which is when the frequency design of eFAST earns its keep.

9.6.5 When Local and Global Analyses Disagree

If local and global sensitivity analyses give different rankings, the disagreement is informative:

- The local analysis may be dominated by behavior near the baseline that is not representative of the full parameter range.
- The global analysis may capture interactions between parameters that the local analysis misses.
- Either analysis may be affected by sampling noise (global) or by the choice of baseline (local).

When the two analyses agree (as for the SIR model), the agreement is a strong statement that the rankings are reliable across the parameter range of interest. When they disagree, additional analysis is warranted: extending the local analysis to a wider range of baselines, increasing M in the global analysis, or using higher-order methods (variance-based global sensitivity, Sobol indices) that can separately quantify first-order and interaction effects, as Section 9.6.4 does for the SIR closed forms.

For the SIR model, the close agreement of the local and global results is a feature of the model's clean algebraic structure. It makes the model an especially informative teaching example: students can see both methods, compare the results, and gain confidence that the rankings are not artifacts of the chosen baseline.

Verify the AI: Local-vs-Global Sensitivity Conflation

The local index $\Upsilon_p^Y = (\partial Y/\partial p)(p/Y)$ is a derivative at one baseline, while a global coefficient such as PRCC is a rank correlation over a sample of parameters. AI expositions often blur the two. A typical passage:

> *"The recovery rate is the most sensitive parameter for $\mathcal{R}_0$, at 1.00, and population mortality the least, at 0.002. These rankings hold robustly across the parameter range."*

The values fit either method, and the passage never says which. The robustness claim needs a global analysis, or local rankings checked at several baselines, because they can flip far from the baseline (Chapter 17, Section 17.5). *Prompt:* "*Is this a local or a global sensitivity? If local, at what baseline? If global, over what range and with what sample size?*" A correct answer names the method and its baseline or range; a local result does not license a global claim, nor the reverse.

9.6.6 Choosing Between Local and Global Analysis

Which analysis to run is a decision about the question being asked, not about which method is more sophisticated. Local analysis is the right tool when the model admits closed-form indices or cheap derivatives, when a single baseline is defensible (a fitted model, an accepted parameter table), and when the question is a ranking of parameters near that operating point. It costs one model evaluation per parameter, or none at all when closed forms exist, and it is the natural

first screen. Global analysis is the right tool when parameter ranges are wide or contested, when nonlinearity or interactions among parameters are suspected, and when the ranking must hold across the plausible parameter region rather than at a point. It costs M model runs for a design of size M (Section 9.6 used $M = 5000$), plus the discipline of choosing defensible ranges, which is itself a modeling decision. The recommended order for a model like SIR is therefore: local first, from the closed forms of Section 9.3, then a confirming LHS-PRCC pass over honest ranges. Agreement between the two, as found here, is evidence that the ranking is a property of the model rather than of the chosen baseline; disagreement is not a failure but a finding, to be run to ground as described in the preceding subsection.

Exercises for this section

1. [**L**] Implement an LHS-PRCC analysis for $\mathcal{R}_0$ in your preferred scientific computing environment. Use $M = 1000$ samples with each parameter varied within a factor of 2 above or below its baseline. Compare your computed PRCC values with the local sensitivity indices.

9.7 The Invasion–Burden Partition: Quantitative Version

This section draws together the closed-form sensitivity indices of Section 9.3 with the global PRCC results of Section 9.6 to give the full quantitative statement of the invasion–burden partition. The partition was introduced informally in the Preface (Figure 2), derived algebraically in Chapter 5, Section 5.7, and illustrated numerically in Chapter 6. Here we give the definitive quantitative version that the rest of the book refers to.

The terminology "invasion–burden partition" was introduced by Hyman and Arriola [2026, Chapter 4] in the context of the time-dependent forward-sensitivity treatment of the SIR model, where it surfaces as the observation that the susceptible contact rate c_S has $\Upsilon_{c_S}^{\mathcal{R}_0} = 0$ identically but $\Upsilon_{c_S}^{i(t)}$ peaks at approximately $+0.64$ near the epidemic peak. The present monograph develops the policy-relevant consequences of the same partition: which parameters drive invasion versus which drive endemic burden, and what this implies for intervention design.

9.7.1 Defining the Partition

Given a threshold θ_{inv} for the sensitivity of $\mathcal{R}_0$ and a threshold θ_{burd} for the sensitivity of i^*, define:

- *Invasion parameters*: those with $|\Upsilon_p^{\mathcal{R}_0}| \geq \theta_{\text{inv}}$ and $|\Upsilon_p^{i^*}| < \theta_{\text{burd}}$.
- *Burden parameters*: those with $|\Upsilon_p^{i^*}| \geq \theta_{\text{burd}}$ and $|\Upsilon_p^{\mathcal{R}_0}| < \theta_{\text{inv}}$.
- *Mixed parameters*: those with $|\Upsilon_p^{\mathcal{R}_0}| \geq \theta_{\text{inv}}$ and $|\Upsilon_p^{i^*}| \geq \theta_{\text{burd}}$.
- *Negligible parameters*: those with both sensitivities below threshold.

The invasion–burden partition is the statement that, for the SIR model, the classification separates the parameters that act on invasion from those that act on burden. The separation is clean only when $\mathcal{R}_0$ is large, because the burden index of c_I and β is $1/(\mathcal{R}_0 - 1)$ (Theorem 5.6). At the book's baseline, $\mathcal{R}_0 \approx 1.92$, it is partial, as the thresholds below show: τ_m acts on burden alone, while c_I, β and τ_R act on both. Where the partition is clean, knowing which parameter is changed tells the modeler which output will respond.

9.7.2 Applying Thresholds to the Baseline

Using $\theta_{\text{inv}} = \theta_{\text{burd}} = 0.5$ (a reasonable threshold for "substantial" sensitivity) and the closed-form indices from Section 9.3 (collected in Table 9.1), the classification at the baseline parameter set is:

Parameter	$\lvert\Upsilon_p^{\mathcal{R}_0}\rvert$	$\lvert\Upsilon_p^{i^*}\rvert$	**Classification**
c_I	1.0	1.09	Mixed
β	1.0	1.09	Mixed
τ_R	1.00	2.1	Mixed (both very high)
τ_m	0.00	1.00	Burden
c_S	0	0.52	Burden
c_R	0	0.52	Burden

With these thresholds, three parameters (c_I, β, τ_R) are mixed and three (τ_m, c_S, c_R) are burden; the i^* values for c_S and c_R come from the general formula (5.21). The partition is partial.

Refining the partition with stricter thresholds. Using $\theta_{\text{inv}} = \theta_{\text{burd}} = 1.1$ (a strict threshold for dominant sensitivity):

Parameter	$\lvert\Upsilon_p^{\mathcal{R}_0}\rvert$	$\lvert\Upsilon_p^{i^*}\rvert$	**Classification**
c_I	1.0	1.09	Invasion (just below threshold)
β	1.0	1.09	Invasion (just below threshold)
τ_R	1.00	2.1	Burden (high burden, not quite dominant invasion)
τ_m	0.00	1.00	Burden (below burden threshold but dominant)
c_S	0	0.52	Negligible (below both thresholds)
c_R	0	0.52	Negligible (below both thresholds)

With stricter thresholds, the partition becomes more informative. At this cutoff a literal reading of the definitions above would place c_I, β, and τ_m in the negligible class, since each index falls just below 1.1; the table's labels instead mark which sensitivity *dominates* for each parameter, which is the operationally useful distinction. c_I and β are invasion-dominant (their $\mathcal{R}_0$ sensitivities are unity and their i^* sensitivities are only modestly above unity at the baseline); τ_R is burden-dominant (its i^* sensitivity of 2.1 is the largest in the table); τ_m is also burden-facing. At this cutoff the two contact rates c_S and c_R fall below both thresholds.

The thresholds are arbitrary cutoffs for visualizing the partition; what matters is the structural observation that τ_m contributes heavily to i^* and negligibly to $\mathcal{R}_0$, while c_I and β contribute heavily to $\mathcal{R}_0$ and only modestly to i^*.

9.7.3 A Worked Numerical Example

To make the partition concrete, consider a 10% perturbation applied to each of the four informative parameters in turn, starting from the baseline set of Table 4.4 in the equal-contact-rate case (for which $\mathcal{R}_0 \approx 1.92$ and $i^*/n^* \approx 0.000785$). For each perturbation we report the resulting $\mathcal{R}_0$, the resulting i^*/n^*, and the percentage changes. The predictions from the local sensitivity indices appear in the last two columns for comparison.

Perturbation	$\mathcal{R}_0$	$\Delta\mathcal{R}_0$ (%)	i^*/n^*	Δi^* (%)	$\Upsilon_p^{\mathcal{R}_0} \cdot 10\%$	$\Upsilon_p^{i^*} \cdot 10\%$
Baseline	1.92	0	0.000785	0	—	—
$c_I \to 1.1c_I$	2.11	+10.0	0.000863	+9.9	+10.0	+10.9
$\beta \to 1.1\beta$	2.11	+10.0	0.000863	+9.9	+10.0	+10.9
$\tau_R \to 1.1\tau_R$	2.11	+10.0	0.000949	+20.9	+10.0	+20.9
$\tau_m \to 1.1\tau_m$	1.92	+0.0	0.000714	−9.1	+0.0	−10.0

The first-order linear approximation from the sensitivity indices is accurate to within about one percentage point for all four perturbations. The partition is visible in the table: perturbing c_I or β changes $\mathcal{R}_0$ substantially and i^* only slightly more; perturbing τ_R changes i^* much more than $\mathcal{R}_0$; perturbing τ_m changes i^* appreciably and $\mathcal{R}_0$ barely at all.

Translating to an operational question: suppose a public-health agency must choose between two interventions, one that reduces c_I by 10% (say, a mild case-isolation policy) and one that reduces τ_R by 10% (say, a mild treatment program that shortens the infectious period). The first intervention reduces $\mathcal{R}_0$ from 1.92 to 1.73, moving the system toward the invasion threshold but not across it. The second intervention reduces $\mathcal{R}_0$ from 1.92 to 1.73 (effectively the same change, since τ_R enters $\mathcal{R}_0$ almost symmetrically with $c_I\beta$) but also reduces the endemic prevalence i^* from $0.000785\,n^*$ to $0.000621\,n^*$, about 21% lower. For outbreak suppression (pushing $\mathcal{R}_0$ below 1), the two interventions are nearly equivalent; for long-term burden reduction, the τ_R intervention is about 1.7 times as effective. The invasion-burden partition makes this trade-off explicit.

The example cannot settle the role of c_S and c_R, because the equal-contact case does not separate them. In the general model a 10% rise in c_S raises i^* by about 5% and a 10% rise in c_R lowers it by about 5%, half the response to c_I; the two cancel when the contacts of susceptible and recovered people change together, as under a general distancing measure. Neither enters $\mathcal{R}_0$, so both can be neglected for invasion, but not for burden.

9.7.4 Why the Partition Holds

The partition's existence is not a numerical accident. It follows from the structure of the two closed-form expressions:

- $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$. Only four parameters enter; the denominator is a sum, so the individual contributions of γ_R and ν_m depend on their ratio.

- $i^* = n^*\nu_m(\mathcal{R}_0 - 1)/[\cdots]$. All six parameters enter (in the general formula); ν_m appears outside the $\mathcal{R}_0$-factor, giving it direct influence on i^* that is not present for $\mathcal{R}_0$.

The parameter τ_m is the crucial one. In $\mathcal{R}_0$, ν_m appears only in the denominator $\gamma_R + \nu_m$, and because ν_m is typically much smaller than γ_R, its sensitivity for $\mathcal{R}_0$ is small. In i^*, ν_m appears both in the denominator (through $\mathcal{R}_0$) and directly through the prefactor $\nu_m \cdot (\mathcal{R}_0 - 1)/(\gamma_R + \nu_m)$, so its sensitivity is substantial.

This structural feature is not specific to the SIR model. Many compartmental models share the property that the demographic rates enter $\mathcal{R}_0$ only as part of a sum with faster epidemiological rates (making their $\mathcal{R}_0$ sensitivity small) but enter the endemic equilibrium directly (giving them substantial i^* sensitivity). The invasion–burden partition extends to generalizations of the SIR model; Chapter 10 touches on this.

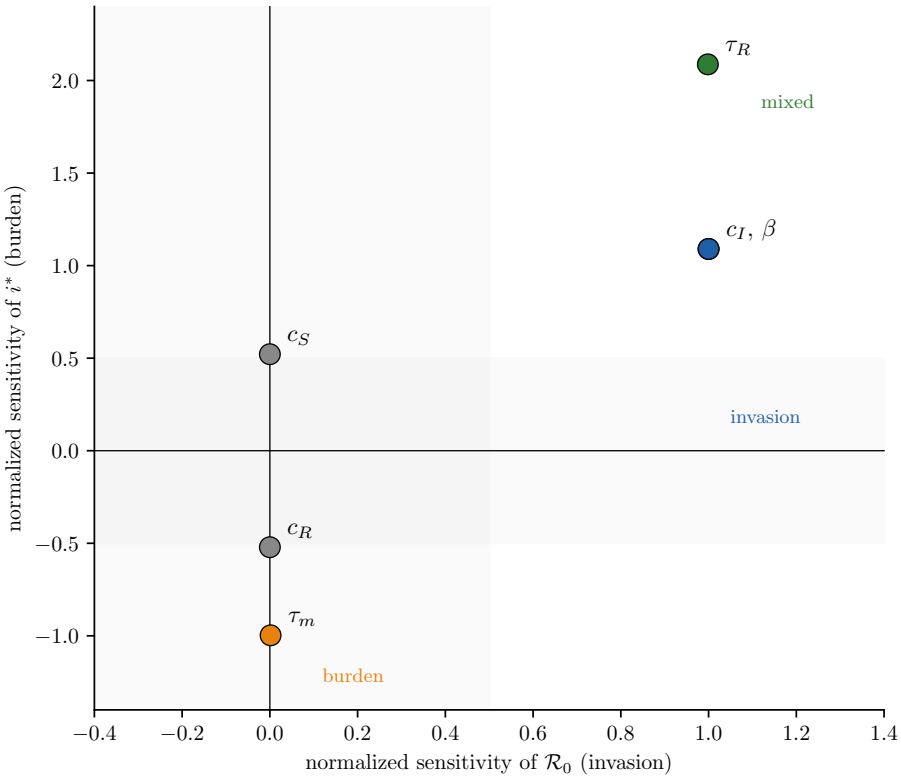


Figure 9.5: Sensitivity-space visualization of the invasion-burden partition at the baseline parameter values. Each parameter is plotted at its normalized sensitivity index for $\mathcal{R}_0$ (horizontal axis) and i^* (vertical axis). The parameters c_I and β (blue) affect $\mathcal{R}_0$ with unit sensitivity and i^* with sensitivity $1/(\mathcal{R}_0 - 1) \approx 1.09$: invasion-dominant at this baseline, and invasion-only as $\mathcal{R}_0$ grows. The parameter τ_m (orange) sits in the burden-only region: it affects i^* substantially but $\mathcal{R}_0$ negligibly. The parameter τ_R (green) affects both. The parameters c_S and c_R (gray) sit on the vertical axis at about ± 0.52 (general formula (5.21)): no effect on $\mathcal{R}_0$, and burden effects of opposite sign that cancel when the two change together. The partition is structural rather than accidental (Section 9.7.4).

9.7.5 Policy Implications

The quantitative version of the invasion–burden partition has concrete implications:

1. Interventions targeting invasion parameters (c_I, β) reduce $\mathcal{R}_0$ with unit elasticity and, near the baseline, reduce i^* by approximately $1/(\mathcal{R}_0 - 1)$ percent per percent. The elasticities do not extrapolate to interventions of realistic size. For $\mathcal{R}_0 = 1.92$, a 50% reduction in c_I takes $\mathcal{R}_0$ to 0.96, across the invasion threshold: the endemic equilibrium ceases to exist, and the actual long-run reduction in i^* is 100% (elimination), not the $\approx 55\%$ the linear index predicts. The threshold-crossing reduction is $\phi^* = 1 - 1/\mathcal{R}_0 \approx 48\%$ at the baseline—the classical critical control effort—and any invasion-parameter reduction beyond ϕ^* eliminates the disease outright (Exercise 7).

2. Interventions that lengthen the burden parameter (τ_m) reduce i^* with approximately unit elasticity and barely affect $\mathcal{R}_0$. Because the elasticity is ≈ -1, i^* scales as $1/\tau_m$ to leading order, and the honest large-perturbation statement is the hyperbolic one: a 50% *increase* in τ_m (slower demographic turnover) multiplies i^* by $\approx 2/3$, a 33% reduction, while changing $\mathcal{R}_0$ by less than 0.1%. The linear extrapolation $-1 \times 50\% = -50\%$ overstates the reduction by 17 percentage points.

3. Interventions targeting the mixed parameter (τ_R) affect both outputs: near the baseline, a 1% reduction in τ_R reduces $\mathcal{R}_0$ by about 1% and i^* by about 2.1%. A 50% reduction again crosses the invasion threshold ($\mathcal{R}_0 \to 0.96$), so the actual reduction in i^* is 100%. The linear prediction is $2.1 \times 50\% = 105\%$, an impossibility: an extrapolated elasticity prediction that exceeds 100% is the index's own announcement that the perturbation has left the linear regime.

These elasticity-based predictions are local approximations. Before applying them to a contemplated intervention of realistic size, check whether the perturbed $\mathcal{R}_0$ crosses 1: if it does, the relevant prediction is elimination, not a percentage read off a linear index—the bifurcation limitation of local analysis stated at the opening of this chapter, encountered here in policy dress. For large perturbations that do not cross the threshold, compute the actual effect from the closed forms or by simulation, and use the indices only for direction and rough magnitude.

A final caution returns to the chapter's epigraph. A sensitivity ranking orders the levers by how far they move the machine; it says nothing about which levers can be reached. A parameter can carry a large index and still be a poor intervention target: β has $\Upsilon_\beta^{\mathcal{R}_0} = +1$, yet no intervention on β can be verified in isolation, because epidemic data constrain only the product $c_I\beta$ (Section 9.4). Other parameters are uncontrollable by any feasible policy, or controllable only by measures that fail on cost, ethics, or public acceptance. The converse also holds: a mid-ranked parameter with a cheap, acceptable, and verifiable control can be the better intervention target. Sensitivity analysis supplies the leverage column of the intervention-design table; measurability, controllability, cost, and acceptability come from outside the model, and a ranking reported without them invites the misreading that the top row is the policy recommendation.

Exercises for this section

1. **[I]** Using the quantitative invasion-burden analysis of Section 9.7, predict the effect on $\mathcal{R}_0$ and i^* of a 30% reduction in c_I. Then verify your prediction by direct simulation.

9.8 Case Study: Uncertainty Quantification for the Influenza Estimate

This section extends the Chapter 7 influenza case study by quantifying the uncertainty in the fitted $\mathcal{R}_0$ estimate that follows from uncertainty in the input parameters.

9.8.1 Input Uncertainties

From the Chapter 7, Section 7.7 analysis, the fitted infected-viewpoint estimate is $\hat{\mathcal{R}}_e \approx 1.40$, the effective reproduction number for that season's partially immune population; since $\hat{\mathcal{R}}_e = \hat{\alpha}/(\gamma_R+\nu_m)$ shares the functional form of the $\mathcal{R}_0$ expression, the sensitivity indices of this chapter apply to it unchanged. The inputs to the computation are:

- Observed incidence time series (assumed exact for this case study; in practice, ascertainment uncertainty would apply).
- Infectious period $\tau_R = 7$ days. Clinical studies suggest a 95% credible interval of $[5, 10]$ days; treating the half-width of that interval as $\approx 1.96\,\sigma$ gives a relative standard deviation of roughly 18% ($\sigma_{\tau_R} \approx 1.3$ days).
- Mean residence time $\tau_m = 7300$ days (≈ 20 years), the demographic-turnover value adopted in the Chapter 7 case study. Because $\nu_m \ll \gamma_R$, it scarcely enters the $\mathcal{R}_0$ denominator; we assume it is known exactly for this exercise.

The dominant uncertainty in $\hat{\mathcal{R}}_e$ is the infectious period τ_R, through the denominator $\gamma_R+\nu_m \approx 1/\tau_R$ of the $\mathcal{R}_0$ formula.

9.8.2 Propagating the Uncertainty

Using the closed-form sensitivity index $\Upsilon^{\mathcal{R}_0}_{\tau_R} \approx 1.00$ at the baseline, an 18% relative standard deviation in τ_R propagates to an approximately 18% relative standard deviation in $\hat{\mathcal{R}}_e$. This is a one-parameter application of the delta-method theorem 9.1: with τ_R the only uncertain parameter, equation (9.18) reduces to $V^{\mathcal{R}_0} = (\Upsilon^{\mathcal{R}_0}_{\tau_R})^2\, V^{\tau_R}$, and the relative standard deviation is

$$\sigma_{\hat{\mathcal{R}}_e}/\hat{\mathcal{R}}_e \;\approx\; |\Upsilon^{\mathcal{R}_0}_{\tau_R}| \cdot \sigma_{\tau_R}/\tau_R \;\approx\; 1.00 \times 0.18 \;\approx\; 0.18.$$

The 95% credible interval for $\hat{\mathcal{R}}_e$ from the infectious-period uncertainty alone is therefore roughly $\hat{\mathcal{R}}_e \cdot (1 \pm 1.96 \times 0.18)$, or $[0.90, 1.90]$ around the point estimate of 1.40. Keeping the σ-to-interval conversion explicit is part of the discipline: quoting the standard deviation as if it were the 95% half-width (or the reverse) silently misstates the interval by a factor of two.

For comparison, the susceptible-viewpoint conversion returned values from 1.75 to 2.80 across a plausible range of F_0 values (Chapter 7, Section 7.7). The two spreads are comparable in width (1.05 versus the 1.00-wide interval above), but they are different in kind. The infected-viewpoint width is statistical: it is driven by a measurable clinical quantity, and better characterization of τ_R shrinks the interval in direct proportion (Exercise 5). The susceptible-viewpoint spread is structural: the incidence data determine the infected-viewpoint estimate directly but determine the corresponding $\mathcal{R}_0$ only through the assumed pre-season susceptible fraction F_0, an assumption the data cannot themselves adjudicate.

This comparison illustrates the book's central fitting claim: the infected-viewpoint fit is conceptually cleaner, and its uncertainty is of the reducible kind—governed by a measurable quantity rather than by an assumption that no amount of incidence data can check.

9.8.3 Formal Sensitivity Analysis Within the Case Study

A formal LHS-PRCC analysis of the influenza case study would treat both τ_R and F_0 (and perhaps other parameters) as random inputs with prior distributions, propagate them through the fitting procedure, and produce a joint posterior on $\hat{\mathcal{R}}_e$. Such an analysis is natural to perform but goes beyond the scope of this book; we note that the sensitivity-index-based propagation above gives a first-order approximation that is adequate for most practical purposes.

The qualitative point is that sensitivity analysis gives a disciplined framework for propagating input uncertainty to output uncertainty. Without it, one is left with a single point estimate whose reliability is not assessed.

Check Your Work: Before Reporting a Sensitivity Analysis

Verifies: sensitivity calculations and interpretations. Before reporting rankings from a tornado plot or a PRCC table, run three checks.

Check 1: realistic ranges. The ranges should reflect real uncertainty, not a default uniform $\pm 50\%$. Spot-check the two or three top-ranked parameters against published 1-σ estimates; a ranking driven by an arbitrary range is an artifact.

Check 2: adequate sample size. For LHS-PRCC use $M \geq 10\,d$ samples for d parameters, and $M \geq 100\,d$ for tight intervals. Recompute on a 50% subsample; if the top ranks change, M was too small, unless the parameters are tied within their sampling error, as c_I, β and τ_R are for $\mathcal{R}_0$.

Check 3: local and global agree. For the SIR model the two should largely agree (Section 9.6). A gap of more than one rank among the top three is informative: document it, and say whether the range strays from the baseline or parameters interact.

A ranking that fails a check needs a caveat that names the failure.

9.9 Summary and Exercises

This chapter has developed sensitivity analysis for the SIR model through both local and global methods.

1. The normalized sensitivity index $\Upsilon_p^q = (p/q)(\partial q/\partial p)$ is the fundamental quantity of local sensitivity analysis. It is dimensionless, comparable across parameters, and interpretable as a relative elasticity.

2. For the SIR model, closed-form sensitivity indices are available for both $\mathcal{R}_0$ and i^*. The $\mathcal{R}_0$ indices are $\Upsilon_{c_I}^{\mathcal{R}_0} = \Upsilon_{\beta}^{\mathcal{R}_0} = 1$, $\Upsilon_{\tau_R}^{\mathcal{R}_0} = \gamma_R/(\gamma_R + \nu_m)$, $\Upsilon_{\tau_m}^{\mathcal{R}_0} = \nu_m/(\gamma_R + \nu_m)$, $\Upsilon_{c_S}^{\mathcal{R}_0} = \Upsilon_{c_R}^{\mathcal{R}_0} = 0$. The i^* indices (in the equal-contact-rate case) are more involved but equally closed-form; in the general model c_S and c_R enter i^* as well, with indices of about ± 0.52 at the baseline.

3. Tornado plots visualize sensitivity indices as ordered horizontal bars, giving an at-a-glance ranking of parameter importance. Side-by-side tornado plots for two outputs (e.g., $\mathcal{R}_0$ and i^*) reveal the structural partition of parameters.

4. Global sensitivity analysis via Latin hypercube sampling and partial rank correlation coefficients integrates over the full parameter uncertainty distribution. For the SIR model, the global rankings closely match the local rankings, validating the local analysis.

5. The invasion–burden partition is quantitative: the dominant contributors to $\mathcal{R}_0$ (c_I, β, and to a lesser extent τ_R) are largely disjoint from the dominant contributor to i^* at large $\mathcal{R}_0$ (τ_m), with τ_R serving as a mixed contributor. This partition has direct implications for intervention design.

6. Applied to the Chapter 7 influenza case study, the sensitivity framework gives an explicit uncertainty bound on the fitted $\mathcal{R}_e$: approximately $[0.90, 1.90]$ at 95% credibility, driven predominantly by uncertainty in the infectious period.

Subsequent chapters extend the sensitivity framework to more complex models. Chapter 10 treats SIR variants (loss of immunity, latent periods, disease-induced mortality); Chapters 11 through 13 treat heterogeneous populations with multiple infectious compartments; Chapters 17–19 work through three end-to-end case studies, synthesized in Chapter 20, in which the sensitivity framework plays a central role.

From $\mathcal{R}_0$ sensitivity to $\mathcal{R}_e$ sensitivity in the case studies. The closed-form indices in this chapter are computed at the disease-free equilibrium (DFE), where the susceptible fraction is $s/n \approx 1$ and the basic reproductive number $\mathcal{R}_0$ governs invasion. In an unfolding outbreak, the appropriate quantity is the effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0 \cdot s(t)/n$, introduced in Chapter 5, Section 5.2. The sensitivity machinery developed here transfers directly: the indices $\Upsilon_p^{\mathcal{R}_e}$ inherit the parameter dependencies of $\mathcal{R}_0$, modulated by whatever factor $s(t)/n$ takes at the time of evaluation. The case-study chapters use this extension in two complementary ways. First, the COVID, dengue, and HIV chapters (Chapters 17–19) work with intervention-modified reproductive numbers labelled by intervention, written as $\mathcal{R}_0^{\text{mask}}$, $\mathcal{R}_0^{\text{isolate}}$, $\mathcal{R}_0^{\text{combo}}$ for COVID-style interventions and as $\mathcal{R}_0^{\text{horiz}}$, $\mathcal{R}_0^{\text{vert}}$ for the HIV horizontal/vertical transmission decomposition; sensitivity indices on these labelled quantities answer "which intervention parameters most reduce $\mathcal{R}_0$ under modification?". Second, the time-varying $\mathcal{R}_e(t)$ produced naturally by the renewal-equation formulation of Chapter 15 (Section 15.5.3) is the surveillance-practice quantity referenced in Chapter 7's "Beyond the early-outbreak fit" discussion; sensitivity of $\mathcal{R}_e(t)$ to its inputs is what permits real-time uncertainty quantification of changing transmission. Both uses preserve the local-sensitivity workflow established in this chapter.

Identifiability scaling as the parameter space grows. The SIR sensitivity framework developed here treats a six-parameter model and enjoys the favorable feature that incidence data, combined with independent estimates of the natural timescales τ_R and τ_m, pins down the one degree of freedom that governs invasion ($\mathcal{R}_0$ or equivalently α). As the extensions of Part IV introduce more biological structure, the parameter count grows and identifiability degrades, not always gracefully. A few patterns are worth recognizing in advance. Mixing-matrix models add independent off-diagonal contact rates per group pair, constrained only by the reciprocity relation (one independent rate per pair rather than two), so a two-group model adds one new parameter but a ten-group model adds nine. Vector-host models add vector demography and an extrinsic incubation period, typically three to five additional parameters, usually estimated from entomological data independent of the human incidence stream. Age- and activity-stratified sexual-transmission models can reach twenty or more parameters before any fitting begins. A recurring operational reality is that the number of

jointly identifiable parameters from incidence data alone rarely exceeds two or three, regardless of the model's dimension. The practical consequence is that most parameters in Part IV extensions are taken from independent data sources (demographic, clinical, entomological) and treated as fixed inputs, not as quantities to be estimated from outbreak incidence. The sensitivity framework then answers a different question: given these fixed inputs, how sensitive are the model's outputs to the values chosen? The case studies of Part V illustrate this pattern concretely: the COVID case study (Chapter 17) fits two or three parameters from incidence while fixing the rest; the HIV case study (Chapter 19) fits perhaps five from combined seroprevalence and incidence data across sex-activity strata, with the remainder from independent clinical and demographic sources.

Sensitivity analysis for chronic-sequelae outputs: an active research frontier. The sensitivity-analysis framework developed here addresses outputs ($\mathcal{R}_0$, i^*, $\mathcal{R}_e(t)$) that are functions of the acute-infection compartmental model. A growing literature uses the same machinery to analyze *chronic-sequelae outputs* that depend on the acute model only indirectly: prevalence of long-COVID or post-acute sequelae of SARS-CoV-2 infection (PASC), prevalence of chronic outcomes after a recovered acute infection, vaccine-impact differentials in chronic-disease populations. Beams et al. [2024] apply the local-sensitivity framework to PASC trajectory predictions and identify a structural partition of parameters that is the chronic-sequelae analog of the invasion–burden partition: some parameters (the duration and equilibrium prevalence of acute infection, the fraction of infections that progress to chronic sequelae) dominate chronic-output trajectories, while others (the duration of immunity, secondary vaccine efficacy against chronic sequelae) contribute only modestly. The methodology transfers without modification; what changes is the operationally-relevant output. The implication is that the sensitivity-discipline framework of this chapter has direct application to chronic-sequelae modeling questions even though the book does not develop the chronic-sequelae compartmental extensions.

Pitfall check

This chapter has developed two of the book's common pitfalls: Pitfall: "Sensitivity Analysis Discipline" (formalized in Section 9.2 as the two commitments any sensitivity analysis must make: choose the right output, choose the right parametrization), and Pitfall: "Invasion versus Equilibrium" (made fully quantitative in Section 9.7 as the structural partition of parameters into invasion-controlling and burden-controlling groups). Before moving to the extension chapters of Part IV, take a moment to confirm the following points are clear in your mind. *Why is the natural sensitivity index computed with respect to τ_R rather than γ_R, even though both parametrizations describe the same model? What is the structural-partition observation, and why does it imply that an intervention designed to reduce $\mathcal{R}_0$ may have a different effect on endemic burden than the reduction in $\mathcal{R}_0$ alone would suggest? How does the local-vs-global sensitivity distinction affect what conclusions one can draw from a tornado plot?* If any of these are unclear, the corresponding pitfall block (or the catalog entry, accessible via the List of Common Pitfalls) is the place to revisit.

Exercises

Four section-level exercises appear above, following Section 9.2, Section 9.3, Section 9.6, and Section 9.7. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab. See Chapter 7 Exercises for the general scheme.*

1. **[C]** State the sensitivity asymmetry $\Upsilon_s^{\hat{\alpha}} = 0$ vs $\Upsilon_s^{\hat{\lambda}} = -1$ in one sentence that a reader who has not seen the derivation would understand. Then explain in one further sentence why this asymmetry is only the sensitivity-theoretic half of the fitting-advantage argument in the chapter's central-result block, and what the other half is.

2. **[D]** Verify equations (9.9) through (9.12) for the sensitivity of i^* by direct differentiation of the equal-contact-rate formula (5.23).

3. **[D]** Derive the sensitivity indices $\Upsilon_p^{i^*}$ for the general-contact-rate case starting from the full expression (5.21). Compare your indices at the baseline parameter set to the equal-contact approximation values given in Section 9.3. By how much do they differ?

4. **[L]** Construct tornado plots for the epidemic peak height i_{peak} and peak time t_{peak} using numerical differentiation. How do the dominant contributors to these outputs compare with those of $\mathcal{R}_0$ and i^*?

5. **[I]** For the influenza case study, suppose the relative standard deviation of τ_R is reduced from 18% to 10% through better clinical data. Compute the resulting reduction in the uncertainty of $\hat{\mathcal{R}}_e$. By what factor does the 95% credible interval shrink?

6. **[D]** At what value of $\mathcal{R}_0$ does the first-order local sensitivity prediction for $\Upsilon_{c_I}^{i^*}$ break down? (Hint: consider values of $\mathcal{R}_0$ very close to 1 where $1/(\mathcal{R}_0 - 1)$ diverges.)

7. **[D]** The Policy Implications list of Section 9.7 warns that elasticity predictions fail when an intervention crosses the invasion threshold. Make the warning exact. (a) Show that reducing an invasion parameter (c_I or β) by the fraction $\phi^* = 1 - 1/\mathcal{R}_0$ brings $\mathcal{R}_0$ exactly to 1, and evaluate ϕ^* at the baseline. (b) Show that for any reduction $\phi > \phi^*$ the endemic equilibrium ceases to exist, so the actual reduction in i^* is 100%, and compare with the linear-elasticity prediction $\Upsilon_{c_I}^{i^*} \cdot \phi$ over $\phi \in (0, 1)$: at what ϕ does the linear prediction first err by more than 10 percentage points? (c) Explain why ϕ^* coincides with the classical critical control effort familiar from vaccination-threshold arguments, and why the coincidence is exact for the invasion parameters but only approximate for τ_R.

8. **[L]** *Spaced retrieval, Chapter 6.* The closed-form sensitivity indices of this chapter are derived analytically. They can also be *measured* numerically by perturbing one parameter at a time and observing the response in $\mathcal{R}_0$ and i^*. Reach back to the Scenario 1 simulation setup of Chapter 6, Section 6.3 (using the baseline parameters of Table 4.4) and execute the following finite-difference verification:

   (a) For each of the six independent parameters $\{c_I, c_S, c_R, \beta, \tau_R, \tau_m\}$, simulate the SIR system at the baseline value and at $1.05\times$ baseline (a 5% perturbation). From the two resulting trajectories, extract the long-time endemic-prevalence value i^* and compute the relative finite-difference sensitivity $\hat{\Upsilon}_p^{i^*} \approx (\Delta i^*/i^*)/(\Delta p/p)$.

   (b) Compare $\hat{\Upsilon}_p^{i^*}$ to the closed-form $\Upsilon_p^{i^*}$ derived in Section 9.3. The two should agree to within the discretization error (which depends on the perturbation size: too small and round-off dominates, too large and the linearization breaks down). For c_S and c_R, compare instead with the general-model indices of Exercise 3.

   (c) Repeat (a) with a 10% perturbation. Confirm that the finite-difference estimate changes for the most-nonlinear parameters (typically c_I and β, which enter i^* through $\mathcal{R}_0 - 1$ in

the denominator) more than for the linearly-acting parameters (typically τ_R; τ_m enters i^* through $1/\tau_m$ and curves as much as c_I).

This exercise turns the analytical results of this chapter into a numerical-verification protocol that any student can execute against their own SIR implementation. It also reinforces that the closed-form sensitivity machinery of this chapter is the algebraic shortcut for what would otherwise require a parameter-perturbation study like the one developed in Chapter 6.

9. **[I]** *Reasoning visibility.* A reviewer of your manuscript writes: "Local sensitivity analysis is a weak technique. The authors should redo their analysis using global sensitivity (LHS-PRCC) before this paper is acceptable." Construct a defense (or concession) of your local-sensitivity approach in three paragraphs:

   (1) Explain why local sensitivity at a well-justified baseline is a valid first pass for SIR-type models, citing the algebraic-tractability and unique-baseline arguments of this chapter (Section 9.3). Identify the specific assumption under which the local indices give the right ranking globally, and state when that assumption is and is not safe.

   (2) Identify two specific failure modes where local sensitivity would mislead and global sensitivity would catch the error: one nonlinearity-driven (the model output depends nonlinearly on a parameter over the uncertainty range, so the local index at the baseline does not generalize), and one interaction-driven (two parameters covary, so changing one and recomputing the local index of the other gives a different ranking than the joint variation). Reference Section 9.6 for the global-method tooling that detects these.

   (3) State the modeler's reasoning rule: *when do you escalate from local to global?* Give two or three observable signs (in either the model structure, the parameter ranges of interest, or the sensitivity-index magnitudes already computed) that should trigger an escalation, and one observable sign that suggests local is sufficient.

   The exercise asks you to articulate, in your own voice, the reasoning that distinguishes a routine sensitivity analysis from an inadequate one. Internalizing this distinction is what separates a student who can compute a tornado plot from one who can use it to defend a modeling conclusion.

10. **[Verify the AI]** *Tornado-plot interpretation audit.* An AI tool produces a tornado plot for the SIR baseline showing τ_R as the most sensitive parameter, with β and c_I following. The AI's accompanying interpretation reads: "The recovery rate τ_R is the highest-priority intervention target, since it has the largest sensitivity. Public-health interventions that shorten the infectious period are predicted to have the largest impact on $\mathcal{R}_0$." Audit this interpretation:

   (a) *Verify the ranking against closed-form indices.* From (9.7) and (9.3), the closed-form indices are $|\Upsilon^{\mathcal{R}_0}_{\tau_R}| \approx 1.00$ and $|\Upsilon^{\mathcal{R}_0}_{c_I}| = |\Upsilon^{\mathcal{R}_0}_{\beta}| = 1.0$ (with c_I and β at exactly 1 and τ_R marginally below). Does the AI's claimed ranking match? Identify the discrepancy.

   (b) *Audit the policy claim.* The AI conflates "most sensitive" with "highest-priority intervention target." These are not the same: an intervention's priority depends on (i) the parameter's sensitivity, (ii) the operational tractability of changing the parameter, and (iii) the parameter's existing uncertainty. State which of (i)–(iii) the AI's interpretation incorporates and which it ignores. For each ignored factor, give an example where it would override the sensitivity ranking.

(c) *Repair the recommendation.* Rewrite the AI's policy paragraph to incorporate all three factors, using the SIR baseline as a worked example. The corrected recommendation should distinguish "the most sensitive parameter at this baseline" from "the most policy-relevant parameter," and should identify which is most useful for which kind of decision.

The exercise practices the audit habit on the most consequential AI failure mode in sensitivity-analysis output: presenting rankings as if they directly translate to policy without flagging the operational and uncertainty caveats.

11. **[AI Lab]** *Tornado-plot generation workflow.* For the SIR baseline parameters of Chapter 4 Table 4.4, execute the three-step audit-and-refine workflow from this chapter's AI Lab box. Specifically:

(a) *Parameter-range elicitation.* Prompt an AI tool: "For the SIR model parameters $\{c_I, c_S, c_R, \beta, \tau_R, \tau_m\}$ in the context of seasonal influenza in a U.S. state population, what is the published 1-σ uncertainty range for each parameter?" Save the response. Audit by spot-checking 2–3 of the AI's claimed ranges against the cited sources (search the cited papers if available; check whether the cited values match the claimed ranges).

(b) *Tornado computation.* Using the AI-elicited ranges from (a), compute the response of $\mathcal{R}_0$ to each parameter at the upper and lower bounds. Generate the tornado plot using your preferred plotting environment. Compare the plot's ranking to the closed-form indices from (9.3)–(9.8): do the rankings match?

(c) *Iteration on disagreement.* If the rankings disagree between the AI-elicited ranges and the closed-form indices, the most likely cause is that the AI's ranges did not represent realistic uncertainty (e.g., uniform ±50% instead of pathogen-specific). Iterate: prompt the AI again with the chapter-specific criterion that ranges should reflect 1-σ uncertainty from the published literature, and recompute. Document whether the iteration resolves the disagreement.

Disclosure: per Appendix D, Section D.4, document the AI system, prompts, dates, and audit findings.

12. **[AI Lab]** *LHS-PRCC verification workflow.* Implement an LHS-PRCC analysis for $\mathcal{R}_0$ in the SIR model and execute the chapter's Check Your Work box as a practiced workflow:

(a) *Setup.* Use the baseline parameters from Chapter 4 Table 4.4 with each parameter varied within a factor of 2 above and below baseline. Run with three sample sizes: $M = 100$, $M = 1000$, $M = 10000$.

(b) *Sample-size adequacy check.* Compute PRCC values for each sample size. Compute the sub-sample stability: split each M in half, recompute on each half, and compare the top-3 rankings. At which sample size does the sub-sample test pass (rankings stable across halves)?

(c) *Local-vs-global comparison.* Compare your large-M PRCC rankings to the closed-form local rankings from Section 9.3. Do they agree on the top-3 parameters? Document any disagreement and identify the cause from the chapter's three options (parameter range too wide; parameter interaction; sampling noise).

(d) *AI-assisted interpretation audit.* Prompt an AI tool to interpret your large-M PRCC results. Audit the interpretation against the chapter's policy-vs-sensitivity distinction

(the previous exercise's lessons): does the AI's interpretation conflate sensitivity rankings with policy priorities, or does it correctly distinguish them?

The exercise practices the full workflow of running a sensitivity analysis with adequate verification, then turning AI tools loose on the interpretation while maintaining audit discipline. Disclosure per Appendix D, Section D.4.

13. **[Open Project]** *Full sensitivity-driven intervention recommendation.* A multi-week capstone project using local and global sensitivity analysis to drive an intervention recommendation for a specific epidemic-modeling problem, with quantified uncertainty in the ranking and a written recommendation memo for a policy audience. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈15–25 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch10-sensitivity-intervention/`.

Part IV

Heterogeneous Mixing: Getting the Contact Structure Right and Interpreting the Reproductive Numbers

Chapter 10

Generalizations of the SIR Model

Start with the simplest problem where you know the answer.

Part IV opens the SIR *framework to the biological variety it must accommodate in practice. The six chapters that follow each extend the framework to a distinct source of heterogeneity: variant compartmental structure (Chapter 10), two-group mixing with interacting subpopulations (Chapter 11), bipartite sexual transmission (Chapter 12), bipartite vector-host transmission (Chapter 13), vertical transmission across generations (Chapter 14), and continuous age or time-since-infection structure via partial differential equations (Chapter 15). A reader whose interest centers on a specific pathogen class can go directly to the relevant chapter: respiratory diseases typically require Chapter 10 (disease-management compartments, asymptomatic infection) plus Chapter 11 (age or activity stratification); sexually-transmitted infections require Chapter 12 and often Chapter 11; vector-borne diseases require Chapter 13; perinatal transmission requires Chapter 14; and analyses that require explicit time-since-infection structure (variable infectiousness profiles, survival-style analyses) require Chapter 15. The framework's three central results (the force-from-infection formulation, the invasion-burden partition, and the infected-viewpoint fitting advantage) carry into each extension without structural change, a uniformity synthesized in Chapter 20.*

The three-compartment SIR model developed in Chapter 4 captures the essential threshold behavior of an epidemic: an outbreak either invades or dies out according to whether $\mathcal{R}_0$ exceeds unity, and if it invades, it approaches a well-defined endemic equilibrium. For many pathogens, the three-compartment structure is enough. For others, additional epidemiological features demand explicit representation in the compartmental structure.

This chapter presents the most common extensions, each corresponding to a well-characterized biological feature: loss of immunity after recovery (Section 10.2), a latent period between exposure and infectiousness (Section 10.3), a disease-management compartment representing quarantine or hospitalization (Section 10.4), disease-induced mortality (Section 10.5), and asymptomatic infection (Section 10.6). Each section gives the biology, the compartmental extension, the new parameters, the adjusted reproductive number, and brief analytical comments. The treatment is brisk by design: each extension deserves a chapter's worth of analysis in its own right, and the specialist literature provides that depth. Our aim is to show how the SIR framework accommodates these extensions without conceptual strain, and to enumerate the common patterns.

A common thread runs through the chapter. Each extension adds compartments (and hence state variables) to the model, but the principles of derivation from first principles, the structure of the reproductive number, and the invasion–burden partition continue to apply. The next-generation

matrix method of Chapter 5, Section 5.3 becomes essential here: the heuristic individual-tracking derivation of $\mathcal{R}_0$ does not generalize cleanly to models with multiple infectious compartments, while the NGM method works mechanically for each case.

10.1 When the Basic SIR Model is Insufficient

Before presenting extensions, we enumerate the specific biological features that typically warrant a more elaborate model.

Loss of immunity. The basic SIR model assumes that recovery confers permanent immunity. This is accurate for some pathogens (measles, mumps, smallpox) but inaccurate for others (seasonal influenza, most coronaviruses, many sexually-transmitted pathogens). When immunity wanes over time, recovered individuals eventually return to the susceptible compartment, and the model requires an $r \to s$ flow. Section 10.2 addresses this.

Latent period. After exposure, individuals are typically not immediately infectious: the pathogen must establish itself and reach a threshold of viral or bacterial load before the host can transmit. This delay, called the latent period (distinct from the incubation period, which is the delay until symptoms), is short (hours to days) for some pathogens and long (weeks to months) for others (tuberculosis, HIV). When the latent period is comparable to or longer than the infectious period, it must be modeled explicitly. Section 10.3 addresses this through the introduction of an exposed compartment E.

Disease-management compartments. Public-health interventions often remove infectious individuals from general circulation: quarantined contacts, hospitalized patients, isolated cases. These individuals remain infectious in some sense (they can transmit to healthcare workers, family members, or other patients) but their contact structure with the general population is dramatically different. Modeling them with a separate compartment is typically warranted. Section 10.4 addresses this through a quarantine compartment Q.

Disease-induced mortality. The basic SIR model aggregates deaths (disease-induced and natural) into the migration-out rate ν_m. For pathogens with non-negligible case-fatality ratios (Ebola, severe influenza strains, pre-antibiotic-era bacterial pneumonia), explicit tracking of disease mortality matters for both the model's population accounting and for matching the data streams of outbreak response, which typically include death counts. Section 10.5 addresses this.

Asymptomatic infection. Some pathogens produce infection without symptoms in a substantial fraction of infectious individuals (norovirus, influenza, SARS-CoV-2). Asymptomatic individuals typically have different contact rates than symptomatic ones (they do not self-isolate) and may have different transmission rates. Explicit modeling of a separate asymptomatic compartment A is often warranted. Section 10.6 addresses this.

Backward bifurcation. The basic SIR model has a *forward* bifurcation at $\mathcal{R}_0 = 1$: the disease-free equilibrium exchanges stability with a newly-appearing endemic equilibrium as $\mathcal{R}_0$ crosses unity from below, and the endemic equilibrium exists only for $\mathcal{R}_0 > 1$. Some epidemiological mechanisms produce the qualitatively different pattern of *backward bifurcation*: a stable endemic equilibrium

exists for a range of $\mathcal{R}_0$ values below 1, coexisting with the stable disease-free equilibrium. Backward bifurcation arises when the model contains a positive-feedback mechanism that amplifies infection at intermediate prevalence; standard sources [Hadeler and van den Driessche, 1997, van den Driessche and Watmough, 2000] include imperfect vaccination that provides only partial protection, treatment saturation in which the per-capita treatment rate drops as caseload rises, differential susceptibility between partially-immune and fully-susceptible compartmentes, and vector-host interactions with saturation effects. The practical consequence is that reducing $\mathcal{R}_0$ below 1 is necessary but not sufficient for elimination; the endemic equilibrium can persist until $\mathcal{R}_0$ is reduced below a lower threshold $\mathcal{R}_0^c < 1$, producing a hysteresis in which the system does not return to the disease-free state by a smooth reversal of the intervention. Chitnis et al. [2006] analyze this phenomenon in a malaria vector-host model and provide a concrete setting in which the two thresholds $\mathcal{R}_0$ and $\mathcal{R}_0^c$ can be identified. This phenomenon is beyond the scope of the basic SIR model but can appear in any of the richer models developed later in the book when the relevant feedback mechanisms are present.

What this chapter does not cover. Several generalizations are deferred to later chapters of the book: heterogeneous populations with multiple distinct groups (Chapter 11), sexually-transmitted infections with sex-specific dynamics (Chapter 12), vector-borne diseases with coupled vector and host populations (Chapter 13), vertical transmission from mother to child (Chapter 14), and age-structured models with time-since-infection (Chapter 15). These are substantive topics in their own right and receive dedicated chapters. The extensions in this chapter are *homogeneous* compartmental extensions: the population remains well-mixed, and the extensions only refine the internal structure of how individuals progress through the disease.

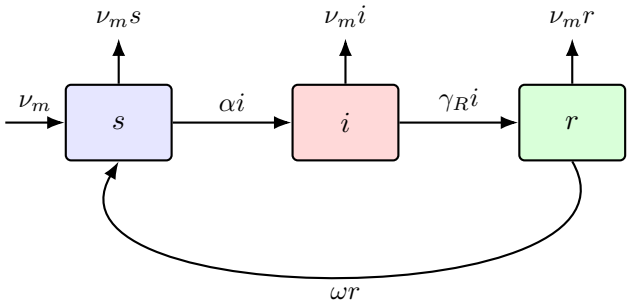


Figure 10.1: The SIRS model adds a waning-immunity flow $r \to s$ at rate ω to the basic SIR structure. Background inflow (ν_m into s) and outflows (rate ν_m from each compartment) model demographic turnover.

10.2 Loss of Immunity: The SIRS Model

If recovered individuals lose their immunity after some characteristic time, they return to the susceptible compartment and may be reinfected. The modification of the SIR model is straightforward: add a flow from r to s at rate ω per recovered individual per day, where $\omega = 1/\tau_{\text{imm}}$ is the waning rate and τ_{imm} is the mean duration of immunity. The compartmental structure is shown in Figure 10.1.

Identifiability of ω from outbreak data. The waning rate ω is not identifiable from early-outbreak incidence data alone: at outbreak onset, the recovered population is small ($r(t) \ll s(t)$), the waning return-flow ωr is therefore small relative to the primary dynamics, and the trajectory is approximately the same as the SIR model with $\omega = 0$. Identifying ω requires data from a regime where $r(t)$ is large enough that the return flow contributes appreciably: post-peak incidence dynamics (where the susceptible pool is replenished by waning rather than only by demographic turnover), longitudinal seroprevalence surveys (which directly measure the depletion of r), or repeat-outbreak data (where the inter-outbreak interval encodes $1/\omega$). Independent estimation from immunological assays (decay of neutralizing-antibody titers, reinfection rates in cohort studies) is the most direct route and is the standard primary source for ω in the literature.

The resulting SIRS model is

$$\dot{s} = \nu_m - \alpha i + \omega r - \nu_m s, \tag{10.1a}$$

$$\dot{i} = \alpha i - \gamma_R i - \nu_m i, \tag{10.1b}$$

$$\dot{r} = \gamma_R i - \omega r - \nu_m r, \tag{10.1c}$$

with $\alpha = c_I \beta P_S$ as before. The new term ωr appears as an outflow of the r compartment and as an inflow to the s compartment.

Parameter interpretation. The new parameter ω (with units day^{-1}) has dimensional inverse time. The mean duration of immunity $\tau_{\text{imm}} = 1/\omega$ is of the order of months for many coronaviruses, years for influenza (where waning is complicated by antigenic drift), and decades for some other pathogens. When $\omega \to 0$ (permanent immunity), the model reduces to the basic SIR; when $\omega \to \infty$ (instantaneous loss of immunity), it reduces to the classical SIS model with no r compartment.

Contact rates for re-susceptibles. The model above uses a single c_S for the susceptible compartment, which tacitly treats the $r \to s$ returnees as behaviorally identical to never-infected susceptibles. Within the compartment-specific contact-rate framework of Chapter 4, waning immunity admits a refinement: returnees may retain contact patterns from the r compartment (c_R) rather than reverting fully to baseline (c_S). For pathogens where prior infection reinforces protective behavior (continued mask use, vaccination uptake, awareness of risk), $c_{S,\text{returnee}}$ may sit closer to c_R than to c_S; for pathogens where waning coincides with behavioral fatigue, the reversion to c_S may be complete. The simplest refinement stratifies the s compartment into s_{naive} and s_{returnee} with distinct contact rates, at the cost of one additional ODE; our uniform-c_S formulation here takes the aggregated baseline as the default.

Reproductive number. Because the $r \to s$ flow occurs entirely within the non-infectious compartments, it does not affect the initial growth of an outbreak: at the disease-free equilibrium, $r = 0$ and the waning term is absent. The heuristic individual-tracking argument of Chapter 5, Section 5.1 applies without modification to the SIRS case, because an exposed-to-infectious transition has not been introduced and the infectious-period residence time $1/(\gamma_R + \nu_m)$ is unchanged from the basic SIR model. The waning rate ω governs how previously-recovered individuals return to the susceptible pool but does not alter the secondary-infection count of a newly-introduced infectious individual. The basic reproductive number of the SIRS model is therefore identical to that of the SIR model:

$$\mathcal{R}_0 \;=\; \frac{c_I \beta}{\gamma_R + \nu_m}. \tag{10.2}$$

The waning rate ω does not appear in $\mathcal{R}_0$. It does, however, affect the endemic equilibrium and the dynamics of the approach to equilibrium.

The endemic equilibrium. In the equal-contact case ($c_S = c_I = c_R$), setting all time derivatives to zero in (10.1) gives

$$i^*_{\text{SIRS}} \;=\; \frac{\nu_m + \omega}{\nu_m + \omega(1 - \gamma_R/(\gamma_R + \nu_m))} i^*_{\text{SIR}},$$

where i^*_{SIR} is the endemic prevalence in the basic SIR model with the same $\mathcal{R}_0$. For $\omega \ll \nu_m$, the correction factor is close to unity and the SIRS equilibrium is close to the SIR equilibrium. For $\omega \gg \nu_m$, the correction factor approaches γ_R/ν_m, which can be substantially larger than 1, giving a much higher endemic prevalence than the basic SIR model predicts. With the book's unequal contact rates the formula is an approximation: at the baseline it falls short of the computed endemic prevalence by less than 3 percent for immunity lasting 180 days or more.

The practical implication: for pathogens with waning immunity, the basic SIR model underestimates the long-term prevalence. Seasonal influenza is a canonical example. Figure 10.7(b) in Section 10.7 shows the effect at the baseline.

Oscillatory dynamics. The SIRS model can exhibit damped oscillations in its approach to equilibrium with a period substantially different from the SIR model. For $\omega \gg \nu_m$, the period is governed by ω rather than ν_m, giving shorter periods (months rather than years). Some seasonal dynamics of influenza can be traced to this mechanism.

Exercises for this section

1. [**D**] Verify by direct computation that the SIRS model reduces to the basic SIR model when $\omega = 0$, and to an SIS model when $\omega \to \infty$. For the intermediate case $\omega = \nu_m$, compute the endemic prevalence using the formula given in Section 10.2 and compare with the SIR endemic prevalence at the same $\mathcal{R}_0$.

2. [**D**] *Backward bifurcation in* SIRS *with imperfect protection.* Consider an extension of the SIRS model in which recovered individuals can be reinfected through contacts with infectious people, with per-contact transmission probability $\beta_R = \theta\beta$, $\theta \geq 0$ ($\theta = 0$ recovers the baseline SIRS; $\theta = 1$ means recovery confers no protection). New infections then arrive at the rate $\beta c_I i\,(c_S s + \theta c_R r)/C$, with $C = c_S s + c_I i + c_R r$. Write the equilibrium condition for i^* and show that for $\theta \leq 1$ no endemic equilibrium exists when $\mathcal{R}_0 \leq 1$, so the bifurcation is forward. Then show that for θ above a critical value $\theta_c > 1$ (recovered individuals more susceptible than naive ones) an endemic equilibrium can exist below $\mathcal{R}_0 = 1$, signalling a *backward bifurcation*: the basic-framework prediction "reduce $\mathcal{R}_0$ below 1 and the disease will die out" can then fail. Find θ_c numerically at the baseline as $\mathcal{R}_0 \to 1^-$, for a mean immunity of 1 year and of 10 years.

10.3 Latent Period: The SEIR Model

Many pathogens have a latent period during which an exposed individual carries the pathogen but is not yet infectious. The natural extension of the SIR model introduces an exposed compartment E between s and i (Figure 10.2):

$$\dot{s} = \nu_m - \alpha i - \nu_m s, \tag{10.3a}$$

$$\dot{E} = \alpha i - \gamma_E E - \nu_m E, \tag{10.3b}$$

$$\dot{i} = \gamma_E E - \gamma_R i - \nu_m i, \tag{10.3c}$$

$$\dot{r} = \gamma_R i - \nu_m r, \tag{10.3d}$$

where $\gamma_E = 1/\tau_E$ is the rate of progression from exposed to infectious, with τ_E the mean latent period. The force from infection α now drives individuals from s into E rather than directly into i.

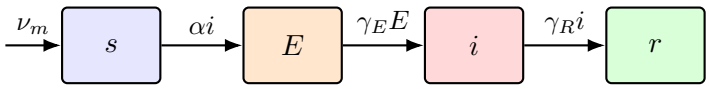


Figure 10.2: The SEIR model inserts an exposed compartment E between s and i to capture the latent period between infection and onset of infectiousness. The force from infection α drives individuals $s \to E$; progression to infectiousness occurs at rate $\gamma_E = 1/\tau_E$. Demographic outflows at rate ν_m from each compartment are omitted for visual clarity.

Parameter interpretation. The latent period τ_E is measured in days or weeks for typical respiratory or gastrointestinal pathogens. For SARS-CoV-2, τ_E is approximately 3–5 days. For measles, τ_E is approximately 10–12 days. For HIV, the latent period (from exposure to detectable viral load) is substantially longer but the model's interpretation of "infectious" shifts because HIV infection has multiple stages.

The SEIR family has seen extensive development for respiratory pathogens. Early applications to seasonal influenza refined the latent-period parameterization against outbreak data [Agyeman et al., 2004], and avian-influenza modelers extended the structure to capture cross-species transmission dynamics [Chen et al., 2012]. During the COVID-19 pandemic, the SIDARTHE model [Giordano et al., 2020] elaborated the SEIR structure with compartments distinguishing detected and undetected infections, illustrating how the base template accommodates disease-management complications without losing its analytical tractability.

Reproductive number via the heuristic individual-tracking argument. The SEIR model has a single infectious compartment (i), with the latent compartment E serving only as a waiting room before an exposed individual can transmit. The heuristic argument of Chapter 5, Section 5.1 applies with one modification: track an average newly-infectious individual from the moment of exposure through the latent period to the infectious period, and accumulate the expected number of secondary infections produced during the infectious period. The modification is that not every exposed individual survives to become infectious: demographic turnover removes a fraction $\nu_m/(\gamma_E+\nu_m)$ of exposed individuals before they progress to i, while the remaining fraction $\gamma_E/(\gamma_E+\nu_m)$ does progress. Given progression, the expected number of secondary infections follows the SIR calculation: contact rate c_I, transmission probability β per contact with a susceptible (and at the disease-free equilibrium almost all contacts are with susceptibles), residence time $1/(\gamma_R+\nu_m)$ in the infectious compartment. Multiplying the progression probability by the conditional number of secondary infections gives

$$\mathcal{R}_0 \;=\; \underbrace{\frac{\gamma_E}{\gamma_E+\nu_m}}_{\text{progression probability}} \cdot \underbrace{\frac{c_I\beta}{\gamma_R+\nu_m}}_{\text{secondary infections given progression}} \;=\; \frac{c_I\beta}{\gamma_R+\nu_m}\cdot\frac{\gamma_E}{\gamma_E+\nu_m}. \tag{10.4}$$

The next-generation matrix calculation below produces the same expression by mechanical computation; we include both because the heuristic argument makes the role of each parameter transparent, and the NGM calculation illustrates the systematic method that extends to models where individual-tracking arguments are harder to construct (Pitfall: "NGM Applicability").

Reproductive number via the next-generation matrix. The SEIR model has two infectious-compartment-adjacent states (E and i), though strictly only i transmits. The next-generation matrix approach of Chapter 5, Section 5.3 applies, and that section works this example in full; we restate the result. Writing the dynamics of (E, i) near the disease-free equilibrium as $\dot{X} = FX - VX$ with

$$F \;=\; \begin{pmatrix} 0 & c_I\beta \\ 0 & 0 \end{pmatrix}, \qquad V \;=\; \begin{pmatrix} \gamma_E+\nu_m & 0 \\ -\gamma_E & \gamma_R+\nu_m \end{pmatrix},$$

we compute

$$V^{-1} \;=\; \frac{1}{(\gamma_E+\nu_m)(\gamma_R+\nu_m)} \begin{pmatrix} \gamma_R+\nu_m & 0 \\ \gamma_E & \gamma_E+\nu_m \end{pmatrix},$$

and the NGM $\boldsymbol{K} = FV^{-1}$ has spectral radius

$$\mathcal{R}_0 \;=\; \frac{c_I\beta}{\gamma_R+\nu_m}\cdot\frac{\gamma_E}{\gamma_E+\nu_m}. \tag{10.5}$$

The first factor is the SIR reproductive number. The second factor, $\gamma_E/(\gamma_E+\nu_m)$, is the probability that an exposed individual completes the latent period and becomes infectious before

migrating out of the population. For $\nu_m \ll \gamma_E$ (latent period much shorter than residence time), this factor is near 1 and $\mathcal{R}_0^{(\mathrm{SEIR})} \approx \mathcal{R}_0^{(\mathrm{SIR})}$. For ν_m comparable to γ_E, the factor is smaller and the reproductive number is reduced: exposed individuals may leave the population before becoming infectious.

Effect on dynamics. The latent period introduces a delay between the introduction of infection and the observation of cases. The early-outbreak growth rate is slower than for the basic SIR model because infectious material must pass through the latent stage before contributing to new infections. The doubling time for SEIR is approximately

$$t_{\mathrm{double}}^{(\mathrm{SEIR})} \approx \frac{\ln 2}{\sigma_2^{(\mathrm{SEIR})}},$$

where $\sigma_2^{(\mathrm{SEIR})}$ is the largest eigenvalue of the Jacobian of the (E, i) subsystem at the DFE. For τ_E small compared to τ_R, $\sigma_2^{(\mathrm{SEIR})} \approx \sigma_2^{(\mathrm{SIR})}$; for τ_E comparable to τ_R, the early growth is noticeably slower.

Every compartment in this chapter inherits the exponential residence-time assumption, which the Erlang chain below relaxes while the mean τ_x stays the independent parameter, as the pitfall The Independence of τ requires (Chapter 16).

Initializing a multi-stage model. The Erlang chain sharpens the initial-condition question already met for the single-compartment model in Chapter 6 (Pitfall: "Natural Initial Conditions"). With k sub-stages, the initial infectious individuals must be distributed *across* the sub-stages, and that distribution is not free. To represent an outbreak already in its growth phase, for instance when starting a simulation from an observed prevalence, the infecteds should be placed in the sub-stages in the proportions the growth phase itself produces: the dominant eigenvector of the linearized sub-stage dynamics, which during exponential growth is weighted toward the earliest sub-stages (for the baseline parameters with $k = 4$ the stage weights are about $0.36, 0.27, 0.21, 0.16$, not a uniform 0.25), because a growing population is dominated by its most recent entrants. Two convenient choices are wrong for this purpose: spreading the infecteds *equally* across the sub-stages (a "balanced" initialization), and lumping them all into the first sub-stage (correct only for a genuine single fresh introduction, where every infected individual has indeed just entered). Either injects a transient while the sub-stage occupancy relaxes toward the eigenvector, biasing the early total-infectious trajectory on which early-outbreak fitting depends. Figure 10.3 shows the effect at the baseline supercritical parameters: the balanced initialization deviates from the correct trajectory by up to about 9%, and the all-in-first-stage initialization by up to about 28%, over the first few weeks. The remedy is to seed the sub-stages with the growth-phase eigenvector, or, when the scenario really is a single introduction, to place the seed in the first sub-stage and let the model relax to the growth phase explicitly.

Exercises for this section

1. **[D]** For the SEIR model, derive the full expression for the early-outbreak growth rate σ_2 as the largest eigenvalue of the Jacobian at the DFE. Compare $\sigma_2^{(\mathrm{SEIR})}$ with $\sigma_2^{(\mathrm{SIR})}$ for $\tau_E = 0, \tau_R/4, \tau_R/2, \tau_R$ (with $\tau_R = 12$ days). How much does the latent period slow the epidemic growth?

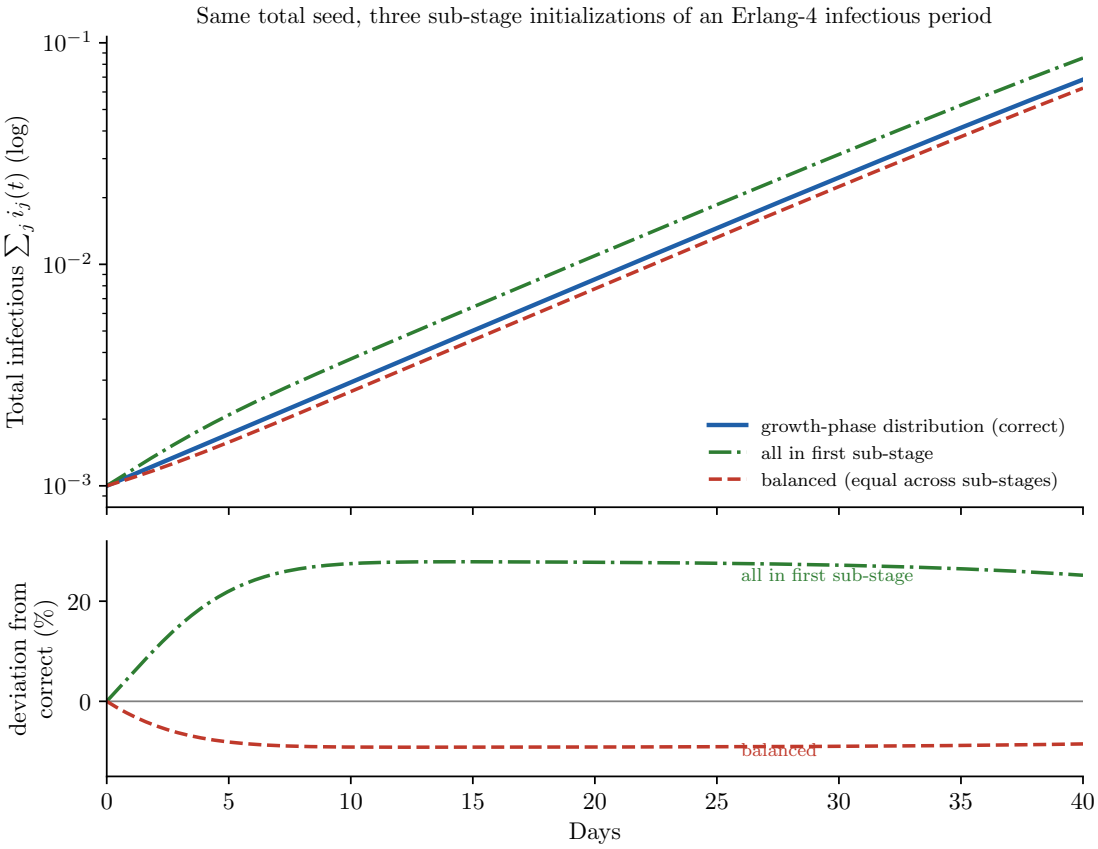


Figure 10.3: Total infectious fraction $\sum_j i_j(t)$ for an Erlang-4 infectious period, initialized three ways with the same total seed $i(0) = 10^{-3}$: the growth-phase eigenvector distribution (correct for an outbreak already growing), a balanced distribution (equal across sub-stages), and all mass in the first sub-stage (correct only for a fresh single introduction). *Upper panel:* the trajectories on a logarithmic scale. *Lower panel:* the relative deviation of each naive initialization from the eigenvector trajectory, reaching about 9% (balanced) and about 28% (first-stage) before the transient decays. Baseline parameters, $c_I = 4$ ($\mathcal{R}_0 \approx 1.92$).

10.4 Disease-Management Compartments: SEIQR and Variants

Public-health interventions typically remove some infectious individuals from general circulation through quarantine, isolation, or hospitalization. These individuals retain infectious potential (they can transmit to healthcare workers or contacts) but their contact rates and probabilities of onward transmission are reduced relative to infectious individuals in the general population.

The natural model adds a quarantine compartment Q with its own contact rate c_Q (typically much smaller than c_I).

The SEIQR model. Combining the latent compartment of Section 10.3 with a quarantine compartment gives the structure shown in Figure 10.4:

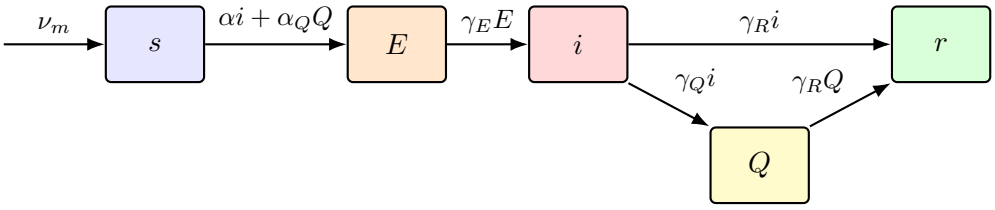


Figure 10.4: The SEIQR model adds a quarantine compartment Q fed by identification and isolation of infectious individuals at rate γ_Q. Quarantined individuals recover into r at rate γ_R (possibly modified by treatment). Transmission from Q enters through the term $\alpha_Q Q$ and is reduced relative to i through the contact rate $c_Q \ll c_I$. Demographic outflows at rate ν_m from each compartment are also omitted.

The governing equations are

$$\dot{s} = \nu_m - (\alpha i + \alpha_Q Q) - \nu_m s, \tag{10.6a}$$
$$\dot{E} = (\alpha i + \alpha_Q Q) - \gamma_E E - \nu_m E, \tag{10.6b}$$
$$\dot{i} = \gamma_E E - (\gamma_R + \gamma_Q + \nu_m) i, \tag{10.6c}$$
$$\dot{Q} = \gamma_Q i - \gamma_R Q - \nu_m Q, \tag{10.6d}$$
$$\dot{r} = \gamma_R (i + Q) - \nu_m r. \tag{10.6e}$$

Here γ_Q is the rate at which infectious individuals are identified and moved to quarantine (with units day^{-1}, interpretable as $1/\tau_Q$ where τ_Q is the mean time from infection onset to quarantine).

Each infectious compartment has its own force from infection,

$$\alpha = c_I \beta P_S^{(I)}, \qquad \alpha_Q = c_Q \beta_Q P_S^{(Q)},$$

where β_Q is the per-contact transmission probability from Q, and $P_S^{(I)}$ and $P_S^{(Q)}$ are the contact-weighted susceptible fractions among the contacts of i and of Q. New infections arrive at the rate $\alpha i + \alpha_Q Q$, with contributions from both the active-infection and quarantine compartments, weighted by their respective contact rates. The full expression for α and α_Q in a multi-compartment model is developed carefully in Chapter 11; for now, note that quarantined individuals typically contribute much less than free-infectious individuals.

Simplification: negligible transmission from quarantine. For many practical purposes, $c_Q \ll c_I$ and the quarantine compartment's contribution to transmission is negligible. In this limit the incidence $\alpha i + \alpha_Q Q$ reduces to αi, with

$$\alpha \;=\; c_I \beta P_S^{(I)} \;=\; c_I \beta \, \frac{c_S s}{c_S s + c_I i + c_R r + c_Q Q}.$$

The denominator still includes $c_Q Q$ for correctness of the probability fraction; only the transmission term $\alpha_Q Q$ is dropped.

Reproductive number. Using the NGM approach on the (E, i, Q) infection-adjacent compartments and assuming $c_Q \ll c_I$,

$$\mathcal{R}_0 \;\approx\; \frac{c_I \beta}{\gamma_R + \gamma_Q + \nu_m} \cdot \frac{\gamma_E}{\gamma_E + \nu_m}. \tag{10.7}$$

The effect of quarantine is to add γ_Q to the denominator, shortening the effective infectious period by the mean time to quarantine. For γ_Q comparable to γ_R, the reduction in $\mathcal{R}_0$ is roughly a factor of two.

The invasion-control interpretation. Quarantine is the prototypical invasion-targeting intervention. It reduces $\mathcal{R}_0$ directly through γ_Q in the denominator. For γ_Q large enough to bring $\mathcal{R}_0 < 1$, the outbreak cannot sustain itself and dies out. This is a different mechanism from reducing c_I (the SIR intervention) but has the same endpoint: moving the system across the invasion threshold. The West African Ebola epidemic provides a detailed illustration: isolation-centered interventions combined with safe-burial practices raised the effective isolation rate high enough to collapse transmission chains, a dynamic captured quantitatively by compartmental models fit to the outbreak data [Webb and Browne, 2016].

Exercises for this section

1. [**C**] The SEIQR model places quarantine rate γ_Q in the denominator of $\mathcal{R}_0$. Why does this make quarantine a directly tunable invasion-control mechanism, in a way that varying γ_R (treatment) or β (vaccination of partial efficacy) is not?

10.5 Disease-Induced Mortality

For pathogens with significant case-fatality ratios, disease-induced deaths should be tracked separately from the background migration-out rate ν_m. The natural modification adds a disease-induced mortality rate μ_d to the infectious compartment's outflow, giving the compartmental structure of Figure 10.5. The SIR equations modify to

$$\dot{s} = \nu_m - \alpha i - \nu_m s, \tag{10.8a}$$
$$\dot{i} = \alpha i - \gamma_R i - (\nu_m + \mu_d) i, \tag{10.8b}$$
$$\dot{r} = \gamma_R i - \nu_m r, \tag{10.8c}$$

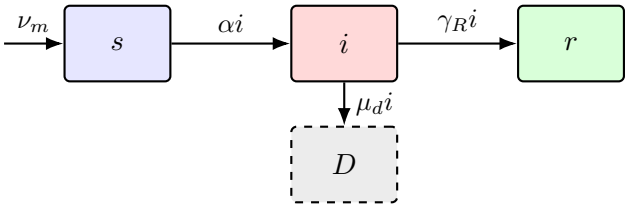


Figure 10.5: Adding disease-induced mortality to the basic SIR model: infectious individuals exit the i compartment at rate μ_d (in addition to the background migration rate ν_m and the recovery rate γ_R) and enter the dead sink D (dashed, not part of the living population). Demographic outflows at rate ν_m from s, i, and r are omitted for clarity.

where μ_d (with units day^{-1}) is the disease-induced mortality rate. The case-fatality ratio (CFR) is

$$\text{CFR} \;=\; \frac{\mu_d}{\mu_d + \gamma_R + \nu_m} \;\approx\; \frac{\mu_d}{\mu_d + \gamma_R} \quad \text{when} \quad \nu_m \ll \gamma_R.$$

For most respiratory infections, CFR is small ($\mu_d \ll \gamma_R$), and CFR $\approx \mu_d/\gamma_R$ (Pitfall: "Conditional versus Marginal Probabilities").

Total population dynamics. The total population $n = s + i + r$ is no longer conserved in the presence of disease-induced mortality:

$$\dot{n} \;=\; \nu_m - \nu_m n - \mu_d i.$$

At equilibrium, $\dot{n} = 0$ gives $n^* = 1 - (\mu_d/\nu_m)i^*$: the population at endemic equilibrium is reduced below the carrying capacity by an amount proportional to the disease burden. For a disease with $\mu_d \sim 0.01\gamma_R$ (CFR of about 1%), this correction is small. For severe pathogens with CFR of 10% or more, the correction is substantial.

Reproductive number. The NGM approach on the infectious compartment gives

$$\mathcal{R}_0 \;=\; \frac{c_I \beta}{\gamma_R + \nu_m + \mu_d}. \tag{10.9}$$

Disease mortality enters the denominator in the same way as recovery and migration: it shortens the effective infectious time by removing infectious individuals from the compartment. For CFRs of 1% or less, the effect on $\mathcal{R}_0$ is a few percent. For severe pathogens (CFR ¿ 10%), the reduction in $\mathcal{R}_0$ due to disease mortality can be substantial.

Data-fitting implication. When both case and death counts are available, the two data streams give independent information. The incidence $j(t) = \alpha i$ constrains one quantity; the death rate $\mu_d i$ constrains another. Their ratio is μ_d/α, a combination that is independent of i and therefore structurally immune to errors in the estimated prevalence. Chapter 7, Section 7.7 discussed the use of multiple data streams briefly; for pathogens with reliable mortality data (Ebola, severe influenza), incorporating the death data significantly tightens parameter estimates.

10.6 Asymptomatic Infection: The SIAR Model

For some pathogens, a substantial fraction of infectious individuals never develop symptoms but remain infectious. These asymptomatic individuals do not self-isolate, maintain their normal contact rates, and may contribute disproportionately to transmission.

The natural modification adds an asymptomatic compartment A alongside the symptomatic infectious compartment i; Figure 10.6 shows the resulting four-compartment structure.

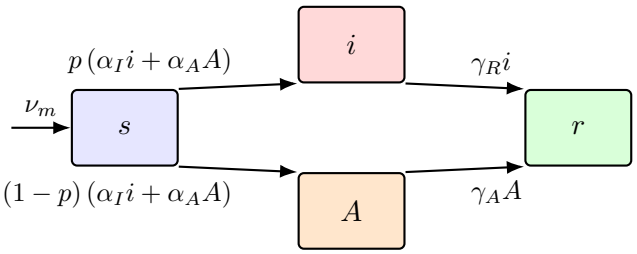


Figure 10.6: The SIAR model splits new infections at the point of infection into a symptomatic track (i, fraction p) and an asymptomatic track (A, fraction $1-p$). Both tracks are infectious, with potentially different contact rates (c_I, c_A) and transmission probabilities (β_I, β_A) feeding the total incidence $\alpha_I i + \alpha_A A$. Recovery rates γ_R (symptomatic) and γ_A (asymptomatic) are typically different. Demographic outflows at rate ν_m from each compartment are omitted for clarity.

The governing equations are most naturally expressed by first defining the per-infectious-individual transmission rates for the two tracks:

$$\alpha_I \;=\; c_I \beta_I P_S, \qquad \alpha_A \;=\; c_A \beta_A P_S, \tag{10.10}$$

where $P_S = c_S s/(c_S s + c_I i + c_A A + c_R r)$ is the contact-weighted susceptible fraction and (β_I, β_A) are the (possibly distinct) per-contact transmission probabilities for symptomatic and asymptomatic transmission. The total rate of new infections is then $\alpha_I i + \alpha_A A$, which splits at the point of infection into a symptomatic fraction p and an asymptomatic fraction $1-p$:

$$\dot{s} = \nu_m - (\alpha_I i + \alpha_A A) - \nu_m s, \tag{10.11a}$$

$$\dot{i} = p\,(\alpha_I i + \alpha_A A) - (\gamma_R + \nu_m) i, \tag{10.11b}$$

$$\dot{A} = (1-p)\,(\alpha_I i + \alpha_A A) - (\gamma_A + \nu_m) A, \tag{10.11c}$$

$$\dot{r} = \gamma_R i + \gamma_A A - \nu_m r, \tag{10.11d}$$

where p is the probability that a newly-infectious individual becomes symptomatic (so $1-p$ is the probability of asymptomatic infection) and γ_A is the recovery rate of asymptomatic individuals (often different from γ_R).

Reproductive number via the heuristic individual-tracking argument. The SIAR model has two infectious compartments (i and A), but a newly-infectious individual enters only one of them: with probability p, the individual becomes symptomatic and joins i; with probability $1-p$, the individual becomes asymptomatic and joins A. The two tracks are mutually exclusive at the individual level. The heuristic argument therefore sums the contributions of the two possible trajectories, weighted by their probabilities:

$$\mathcal{R}_0 \;=\; p \cdot \underbrace{\frac{c_I \beta_I}{\gamma_R + \nu_m}}_{\mathcal{R}_0^{(I)}\text{: if symptomatic}} + (1-p) \cdot \underbrace{\frac{c_A \beta_A}{\gamma_A + \nu_m}}_{\mathcal{R}_0^{(A)}\text{: if asymptomatic}}. \tag{10.12}$$

Each term applies the basic SIR calculation to the corresponding branch: the symptomatic individual produces $c_I\beta_I/(\gamma_R+\nu_m)$ secondary infections during the infectious period; the asymptomatic individual produces $c_A\beta_A/(\gamma_A+\nu_m)$. The probability-weighted sum is the per-individual expected number of secondary infections, which is $\mathcal{R}_0$. The next-generation matrix calculation below produces the same expression by a different route: the matrix FV^{-1} turns out to have rank one (the two tracks are mutually exclusive, so the two columns of F are proportional in the sense of producing the same branching distribution $(p, 1-p)$), and the unique non-zero eigenvalue of a rank-one matrix is its trace, giving the same result. The heuristic and NGM routes agree.

Reproductive number via the next-generation matrix. Linearizing the (i, A) subsystem at the disease-free equilibrium $(s, i, A) = (1, 0, 0)$, where $P_S \to 1$ and therefore $\alpha_I \to c_I\beta_I$ and $\alpha_A \to c_A\beta_A$, the NGM decomposition $\dot{X} = FX - VX$ gives

$$F \;=\; \begin{pmatrix} p\,c_I\beta_I & p\,c_A\beta_A \\ (1-p)\,c_I\beta_I & (1-p)\,c_A\beta_A \end{pmatrix}, \qquad V \;=\; \begin{pmatrix} \gamma_R+\nu_m & 0 \\ 0 & \gamma_A+\nu_m \end{pmatrix}.$$

The product is

$$FV^{-1} \;=\; \begin{pmatrix} \dfrac{p\,c_I\beta_I}{\gamma_R+\nu_m} & \dfrac{p\,c_A\beta_A}{\gamma_A+\nu_m} \\ \dfrac{(1-p)\,c_I\beta_I}{\gamma_R+\nu_m} & \dfrac{(1-p)\,c_A\beta_A}{\gamma_A+\nu_m} \end{pmatrix}.$$

This matrix has rank one (the two rows are proportional: the second equals $(1-p)/p$ times the first). Its non-zero eigenvalue is therefore the trace, giving

$$\mathcal{R}_0 \;=\; p\,\frac{c_I\beta_I}{\gamma_R+\nu_m} \;+\; (1-p)\,\frac{c_A\beta_A}{\gamma_A+\nu_m} \;=\; p\,\mathcal{R}_0^{(I)} + (1-p)\,\mathcal{R}_0^{(A)}, \tag{10.13}$$

where $\mathcal{R}_0^{(I)}$ and $\mathcal{R}_0^{(A)}$ are the reproductive contributions from a fully symptomatic or fully asymptomatic outbreak respectively. The basic reproductive number of the mixed model is the $(p, 1-p)$-weighted average. In the special case of equal recovery rates $\gamma_R = \gamma_A$, this simplifies to

$$\mathcal{R}_0 \;=\; \frac{p\,c_I\beta_I + (1-p)\,c_A\beta_A}{\gamma_R+\nu_m}, \tag{10.14}$$

a direct weighted average of the per-contact transmission parameters. The asymptomatic fraction $(1-p)$ can dominate the expression when $c_A\beta_A$ is large (asymptomatic individuals are not self-isolated and may transmit efficiently).

The asymptomatic reservoir problem. To make the policy implications concrete, consider the simplified case $\gamma_R = \gamma_A$ of (10.14). If asymptomatic individuals have $c_A \approx c_S$ (no self-isolation) while symptomatic have $c_I < c_S$, and if $\beta_I = \beta_A = \beta$ (equal per-contact transmission), then

$$\mathcal{R}_0 \;=\; \frac{pc_I\beta + (1-p)c_S\beta}{\gamma_R+\nu_m}.$$

For $p = 0.5$ and $c_I = c_S/2$, the asymptomatic contribution dominates:

$$\mathcal{R}_0 \;=\; \frac{0.5\cdot 0.5c_S\beta + 0.5\cdot c_S\beta}{\gamma_R+\nu_m} \;=\; \frac{0.75c_S\beta}{\gamma_R+\nu_m}.$$

Asymptomatic transmission contributes 67% of the reproductive number in this scenario. This observation explains why case-isolation policies that target symptomatic individuals can fail to halt outbreaks of pathogens with substantial asymptomatic spread: the asymptomatic reservoir is untouched by the intervention.

Data-fitting implications. Asymptomatic cases are typically undetected, so case-count data measure only the symptomatic fraction. The ratio of symptomatic to asymptomatic cases, $p/(1-p)$, must be estimated independently, typically from serological surveys or contact-tracing studies that identify asymptomatic contacts. Chapter 7's infected-viewpoint fit $\hat{\alpha} = j/i$ requires i (observed prevalence of *symptomatic* infection); the asymptomatic compartment contributes to dynamics but not to observed incidence.

> **Caution: match the populations, and convert with the right generation interval.** The estimator $\hat{\alpha} = j_{\text{obs}}/i_{\text{obs}}$, computed from symptomatic incidence and symptomatic prevalence, keeps the infected-viewpoint immunity to s (Chapter 7) and measures the growth rate. Dividing symptomatic incidence by total prevalence would bias it low by about the factor p. When the two tracks share the infectious period, converting the growth rate to $\mathcal{R}_0$ does not need p. What the estimator cannot reveal is the split $p\,\mathcal{R}_0^{(I)}$ against $(1-p)\,\mathcal{R}_0^{(A)}$ in (10.13). That needs p and the asymptomatic track's relative transmission, both from outside the case counts. Chapter 17 shows both halves.

10.7 A Common Pattern

The five extensions discussed in this chapter share a common structural pattern. Each extension adds one or more compartments to the basic SIR model and modifies the flow equations accordingly. In each case:

1. The compartmental extension has a clear biological motivation.
2. The reproductive number $\mathcal{R}_0$ receives a structural modification that reflects the new compartment's role: new factors in the numerator if new infectious compartments appear (SIAR), or new terms in the denominator if new outflow channels from existing infectious compartments appear (mortality, quarantine).
3. Both methods introduced in Chapter 5 recover $\mathcal{R}_0$: the heuristic individual-tracking argument gives a transparent interpretation, parameter by parameter, of how each term in $\mathcal{R}_0$ arises from the biology (progression probabilities, contact rates, residence times); the next-generation matrix method gives $\mathcal{R}_0$ mechanically through $\rho(FV^{-1})$ without requiring an individual-tracking narrative. For the models in this chapter, both methods produce the same expression, and we have exhibited both to show that the results agree.
4. The invasion-burden partition extends: parameters entering $\mathcal{R}_0$ remain invasion-facing, while parameters affecting only the steady-state distribution of individuals across compartments remain burden-facing.
5. The fitting advantage of the infected-viewpoint formulation persists: in all the extensions discussed, $\hat{\alpha} = j/i$ (where i is the observable symptomatic prevalence) is the natural fit quantity, and it retains its structural immunity to uncertainty in the susceptible population.

The table repays reading by column. The third column is item 2 of the enumeration above, made checkable against the equations rather than asserted: quarantine and disease-induced mortality are different biology that enters $\mathcal{R}_0$ through the same denominator, and SIAR is the only extension whose $\mathcal{R}_0$ is a sum rather than a product. The fourth and fifth columns are item 4: ω is the chapter's clearest burden-facing parameter, appearing in neither next-generation matrix and therefore, by Proposition 10.2(i), leaving $\mathcal{R}_0$ identically unchanged while it governs the endemic level.

Table 10.1: The five extensions of this chapter, their effect on $\mathcal{R}_0$, and the resulting invasion–burden classification. The base model is $\mathcal{R}_0 = c_I\beta/(\gamma_R + \nu_m)$. Reading down the third column shows the pattern the enumeration below describes: a new compartment either multiplies $\mathcal{R}_0$ by a probability of reaching the infectious stage, adds a removal channel to the denominator, splits the count across mutually-exclusive tracks, or—in the one case that leaves the next-generation matrices untouched—does nothing to $\mathcal{R}_0$ at all.

Model	New compartment	Effect on $\mathcal{R}_0$	Enters	Class
SIR	—	base	—	—
SIRS	waning $r \to s$, rate ω	unchanged, (10.2)	neither	burden-facing
SEIR	latent E	$\times\, \gamma_E/(\gamma_E + \nu_m)$, (10.5)	V	invasion-facing
SEIQR	quarantine Q	latent factor, and γ_Q added to the denominator, (10.7)	V	invasion-facing
mortality	outflow μ_d from i	μ_d added to the denominator, (10.9)	V	invasion-facing
SIAR	asymptomatic A, fraction $1-p$	$p\,\mathcal{R}_0^{(I)} + (1-p)\,\mathcal{R}_0^{(A)}$, (10.13)	F, V	invasion-facing

What $\mathcal{R}_0$ does not fix. Item 4 separates the parameters that set invasion from those that set burden. Figure 10.7 shows how much the second group matters. Panel (a) runs three models with the same $\mathcal{R}_0 = 1.92$: the basic SIR model, the SEIR model with a mean latent period of 5 days, and the SIR model with an Erlang-4 infectious period of the same mean. They share a threshold and little else. The latent period delays the peak from day 85 to day 140 and lowers it from 0.17 to 0.11. The Erlang period does the opposite: the peak comes at day 62 and reaches 0.31, because a less variable generation interval grows faster at the same $\mathcal{R}_0$ (Chapter 15, Figure 15.3). Even the fraction ever infected differs, from 0.81 to 0.87 by day 400. In the classic model with equal contact rates and a closed population the three final sizes coincide. Here they do not, because P_S weights each infectious compartment by its own contact rate. Panel (b) holds $\mathcal{R}_0$ fixed and varies the duration of immunity. The first outbreak is the same, but waning immunity raises the endemic prevalence from 0.0008 to 0.004 with immunity of 5 years, and to 0.016 with immunity of 1 year. A model chosen for its $\mathcal{R}_0$ alone can be right about invasion and wrong about everything that follows.

When the heuristic argument works and when it does not. The heuristic individual-tracking argument works cleanly for the extensions of this chapter because each new compartment added to the SIR base either (a) precedes the infectious stage and contributes a progression-probability factor (latent period E, exposed-to-infectious transit in SEIR), (b) follows the infectious stage and does not contribute to secondary infections (quarantine Q after symptom-driven removal, recovery r, waning flow $r \to s$), or (c) represents a probabilistic branching into one of several mutually-exclusive infectious tracks (i versus A in SIAR). In all three cases, the heuristic argument computes $\mathcal{R}_0$ as a probability-weighted sum over single-individual trajectories, and the NGM calculation produces the same expression because the matrix FV^{-1} is either scalar, triangular, or rank-one in the relevant subspace.

The heuristic argument is more limited than the NGM method, and the limit matters for the chapters ahead. The heuristic tracks a single newly-infectious individual through the single track that individual follows and counts secondary infections produced along that track. This works when the infection process is ultimately one-dimensional: there is a unique type of "transmitter" (whatever stage the individual passes through), and the "secondary infection produced" is of the same ultimate type. The models of this chapter all have this structure.

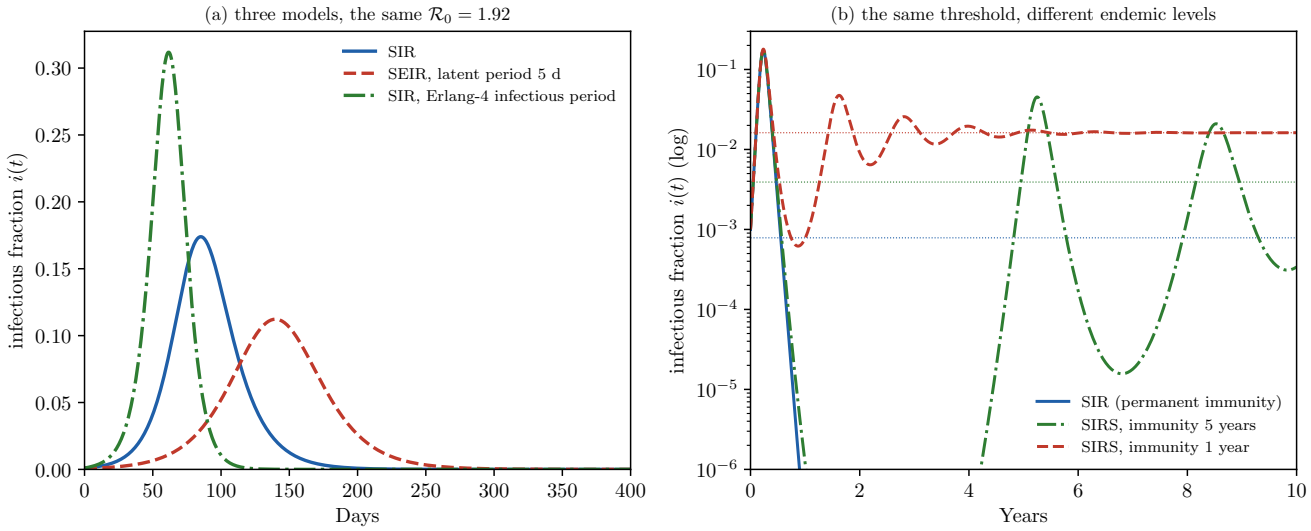


Figure 10.7: The same $\mathcal{R}_0$, different epidemics, at the baseline parameters of Table 4.4. (a) The infectious fraction for the basic SIR model, the SEIR model (mean latent period 5 days; exposed individuals keep the susceptible contact rate c_S) and the SIR model with an Erlang-4 infectious period of the same mean. Each model's transmission factor is rescaled so that $\mathcal{R}_0 = 1.92$; the seed is $i(0) = 10^{-3}$ in the first infectious stage. (b) The SIR model and the SIRS model with mean immunity of 5 years and of 1 year, with the same $\mathcal{R}_0$ and seed, on a logarithmic scale over ten years. Thin dotted lines mark each model's endemic prevalence.

Bipartite and multi-type transmission systems, however, do not. In the two-group model of Chapter 11, a susceptible in group 1 can only be infected by an infectious individual in group 2 (under the simplest cross-group-only assumption), and vice versa; a newly-infectious individual in group 1 produces secondary infections only in group 2, whose secondary infections in turn are in group 1. The natural "secondary infections" count is of a different type than the original individual, and two generations must be completed to return to the same type. The heuristic argument, if naively applied, would compute one generation's worth of secondary infections in the wrong group and would miss the square-root dampening structure that is the signature of bipartite transmission. The NGM method handles this automatically, because the spectral radius of a bipartite FV^{-1} is the geometric mean of the two directional components, giving $\mathcal{R}_0 \propto \sqrt{(\text{group-1-to-2 rate}) \cdot (\text{group-2-to-1 rate})}$. Chapter 11, Section 11.6 formalizes this observation as a general stage/cyclic/geometric-mean framework for simple cyclic multistage models. The same structural distinction applies to the host-vector coupling of Chapter 13 (human-to-mosquito and mosquito-to-human are different types) and to the male-female sexual transmission of Chapter 12 (cross-sex infection creates two mutually-dependent populations). For multigroup models whose structure is not a simple cycle, such as the SIAR asymptomatic-symptomatic split in Section 10.6 above, $\mathcal{R}_0 = \rho(\boldsymbol{K})$ remains the threshold quantity but does not reduce to the nth root of a single cycle product.

Remark 10.1 (Which derivation to use). The practical rule: use the heuristic argument whenever the transmission process admits a clean one-dimensional narrative (a single newly-infectious individual produces a computable number of secondary infections of the same eventual type). Use the NGM method whenever the transmission process is bipartite or multi-type, where secondary infections are of a different type than the primary infection and two or more generations must complete before the counting closes. The scalar and branching cases of this chapter are one-dimensional in this

sense; the bipartite cases of Chapters 11, 12, and 13 are not, and in those chapters we use the NGM method exclusively.

The common pattern is what makes the SIR framework usable across a wide range of disease-specific models. Rather than re-deriving the framework for each disease, the modeler adds compartments corresponding to the disease's specific biology and inherits the analytical and fitting machinery from the base SIR model. Proposition 10.2 below gives the general argument; the extensions in this chapter are concrete instances of the pattern. Modelers coming from the SEIRS literature [Huo et al., 2016] or the respiratory-mortality literature [Gumel et al., 2006] will recognize their own models as instances of the same template.

Item 4 of the enumeration deserves a precise statement, since it is the structural claim that makes the framework carry over. We give it the form of a proposition that specializes Theorem 5.6 to the family of generalized models of this chapter.

Proposition 10.2 (Invasion–burden partition for generalized SIR models)**.** *Let $\mathcal{M}$ be any compartmental epidemic model built from the* SIR *framework by the operations of this chapter (adding waning-immunity flow, latent compartments, quarantine compartments, disease-induced mortality, or asymptomatic branching). Write the dynamics of the infection-adjacent compartments near the disease-free equilibrium as $\dot{X} = FX - VX$, with F and V the standard NGM matrices, and let $\mathcal{R}_0 = \rho(FV^{-1})$. Let $\Pi_{\mathcal{M}}$ denote the full parameter set of $\mathcal{M}$. Then:*

(i) If a parameter $p \in \Pi_{\mathcal{M}}$ appears in neither F nor V, then $\partial\mathcal{R}_0/\partial p = 0$ identically; such parameters are called burden-facing *and affect only the endemic equilibrium distribution. The converse requires a non-cancellation hypothesis: a parameter appearing in both factors may still leave $\mathcal{R}_0$ unchanged, as when p scales F and V equally, $F = pF_0$ and $V = pV_0$, so that $FV^{-1} = F_0V_0^{-1}$ carries no dependence on p. For the models of this chapter no such cancellation occurs, and the converse holds; for an extension outside this family it must be checked rather than assumed.*

(ii) Suppose p satisfies the non-cancellation hypothesis*: p does not multiply F and V by a common factor. Then p enters $\mathcal{R}_0$ with leading-order sensitivity $\Upsilon_p^{\mathcal{R}_0} = O(1)$ precisely when p enters F directly, or enters V through an infection-compartment removal rate comparable to or exceeding ν_m; such parameters are called* invasion-facing*. The hypothesis cannot be dropped. If p scales both factors equally, $F \mapsto pF$ and $V \mapsto pV$, then $(pF)(pV)^{-1} = FV^{-1}$, so $\mathcal{R}_0$ is unchanged even though p appears in both—a parameter can be present throughout the next-generation construction and still be burden-facing.*

(iii) For any burden-facing parameter p, the relative sensitivity $|\Upsilon_p^{i^*_{\mathcal{M}}}|$ of the endemic prevalence to p is $O(1)$ independent of $\mathcal{R}_0$; for any invasion-facing parameter p, $|\Upsilon_p^{i^*_{\mathcal{M}}}|$ decays as $1/(\mathcal{R}_0 - 1)$ in the large-$\mathcal{R}_0$ regime.

Proof sketch. The forward direction of statement (i) is immediate from the NGM decomposition: $\mathcal{R}_0$ is the spectral radius of FV^{-1} and therefore depends on p only through its appearance in F or V, so a parameter absent from both cannot affect it. The converse is not automatic, since appearance in F or V does not by itself imply a nonzero derivative of the spectral radius; it is verified for the models of this chapter by inspecting the explicit forms, in each of which the parameter enters FV^{-1} without cancelling. Statement (ii) follows from the explicit forms of F and V for each model in this chapter; the leading-order calculation mirrors the one behind (5.31). Statement (iii) is the generalization of the asymptotic row structure in (5.31): a burden-facing parameter produces a -1 term from the direct demographic-inflow dependence of $i^*_{\mathcal{M}}$ on ν_m that does not vanish as $\mathcal{R}_0$ grows, while an

invasion-facing parameter produces only indirect dependence through $\mathcal{R}_0$ whose contribution scales as $\mathcal{R}_0/(\mathcal{R}_0-1)\cdot\Upsilon_p^{\mathcal{R}_0} \to \Upsilon_p^{\mathcal{R}_0}/(\mathcal{R}_0-1)$ asymptotically. The full calculation parameter-by-parameter is mechanical and is developed in Chapter 9 for each extension listed above. □

Proposition 10.2 is the reason the invasion-burden distinction is a property of the framework, not of the basic SIR model alone. Every disease-specific extension inherits the partition, and the same fitting and policy conclusions apply. This is what justifies the investment in developing the framework carefully at the level of the basic model: the payoff is a structural result that carries forward across the dozens of disease-specific models that clinicians and public-health scientists actually use.

Exercises for this section

1. **[C]** The chapter argues that all five extensions "fit within a single conceptual framework." State the five ingredients of this common pattern (Section 10.7) and explain why preserving the invasion-burden partition is the conceptually most important of them.

10.8 Summary and Exercises

This chapter has presented five standard extensions of the SIR model:

1. **SIRS**: Loss of immunity at rate ω. The reproductive number is unchanged, but the endemic prevalence is increased, especially for fast-waning ($\omega \gg \nu_m$) cases.

2. **SEIR**: Latent period with rate $\gamma_E = 1/\tau_E$. The reproductive number acquires a factor $\gamma_E/(\gamma_E+\nu_m)$ reflecting the probability of surviving the latent stage; for $\tau_E \ll \tau_m$ this factor is near 1.

3. **SEIQR**: Quarantine of symptomatic individuals at rate γ_Q. The reproductive number acquires γ_Q in its denominator, giving a direct mechanism for interventions to target invasion.

4. **Disease-induced mortality**: An additional outflow μ_d from the infectious compartment. $\mathcal{R}_0$ has μ_d in its denominator; for severe pathogens, this is a substantial correction.

5. **SIAR**: Asymptomatic compartment with its own contact rate and recovery. The reproductive number is a p-weighted combination of symptomatic and asymptomatic contributions.

The common pattern enumerated in Section 10.7 shows that all five extensions fit within a single conceptual framework: new compartments, NGM-computed $\mathcal{R}_0$, preserved invasion-burden partition, preserved fitting advantage.

Chapters 11 through 13 extend the framework further to heterogeneous populations (multiple interacting groups), Chapter 14 to vertical transmission, Chapter 15 to age-structured and time-since-infection formulations, and Chapters 17–20 to disease-specific case studies and their cross-cutting synthesis, integrating the tools developed so far.

Pitfall check

This chapter has extended Pitfall: "The Independence of τ" to the Erlang-chain construction (in Section 10.3, where the exponential-residence-time assumption can be relaxed by replacing a single compartment with a chain of k exponential sub-compartments), and has exercised Pitfall: "Short-Term Simulation Assumptions" (in Section 10.5, where the explicit treatment of disease deaths and the population-conservation question reveal which simulation-horizon assumptions are being relaxed). Before moving to the heterogeneous-population extensions of Chapter 11, take a moment to confirm the following points are clear in your mind. *Why does the Erlang chain preserve τ_x as the independent parameter even as it adds the shape parameter k? What is the relationship between the coefficient of variation of the residence-time distribution and the integer k, and how does the operational consequence (timing of secondary infections, contact-tracing windows) change as k grows? Which of the chapter's five model extensions force the modeler to relax the conserved-population simulation-horizon assumption, and which can be analyzed within it?* If any of these are unclear, the corresponding pitfall block (or the catalog entry, accessible via the List of Common Pitfalls) is the place to revisit.

Exercises

*Five section-level exercises appear above, following Section 10.2, Section 10.3, Section 10.4, and Section 10.7. The exercises below integrate across the chapter's five model extensions. Exercises are tagged by type: **C** = Concept check, **D** = Derivation, **I** = Interpretation / modeling judgment, **L** = Computational lab.*

1. **[C]** Read the third and fifth columns of Table 10.1: which of the five extensions leave $\mathcal{R}_0$ unchanged from the basic SIR value, and which modify it? The table gives the classification; in one sentence, explain *why* it comes out that way, in terms of the distinction between invasion-controlling and burden-controlling parameters.

2. **[D]** Using the NGM approach, derive the expression for $\mathcal{R}_0$ in the SEIQR model (10.6). Verify the approximate result (10.7) in the limit $c_Q \to 0$.

3. **[D]** For the disease-mortality model (10.8), derive the endemic equilibrium and show that the equilibrium total population n^* is reduced below N^c by $(\mu_d/\nu_m)i^*$.

4. **[D]** For the SIAR model with $p = 0.5$, $\gamma_R = \gamma_A$, $\beta_I = \beta_A = 0.040$, $c_A = c_S = 20$, $c_I = 4$ (symptomatic self-isolation), compute $\mathcal{R}_0$ and verify the conclusion that asymptomatic transmission dominates.

5. **[D]** Consider a two-type variant of the SIAR model in which symptomatic individuals infect only asymptomatic ones and asymptomatic only symptomatic. Attempt the heuristic individual-tracking argument of Chapter 5 on this model: track one newly infectious symptomatic individual, count the secondary infections it produces, and write down the expression you obtain. Now compute $\rho(\boldsymbol{K})$ for the same model and compare. Identify the step at which the heuristic argument goes wrong, then read Remark 10.1 and check that the failure you found is the one it predicts.

6. **[I]** Consider an intervention that reduces c_I by 75% (aggressive case-isolation) but does not affect c_A. For the SIAR parameters of Exercise 4, compute the new $\mathcal{R}_0$. By what fraction has

the intervention reduced $\mathcal{R}_0$? Contrast with the same intervention in an SIR model where there is no asymptomatic compartment.

7. **[D]** For a pathogen with CFR $= 0.05$ and $\tau_R = 14$ days, compute μ_d and the effect on $\mathcal{R}_0$. What is the relative change compared to the same pathogen with CFR $= 0$?

8. **[L]** Simulate the disease-mortality model (10.8) twice, holding $c_I\beta$, γ_R and ν_m fixed throughout at values for which $\mathcal{R}_0 = 2$ when $\mu_d = 0$: once with $\mu_d = 0$, and once with μ_d chosen so that the case-fatality ratio is 0.25. Report $\mathcal{R}_0$, the peak prevalence, the final size among survivors, and the cumulative deaths for each run. Which of the three quantities does μ_d change most, and does that match where μ_d appears in (10.9)? Use your answer to say, in one sentence, what $\mathcal{R}_0$ does and does not govern.

9. **[I]** Propose a compartmental extension combining two features from this chapter (for example, SEIRS with waning immunity, or SEIQR with disease mortality). Draw the block diagram, write the equations, and derive $\mathcal{R}_0$ via NGM.

10. **[L]** *Spaced retrieval, Chapter 9.* Use the LHS-PRCC machinery introduced in Chapter 9, Section 9.6 to compute the global sensitivity of $\mathcal{R}_0$ in the SEIQR model (10.6) of Section 10.4. Vary the four parameters $\{c_I, \beta, \tau_R, c_Q\}$ within a factor of 2 above and below their baseline values from Table 4.4 of Chapter 4. The book tabulates no baseline for c_Q; choose one satisfying $c_Q \ll c_I$, state the value you chose, and report in part (c) whether the ranking is sensitive to it. Use $M = 2000$ samples and uniform multiplicative variation. Report:

    (a) PRCC estimates with 95% CIs, ranked by absolute magnitude.

    (b) A side-by-side comparison with the local sensitivity-index ranking obtained by partial differentiation of the closed-form expression (10.7).

    (c) Whether the rank order of the four parameters is preserved between the local and global methods, or whether the global analysis surfaces an interaction effect that the local analysis missed.

    The exercise applies the global-sensitivity workflow of Chapter 9 to a model derived in this chapter, reinforcing both the LHS-PRCC method and the structural similarity between the original SIR and its extensions.

11. **[L]** *Spaced retrieval, Chapter 7.* Apply the infected-viewpoint fitting workflow of Chapter 7, Section 7.4.3 to the SIRS extension of Section 10.2 (the waning-immunity model). Generate synthetic case time series by simulating the SIRS model forward over 5 years with the baseline SIR parameters of Table 4.4 plus a waning rate $\omega = 1/365$ day^{-1} (1-year mean immune duration). Add multiplicative log-normal noise with $\sigma = 0.10$ to both $j(t)$ and $i(t)$. Then:

    (a) Apply the infected-viewpoint estimator $\hat{\alpha}(t) = j(t)/i(t)$ from (7.12) of Chapter 7 to the synthetic data. Compare $\hat{\alpha}$ at the early-time plateau (where the dynamics are dominated by the SIR-like initial growth) with the late-time behavior (average $\hat{\alpha}$ over the fifth year; the approach to the new endemic equilibrium is a damped oscillation).

    (b) Use the early-time $\hat{\alpha}$ to recover an estimate $\hat{\mathcal{R}}_0$ via the workflow of Section 7.4.3. Compare with the true $\mathcal{R}_0$ used to generate the data.

(c) Identify the modification to the workflow needed to recover ω separately from the standard SIR parameters: what additional data feature (e.g., the late-time prevalence relative to early-time prevalence, or the time to reach endemic equilibrium) carries the ω-information that the early-time $\hat{\alpha}$ alone does not?

The exercise demonstrates that the fitting workflow of Chapter 7 extends to chapter-extension models with one additional model-specific parameter, and identifies which features of the data carry the additional identifiability.

12. **[I]** *Reasoning visibility.* A colleague describes a new pathogen and asks: should they use the baseline SIR model of Chapters 4–5, or one of the five extensions of this chapter? Construct an ordered diagnostic checklist (5–7 items) that they can run through to make the decision. For each item, state:

    (1) The biological feature being checked (e.g., "does the pathogen have a clinically meaningful asymptomatic compartment?").

    (2) The empirical or epidemiological evidence that would settle the check (e.g., "serological surveys identifying infectious individuals who never developed symptoms, or contact-tracing studies finding undetected sources of secondary cases").

    (3) Which extension of this chapter is triggered if the check returns "yes" (e.g., "Section 10.6 SIAR"), and what additional parameter the extension introduces.

    (4) Whether the check is independent of the others or whether ordering matters (e.g., asymptomatic-screening should logically precede waning-immunity questions because the asymptomatic compartment changes the basic transmission structure).

    End your checklist with a paragraph stating the rule for combining triggered extensions: when two checks both come back "yes" (e.g., the pathogen has both a latent period and disease-induced mortality), how do you decide whether to combine the extensions in a single model or whether to model them separately and compare? Your answer should reference the common-pattern observation of Section 10.7 of this chapter. The exercise externalizes the model-selection reasoning that is otherwise buried in the modeler's craft, and produces a reusable diagnostic that the student can apply to a novel pathogen on the first pass.

13. **[Open Project]** *A sixth extension through the common pattern.* A multi-week capstone project choosing a compartmental extension this chapter does not treat—waning vaccine-derived immunity, a treated compartment, or a second infectious stage—deriving its reproductive number through the common pattern of Section 10.7, and reporting whether the pattern holds or where it breaks.

Chapter 11

Two-Group Mixing

First write what must be true. Then admit what you are assuming.

The models of Chapters 4 through 10 assume the population is homogeneous: every individual has the same contact rate, transmission probability, and recovery rate as every other individual in the same compartment. Real populations are heterogeneous in epidemiologically important ways. Children have different contact patterns than adults; rural populations differ from urban; different occupational groups have different exposure risks. This chapter develops the two-group generalization of the SIR model, which is the simplest non-trivial heterogeneous-population model and the entry point to the matrix techniques that appear in Chapters 12 through 13.

Section 11.1 motivates the treatment. Section 11.2 introduces the mixing matrix that encodes between-group contact patterns. Section 11.3 writes out the full six-compartment SIR system and computes its equilibria. Section 11.4 derives the basic reproductive number via the next-generation matrix, which here is a 2×2 matrix whose spectral radius gives $\mathcal{R}_0$. Section 11.5 examines assortative versus disassortative mixing and the limiting cases. Section 11.7 revisits herd immunity in heterogeneous populations. Section 11.8 works through a brief numerical case study. Section 11.9 closes with a summary and exercises.

Advanced material note. The bipartite next-generation matrix proof details in Section 11.4 can be read at two levels: the main-path reader needs only the resulting square-root formula for $\mathcal{R}_0$; readers seeking the full algebraic justification can work through the detailed derivation.

11.1 Why Heterogeneity Matters

The homogeneous-mixing approximation treats every individual as equivalent to every other in the same compartment. This is a substantial simplification: it says that a child and an adult have the same contact rate, that a healthcare worker and an office worker have the same exposure risk, and that an urban resident and a rural resident have the same probability of encountering an infectious individual. Whether this approximation is adequate depends on the question being asked.

For estimating $\mathcal{R}_0$ during the exponential-growth phase of an outbreak, the homogeneous approximation is often adequate: the outbreak's initial growth rate is driven by the overall rate of transmission in the population, averaged over whatever heterogeneity is present. For estimating the endemic prevalence in each subgroup, or for designing targeted interventions (e.g., vaccinating children preferentially), heterogeneity matters directly and the homogeneous model gives misleading

answers. The earliest systematic treatment of heterogeneous transmission in mathematical epidemiology appeared in Lajmanovich and Yorke [1976], which established threshold conditions for deterministic models with multiple interacting subpopulations; the framework was elaborated for a wide range of mixing patterns by Castillo-Chavez et al. [1989] and Busenberg and Castillo-Chavez [1991], who provided the general theoretical scaffolding we rely on below.

A key practical distinction: the herd immunity threshold. The most pedagogically important consequence of heterogeneity is that the herd immunity threshold differs between homogeneous and heterogeneous models. In a homogeneous population, the threshold is $1 - 1/\mathcal{R}_0$ of the population. In a population with high-contact and low-contact groups, the threshold depends on how immunity is distributed across groups: immunizing the high-contact group preferentially achieves herd immunity at a lower total vaccination fraction than uniform immunization. Section 11.7 develops this quantitatively.

Why start with two groups? Any treatment of heterogeneity must begin with a finite-group approximation. The two-group model is the simplest heterogeneous extension that captures the qualitative effects of non-uniform mixing. Its 2×2 next-generation matrix has a closed-form spectral radius that reveals the structure of $\mathcal{R}_0$ without obscuring algebra. Chapter 12 extends the techniques here to sexual transmission, including a multigroup model (Section 12.5), and Chapter 13 to vector-host models, each a specialization of the general heterogeneous framework.

11.2 The Mixing Matrix

Label the two groups by indices $j, k \in \{1, 2\}$. For each group j, let N_j^{c} be the carrying capacity of the group, and let (S_j, I_j, R_j) denote the compartmental state variables for group j. The total carrying capacity is $N^{\mathrm{c}} = N_1^{\mathrm{c}} + N_2^{\mathrm{c}}$.

The contact structure between the two groups is encoded in a mixing matrix. Before writing down the full heterogeneous model, it is worth pausing over a basic bookkeeping fact that students often find unintuitive the first time they see it. In a two-group model, the fractions of contacts need not be symmetric, but the *total number of cross-group contacts* must be. This is the balance equation.

A contact is counted from two sides. Let M_{jk} denote the total number of contacts per day between group-j individuals and group-k individuals, counted from the group-j side. For within-group contact, M_{11} and M_{22} count contacts internal to each group. For between-group contact, M_{12} and M_{21} refer to the same physical events seen from opposite sides. Each contact between a group-1 individual and a group-2 individual is one event involving two people, so the total number of such contacts counted from group 1 must equal the total number counted from group 2:

$$M_{12} \;=\; M_{21}. \tag{11.1}$$

We will call this the *balance equation*. Some authors refer to the same condition as reciprocity.

From total contacts to per-person contact rates. The quantity that enters the model equations is not the total number of contacts in the whole population, but the number of contacts made by a single individual. In the "who-acquires-contact-from-whom" convention, C_{jk} is the number

of contacts per day that a group-j individual has with group-k individuals. Multiplying by the number of group-j individuals gives the total number of such contacts:

$$M_{jk} \;=\; C_{jk}N_j^{\mathrm{c}}.$$

Substituting this into (11.1) gives

$$C_{12}N_1^{\mathrm{c}} \;=\; C_{21}N_2^{\mathrm{c}}. \tag{11.2}$$

Equation (11.2) is the form of the balance equation that appears most often in two-group epidemic models. It says that the off-diagonal entries of the mixing matrix are not independent: once C_{12} and the group sizes are fixed, C_{21} is determined. This convention has the additional advantage that $\sum_k C_{jk} = c_j$ is the total contact rate of a group-j individual.

Why the off-diagonal entries need not be equal. Students often expect $C_{12} = C_{21}$ because contact is symmetric. That is not correct unless the two groups have the same size. The symmetric quantity is the *total* number of contacts, not the per-person rates. For example, suppose $N_1^{\mathrm{c}} = 900$ and $N_2^{\mathrm{c}} = 100$, and each group-1 individual has $C_{12} = 2$ contacts per day with group 2. Then the total number of group-1–group-2 contacts per day is

$$C_{12}N_1^{\mathrm{c}} \;=\; 2 \times 900 \;=\; 1800.$$

To satisfy the balance equation, group 2 must generate the same total number of cross-group contacts, so

$$C_{21} \;=\; \frac{C_{12}N_1^{\mathrm{c}}}{N_2^{\mathrm{c}}} \;=\; \frac{1800}{100} \;=\; 18.$$

Each group-2 individual has many more contacts with group 1 than each group-1 individual has with group 2. The totals are symmetric even though the fractions are not.

The balance equation (11.2) is a conservation law for contacts, and a mixing matrix that violates it biases $\mathcal{R}_0$; checking it is the pitfall Heterogeneous-Mixing Balance Conditions (Chapter 16).

Verbalize each entry, not only the balance. The balance equation is a *consistency* check: it confirms that the two off-diagonal entries agree with each other, not that either is the right size. A complementary and independent test is to read each entry aloud as a fraction of contacts, the plausibility habit recommended for parameters generally in Chapter 1: the statement "group j directs the fraction C_{jk}/c_j of its contacts at group k" should be weighed against what is known about how the two groups actually mix. A matrix in which group j makes 44% of its contacts with a group it is thought to encounter only rarely is biologically implausible even when it satisfies balance exactly—balance constrains the off-diagonals relative to *each other* and says nothing about their size relative to the social structure being modeled (Pitfall: "Dimensional Consistency").

The mixing matrix. We use the per-person contact-rate convention throughout the chapter because it keeps the force-of-infection and force-from-infection formulas close to the notation of the earlier homogeneous chapters. The mixing matrix is

$$\mathbf{C} \;=\; \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \tag{11.3}$$

where the diagonal entries represent within-group contact and the off-diagonal entries represent between-group contact. Because of the balance equation (11.2), the matrix has four entries but only three independent ones. Figure 11.1 visualizes the matrix for a typical assortative pattern in which individuals of each group contact their own group more often than the other group.

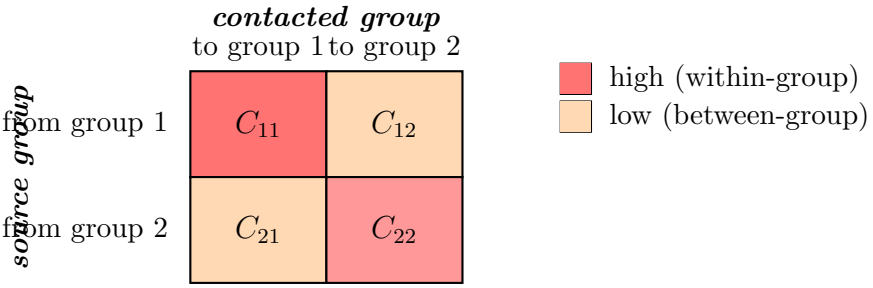


Figure 11.1: The 2×2 mixing matrix $\mathbf{C}$ visualized as a heatmap for a typical assortative pattern: individuals of each group have more contacts within their own group (dark diagonal cells C_{11}, C_{22}) than with the other group (light off-diagonal cells C_{12}, C_{21}). The balance equation $C_{12}N_1^{\mathrm{c}} = C_{21}N_2^{\mathrm{c}}$ links the off-diagonal cells when group sizes differ.

How balance is enforced in this book. For the main development of this chapter and the bipartite applications in Chapters 12 and 13, we use the simplest approach: specify one off-diagonal rate and determine the other from (11.2). This keeps the algebra transparent and emphasizes the main epidemiological point that heterogeneous mixing is constrained by population size. In larger or empirically-calibrated models, one may instead reconcile incompatible contact estimates using more elaborate balancing procedures such as harmonic-mean symmetrization, iterative proportional fitting, or entropy-based adjustments. Those methods are useful in practice, but they are not needed to understand the core ideas developed here.

Compartment-specific contact rates. In our SIR framework, contact rates may differ by epidemiological status within each group. We generalize the mixing matrix to compartment-specific forms. Let $c_{j,S}, c_{j,I}, c_{j,R}$ denote the contact rates for susceptible, infectious, and recovered individuals in group j. The full contact structure is then specified by three matrices:

$$\begin{aligned} \mathbf{C}^S &= \left(c^S_{jk}\right)_{j,k=1,2} \qquad \text{(susceptible-to-any contact)}, \\ \mathbf{C}^I &= \left(c^I_{jk}\right)_{j,k=1,2} \qquad \text{(infectious-to-any contact)}, \\ \mathbf{C}^R &= \left(c^R_{jk}\right)_{j,k=1,2} \qquad \text{(recovered-to-any contact)}. \end{aligned}$$

To keep notation manageable, we sometimes suppress the superscript and write C_{jk} for the dominant-compartment value, usually the susceptible contact rate, adjusting as needed for context. Empirical mixing matrices calibrated from survey and synthetic-population data show substantial off-diagonal structure in age-stratified populations and serve as the quantitative counterpart to the abstract formalism here; see Del Valle et al. [2007] for a representative construction from a large-scale synthetic urban population.

Total contact balance. Define the pair pool

$$C_{\mathrm{total},kj} \;=\; c^S_{kj}S_k + c^I_{kj}I_k + c^R_{kj}R_k,$$

the contacts group-k people direct at group j. The contact-weighted probability that such a contact is made by a susceptible individual of group k is

$$P^{(k)}_{S,j} \;=\; \frac{c^S_{kj}S_k}{C_{\mathrm{total},kj}}.$$

Similar expressions hold for $P^{(k)}_{I,j}$ and $P^{(k)}_{R,j}$. For each pair of groups the three sum to unity: $P^{(k)}_{S,j} + P^{(k)}_{I,j} + P^{(k)}_{R,j} = 1$.

Exercises for this section

1. **[D]** Verify the reciprocity constraint $C_{12}N^c_1 = C_{21}N^c_2$ by showing that the total number of group-1–group-2 contacts per day counted from each side must be equal.

11.3 The Two-Group SIR Dynamics

With the contact structure specified, we can write the dynamics. The force from infection (Pitfall: "Force of Infection versus Force from Infection") for an infectious individual in group j is

$$\alpha_j \;=\; \sum_k c^I_{jk}\beta_{jk}\,\frac{c^S_{kj}S_k}{C_{\text{total},kj}} \;=\; \sum_k c^I_{jk}\beta_{jk}\,P^{(k)}_{S,j}. \tag{11.4}$$

We have kept the transmission factor β_{jk} potentially different for between-group contacts; for most applications, $\beta_{jk} = \beta$ is a single number. With $\beta_{jk} = \beta$ and equal contact rates ($c^S_{jk} = c^I_{jk} = c^R_{jk} = C_{jk}$), the pool is $C_{\text{total},kj} = C_{kj}N^c_k$, so $P^{(k)}_{S,j} = S_k/N^c_k$ exactly, and the force from infection for a group-j infectious individual simplifies to

$$\alpha_j \;=\; \sum_k C_{jk}\beta\,\frac{S_k}{N^c_k}. \tag{11.5}$$

The full two-group SIR dynamics are a six-compartment system. We write the dynamics in target-form: each compartment's gain or loss is expressed in terms of the new-infection flux *into* the target group. Define the target-form per-group new-infection rate

$$\Phi_k \;=\; \sum_j c^I_{jk}\,\beta_{jk}\,P^{(k)}_{S,j}\,I_j, \tag{11.6}$$

which is the rate at which group-k susceptibles get infected (sum over source groups j of the per-source flux). The full dynamics are

$$\dot S_k = \nu_m N^c_k - \Phi_k - \nu_m S_k, \tag{11.7a}$$
$$\dot I_k = \Phi_k - \gamma_R I_k - \nu_m I_k, \tag{11.7b}$$
$$\dot R_k = \gamma_R I_k - \nu_m R_k, \tag{11.7c}$$

for $k = 1, 2$, with Φ_k given by (11.6).

The relationship between the source-form quantity α_j in equation (11.4) and the target-form flux (11.6) is

$$\alpha_j\,I_j \;=\; \sum_k c^I_{jk}\,\beta_{jk}\,P^{(k)}_{S,j}\,I_j \;=\; \sum_k \frac{\Phi_k\big|_{\text{from source } j}}{1},$$

that is, $\alpha_j I_j$ is the total new-infection rate *from* source group j, distributed across target groups by the contact-and-conversion structure. The source decomposition $\sum_j \alpha_j I_j$ and the target decomposition $\sum_k \Phi_k$ give the same total new-infection rate but split it differently across the four indices

(j, k). The single-compartment SIR form $\dot{I} = \alpha I$ of Chapter 4 sits at the boundary case where source and target coincide trivially; in the multi-group setting they do not, and the target-form ODE (11.7) above is the correct generalization.

The system has six state variables $(S_1, S_2, I_1, I_2, R_1, R_2)$ and, in general, a larger parameter set than the single-group model: two recovery rates γ_R (assumed equal here for simplicity), two migration rates ν_m (similarly assumed equal; Pitfall: "Short-Term Simulation Assumptions"), the full mixing matrix $\mathbf{C}$ (three independent entries after reciprocity), and the transmission factor β.

Disease-free equilibrium. Setting all time derivatives to zero with $I_j = 0$ for $j = 1, 2$ gives

$$(S_1^0, S_2^0, I_1^0, I_2^0, R_1^0, R_2^0) \;=\; (N_1^c, N_2^c, 0, 0, 0, 0).$$

Existence and uniqueness are immediate.

Endemic equilibrium. The endemic equilibrium is more involved. For $I_k^* > 0$ for both groups, setting $\dot{I}_k = 0$ in (11.7) gives $\Phi_k^* = (\gamma_R + \nu_m)\, I_k^*$ for each k, equivalently the per-capita target-form force of infection Φ_k^*/I_k^* equals the removal rate $\gamma_R + \nu_m$. This is two conditions. Combined with the four balance equations (one for each S_k and each R_k), we have six conditions on six unknowns. The resulting system is generally nonlinear and does not admit a closed-form solution in the fully general case.

In the special case of equal contact rates within each group ($c_j^S = c_j^I = c_j^R$) and symmetric within-group parameters, explicit expressions can be derived; we defer the details to Exercise 1.

Remark 11.1 (Coupling is instantaneous: a continuum artifact at first contact). The between-group terms in (11.7) couple the two populations *instantaneously*: the instant group 1 carries any infection, the cross term Φ_2 is positive and seeds group 2, however small I_1 may be. Generalized to a metapopulation or network of many communities—each community a node, each link a mixing entry—these terms light up every reachable node the moment the source node is infected, at infinitesimal fractional-individual levels. An individual-based or network model behaves differently: a node acquires infection only when the first infectious *person* actually arrives, so the pathogen spreads as a sequence of discrete introductions with real waiting times between them, not as an instantaneous wash across the network. Like the temporal fade-out of Pitfall: "The Stochastic-to-Deterministic Regime Transition", this deterministic coupling is the mean-field skeleton of a process that is, at first contact between populations, fundamentally stochastic—reliable once each node's outbreak is established, and misleading in the earliest stage, where the timing of the first cross-population case can turn on a single traveler. Network and metapopulation epidemic models are developed by Brauer et al. [2019, Sec. 4.2] and Martcheva [2015, Ch. 15].

11.4 The Basic Reproductive Number via the Next-Generation Matrix

The two-group model is the first setting in which $\mathcal{R}_0$ is unambiguously a matrix-derived quantity. The heuristic individual-tracking derivation of Chapter 5, Section 5.1, and its generalizations to the scalar and branching extensions of Chapter 10, Section 10.7, does not extend to the two-group setting. The reason is structural, not technical: in a bipartite transmission system a newly-infectious individual in group 1 produces secondary infections in group 2, and those group-2 individuals produce their own secondary infections back in group 1, so a single generation of secondary infections

counts individuals of the opposite type than the one that started the chain. The "single typical infectious individual" whose secondary-infection count we summed in Chapter 5, Section 5.1 no longer exists: group-1 and group-2 infectious individuals produce different numbers of secondary infections of different types, and no single-individual narrative closes the counting loop in one generation. The correct $\mathcal{R}_0$ is the geometric mean of the two directional rates, a result that falls out of the spectral radius of a 2×2 matrix with zeros on the diagonal but requires a second-generation argument if derived heuristically. The next-generation matrix (NGM) method of van den Driessche and Watmough [2002], introduced in Section 5.3 of Chapter 5, handles the bipartite structure mechanically through $\rho(\boldsymbol{F}\boldsymbol{V}^{-1})$; we use it exclusively in this chapter and in the sexual-transmission (Chapter 12) and vector-host (Chapter 13) chapters that share the same bipartite structure.

11.4.1 The NGM for Two Groups

At the disease-free equilibrium $(N_1^{\mathrm{c}}, 0, N_2^{\mathrm{c}}, 0, 0, 0)$, the dynamics of the infected compartments (I_1, I_2) linearize to

$$\begin{pmatrix} \dot{I}_1 \\ \dot{I}_2 \end{pmatrix} \;=\; (\boldsymbol{F} - \boldsymbol{V}) \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}, \tag{11.8}$$

with $\boldsymbol{F}$ the matrix of new-infection rates and $\boldsymbol{V}$ the matrix of all other transitions out of the infected compartments.

Computing $\boldsymbol{F}$. At the DFE, group-k is entirely susceptible: $S_k = N_k^{\mathrm{c}}, I_k = R_k = 0$. A single infectious individual in group j produces new infections in group k at rate $c_{jk}^I \beta$, provided the contact is with a susceptible (which it is, at the DFE, with probability near 1). The rate of new group-k infections from group-j infectious is therefore $c_{jk}^I \beta$, and the matrix element is

$$F_{kj} \;=\; c_{jk}^I \beta \qquad \text{(rate of new group-}k\text{ infection per group-}j\text{ infectious).}$$

Here we have written F_{kj} with k as the "new-infection" row and j as the "source" column, consistent with the convention that $\boldsymbol{F}$ acts on the vector (I_1, I_2) to produce $(\dot{I}_1, \dot{I}_2)$ at rates arising from new infections.

Assuming $\beta_{jk} = \beta$ (single transmission factor),

$$\boldsymbol{F} \;=\; \beta \begin{pmatrix} c_{11}^I & c_{21}^I \\ c_{12}^I & c_{22}^I \end{pmatrix} \;=\; \beta\, (\mathbf{C}^I)^T. \tag{11.9}$$

The transpose appears because of the indexing convention: $\mathbf{C}_{jk}^I$ is the rate at which group-j infectious individuals contact group-k individuals, while $\boldsymbol{F}$ needs rows indexed by "new-infection target."

Computing $\boldsymbol{V}$. The matrix $\boldsymbol{V}$ captures the loss of infectious individuals through recovery and migration. With equal rates across groups,

$$\boldsymbol{V} \;=\; (\gamma_R + \nu_m)\mathbf{I}, \tag{11.10}$$

where $\mathbf{I}$ is the 2×2 identity matrix. This is diagonal because recovery and migration do not mix groups: an infectious individual in group j stays in group j until it recovers or migrates out.

The NGM. The next-generation matrix is $\boldsymbol{K} = \boldsymbol{F}\boldsymbol{V}^{-1}$:

$$\boldsymbol{K} \;=\; \frac{\beta}{\gamma_R + \nu_m}\begin{pmatrix} c_{11}^I & c_{21}^I \\ c_{12}^I & c_{22}^I \end{pmatrix} \;=\; \frac{\beta}{\gamma_R + \nu_m}(\mathbf{C}^I)^T. \tag{11.11}$$

This is a 2×2 matrix whose (k, j) entry is the expected number of new group-k infections generated by a single group-j infectious individual over its entire infectious period.

$\mathcal{R}_0$ as the spectral radius. The basic reproductive number of a multi-compartment model is the spectral radius (largest absolute eigenvalue) of $\boldsymbol{K}$:

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}). \tag{11.12}$$

For the 2×2 case, $\rho(\boldsymbol{K})$ is an eigenvalue of the matrix, computed from the characteristic polynomial

$$\det(\boldsymbol{K} - \mathcal{R}\mathbf{I}) \;=\; 0.$$

Expanding,

$$\mathcal{R}^2 - \operatorname{tr}(\boldsymbol{K})\mathcal{R} + \det(\boldsymbol{K}) \;=\; 0,$$

so the two eigenvalues are

$$\mathcal{R}_\pm \;=\; \frac{\operatorname{tr}(\boldsymbol{K}) \pm \sqrt{\operatorname{tr}(\boldsymbol{K})^2 - 4\det(\boldsymbol{K})}}{2},$$

and $\mathcal{R}_0 = \mathcal{R}_+$, the larger root.

Example: symmetric groups with purely within-group contact. For $C_{11} = C_{22} = c$ and $C_{12} = C_{21} = 0$ (no between-group contact), $\boldsymbol{K}$ is diagonal with $\boldsymbol{K}_{11} = \boldsymbol{K}_{22} = c\beta/(\gamma_R + \nu_m)$. The spectral radius is $c\beta/(\gamma_R + \nu_m)$, which is the single-group SIR value as expected.

Example: symmetric groups with pure between-group contact. For $C_{11} = C_{22} = 0$ and $C_{12} = C_{21} = c$ (purely cross-group contact), $\boldsymbol{K}$ has trace 0 and determinant $-c^2\beta^2/(\gamma_R+\nu_m)^2$. The characteristic polynomial is $\mathcal{R}^2 - c^2\beta^2/(\gamma_R + \nu_m)^2 = 0$, giving $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$. Same value as the within-group-only case.

Example: asymmetric groups with mixed contact. For generic C_{jk}, the spectral radius depends on all three independent mixing-matrix entries. We develop this in the next section, distinguishing assortative from disassortative mixing.

From $\mathcal{R}_0$ to $\mathcal{R}_e(t)$ in the two-group setting. The NGM derivation above produces $\mathcal{R}_0$ at the disease-free equilibrium, where both groups are fully susceptible: $S_1/N_1^c = S_2/N_2^c = 1$. As an outbreak progresses and each group's susceptible fraction depletes, the operationally-relevant quantity is the time-varying effective reproductive number $\mathcal{R}_e(t)$, introduced in Chapter 5, Section 5.2 for the homogeneous case and derived from the renewal-equation perspective in Chapter 15, Section 15.5.3. In the two-group setting, the construction generalizes mechanically: the new-infection matrix at time t, written $\boldsymbol{F}(t)$, is built by replacing each entry F_{kj} of $\boldsymbol{F}$ with $F_{kj}\, S_k(t)/N_k^c$. The index convention is the one fixed above: the row k is the group acquiring infection and the column j is the group transmitting it, so the susceptible fraction that enters is the *target* group's, S_k/N_k^c, not the

source group's. The transition matrix $\boldsymbol{V}$ is unchanged, since depletion affects who can be infected rather than how long an infection lasts. The effective reproductive number is then

$$\mathcal{R}_e(t) \;=\; \rho\big(\boldsymbol{F}(t)\,\boldsymbol{V}^{-1}\big),$$

the spectral radius of the time-dependent next-generation matrix $\boldsymbol{K}(t) = \boldsymbol{F}(t)\boldsymbol{V}^{-1}$. Note that $\boldsymbol{V}^{-1}$ is applied once: it is $\boldsymbol{F}$, not $\boldsymbol{K}$, that carries the time dependence. The same threshold interpretation applies: $\mathcal{R}_e(t) > 1$ indicates currently-growing transmission averaged over the two groups; $\mathcal{R}_e(t) < 1$ indicates currently-declining transmission. Two structural points are worth noting. First, $\mathcal{R}_e(t)$ in the heterogeneous setting does not factor cleanly as $\mathcal{R}_0 \cdot \bar{S}(t)/N$ for any scalar averaging $\bar{S}$ unless mixing is homogeneous; the depletion factor enters per-group and the spectral radius of the modified matrix may not be a simple multiple of $\mathcal{R}_0$. Second, the eigenvector of $\boldsymbol{K}(t)\boldsymbol{V}^{-1}$ at the threshold $\mathcal{R}_e(t) = 1$ identifies which mixture of group-1 and group-2 infections constitutes the marginal transmission generation; this is the heterogeneous analogue of the single-population threshold condition. The Chapter 13 treatment of bipartite vector-host transmission applies the same construction to the human-mosquito setting, and the COVID-19 case study of Chapter 17 uses the labelled-superscript convention $\mathcal{R}_0^{\text{intervention}}$ for intervention-modified reproductive numbers in heterogeneous populations.

Exercises for this section

1. **[C]** The reproductive number for the two-group model is the spectral radius of a 2×2 NGM. Explain in plain English why a single eigenvalue captures the threshold for invasion in a multi-group system, and what biological information the corresponding eigenvector encodes.

2. **[I]** The text states that $\mathcal{R}_e(t)$ in a heterogeneous setting does not factor cleanly as $\mathcal{R}_0 \cdot \bar{S}(t)/N$ for any scalar averaging $\bar{S}$ unless mixing is homogeneous. Construct a concrete two-group example: pick numerical values for $C_{11}, C_{12}, C_{21}, C_{22}, \beta, \gamma_R, \nu_m$ and a group-specific susceptible-fraction depletion $S_1(t)/N_1^c = 0.7$ versus $S_2(t)/N_2^c = 0.3$. Compute $\mathcal{R}_0$ and $\mathcal{R}_e(t)$, then check whether any single weighted average of the depletion factors reproduces the ratio $\mathcal{R}_e(t)/\mathcal{R}_0$. Discuss what this implies for surveillance practice in heterogeneous populations.

11.5 Assortative and Disassortative Mixing

The mixing matrix $\mathbf{C}$ encodes the between-group contact structure. Certain limiting structures are particularly informative because they reveal the qualitative effect of heterogeneity on disease spread.

11.5.1 Assortative Mixing

Mixing is *assortative* when within-group contact exceeds between-group contact: $C_{11} > C_{12}$ and $C_{22} > C_{21}$. Individuals preferentially contact members of their own group.

A common parameterization uses a single assortativity parameter $\epsilon \in [0, 1]$:

$$\mathbf{C} \;=\; (1-\epsilon)\mathbf{C}_{\text{random}} + \epsilon\,\mathbf{D}, \tag{11.13}$$

where $\mathbf{C}_{\text{random}}$ is the random-mixing matrix (proportional to group sizes) and $\mathbf{D}$ is a diagonal "pure-within-group" matrix. As $\epsilon \to 0$, mixing is random; as $\epsilon \to 1$, the two groups are isolated from each

other. The group-size form of $\mathbf{C}_{\text{random}}$ satisfies the balance equation (11.2) only when $c_1 = c_2$. When the contact rates differ it violates balance (Pitfall: "Heterogeneous-Mixing Balance Conditions"), so the case study in Section 11.8 sets its second row by reciprocity instead.

Effect on $\mathcal{R}_0$. In the isolated-groups limit ($\epsilon \to 1$), $\mathbf{C}$ is diagonal with entries c_1^W, c_2^W (the within-group contact rates). The NGM is diagonal with entries $c_j^W \beta/(\gamma_R + \nu_m)$, and

$$\mathcal{R}_0 \;=\; \max_j \frac{c_j^W \beta}{\gamma_R + \nu_m}.$$

The epidemic is driven by the group with the higher within-group contact rate. The other group has its own, smaller reproductive number, and if that number is below unity, the epidemic in that group dies out while the other group sustains it.

For intermediate ϵ, between-group contact couples the groups and $\mathcal{R}_0$ reflects a weighted average.

11.5.2 Disassortative Mixing

Mixing is *disassortative* when between-group contact exceeds within-group contact: $C_{12} > C_{11}$ and $C_{21} > C_{22}$. Individuals preferentially contact members of the *other* group. This is the dominant pattern for sexually-transmitted infections (with heterosexual transmission, where C_{12} is male-female and C_{21} is female-male, both potentially much larger than within-sex contact). Chapter 12 develops the sexually-transmitted infection case in detail.

Effect on $\mathcal{R}_0$. In the pure-disassortative limit ($C_{11} = C_{22} = 0$, $C_{12}, C_{21} > 0$), the NGM has zero diagonal and non-zero off-diagonal. Its spectral radius is

$$\mathcal{R}_0 \;=\; \frac{\beta}{\gamma_R + \nu_m}\sqrt{C_{12}C_{21}}. \tag{11.14}$$

The characteristic value is the geometric mean of the two between-group contact rates. Note that $\mathcal{R}_0$ here involves $\sqrt{\cdot}$ of contact rates, not contact rates themselves. This is qualitatively different from the homogeneous or assortative cases. The square-root structure reflects a general feature of bipartite and cyclic transmission models: the spectral radius of the next-generation matrix equals the geometric mean of directional reproductive numbers, a relationship formalized in Section 11.6 below. Chapters 12 and 13 apply the formalism to sexually-transmitted and vector-borne diseases.

Verify the AI

The mixing-balance arithmetic trap. Asked for a mixing matrix $\boldsymbol{C}$ for two unequal groups, AI assistants routinely compute the off-diagonal entries independently and break the conservation law (11.2), $C_{12}N_1^c = C_{21}N_2^c$; the matrix looks plausible and the bookkeeping fails.
Ask: *"Generate a 2×2 mixing matrix $\boldsymbol{C}$ for two groups of sizes $N_1^c = 900$ and $N_2^c = 100$, with $C_{12} = 2$ contacts per group-1 individual per day with group 2. State the value of C_{21} and verify the balance equation."* A correct answer derives $C_{21} = C_{12}N_1^c/N_2^c = 18$ and checks $1800 = 1800$. The symmetric guess $C_{21} = 2$ misses by nearly an order of magnitude, and rounding 18 to 20 gives 1800 against 2000: small enough to pass a glance, large enough to bias $\mathcal{R}_0$.
Structural verification. Check the balance for any proposed matrix: in the pure-disassortative case (11.14) a ten-percent violation biases $\mathcal{R}_0$ by about five percent. Take the better-supported entry from data and derive the other from the constraint.

11.5.3 Proportional Mixing (Random)

Random or proportional mixing is the intermediate case: individuals contact each other at rates proportional to the other group's size. The mixing matrix is

$$C_{jk} = \frac{c\, N_k^{\mathrm{c}}}{N^{\mathrm{c}}},$$

where c is a single overall contact rate per person. The reciprocity constraint (11.2) is automatically satisfied. In this case, the NGM reduces to a rank-1 matrix:

$$\boldsymbol{K} = \frac{c\beta}{\gamma_R + \nu_m} \begin{pmatrix} N_1^{\mathrm{c}}/N^{\mathrm{c}} & N_1^{\mathrm{c}}/N^{\mathrm{c}} \\ N_2^{\mathrm{c}}/N^{\mathrm{c}} & N_2^{\mathrm{c}}/N^{\mathrm{c}} \end{pmatrix},$$

whose spectral radius is $c\beta/(\gamma_R + \nu_m)$ (the single non-zero eigenvalue; the other eigenvalue is 0). Under random mixing, the two-group model gives the same $\mathcal{R}_0$ as the homogeneous one-group model.

This observation generalizes: random mixing produces no net heterogeneity effect. The two-group model differs from the one-group model only to the extent that mixing departs from random. The proportionate-mixing ansatz was introduced as an analytical simplification by Dietz and Schenzle [1985] and remains a standard reference point; activity-level heterogeneity extensions of this ansatz were developed in depth by Hyman and Stanley [1989] and in the textbook treatment of Anderson and May [1991, Ch. 11].

Exercises for this section

1. [C] Distinguish in one paragraph between assortative and disassortative mixing. For each, give one real-world example (e.g., age-structured respiratory disease vs. heterosexually transmitted infection) and state which of the two patterns is dominant.

11.6 Stage, Cyclic, and Geometric-Mean Reproductive Numbers

The square-root formula (11.14) that emerged for pure-disassortative two-group mixing is not an accident of that special case. It reflects a general structural property of bipartite and cyclic transmission models. To keep the notation precise in such models, we distinguish three related reproductive-number quantities. The book-wide threshold quantity $\mathcal{R}_0$ remains the spectral radius of the next-generation matrix throughout; the new quantities are directional and cycle-level companions that clarify what $\mathcal{R}_0$ is averaging.

Definition 11.2 (Stage reproductive number)**.** For a multistage transmission model with infected stages

$$I_1 \to I_2 \to \cdots \to I_N \to I_1,$$

the *stage reproductive number* $\mathcal{R}_0^{\ell\to\ell+1}$ is the expected number of newly infectious individuals in stage $\ell+1$ generated by one infectious individual in stage ℓ when the population of stage $\ell+1$ is fully susceptible. Its effective counterpart is denoted $\mathcal{R}_e^{\ell\to\ell+1}(t)$.

Definition 11.3 (Cyclic reproductive number)**.** The *cyclic reproductive number* based at stage ℓ is the expected number of new infections in the same stage ℓ produced after one full return cycle through all stages. For a simple N-stage cycle,

$$\mathcal{C}_0^{(\ell)} \;=\; \mathcal{R}_0^{\ell\to\ell+1}\,\mathcal{R}_0^{\ell+1\to\ell+2}\cdots\mathcal{R}_0^{\ell-1\to\ell}, \tag{11.15}$$

where stage indices are understood cyclically. The effective cyclic reproductive number is denoted $\mathcal{C}_e^{(\ell)}(t)$.

Relation to the next-generation matrix. The book reserves the notation $\mathcal{R}_0 = \rho(\boldsymbol{K})$ for the standard next-generation-matrix quantity. For a simple cyclic multistage model in which transmission flows from each stage only to the next and returns to the origin after L steps,

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}) \;=\; \big(\mathcal{C}_0^{(\ell)}\big)^{1/L}. \tag{11.16}$$

That is, the next-generation-matrix reproductive number is the geometric mean of the stage reproductive numbers, equivalently the Lth root of the cyclic reproductive number. Identity (11.16) is the formal content of the square-root structure seen for two-group disassortative mixing: with $L = 2$ stages (group 1 and group 2), the geometric mean becomes a square root, and $\mathcal{R}_0 = \sqrt{\mathcal{R}_0^{1\to2}\,\mathcal{R}_0^{2\to1}}$.

Remark 11.4 (Scope of the cycle-root identity). Equation (11.16) is a special property of simple cyclic multistage models. In a more general multigroup model with multiple infection pathways, branching, feedback loops, or several infected compartments per stage, the spectral radius $\rho(\boldsymbol{K})$ need not reduce to the Lth root of a single cycle product. In those settings, $\mathcal{R}_0 = \rho(\boldsymbol{K})$ remains the threshold quantity, but stage-level and cycle-level quantities must be interpreted with more care; the HIV three-group model of Chapter 19, Section 19.2 and the SIAR asymptomatic-symptomatic split of Chapter 10, Section 10.6 are examples where the simple-cycle identity does not apply.

Why this framework matters. The three-level terminology separates three questions that are easy to conflate in bipartite and cyclic models:

(i) How many infections does one stage produce in the next stage? This is a stage reproductive number $\mathcal{R}_0^{\ell\to\ell+1}$, a per-direction quantity that is generally asymmetric in ℓ.

(ii) How many same-stage infections accumulate after one complete return cycle? This is the cyclic reproductive number $\mathcal{C}_0^{(\ell)}$.

(iii) What scalar from the next-generation matrix determines the threshold? This is $\mathcal{R}_0 = \rho(\boldsymbol{K})$, which for a simple cycle equals the geometric mean of the stage reproductive numbers.

$\mathcal{R}_0$ and $\mathcal{C}_0^{(\ell)}$ cross the threshold together ($\mathcal{C}_0^{(\ell)} > 1 \iff \mathcal{R}_0 > 1$ for simple cycles), so either determines invasion or decline. The stage numbers do not: $\mathcal{R}_0$ is their geometric mean, so one stage number can sit below 1 while the infection invades. Their *numerical* values differ, and the book consistently reports $\mathcal{R}_0 = \rho(\boldsymbol{K})$; readers encountering other sources should check which quantity is being quoted.

Connection with older terminology. Earlier literature sometimes distinguishes only a *per-cycle* reproductive number (corresponding to $\mathcal{C}_0^{(\ell)}$) from a *per-generation* reproductive number (corresponding to $\mathcal{R}_0 = \rho(\boldsymbol{K})$ in the simple-cyclic case). The stage/cyclic/geometric-mean terminology developed here is strictly more precise: it names the directional quantities whose product forms the cyclic value, rather than collapsing them into a single per-generation scalar. Chapters 12 and 13 use the framework in concrete bipartite settings.

11.7 Herd Immunity in Heterogeneous Populations

The herd immunity threshold was introduced in Chapter 5, Section 5.1: for a homogeneous population, a fraction $1 - 1/\mathcal{R}_0$ of the population must be immunized (or equivalently removed from the susceptible compartment) to bring the effective reproductive number below 1. This section examines how that threshold changes in a heterogeneous population.

Uniform immunization. Suppose a fraction f of each group is immunized, independent of group membership. The susceptible fraction is reduced by $(1-f)$ in every group, so the force-from-infection scales by $(1 - f)$. The effective reproductive number becomes $\mathcal{R}_e = (1 - f)\mathcal{R}_0$. Setting $\mathcal{R}_e = 1$ gives the uniform-immunization threshold

$$f^{\text{uniform}} \;=\; 1 - \frac{1}{\mathcal{R}_0}.$$

This matches the homogeneous result.

Group-targeted immunization. Now let the coverage differ between the groups, immunizing one group preferentially. Let f_1, f_2 be the immunization fractions of groups 1 and 2. The effective NGM is

$$\boldsymbol{K}^{\text{eff}} \;=\; \begin{pmatrix} (1 - f_1) & 0 \\ 0 & (1 - f_2) \end{pmatrix} \boldsymbol{K}.$$

The effective reproductive number is the spectral radius of $\boldsymbol{K}^{\text{eff}}$. For the special case of symmetric groups with pure between-group contact (pure disassortative mixing from Section 11.5),

$$\mathcal{R}_e \;=\; \sqrt{(1 - f_1)(1 - f_2)}\,\mathcal{R}_0.$$

Setting $\mathcal{R}_e = 1$ gives the constraint

$$(1 - f_1)(1 - f_2) \;=\; 1/\mathcal{R}_0^2.$$

There is a continuum of (f_1, f_2) pairs satisfying this constraint, and it is worth being careful about which of them is cheapest, because the intuitive answer is wrong. Write $u = 1-f_1$ and $v = 1-f_2$, so the constraint is $uv = 1/\mathcal{R}_0^2$ and the total immunized fraction (for equal group sizes) is $(2-u-v)/2$. Minimizing the total therefore means *maximizing* $u + v$ subject to a fixed product—and by the arithmetic–geometric mean inequality $u + v \geq 2\sqrt{uv}$, with equality exactly when $u = v$. Equal allocation $f_1 = f_2 = 1 - 1/\mathcal{R}_0$ is thus the allocation that requires the *most* total vaccination, not the least. The minimum sits at the opposite extreme, on the boundary: for $\mathcal{R}_0 = 2$, equal allocation immunizes half the population, while immunizing group 2 alone to $f_2 = 1 - 1/\mathcal{R}_0^2 = 0.75$ meets the same constraint at a total of 0.375. Pushing this to its limit, if we can achieve $f_1 = 1$ (fully immunize group 1), then $f_2 = 0$ suffices regardless of $\mathcal{R}_0$: the epidemic cannot be sustained by group 1 alone because group-1 individuals contact only group-2 individuals, and all group-1 individuals are immune. Figure 11.3(a) draws this boundary with the uniform and the cheapest allocations marked.

This extreme case illustrates the principle: targeted immunization of the high-contact group achieves herd immunity at a lower total vaccination fraction than uniform immunization. The principle extends to more realistic mixing patterns and is the theoretical basis for "ring vaccination" and other targeted immunization strategies.

Coverage weighted by contact rate. Between uniform coverage and full targeting lies a middle course: vaccinate each group in proportion to its contact rate, $f_j = k\,c_j$, so that people who make more contacts are more likely to be immunized. Under proportional mixing the next-generation matrix has rank one, with entries proportional to $c_j N_j^{\mathrm{c}} c_k$, and the coverage scales its rows. The effective reproductive number is then

$$\mathcal{R}_e \;=\; \mathcal{R}_0 \left(1 - k\,\frac{\langle c^3\rangle}{\langle c^2\rangle}\right),$$

where $\langle c^n\rangle$ is the nth moment of the contact-rate distribution. Setting $\mathcal{R}_e = 1$ and adding up the doses gives the total coverage

$$F^{\text{contact}} \;=\; \Big(1 - \frac{1}{\mathcal{R}_0}\Big)\frac{\langle c\rangle\langle c^2\rangle}{\langle c^3\rangle}.$$

Since c and c^2 rise together, Chebyshev's sum inequality gives $\langle c\rangle\langle c^2\rangle \le \langle c^3\rangle$, with equality only when every contact rate is the same. So $F^{\text{contact}} \le f^{\text{uniform}}$: weighting coverage by contact rate never needs more vaccine than uniform coverage. For two groups of equal size with $c_2 = 2c_1$ and $\mathcal{R}_0 = 2$, uniform coverage needs 0.5 and contact-weighted coverage needs 0.417. The gap widens as the contact rates spread apart.

Minimum-total-vaccinations targeted immunization. A different optimization objective (minimize total vaccinations subject to $\mathcal{R}_e \le 1$) produces the targeted-immunization principle stated above: vaccinate the highest-contact group preferentially [Anderson and May, 1985]. For the bipartite case with the constraint $(1 - f_1)(1 - f_2) = 1/\mathcal{R}_0^2$ and unequal group sizes $N_1^{\mathrm{c}}, N_2^{\mathrm{c}}$, the minimum-total objective $\min(f_1 N_1^{\mathrm{c}} + f_2 N_2^{\mathrm{c}})$ has a corner solution at $f_1 = 1 - 1/\mathcal{R}_0^2$, $f_2 = 0$ (or its symmetric variant) when feasible: vaccinating the small high-contact group to a high rate, leaving the large low-contact group unvaccinated. The minimum-total threshold can be substantially lower than uniform when the high-contact group is small in absolute size.

The three thresholds answer different questions. f^{uniform} asks what equal coverage in every group achieves $\mathcal{R}_e = 1$; it matches the practical default of programs that reach people roughly in proportion to population. F^{contact} asks the same of coverage that follows contact rates, which needs a way to reach people according to how much they mix. $f^{\text{min-total}}$ asks for the smallest total fraction that achieves $\mathcal{R}_e = 1$ under any allocation; it is the benchmark when administrative capacity allows targeted vaccination of identifiable high-contact subgroups (essential workers, healthcare workers, hub-occupation members).

The practical implication is that heterogeneity in contact patterns enables targeted-allocation strategies to achieve herd immunity with fewer total vaccinations than uniform coverage, a resource-saving observation that has motivated targeted immunization programs in public-health practice.

Exercises for this section

1. **[D]** Under proportional mixing, take three groups with equal group sizes $N_k^{\mathrm{c}} = N^{\mathrm{c}}/3$ and contact rates $c_1 = 20, c_2 = 10, c_3 = 5$. Compute the contact-weighted threshold F^{contact} for $\mathcal{R}_0 = 3$ and compare it with the uniform threshold and with the smallest total coverage under any allocation.

11.8 A Numerical Case Study

Consider a two-group population in which group 1 represents "essential workers" (high contact rate due to occupation) and group 2 represents "other adults" (lower contact rate). Figure 11.2 shows the contact structure before specifying the parameters.

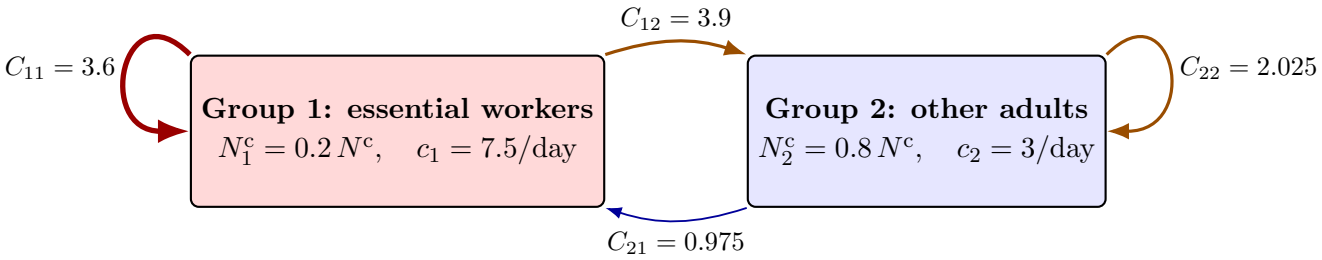


Figure 11.2: The two-group case study. Arrow thickness indicates contact-rate magnitude; within-group loops (curved self-arrows) represent the diagonal entries of the mixing matrix, and between-group arrows represent the off-diagonal entries. The asymmetry $C_{12} = 3.9 \neq C_{21} = 0.975$ reflects group-size differences under the reciprocity constraint $C_{12}N_1^c \approx C_{21}N_2^c$. Group 1 workers have a higher total contact rate and higher within-group mixing than Group 2.

Hypothetical parameter values:

- Group 1 size: $N_1^c = 0.2N^c$ (20% of population).
- Group 2 size: $N_2^c = 0.8N^c$.
- Contact rates: $c_1 = 7.5$ contacts/day (group 1), $c_2 = 3$ contacts/day (group 2).
- Between-group mixing: $\epsilon = 0.35$ assortativity parameter (moderate assortative).
- Transmission factor: $\beta = 0.040$.
- Recovery rate: $\gamma_R = 0.083$ day^{-1} ($\tau_R = 12$).
- Migration rate: $\nu_m \approx 0.000137$ day^{-1} ($\tau_m = 7300$, i.e., ≈ 20 years).

The full mixing matrix, with row j being "group-j contacts with group-k," is approximately

$$\mathbf{C} \approx \begin{pmatrix} 3.6 & 3.9 \\ 0.975 & 2.025 \end{pmatrix}$$

(the reciprocity constraint $C_{12}N_1^c = C_{21}N_2^c$ is satisfied exactly: $3.9 \times 0.2 = 0.78 = 0.975 \times 0.8$. The off-diagonal entries are not independent—once C_{12} and the group sizes are fixed, reciprocity determines $C_{21} = C_{12}N_1^c/N_2^c$, and the diagonal entries follow from the group contact rates $c_1 = 7.5$ and $c_2 = 3$. Calibrating a mixing matrix means enforcing this balance, not fitting all four entries freely).

Computing $\mathcal{R}_0$. The NGM, with $\gamma_R + \nu_m$ rounded to 0.0835, is

$$\boldsymbol{K} \;=\; \frac{0.040}{0.0835}\begin{pmatrix} 3.6 & 0.975 \\ 3.9 & 2.025 \end{pmatrix} \;\approx\; \begin{pmatrix} 1.72 & 0.47 \\ 1.87 & 0.97 \end{pmatrix}.$$

The trace is 2.69 and the determinant is $1.67 - 0.87 = 0.80$. The eigenvalues solve

$$\mathcal{R}^2 - 2.69\,\mathcal{R} + 0.80 \;=\; 0,$$

giving $\mathcal{R} \in \{2.35, 0.34\}$. The basic reproductive number is

$$\mathcal{R}_0 \;=\; 2.35.$$

For comparison, the single-group "population-average" computation using the population-averaged contact rate $\bar{c} = 0.2 \times 7.5 + 0.8 \times 3 = 3.9$ gives $\mathcal{R}_0^{\text{avg}} = 3.9 \times 0.040/0.0835 \approx 1.87$. The heterogeneity-aware computation gives a 26% higher estimate.

The source of the difference. The heterogeneity-aware computation weights contacts by the group making them. Group 1's 7.5 contacts/day are weighted heavily because they contribute strongly to $\mathcal{R}_0$; the population average under-weights them and over-weights group 2's low-impact contacts. The lesson, stated with the qualifier it needs: computing $\mathcal{R}_0$ from a population-averaged contact rate underestimates the true reproductive number when high-contact individuals also contact each other preferentially, as under proportional mixing (Section 11.5). The direction is general when the mixing matrix balances and β and the removal rate are common to the groups: the averaged rate never overestimates $\mathcal{R}_0$, and matches it only when every group makes the same number of contacts. To see why, write $A_{jk} = C_{jk}N_j^{\text{c}}$ for the contacts per day between groups j and k; balance makes $\mathbf{A}$ symmetric, so $\mathbf{C}$ is similar to the symmetric matrix $\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$, with $\mathbf{D} = \text{diag}(N_j^{\text{c}})$, and $\rho(\mathbf{C})$ is the largest value of its Rayleigh quotient. At the trial vector $x_j = (N_j^{\text{c}}/N^{\text{c}})^{1/2}$ the quotient equals the population-average contact rate $\bar{c} = \sum_j N_j^{\text{c}} c_j / N^{\text{c}}$, where $c_j = \sum_k C_{jk}$, so $\rho(\mathbf{C}) \geq \bar{c}$, with equality only when that vector is an eigenvector, that is, when every c_j is the same. The disassortative case of Section 11.5 is no exception: with groups of 20% and 80% and $C_{12} = 4$, $C_{21} = 1$, the geometric mean is 2 and the population average 1.6. What the average discards is the covariance between how often people mix and whom they reach.

Herd immunity threshold. We compute the three herd immunity thresholds of Section 11.7: uniform coverage, coverage weighted by contact rate, and the smallest total.

The uniform-immunization threshold is

$$f^{\text{uniform}} \;=\; 1 - 1/2.35 \;\approx\; 0.575.$$

This population mixes assortatively rather than proportionally, so the closed form for F^{contact} does not apply, and the threshold must be computed on the case study's own $\boldsymbol{K}$. Coverage $f_j = k\,c_j$, with $c_1 = 7.5$ and $c_2 = 3$, brings $\mathcal{R}_e$ to 1 at $f_1 \approx 0.73$ and $f_2 \approx 0.29$, a total of $0.2 \times 0.73 + 0.8 \times 0.29 \approx 0.38$. Weighting by contact rate saves about a third of the uniform coverage.

The minimum-total-vaccinations threshold goes further, and it too must be computed on this population's own boundary. The constraint $(1 - f_1)(1 - f_2) = 1/\mathcal{R}_0^2$ of Section 11.7 holds only for symmetric groups with purely between-group contact. This population mixes assortatively, so the boundary is the set of coverages with $\rho\big(\text{diag}(1 - f_1, 1 - f_2)\,\boldsymbol{K}\big) = 1$, where coverage f_j scales row j of $\boldsymbol{K}$ (Figure 11.3(b)). On that boundary the cheapest allocation is again a corner: vaccinate

none of group 2 and $f_1 \approx 0.97$ of group 1, a total of $0.2 \times 0.97 \approx 0.19$, about a third of the uniform 0.575. The bipartite constraint would have put the corner at $f_1 = 1 - 1/\mathcal{R}_0^2 \approx 0.82$, a total of 0.164, but that allocation leaves $\mathcal{R}_e \approx 1.16$, and transmission continues. Vaccinating all of group 1 ($f_1 = 1$, a total of 0.20) leaves $\mathcal{R}_e = K_{22} \approx 0.97$, just below threshold, because group 2 cannot sustain transmission on its own.

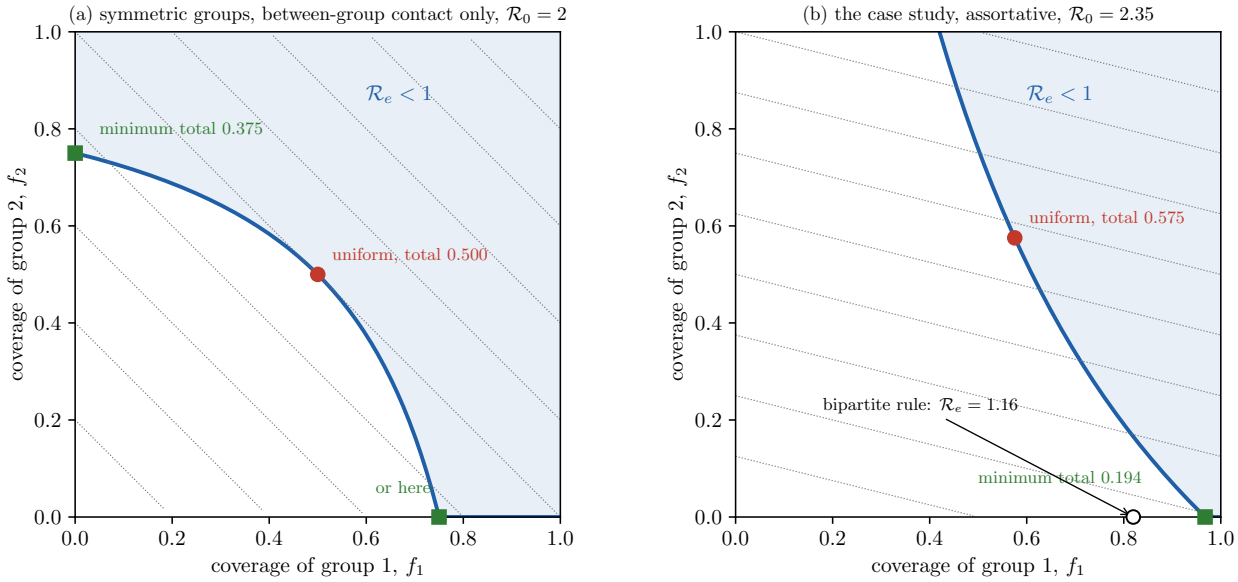


Figure 11.3: The herd-immunity boundary $\rho\big(\mathrm{diag}(1-f_1, 1-f_2)\,\boldsymbol{K}\big) = 1$ in the plane of the two coverages; above and to the right of the blue curve $\mathcal{R}_e < 1$. Dotted lines join allocations with the same total coverage. (a) Symmetric groups of equal size with purely between-group contact, $\mathcal{R}_0 = 2$: the boundary is $(1-f_1)(1-f_2) = 1/\mathcal{R}_0^2$, uniform coverage needs a total of 0.5, and vaccinating one group alone to 0.75 needs 0.375. (b) The case study of this section: uniform coverage needs 0.575; the cheapest allocation vaccinates 97% of group 1 and none of group 2, a total of 0.194. The open circle is the corner the bipartite constraint predicts ($f_1 = 0.82$), which leaves $\mathcal{R}_e = 1.16$.

This corner solution is the practical content of the targeted-immunization principle: when administrative capacity allows identification and preferential coverage of the small high-contact group, total vaccinations needed for herd immunity drop sharply.

In real outbreak response, neither extreme is operationally attainable: uniform coverage requires no targeting capability, and the corner solution requires perfect identification and full-coverage compliance in the targeted group. Practical strategies fall between the two, and the chapter's three thresholds ($f^{\text{uniform}} \approx 0.575$, $F^{\text{contact}} \approx 0.38$, $f^{\text{min-total}} \approx 0.19$) bracket the operational range.

11.9 Summary and Exercises

This chapter has introduced the two-group SIR model as the simplest non-trivial heterogeneous-population extension. The main results are:

1. The mixing matrix $\mathbf{C}$ encodes between-group contact patterns, with three independent entries after the reciprocity constraint.

2. The two-group SIR dynamics are a six-compartment system (11.7) with force from infection given by (11.4).

3. The basic reproductive number is the spectral radius of the 2×2 next-generation matrix (11.11). For a general mixing matrix, the characteristic polynomial is quadratic and the spectral radius is a closed-form expression in the entries of $\mathbf{C}$.

4. Assortative mixing ($C_{jj} > C_{jk}$) produces a characteristic group-specific epidemic pattern; disassortative mixing ($C_{jk} > C_{jj}$) produces a square-root-structured $\mathcal{R}_0$ that is the geometric mean of directional stage reproductive numbers (Section 11.6); proportional random mixing reduces to the homogeneous one-group case.

5. Uniform coverage needs $1 - 1/\mathcal{R}_0$ whatever the mixing. When contact rates differ, targeting needs less: coverage weighted by contact rate never needs more than uniform coverage, and the cheapest allocation concentrates on the high-contact group.

6. A numerical case study with essential-workers and other-adults groups shows that heterogeneity-aware computation of $\mathcal{R}_0$ can give substantially different answers from population-averaged computation.

Chapter 12 extends the two-group framework to sexually-transmitted infections, where the natural group structure is male-female with strong disassortative mixing. Chapter 13 extends to vector-host models, a different kind of two-group structure. Chapter 15 extends to continuous-age structure.

Framework use in two-group fitting. The two-group extension inherits the infected-viewpoint fitting advantage per group: the estimator $\hat{\alpha}_k = J_k/I_k$ for group k remains structurally immune to uncertainty in the group-specific susceptible population S_k, as established in the sensitivity-analysis paper of Hyman [2026d]. Two-group fits to heterogeneous-population outbreak data therefore inherit the framework's structural-immunity property at the per-group level, with the total reproductive number then computed from the 2×2 next-generation matrix as in this chapter. The practical consequence is that for an outbreak with reliable group-stratified incidence and prevalence data (for example, age-stratified COVID-19 case counts), the two-group $\mathcal{R}_0$ estimate is structurally immune to uncertainty in the denominator populations for each group.

Scope note: per-capita formulation across geographic scales. The two-group framework as developed here treats each group as a single well-mixed subpopulation. When the same physical system is modeled at several geographic resolutions (a metropolitan area treated as one mixing pool, say, or as a network of neighborhood-level pools), the choice of infection-rate formulation decides whether the results agree across resolutions. Conrad et al. [2025] test common spatial modeling assumptions mathematically and numerically. Under gravity-weighted mixing, which makes the mixing matrix asymmetric, the standard per-capita infection rate fails their consistency tests for heterogeneous populations: results such as the final epidemic size change with the grid resolution, and the errors are largest where symmetry is lowest, as in sparse geographies. An averaged infection rate passes the same tests, and a symmetric mixing matrix serves as a first check. Modelers working at a single resolution, as this book does throughout, need not address the issue; modelers who compare results across resolutions, or use outputs at one resolution to inform decisions at another, should check which formulation their model uses.

Pitfall check

This chapter has turned the catalogue's mixing entries into checks on a contact matrix. Under Pitfall: "Heterogeneous-Mixing Balance Conditions", the contacts group 1 directs at group 2 must match those group 2 receives, or contacts vanish from one side of the bookkeeping. Pitfall: "Dimensional Consistency" adds what balance cannot: the entries' sizes must fit the social structure being modeled (Section 11.2). The force from infection carries over group by group (Pitfall: "Force of Infection versus Force from Infection"), equal recovery and migration rates were assumed for simplicity (Pitfall: "Short-Term Simulation Assumptions"), and the coupled model is the mean-field skeleton of a stochastic process (Pitfall: "The Stochastic-to-Deterministic Regime Transition"); all three appear in Section 11.3. The case study is the home of Pitfall: "Contact-Rate Averaging": the population's mean contact rate gives $\mathcal{R}_0 \approx 1.87$, and the two-group computation 2.35, 26 percent higher (Section 11.8). Before trusting a heterogeneous model, check that its matrix balances, that its entries have plausible sizes, and that no rate in it is a mean over people who mix differently.

Exercises

Five section-level exercises appear above, following Section 11.2, Section 11.4, Section 11.5, and Section 11.7. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[D]** For the case of symmetric groups ($N_1^c = N_2^c = N^c/2$) with equal-contact-rates ($c^S = c^I = c^R$) and mixing matrix $\mathbf{C}$ with $C_{11} = C_{22} = w$ (within-group) and $C_{12} = C_{21} = b$ (between-group), derive the closed-form expression for the endemic equilibrium.

2. **[D]** Compute $\mathcal{R}_0$ for the following mixing matrices, assuming $\beta = 0.040$ and $\gamma_R + \nu_m = 0.0835$:

   (a) Pure assortative: $\mathbf{C} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$.

   (b) Pure disassortative: $\mathbf{C} = \begin{pmatrix} 0 & 5 \\ 5 & 0 \end{pmatrix}$.

   (c) Random proportional (equal groups): $\mathbf{C} = \begin{pmatrix} 2.5 & 2.5 \\ 2.5 & 2.5 \end{pmatrix}$.

   Are the values the same? Explain why or why not.

3. **[I]** For the case study of Section 11.8, what happens to $\mathcal{R}_0$ if we raise the assortativity parameter to $\epsilon = 0.6$ while keeping all other quantities fixed? Interpret.

4. **[D]** For a sexually-transmitted infection with pure disassortative mixing ($C_{12} = 3, C_{21} = 2$) and transmission probability β, use the framework of Section 11.6 to compute (a) the two stage reproductive numbers $\mathcal{R}_0^{1\to 2}$ and $\mathcal{R}_0^{2\to 1}$, (b) the cyclic reproductive number $\mathcal{C}_0^{(1)}$, and (c) the next-generation-matrix quantity $\mathcal{R}_0 = \rho(\boldsymbol{K})$. Verify identity (11.16) for this example, and confirm that $\mathcal{R}_0$ and $\mathcal{C}_0^{(1)}$ cross 1 together as β varies, while the two stage numbers do not. Find the range of β in which the infection invades although one stage number is below 1.

5. **[I]** In the two-group case study, suppose group 1 (essential workers) is fully immunized ($f_1 = 1$) while group 2 is not immunized ($f_2 = 0$). Compute the resulting effective reproductive number and determine whether the epidemic is contained.

6. **[D]** For a two-group model with $N_1^c = N_2^c$ and $C_{11} = C_{22} = 0$, $C_{12} = C_{21} = 5$, suppose $\beta_{12} \neq \beta_{21}$ (male-to-female and female-to-male transmission differ). Derive $\mathcal{R}_0$ in terms of β_{12} and β_{21}.

7. **[L]** Implement the two-group SIR system (11.7) and reproduce the case study of Section 11.8. Then, holding the other parameters fixed, sweep the assortativity parameter ϵ from 0 (group 1 mixing in proportion to group size) to 0.9 (highly assortative) in steps of 0.1. Plot the resulting $\mathcal{R}_0$ and the per-group endemic prevalences I_1^*, I_2^* as functions of ϵ. Identify the value of ϵ at which the two endemic prevalences differ by a factor of two.

8. **[L]** *Spaced retrieval, Chapter 7.* Apply the infected-viewpoint estimator of Chapter 7, Section 7.4.3 (recall $\hat{\alpha} = J/I$ from (7.12)) to a two-group setting. Generate synthetic per-group case time series by simulating the case-study parameters of Section 11.8 forward over 200 days and adding multiplicative log-normal noise with $\sigma = 0.10$ to both $J_j(t)$ and $I_j(t)$ for $j \in \{1, 2\}$. Then, treating each group's time series as if it came from a single homogeneous population, compute $\hat{\alpha}_j(t) = J_j(t)/I_j(t)$ over the early plateau. Show that over the early plateau the two estimates agree, and explain why: what does each group's $\hat{\alpha}_j$ estimate? Then find where in the data the mixing matrix does leave a trace. Discuss whether the fitting procedure of Chapter 7 can be applied to each group independently, or whether the cross-group coupling forces a joint fit. The exercise tests two ideas at once: (a) the infected-viewpoint estimator extends naturally from the homogeneous setting of Chapter 7 to the heterogeneous setting of this chapter; (b) the per-group $\hat{\alpha}_j$ are equal early, so the information about the mixing matrix sits in the groups' shares of cases and in the depletion phase, which a homogeneous fit does not use.

9. **[Open Project]** *Full two-group* SIR *fitting and intervention recommendation.* A multi-week capstone project fitting a two-group SIR model to an outbreak with observable group-level heterogeneity (suggested options include age-stratified COVID-19, risk-group-stratified HIV, school-age-stratified influenza, or vaccination-stratified measles), and producing an intervention recommendation that accounts for the heterogeneity. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈20–30 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch12-two-group-fitting/`.

Chapter 12

Sexual Transmission

Separate the law from the closure. The law tells you what must balance; the closure tells you what you believe.

Sexually transmitted infections present three distinctive modeling features that warrant dedicated treatment. First, transmission is predominantly between demographic groups (heterosexual: male-female; MSM: within a concentrated sexual network), giving a strongly disassortative mixing structure that pushes the matrix techniques of Chapter 11 to their natural application. Second, many sexually-transmitted pathogens do not confer lasting immunity, requiring SIS or SIR-without-immunity model structures rather than the SIR framework. Third, the infectious period for some sexually-transmitted pathogens is long (years to lifetime for HIV before antiretroviral therapy), comparable to or longer than the reproductive lifetime, which alters the interpretation of migration and demographic turnover.

This chapter develops the modeling approach for these features. Section 12.1 motivates the distinctions. Section 12.2 writes out the bipartite heterosexual model. Section 12.3 treats the SIS variant common for curable bacterial infections. Section 12.4 derives the reproductive number and discusses the per-cycle vs. per-generation convention. Section 12.5 extends the bipartite structure to a combined heterosexual and MSM population, where the core-group concentration of transmission becomes visible. Section 12.6 discusses interventions (condom use, partner notification, antibiotic treatment) and their effect on $\mathcal{R}_0$. Section 12.7 provides a case study (gonorrhea, for the SIS structure). Section 12.8 summarizes and provides exercises.

Advanced material note. The stratified bipartite NGM derivation (see Section 12.4 and the multi-group extension in Section 12.5) uses the same bipartite technique as Chapter 11, now applied to activity levels rather than demographic groups. The main-path reader can extract the structural result; full algebraic steps are presented for readers wanting them.

12.1 Distinctive Features of Sexual Transmission

Bipartite structure. Most sexually-transmitted pathogens spread through well-defined partnerships. For heterosexual transmission, the two-group structure is male (M) and female (F), with male-female contacts dominating and M-M and F-F contacts either absent (for strictly heterosexual dynamics) or modeled as a distinct MSM subgroup requiring a third group. For transmission within a tightly-connected sexual network (as for some MSM community dynamics), the relevant group structure is defined by sexual-activity level rather than by biological sex.

The mixing matrix for a strictly heterosexual model has the form

$$\mathbf{C} = \begin{pmatrix} 0 & C_{MF} \\ C_{FM} & 0 \end{pmatrix},$$

with diagonal zeros. This is the extreme-disassortative limit of Chapter 11 (all contacts fall between the two groups and none within either), and the book's analytical techniques apply directly.

No immunity for many STIs. Several important sexually-transmitted pathogens do not confer lasting immunity:

- Gonorrhea: reinfection possible immediately after treatment.
- Chlamydia: similar; repeat infections common.
- Syphilis: some immunity after treated primary infection, but not reliable.
- HIV: no recovery from infection without antiretroviral therapy; antiretroviral therapy reduces transmission but does not clear infection.

For these pathogens, the R compartment is either absent (HIV, where infectious individuals never recover) or becomes a short-lived transit compartment before individuals return to S (bacterial STIs with antibiotic treatment returning individuals to susceptible status).

Long infectious periods. For HIV without antiretroviral therapy, the infectious period from seroconversion to AIDS is approximately 10 years. Untreated syphilis has infectious periods of years. Gonorrhea without treatment resolves in weeks to months, but with substantial untreated-infectious-period variability.

These long infectious periods interact with demographic turnover in ways that the book's ν_m parameter (mean residence time in the modeled population) becomes central. For HIV with $\tau_R \sim 10$ years and τ_m comparable (mean time in the modeled sexual-active population), the two rates are both non-negligible in the $\mathcal{R}_0$ formula.

Partner-formation dynamics. Real sexual contact occurs in the context of partnerships, which form and dissolve over time. The "contact rate" language of the book's homogeneous-mixing framework is an approximation that averages over partnership dynamics. For pathogens with short contact-to-transmission windows and partners that have many encounters, the approximation is often adequate. For pathogens with low per-contact transmission but long partnership durations, partnership-formation models may give meaningfully different predictions; we do not develop these models here, noting their existence and pointing to Brauer et al. [2019] for a treatment.

Exercises for this section

1. [**C**] Many STIs are modeled with SIS rather than SIR structure. State the biological reason for this choice, and identify one STI for which an SIR structure would be more appropriate.

12.2 The Bipartite Heterosexual Model

We develop the model for heterosexual transmission with strict male-female mixing. The generalization to MSM or mixed-orientation populations is a multi-group extension that we outline at the end of the section.

State variables. For each sex, three compartments: susceptible, infectious, recovered. We use subscripts M for male and F for female:

$$(S_M, I_M, R_M), \qquad (S_F, I_F, R_F),$$

with total populations $N_M = S_M + I_M + R_M$ and $N_F = S_F + I_F + R_F$, approaching carrying capacities N_M^c and N_F^c respectively.

Contact structure. The mixing matrix is

$$\mathbf{C} = \begin{pmatrix} 0 & c_{MF} \\ c_{FM} & 0 \end{pmatrix},$$

where c_{MF} is the rate at which males have contacts with females, and c_{FM} is the reciprocal rate for females with males. The balance equation from Chapter 11, Section 11.2 applies directly (Pitfall: "Heterogeneous-Mixing Balance Conditions"): the total number of male-to-female contacts per day must equal the total number counted from the female side, so

$$c_{MF} N_M = c_{FM} N_F.$$

This is the same principle as in the two-group model of Chapter 11, now specialized to the zero-diagonal bipartite case. For populations with approximately equal sizes ($N_M^c \approx N_F^c$), $c_{MF} \approx c_{FM}$; for populations with sex-imbalance, the rates differ inversely to the size ratio. The balance equation must hold whatever we assume about behavior; the zero diagonal is the assumption, and the chapter's conclusions rest on it.

The bipartite structure with zero diagonal is shown schematically in Figure 12.1: the male and female subpopulations are treated as disjoint nodes connected only by cross-sex transmission, with no within-sex transmission path.

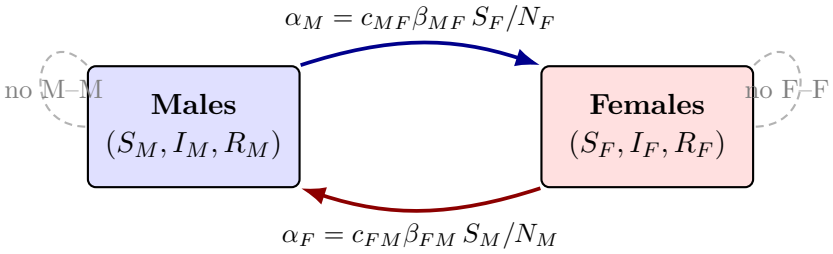


Figure 12.1: The bipartite heterosexual model. Transmission occurs only between males and females; within-sex contact is excluded by construction (dashed gray loops indicate the zero within-sex mixing). The male force from infection α_M depends on the *female* susceptible fraction S_F/N_F, and vice versa: the hallmark of bipartite dynamics. Transmission probabilities per contact β_{MF} and β_{FM} are generally distinct (for HIV, roughly twofold different due to anatomical asymmetry).

Transmission factors. Male-to-female and female-to-male transmission probabilities per unprotected contact differ biologically. For HIV, per-contact female-to-male transmission is about 1 per 1000 contacts, while per-contact male-to-female transmission is higher (roughly 1 per 500), due to anatomical factors. The transmission factors are generally asymmetric:

$$\beta_{MF} \neq \beta_{FM}.$$

For chlamydia and gonorrhea, the per-contact transmission probabilities are higher (roughly 0.2 to 0.5) and less asymmetric.

Force from infection. For an infectious male, contacts occur at rate c_{MF} (all with females, given strict heterosexual mixing). The fraction of those contacts that are with susceptible females is

$$P_{S,M}^{(F)} \;=\; \frac{c_{FM}^{S} S_F}{C_{\text{total},FM}} \;=\; \frac{S_F}{N_F},$$

with $C_{\text{total},FM}$ the pair pool of Chapter 11, Section 11.2; the equality is exact when contact rates are compartment-independent within each sex. The force from infection for an infectious male is

$$\alpha_M \;=\; c_{MF}\beta_{MF}\,\frac{S_F}{N_F},$$

with each contact with a susceptible female transmitting with probability β_{MF}. Similarly,

$$\alpha_F \;=\; c_{FM}\beta_{FM}\,\frac{S_M}{N_M}.$$

Compartment-specific extension. The uniform c_{MF}, c_{FM} above is the classical approximation. The SIR framework of Chapter 4 and the compartment-specific contact matrices $\mathbf{C}^S, \mathbf{C}^I, \mathbf{C}^R$ of Chapter 11 apply directly in the bipartite setting: the male-to-female rate admits compartment-specific forms $c_{MF}^S, c_{MF}^I, c_{MF}^R$ (and analogously $c_{FM}^S, c_{FM}^I, c_{FM}^R$ for females), with the uniform case $c_{MF}^S = c_{MF}^I = c_{MF}^R = c_{MF}$ recovered as the baseline used throughout this chapter. The refinement matters most for curable STIs where diagnosis triggers rapid behavioral change: patients diagnosed with gonorrhea typically abstain or use barrier methods during the treatment period, giving $c_{MF}^I \ll c_{MF}^S$ on the treatment timescale. For HIV, where infection is chronic, the compartment distinction acts on a slower timescale but remains operationally relevant for treatment-as-prevention (Chapter 19). The intervention analysis of Section 12.6 develops this mechanism explicitly under Section 12.6.

Model equations. The bipartite dynamics are written in target-form: the new-infection flux into each sex's I compartment is generated by the *other* sex's infectious population. Defining the target-form fluxes

$$\Phi_M \;=\; \alpha_F\, I_F \qquad \text{(rate of new male infections, generated by infectious females)}, \tag{12.1}$$

$$\Phi_F \;=\; \alpha_M\, I_M \qquad \text{(rate of new female infections, generated by infectious males)}, \tag{12.2}$$

the full system reads

$$\dot{S}_M = \nu_m N_M^{\mathrm{c}} - \Phi_M - \nu_m S_M + (\text{recovery terms, if any}), \tag{12.3a}$$
$$\dot{I}_M = \Phi_M - \gamma_R I_M - \nu_m I_M, \tag{12.3b}$$
$$\dot{R}_M = \gamma_R I_M - \nu_m R_M, \tag{12.3c}$$
$$\dot{S}_F = \nu_m N_F^{\mathrm{c}} - \Phi_F - \nu_m S_F + (\text{recovery terms}), \tag{12.3d}$$
$$\dot{I}_F = \Phi_F - \gamma_R I_F - \nu_m I_F, \tag{12.3e}$$
$$\dot{R}_F = \gamma_R I_F - \nu_m R_F, \tag{12.3f}$$

where the recovery terms differ between the immunizing (SIR) and non-immunizing (SIS) variants (Section 12.3).

Distinctive feature of the bipartite force expressions. Note that new *male* infections $\Phi_M = \alpha_F I_F$ involve S_M through the female-source force $\alpha_F = c_{FM}\beta_{FM}\, S_M/N_M$, and new *female* infections $\Phi_F = \alpha_M I_M$ involve S_F through the male-source force $\alpha_M = c_{MF}\beta_{MF}\, S_F/N_F$. Cross-coupling between the sexes is the hallmark of bipartite transmission and is encoded in the fact that the new-infection flux into each sex's I compartment is generated by the *other* sex's infectious population. The single-sex SIR form $\dot{I} = \alpha I$ of Chapter 4 sits at the boundary case where source and target coincide trivially; here they do not, and the target-form ODE (12.3) above is the correct generalization.

Convention warning. The subscript on α here names the *source* of infection: α_M is the rate at which one infectious male generates new infections, so it is paired with I_M in $\Phi_F = \alpha_M I_M$. This is the opposite of the convention used in the vector-borne model of Chapter 13 and the HIV model of Chapter 19, where the subscript names the *target* and the force is paired with a susceptible population, as in $\alpha_M^H S_M^H$. Both conventions are standard. Each is natural in its own setting: per-source when the question is how much transmission an infectious individual generates, per-target when it is what hazard a susceptible individual faces. They are not interchangeable, and a reader moving between these chapters should check which is in force. The Notation and Conventions summary states the per-target convention for Chapters 13 and 19; this chapter is the per-source exception.

Exercises for this section

1. **[D]** For a bipartite model with sex-imbalance $N_M^c = 0.6N^c$, $N_F^c = 0.4N^c$, and reciprocity $c_{MF}N_M^c = c_{FM}N_F^c$, derive the relationship between c_{MF} and c_{FM}. If $c_{MF} = 1.0$, compute c_{FM}.

12.3 The SIS Structure for Curable STIs

For bacterial STIs like gonorrhea and chlamydia, antibiotic treatment cures the infection and returns the individual to the susceptible compartment. There is no lasting immunity, so recovered individuals re-enter S directly rather than transiting through R. The model simplifies to a four-compartment SIS structure (with the same target-form fluxes $\Phi_M = \alpha_F I_F$ and $\Phi_F = \alpha_M I_M$ as in Section 12.2):

$$\dot{S}_M = \nu_m N_M^c - \Phi_M + \gamma_R I_M - \nu_m S_M, \tag{12.4a}$$
$$\dot{I}_M = \Phi_M - \gamma_R I_M - \nu_m I_M, \tag{12.4b}$$
$$\dot{S}_F = \nu_m N_F^c - \Phi_F + \gamma_R I_F - \nu_m S_F, \tag{12.4c}$$
$$\dot{I}_F = \Phi_F - \gamma_R I_F - \nu_m I_F, \tag{12.4d}$$

with α_M and α_F as in Section 12.2. The R compartment is dropped; upon recovery, individuals return to S directly. The rate γ_R here represents treatment-assisted clearance of infection rather than natural immunity.

Key difference from the SIR variant. In the SIR variant, a wave of infection depletes the susceptible population as individuals move to R. In the SIS variant, the susceptible population is

continuously replenished by recovered individuals. The endemic prevalence can be substantially different, and the dynamics lack the depleting transient of an SIR outbreak: the trajectory approaches endemic equilibrium monotonically rather than with a peak-and-decline structure.

The SIS structure also simplifies the sensitivity analysis: without an R compartment, the state space is reduced and some quantities have simpler formulas.

Reproductive number for the SIS bipartite system. At the disease-free equilibrium, $(S_M, I_M, S_F, I_F) = (N_M^c, 0, N_F^c, 0)$. The infected-compartment dynamics linearize around this equilibrium with

$$\boldsymbol{F} \;=\; \begin{pmatrix} 0 & c_{MF}\beta_{MF} \\ c_{FM}\beta_{FM} & 0 \end{pmatrix}, \qquad \boldsymbol{V} \;=\; (\gamma_R + \nu_m)\mathbf{I}.$$

(The indexing convention: row j is "new infection in sex j"; column k is "infectious in sex k.") The NGM is

$$\boldsymbol{K} \;=\; \boldsymbol{F}\boldsymbol{V}^{-1} \;=\; \frac{1}{\gamma_R + \nu_m}\begin{pmatrix} 0 & c_{MF}\beta_{MF} \\ c_{FM}\beta_{FM} & 0 \end{pmatrix}. \tag{12.5}$$

12.4 The Basic Reproductive Number

The NGM (12.5) has zero diagonal and non-zero off-diagonal entries. Its characteristic polynomial is

$$\det(\boldsymbol{K} - \mathcal{R}\mathbf{I}) \;=\; \mathcal{R}^2 - \frac{c_{MF}\beta_{MF}c_{FM}\beta_{FM}}{(\gamma_R + \nu_m)^2} \;=\; 0.$$

The eigenvalues are

$$\mathcal{R} \;=\; \pm\frac{\sqrt{c_{MF}\beta_{MF}c_{FM}\beta_{FM}}}{\gamma_R + \nu_m}.$$

The spectral radius is the absolute value of the larger eigenvalue:

$$\mathcal{R}_0 \;=\; \frac{\sqrt{c_{MF}\beta_{MF}c_{FM}\beta_{FM}}}{\gamma_R + \nu_m}. \tag{12.6}$$

12.4.1 Stage, Cyclic, and Geometric-Mean Reproductive Numbers for Bipartite STIs

The bipartite $\mathcal{R}_0$ formula (12.6) is an instance of the stage/cyclic/geometric-mean framework of Chapter 11, Section 11.6. With the male and female infected classes as the two stages I_M and I_F, the stage reproductive numbers (Definition 11.2) are

$$\mathcal{R}_0^{M\to F} \;=\; \frac{c_{MF}\,\beta_{MF}}{\gamma_R + \nu_m}, \qquad \mathcal{R}_0^{F\to M} \;=\; \frac{c_{FM}\,\beta_{FM}}{\gamma_R + \nu_m}, \tag{12.7}$$

each reading the expected number of secondary infections one infectious individual produces in the other sex. The cyclic reproductive number (Definition 11.3) based at the male stage is the product

$$\mathcal{C}_0^{(M)} \;=\; \mathcal{R}_0^{M\to F}\,\mathcal{R}_0^{F\to M} \;=\; \frac{c_{MF}\,\beta_{MF}\,c_{FM}\,\beta_{FM}}{(\gamma_R + \nu_m)^2}, \tag{12.8}$$

counting the expected number of male secondary infections produced after one full male-to-female-and-back cycle. The book-wide quantity $\mathcal{R}_0 = \rho(\boldsymbol{K})$ is then, by identity (11.16) applied at $L = 2$, the square root of the cyclic value:

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}) \;=\; \sqrt{\mathcal{C}_0^{(M)}} \;=\; \sqrt{\mathcal{R}_0^{M\to F}\,\mathcal{R}_0^{F\to M}}. \tag{12.9}$$

This is identity (12.6) rewritten in the three-level language: $\mathcal{R}_0$ is the geometric mean of the directional stage reproductive numbers, equivalently the square root of the cyclic reproductive number.

> **Reader note: conventions in the STI literature.**
>
> $\mathcal{R}_0$ and $\mathcal{C}_0^{(M)}$ cross the threshold together, so either determines invasion. The two stage reproductive numbers do not: $\mathcal{R}_0$ is their geometric mean, so one can sit below 1 while the infection invades. Their *numerical* values differ. The book consistently reports $\mathcal{R}_0 = \rho(\boldsymbol{K})$, the geometric mean. Earlier literature often distinguishes only a *per-cycle* quantity (corresponding to the cyclic $\mathcal{C}_0^{(M)}$ here) from a *per-generation* quantity (corresponding to $\mathcal{R}_0$ here). When reading the STI literature, check which quantity a source reports: a per-cycle value is the square of the per-generation one. This chapter's untreated gonorrhea has $\mathcal{R}_0 \approx 3.3$ per generation and $\mathcal{C}_0^{(M)} \approx 11$ per cycle.

Geometric interpretation. The geometric-mean characterization (12.9) has a transparent reading. A male case generates $c_{MF}\beta_{MF}/(\gamma_R + \nu_m) = \mathcal{R}_0^{M\to F}$ female cases, on average, during his infectious period. A female case generates $\mathcal{R}_0^{F\to M}$ male cases. Neither per-direction count alone determines invasion: one must travel both legs of the cycle. The geometric mean of the two directional counts is the cycle-normalized scalar that sets the threshold under the NGM convention, and it reduces to the homogeneous formula $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$ when both stage reproductive numbers are equal.

12.4.2 Example: Gonorrhea

For untreated gonorrhea with typical parameter values:

- $c_{MF} \approx c_{FM} \approx 1.5$ contacts per person per week (≈ 0.2 per day).
- $\beta_{MF} \approx 0.35$, $\beta_{FM} \approx 0.25$ (per-contact transmission probabilities).
- $\tau_R \approx 8$ weeks without treatment ($\gamma_R \approx 0.018$ per day).
- $\nu_m \approx 0$ (migration time years, so negligible relative to γ_R).

Using (12.6):

$$\mathcal{R}_0 \;=\; \frac{\sqrt{0.2 \times 0.35 \times 0.2 \times 0.25}}{0.018} \;=\; \frac{\sqrt{0.0035}}{0.018} \;=\; \frac{0.0592}{0.018} \;\approx\; 3.3.$$

For untreated gonorrhea at the case-study parameter set, the basic reproductive number is approximately 3.3 as the calculation above shows; published estimates span roughly 2 to 4 on the per-generation $\mathcal{R}_0 = \rho(\boldsymbol{K})$ scale, with the variation reflecting different assumed contact rates and per-direction transmission probabilities across populations. (The case study in Section 12.7 below

uses the same per-direction values with a contact rate of 0.21 rather than 0.20 per day and gives $\mathcal{R}_0 \approx 3.45$; both are within the literature range.) With effective antibiotic treatment (reducing τ_R to 1 week, $\gamma_R \approx 0.14$ per day), the same contact parameters give $\mathcal{R}_0 \approx 0.41$, well below threshold and consistent with the control of gonorrhea prior to the rise of antibiotic resistance.

12.4.3 Example: HIV

For untreated HIV:

- $c_{MF} \approx c_{FM} \approx 0.1$ contacts per week (≈ 0.014 per day).
- $\beta_{MF} \approx 0.002$, $\beta_{FM} \approx 0.001$.
- $\tau_R \approx 10$ years ≈ 3650 days ($\gamma_R \approx 0.00027$ per day).
- ν_m comparable: $\tau_m = 30$ years ($\nu_m \approx 0.000091$ per day), the typical duration of sexual activity.

$$\mathcal{R}_0 = \frac{\sqrt{0.014 \times 0.002 \times 0.014 \times 0.001}}{0.000274 + 0.000091} = \frac{\sqrt{3.92 \times 10^{-10}}}{0.000365} = \frac{1.98 \times 10^{-5}}{0.000365} \approx 0.054.$$

This per-generation number is far below the invasion threshold. HIV in this scenario cannot sustain an epidemic, yet HIV has demonstrably spread globally. The resolution: the parameters above are for a low-activity general population. In specific subpopulations with substantially higher contact rates (commercial sex work, concurrent partnership networks), the contact parameters can be 50–100 times higher, pushing $\mathcal{R}_0$ above 1 within those subpopulations. The overall epidemic dynamics emerge from these high-risk subpopulations serving as sources for the broader population.

This observation motivates models with multiple heterogeneous subpopulations (Section 12.5) and explains why homogeneous-mixing models often give misleading answers for STI dynamics. Formal analysis of this multi-group architecture for HIV, with infectivity varying across disease stages, was developed in Hyman et al. [1994] and Hyman et al. [1999]; the staged-progression framework remains one of the most widely deployed STI-modeling templates.

Exercises for this section

1. **[C]** Explain in plain English why the bipartite $\mathcal{R}_0$ takes a square-root form: $\mathcal{R}_0 = \sqrt{c_{MF}\beta_{MF}c_{FM}\beta_{FM}}/(\gamma_R + \nu_m)$. Relate your answer to the geometric-mean identity (11.16): what cycle is being averaged, and why is the geometric mean the right averaging operation?

12.5 Multi-Group Extension: Heterosexual + MSM Populations

The bipartite male-female structure is a simplification: real sexual networks include MSM (men who have sex with men) subpopulations with distinct contact patterns, and within-sex assortative mixing across activity levels. A more complete model stratifies the population into more than two groups. Stratification has a cost the data must be able to pay: incidence alone does not pin down both the size of each activity stratum and the mixing between strata (Pitfall: "Identifiability"); Chapter 19 shows how the infected-viewpoint estimator sidesteps the stratum-size half of that problem.

12.5.1 A Three-Group Model

Consider three groups: heterosexual males (M_H), heterosexual females (F), and MSM (M_M). The within-MSM subpopulation mixes strongly with itself; between-group contact occurs between F and M_H (dominant), and between M_H and M_M (for bisexually-active males). The mixing matrix is 3×3 with structural zeros (female-female, female-MSM if the MSM group is strictly same-sex) and nonzero entries elsewhere. Figure 12.2 visualizes this sparse contact structure.

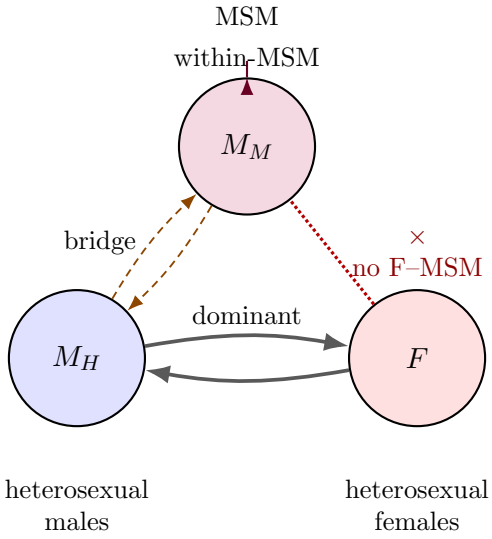


Figure 12.2: The three-group MSM extension. Solid gray arrows mark the dominant $M_H \leftrightarrow F$ heterosexual transmission path. Dashed orange arrows mark the weaker $M_H \leftrightarrow M_M$ bridge, maintained by bisexually-active males. The purple self-loop on M_M represents strong within-MSM mixing (the core-transmission mechanism). The red dotted segment marks the structural zero F–M_M (no direct female-MSM contact under the strict-orientation assumption). The sparse structure captures the historical pattern of U.S. and European HIV transmission: initial concentration in MSM, bridging to heterosexual populations through bisexual contacts.

Structural features. The three-group structure captures two mechanisms missed by the bipartite model:

- *Bridge populations.* Bisexually active individuals connect the MSM subnetwork to the heterosexual network, transmitting infections between otherwise-disjoint communities. Historical HIV epidemics in the United States and Europe followed this pattern: initial MSM-concentrated spread bridged to heterosexual populations through bisexual contacts.

- *Concentrated high-activity subnetworks.* Within MSM, activity-level stratification (low-activity, medium-activity, high-activity) produces dynamics where the high-activity subgroup sustains transmission that the lower-activity subgroups alone could not. The high-activity subgroup acts as a reservoir that maintains $\mathcal{R}_0 > 1$ for the full population even when the overall average is below threshold.

Reproductive number for multi-group models. The NGM extends to larger matrices, and $\mathcal{R}_0$ is computed as the spectral radius of the corresponding matrix. For a 3-group model, the

characteristic polynomial is cubic; for larger groups, numerical eigenvalue computation is typical. The structural features just listed emerge from the spectral-radius computation: a cubic polynomial can have one large eigenvalue (corresponding to the high-activity group) and smaller subdominant eigenvalues, leading to a $\mathcal{R}_0$ dominated by the high-activity subgroup.

The core-transmission concept. The "core group" theory of Hethcote and Yorke [1984] formalizes this: in a model with a high-activity core and a low-activity peripheral population, the core-group-only reproductive number must exceed unity for the epidemic to sustain, while the overall population average may be well below 1. This is one of the fundamental insights of STI epidemiology.

12.6 Interventions and Their Effect on $\mathcal{R}_0$

Three intervention categories dominate STI control:

Condom use (reduces β). Condom use during sexual contact reduces the per-contact transmission probability. A condom-use fraction f_c with condom efficacy e_c gives effective transmission probability

$$\beta_{\text{eff}} = \beta(1 - f_c e_c).$$

For condoms with $e_c \approx 0.85$ and $f_c = 0.5$ (half of contacts protected), $\beta_{\text{eff}} = 0.575\beta$, a 43% reduction. Because the bipartite $\mathcal{R}_0$ scales as the square root of $\beta_{MF}\beta_{FM}$, a reduction of 43% in both transmission factors reduces $\mathcal{R}_0$ by 43% (same factor, since the square root scales linearly in a proportional reduction).

Partner notification and treatment (reduces τ_R). Partner notification identifies sexual contacts of an infectious individual and offers them treatment. This accelerates the identification and treatment of previously-undetected infections, effectively reducing the mean infectious period τ_R. A reduction of τ_R from 8 weeks to 2 weeks (a 4-fold reduction) for untreated gonorrhea translates to a 4-fold reduction in $\mathcal{R}_0$.

The policy implication: aggressive partner notification is particularly effective for pathogens where the untreated infectious period is long. For gonorrhea, each week of infectious duration reduction has approximately linear effect on $\mathcal{R}_0$ through the denominator $\gamma_R + \nu_m$.

Antiretroviral therapy (reduces β). For HIV, antiretroviral therapy suppresses viral replication and reduces per-contact transmission probability to near zero. A population with high antiretroviral coverage has $\beta_{\text{eff}} \ll \beta$, and $\mathcal{R}_0$ reduces correspondingly. The "treatment-as-prevention" strategy relies on this mechanism: reducing the effective transmissibility of HIV by reducing viral load through widespread antiretroviral use.

Behavior change among infected hosts (reduces c^I_{MF}, c^I_{FM}). Interventions acting specifically on diagnosed-infectious individuals reduce the compartment-specific contact rates c^I_{MF} (infected males' contact rate with females) and c^I_{FM} (infected females' contact rate with males) from the compartment-specific extension on p. 270. Mechanisms include counseling at diagnosis, partner-notification programs that incentivize abstinence during treatment, and the structural effect of symptomatic presentation itself: patients with acute gonorrhea or chlamydia often withdraw from

sexual activity during symptomatic periods. Applied symmetrically to both sexes, $c^I_{MF} \to (1-\chi_I)c^I_{MF}$ and $c^I_{FM} \to (1-\chi_I)c^I_{FM}$ yield

$$\mathcal{R}_0 \;\to\; (1-\chi_I)\,\mathcal{R}_0 \quad \text{(unit sensitivity in } \chi_I\text{)},$$

because each direction of the bipartite cycle carries an infected-host contact rate in (12.6). If the intervention reaches only one sex (for instance, symptomatic males reducing contacts while asymptomatic females continue unchanged), the reduction becomes half-sensitivity $\mathcal{R}_0 \to \mathcal{R}_0\sqrt{1-\chi_I}$, paralleling the vector-borne case in Chapter 13, Section 13.5.3. The gonorrhea case study (Section 12.7) illustrates the symmetric case: prompt treatment of both diagnosed males and diagnosed females, combined with behavioral change during the treatment window, can drive $\mathcal{R}_0$ below threshold at coverage levels where either mechanism alone would be insufficient.

Combination interventions. Interventions stack multiplicatively on $\mathcal{R}_0$ to the extent they act on independent components:

$$\mathcal{R}_0^{\text{intervention}} \;=\; \mathcal{R}_0 \cdot f_{\text{condom}} \cdot f_{\text{partner-notification}} \cdot f_{\text{ART}},$$

where each f_ℓ is a reduction factor between 0 and 1 attributable to that intervention. Reducing $\mathcal{R}_0$ below 1 by combination intervention is often more realistic than relying on any single intervention; this is the underlying rationale for comprehensive STI prevention programs. When a prophylactic vaccine becomes available, the same multiplicative structure extends naturally; Kribs-Zaleta and Martcheva [2002] analyze the resulting dynamical systems and derive conditions under which vaccination alone can eliminate a sexually transmitted infection.

Exercises for this section

1. **[I]** Consider a combination intervention with 60% condom use (efficacy 85%), reducing τ_R from 8 weeks to 4 weeks through partner notification. Compute the combined effect on $\mathcal{R}_0$ for untreated gonorrhea parameters. Is this sufficient to bring $\mathcal{R}_0$ below 1?

12.7 Case Study: Gonorrhea Dynamics

We apply the bipartite SIS framework to gonorrhea, illustrating how the model parameters produce qualitatively different dynamics for this pathogen than for influenza (Chapters 7 and 9) or respiratory SIR variants.

12.7.1 Parameter Values

- $N^c_M = N^c_F = N^c/2$: approximately equal male and female sexually-active populations.
- Contact rates: $c_{MF} = c_{FM} = 1.5$ contacts per week between opposite sexes, or 0.21 per day.
- Transmission factors: $\beta_{MF} = 0.35$, $\beta_{FM} = 0.25$ (male-to-female slightly higher).
- Untreated infectious period: $\tau_R = 8$ weeks ≈ 56 days; $\gamma_R = 1/56 \approx 0.018$ per day.
- Migration time: $\tau_m = 30$ years (typical sexually-active period); $\nu_m = 1/(30\times 365) \approx 9\times 10^{-5}$ per day, negligible relative to γ_R.

12.7.2 Reproductive Number and Endemic Equilibrium

From equation (12.6):

$$\mathcal{R}_0 \;=\; \frac{\sqrt{0.21 \times 0.35 \times 0.21 \times 0.25}}{0.018} \;\approx\; \frac{0.0621}{0.018} \;\approx\; 3.45.$$

The reproductive number is about 3.5, well above the invasion threshold.

The endemic prevalence in the bipartite SIS model, with symmetric population sizes and the asymmetry in β_{MF}/β_{FM}, requires solving a coupled nonlinear system. For the symmetric-β case ($\beta_{MF} = \beta_{FM} = \beta$), the prevalence in each sex satisfies

$$\frac{I_M^*}{N_M} \;=\; \frac{I_F^*}{N_F} \;=\; 1 - \frac{1}{\mathcal{R}_0}.$$

With equal β this would give $1 - 1/3.45 \approx 0.71$ in each sex. The case study's unequal β splits it: solving the coupled system (Exercise 6) gives about 68% of males and 74% of females infected. This is a higher number than observed in practice because our parameter set represents the pre-treatment scenario; contemporary populations have a range of behavioral and treatment factors reducing observed prevalence.

12.7.3 Effect of Treatment

A contemporary public-health response to gonorrhea reduces the average time to treatment significantly. With effective screening and case management, τ_R is reduced from 8 weeks to approximately 2 weeks. Using $\tau_R = 14$ days, $\gamma_R = 0.071$ per day:

$$\mathcal{R}_0^{\text{treated}} \;=\; \frac{0.0621}{0.071} \;\approx\; 0.87.$$

The reproductive number falls below 1, and the epidemic cannot sustain itself: treatment alone, without behavioral change, is sufficient to control the spread.

This was the situation in the 20th century when antibiotics were effective. Unfortunately, antimicrobial resistance has reduced the effectiveness of standard antibiotic regimens in recent decades, and treatment failure has become more common. The modeling framework captures this: if τ_R is effectively restored to its untreated value by resistance, $\mathcal{R}_0$ returns above 1, and transmission resurges.

12.7.4 Sensitivity of $\mathcal{R}_0$ to Contact Rate

Read off Equation (12.6), the sensitivity index of Chapter 9 is $\Upsilon_{c_{MF}}^{\mathcal{R}_0} = 1/2$. But c_{MF} cannot move alone. The balance equation of Section 12.2 makes every male-female contact also a female-male contact, so with the group sizes fixed, halving c_{MF} halves c_{FM}, and $\mathcal{R}_0$ falls by half: the sensitivity to the contact rate is 1, as in the homogeneous model. The square root halves the sensitivity only to a parameter that can change on one side of the cycle: a per-contact probability β_{MF} or β_{FM}, or the contact rate of infected hosts of one sex (Section 12.6).

12.8 Summary and Exercises

This chapter developed the bipartite framework for sexually-transmitted infection models:

1. Sexual transmission motivates bipartite mixing (male-female for heterosexual; activity-stratified for MSM), which is the extreme-disassortative limit of Chapter 11. The mixing matrix has structural zeros and non-zero off-diagonal entries.

2. Many STIs lack lasting immunity, giving an SIS model structure rather than SIR. Recovered individuals return directly to susceptible upon treatment, and dynamics approach endemic equilibrium monotonically.

3. The SIR-to-SIS simplification drops the R compartment; the transmission-related results (force from infection, reproductive number via NGM) transfer directly.

4. The bipartite $\mathcal{R}_0$ is the spectral radius of a 2×2 NGM with zeros on the diagonal, yielding $\mathcal{R}_0 = \sqrt{c_{MF}\beta_{MF}c_{FM}\beta_{FM}}/(\gamma_R + \nu_m)$. The square root halves $\mathcal{R}_0$'s sensitivity to each per-contact probability; a change in the contact rate moves both directions through the balance equation, and $\mathcal{R}_0$ with it.

5. The stage/cyclic/geometric-mean framework of Chapter 11, Section 11.6 applies cleanly to bipartite STIs: stage reproductive numbers $\mathcal{R}_0^{M\to F}$ and $\mathcal{R}_0^{F\to M}$ quantify per-direction transmission; their product is the cyclic reproductive number $\mathcal{C}_0^{(M)}$; and the book's threshold quantity $\mathcal{R}_0 = \rho(\boldsymbol{K})$ is the geometric mean of the two stage reproductive numbers. $\mathcal{R}_0$ and $\mathcal{C}_0^{(M)}$ cross 1 together; the stage numbers need not. Earlier per-cycle/per-generation language corresponds to $\mathcal{C}_0^{(M)}$ and $\mathcal{R}_0$ respectively.

6. Multi-group extensions (heterosexual + MSM + activity-stratified) capture core-group dynamics missing from the bipartite model. In these models, high-activity subgroups sustain transmission even when the population average $\mathcal{R}_0$ is below 1.

7. Interventions (condom use, partner notification, antiretroviral therapy) act multiplicatively on $\mathcal{R}_0$. Combining interventions is often required to push $\mathcal{R}_0$ below 1.

8. Case study of gonorrhea: untreated $\mathcal{R}_0 \approx 3.5$; aggressive treatment reduces to $\mathcal{R}_0 \approx 0.9$, below threshold. Resistance-driven treatment failure restores the threat.

Chapter 13 applies the stage/cyclic/geometric-mean framework to vector-borne diseases (mosquito-host), where the same bipartite structure reappears with the vector and host as the two stages.

Framework use in sexual-transmission fitting. For curable STIs modeled with SIS structure at endemic equilibrium, both viewpoints remain practical because the prevalence I is directly measurable through cross-sectional screening. For the bipartite case (SIR with two groups or an SIR-like flow), the per-group estimator $\hat{\alpha}_k = J_k/I_k$ inherits the framework's structural-immunity property [Hyman, 2026d]: the group-specific fit does not depend on a committed value for the group-specific susceptible population. The gonorrhea case study of Section 12.7 uses this viewpoint implicitly in its per-group analysis; explicit framing of the fit follows the per-group pattern established in Chapter 11. The HIV high-activity subpopulation application in Chapter 19 demonstrates the framework use in a concrete fitted context.

Exercises

Four section-level exercises appear above, following Section 12.1, Section 12.2, Section 12.4, and Section 12.6. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[D]** Verify that the bipartite $\mathcal{R}_0$ formula (12.6) is the spectral radius of (12.5) by computing eigenvalues directly and taking the larger absolute value.

2. **[D]** For untreated gonorrhea (Section 12.7 parameter values), compute the stage reproductive numbers $\mathcal{R}_0^{M\to F}$ and $\mathcal{R}_0^{F\to M}$, the cyclic reproductive number $\mathcal{C}_0^{(M)}$, and the book's $\mathcal{R}_0 = \rho(\boldsymbol{K})$. Verify that $\mathcal{R}_0 = \sqrt{\mathcal{C}_0^{(M)}}$ numerically and compare the numerical value of $\mathcal{R}_0$ with literature quotes that use the older per-cycle convention.

3. **[I]** For HIV with the Section 12.4 parameters (general population), the per-generation $\mathcal{R}_0 \approx 0.054$ is well below threshold. Show that for a high-activity subpopulation with $c_{MF} = c_{FM} = 0.5$ per day (about 35 times higher), $\mathcal{R}_0$ exceeds 1. What intervention would be needed to bring $\mathcal{R}_0$ below 1 for this subpopulation?

4. **[D]** Derive the sensitivity index $\Upsilon_{c_{MF}}^{\mathcal{R}_0}$ for the bipartite model with every other parameter fixed, then again under the balance equation with N_M and N_F fixed. Which one measures the effect of a change in sexual contact, and why?

5. **[D]** For a three-group model with heterosexual males, heterosexual females, and MSM, with bridge contacts between MSM and heterosexual males, write the NGM and discuss how the eigenvalue structure depends on the strength of the bridge contact.

6. **[L]** For the bipartite SIS endemic equilibrium with asymmetric β values, set up the coupled nonlinear system for I_M^* and I_F^*. Solve numerically for the gonorrhea parameter set.

7. **[I]** *Spaced retrieval, Chapter 11.* The bipartite male-female model of this chapter is a special case of the two-group framework of Chapter 11, in the limit where the off-diagonal entries of the mixing matrix $\mathbf{C}$ dominate the diagonal entries. Identify the explicit limit on $\mathbf{C}$ and the explicit specialization of β under which the two-group $\mathcal{R}_0$ formula (11.12) of Chapter 11, Section 11.4 reduces to this chapter's bipartite formula (12.6). Show the algebra of the reduction in three or four lines. Then explain in one short paragraph (i) why the bipartite case is structurally simpler than the general two-group case (which entry of $\mathbf{C}$ vanishes and what dimensional reduction results) and (ii) what additional structure (for example, sex-asymmetric per-partnership transmission) must be carried separately in the bipartite case that is automatic in the symmetric two-group case. The exercise reinforces that the bipartite framework of this chapter is a specialization, not a generalization, of the heterogeneous framework of Chapter 11.

8. **[Open Project]** *Core-group targeting in a multi-group model.* A multi-week capstone project extending the bipartite structure to three or more activity groups for a chosen sexually transmitted infection, and testing whether targeting the core group outperforms uniform coverage at equal cost.

Chapter 13

Vector-Borne Diseases

A hidden assumption is a theorem waiting to break.

Vector-borne diseases transmit through an intermediate host, typically an arthropod, rather than by direct contact between human hosts. The arthropod (mosquito, tick, sandfly, tsetse fly) acquires the pathogen by biting an infectious human, carries it for a period called the extrinsic incubation period (EIP), and then transmits it to susceptible humans through subsequent bites. This indirect transmission structure gives vector-borne diseases a modeling framework distinct from the direct-contact models developed so far. The bipartite vector-host structure shares formal features with the sexual-transmission models of Chapter 12, but the vector-specific biology (short vector lifespan, long extrinsic incubation, biting-driven transmission) produces a reproductive-number formula with distinctive structure.

Malaria has been the canonical vector-borne disease for mathematical modeling since Ross's 1911 treatise, which established the first quantitative threshold for disease eradication: if the mosquito density falls below a critical value, malaria transmission cannot sustain itself. Macdonald's refinement in the 1950s gave the modern Ross-Macdonald formula, which remains the framework underlying most contemporary vector-borne modeling; a textbook development within the same modeling tradition is given by Martcheva [2015, Ch. 4]. This chapter develops the Ross-Macdonald framework within the book's notation and extends it to include modern modeling features (intervention strategies, heterogeneous host exposure).

Section 13.1 motivates the treatment. Section 13.2 summarizes vector biology relevant for modeling. Section 13.3 writes out the vector-host bipartite SIR model. Section 13.4 derives the reproductive number via the next-generation matrix and presents the Ross-Macdonald formula. Section 13.5 discusses vector-control interventions. Section 13.6 works through a malaria case study. Section 13.7 summarizes and provides exercises.

Advanced material note. The spectral-radius proof details in Section 13.4 are presented as a natural extension of the Chapter 11 construction. The main-path reader can work with the resulting formula $\mathcal{R}_0 = \sqrt{\mathcal{R}_0^{v\to h} \cdot \mathcal{R}_0^{h\to v}}$ directly without needing the full spectral argument.

13.1 Why Vector-Borne Diseases Warrant Dedicated Treatment

The indirect transmission structure. Direct-transmission pathogens (measles, influenza, SARS-CoV-2) spread through host-to-host contact. Vector-borne pathogens (malaria, dengue, Zika, Lyme disease, West Nile virus) spread through a two-step cycle: human-to-vector transmission, then vector-to-human transmission. The two steps have different rates, different probabilities, and different time delays, and neither alone sustains the pathogen. Both must succeed for sustained transmission.

The cycle structure means that a single infectious human does not directly produce new human infections. Instead, a single infectious human produces some number of newly-infectious vectors, each of which produces some number of newly-infected humans. The overall reproductive number is the product of these two per-stage multiplicities, mediated by the probabilities of each transition.

Vector biology matters. Vector populations have dynamics distinct from human populations. Mosquitoes live for days to weeks (much shorter than human infectious periods of weeks for malaria); they are affected by temperature, humidity, and breeding-site availability; their populations can be manipulated by insecticides, bed nets, and environmental management. These biological features directly enter the reproductive-number formula through vector-specific parameters.

Intervention portfolio is distinctive. Direct-transmission interventions (case isolation, contact reduction) have analogs for vector-borne diseases, but vector-borne diseases also admit interventions targeting the vector itself: larvicides to reduce vector population, insecticide-treated bed nets to reduce human-vector contact, and environmental modifications to eliminate breeding sites. These vector-targeted interventions have no direct analog in direct-transmission disease control. Comparative studies of co-circulating arboviruses (dengue, chikungunya, Zika) show that the same *Aedes* population can sustain outbreaks of multiple pathogens with qualitatively different dynamics, reinforcing the importance of vector-targeted rather than pathogen-specific controls [Manore et al., 2014, 2015].

Connection to Chapter 12. The bipartite structure of vector-borne models closely parallels the bipartite structure of Chapter 12's sexually-transmitted-infection models. Both have zero-diagonal mixing matrices and square-root $\mathcal{R}_0$ formulas. The stage/cyclic/geometric-mean framework of Chapter 11, Section 11.6 applies here with the host and vector as the two stages of the transmission cycle: Section 13.4.1 develops this in full. The original Ross-Macdonald formula reports what is now called the cyclic reproductive number $\mathcal{C}_0^{(h)}$; the book's $\mathcal{R}_0$ is its square root, equivalently the geometric mean of the host-to-vector and vector-to-host stage reproductive numbers. Both quantities give the same threshold.

13.2 Vector Biology

The vector population has its own compartmental structure. For an SI treatment of vectors (vectors do not recover during their short lives, so there is no return flow to the susceptible class), the natural compartments are susceptible and infectious:

$$(S_v, I_v),$$

with total vector population $N_v = S_v + I_v$ approaching a carrying capacity N_v^{c} set by the environment (breeding sites, food sources, climate).

13.2.1 Vector Mortality and Birth

Vector populations turn over rapidly. Mosquitoes live on the order of $\tau_v^m \approx$ 10-20 days in the wild. The mortality rate is

$$\mu_v \;=\; 1/\tau_v^m \;\approx\; 0.05\text{-}0.1 \text{ per day}.$$

Comparing to human mortality $\nu_m \sim 10^{-5}$ per day, the vector mortality rate is three to four orders of magnitude larger. This large disparity between human and vector demographic rates is the defining feature of vector-borne modeling (Pitfall: "Short-Term Simulation Assumptions").

The vector population receives a constant birth rate matching mortality to maintain carrying capacity:

$$\text{vector birth rate} \;=\; \mu_v N_v^{\mathrm{c}}.$$

Newly-emerged vectors are always susceptible (vertical transmission is negligible for most vectors), so the birth rate enters the S_v compartment directly.

13.2.2 Extrinsic Incubation Period

After a vector bites an infectious human and acquires the pathogen, the pathogen must develop inside the vector before the vector becomes infectious to subsequent hosts. This extrinsic incubation period (EIP) is organism-specific: for *Plasmodium* in *Anopheles* mosquitoes, EIP is approximately $\tau_v^E \approx$ 10-14 days; for dengue virus in *Aedes* mosquitoes, $\tau_v^E \approx$ 8-10 days.

The EIP is critical because it is comparable to the vector lifespan. A mosquito that acquires malaria might die before completing the EIP, never becoming infectious. The probability that a vector survives the EIP is

$$p_{\mathrm{EIP}} \;=\; e^{-\mu_v \tau_v^E}.$$

For $\mu_v = 0.1$/day and $\tau_v^E = 10$ days, $p_{\mathrm{EIP}} = e^{-1} \approx 0.37$. Only about a third of vectors that acquire infection survive to become infectious. This probability appears directly in the vector-borne $\mathcal{R}_0$ formula.

The exposed vector compartment can be modeled explicitly:

$$(S_v, E_v, I_v),$$

with flow $S_v \to E_v$ upon infection acquisition, and $E_v \to I_v$ at rate $\gamma_v^E = 1/\tau_v^E$. Many analyses simplify by removing the explicit exposed compartment and multiplying the vector-to-host transmission term by a survival probability instead. The two devices are *not* equivalent, and the difference is large enough to matter. A fixed extrinsic incubation period of duration τ_v^E—the assumption behind a delay-differential formulation of the vector dynamics—gives the survival probability $p_{\mathrm{EIP}}^{\mathrm{delay}} = e^{-\mu_v \tau_v^E}$ derived above, whereas an exponentially distributed sojourn in E_v at rate γ_v^E gives $p_{\mathrm{EIP}}^{\mathrm{exp}} = \gamma_v^E/(\gamma_v^E + \mu_v) = 1/(1 + \mu_v \tau_v^E)$, the form derived in Section 13.3. At the chapter's baseline $\mu_v \tau_v^E = 1$ these are 0.37 and 0.50: the exponential compartment predicts that half again as many infected vectors survive to become infectious, and that factor passes directly into $\mathcal{R}_0$. The discrepancy is the same one Chapter 15 makes precise—an exponential sojourn is not a deterministic delay, and no choice of rate makes it one. Which model to use is a modeling decision about mosquito biology rather than a matter of convenience; what is not optional is using one of them consistently and reporting which.

The empirical evidence settles the question, and it settles it against both endpoints. Chan and Johansson [2012] pooled 146 dengue EIP observations from eight studies and fitted a lognormal model with temperature as a covariate, obtaining a mean of about 15 days at 25 °C with a middle-95% range of roughly 5–33 days, and a mean of 6.5 days at 30 °C with range 2.4–15 days. Those ranges imply a coefficient of variation near 0.51 and 0.49 respectively—dispersed, but far from the CV = 1 that an exponential compartment assumes. Independently, Robert et al. [2019] fitted the dengue EIP with a gamma distribution of shape $\theta = 3$, and CV $= 1/\sqrt{3} \approx 0.58$ agrees. Winokur et al. [2020] report the same qualitative picture for Zika measured by direct transmission, with EIP_{50} falling from 24.2 days at 21 °C to 5.1 days at 30 °C.

Two structural conclusions follow. First, the shape parameter belongs in the model: an Erlang chain of $k = 3$ sub-stages, $E_v^{(1)} \to E_v^{(2)} \to E_v^{(3)} \to I_v$, reproduces the observed dispersion using machinery this book already develops in Chapter 10 (Pitfall: "The Independence of τ"). Its survival factor is $p_{\text{EIP}} = (1 + \mu_v \tau_v^E/3)^{-3}$, which at $\mu_v \tau_v^E = 1$ gives 0.42—between the 0.50 of the single exposed compartment and the 0.37 of the fixed delay. Since $\mathcal{R}_0 \propto \sqrt{p_{\text{EIP}}}$ here, the two limiting forms misstate $\mathcal{R}_0$ by about +9% and −7% respectively: they bracket the supported value, roughly symmetrically, and neither is a safe default.

Second, and less obvious, the dispersion is nearly temperature-invariant while the mean is not. In the Chan and Johansson [2012] fits the coefficient of variation moves only from 0.51 to 0.49 between 25 and 30 °C, while the mean more than halves. That separation licenses the modeling structure: treat k as a fixed shape parameter and let $\tau_v^E(T)$ carry the temperature dependence, rather than absorbing an environmental driver into a dispersion parameter where it does not belong. Absorbing a systematic environmental driver into a dispersion parameter would misattribute a predictable seasonal signal to random variation, and would leave the model unable to respond to the temperature forecasts that make vector-borne prediction useful in the first place.

Three cautions travel with these numbers. The endpoint matters: Robert et al. [2019] measure disseminated infection rather than transmission, which is part of why their fitted means run longer than Chan and Johansson [2012], so a fitted $\omega(T)$ curve should not be treated as a universal dengue relationship. Extrapolation beyond the experimental temperature range is unsafe. And constant-temperature laboratory fits should be checked against fluctuating-temperature conditions before being used for field prediction [Tjaden et al., 2013, Delrieu et al., 2023]. For chikungunya specifically, Christofferson et al. [2016] caution that a transmission fraction measured at a single sampling time is not an EIP estimate, and that the sampling day should not be inserted directly as a latent period.

13.2.3 Biting Rate

Vectors make contacts with hosts through biting. The biting rate per vector, a, is measured in bites per vector per day. Typical values: $a \approx 0.3$/day for *Anopheles* (once every three days); $a \approx 0.5$/day for *Aedes* (once every two days).

The biting rate is the vector-borne analog of the contact rate in direct-transmission models. The total bites delivered to the human population per day is

$$aN_v,$$

distributed across the N_h human hosts. Each human receives on average $aN_v/N_h = am$ bites per day, where $m = N_v/N_h$ is the vector-to-host ratio (Pitfall: "Heterogeneous-Mixing Balance Conditions").

Compartment-specific biting exposure. The expression am bites per host per day treats all hosts as equally available to vectors. Within the SIR framework this is the uniform special case; host behavior and host physiology can make susceptible, infectious, and recovered individuals differentially available. Let $\eta_S, \eta_I, \eta_R \geq 0$ denote relative biting-exposure weights for the three host classes, normalized so that $\eta_S = 1$. The ratio η_I/η_S captures, for example, reduced biting exposure among symptomatic individuals who stay indoors, increase bed-net use following symptom onset, or seek treatment. With compartment-specific weights, the per-host bite rate for class X becomes

$$a_X \;=\; a\,\eta_X\,\frac{N_v}{\eta_S S_h + \eta_I I_h + \eta_R R_h},$$

which reduces to am when $\eta_S = \eta_I = \eta_R = 1$. The body of this chapter develops the classical Ross-Macdonald theory in the uniform case; the compartment-specific extension is taken up in Section 13.5 as a target of intervention, and Xue et al. [2018] treats it operationally for arboviral control programs combining Wolbachia release with behavior-change interventions among infected hosts.

13.2.4 Density-Dependent Biting and the Baseline Calibration

The constant biting rate a adopted above is appropriate when vector and host populations remain near a stable baseline. It is less appropriate when N_v or N_h vary over a wide range, as they do under strong seasonal forcing, climate variability, or vector-control interventions that substantially alter mosquito abundance. In those regimes, the realized biting rate saturates at one of two biological limits depending on which side of the vector-host encounter is rate-limiting, and a constant a cannot capture both limits at once.

Two biological limits. Each mosquito has a *desired* number of blood meals per day, σ_v, set by its reproductive physiology. Each host can *tolerate* a finite number of bites per day, σ_h, set by host biting-defensive behavior and, in high-density settings, by sheer physical feasibility. The realized per-vector biting rate is the smaller of what the two sides permit:

$$\text{mosquito-limited:}\quad \sigma \to \sigma_v \quad \text{as } N_v/N_h \to 0, \qquad \text{host-limited:}\quad \sigma \to \sigma_h N_h/N_v \quad \text{as } N_v/N_h \to \infty.$$

In the mosquito-limited regime, each mosquito gets all the bites it wants and the realized rate equals its desired rate. In the host-limited regime, hosts are the bottleneck and bites must be shared across an abundance of mosquitoes, so the per-vector rate falls as $1/N_v$ while the per-host rate saturates at σ_h.

A biting law that interpolates smoothly between the two limits is the saturation form developed by Chitnis et al. [2006] and in related work on Anopheles-host contact,

$$\sigma(N_v, N_h) \;=\; \frac{\sigma_v\,\sigma_h\,N_h}{\sigma_v\,N_v + \sigma_h\,N_h}. \tag{13.1}$$

By inspection, $\sigma \to \sigma_v$ when $\sigma_v N_v \ll \sigma_h N_h$ (mosquito-limited), and $\sigma \to \sigma_h N_h/N_v$ when $\sigma_v N_v \gg \sigma_h N_h$ (host-limited). The functional form encodes the correct asymptotic behavior at both ends of the vector-host-ratio spectrum.

The Ross-Macdonald biting rate as a baseline calibration. The constant Ross-Macdonald biting rate a is most usefully read as the saturation formula (13.1) evaluated at a *baseline* population pair $(N_v^{\mathrm{c}}, N_h^{\mathrm{c}})$ and frozen at that value:

$$a \;=\; \sigma(N_v^{\mathrm{c}}, N_h^{\mathrm{c}}) \;=\; \frac{\sigma_v\,\sigma_h\,N_h^{\mathrm{c}}}{\sigma_v\,N_v^{\mathrm{c}} + \sigma_h\,N_h^{\mathrm{c}}}. \tag{13.2}$$

Two biological constants (σ_v, σ_h) and two baseline populations (N_v^{c}, N_h^{c}) jointly determine a, but not independently. For calibration, it is convenient to introduce the dimensionless *baseline supply ratio*

$$\rho \;=\; \frac{\sigma_h\,N_h^{\mathrm{c}}}{\sigma_v\,N_v^{\mathrm{c}}}, \tag{13.3}$$

which measures the host-side bite supply relative to the mosquito-side demand at baseline. Equation (13.2) then takes the compact form

$$a \;=\; \sigma_v\,\frac{\rho}{1+\rho},$$

showing that $\rho \gg 1$ places the baseline in the mosquito-limited regime with $a \approx \sigma_v$, while $\rho \ll 1$ places it in the host-limited regime with $a \approx \sigma_h N_h^{\mathrm{c}}/N_v^{\mathrm{c}}$. Most real-world malaria and dengue settings operate in an intermediate regime, ρ of order 1, where neither side is individually rate-limiting.

When the constant-rate simplification is safe, and when it is not. If (N_v, N_h) stays close to $(N_v^{\mathrm{c}}, N_h^{\mathrm{c}})$, the Taylor-expansion of (13.1) around the baseline is flat to leading order, and the constant a from (13.2) is a good approximation. This is the regime in which the classical Ross-Macdonald framework of Section 13.3 onward is sound, and it is the regime in which the case studies of Chapter 18 and the malaria application later in this chapter operate.

If mosquito abundance varies substantially across the simulation interval, the constant-a assumption breaks down in ways that distort transmission asymmetrically. In a winter collapse (N_v falls by an order of magnitude), the constant-a model continues to report the baseline biting rate per vector even though, at much lower N_v, each mosquito is effectively mosquito-limited and should bite at its maximum desired rate $\sigma_v > a$; the constant-a model under-predicts per-vector transmission. In a summer explosion (N_v rises by an order of magnitude), the constant-a model continues to report the baseline rate per vector even though, at much higher N_v, the system becomes host-limited and each mosquito should bite less often; the constant-a model over-predicts per-vector transmission. Because biting rate is the most sensitive parameter in the reproductive-number formula, these distortions propagate directly into seasonal $\mathcal{R}_e(t)$ estimates and into the evaluation of density-altering interventions.

Recommendation for readers. For the main development of this chapter and for the steady-state case studies of Chapter 18 and the malaria case study later in this chapter, the constant Ross-Macdonald a is adequate and is retained. For simulations that run across regimes of substantially different N_v/N_h ratios (seasonal forcing studies, climate-shift scenarios, evaluations of interventions that reduce vector density by an order of magnitude or more), the density-dependent form (13.1) should replace the constant a in the forces of infection of Section 13.3. The calibration proceeds by fixing σ_v and σ_h from entomological measurements (or published species-level values), measuring or estimating the baseline populations $(N_v^{\mathrm{c}}, N_h^{\mathrm{c}})$, and computing a at baseline via (13.2); the supply ratio ρ of (13.3) then indicates whether the baseline is in the mosquito-limited, host-limited, or

intermediate regime and sets expectations for how a will move as N_v or N_h change. The companion repository (Appendix G, The Companion Repository) will provide a reference implementation of the density-dependent biting law for readers using the book's simulation code in seasonal or density-varying contexts.

13.2.5 Transmission Efficiencies

Each bite has two probabilities of transmission:

- b_v: probability that an infectious human infects a susceptible vector through a single bite. This depends on the density of pathogens in the human's blood and the biology of the vector's gut.
- b_h: probability that an infectious vector infects a susceptible human through a single bite. This depends on the density of pathogens in the vector's salivary glands and the immune response of the host.

These two probabilities are generally different. For *Plasmodium falciparum*, $b_v \approx 0.1$ and $b_h \approx 0.5$: humans infect vectors less efficiently than vectors infect humans. This asymmetry has implications for the reproductive-number calculation. Temperature, humidity, and seasonal breeding site availability make every one of these parameters context-dependent; Chitnis et al. [2013] develop a framework for incorporating such environmental drivers directly into the transmission model, and Aron and May [1982] give the foundational treatment of mosquito population dynamics with acquired immunity effects.

Exercises for this section

1. **[D]** Derive the expression for the EIP survival probability $p_{\text{EIP}} = e^{-\mu_v \tau_v^E}$ from the assumption that vector mortality is an exponential process with rate μ_v and the EIP is a fixed duration τ_v^E. How does the expression change if the EIP is itself exponentially distributed with rate γ_v^E?

13.3 The Vector-Host SIR Model

The vector-host system has two compartmental subsystems coupled through cross-species transmission. Figure 13.1 shows the overall structure: hosts and vectors each carry their own SIR (or SI for vectors) compartments, and infection moves between species through biting events at rate a per vector.

Combining the human and vector compartmental structures, the full vector-host model is

$$\dot{S}_h = \nu_m N_h^{\text{c}} - \alpha_h S_h - \nu_m S_h, \tag{13.4a}$$

$$\dot{I}_h = \alpha_h S_h - \gamma_R I_h - \nu_m I_h, \tag{13.4b}$$

$$\dot{R}_h = \gamma_R I_h - \nu_m R_h, \tag{13.4c}$$

$$\dot{S}_v = \mu_v N_v^{\text{c}} - \alpha_v S_v - \mu_v S_v, \tag{13.4d}$$

$$\dot{I}_v = \alpha_v S_v - \mu_v I_v, \tag{13.4e}$$

where α_h is the host force of infection (rate at which a single susceptible host acquires infection) and α_v is the vector force of infection (rate at which a single susceptible vector acquires infection).

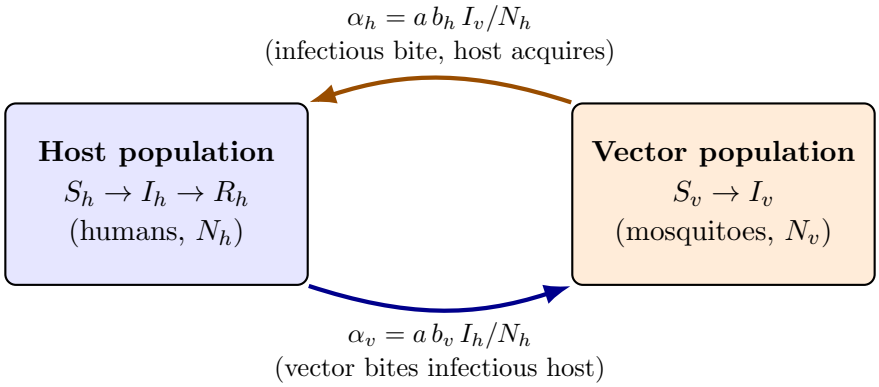


Figure 13.1: The vector-host transmission cycle. Infection passes between two distinct populations through biting: an infectious vector biting a susceptible host transmits with probability b_h, while a susceptible vector biting an infectious host acquires infection with probability b_v. Each population carries its own compartmental dynamics; the coupling between them is through the forces α_h and α_v. The biting rate a (bites per vector per day) appears in both forces because it is the rate-limiting step for both transmission directions.

This chapter therefore carries three distinct objects under the symbol α, and it is worth tabulating them once because a reader arriving from Chapters 3–9 will expect only the third. The per-target forces α_h and α_v defined here multiply a *susceptible* population, so in the book's primary notation they are forces *of* infection and would be written λ; the vector-borne literature writes them with α and this chapter follows that convention. The disease-free per-source rates $\alpha_h^{(0)}$ and $\alpha_v^{(0)}$ introduced in Section 13.4.1 are the opposite kind: $\alpha_h^{(0)} = ab_vm$ is the rate at which one infectious *host* generates new vector infections. And the fitting estimators $\hat{\alpha}_h^{\text{inf}} = J_h/I_v$ and $\hat{\alpha}_v^{\text{inf}} = J_v/I_h$ are per-source in the book's sense, dividing incidence by the population that generated it. Only the last of these matches the α of Chapter 3; the superscripts are what distinguish them, and they are not decorative. Note: in this chapter, following vector-borne modeling convention, we use the force-of-infection (susceptible-viewpoint) form for both hosts and vectors, because the host-side and vector-side rates appear naturally in this form in the Ross-Macdonald derivation. The force-from-infection form (Chapter 3) gives the same dynamics via the equivalence of Proposition 3.1; the multi-population conventions subsection of Appendix F (p. 484) collects the per-source/per-target distinction and the bridge identity between the per-source α of Chapter 3 and the per-target α_h, α_v used here. The convention-independent population-level new-infection rates

$$J_h(t) \;=\; \alpha_h\, S_h, \qquad J_v(t) \;=\; \alpha_v\, S_v,$$

are consistent with the $J(t)$ defined in Chapter 3 equation (3.2), and the per-direction infected-viewpoint estimators $\hat{\alpha}_h^{\text{inf}} = J_h/I_v$ and $\hat{\alpha}_v^{\text{inf}} = J_v/I_h$ (using the cross-population infectious counts as denominators) recover the per-source rates if needed.

The two forces. The host force of infection is

$$\alpha_h \;=\; a\, b_h\, \frac{I_v}{N_h}. \tag{13.5}$$

Each host receives $aN_v/N_h = am$ bites per day from the vector population, of which a fraction I_v/N_v are from infectious vectors, and each such bite infects with probability b_h. Combining,

$\alpha_h = am(I_v/N_v)b_h = ab_hI_v/N_h$. Note that N_v cancels in this simplification: the force depends on the *absolute number* of infectious vectors, not the vector-to-host ratio directly.

The vector force of infection is

$$\alpha_v \;=\; a\,b_v\,\frac{I_h}{N_h}. \tag{13.6}$$

Each vector bites at rate a, of which a fraction I_h/N_h are bites on infectious humans, and each such bite infects the vector with probability b_v.

Structural observation. Both α_h and α_v depend on $1/N_h$. This reflects a biological fact about vectors: the number of bites a vector delivers per day (a) is determined by vector biology, not by host density, so as N_h increases (with vector population fixed), the per-host force of infection decreases. This "bite-dilution" effect has been studied extensively and is the basis for some vector-borne-disease management strategies (e.g., livestock as alternative blood meal sources for mosquitoes).

Compartment-specific form of the two forces. The host and vector forces of infection above treat all human compartments as equally available to vectors. With the biting-exposure weights η_S, η_I, η_R introduced in Section 13.2.3, the forces become

$$\alpha_h \;=\; \frac{a\,\eta_S\,b_h\,I_v}{\eta_S S_h + \eta_I I_h + \eta_R R_h}, \qquad \alpha_v \;=\; \frac{a\,\eta_I\,b_v\,I_h}{\eta_S S_h + \eta_I I_h + \eta_R R_h}. \tag{13.7}$$

The first numerator carries η_S because the force acting on susceptibles is weighted by their relative biting attractiveness; the second carries η_I because only bites on infected hosts transmit to vectors. The common denominator is the total biting-weighted host abundance. Setting $\eta_S = \eta_I = \eta_R = 1$ recovers (13.5) and (13.6) exactly. We develop the Ross-Macdonald theory below in the uniform case ($\eta \equiv 1$), which is the classical baseline; the compartment-specific forms (13.7) extend each result with a single substitution, taken up in Section 13.5.

The exposed-vector extension. Incorporating the EIP via an explicit E_v compartment adds one equation:

$$\begin{aligned}
\dot{S}_v &= \mu_v N_v^{\mathrm{c}} - \alpha_v S_v - \mu_v S_v,\\
\dot{E}_v &= \alpha_v S_v - \gamma_v^E E_v - \mu_v E_v,\\
\dot{I}_v &= \gamma_v^E E_v - \mu_v I_v,
\end{aligned}$$

with $\gamma_v^E = 1/\tau_v^E$. The probability that an exposed vector becomes infectious (rather than dying first) is

$$p_{\mathrm{EIP}} \;=\; \frac{\gamma_v^E}{\gamma_v^E + \mu_v} \;=\; \frac{1/\tau_v^E}{1/\tau_v^E + \mu_v} \;=\; \frac{1}{1 + \mu_v \tau_v^E}.$$

For $\mu_v\tau_v^E \gtrsim 1$ (typical for *Anopheles-Plasmodium*), p_{EIP} is substantially less than 1.

Alternatively, one can absorb the EIP into an effective vector-to-host transmission probability: replace b_h with $b_h^{\mathrm{eff}} = b_h \cdot p_{\mathrm{EIP}}$. This approach maintains the two-compartment vector model and simply adjusts b_h. We adopt the latter convention for the remainder of this chapter to keep the algebra transparent, with the fixed-delay survival $p_{\mathrm{EIP}} = e^{-\mu_v\tau_v^E}$ of Section 13.2 unless stated otherwise.

AI Lab: Vector-Host Parameter Estimation Workflow

The vector-host model has more parameters than the SIR model (a, b_h, b_v, m, μ_v, γ_v^E), and its data are lopsided: host case counts are common, while vector-density and feeding data come from sparse field studies. The workflow is a *three-step audit-and-refine*.

Step 1: data inventory. Sort each parameter as directly observable, identifiable only in combination, or needing an outside assumption, and ask the AI: "*For a vector-host model of [pathogen X] in [setting Y] with data [list sources], which parameters are identifiable alone, which only as products or ratios, and which need values from the entomological literature?*" A correct answer says that host case counts identify $\mathcal{R}_0$ (Section 13.4) but not a, b_h, b_v and m separately.

Step 2: entomological values. For the third group, ask for published values from settings like yours, with ranges and citations. Check every citation against its source; invented or misattributed citations are the failure to catch.

Step 3: sensitivity propagation. $\mathcal{R}_0$ is a geometric mean of its factors (Section 13.4.1), so their uncertainties combine multiplicatively. Ask the AI to propagate the ranges of Step 2 into an interval for $\mathcal{R}_0$. A correct answer propagates on the log scale; additive Gaussian propagation understates the tails.

Disclose AI assistance, including AI-supplied values and their sources (Appendix D, Section D.4).

Exercises for this section

1. **[D]** *Direct vector-host model derivation.* Following the infected-viewpoint derivation of Chapter 3, derive the coupled vector-host system from first principles. For the host side, the rate of new host infections per unit time is the number of infectious-vector bites that land on susceptible hosts: $J_h = a(S_h/N_h^c) \cdot b_h \cdot I_v$, where a is the per-vector biting rate, S_h/N_h^c is the probability a randomly bitten host is susceptible, and b_h is the per-bite transmission probability vector-to-host. Show that the host-side force of infection is $\alpha_h = ab_hI_v/N_h^c$, and derive the corresponding $\dot{I}_h$ equation including recovery γ_R and demographic turnover ν_m. For the vector side, repeat the derivation with the roles of host and vector swapped (vectors bite hosts, not the reverse): show that the rate of new vector infections is $J_v = ab_vI_hS_v/N_h^c$ (note: N_h^c in the denominator, not N_v^c, because the per-bite host-encounter probability is $1/N_h^c$). Hence the vector-side force of infection is $\alpha_v = ab_vI_h/N_h^c = ab_vmI_h/N_v^c$, where $m = N_v^c/N_h^c$ is the vector-to-host ratio. Combine these to write out the full coupled system (13.4) and verify that it matches the form presented in Section 13.3. (This exercise fills in the derivation step that the chapter presents in summarized form.)

13.4 The Ross-Macdonald Reproductive Number

We derive the basic reproductive number of the vector-host model via the next-generation matrix, following Chapter 5, Section 5.3. The infected compartments are (I_h, I_v); the model linearized at the disease-free equilibrium $(S_h, I_h, S_v, I_v) = (N_h^c, 0, N_v^c, 0)$ yields

$$\boldsymbol{F} = \begin{pmatrix} 0 & ab_h/N_h^c \cdot N_h^c \\ ab_v \cdot N_v^c/N_h^c & 0 \end{pmatrix} = \begin{pmatrix} 0 & ab_h \\ ab_vm & 0 \end{pmatrix},$$

where $m = N_v^c/N_h^c$ is the vector-to-host ratio at the DFE. The factor structure: row 1 (new host infection) comes only from infectious vectors with rate ab_h per infectious vector; row 2 (new vector

infection) comes only from infectious hosts with rate $ab_v m$ per infectious host (because each host is bitten at rate am, and each bite can infect the biting vector with probability b_v).

The transition matrix is diagonal:

$$\boldsymbol{V} \;=\; \begin{pmatrix} \gamma_R + \nu_m & 0 \\ 0 & \mu_v \end{pmatrix}.$$

The upper-left entry is the exit rate from I_h (recovery plus host mortality); the lower-right is the exit rate from I_v (vector mortality).

The NGM and its spectral radius. Computing $\boldsymbol{K} = \boldsymbol{F}\boldsymbol{V}^{-1}$:

$$\boldsymbol{K} \;=\; \begin{pmatrix} 0 & ab_h/\mu_v \\ ab_v m/(\gamma_R + \nu_m) & 0 \end{pmatrix}.$$

The characteristic polynomial is

$$\mathcal{R}^2 \;=\; \frac{a^2 b_h b_v m}{\mu_v(\gamma_R + \nu_m)},$$

and the spectral radius is

$$\mathcal{R}_0 \;=\; \sqrt{\frac{a^2 b_h b_v m}{\mu_v(\gamma_R + \nu_m)}} \;=\; a\sqrt{\frac{b_h b_v m}{\mu_v(\gamma_R + \nu_m)}}. \tag{13.8}$$

This is the next-generation-matrix Ross-Macdonald reproductive number. Chapter 11, Section 11.6 introduced a three-level framework for reproductive numbers in bipartite and cyclic models; this section applies the framework to the vector-host setting, where the two stages of the transmission cycle are the host and the vector.

13.4.1 Stage, Cyclic, and Geometric-Mean Reproductive Numbers for the Vector-Host Model

The stage reproductive numbers (Definition 11.2) read directly off the spectral-radius calculation above:

$$\mathcal{R}_0^{h\to v} \;=\; \frac{a\, b_v\, m}{\gamma_R + \nu_m}, \qquad \mathcal{R}_0^{v\to h} \;=\; \frac{a\, b_h}{\mu_v}. \tag{13.9}$$

The first counts the number of vectors a single infectious host infects during its infectious period; the second counts the number of hosts a single infectious vector infects during its infectious lifespan. Their product is the cyclic reproductive number (Definition 11.3) based at the host stage:

$$\mathcal{C}_0^{(h)} \;=\; \mathcal{R}_0^{h\to v}\,\mathcal{R}_0^{v\to h} \;=\; \frac{a^2\, b_h\, b_v\, m}{\mu_v\,(\gamma_R + \nu_m)}, \tag{13.10}$$

which counts the expected number of new host infections produced after one full host-to-vector-and-back cycle. The book's threshold quantity, by identity (11.16) at $L = 2$, is the square root of the cyclic value:

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}) \;=\; \sqrt{\mathcal{C}_0^{(h)}} \;=\; \sqrt{\mathcal{R}_0^{h\to v}\,\mathcal{R}_0^{v\to h}}. \tag{13.11}$$

This is identity (13.8) rewritten in the three-level language. The original Ross-Macdonald formula quotes $\mathcal{C}_0^{(h)}$ (the cyclic quantity) as its "reproductive number"; the book's $\mathcal{R}_0$ is its square root,

the geometric mean of the two stage reproductive numbers. $\mathcal{R}_0$ and $\mathcal{C}_0^{(h)}$ cross 1 together; the two stage numbers need not, and at the case study the vector-to-host number is below 1 while the infection invades. Modern extensions of the Ross-Macdonald framework refine this calculation with stage-structured vector biology, seasonally varying parameters, and spatial heterogeneity; Chitnis et al. [2008] derive $\mathcal{R}_0$ for a full mosquito-life-cycle malaria model using the same NGM machinery, and Chitnis et al. [2006] analyze the backward-bifurcation phenomena that can arise when vector-host interactions include saturation effects.

13.4.2 Dissection of the Formula

The formula (13.8) has a clean interpretation as the geometric mean of two rate ratios:

$$\mathcal{R}_0 \;=\; \sqrt{\mathcal{R}_0^{h\to v}\cdot \mathcal{R}_0^{v\to h}},$$

with

$$\mathcal{R}_0^{h\to v} \;=\; \frac{ab_v}{\gamma_R+\nu_m}\cdot m \qquad \text{(vectors infected per host)},$$

$$\mathcal{R}_0^{v\to h} \;=\; \frac{ab_h}{\mu_v} \qquad \text{(hosts infected per vector)}.$$

The first factor counts how many vectors a single infectious host infects during its infectious period; it depends on the biting rate a, the human-to-vector transmission probability b_v, the host infectious duration $1/(\gamma_R+\nu_m)$, and the number of vectors available to bite per host m. The second factor counts how many hosts a single infectious vector infects during its infectious lifespan; it depends on a, b_h, and the vector lifespan $1/\mu_v$. The book's $\mathcal{R}_0$ is the geometric mean of these two stage reproductive numbers.

Biologically transparent factoring. The directional factoring above counts secondary infections per infectious individual, but a cleaner reading follows from the three-factor form introduced by Xue et al. [2018] for arboviral modeling, which isolates the three biological steps that together determine each directional reproductive number.

Proposition 13.1 (Biologically transparent factoring of Ross–Macdonald $\mathcal{R}_0$)**.** *The Ross–Macdonald vector-host reproductive number admits the per-direction three-factor form*

$$\mathcal{R}_0^{d\to d'} \;=\; P_d\cdot \alpha_d^{(0)}\cdot \tau_{id},$$

where P_d is the probability that a newly-infectious individual in compartment d survives the relevant latent period (EIP for vectors, latent period for hosts when present), $\alpha_d^{(0)}$ is the rate at which an infectious d-individual generates new d'-infections at the disease-free equilibrium, and τ_{id} is the mean infectious residence time in compartment d. The book's $\mathcal{R}_0$ is the geometric mean

$$\mathcal{R}_0 \;=\; \sqrt{\mathcal{R}_0^{v\to h}\cdot \mathcal{R}_0^{h\to v}} \;=\; \sqrt{P_vP_h\cdot \alpha_v^{(0)}\alpha_h^{(0)}\cdot \tau_{iv}\tau_{ih}}.$$

The three-factor breakdown organizes the seven model parameters into three biologically meaningful groups, each with a direct operational reading. For the vector-to-host direction, a newly-infected vector produces secondary human infections only if it (a) survives the extrinsic incubation period,

(b) infects susceptibles at some daily rate during its infectious period, and (c) remains alive long enough for the rate to integrate into a meaningful count:

$$\mathcal{R}_0^{v\to h} = \underbrace{P_v}_{\substack{\text{EIP}\\ \text{survival}}} \cdot \underbrace{\alpha_v^{(0)}}_{\substack{\text{daily rate of}\\ \text{secondary human}\\ \text{infections}}} \cdot \underbrace{\tau_{iv}}_{\substack{\text{vector}\\ \text{infectious}\\ \text{lifespan}}}, \tag{13.12}$$

where $P_v = \gamma_v^E/(\gamma_v^E + \mu_v)$ is the probability that a newly-infected mosquito survives the EIP to become infectious (Section 13.2), $\alpha_v^{(0)} = ab_h$ is the disease-free-equilibrium force from a single infectious vector onto the susceptible human population, and $\tau_{iv} = 1/\mu_v$ is the mean residual vector lifespan. When the EIP is absorbed into an effective transmission probability $b_h^{\text{eff}} = b_h \cdot P_v$ as in the rest of this chapter, the survival factor P_v moves into the rate factor and the three-factor form reduces to the two-factor $\mathcal{R}_0^{v\to h} = ab_h^{\text{eff}}/\mu_v$ used earlier. We state the three-factor form here because it makes the biological role of each term explicit.

The host-to-vector direction has the same three-factor structure:

$$\mathcal{R}_0^{h\to v} = \underbrace{P_h}_{\substack{\text{latent-period}\\ \text{survival}}} \cdot \underbrace{\alpha_h^{(0)}}_{\substack{\text{daily rate of}\\ \text{secondary vector}\\ \text{infections per host}}} \cdot \underbrace{\tau_{ih}}_{\substack{\text{host}\\ \text{infectious}\\ \text{duration}}}, \tag{13.13}$$

where $P_h = 1$ for the SIR-host model of this chapter (no latent compartment on the host side), $\alpha_h^{(0)} = ab_v m$ is the daily rate at which an infectious host generates new vector infections at the disease-free equilibrium (the factor m reflects that each host is bitten by m vectors on average), and $\tau_{ih} = 1/(\gamma_R + \nu_m)$ is the mean duration of host infectiousness. For an SEIR-host model, P_h reduces to $\gamma_h^E/(\gamma_h^E + \nu_m)$ as in Chapter 10, Section 10.3.

The book's reproductive number is the geometric mean

$$\mathcal{R}_0 = \sqrt{\mathcal{R}_0^{v\to h} \cdot \mathcal{R}_0^{h\to v}} = \sqrt{P_v P_h \cdot \alpha_v^{(0)}\alpha_h^{(0)} \cdot \tau_{iv}\tau_{ih}}, \tag{13.14}$$

which organizes the seven parameters of the model into three biologically meaningful groups: latent-period survival probabilities (one per direction), daily transmission rates at the DFE (one per direction), and infectious residence times (one per direction). Each group has a direct operational reading. Interventions that shorten the vector lifespan (τ_{iv}) target the third group. Interventions that reduce vector competence for the pathogen (b_h, b_v) target the second group through the transmission rates. Interventions that extend the EIP relative to the vector lifespan (such as environmental cooling of vector microhabitats, Section 13.5) target the first group through P_v. The factored form (13.14) makes these operational targets visible on the face of the formula. Chapter 18, Section 18.5 applies this reading directly: each dengue-control scenario is labeled by which factor in (13.14) it modifies, and the effect on $\mathcal{R}_0$ follows from the geometric-mean dampening by a factor of $\sqrt{(\text{parameter change})}$ when only one direction is affected.

Sensitivity indices read directly from the factored form. The square-root structure of (13.14) gives immediate sensitivity indices by differentiation:

$$\Upsilon_{P_v}^{\mathcal{R}_0} = \Upsilon_{P_h}^{\mathcal{R}_0} = \Upsilon_{\alpha_v^{(0)}}^{\mathcal{R}_0} = \Upsilon_{\alpha_h^{(0)}}^{\mathcal{R}_0} = \Upsilon_{\tau_{iv}}^{\mathcal{R}_0} = \Upsilon_{\tau_{ih}}^{\mathcal{R}_0} = \tfrac{1}{2},$$

so each of the six biological factors contributes equally in log-derivative terms: a 10% increase in any one factor produces a 5% increase in $\mathcal{R}_0$. The biting rate a is the exception: it enters *both* $\alpha_v^{(0)} = ab_h$

and $\alpha_h^{(0)} = ab_v m$, so its total sensitivity is $\Upsilon_a^{\mathcal{R}_0} = \Upsilon_{\alpha_v^{(0)}}^{\mathcal{R}_0} + \Upsilon_{\alpha_h^{(0)}}^{\mathcal{R}_0} = 1$. This is the mathematical origin of the biting-rate unit-sensitivity result stated earlier in this section and visualized in Figure 13.2: biting-rate interventions act on both directions of the bipartite transmission cycle, while most other interventions act on only one direction.

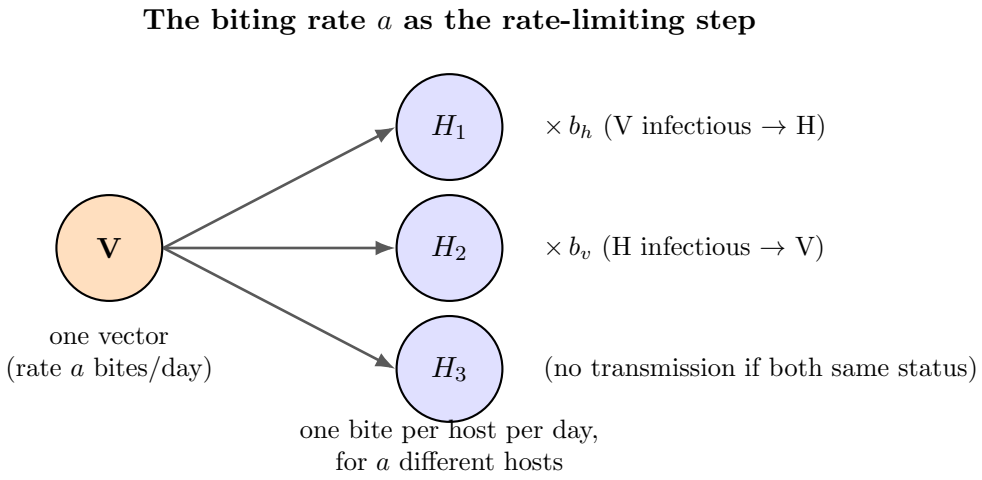


Figure 13.2: The biting rate a couples both transmission directions in the Ross-Macdonald model. Each vector delivers a bites per day; each bite is potentially a transmission event in one direction or the other depending on which side is infectious. The per-bite transmission probabilities b_h (vector → host) and b_v (host → vector) are biological constants of the pathogen-vector pairing. Because a multiplies both stage reproductive numbers, the cyclic reproductive number $\mathcal{C}_0^{(h)}$ scales as a^2, and the book's $\mathcal{R}_0 = \sqrt{\mathcal{C}_0^{(h)}}$ scales linearly in a (13.8). This is the rate-amplification mechanism that makes biting-rate reduction (bed nets, repellents) one of the two strongest levers on $\mathcal{R}_0$, with adult vector mortality.

Square-root dampening. As in Chapter 12's bipartite $\mathcal{R}_0$, the sensitivity of $\mathcal{R}_0$ to any single parameter is half of the corresponding cyclic-reproductive-number sensitivity. A doubling of the vector-to-host ratio m produces a factor-of-$\sqrt{2}$ increase in $\mathcal{R}_0$, not a factor of 2.

However, the biting rate a appears quadratically under the square root (once for $h \to v$ and once for $v \to h$), giving it a unit sensitivity: a doubling of a doubles $\mathcal{R}_0$. This is a strong lever: reducing biting rates (through bed nets, repellents, or housing modifications) has a larger effect on $\mathcal{R}_0$ than reducing m by the same factor. Only vector mortality, which also governs survival through the EIP, is stronger (Section 13.6).

Extension to compartment-specific biting exposure. With the biting-exposure weights η_S, η_I, η_R from (13.7), the same NGM derivation gives

$$\mathcal{R}_0 \;=\; a\sqrt{\frac{b_h b_v m\,\eta_I}{\mu_v(\gamma_R + \nu_m)}}, \tag{13.15}$$

where $\eta_S = 1$ is fixed by convention and η_R does not appear because recovered individuals do not contribute to transmission. The classical formula (13.8) is the case $\eta_I = 1$. The weight η_I enters $\mathcal{R}_0^2$ once rather than twice, because reducing infected-host biting exposure only affects the $h \to v$

direction of transmission. The sensitivity indices follow: $\Upsilon_{\eta_I}^{\mathcal{R}_0} = 1/2$. Because the weights are relative, sharing out a fixed a bites per vector, lowering every weight by the same factor leaves $\mathcal{R}_0$ unchanged; only a reduction in a itself, bites prevented rather than redirected, has unit sensitivity.

13.4.3 The Stability-Indicator: Vectorial Capacity

A related quantity, the *vectorial capacity*, is defined as the daily rate of secondary host infections that a single host case would generate assuming the vector population is not depleted:

$$V = \frac{a^2 b_h b_v m}{\mu_v} \cdot p_{\text{EIP}} \approx \frac{a^2 b_h b_v m \cdot e^{-\mu_v \tau_v^E}}{\mu_v}.$$

This is closely related to $\mathcal{R}_0^2$: $\mathcal{R}_0^2 = V/(\gamma_R + \nu_m)$, so vectorial capacity equals $\mathcal{R}_0^2$ *divided by* the mean host infectious duration, $V = \mathcal{R}_0^2(\gamma_R + \nu_m) = \mathcal{R}_0^2/\tau_I$ with $\tau_I = 1/(\gamma_R + \nu_m)$. The units confirm the direction: V is a rate (secondary host infections per host case per day), while $\mathcal{R}_0^2$ is dimensionless. Vectorial capacity provides a summary statistic for the vector component of transmission potential, independent of host recovery rates, and is widely used in entomological studies where the human infectious duration is not the primary variable.

From $\mathcal{R}_0$ to $\mathcal{R}_e(t)$ in the bipartite vector-host setting. The NGM-derived $\mathcal{R}_0$ above is computed at the disease-free equilibrium, where both the host and the vector populations are fully susceptible: $S_h/N_h^c = S_v/N_v^c = 1$. As an outbreak progresses, the operationally-relevant quantity is the time-varying effective reproductive number $\mathcal{R}_e(t)$ (Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$"), generalizing the homogeneous construction of Chapter 5, Section 5.2 and the renewal-equation derivation of Chapter 15, Section 15.5.3, and parallel to the heterogeneous-population construction of Chapter 11. The bipartite setting admits a particularly clean form. Because the geometric-mean structure $\mathcal{R}_0 = \sqrt{\mathcal{R}_0^{h\to v} \cdot \mathcal{R}_0^{v\to h}}$ factors the cycle into a host-side and a vector-side component, the depletion factors enter symmetrically:

$$\mathcal{R}_e(t) = \sqrt{\mathcal{R}_0^{h\to v} \cdot \mathcal{R}_0^{v\to h}} \cdot \sqrt{\frac{S_h(t)}{N_h^c} \cdot \frac{S_v(t)}{N_v^c}} = \mathcal{R}_0 \cdot \sqrt{\frac{S_h(t)}{N_h^c} \cdot \frac{S_v(t)}{N_v^c}}.$$

The square-root structure reflects the bipartite cycle: a transmission generation alternates between host and vector, so the depletion factor that enters $\mathcal{R}_e(t)$ is the geometric mean of the two susceptible fractions rather than either one alone. Two operational consequences follow. First, vector-control interventions and host-side interventions are not interchangeable for changing $\mathcal{R}_e(t)$: reducing $S_v(t)/N_v^c$ (e.g., by killing vectors) reduces $\mathcal{R}_e$ by the same factor as reducing $S_h(t)/N_h^c$ (e.g., by vaccination), but only after the square-root averaging. Second, the threshold $\mathcal{R}_e(t) = 1$ in a bipartite system can be crossed in either direction without either susceptible fraction reaching zero, which is the operational reading of why combined host-vector interventions are typically more effective than single-mode interventions of equivalent total cost. The dengue case study of Chapter 18 exploits this structure quantitatively; the labelled-superscript convention $\mathcal{R}_0^{\text{intervention}}$ developed in Chapter 17 for COVID-19 applies here in directly analogous form, e.g., $\mathcal{R}_0^{\text{IRS}}$ for indoor residual spraying or $\mathcal{R}_0^{\text{ITN}}$ for insecticide-treated nets, each evaluated at the disease-free equilibrium and connected to $\mathcal{R}_e(t)$ by the same square-root depletion factor.

Exercises for this section

1. [**C**] The Ross–Macdonald $\mathcal{R}_0$ has unit sensitivity to the biting rate a and one-half sensitivity to the per-bite probabilities and to m (Proposition 13.1); vector mortality acts through both the vector's lifespan and its survival of the EIP. Explain in plain English why the biting rate and vector mortality are stronger levers than the others, and what this implies for prioritizing intervention targets.

2. [**I**] The bipartite $\mathcal{R}_e(t)$ formula derived above shows that $\mathcal{R}_e(t) = \mathcal{R}_0 \cdot \sqrt{(S_h/N_h^c)(S_v/N_v^c)}$. Suppose two interventions reduce $S_h(t)/N_h^c$ from 1 to 0.5 and $S_v(t)/N_v^c$ from 1 to 0.5, respectively, applied independently. Compute $\mathcal{R}_e/\mathcal{R}_0$ for each intervention applied alone, and for both applied together. Compare with the analogous calculation in a homogeneous (non-bipartite) population, where $\mathcal{R}_e/\mathcal{R}_0 = S(t)/N$ enters linearly. Explain in plain English why the bipartite structure produces a different combined effect than the homogeneous structure, and what this implies for cost-benefit analysis of host-side versus vector-side interventions.

13.5 Interventions and Their Effect on $\mathcal{R}_0$

The Ross-Macdonald formula (13.8) decomposes the vector-borne reproductive number into a small set of biologically-interpretable parameters, each of which corresponds to a specific intervention strategy. This decomposition has guided vector-borne disease control policy for nearly a century.

13.5.1 Reducing Vector Population (m)

Insecticides, larvicides, and environmental management of breeding sites reduce the vector population. Inside the Ross-Macdonald formula, this affects $m = N_v/N_h$, with $\mathcal{R}_0 \propto \sqrt{m}$. A doubling of vector population produces a $\sqrt{2} \approx 1.41$-fold increase in $\mathcal{R}_0$; halving the population reduces $\mathcal{R}_0$ by a factor of $\sqrt{2}$.

The square-root dependence makes vector-population-reduction interventions *less* effective than the intuition from direct-transmission models suggests. In a direct-transmission SIR model, halving contact rates halves $\mathcal{R}_0$; in a vector-borne model, halving vector population reduces $\mathcal{R}_0$ by only 29%.

Threshold for elimination. The elimination threshold is $\mathcal{R}_0 < 1$. Using the formula, this requires

$$m \;<\; \frac{\mu_v(\gamma_R + \nu_m)}{a^2 b_h b_v}.$$

For typical malaria parameters (developed in the case study), the threshold vector-to-host ratio is order 0.1 to 1.0: bringing mosquitoes to less than one per person in the population. This is challenging to sustain.

13.5.2 Reducing Biting Rate (a)

Insecticide-treated bed nets (ITNs) and indoor residual spraying (IRS) reduce the effective biting rate on humans. Because a appears twice in $\mathcal{R}_0^2 = a^2 b_h b_v m/[\mu_v(\gamma_R + \nu_m)]$, once for each direction of transmission, the sensitivity of $\mathcal{R}_0$ to a is unity:

$$\Upsilon_a^{\mathcal{R}_0} \;=\; 1.$$

A 50% reduction in biting rate reduces $\mathcal{R}_0$ by 50%, a larger effect than any other single parameter except vector mortality (Section 13.6), which is one reason modern malaria control programs give bed nets priority.

Combined with vector mortality. Insecticide-treated bed nets do more than reduce biting: they kill biting mosquitoes. The net effect is to simultaneously reduce a (bites) and increase μ_v (vector mortality). Both effects reduce $\mathcal{R}_0$. The combined reduction is larger than either component alone.

13.5.3 Behavior Change Among Infected Hosts (η_I)

Interventions that reduce biting exposure specifically among infected hosts operate on η_I in (13.15). Operational mechanisms include symptomatic individuals staying indoors, increasing bed-net use after symptom onset, seeking treatment and being removed from community contact, and clinical case-management programs that provide housing support during viremic periods. Through (13.15), a reduction $\eta_I \to (1-\chi_I)\eta_I$ produces

$$\mathcal{R}_0 \;\to\; \mathcal{R}_0\sqrt{1-\chi_I},$$

with sensitivity $\Upsilon_{\eta_I}^{\mathcal{R}_0} = 1/2$. A 50% reduction in infected-host biting exposure yields a $\sqrt{0.5} \approx 0.71$ factor, i.e. a 29% reduction in $\mathcal{R}_0$; a 30% reduction gives a 16% reduction in $\mathcal{R}_0$.

This is smaller than the unit-sensitivity effect of a uniform biting-rate reduction. The reason is structural: infected hosts are present in the population only in proportion to the current infectious prevalence, so reducing their exposure interrupts only the $h \to v$ half of the transmission cycle, not both halves as a uniform a-reduction would. A program that covers susceptibles as well as infecteds recovers unit sensitivity only if it prevents bites, lowering a. A program that only redirects bites changes the weights, and protecting susceptibles alone would send more bites to infected hosts. Operationally, bed-net programs that distribute nets without regard to epidemiological status act on a itself (unit sensitivity); programs targeted at confirmed cases or symptomatic individuals act on η_I only (half sensitivity).

Combined with other control mechanisms, behavior-change interventions in infected hosts contribute to $\mathcal{R}_0$ reduction at lower coverage than vector-population reduction of equivalent magnitude; Xue et al. [2018] analyzes this mechanism in the context of arboviral disease (chikungunya, dengue, and Zika) under combined Wolbachia release and infected-host behavior change.

13.5.4 Reducing Transmission Probabilities (b_h, b_v)

Vaccines, prophylactic drugs, and treatment of infectious humans reduce the per-bite transmission probabilities. For malaria, partially-effective vaccines (like the RTS,S vaccine) reduce b_h (vector-to-human probability) by perhaps 30%. Treatment of infectious hosts reduces b_v (human-to-vector probability) by clearing parasites from the blood, and also reduces I_h directly.

Because b_h and b_v each appear once under the square root, $\mathcal{R}_0 \propto \sqrt{b_h b_v}$. A 30% reduction in b_h (e.g., 30% vaccine efficacy) reduces $\mathcal{R}_0$ by $\sqrt{1-0.3} \approx 0.84$, or a 16% reduction. This is a modest but meaningful effect; combined with other interventions it contributes to the overall reduction of $\mathcal{R}_0$ below 1.

13.5.5 The Extrinsic Incubation Period as Intervention Target

Because $p_{\text{EIP}} = e^{-\mu_v \tau_v^E}$ enters the effective b_h, and τ_v^E depends on ambient temperature (the parasite develops faster at warmer temperatures), climate affects $\mathcal{R}_0$ through the EIP. In cool climates, τ_v^E is longer, p_{EIP} is smaller, and $\mathcal{R}_0$ is correspondingly reduced. This is one mechanism by which malaria historically was a geographically-bounded disease (Northern climates too cool for efficient transmission), and it is one of the mechanisms by which climate change is expected to expand malaria's geographic range.

From a policy perspective, environmental management of mosquito microhabitats can affect effective temperature and hence τ_v^E: shaded areas have lower temperatures, longer EIP, and reduced transmission relative to sunny areas with equivalent vector populations. This effect is usually minor relative to direct vector-control interventions but contributes to the overall epidemiological picture.

13.5.6 Combination Interventions

Like the STI interventions of Chapter 12, vector-borne interventions stack multiplicatively on $\mathcal{R}_0$. A combination of 50% bed-net coverage (halving a and additional vector mortality from net-killed mosquitoes), 30% reduction in vector population through larvicides ($m \to 0.7m$), and a partially-effective vaccine (30% reduction in b_h) gives combined

$$\mathcal{R}_0^{\text{intervention}} \;\approx\; \mathcal{R}_0 \cdot 0.5 \cdot \sqrt{0.7} \cdot \sqrt{0.7} \;=\; 0.35\,\mathcal{R}_0.$$

If the baseline $\mathcal{R}_0$ is 2.5, this combination brings $\mathcal{R}_0$ to about 0.87, below the elimination threshold. Sustained combination intervention is required for elimination; reversion upon relaxation of any component can return $\mathcal{R}_0$ above 1, a phenomenon observed repeatedly in malaria-elimination campaigns.

Exercises for this section

1. **[I]** Consider an intervention that doubles vector mortality ($\mu_v \to 2\mu_v$, as might occur with intensive insecticide spraying). Compute the resulting reduction in $\mathcal{R}_0$ for the malaria case study. Compare with the effect of halving biting rate.

13.6 Case Study: Malaria

13.6.1 Parameter Values

Typical parameter values for *Plasmodium falciparum* malaria transmitted by *Anopheles gambiae* in a high-transmission African setting:

- Vector biting rate: $a \approx 0.3$/day.
- Human-to-vector transmission probability: $b_v \approx 0.1$ per bite.
- Vector-to-human transmission probability: $b_h \approx 0.5$ per bite.
- Vector mortality: $\mu_v \approx 0.1$/day ($\tau_v^m \approx 10$ days).
- Extrinsic incubation period: $\tau_v^E \approx 12$ days.

- EIP survival probability: $p_{\text{EIP}} = e^{-0.1\times 12} = e^{-1.2} \approx 0.30$.
- Effective vector-to-human: $b_h^{\text{eff}} = 0.5 \times 0.30 = 0.15$.
- Human recovery rate: $\gamma_R \approx 0.01$/day ($\tau_R \approx 100$ days; untreated malaria persists for months).
- Human mortality: $\nu_m \approx 10^{-5}$/day (negligible relative to γ_R).
- Vector-to-host ratio: $m \approx 5$ (five mosquitoes per person in high-transmission settings).
- Biting-exposure weight on infected hosts: $\eta_I = 1$, the classical equal-exposure assumption, with $\eta_S = 1$ fixed by convention. This choice recovers the Ross–Macdonald form (13.8) from the compartment-specific form (13.15); a case study with evidence for status-dependent biting exposure would carry $\eta_I \neq 1$ and use the latter.

13.6.2 Computing $\mathcal{R}_0$

Using (13.8) with the effective $b_h^{\text{eff}} = 0.15$:

$$\begin{aligned}\mathcal{R}_0^2 &= \frac{a^2 b_h^{\text{eff}} b_v m}{\mu_v(\gamma_R + \nu_m)} \\ &= \frac{0.09 \times 0.15 \times 0.1 \times 5}{0.1 \times 0.01} \\ &= \frac{0.00675}{0.001} \\ &= 6.75.\end{aligned}$$

Taking the square root:

$$\mathcal{R}_0 \approx 2.6.$$

The corresponding cyclic reproductive number is $\mathcal{C}_0^{(h)} = \mathcal{R}_0^2 \approx 6.75$. Published per-cycle estimates for African *P. falciparum* span from about 1 to over 3,000 [Smith et al., 2007], so the case study sits at the low end of the range.

The reproductive number is well above the elimination threshold, consistent with the observed pattern of sustained transmission.

13.6.3 Sensitivity Analysis

Applying the sensitivity framework of Chapter 9 to (13.8), and to (13.15) for the biting-exposure weight:

- $\Upsilon_a^{\mathcal{R}_0} = 1.0$ (unit sensitivity: a dominates).
- $\Upsilon_{b_h}^{\mathcal{R}_0} = \Upsilon_{b_v}^{\mathcal{R}_0} = 1/2$ (square-root sensitivity).
- $\Upsilon_m^{\mathcal{R}_0} = 1/2$.
- $\Upsilon_{\mu_v}^{\mathcal{R}_0} = -\frac{1}{2}(1 + \mu_v \tau_v^E) \approx -1.1$ (vector mortality enters twice: through the vector's infectious lifespan and through EIP survival $p_{\text{EIP}} = e^{-\mu_v \tau_v^E}$).
- $\Upsilon_{\tau_v^E}^{\mathcal{R}_0} = -\frac{1}{2}\mu_v \tau_v^E \approx -0.6$ (a longer EIP lowers survival to infectiousness).

- $\Upsilon^{\mathcal{R}_0}_{\gamma_R} = -1/2 \cdot \gamma_R/(\gamma_R + \nu_m) \approx -1/2$ (treatment shortens infectious period).
- $\Upsilon^{\mathcal{R}_0}_{\eta_I} = 1/2$ (compartment-specific biting exposure on infected hosts, from (13.15)).

The most sensitive parameter is vector mortality, then the biting rate a. Interventions that kill adult mosquitoes, such as insecticide-treated nets and indoor residual spraying, act on the most sensitive parameter, and nets act on the biting rate as well.

13.6.4 Intervention Scenarios

We evaluate three intervention scenarios using the baseline parameters:

Scenario 1: Bed nets (50% biting reduction). Reduce a from 0.3 to 0.15. Then $\mathcal{R}_0^2 = 0.0225 \times 0.15 \times 0.1 \times 5/(0.1 \times 0.01) = 1.69$, giving $\mathcal{R}_0 = 1.3$. Still above threshold but significantly reduced; combined with other interventions could reach elimination.

Scenario 2: Treatment (5-fold recovery-rate increase). Reduce τ_R from 100 days to 20 days (prompt antimalarial treatment), giving $\gamma_R = 0.05$/day. Then $\mathcal{R}_0^2 = 0.00675/(0.1 \times 0.05) = 1.35$, giving $\mathcal{R}_0 = 1.16$. Close to threshold but not quite below.

Scenario 3: Combination (bed nets + treatment + partial vaccine). Combining 50% bed net coverage ($a = 0.15$), prompt treatment ($\gamma_R = 0.05$), and a partial vaccine reducing b_h^{eff} by 30% ($b_h^{\text{eff}} = 0.105$):

$$\mathcal{R}_0^2 = \frac{0.0225 \times 0.105 \times 0.1 \times 5}{0.1 \times 0.05} = \frac{0.00118}{0.005} = 0.236,$$

giving $\mathcal{R}_0 = 0.49$. Well below threshold; elimination achievable if sustained.

This progression illustrates the core lesson of vector-borne disease control: single interventions are rarely sufficient, but combinations bring $\mathcal{R}_0$ below 1 reliably. Sustaining the combination is the persistent challenge for malaria-elimination campaigns.

Exercises for this section

1. [**I**] Compute the elimination-threshold vector-to-host ratio m_{crit} for the malaria case study. If the baseline $m = 5$, how much must vector population be reduced to reach $m < m_{\text{crit}}$ with all other parameters fixed? Is this feasible in practice?

13.7 Summary and Exercises

This chapter developed the vector-borne disease framework:

1. Vector-borne diseases transmit through an arthropod intermediate. The bipartite structure parallels the sexual-transmission models of Chapter 12 with vector-specific biology (short vector lifespan, extrinsic incubation, biting-driven transmission).

2. The vector-host model (13.4) is a five-compartment system with distinct dynamics for hosts (S_h, I_h, R_h) and vectors (S_v, I_v). An explicit exposed-vector compartment E_v can be added to model the EIP, or the EIP can be absorbed into an effective transmission probability $b_h^{\text{eff}} = b_h \cdot p_{\text{EIP}}$.

3. The Ross-Macdonald reproductive number in the book's convention is

$$\mathcal{R}_0 \;=\; a\sqrt{\frac{b_h b_v m}{\mu_v(\gamma_R+\nu_m)}},$$

the geometric mean of the host-to-vector and vector-to-host stage reproductive numbers. The cyclic reproductive number $\mathcal{C}_0^{(h)} = \mathcal{R}_0^2$ is the original Ross-Macdonald form.

4. Biting rate a has unit sensitivity in $\mathcal{R}_0$, and vector mortality, counted through EIP survival, more: $-\frac{1}{2}(1+\mu_v\tau_v^E)$. The per-bite probabilities and m have one-half sensitivity.

5. Interventions combine multiplicatively. Vector-control (bed nets, larvicides), treatment (shortening τ_R), and vaccines (b_h reduction) can jointly push $\mathcal{R}_0$ below 1 in settings where no single intervention suffices.

6. Malaria case study: baseline $\mathcal{R}_0 \approx 2.6$ in high-transmission African settings. Bed nets alone reduce to $\mathcal{R}_0 \approx 1.3$; combination with prompt treatment and partial vaccine reaches $\mathcal{R}_0 \approx 0.5$, achieving elimination conditions.

Chapter 14 addresses vertical (mother-to-child) transmission, another distinctive transmission mode. Chapter 15 develops the PDE formulation for age-structured and time-since-infection models. Chapter 18 returns to vector-borne diseases with a detailed case study integrating fitting, sensitivity analysis, and intervention evaluation.

Framework use in vector-borne fitting. The bipartite Ross-Macdonald structure inherits the infected-viewpoint fitting advantage established in the sensitivity-analysis paper of Hyman [2026d]: the estimators $\hat{\alpha}_h = J_h/I_h$ (host side) and $\hat{\alpha}_v = J_v/I_v$ (vector side) are computable without any estimate of the corresponding susceptible populations S_h and S_v. Each is a per-target-population rate rather than a per-source one, since hosts do not infect hosts: the source-side factor is absorbed into the estimated quantity rather than removed from it (Chapter 20, Section 20.5). What the estimator avoids is the need to *know* S_v, not a dependence of the estimand on it. In vector-borne outbreak response, where S_v (the susceptible mosquito population) is typically the most poorly known quantity, this structural immunity is directly useful: field surveys of vector density are expensive and subject to seasonal variation, while host-side incidence data are routinely collected through public-health surveillance. The malaria case study of Section 13.6 uses host-side incidence data to fit $\hat{\alpha}_h$ without requiring a committed value for S_v; vector-side parameters are taken from independent entomological sources. The dengue case study of Chapter 18 applies the same pattern in a fitted context with quantified uncertainty bounds.

Exercises

Six section-level exercises appear above, following Section 13.2, Section 13.3, Section 13.4, Section 13.5, and Section 13.6. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[C]** The chapter offers two ways of incorporating the extrinsic incubation period (EIP): an explicit E_v compartment and an effective transmission probability $b_h^{\text{eff}} = b_h \cdot p_{\text{EIP}}$. State the trade-off between the two approaches in one sentence each.

2. [**D**] Verify the Ross-Macdonald $\mathcal{R}_0$ formula (13.8) by computing the spectral radius of the NGM $\boldsymbol{K}$ directly.

3. [**D**] For the malaria case study parameters, compute the vectorial capacity V and verify the relation $\mathcal{R}_0^2 = V/(\gamma_R + \nu_m)$.

4. [**D**] Derive the relationship between the explicit-E_v-compartment model and the effective-b_h approximation. Show that they give the same $\mathcal{R}_0$ when b_h^{eff} uses the exponential-sojourn survival $1/(1+\mu_v\tau_v^E)$, and compute the difference at the case study, which uses $e^{-\mu_v\tau_v^E}$.

5. [**I**] For a vector-borne disease with temperature-dependent EIP, how does a 2°C warming (reducing τ_v^E by 20%) change $\mathcal{R}_0$? Derive the sensitivity of $\mathcal{R}_0$ to τ_v^E through p_{EIP}.

6. [**I**] Consider a setting with seasonal variation in vector population ($m(t)$ varying by factor 10 between dry and wet seasons). How would the seasonal dynamics of $\mathcal{R}_0$ manifest in case counts? Sketch a simulation approach using the SIR framework of Chapter 6.

7. [**L**] Implement the vector-host system (13.4) and reproduce the malaria case study of Section 13.6 at baseline $\mathcal{R}_0 \approx 2.6$. Then evaluate three intervention scenarios in turn: (a) bed nets that reduce a by 50%; (b) prompt treatment that halves τ_R; (c) a combination of (a) and (b). Compute the resulting $\mathcal{R}_0$ for each scenario and report the relative reductions. Verify numerically that the combination intervention reduces $\mathcal{R}_0$ multiplicatively, consistent with the chapter's intervention-arithmetic claim.

8. [**I**] *Spaced retrieval, Chapter 11.* The vector-host model of this chapter and the two-group framework of Chapter 11 are both bipartite systems: each has two "populations" that interact through a coupling matrix, and the basic reproductive number is computed as the spectral radius of a 2×2 next-generation matrix. Compare the two side by side:

   (a) Map the entries of the host-vector NGM (computed in Section 13.4) onto the form of the two-group $\mathcal{R}_0$ formula (11.12) of Chapter 11, Section 11.4. Identify what plays the role of the contact-matrix entries $C_{\ell j}$, what plays the role of the per-partnership transmission probability β, and what plays the role of the residence-time terms in each population.

   (b) Identify the structural specialization that makes the vector-host model simpler: which entries of a fully general two-group mixing matrix vanish in the host-vector setting, and what assumption about the populations' intra-group transmission underlies the vanishing? (Hint: hosts do not infect other hosts directly through this mechanism, and similarly for vectors.)

   (c) Identify a feature *present* in the vector-host model that is *absent* from the two-group framework as developed in Chapter 11: the EIP survival factor $p_{\text{EIP}} = e^{-\mu_v\tau_v^E}$ enters the effective vector-to-host transmission probability b_h^{eff} but has no analogue in the heterosexual bipartite model of Chapter 12 or the activity-stratified model of Chapter 11. State in one sentence why vector-borne diseases require this additional factor and most direct-transmission diseases do not.

   The exercise reinforces that vector-borne and heterogeneous direct-transmission models share a common mathematical skeleton (bipartite NGM, square-root reproductive number) while differing in the biological specifics that govern individual matrix entries. Recognizing the shared skeleton is the route to applying $\mathcal{R}_0$-derivation techniques developed in one setting to a structurally similar but biologically distinct setting.

9. **[I]** *Reasoning visibility.* A collaborator suggests simplifying the vector-host model by setting the vector mortality rate μ_v to zero, arguing that "vectors die so slowly compared to the host disease cycle that we can ignore vector turnover." Walk through the reasoning steps you would use to evaluate whether this simplification is acceptable. Your answer should be structured as five numbered steps:

   (1) State the simplification mathematically: which terms in the vector equations of system (13.4) vanish or get replaced when $\mu_v = 0$, and which auxiliary quantities (e.g., p_{EIP}) lose their dependence on μ_v. Identify what p_{EIP} becomes in the limit $\mu_v \to 0$.

   (2) Quantify the timescale comparison: state the actual ratio of vector lifespan to host infectious period for typical *Anopheles*-malaria parameters (vector lifespan $\tau_v^m \sim 14$ days vs. host infectious period $\tau_R \sim 7$–30 days). Is the collaborator's claim numerically defensible? State explicitly the bound on τ_v^m/τ_R below which the simplification is dangerous.

   (3) Compute the effect on $\mathcal{R}_0$: with $\mu_v = 0$ in the formula (13.8), $\mathcal{R}_0$ would diverge. State why this happens (a vector that never dies has infinite expected onward transmission) and what this divergence reveals about the role of μ_v in the model.

   (4) Assess what data or comparisons you would consult to decide: would a side-by-side simulation comparing μ_v at its true value and at half-its-value (a perturbation, not a limit) localize the sensitivity? What entomological data product would be most relevant?

   (5) State your reasoned recommendation to the collaborator: accept, reject, or provisionally accept with caveats. Provide one sentence of justification, and one sentence on what would change your answer if the pathogen were one with a fast-dying vector (e.g., *Aedes aegypti* with $\tau_v^m \sim 10$ days) versus a long-lived one (e.g., a tick vector with $\tau_v^m \sim 1$ year).

   The exercise asks you to externalize the diagnostic reasoning that distinguishes a defensible model simplification from one that destroys an essential structural feature. Vector mortality is in fact the unit-sensitivity dominant parameter for many vector-borne diseases (the bipartite EIP-survival channel), so this collaborator's instinct, while not unreasonable, is wrong. Walking through the reasoning makes the wrongness visible.

10. **[AI Lab]** *Vector-host parameter estimation workflow.* Choose a vector-borne pathogen for which you have access to host case-count data (or simulate a synthetic dataset following Section 13.3 parameters). Execute the chapter's three-step audit-and-refine workflow:

   (a) *Step 1: Data inventory.* Classify the model parameters $\{a, b_h, b_v, m, \mu_v, \tau_v^E, \gamma_R, \nu_m\}$ into three categories: directly observable from case-count data; jointly identifiable as products or ratios; requiring auxiliary entomological assumption. Prompt an AI tool: *"For a vector-host model of [pathogen X] with available [data sources], which parameters are individually identifiable from incidence data, which are jointly identifiable as composites, and which require auxiliary entomological data?"* Audit the AI's classification against the chapter's identifiability arguments: a chapter-aware response will identify $\mathcal{R}_0$ as a composite that is identifiable from host-incidence data alone, but will note that the individual factors a, b_h, b_v, m are not separately identifiable without entomological surveys. Document any disagreement between the AI's classification and the chapter's structure.

   (b) *Step 2: Entomological-value elicitation.* For the parameters in category (iii), prompt the AI tool: *"For [vector species X] in [climatic setting Y], what are the published values of the per-vector biting rate a, vector mortality μ_v, and EIP duration τ_v^E, with uncertainty*

ranges and citations?" Audit the AI's claimed values: (i) spot-check 2–3 of the cited sources (do they exist? do their reported values match the AI's claims?); (ii) check whether the AI's uncertainty ranges reflect the substantial between-setting variability that the entomological literature reports; (iii) document any AI hallucinations (invented citations, misattributed values).

(c) *Step 3: Sensitivity propagation.* Using the AI-elicited values from Step 2 with the audit corrections, propagate the uncertainty through the $\mathcal{R}_0$ formula of Section 13.4. Compute the resulting CI on $\mathcal{R}_0$ using both (a) multiplicative-error propagation (correct for the geometric-mean structure) and (b) additive Gaussian propagation (the typical AI default). Compare the two CIs: which is wider? Is the difference operationally significant for an intervention recommendation?

The exercise practices the full vector-host parameter-estimation workflow with explicit AI-audit at each step. Disclosure: per Appendix D, Section D.4, document the AI system, all prompts, the AI-claimed entomological values, the citation-spot-check findings, and any hallucinations identified. The disclosure is operationally important for any vector-borne-disease modeling work in which the entomological parameters were AI-elicited.

11. **[Verify the AI]** *Auditing an AI-generated Ross-Macdonald derivation.* Prompt an AI tool: *"Derive the basic reproductive number for malaria from a vector-host transmission model. Show the next-generation-matrix construction, and give a numerical estimate using typical malaria parameters in sub-Saharan Africa."* Save the response. Audit the derivation against this chapter's three-error structure for vector-borne models:

    (a) *Per-bite-vs-per-day audit.* Does the AI's derivation use the biting rate a (units: bites per mosquito per day) consistently? Vector-borne models are particularly susceptible to a units error in which a is taken to be a probability (per-bite, dimensionless) rather than a rate (per-day). The diagnostic question: *"What are the units of a in your formula? Verify by dimensional analysis: $a^2 b_h b_v m/(\mu_v(\gamma_R+\nu_m))$ should be dimensionless because $\mathcal{R}_0$ is."* An AI that has confused the per-bite probability with the per-day rate will produce a $\mathcal{R}_0$ formula off by a factor with units of time, often manifesting as a $\mathcal{R}_0$ value 365 times too large or too small.

    (b) *EIP-survival audit.* Does the AI's derivation include the extrinsic-incubation-period survival factor $p_{\text{EIP}} = e^{-\mu_v \tau_v^E}$ in the effective vector-to-host transmission probability b_h^{eff}? Omitting p_{EIP} overestimates $\mathcal{R}_0$ by a factor of $1/\sqrt{p_{\text{EIP}}} \approx 1.6$ (and the cyclic number by $1/p_{\text{EIP}} \approx 2.7$) for typical malaria parameters ($\mu_v \approx 0.1$/day, $\tau_v^E \approx 10$ days, so $p_{\text{EIP}} \approx 0.37$). The diagnostic question: *"Does your b_h^{eff} include the probability that an infected mosquito survives the extrinsic incubation period? If not, your $\mathcal{R}_0$ is overstated by approximately a factor of $1/\sqrt{p_{\text{EIP}}}$."* Verify whether the AI's response presents the explicit E_v-compartment model, the b_h^{eff} approximation, or omits both.

    (c) *Bipartite-vs-direct audit.* Does the AI's $\mathcal{R}_0$ formula take the square-root form characteristic of bipartite transmission (i.e., $\mathcal{R}_0 \propto \sqrt{a^2 b_h^{\text{eff}} b_v m/(\mu_v(\gamma_R+\nu_m))}$) or a non-square-root form? A non-square-root $\mathcal{R}_0$ for a vector-borne disease indicates that the AI has mistakenly applied a direct-transmission framework where the bipartite cycle is required. The diagnostic question: *"In your $\mathcal{R}_0$ formula, are the host-side and vector-side parameters under a square root or in a single product? Justify by referring to the bipartite next-generation matrix."* The expected square-root structure follows from the spectral radius of a 2×2 NGM with zeros on the diagonal, derived in Section 13.4.

The exercise practices the audit habit on the three highest-impact failure modes in AI-generated vector-host derivations: dimensional confusion in a, omission of EIP survival, and bipartite-structure misapplication. Each can produce $\mathcal{R}_0$ estimates wrong by factors ranging from 2 to several hundred. Disclosure: per Appendix D, Section D.4, document the AI system, prompts, and audit findings.

12. [**Open Project**] *Full vector-host implementation and fitting project.* A multi-week capstone project implementing a vector-host model for a chosen vector-borne pathogen (suggested options include dengue, malaria, Zika, and West Nile), fitting the model to outbreak data, and producing intervention recommendations weighted by the vector-host parameter sensitivities. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈20–30 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch14-vector-host-fitting/`.

Chapter 14

Vertical Transmission

Before solving a model, ask which assumptions might not be true.

The transmission modes considered so far have been horizontal: pathogens spread between contemporaries through respiratory contact, sexual contact, or vector-mediated routes. Several important pathogens also transmit vertically, from mother to child either in utero, during birth, or through breastfeeding. Vertical transmission introduces new structural features: the rate of new infections depends on the birth rate rather than on contact rates between current population members, and the epidemiological burden may extend across generations through congenital effects that persist long after the original infection.

This chapter develops the vertical-transmission framework in five steps. Section 14.1 characterizes the diseases for which vertical transmission matters and describes the biological mechanisms. Section 14.2 writes out the modified compartmental model with explicit birth-rate-driven new infections. Section 14.3 derives the modified reproductive number that incorporates vertical transmission alongside horizontal. Section 14.4 surveys the four most important vertically transmitted pathogens (HIV, HBV, Zika, rubella). Section 14.5 discusses interventions targeting the vertical pathway. Section 14.6 summarizes and provides exercises.

The chapter is brief by design. Vertical transmission does not, for most pathogens, change the threshold structure of the model: $\mathcal{R}_0$ from horizontal transmission alone usually dominates, and vertical transmission adds a small correction. Where vertical transmission dominates the dynamics (as for some congenital infections), the structure is closer to a demographic model than to an epidemic model and requires its own specialized treatment outside our scope.

Advanced material note. The expression for $\mathcal{R}_0$ in the presence of both horizontal and vertical transmission, derived in Section 14.3, follows a per-individual integration argument that the main-path reader can take on faith from the resulting formula (14.2). The detailed derivation is presented for readers who want to verify the construction or extend it to settings with sex-asymmetric vertical transmission (Exercise 3).

14.1 When Vertical Transmission Matters

A pathogen transmits vertically when an infected mother passes the pathogen to her offspring. The mechanisms vary by pathogen:

- *Transplacental* (in utero): the pathogen crosses the placenta and infects the fetus during pregnancy. Examples: rubella, cytomegalovirus, Zika virus, syphilis.

- *Perinatal* (during birth): the pathogen transfers during passage through the birth canal or through contact with maternal blood and fluids. Examples: HIV, hepatitis B, herpes simplex virus.

- *Postnatal via breastfeeding*: the pathogen transfers in breast milk. Examples: HIV, hepatitis B, HTLV-1.

The probability that an infected mother transmits to her child varies widely. For HIV without intervention, the per-pregnancy transmission probability is approximately 25–35% (combining in-utero, intrapartum, and breastfeeding routes). For hepatitis B, 70–90% from mothers with high viral load. For Zika, about 25–47% by trimester of exposure [Ades et al., 2021], with congenital Zika syndrome in a smaller fraction. For rubella, approximately 80% if maternal infection occurs in the first twelve weeks; that figure is the probability of congenital rubella syndrome given infection, not a per-birth transmission probability (see the note following Table 14.1).

Why model vertical transmission separately. Three factors distinguish vertical from horizontal transmission and warrant separate modeling treatment:

1. *Rate driven by demographics, not contact.* Horizontal transmission rate is $\alpha i = c_I \beta P_S \cdot i$, depending on contact rates and the current susceptible population. Vertical transmission rate is approximately $\nu \cdot p_{\text{MTCT}} \cdot i_F$, where ν is the per-capita birth rate (births per woman per year), p_{MTCT} is the mother-to-child transmission probability, and i_F is the number of infected women of reproductive age. The rate depends on demographic parameters and the size of the infected female population, not on contact patterns.

2. *Generational persistence.* Vertical transmission produces infections in newborns who carry the pathogen forward for years (or for life, in the case of HIV without antiretroviral therapy). Even if horizontal transmission is brought below threshold, vertical transmission can sustain the pathogen across generations.

3. *Congenital outcomes.* For some pathogens (rubella, Zika, syphilis), the fetal infection produces severe lifelong outcomes that constitute the dominant disease burden. The model's purpose shifts from predicting case counts to predicting birth defects, requiring careful treatment of when in pregnancy infection occurs.

Pathogens where vertical transmission is structurally central. For most pathogens we have considered (influenza, COVID-19, gonorrhea, malaria), vertical transmission is negligible or rare and can be ignored without loss. For HIV, hepatitis B, and rubella, vertical transmission is a substantial component of the epidemiological picture and the model must include it. The remainder of this chapter focuses on these latter cases. The general theory of models with vertical transmission, including existence of endemic equilibria and threshold conditions when both horizontal and vertical routes operate, was developed by Busenberg and Castillo-Chavez [1991] and Cha et al. [1998]; Martcheva and Thieme [2003] extended the framework to include disease-stage progression within vertically infected cohorts.

Exercises for this section

1. [C] Explain in plain English why vertical transmission "rarely changes the qualitative threshold structure" of a disease's invasion potential. Under what numerical regime does this conclusion fail?

14.2 The Vertical-Transmission Model

We extend the basic SIR model to include vertical transmission. The simplest formulation treats men and women symmetrically and assumes a single per-pregnancy transmission probability; refinements for sex-asymmetric models, age structure, and timing of infection within pregnancy are noted at the end of the section.

14.2.1 The Modified Equations

Let ν_b denote the per-capita birth rate (with units year^{-1}, normalized to give a stable population at carrying capacity when integrated over the female reproductive lifespan). Births enter the susceptible compartment unless the mother is infected and vertical transmission occurs.

Caution: unit conventions for vertical-transmission rates. Vertical-transmission and demographic rates in the published literature are reported in different time units than horizontal-transmission rates. Birth rates ν_b are typically reported per year (or per live birth, when expressed cumulatively over a reproductive lifetime); vertical-transmission probabilities p_{MTCT} are dimensionless. By contrast, recovery and mortality rates (γ_R, ν_m) and contact rates (c_I, c_S) are typically reported per day for acute infections. Before computing $\mathcal{R}_0$ or simulating the system, all rates must be converted to a single common time unit. The convention used in this book is per-day: a published $\nu_b = 0.012$ per year (an aggregate U.S. crude birth rate) becomes $\nu_b = 0.012/365 \approx 3.3 \times 10^{-5}$ per day before substitution into the model. Mixing units is the most common arithmetic error in vertical-transmission models and produces $\mathcal{R}_0$ estimates wrong by a factor on the order of 365.

The fraction of births to infected mothers is approximately i/n (assuming birth rate is independent of infection status, a simplification we discuss below). Of these births, a fraction p_{MTCT} become infected at birth and enter the i compartment directly. The remaining fraction enter s. Figure 14.1 shows the branching graphically.

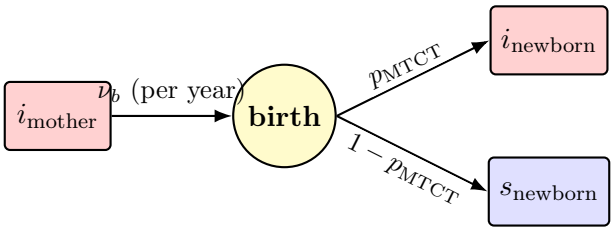


Figure 14.1: Vertical transmission as a branching event at birth. Each newborn of an infectious mother becomes infected with probability p_{MTCT} and enters the i compartment directly, or remains uninfected (probability $1 - p_{\mathrm{MTCT}}$) and enters s. The transmission probability is pathogen-specific and is dramatically reduced by available interventions: antiretroviral therapy for HIV reduces p_{MTCT} from ∼30% to under 2%; neonatal hepatitis-B vaccination reduces it from ∼80% to under 10%.

The modified SIR equations are

$$\dot{s} = \nu_b n \left(1 - p_{\mathrm{MTCT}} \frac{i}{n}\right) - \alpha i - \nu_m s, \tag{14.1a}$$

$$\dot{i} = \nu_b n \cdot p_{\mathrm{MTCT}} \frac{i}{n} + \alpha i - \gamma_R i - \nu_m i, \tag{14.1b}$$

$$\dot{r} = \gamma_R i - \nu_m r. \tag{14.1c}$$

Here we have replaced the previous notation ν_m for the inflow to s with the more explicit $\nu_b n \cdot (1 - p_{\mathrm{MTCT}} i/n)$, separating the demographic birth rate from the disease-induced reduction in susceptible births (Pitfall: "Short-Term Simulation Assumptions"). At steady population ($\nu_b = \nu_m$), this reduces to the previous formulation when vertical transmission is absent ($p_{\mathrm{MTCT}} = 0$).

The vertical-transmission term in $\dot{i}$, $\nu_b n \cdot p_{\mathrm{MTCT}} i/n = \nu_b p_{\mathrm{MTCT}} i$, has the same structure as the horizontal force term αi but with the birth rate ν_b replacing the contact rate c_I and the per-pregnancy transmission probability p_{MTCT} replacing the per-contact transmission β.

Interpreting ν_b as an effective rate. The simplified scalar formulation above treats all infectious individuals as birth-capable. In a sex- and age-structured population, only infected females of reproductive age contribute to vertical transmission, so ν_b in the scalar model should be read as an *effective birth rate per infectious individual*, averaged over sex and reproductive status. Operationally, if ν_b^F is the per-female birth rate and f_{repro} is the fraction of infectious individuals who are females of reproductive age, then $\nu_b^{\mathrm{eff}} = \nu_b^F \cdot f_{\mathrm{repro}}$ is the quantity to substitute into the scalar equation. For pathogens where the infection prevalence is approximately uniform across demographic groups, f_{repro} is the population fraction of females of reproductive age (typically 0.20–0.25); for pathogens with strongly age- or sex-asymmetric prevalence (HIV in some settings, hepatitis B in others), the effective rate must be computed from the infection-stratified demographics. The sex-stratified model in Section 14.2.2's "Sex-asymmetric models" refinement makes the demographic averaging explicit; the simplified scalar form is adequate when the demographic averaging factors are approximately constant on the timescale of the model.

Combining horizontal and vertical forces. The total force from infection is

$$\alpha_{\mathrm{total}} \;=\; \alpha + \nu_b p_{\mathrm{MTCT}} \;=\; c_I \beta P_S + \nu_b p_{\mathrm{MTCT}}.$$

The two terms have different functional dependence: α depends on P_S (which is near unity early but decreases as the susceptible pool depletes), while the vertical term is constant in P_S. For long-term endemic dynamics, both terms contribute; for early outbreak dynamics where $P_S \approx 1$, both terms are at their maximum and add directly. Figure 14.2 shows the two parallel pathways feeding the i compartment.

14.2.2 Refinements

Sex-asymmetric models. The simple formulation above treats men and women symmetrically. In reality, vertical transmission depends on the female infectious population only. A sex-stratified model (in the style of Chapter 12's bipartite framework) is more accurate:

$$\dot{i}_{\mathrm{newborn}} \;=\; \nu_b^F \cdot n_F \cdot p_{\mathrm{MTCT}} \cdot \frac{i_F}{n_F},$$

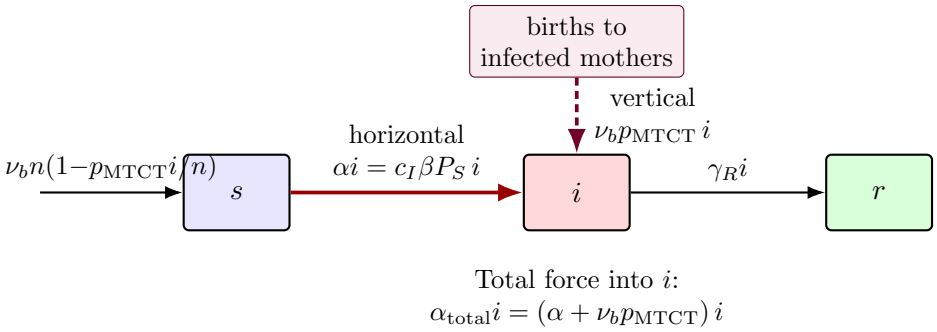


Figure 14.2: Two parallel pathways feed the i compartment in a model with both horizontal and vertical transmission. Horizontal transmission (solid red arrow, rate $\alpha i = c_I \beta P_S i$) depends on the susceptible fraction P_S and decreases as the epidemic depletes the susceptible pool. Vertical transmission (dashed purple arrow, rate $\nu_b p_{\mathrm{MTCT}} i$) is independent of P_S and contributes a constant per-infectious-individual rate set by the birth rate ν_b and the per-pregnancy transmission probability p_{MTCT}. The two pathways add in the total force into i, leading to an endemic prevalence that can be sustained even when horizontal transmission alone would die out. The vertical arrow carries far less traffic: at the parameters of Section 14.3, births arrive some 73,000 times less often than contacts. Each birth transmits with a much higher probability, so neither rate alone predicts which pathway matters.

where ν_b^F is the per-female birth rate and i_F is the female infectious population. For pathogens where female and male infection prevalences differ (HIV in some settings has higher female prevalence in young adults), this distinction matters.

Trimester-of-infection effects. For congenital rubella and Zika, the timing of maternal infection within pregnancy strongly affects the probability of severe fetal outcomes. Models for these pathogens typically include the trimester of infection as an additional state variable, increasing model complexity substantially. We do not develop these models here.

Antiretroviral and antiviral therapy. Effective interventions reduce p_{MTCT} substantially. For HIV with antiretroviral therapy during pregnancy and breastfeeding, p_{MTCT} falls from 25–35% to under 2%. For hepatitis B with neonatal vaccination at birth, p_{MTCT} falls from 70–90% to under 10%. These large reductions are the single most-effective intervention for vertically transmitted pathogens.

14.2.3 Uncertainty in the Vertical Transmission Probability

Every number in $\nu_b\, p_{\mathrm{MTCT}}$ carries uncertainty, and the three sources behind p_{MTCT} do not combine the way a single reported percentage suggests (Pitfall: "Estimator Honesty"). Chapters 7 through 9 supply the machinery; what follows is which piece of it each source needs.

Parameter uncertainty. A published p_{MTCT} is an estimate from a cohort, and vertical-transmission cohorts are small: a few hundred pregnancies is a large study. The HIV range quoted above, 25–35%, is not a modeling choice between two values but the width of what the data support. Carry it as an interval and propagate it, rather than fitting the midpoint and reporting a single vertical contribution. The delta method of Section 9.3.5 does this in one line, since the

vertical contribution to $\mathcal{R}_0$ is a product: its relative standard error is at least the relative standard error of p_{MTCT} itself.

Trimester variability. Where transmission depends on timing, p_{MTCT} is not a constant but a mixture,

$$p_{\mathrm{MTCT}} = \sum_{k=1}^{3} w_k \, p_k,$$

with p_k the probability of the outcome of interest when maternal infection falls in trimester k, and w_k the share of maternal infections that fall there. Two cautions come with this expression, and the second is the one modelers get wrong.

Say which outcome p_k measures. Congenital infection and congenital disease are different events with different gestational profiles, and the literature quotes both. For rubella, Miller et al. [1982] followed more than a thousand confirmed maternal infections and found congenital *infection* in over 80% of cases in the first twelve weeks, 54% at 13–14 weeks and 25% at the end of the second trimester, after which the infection rate *rises again* toward the end of pregnancy; congenital *defects*, by contrast, appeared in every infant infected before the eleventh week and in about a third of those infected at 13–16 weeks, and are rare after week 18. The transmission probability is therefore U-shaped in gestational age while the disease probability is not, and a model that assumes a monotone decline gets the third trimester wrong. The $\sim 80\%$ quoted for rubella in Section 14.4 is a disease probability conditional on first-trimester infection: it is p_1, not p_{MTCT}, and substituting it for the mixture overstates the vertical disease burden by roughly $1/w_1$ (Pitfall: "Conditional versus Marginal Probabilities"). Zika shows the same split. Ades et al. [2021] estimate vertical transmission at 47%, 28% and 25% by trimester of exposure, against congenital Zika syndrome in 9%, 3% and 1% of deliveries.

The weights belong to the epidemic, not to the pathogen. For a given birth cohort, w_k is fixed by where that cohort sat in gestation when the wave passed, so it is the offset between the epidemic curve and conception that sets the mixture, and the resulting burden appears in births lagged by up to nine months. This is how the Zika epidemic was read: Johansson et al. [2016] aligned successive Brazilian birth cohorts against trimester-specific infection rates and found microcephaly risk tied to first-trimester exposure and negligibly to the later two. For infections at an instant, rather than in a birth cohort, the weights instead track the gestational composition of the pregnant population, which is close to the duration shares when the hazard does not depend on gestational stage. Either way, holding w_k fixed while the model runs an outbreak is a short-term assumption of exactly the kind this book asks to be named (Pitfall: "Short-Term Simulation Assumptions"). It is also a hard thing to estimate after the fact: Hay et al. [2018] fitted gestational-age-varying risk to incidence data from Brazil and Colombia, recovered different risk profiles in different places, and concluded that routine surveillance data are not sufficient to pin the profile down.

Surveillance under-ascertainment. The estimator $\hat{p}_v = j_{\mathrm{newborn,\,infected}}/i_{\mathrm{maternal,\,exposed}}$ has a numerator and a denominator that are missed at different rates. Congenital infection is often detected only when it is symptomatic, while maternal infection is caught by antenatal screening that may be near-universal or near-absent depending on the setting. Equal under-ascertainment cancels in the ratio, which is the population-invariance property the estimator inherits from $\hat{\alpha} = j/i$; unequal under-ascertainment does not, and it biases $\hat{p}_v$ toward the more completely observed side (Pitfall: "Differential Ascertainment in a Ratio Estimator"). Chapter 8, Section 8.5 gives the

multiplier correction, and the identifiability caution of Section 8.3 applies with force here: a single-multiplier model cannot separate a low transmission probability from good maternal case-finding.

Worse, the detection probability on the numerator can vary with the very quantity being estimated. Ades et al. [2021] put the sensitivity of markers of congenital Zika infection at 42% for first-trimester exposure against 85% and 80% for the second and third, so the least detectable infections are the most dangerous ones and a naive ratio understates exactly the cases that matter.

What to report. Which outcome p_k measures, an interval on p_{MTCT}, the trimester weights used and where they came from, and the ascertainment assumption on each side of the ratio. A vertical contribution reported without these is a point estimate of a quantity the data rarely pin down.

Exercises for this section

1. [C] The chapter argues that the vertical-transmission analogue of the infected-viewpoint estimator is $\hat{p}_v = j_{\text{newborn, infected}}/i_{\text{maternal, exposed}}$. Explain why this estimator inherits the same population-invariance property as $\hat{\alpha} = j/i$ from Chapter 7.

14.3 The Reproductive Number with Vertical Transmission

The reproductive number derivation extends straightforwardly. At the disease-free equilibrium, $s = 1, i = r = 0$. A single infectious individual generates secondary infections through two channels: horizontal (via contact at rate c_I, with each contact transmitting with probability β) and vertical (via births at rate ν_b if the individual is female and of reproductive age, with each birth transmitting with probability p_{MTCT}).

For a single infectious individual, integrating these rates over the infectious lifetime $\tau_R^* = 1/(\gamma_R + \nu_m)$ gives

$$\mathcal{R}_0 \;=\; \frac{c_I\beta + \nu_b p_{\text{MTCT}}}{\gamma_R + \nu_m}. \tag{14.2}$$

The two transmission channels add in the numerator: each contributes its own per-individual generation rate. The $\mathcal{R}_0$ extends the horizontal-only formula by a vertical-contribution term $\nu_b p_{\text{MTCT}}/(\gamma_R + \nu_m)$.

Magnitude of the vertical contribution. For a typical sexually-active adult population, $\nu_b \approx 0.02$ year^{-1} ($\approx 5.5 \times 10^{-5}$ day^{-1}). For HIV without antiretroviral therapy, $p_{\text{MTCT}} \approx 0.30$. The vertical contribution to $\mathcal{R}_0$ is

$$\frac{\nu_b p_{\text{MTCT}}}{\gamma_R + \nu_m} \;\approx\; \frac{5.5 \times 10^{-5} \times 0.30}{0.0001 + 0.0001} \;=\; \frac{1.65 \times 10^{-5}}{2 \times 10^{-4}} \;\approx\; 0.08.$$

For an HIV epidemic with horizontal $\mathcal{R}_0 \approx 2$ (in a high-risk subpopulation), the vertical contribution adds approximately 4% to $\mathcal{R}_0$. This is small but not negligible: vertical transmission alone is far below threshold ($\mathcal{R}_0^{\text{vert}} = 0.08 < 1$), but combined with horizontal, it pushes the total slightly higher.

Why a rare event still moves $\mathcal{R}_0$

Compare the two channels at the same units, per day. An infectious adult makes contacts at $c_I \approx 4\,\text{day}^{-1}$ and gives birth at $\nu_b \approx 5.5 \times 10^{-5}\,\text{day}^{-1}$. Transmission opportunities arrive through the vertical channel about 73,000 times less often.

Each opportunity is worth far more. With horizontal $\mathcal{R}_0 \approx 2$ and $\gamma_R + \nu_m = 2 \times 10^{-4}\,\text{day}^{-1}$, the per-contact probability is $\beta = 10^{-4}$, against $p_{\text{MTCT}} \approx 0.30$ for a birth: a factor of 3,000 the other way.

The two factors nearly cancel. What survives is their ratio, 73,000/3,000 ≈ 24, so the vertical channel contributes about 1/24 of the horizontal one — the 4% computed above. A channel that operates seventy thousand times more rarely than contact is not thereby negligible, and a channel that transmits three thousand times more reliably is not thereby dominant. Only the product of frequency and per-event probability decides, which is why both must be carried in the same units to the end of the calculation.

For hepatitis B, the calculation differs because the chronic-carriage state provides a lifetime of infectiousness ($\tau_R^* \to \tau_m = 1/\nu_m$ for chronic carriers, as $\gamma_R \to 0$). The vertical contribution can dominate in populations where chronic-carriage prevalence in women of reproductive age is substantial.

Transplacental routes. The formula counts births to mothers who are infectious at delivery, which fits perinatal and breastfeeding transmission. For a transplacental pathogen the opportunity is a pregnancy at the time of infection: a woman infected early in pregnancy can deliver an infected child months after her own infection has cleared. The vertical contribution is then about $\nu_b \tau_{\text{gest}} \bar{p}_{\text{MTCT}}$, with $\tau_{\text{gest}} \approx 0.75$ year and $\bar{p}_{\text{MTCT}}$ the transmission probability averaged over the trimester of infection, and does not depend on τ_R^*. For a short infection such as Zika's it is some forty times the scalar form, and still small next to horizontal transmission.

Threshold implication. If horizontal $\mathcal{R}_0$ alone is well above 1, vertical transmission rarely changes the threshold conclusion: the epidemic is sustained by horizontal transmission, and vertical adds an incremental contribution. If horizontal $\mathcal{R}_0$ is just below 1 (as during a successful intervention campaign), vertical transmission can be enough to keep the total above 1, sustaining the pathogen across generations even as direct transmission is suppressed. This phenomenon explains why elimination of vertically-transmitted pathogens often requires concurrent intervention on both transmission modes.

Exercises for this section

1. **[D]** Verify equation (14.2) by an individual-tracking argument: a single infectious individual generates secondary infections through horizontal contacts at rate $c_I \beta$ and through births at rate $\nu_b p_{\text{MTCT}}$ (assuming the individual is female and of reproductive age); integrate over the infectious lifetime $1/(\gamma_R + \nu_m)$.

14.4 Surveys of Specific Pathogens

We briefly summarize the four most epidemiologically important vertically transmitted pathogens, highlighting the distinctive modeling features of each.

Table 14.1: Vertical-transmission characteristics of the four pathogens surveyed in this section. All values are those stated in the subsections below; ranges reflect the spread reported across settings rather than parameter uncertainty in any single study. The final column names the feature that most constrains how each pathogen must be modeled.

	HIV	**Hepatitis B**	**Zika**	**Rubella**
Horizontal route	sexual, parenteral	sexual, parenteral, household	vector-borne, sexual	respiratory droplets
Vertical route	transplacental, perinatal, breastfeeding	perinatal	transplacental	transplacental (1st trimester)
p_{MTCT}, untreated	25–35%	$\sim 80\%$ (HBeAg+) $\sim 10\%$ (HBeAg−)	25–47% (by trimester)	see note
p_{MTCT}, treated	$< 2\%$ (ART)	$< 10\%$ (HBIG + vaccine)	—	vaccination (elimination)
Vertical share of $\mathcal{R}_0$	$\sim 5\%$	substantial	—	small
Modeling constraint	lifelong-care burden	chronic carriage; HBeAg stratification	trimester timing; vector coupling	post-elimination importation

Note on rubella. The figure most often quoted for rubella, $\sim 80\%$, is the probability of congenital rubella syndrome *given* maternal infection in the first twelve weeks of pregnancy. It is not a per-birth p_{MTCT} and must not be substituted into $\nu_b p_{\mathrm{MTCT}}$ without first conditioning on the timing of maternal infection (Section 14.2.3). Rubella combines a high conditional transmission probability with a small $\nu_b p_{\mathrm{MTCT}}$ relative to its high horizontal $\mathcal{R}_0$, which is why its vertical contribution is small despite the alarming conditional figure.

14.4.1 HIV

HIV transmits both horizontally (sexual, parenteral) and vertically (transplacental, perinatal, breastfeeding). Without intervention, $p_{\mathrm{MTCT}} \approx 25$–$35\%$, with the largest fraction occurring during late pregnancy and birth. With antiretroviral therapy during pregnancy and labor, $p_{\mathrm{MTCT}} < 2\%$.

The vertical contribution to $\mathcal{R}_0$ is small relative to horizontal ($\sim 5\%$ in typical populations), but the vertically-acquired infections produce a generation of children with HIV who require lifelong care. Modern global health goals for HIV elimination prioritize prevention of mother-to-child transmission as a high-impact, achievable target distinct from the longer-term goal of horizontal-transmission elimination. Martcheva and Li [2013] develop a multiscale framework that links the within-host dynamics governing maternal viral load to the population-level vertical transmission probability, providing a mechanistic basis for how antiretroviral treatment reduces p_{MTCT} through its effect on maternal viremia.

14.4.2 Hepatitis B: A Worked Fitting Example

Hepatitis B virus (HBV) transmits horizontally (sexual, parenteral, household) and vertically (perinatal). We use HBV as a worked example of fitting vertical-transmission parameters from observable data. HBV in a high-prevalence setting is also the book's clearest case of a fit that must begin at an endemic state rather than at a near-disease-free seed (Pitfall: "Natural Initial Conditions"): the

surveillance record describes a population in which the pathogen has been endemic for generations, and a model started near the disease-free equilibrium fits a transient that never happened.

The biology and the model. The vertical transmission probability for HBV is strongly stratified by maternal hepatitis-B-e-antigen (HBeAg) status. HBeAg-positive mothers transmit the virus to their newborns with probability approximately $p_v \approx 0.9$ in the absence of intervention; HBeAg-negative mothers transmit with probability approximately $p_v \approx 0.1$. This stratification matters clinically because the consequence, chronic HBV infection in the newborn, carries lifetime risk of cirrhosis and hepatocellular carcinoma. Before universal neonatal vaccination, vertical transmission dominated the global HBV burden in endemic regions.

Fitting the vertical transmission probability. The vertical transmission probability p_v is naturally estimated as a ratio of observed quantities:

$$\hat{p}_v \;=\; \frac{j_{\text{newborn, infected}}}{i_{\text{maternal, exposed}}}, \tag{14.3}$$

where the numerator is the rate of newborns testing positive for HBV surface antigen at 6 months and the denominator is the count of newborns whose mothers were HBV-positive at delivery. This is a perinatal analogue of the infected-viewpoint estimator $\hat{\alpha} = j/i$ of Chapter 7: both are ratios of directly observable quantities, and neither requires committing to a value for a susceptible-population size. The sensitivity-analysis paper of Hyman [2026d] establishes that this structural-immunity property extends to perinatal transmission: the fitted $\hat{p}_v$ does not scale with the assumed size of the susceptible-newborn cohort.

HBeAg stratification and pooled estimates. A naive pooled estimate of p_v mixes HBeAg-positive and HBeAg-negative mothers and gives a weighted average reflecting the HBeAg prevalence in the mother population. In a population with 30% HBeAg-positive mothers, the pooled $\hat{p}_v \approx 0.3(0.9) + 0.7(0.1) = 0.34$; in a population with 10% HBeAg-positive mothers, the pooled value is 0.18. Both pooled estimates are correct for their respective populations but mislead when applied across populations with different HBeAg prevalence. Stratified fitting, with separate estimators for each HBeAg stratum, gives population-invariant parameter values.

Policy implication: the HBIG + vaccine protocol. Hepatitis B immune globulin (HBIG) administered within 12 hours of birth, combined with the first dose of HBV vaccine, reduces p_v from 0.9 (HBeAg-positive mothers, no intervention) to approximately 0.03 (full protocol, tested in large cohort studies). The fitted-estimator framework quantifies this intervention's effect directly:

$$\hat{p}_v^{\text{no intervention}} \approx 0.9, \qquad \hat{p}_v^{\text{HBIG + vaccine}} \approx 0.03.$$

The reduction factor is 30. Combined with universal neonatal HBV vaccination at birth, the protocol has brought perinatal HBV transmission under control in settings with reliable medical infrastructure. The model framework here predicts the eradication trajectory: with high vaccination coverage, the vertical contribution to $\mathcal{R}_0$ falls to near zero, and the residual burden becomes determined by horizontal transmission alone.

The Hepatitis B case as a framework demonstration. The HBV example illustrates the framework's practical reach beyond standard respiratory and vector-borne models. The key observation is that the estimator structure $\hat{p}_v = j_{\text{newborn}}/i_{\text{maternal}}$ is the same as $\hat{\alpha} = j/i$ in Chapter 7: both are ratios of directly observable quantities chosen so that the modeler's uncertainty about the denominator population is immaterial to the estimate. This structural invariance is the book's central practical claim, here instantiated for perinatal transmission.

14.4.3 Zika

Zika virus transmits horizontally (vector-borne, sexual) and vertically (transplacental). The vertical transmission probability is about 25–47% by trimester of exposure [Ades et al., 2021], with congenital Zika syndrome in a smaller fraction. That consequence (microcephaly and other neurodevelopmental outcomes) is severe, and the timing of maternal infection within pregnancy is critical: first-trimester infection has the worst outcomes.

The trimester dependence makes the mixture of Section 14.2.3 unavoidable here: a single p_{MTCT} for Zika is a weighted average whose weights shift with the outbreak.

The Zika modeling framework necessarily incorporates both vector dynamics (Chapter 13) and vertical transmission. The combined model is one of the more complex applications of the framework developed in this book, and recent modeling efforts have addressed it specifically.

14.4.4 Rubella

Rubella transmits horizontally (respiratory droplets) and vertically (transplacental, predominantly first trimester). Maternal rubella infection in early pregnancy produces congenital rubella syndrome with high probability ($\sim 80\%$ if infection occurs in the first 12 weeks); that figure is the first-trimester conditional probability, not a per-birth p_{MTCT} (Section 14.2.3).

Rubella vaccination has eliminated the disease in many countries. The model's purpose for rubella is twofold: predicting the post-elimination probability of importation-driven outbreaks and assessing vaccination policy in regions where coverage is incomplete. The vertical contribution to $\mathcal{R}_0$ is small (rubella has high p_{MTCT} but low $\nu_b p_{\text{MTCT}}$ relative to its high horizontal $\mathcal{R}_0 \approx 6$), so eliminating rubella is achieved primarily through horizontal-transmission control via vaccination.

Exercises for this section

1. **[D]** *HIV vertical contribution under ART rollout.* For HIV in a high-prevalence subpopulation with horizontal $\mathcal{R}_0^{\text{horiz}} = 1.6$, $\nu_b = 0.025$ year^{-1}, baseline $p_{\text{MTCT}} = 0.30$ (without antiretroviral therapy), and infectious duration $\tau_R^* = 10$ years, compute the vertical-transmission contribution $\mathcal{R}_0^{\text{vert}} = \nu_b p_{\text{MTCT}} \tau_R^*$ at baseline. Then track $\mathcal{R}_0^{\text{vert}}$ through three ART-coverage scenarios: (a) 25% population coverage with the residual untreated p_{MTCT} at 0.30 and the treated value at 0.02 (effective $\bar{p}_{\text{MTCT}} = 0.75 \cdot 0.30 + 0.25 \cdot 0.02$); (b) 60% coverage; (c) 90% coverage. For each scenario, compute the total $\mathcal{R}_0 = \mathcal{R}_0^{\text{horiz}} + \mathcal{R}_0^{\text{vert}}$ assuming ART simultaneously reduces $\mathcal{R}_0^{\text{horiz}}$ by 90% of the coverage fraction. At what ART coverage does $\mathcal{R}_0$ cross 1?

2. **[D]** *Zika vertical contribution with trimester stratification.* Zika exhibits a strong trimester-of-infection effect: p_{MTCT} for fetal infection is approximately 0.47 in the first trimester, 0.28 in the second, and 0.25 in the third [Ades et al., 2021]. Construct the trimester-averaged p_{MTCT} assuming uniform infection probability across trimesters (each weighted by its 13-week

duration). Using the resulting $\bar{p}_{\mathrm{MTCT}}$ together with $\nu_b = 0.012$ year^{-1} and a horizontal-transmission infectious period $\tau_R^* = 7$ days $= 0.019$ year (standard Zika duration), compute the vertical contribution to $\mathcal{R}_0$. Compare with the horizontal contribution at $\mathcal{R}_0^{\mathrm{horiz}} = 2$ for an outbreak in a vector-suitable area. Why is the vertical contribution effectively negligible at the population level for Zika, despite the clinically severe outcomes the trimester-1 infection produces? (Hint: compare $\nu_b \tau_R^*$, the births during a 7-day infection, with $\nu_b \tau_{\mathrm{gest}}$, the chance of being pregnant when infected, and say which one counts transplacental transmission. The contribution is small for Zika because few of the infected are pregnant.)

14.5 Interventions Targeting Vertical Transmission

Vertical-transmission interventions act on p_{MTCT} rather than on contact rates or transmission factors. Three intervention categories dominate.

Antiretroviral therapy during pregnancy. For HIV, antiretroviral therapy during the third trimester and through breastfeeding reduces maternal viral load and dramatically reduces p_{MTCT}, from baseline 25–35% to under 2% with optimal regimens. The effect is multiplicative on the vertical contribution to $\mathcal{R}_0$:

$$\mathcal{R}_0^{\mathrm{vert,ART}} \;=\; \mathcal{R}_0^{\mathrm{vert}} \cdot \frac{p_{\mathrm{MTCT}}^{\mathrm{ART}}}{p_{\mathrm{MTCT}}^{\mathrm{baseline}}} \;\approx\; \mathcal{R}_0^{\mathrm{vert}} \cdot \frac{0.02}{0.30} \;\approx\; 0.07 \mathcal{R}_0^{\mathrm{vert}}.$$

Both probabilities in that ratio are cohort estimates, so the reduction factor has an interval of its own; Section 14.2.3 says how to carry it. For populations where ART coverage is high, the vertical contribution is reduced by more than 90%.

Neonatal vaccination. For hepatitis B, vaccination at birth (with optional hepatitis B immunoglobulin for high-risk newborns) prevents establishment of chronic infection in the newborn, reducing $p_{\mathrm{MTCT}}^{\mathrm{effective}}$ from 70–90% to under 10%. This intervention has been broadly implemented in much of the world and has substantially reduced the global hepatitis B burden among children.

Maternal screening and prophylactic measures. For Zika and rubella, no specific perinatal intervention exists; prevention focuses on maternal vaccination (rubella, before pregnancy) or maternal exposure avoidance (Zika, through vector control during pregnancy). These interventions act on $\mathcal{R}_0$ through the maternal infection probability rather than on p_{MTCT}.

Combined interventions. For pathogens with both substantial horizontal and vertical components, combined interventions targeting both modes are required for elimination. The relative effort allocation depends on the relative magnitudes of the two contributions to $\mathcal{R}_0$. For HIV in modern settings with antiretroviral access, vertical contribution is small and effort is concentrated on horizontal-transmission interventions. For hepatitis B in pre-vaccination settings, vertical was dominant and effort was concentrated on neonatal vaccination.

Exercises for this section

1. **[I]** Suppose horizontal $\mathcal{R}_0$ is brought to exactly 0.95 by behavioral interventions, just below the elimination threshold. For what value of vertical $\mathcal{R}_0$ contribution does the total $\mathcal{R}_0$ remain below 1? At what point would elimination be lost?

14.6 Summary and Exercises

This chapter has developed the vertical-transmission framework as an extension of the SIR model. The main results:

1. Vertical transmission introduces a new term in the infection-equation flow: $\nu_b p_{\mathrm{MTCT}} i$, with rate driven by demographic birth rate ν_b and per-pregnancy transmission probability p_{MTCT}.

2. The total reproductive number (14.2) adds a vertical contribution $\nu_b p_{\mathrm{MTCT}}/(\gamma_R + \nu_m)$ to the horizontal $\mathcal{R}_0$. For most pathogens, the vertical contribution is small (a few percent); for hepatitis B and certain settings of HIV, it can be substantial.

3. Vertical transmission rarely changes the qualitative threshold structure: if horizontal $\mathcal{R}_0$ is well above 1, the disease invades regardless of the vertical contribution; if horizontal $\mathcal{R}_0$ is well below 1, the vertical contribution cannot push the total above 1 in most realistic settings.

4. Vertical transmission interventions (ART for HIV, neonatal vaccination for HBV) act on p_{MTCT} and can reduce the vertical contribution by 90% or more.

5. Four pathogens (HIV, HBV, Zika, rubella) illustrate the range of vertical-transmission roles: horizontally-dominated with vertical as a secondary concern (HIV, rubella), formerly vertical-dominated and now controlled through neonatal vaccination (HBV), and a vector-borne pathogen with vertical as the principal route to congenital outcomes (Zika).

Chapter 15 extends the framework further to age-structured models, where the population is stratified by continuous age and the dynamics are described by partial differential equations. Age structure becomes essential when the vertical-transmission probability or the disease-acquisition probability varies strongly with age.

Which extension does your problem need? The scalar model of this chapter treats the infectious population as one pool with one effective birth rate. Four features break that assumption, and each has its own chapter. Take them in the order the question arises.

1. *Does the infection itself have more stages than infectious and recovered?* A latent period, a quarantine or hospitalization compartment, waning immunity, asymptomatic infection or disease-induced mortality all add compartments without adding groups. Chapter 10. For vertical transmission this matters most in chronic-carriage pathogens such as hepatitis B, where the infectious lifetime is what makes the vertical contribution large.

2. *Do different groups transmit at different rates, or to each other?* Once the population splits by sex, by risk or by age band, the contacts produced by one group toward another must equal the contacts received, and that constraint does real work. Chapter 11. This is the chapter behind the effective birth rate $\nu_b^{\mathrm{eff}} = \nu_b^F f_{\mathrm{repro}}$ of Section 14.2: the scalar ν_b is a two-group model already, averaged.

3. *Is the transmission route itself structured?* Sexual transmission, where partnerships rather than contacts carry the pathogen, is Chapter 12; vector-borne transmission, where a second host interposes, is Chapter 13. Both matter here because HIV and Zika reach the mother by those routes before reaching the child by this one.

4. *Does the risk vary continuously with age rather than by group?* Then the sums become integrals and the model becomes a partial differential equation. Chapter 15, and the criterion is the one stated just above.

A model needs the extension whose assumption it violates, not every extension available. The four pathogens of Section 14.4 each fail a different one.

Framework use in vertical-transmission fitting. Vertical transmission parameters enter the reproductive number through the vertical-transmission probability p_v. This parameter is typically estimated from perinatal surveillance data using the ratio estimator

$$\hat{p}_v \;=\; \frac{j_{\text{newborn, infected}}}{i_{\text{maternal, exposed}}}$$

developed in Section 14.4.2 for the hepatitis-B example. This is the perinatal analogue of the infected-viewpoint estimator $\hat{\alpha} = j/i$ of Chapter 7: both are ratios of directly observable quantities, and the fitting structural-immunity result of the sensitivity-analysis paper of Hyman [2026d] applies in both cases. The perinatal estimator does not require committing to a susceptible-newborn-cohort size, which is typically the most uncertain quantity in vertical-transmission studies. When HBeAg status or other clinical stratifiers are available, stratified fitting gives population-invariant parameter values as demonstrated in Section 14.4.2. The framework's structural claim (that ratios of observable quantities are structurally immune to uncertainty in poorly-constrained denominators) extends to vertical transmission with no modification.

Exercises

Six section-level exercises appear above, following Section 14.1, Section 14.2, Section 14.3, Section 14.4, and Section 14.5. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[D]** For HIV with horizontal $\mathcal{R}_0 = 2.0$ in a high-risk subpopulation, $\nu_b = 0.02$ year^{-1}, $p_{\text{MTCT}}^{\text{baseline}} = 0.30$, and $\tau_R^* = 10$ years, compute the vertical contribution to $\mathcal{R}_0$ and the total $\mathcal{R}_0$ before and after antiretroviral therapy reduces p_{MTCT} to 0.02.

2. **[D]** For hepatitis B with chronic carriage, the infectious-lifetime is essentially the residence time $\tau_m \approx 50$ years. With $\nu_b = 0.02$ year^{-1} and $p_{\text{MTCT}} = 0.85$ (without vaccination), compute the vertical $\mathcal{R}_0$ contribution. How does it change after universal neonatal vaccination reduces effective p_{MTCT} to 0.05?

3. **[D]** For a sex-asymmetric model with separate male and female compartments, derive the expression for $\mathcal{R}_0$ analogous to (14.2), accounting for the fact that vertical transmission contributes only through the female infectious population.

4. **[D]** Model the trimester-of-infection effect for Zika: stratify the female population by trimester of pregnancy (s_F^1, s_F^2, s_F^3 for first, second, third trimester) and infection status. Write the equations and identify the additional parameters required.

5. **[D]** For rubella with horizontal $\mathcal{R}_0 = 6$, $\nu_b = 0.012$ year^{-1} and a 7-day infectious period, take a congenital-infection probability of 0.80 for maternal infection in the first trimester

and, as a simplification, zero later (Section 14.2.3 reports a late rise). Compute the vertical $\mathcal{R}_0$ contribution in the scalar form of Equation (14.2) and in the pregnancy-window form of Section 14.3. Why is either small next to 6, and why does the note after Table 14.1 warn against using 0.80 as p_{MTCT}?

6. [**I**] Compare the relative magnitudes of horizontal and vertical $\mathcal{R}_0$ contributions for the four diseases of Section 14.4 (HIV, HBV, Zika, rubella) using best-estimate parameter values. Which diseases have vertical contribution exceeding 25% of horizontal?

7. [**Open Project**] *Where the vertical contribution actually matters.* A multi-week capstone project quantifying, for one of this chapter's four pathogens, the share of burden carried vertically across plausible parameter ranges, and identifying the regime in which vertical intervention outperforms horizontal at the same coverage.

Chapter 15

Continuous-Structure Models: PDEs and Integral Equations

Nondimensionalize first. The equation often tells you what matters after you strip off the units.

The compartmental models developed throughout the book treat individuals as belonging to discrete categories: susceptible, infectious, recovered. Each individual's infectious history is collapsed into membership in one compartment. Two important biological features escape this discretization: *age structure*, where the population's behavior and disease susceptibility vary with chronological age, and *time-since-infection structure*, where an individual's infectiousness varies with how long they have been infected.

The natural mathematical framework for these features is partial differential equations: the population density becomes a function of both time and an additional continuous variable (age, time since infection), and the dynamics are described by transport-style equations. The framework is more powerful than the compartmental approach but also more demanding. This chapter develops the essential apparatus. The cost of retaining the exponential-distribution assumption when it does not match the underlying biology has been documented quantitatively for COVID-19: Hong et al. [2024] show that the history-independent exponential assumption introduces substantial bias in $\mathcal{R}_0$ estimates from early-pandemic data, and that gamma-distributed latent and infectious periods reduce it. Their result is the empirical anchor for the renewal-equation generalization developed in this chapter.

Section 15.1 motivates the continuous-structure approach. Section 15.2 develops the age-structured population model and the von Foerster equation. Section 15.3 develops the time-since-infection structure, where infectiousness varies through the course of an infection. Section 15.4 derives the Kermack-McKendrick integral equation, which is both the historical foundation of mathematical epidemiology and the tightest connection between continuous-structure and compartmental formulations. Section 15.5 derives the basic reproductive number from the renewal equation, recovering the compartmental formulas as special cases. Section 15.6 discusses numerical methods. Section 15.7 works through a brief example. Section 15.8 summarizes and provides exercises.

This chapter is more technically demanding than the rest of the book. We assume familiarity with first-order PDEs, the method of characteristics, and basic measure-theoretic notions (no formal measure theory, but the integral notation should not be a barrier). Readers seeking a less technical

entry point can read Sections 15.1 and 15.4 for the conceptual content and skip the detailed PDE derivations.

Advanced material note. This chapter uses mathematical tools (first-order PDEs, integral equations, method of characteristics) that exceed the prerequisites stated for the rest of the book. Readers may treat this chapter as optional: the central results about $\mathcal{R}_0$ and invasion thresholds carry over from the compartmental chapters, and the renewal-equation framework is presented here primarily for readers who need it or plan to consult the primary literature on age-structured epidemic models. Sections 15.4 and 15.5 carry the core material; the remaining sections provide mathematical apparatus for readers wanting deeper treatment.

15.1 Why Continuous Structure?

Age-dependent contact and disease parameters. Several diseases have epidemiologically important age structure:

- Childhood infections (measles, mumps, rubella, pertussis, varicella) have age-specific contact patterns: school-age children mix predominantly with other children, producing transmission patterns that are quite different from adult-mixing patterns.
- Severity of disease often varies with age. For seasonal influenza, mortality is concentrated in the very young and the elderly. For COVID-19, age-specific mortality varies by orders of magnitude. For dengue, severe forms are more common in those with prior exposure.
- Vaccination programs are administered at specific ages (typically childhood for live-attenuated vaccines, periodically through life for boosters). Modeling vaccination requires age structure.
- For demographic transition: in long-running endemic situations, the age structure of the population at endemic equilibrium reflects the interaction of disease and demography.

Variable infectiousness over the course of infection. The compartmental treatment assumes infectiousness is constant during the time an individual is in the i compartment. This is a substantial simplification:

- Influenza has a sharp peak of viral shedding around days 2–3 of infection, declining rapidly thereafter.
- HIV has high infectiousness during acute infection (first weeks), low infectiousness during chronic asymptomatic infection (years), and rising infectiousness during symptomatic AIDS (months).
- Tuberculosis has variable infectiousness depending on whether infection is active or latent, with reactivation possible decades after initial exposure.

Capturing these patterns requires tracking time since infection as a continuous variable, which the compartmental approach handles only crudely (through additional compartments like E for exposed or staged compartments like i_1, i_2, i_3).

The trade-off. Continuous-structure models offer more biological fidelity but demand more mathematical work. The PDE formulation generalizes the compartmental approach: every compartmental model is recoverable as a special case of the corresponding PDE model with appropriate distributional assumptions. The reverse is not true: many PDE models cannot be reduced to a finite compartmental approximation without losing essential features.

For research-level work on age-structured or time-since-infection-structured pathogens, continuous-structure models are often preferred. For pedagogical and operational work, compartmental models remain dominant because of their tractability and the directness with which their parameters connect to observable quantities. This chapter introduces the continuous-structure framework so that readers can move between the two approaches as needed.

15.2 Age-Structured Population Models

We begin with the simplest continuous-structure setting: a population stratified by chronological age, with no disease yet introduced. The development is the von Foerster equation for age-structured demography, which underlies all later age-structured epidemic models.

The population density. Let $n(t, a)$ denote the population density at time t at age a, with units of individuals per unit age. The total population at time t is

$$N(t) \;=\; \int_0^\infty n(t,a)\,da,$$

and the number of individuals in any age interval $[a_1, a_2]$ at time t is $\int_{a_1}^{a_2} n(t,a)\,da$.

The density $n(t, a)$ evolves according to two competing processes: aging (every individual ages at one year per year) and mortality (individuals die at an age-dependent rate $\mu(a)$).

15.2.1 The von Foerster Equation

Consider how $n(t, a)$ changes during a small time increment Δt. An individual at age a at time t is at age $a + \Delta t$ at time $t + \Delta t$, provided they survive. The conservation statement is

$$n(t+\Delta t, a+\Delta t) \;=\; n(t,a)\cdot\big(1-\mu(a)\Delta t + O(\Delta t^2)\big).$$

Expanding the left side in Taylor series and collecting terms,

$$n(t,a) + \Delta t\big(\partial_t n + \partial_a n\big) + O(\Delta t^2) \;=\; n(t,a) - \mu(a) n(t,a)\Delta t + O(\Delta t^2).$$

Equating the $O(\Delta t)$ terms,

$$\frac{\partial n}{\partial t} + \frac{\partial n}{\partial a} \;=\; -\mu(a)\,n(t,a). \tag{15.1}$$

This is the *von Foerster equation*. It is a first-order linear PDE with characteristic curves of the form $a - t = \text{const}$. Figure 15.1 visualizes these characteristics in the (t, a) plane: each cohort follows a diagonal line, with population thinning along the line at the local mortality rate $\mu(a)$. The equation originated in cellular-population biology [von Foerster, 1959] and was imported into mathematical demography and epidemiology by Hoppensteadt [1975]; its modern treatment in the context of structured epidemic models is developed comprehensively in the monograph of Iannelli [1995].

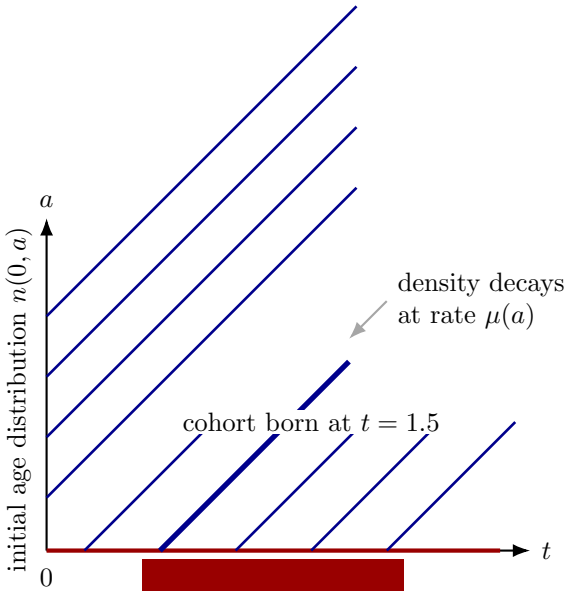


Figure 15.1: Characteristics of the von Foerster equation (15.1) in the (t, a) plane. Each cohort moves along a diagonal line of slope one, with population density thinning along the line at the age-specific mortality rate $\mu(a)$. Characteristics emanate from two boundaries: the initial age distribution along $t = 0$ (cohorts already alive at the start) and the boundary $a = 0$ along $t \geq 0$ (newly-born cohorts entering through the birth boundary condition $n(t, 0) = B(t)$). The highlighted line traces the cohort born at $t = 1.5$.

The boundary condition. The von Foerster equation requires a boundary condition at $a = 0$ specifying the rate of new births:

$$n(t, 0) \;=\; B(t), \tag{15.2}$$

where $B(t)$ is the total birth rate at time t. In a population with age-specific fertility rate $b(a)$,

$$B(t) \;=\; \int_0^\infty b(a)\, n(t, a)\, da.$$

This makes the system self-referential: the birth rate depends on the current age distribution, which itself evolves under the von Foerster equation. The combined system is the McKendrick-von Foerster system of demography.

A dimensional caution. The boundary condition (15.2) equates two quantities of different kind: $n(t, 0)$ is a *density* (individuals per unit age), whereas the birth rate $B(t)$ is a *rate* (individuals per unit time). They agree because age is itself a time and cohorts age at unit speed, so the flux of individuals across the $a = 0$ boundary is $n(t, 0)\,(da/dt) = n(t, 0)$; the equality silently uses $da/dt = 1$. This matters the moment the same form is carried to a structuring variable that is not a time. In a spatial generalization—mosquitoes measured in individuals per square meter, say—the analogue of (15.2) is not a density set equal to a rate but a *flux* condition carrying a diffusion coefficient or a velocity ($-D\,\partial_x n$ or $v\,n$), whose units the naive port silently drops. Boundary and source terms are, for this reason, the single most frequent place the count-to-density transition is mishandled (Pitfall: "Dimensional Consistency", Pitfall: "Density vs. Frequency Dependence").

15.2.2 The Stable Age Distribution

For constant fertility and mortality (independent of t), the system admits a separable solution

$$n(t,a) \;=\; e^{\sigma_{\mathrm{d}} t}\, w(a),$$

where σ_{d} is a growth rate and $w(a)$ is the stable age distribution. Substituting into (15.1) and solving the resulting first-order ODE for w (Exercise 1) gives

$$w(a) \;=\; w(0)\exp\left(-\sigma_{\mathrm{d}} a - \int_0^a \mu(a')\,da'\right) \;=\; w(0)\,\ell(a)\,e^{-\sigma_{\mathrm{d}} a}, \tag{15.3}$$

where $\ell(a) = e^{-\int_0^a \mu(a')\,da'}$ is the survival probability from birth to age a. Imposing the birth boundary condition gives the *Lotka equation*,

$$1 \;=\; \int_0^\infty b(a)\,\ell(a)\,e^{-\sigma_{\mathrm{d}} a}\,da, \tag{15.4}$$

a transcendental equation for the growth rate σ_{d} in terms of the demographic functions $b(a)$ and $\mu(a)$. For a given demographic regime, σ_{d} is the unique real solution; the population grows ($\sigma_{\mathrm{d}} > 0$), shrinks ($\sigma_{\mathrm{d}} < 0$), or remains stable ($\sigma_{\mathrm{d}} = 0$). The stable age distribution $\ell(a)e^{-\sigma_{\mathrm{d}} a}$ is the long-term shape of the age distribution and constitutes the background population structure on which disease dynamics unfold in the age-structured SIR model below.

15.2.3 Age-Structured SIR Model

Stratifying the SIR model by age gives separate densities $s(t,a)$, $i(t,a)$, $r(t,a)$, with $n(t,a) = s+i+r$ the total density; the age-structured formulation is developed at length by Martcheva [2015, Ch. 12]. The disease equations become (suppressing the (t,a) arguments):

$$\partial_t s + \partial_a s = -\alpha(t,a)\,s - \mu(a)s, \tag{15.5a}$$
$$\partial_t i + \partial_a i = \alpha(t,a)\,s - (\gamma_R + \mu(a))i, \tag{15.5b}$$
$$\partial_t r + \partial_a r = \gamma_R i - \mu(a)r, \tag{15.5c}$$

where the force of infection $\alpha(t,a)$ depends on contacts with infectious individuals across all ages:

$$\alpha(t,a) \;=\; \int_0^\infty C(a,a')\beta\,\frac{i(t,a')}{n(t,a')}\,da'. \tag{15.6}$$

Here $C(a,a')$ is the age-mixing kernel: the contact rate between a person of age a and a person of age a'. This generalizes the discrete mixing matrix of Chapter 11 to a continuous age variable.

Notation note. In this chapter, $\alpha(t,a)$ denotes a per-susceptible-of-age-a force of infection (the susceptible-viewpoint convention used in the renewal-equation literature; see Kermack and McKendrick 1927, Diekmann and Heesterbeek 2000). It is the natural object in PDE settings because the boundary condition $i(t,0) = \alpha\, s$ in the time-since-infection PDE (eq. (15.7) below) requires a per-susceptible rate to multiply s. The book's primary single-compartment convention reserves α for the per-source-infectious rate $c_I\beta P_S$ (Chapter 3, Section 3.2); the population-level new-infection rate

$$j(t,a) \;=\; \alpha(t,a)\,s(t,a)$$

is the convention-independent quantity, consistent with the $J(t)$ defined in Chapter 3 equation (3.2). The renewal-equation analysis later in this chapter works directly with $J(t)$ as the central object, sidestepping the source-vs-target distinction entirely.

The boundary condition is at $a = 0$: newborns are typically all susceptible (vertical transmission excluded from this discussion):

$$s(t,0) \;=\; B(t), \qquad i(t,0) \;=\; 0, \qquad r(t,0) \;=\; 0.$$

The system (15.5) together with the boundary conditions and an initial age distribution constitutes the age-structured SIR model.

Remark 15.1 (From counts to densities: a dimensional transition that hides formulation errors). Generalizing a compartmental ODE to a structured model silently changes what each state variable *is*: in the ODE, S, I, and R are counts, whereas here they are *densities*—individuals per unit age—and a spatial generalization would make them densities per unit area, so that a two-dimensional mosquito model measures the vector population in mosquitoes per square meter. Every term's dimensions change with them, and the safe check is the one used above: integrating the density over the structuring variable must return the ODE count, $\int_0^\infty n(t,a)\,da = N(t)$. The transmission term is one place this goes wrong; the boundary and source conditions, subject to the caution at (15.2), are another and often the more treacherous. Written as a *ratio*—as the force of infection (15.6) is, through $i(t,a')/n(t,a')$—it is dimensionless and survives the count-to-density shift untouched; a density-dependent alternative, one proportional to the infectious density i itself, does not, and rescaling the spatial domain then moves $\mathcal{R}_0$ and the equilibria by a factor of the structuring scale. This is why the Ross–Macdonald threshold of Chapter 13, built on the dimensionless ratio $m = n_v/n_h$, carries cleanly over to a spatial mosquito model with the per-area factors cancelling, whereas a formulation written in raw densities must be re-dimensionalized before its threshold can be trusted. The transition trips age-structured and spatial generalizations precisely because the variables look identical on the page (Pitfall: "Dimensional Consistency", Pitfall: "Density vs. Frequency Dependence").

Exercises for this section

1. **[D]** Verify by direct calculation that, with no mortality ($\mu(a) = 0$), the von Foerster equation (15.1) changes the total $N(t) = \int_0^\infty n(t,a)\,da$ only through births, $dN/dt = B(t)$, so that N is constant when $B = 0$.

15.3 Time-Since-Infection Structure

A different continuous variable is the time elapsed since an individual became infected. Let τ denote this time-since-infection variable, sometimes called *infection age* (distinct from chronological age a). For an individual infected at calendar time $t - \tau$, the time since infection at calendar time t is exactly τ.

15.3.1 Time-Since-Infection Density

Define $i(t,\tau)$ as the density of infectious individuals at time t who were infected τ time units ago. The time-since-infection (age-of-infection) formulation is developed by Brauer et al. [2019, Sec. 4.5]

and Martcheva [2015, Ch. 13]. The total infectious population at time t is

$$I(t) \;=\; \int_0^\infty i(t,\tau)\,d\tau.$$

The dynamics of $i(t,\tau)$ are governed by aging in τ (each infectious individual's τ increases at one unit per unit time) and by exit from the infectious compartment (recovery, mortality, or other removal):

$$\frac{\partial i}{\partial t} + \frac{\partial i}{\partial \tau} \;=\; -\gamma(\tau)\,i(t,\tau), \tag{15.7}$$

where $\gamma(\tau)$ is the rate of exit from the infectious compartment for an individual whose time since infection is τ. Common choices include $\gamma(\tau) = \gamma_R$ (constant exit rate, equivalent to an exponentially distributed infectious period; Pitfall: "The Independence of τ") or a deterministic infectious period of duration τ_R, equivalent to a delay differential equation. The deterministic case is worth stating carefully, because the obvious shorthand is wrong. Survival to infection age τ is $\mathcal{F}(\tau) = \exp\big(-\int_0^\tau \gamma(u)\,du\big)$, so setting $\gamma(\tau) = \delta(\tau - \tau_R)$ contributes only unit mass to the integral and leaves $\mathcal{F}(\tau) = e^{-1} \approx 0.37$ for $\tau > \tau_R$: better than a third of the cohort is still infectious after the supposedly deterministic period ends. A deterministic period is obtained by specifying the survival function directly, $\mathcal{F}(\tau) = \mathbf{1}\{\tau < \tau_R\}$, which corresponds to a hazard that diverges at τ_R rather than to a finite-mass atom there.

The boundary condition at $\tau = 0$ is the rate of new infections:

$$i(t,0) \;=\; \alpha(t)\,S(t),$$

where $\alpha(t)$ is the (population-level) force of infection at time t and $S(t)$ is the susceptible population.

15.3.2 The Survival Function and Infectiousness Profile

The probability that an individual still belongs to the infectious compartment at time-since-infection τ, given that they were infectious at time 0, is

$$F(\tau) \;=\; \exp\left(-\int_0^\tau \gamma(\tau')\,d\tau'\right).$$

For exponential exit $\gamma = \gamma_R$, $F(\tau) = e^{-\gamma_R \tau}$, the standard exponential survival.

In the time-since-infection framework, individuals can also have variable infectiousness. Let $A(\tau)$ denote the per-capita rate at which an infectious individual generates new infections at time-since-infection τ. Then the contribution of all infectious individuals at time t to the total force of infection is

$$\text{total new-infection rate at } t \;=\; \int_0^\infty A(\tau)\,i(t,\tau)\,d\tau. \tag{15.8}$$

Decomposition of $A(\tau)$ into contact and transmission components. The generation rate decomposes as $A(\tau) = c_I(\tau)\,\beta(\tau)\,P_S$, where $c_I(\tau)$ is the infected host's contact rate at time-since-infection τ and $\beta(\tau)$ is the per-contact transmission probability at τ. The two components capture distinct mechanisms: behavior change after symptom onset acts on $c_I(\tau)$; staged viral-load dynamics act on $\beta(\tau)$. The composite $A(\tau)$ profile is analytically convenient but bundles the two effects; intervention targeting analyses may need to disentangle them. The rest of this chapter works with the composite $A(\tau)$.

Figure 15.2 illustrates the two profiles $F(\tau)$ and $A(\tau)$ together: the survival function declines monotonically from unity, while the infectiousness profile typically rises through a latent period, peaks during the symptomatic phase, and tails off as recovery approaches. The product $A(\tau)F(\tau)$ gives the expected per-capita generation rate at time-since-infection τ for an individual who was infectious at $\tau = 0$.

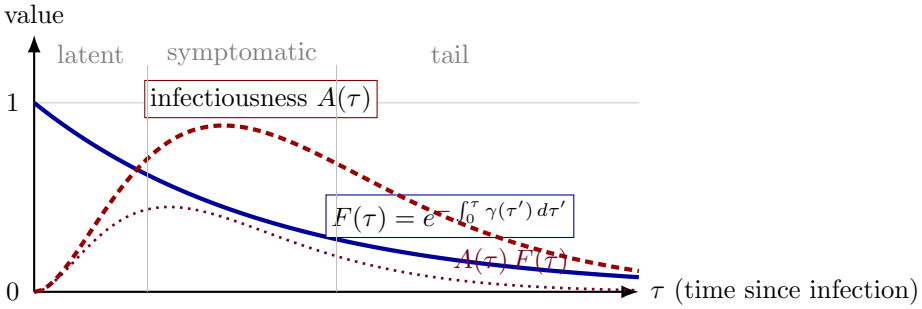


Figure 15.2: The two profiles that shape time-since-infection dynamics. The survival function $F(\tau) = \exp(-\int_0^\tau \gamma(\tau')\,d\tau')$ (solid blue) gives the probability that an individual is still in the infectious compartment at time-since-infection τ; for constant exit rate γ_R this is the standard exponential decay. The infectiousness profile $A(\tau)$ (dashed red) gives the per-capita rate at which an infectious individual generates new infections at time-since-infection τ; for many pathogens it rises through a latent period, peaks during symptomatic shedding, and tails off as the individual approaches recovery. Their product $A(\tau)F(\tau)$ (dotted purple) integrated over τ gives the expected number of secondary infections per index case, the basic reproductive number (15.13).

Setting the population force of infection $\alpha(t) = \frac{1}{S(t)} \int A(\tau) i(t,\tau) d\tau$ (so that αS recovers the new-infection rate), the boundary condition at $\tau = 0$ becomes self-referential:

$$i(t,0) \;=\; \int_0^\infty A(\tau)\, i(t,\tau)\, d\tau.$$

This is a homogeneous integral equation for $i(t,\tau)$, whose long-term solution structure determines the dynamics. We develop this equation in the next section as the Kermack-McKendrick integral equation.

Recovering the compartmental model. To see how the time-since-infection PDE reduces to the compartmental SIR model, take $A(\tau) = c_I \beta P_S$ (constant, independent of τ) and $\gamma(\tau) = \gamma_R + \nu_m$ (constant exit rate). Then $F(\tau) = e^{-(\gamma_R+\nu_m)\tau}$, and the compartmental $I(t) = \int_0^\infty i(t,\tau) d\tau$ satisfies

$$\dot{I}(t) \;=\; \frac{d}{dt}\int_0^\infty i(t,\tau)d\tau \;=\; i(t,0) - \int_0^\infty (\gamma_R+\nu_m) i(t,\tau) d\tau \;=\; \alpha S - (\gamma_R+\nu_m) I.$$

This recovers the compartmental SIR equation. The compartmental model is a special case of the time-since-infection PDE corresponding to constant infectiousness and exponential infectious-period distribution.

Exercises for this section

1. **[D]** Use the method of characteristics to solve the time-since-infection PDE (15.7) with the boundary condition $i(t,0) = \alpha(t)S(t)$, for $\tau < t$ and for $\tau > t$. Section 15.4 uses the first case.

15.4 The Kermack-McKendrick Integral Equation

The 1927 paper of Kermack and McKendrick formulated epidemic dynamics as an integral equation, predating the compartmental ODE formulation by decades. Their formulation is more general than the standard SIR ODE and remains relevant for variable-infectiousness models. We derive the Kermack-McKendrick equation from the time-since-infection PDE of the previous section.

15.4.1 Derivation

Begin with the time-since-infection density $i(t,\tau)$ satisfying (15.7) with boundary condition $i(t,0) = \alpha(t)S(t)$. We solve for $i(t,\tau)$ via the method of characteristics: along a characteristic curve $t-\tau = \text{const}$, the PDE reduces to an ODE in τ:

$$\frac{di}{d\tau} \;=\; -\gamma(\tau)i,$$

with solution $i = i_0 \exp\bigl(-\int_0^\tau \gamma(\tau')\,d\tau'\bigr) = i_0 F(\tau)$, where i_0 is the initial value at $\tau = 0$.

For the characteristic that starts at $\tau = 0$ at calendar time $t-\tau$, the initial value is the boundary value $i(t-\tau,0) = \alpha(t-\tau)S(t-\tau)$. Therefore

$$i(t,\tau) \;=\; \alpha(t-\tau)\,S(t-\tau)\,F(\tau). \tag{15.9}$$

This expresses the current density of infectious individuals with time-since-infection τ in terms of the past force of infection.

The total force of infection. Substituting (15.9) into the total-force expression (15.8):

$$\alpha(t)S(t) \;=\; \int_0^\infty A(\tau)i(t,\tau)d\tau \;=\; \int_0^\infty A(\tau)F(\tau)\alpha(t-\tau)S(t-\tau)d\tau.$$

Define the kernel

$$K(\tau) \;=\; A(\tau)F(\tau). \tag{15.10}$$

Here K is evaluated in a fully susceptible population, $K(\tau) = c_I(\tau)\,\beta(\tau)\,F(\tau)$, the value of $A(\tau)F(\tau)$ at $P_S = 1$. Only a fraction $S(t)/N$ of contacts reach a susceptible, so the equation becomes

$$\alpha(t)S(t) \;=\; \frac{S(t)}{N}\int_0^\infty K(\tau)\,\alpha(t-\tau)S(t-\tau)\,d\tau. \tag{15.11}$$

This is the *Kermack-McKendrick renewal equation* for the new-infection rate $J(t) = \alpha(t)S(t)$. Defining $J(t) = \alpha(t)S(t)$ explicitly:

$$J(t) \;=\; \frac{S(t)}{N}\int_0^\infty K(\tau)\,J(t-\tau)\,d\tau. \tag{15.12}$$

The new-infection rate at time t is a weighted sum of past new-infection rates, with weight $K(\tau)$ given by the product of infectiousness profile and survival, scaled by the susceptible fraction $S(t)/N$.

15.4.2 Interpretation of the Kernel

The kernel $K(\tau) = A(\tau)F(\tau)$ has a direct epidemiological interpretation. $A(\tau)$ is the rate at which an infectious individual generates new infections at time-since-infection τ. $F(\tau)$ is the probability that an individual originally infected is still in the infectious compartment at time-since-infection τ. The product $K(\tau)$ is the per-capita rate at which an originally-infectious individual generates new infections at time-since-infection τ.

The total expected number of secondary infections from a single index case is

$$\mathcal{R}_0 \;=\; \int_0^\infty K(\tau)\,d\tau \;=\; \int_0^\infty A(\tau)F(\tau)d\tau. \tag{15.13}$$

This is the basic reproductive number expressed as an integral over the time-since-infection distribution. It is the most general version of $\mathcal{R}_0$ and reduces to all the compartmental formulas as special cases.

Connection to the compartmental formula. For the compartmental case with $A(\tau) = c_I\beta$ (constant infectiousness) and $F(\tau) = e^{-(\gamma_R+\nu_m)\tau}$ (exponential exit),

$$\mathcal{R}_0 \;=\; \int_0^\infty c_I\beta \cdot e^{-(\gamma_R+\nu_m)\tau}d\tau \;=\; \frac{c_I\beta}{\gamma_R+\nu_m},$$

recovering the formula from Chapter 5.

For variable infectiousness (e.g., influenza with sharp peak around days 2–3), the integral (15.13) computes $\mathcal{R}_0$ from the actual infectiousness profile, which can differ substantially from the compartmental approximation.

15.4.3 The General Form of the Kermack-McKendrick Equation

The full Kermack-McKendrick formulation also includes a susceptible-population equation, since the new-infection rate depletes the susceptible pool:

$$\dot{S}(t) \;=\; -J(t) \;=\; -\frac{S(t)}{N}\int_0^\infty K(\tau)J(t-\tau)d\tau. \tag{15.14}$$

Coupled with (15.12), this gives a closed system for $J(t)$ and $S(t)$ as functionals of past behavior. The system reduces to the standard SIR ODE under the constant-rate assumptions, and admits richer behavior (oscillations, complex transients) for general kernels.

The historical Kermack-McKendrick treatment showed the existence of an epidemic threshold and gave conditions on $K(\tau)$ for outbreak versus extinction, predating modern compartmental analyses. The renewal-equation approach remains the most flexible framework for epidemic models with realistic infectiousness profiles, and has been the basis for many contemporary analyses of COVID-19, influenza, and Ebola dynamics. A rigorous treatment of the threshold theory for age-structured epidemic models in this tradition is given by Inaba [1990], and Hethcote [1997] reviews how the age-structured framework connects to compartmental approximations used in applied work.

Exercises for this section

1. **[C]** The renewal equation (15.12) reduces to the standard SIR ODE under one specific assumption about the infectiousness kernel $K(\tau)$. State this assumption, and explain in one sentence why it is the assumption that lets the time-since-infection variable be eliminated.

15.5 The Basic Reproductive Number from the Renewal Equation

The Kermack-McKendrick equation (15.12) is a linear renewal equation while $S(t) \approx N$:

$$J(t) \;\approx\; \int_0^\infty K(\tau) J(t-\tau) d\tau.$$

The behavior of this renewal equation is determined by the integral kernel $K(\tau)$ alone, and the basic reproductive number (15.13) is the key threshold parameter. Readers familiar with discrete-time statistical models will recognize the linear renewal equation as the continuous-time analog of an autoregressive AR(∞) process: discretizing time as $t_k = k\Delta\tau$ with $\Delta\tau$ small enough that K varies slowly over $\Delta\tau$, the equation becomes $J_{t_k} = \sum_{j=1}^{\infty} K(j\Delta\tau)\Delta\tau \cdot J_{t_{k-j}}$, an AR process whose lagged-coefficient sequence is the discretized infectiousness kernel and whose coefficient sum is a Riemann sum of the kernel and so approximates $\mathcal{R}_0$. Truncation to a finite memory of L lags gives the AR(L) approximation often used in time-series modeling of incidence data.

Verify the AI

The integral-kernel manipulation trap. Asked for $\mathcal{R}_0$ from a renewal equation, AI assistants routinely confuse three quantities: the infectiousness profile $A(\tau)$, a rate; the survival function $F(\tau)$, a probability; and the kernel $K(\tau) = A(\tau)\,F(\tau)$, whose integral is $\mathcal{R}_0$. The usual error integrates A or F alone; the answer can look reasonable because its magnitude is often right.
Ask: *"With $A(\tau) = A_0$ on $[0, T]$ and zero outside, and $F(\tau) = e^{-\gamma\tau}$, what is $\mathcal{R}_0$?"* A correct answer integrates the product, $\mathcal{R}_0 = (A_0/\gamma)(1 - e^{-\gamma T})$, which tends to A_0/γ as $T \to \infty$ and to $A_0 T$ as $\gamma T \to 0$. Integrating A alone gives $A_0 T$, with no recovery in it; integrating F alone gives $1/\gamma$, with no infectiousness in it.
Structural verification. Units catch one error: F is a probability, so $\int F\,d\tau$ has units of time, not a pure number. Limits catch the other. $\mathcal{R}_0$ must vanish as $A \to 0$, which $\int F$ fails, and must vanish as recovery becomes instantaneous, $\gamma \to \infty$, which $\int A = A_0 T$ fails, since it ignores recovery altogether.

15.5.1 Threshold Analysis via the Laplace Transform

To determine whether solutions of the renewal equation grow or decay, we look for exponential solutions $J(t) = J_0 e^{\sigma t}$. Substituting,

$$J_0 e^{\sigma t} \;=\; \int_0^\infty K(\tau) J_0 e^{\sigma(t-\tau)} d\tau \;=\; J_0 e^{\sigma t} \int_0^\infty K(\tau) e^{-\sigma\tau} d\tau.$$

Cancelling and defining $\widehat{K}(\sigma) = \int_0^\infty K(\tau) e^{-\sigma\tau} d\tau$ (the Laplace transform of the kernel), the existence of a non-trivial exponential solution requires

$$\widehat{K}(\sigma) \;=\; 1. \tag{15.15}$$

This is the characteristic equation. Its real solutions σ are the exponential growth rates of the renewal equation.

The threshold condition. At $\sigma = 0$, $\widehat{K}(0) = \int K(\tau)d\tau = \mathcal{R}_0$. The function $\widehat{K}(\sigma)$ is monotonically decreasing in σ for $\sigma > 0$ (assuming $K \geq 0$), starting from $\mathcal{R}_0$ at $\sigma = 0$.

If $\mathcal{R}_0 > 1$: the line $\widehat{K}(\sigma) = 1$ crosses with positive σ, giving exponential growth. If $\mathcal{R}_0 < 1$: $\widehat{K}(0) < 1$, and the only solution to $\widehat{K}(\sigma) = 1$ has negative σ, giving exponential decay. If $\mathcal{R}_0 = 1$: marginal case; the dominant solution has $\sigma = 0$.

This recovers the threshold structure of compartmental models from the more general renewal-equation framework. The basic reproductive number $\mathcal{R}_0 = \widehat{K}(0)$ is the threshold parameter, and the inequality $\mathcal{R}_0 > 1$ is necessary and sufficient for invasion.

15.5.2 The Mean Generation Time

A useful auxiliary quantity is the mean generation time, defined as

$$T_g \;=\; \frac{\int_0^\infty \tau K(\tau)d\tau}{\int_0^\infty K(\tau)d\tau} \;=\; \frac{1}{\mathcal{R}_0}\int_0^\infty \tau K(\tau)d\tau. \tag{15.16}$$

This is the mean time between an individual's infection and the infections they generate, weighted by the per-time-since-infection rate $K(\tau)$.

The mean generation time relates the doubling time of the epidemic to $\mathcal{R}_0$ via

$$\sigma \;\approx\; \frac{\mathcal{R}_0 - 1}{T_g} \quad \text{(small-}\mathcal{R}_0 - 1\text{)},$$

matching the compartmental result $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ from Chapter 5 when $T_g = 1/(\gamma_R + \nu_m)$.

Remark 15.2 (When the two growth-rate expressions part company). The agreement noted above is not a coincidence but a special case, and it is worth knowing where it ends. The renewal expression is the small-$(\mathcal{R}_0 - 1)$ limit of the Euler–Lotka relation and depends on the generation-interval distribution only through its mean, so it holds whatever that distribution is. The compartmental $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ is exact only for an exponentially distributed infectious period, because that is the case in which $T_g = 1/(\gamma_R + \nu_m)$. When the residence time is Erlang or near-deterministic—the situations this chapter and Chapter 10 exist to describe—the mean generation time is no longer the reciprocal of the exit rate, and using σ_2 imports an exponential assumption that the rest of the model has abandoned (Pitfall: "Generation-Time-Distribution Misspecification").

Practical importance. The mean generation time sets the timescale on which an outbreak unfolds, but far from $\mathcal{R}_0 = 1$ it does not fix the doubling time by itself. The Euler–Lotka relation $1 = \mathcal{R}_0 \int_0^\infty g(\tau)\, e^{-\sigma\tau}\, d\tau$, with g the generation-interval density, gives the early growth rate σ and the doubling time $\ln 2/\sigma$. An exponential interval gives $\sigma = (\mathcal{R}_0 - 1)/T_g$ exactly; a fixed interval gives $\sigma = \ln \mathcal{R}_0/T_g$. Among gamma-shaped intervals with shape at least one, these two are the extremes. For SARS-CoV-2 with $T_g \approx 5$ days and $\mathcal{R}_0 \approx 2.5$, the doubling time lies between 2.3 days (exponential) and 3.8 days (fixed). For measles with $T_g \approx 12$ days and $\mathcal{R}_0 \approx 15$, it lies between 0.6 and 3.1 days, a factor of five. The small-$(\mathcal{R}_0 - 1)$ formula above cannot choose between them: measles lies far outside its range, and the answer depends on the interval's shape (Pitfall: "Generation-Time-Distribution Misspecification").

15.5.3 The Effective Reproductive Number $\mathcal{R}_e(t)$ in the Renewal-Equation Framework

The basic reproductive number $\mathcal{R}_0 = \widehat{K}(0)$ derived in Section 15.5 is the threshold quantity at the disease-free equilibrium, where the susceptible fraction is approximately one. As an outbreak progresses and the susceptible pool depletes, the operationally-relevant threshold becomes the time-varying *effective* reproductive number $\mathcal{R}_e(t)$, which measures the expected number of secondary cases generated by a typical infectious individual at time t in the current population. Chapter 5, Section 5.2 introduces this quantity in the compartmental setting, where it takes the simple form $\mathcal{R}_e(t) = \mathcal{R}_0 \cdot S(t)/N$. The renewal-equation framework lets us derive the same relationship from the integral-equation perspective, and clarifies how $\mathcal{R}_e(t)$ couples to the infectiousness kernel.

Returning to the full Kermack-McKendrick equation (15.12), without the linearization $S(t) \approx N$ used to derive $\mathcal{R}_0$, the new-infection rate is

$$J(t) \;=\; \frac{S(t)}{N} \int_0^\infty K(\tau)\, J(t-\tau)\, d\tau, \tag{15.17}$$

where the kernel $K(\tau) = A(\tau)F(\tau)$ retains its pathogen-specific interpretation as the per-capita rate of secondary infection generation by a typical infector at time-since-infection τ, and the factor $S(t)/N$ accounts for the fraction of contacts that land on susceptibles. The total expected number of secondary infections generated by a single infector introduced at time t into the current population is

$$\mathcal{R}_e(t) \;=\; \frac{S(t)}{N} \int_0^\infty K(\tau)\, d\tau \;=\; \mathcal{R}_0 \cdot \frac{S(t)}{N}. \tag{15.18}$$

The integral $\int_0^\infty K(\tau)\, d\tau = \widehat{K}(0) = \mathcal{R}_0$ is the basic reproductive number, fixed by the pathogen and the susceptible-population's contact behavior. The factor $S(t)/N$ is the depletion factor, fixed by the trajectory of the outbreak. The product is the operational quantity surveillance practice attempts to estimate in real time.

Equation (15.18) reveals two facts that the compartmental derivation of Chapter 5 cannot make explicit. First, $\mathcal{R}_e(t)$ inherits its functional form from $K(\tau)$ via $\mathcal{R}_0$, which means that the kernel's shape (the infectiousness profile) does not appear in $\mathcal{R}_e(t)$ except through its normalization. Pathogens with very different time-since-infection profiles but identical $\mathcal{R}_0$ produce identical $\mathcal{R}_e(t)$ trajectories, modulo their effect on $S(t)$ through the dynamics. Second, the threshold $\mathcal{R}_e(t) = 1$ is the instantaneous condition for the new-infection rate to be momentarily steady (in the sense that $J(t) \approx J(t - T_g)$ on the generation-time scale); it is the natural diagnostic for whether transmission is currently growing, declining, or stationary.

Empirical estimation of $\mathcal{R}_e(t)$. Surveillance practice during the COVID-19 pandemic used (15.18), or close variants, to produce continuously-updated $\mathcal{R}_e(t)$ estimates from observed incidence series. The standard approach inverts the renewal equation directly: given an observed incidence trajectory $\widehat{J}(t)$ and a pathogen-specific generation-interval distribution $w(\tau) = K(\tau)/\mathcal{R}_0$, the estimator

$$\widehat{\mathcal{R}}_e(t) \;=\; \widehat{J}(t) \Big/ \int_0^\infty w(\tau)\, \widehat{J}(t-\tau)\, d\tau$$

recovers $\mathcal{R}_e(t)$ without requiring an explicit estimate of $S(t)/N$ [Cori et al., 2013, Thompson et al., 2019]. This is one of the most operationally important applications of the renewal-equation framework, and the reason the framework remains central to applied infectious-disease epidemiology even when the underlying compartmental model is the simpler SIR (and even when the modeller

never writes down the renewal equation explicitly). Two recent extensions widen the framework's applicability. Green et al. [2022] develop a multi-type renewal equation for populations heterogeneous in transmissibility (e.g., asymptomatic and symptomatic infectors with distinct generation-time distributions) and show that using the single-type equation (15.18) when heterogeneity is present introduces bias in $\widehat{\mathcal{R}}_e(t)$ that grows with the magnitude of the between-group difference in generation-time distribution. Pang et al. [2025] fuse the renewal-equation framework with generalized additive models to incorporate day-of-the-week effects in surveillance data directly into the estimation pipeline, a recent operational improvement that connects this chapter's mathematics to the data-preprocessing pitfalls developed in Chapter 7. Chapter 7, Section 7.8 discusses how this surveillance quantity connects to the chapter's $\hat{\alpha} = J/I$ fitting machinery; the case studies of Chapters 17 and 19 illustrate the operational use of $\mathcal{R}_e(t)$ for COVID-19 and HIV respectively.

Exercises for this section

1. **[C]** The chapter notes that two infectiousness profiles can have the same $\mathcal{R}_0$ but different mean generation times T_g. Explain in plain English why these are independent quantities, and why a public-health response timed by $\mathcal{R}_0$ alone (without considering T_g) may misjudge the urgency of an outbreak.

15.6 Numerical Methods for PDE Models

The PDE and integral-equation formulations of this chapter require their own numerical machinery. The discrete-time methods of Chapter 6 (Euler, Runge-Kutta) do not directly apply: the equations involve continuous distributions of times-since-infection or ages.

15.6.1 Discretization in Time-Since-Infection

The simplest approach replaces continuous time-since-infection τ with a discrete grid $\tau_k = k\Delta\tau$ for $k = 0, 1, 2, \ldots, K$. The PDE (15.7) becomes a system of ordinary differential equations:

$$\frac{di_k}{dt} \;=\; -\gamma_k i_k - \frac{i_k - i_{k-1}}{\Delta\tau} \;\approx\; -\big(\gamma_k + \tfrac{1}{\Delta\tau}\big) i_k + \tfrac{1}{\Delta\tau} i_{k-1},$$

where $i_k(t) = i(t, \tau_k)\Delta\tau$ is the discretized population in time-since-infection bin k. The boundary condition at $\tau_0 = 0$ couples i_0 to the new-infection rate.

This is essentially a system of compartments with a fixed transition structure: bin k transitions to bin $k+1$ at rate $1/\Delta\tau$. With sufficiently fine discretization, the discrete system approximates the PDE arbitrarily well.

Stability constraint. The discretization is stable for $\Delta t \leq \Delta\tau$ (the CFL condition for first-order upwind discretization). For typical $\Delta\tau = 0.5$ days and time stepping with $\Delta t = 0.1$ days, this is easily satisfied.

15.6.2 Direct Solution of the Renewal Equation

The Kermack-McKendrick integral equation (15.12) can be solved directly by discretizing time and approximating the integral by quadrature:

$$J(t_n) \;\approx\; \Delta\tau \sum_{k=0}^{K} K(\tau_k) J(t_n - \tau_k),$$

with $\tau_k = k\Delta\tau$ and $t_n = n\Delta t$ on a regular time grid. For $\Delta t = \Delta\tau$, the formula becomes a simple convolution.

Iterating from initial data $J(t)$ for $t < 0$ (the pre-outbreak history, typically zero), the formula generates $J(t)$ at successive time points. The susceptible population is updated separately by integrating $\dot{S} = -J$:

$$S(t_{n+1}) \;=\; S(t_n) - J(t_n)\Delta t.$$

This direct approach has the appeal that it works with the kernel $K(\tau)$ directly, without requiring a discretization of the time-since-infection variable. For variable-infectiousness modeling where the kernel is known empirically (e.g., from viral-load measurements), this is often the natural numerical approach.

15.6.3 Age-Structured PDE

Advanced material. This subsection treats the discretization of the age-structured SIR PDE (15.5) in both time and age. The main-path reader, who needs only the renewal-equation methods of the previous two subsections, can skip directly to Section 15.7; the discretization here is presented for readers planning to implement age-structured simulations.

The age-structured SIR PDEs (15.5) require discretization in both t and a. The simplest scheme is upwind finite differences:

$$\frac{s_j^{n+1} - s_j^n}{\Delta t} + \frac{s_j^n - s_{j-1}^n}{\Delta a} \;=\; -\alpha_j^n s_j^n - \mu_j s_j^n,$$

with $s_j^n \approx s(t_n, a_j)$. The CFL condition is $\Delta t \le \Delta a$. The age-mixing kernel $C(a, a')$ is discretized to a matrix C_{jk}, recovering the compartmental two-group structure of Chapter 11 when only a few age groups are used.

For practical applications with 100 age bins (1-year resolution from 0 to 99) and 100-day simulations at $\Delta t = 0.1$ days, the discrete system has 300 state variables advanced over 1000 time steps; a dense mixing sum over the 100 age bins multiplies the work, to roughly $300 \times 100 \times 1000 = 3 \times 10^7$ operations. This is feasible on a laptop but warrants attention to efficient sparse-matrix methods.

Exercises for this section

1. **[L]** *Numerical scheme comparison: upwind versus Lax–Wendroff for age-structured* SIR. Implement two finite-difference schemes for the linear advection part of the von Foerster equation (15.1) on a 100-bin age grid (1-year resolution, $a \in [0, 99]$):

   (a) *Upwind:* $n_j^{n+1} = n_j^n - (\Delta t/\Delta a)(n_j^n - n_{j-1}^n) - \mu_j \Delta t \cdot n_j^n$.

   (b) *Lax–Wendroff:* $n_j^{n+1} = n_j^n - \tfrac{1}{2}(\Delta t/\Delta a)(n_{j+1}^n - n_{j-1}^n) + \tfrac{1}{2}(\Delta t/\Delta a)^2(n_{j+1}^n - 2n_j^n + n_{j-1}^n) - \mu_j \Delta t \cdot n_j^n$.

Both schemes should be initialized with a sharply-peaked initial age density (e.g., $n(0,a) = \delta_{a,30}$, all population at age 30) and run forward 50 years with constant mortality $\mu = 0.012$/year. Plot $n(t,a)$ at $t = 0, 10, 25, 50$ years for both schemes. Identify (1) which scheme exhibits more numerical diffusion (broadening of the initially sharp peak), (2) which exhibits more numerical dispersion (oscillations or spurious wiggles near the peak edges), and (3) the trade-off between the two phenomena. Verify that both schemes preserve total mass $\int n(t,a)da$ to within numerical precision when $\mu = 0$ is used. The companion repository provides reference Python notebooks (Colab-executable) for this exercise. (This exercise illustrates the practical compromises inherent in solving hyperbolic PDEs numerically; the dispersive Lax–Wendroff is higher-order accurate but introduces unphysical oscillations, while the diffusive upwind smears sharp features but preserves monotonicity.)

15.7 An Example: Variable-Infectiousness Model

We apply the renewal-equation framework to a stylized variable-infectiousness pathogen, illustrating the quantitative difference from the compartmental approximation. The example is calibrated to seasonal influenza, where the infectiousness profile is among the best characterized empirically: viral shedding begins on day 1 post-infection, peaks around day 2 to 3, and declines steeply thereafter, with effective infectiousness essentially zero beyond day 7. The empirical $A(\tau)$ for influenza combines both decomposition components: $\beta(\tau)$ tracks viral-load dynamics (rising to peak at days 2–3, declining thereafter), while $c_I(\tau)$ reflects reduced contacts during the symptomatic days through self-isolation and fever-driven withdrawal. The combined effect is the $A(\tau)$ modeled below.

The infectiousness profile. We model the per-capita generation rate $A(\tau)$ as

$$A(\tau) \;=\; A_0\,\tau^2 e^{-\tau/\tau_p}, \tag{15.19}$$

which rises from zero at $\tau = 0$, peaks at $\tau = 2\tau_p$, and decays exponentially. For influenza, $\tau_p = 1.25$ days puts the peak at 2.5 days, matching the empirical viral-shedding peak. Recovery is modeled as exponential with rate $\gamma_R = 0.2$/day (mean infectious time of 5 days), giving survival function $F(\tau) = e^{-\gamma_R \tau}$. The renewal-equation kernel is

$$K(\tau) \;=\; A(\tau)\,F(\tau) \;=\; A_0 \tau^2 e^{-\tau(1/\tau_p + \gamma_R)}.$$

Computing the renewal-equation $\mathcal{R}_0$ and generation time. With $\tau_p = 1.25$ and $\gamma_R = 0.2$, the exponent rate is $\xi = 1/1.25 + 0.2 = 1.0$/day. Using $\int_0^\infty \tau^2 e^{-\xi\tau}\,d\tau = 2/\xi^3$,

$$\mathcal{R}_0^{\text{renewal}} \;=\; \int_0^\infty K(\tau)\,d\tau \;=\; A_0 \cdot \frac{2}{\xi^3} \;=\; 2A_0.$$

Calibrating to the empirical value $\mathcal{R}_0 = 1.3$ for seasonal influenza gives $A_0 = 0.65$. The mean generation time is

$$T_g^{\text{renewal}} \;=\; \frac{\int_0^\infty \tau K(\tau)\,d\tau}{\mathcal{R}_0} \;=\; \frac{A_0 \cdot 6/\xi^4}{2A_0} \;=\; \frac{3}{\xi} \;=\; 3.0 \text{ days}.$$

Compartmental approximation with the same $\mathcal{R}_0$. The compartmental SIR model with constant per-capita generation rate A_{const} and exponential exit at rate γ_R gives

$$\mathcal{R}_0^{\text{compartmental}} = \frac{A_{\text{const}}}{\gamma_R}, \qquad T_g^{\text{compartmental}} = \frac{1}{\gamma_R}.$$

Matching $\mathcal{R}_0 = 1.3$ with $\gamma_R = 0.2$ requires $A_{\text{const}} = 0.26$; the corresponding mean generation time is $T_g = 1/0.2 = 5$ days. The two models agree on $\mathcal{R}_0$ by calibration but disagree on T_g.

Numerical comparison. Table 15.1 summarizes the comparison between the two models, both calibrated to $\mathcal{R}_0 = 1.3$.

Table 15.1: Comparison of the compartmental SIR model and the renewal-equation model for an influenza-like infectiousness profile, both calibrated to $\mathcal{R}_0 = 1.3$. The two models agree on the threshold but differ by a factor of 5/3 in the timing of secondary-infection generation, with concrete consequences for outbreak-response timing and early-growth-rate interpretation.

Quantity	Compartmental SIR	Renewal equation
$\mathcal{R}_0$	1.30 (by construction)	1.30 (by construction)
Mean generation time T_g	5.0 days	3.0 days
Time to peak infectiousness	0 days (constant)	2.5 days
Effective infectiousness at day 7	$A_{\text{const}} \cdot F(7) = 0.064$	$K(7) = 0.029$
Early growth rate σ	$\gamma_R(\mathcal{R}_0 - 1) = 0.060$/day	0.091/day (via Laplace)
Early doubling time	11.6 days	7.6 days

The two rows that matter most for outbreak-response decisions are the last two: the compartmental approximation gives an early growth rate of about 0.06/day (doubling time $\sim$ 12 days), while the renewal-equation model with the same $\mathcal{R}_0$ gives an early growth rate of 0.09/day (doubling time $\sim$ 8 days). The roughly 50% acceleration comes from the shorter mean generation time, 3 days against 5, not from the delayed peak. At an equal mean of 3 days, an exponential interval would give $\sigma = (\mathcal{R}_0 - 1)/T_g = 0.100$/day, faster than the renewal model's 0.091/day: the delayed, concentrated profile slows early growth slightly. The compartmental model errs because calibrating γ_R to the infectious period also fixes $T_g = 1/\gamma_R$, which overstates the generation time when infectiousness is concentrated in the first few days. An early-warning system that makes this assumption over-predicts the doubling time, here by a factor of about 1.5 (11.6 against 7.6 days).

Reading $\mathcal{R}_0$ from an observed growth rate. Outbreak analysts usually work the other way: they observe the early growth rate and infer $\mathcal{R}_0$. The Euler–Lotka relation gives $\mathcal{R}_0 = 1/M(-\sigma)$, where M is the moment generating function of the generation-interval density [Wallinga and Lipsitch, 2007]. For a gamma-distributed interval with mean T_g and shape k, this is $\mathcal{R}_0 = (1 + \sigma T_g/k)^k$. The exponential interval is the case $k = 1$, which gives $\mathcal{R}_0 = 1 + \sigma T_g$. A fixed interval is the limit $k \to \infty$, which gives $\mathcal{R}_0 = e^{\sigma T_g}$, the upper bound for a given mean [Wallinga and Lipsitch, 2007]. The kernel of this example, $K(\tau) \propto \tau^2 e^{-\tau}$, is the gamma case $k = 3$ with $T_g = 3$ days. Figure 15.3 draws four readings of the same growth rate. At the example's $\sigma = 0.091$/day, the correct shape returns $\mathcal{R}_0 = 1.30$. An exponential or a fixed interval with the same mean gives 1.27 or 1.32. The compartmental reading, which takes $T_g = 1/\gamma_R = 5$ days, gives 1.46. Here the wrong mean does most of the damage; the shape matters more as the growth rate rises.

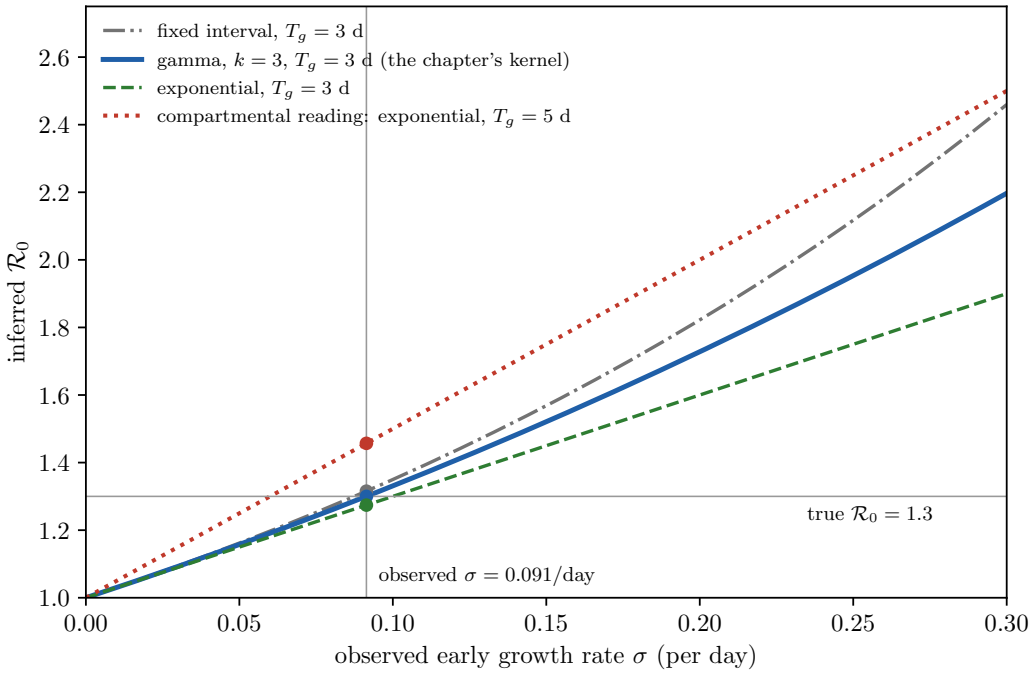


Figure 15.3: The basic reproductive number inferred from an observed early growth rate σ depends on the generation interval assumed. Solid: the gamma interval of the worked example ($k = 3$, $T_g = 3$ days), which returns the true $\mathcal{R}_0 = 1.3$ at the example's $\sigma = 0.091$/day. Dashed and dash-dotted: exponential and fixed intervals with the same mean, the lower and upper readings. Dotted: the compartmental reading, an exponential interval with $T_g = 1/\gamma_R = 5$ days, which gives 1.46.

When the compartmental approximation is adequate. If the infectiousness profile is close to exponential (monotonically decreasing from birth with no rising phase), the compartmental approximation is adequate and the two models produce similar dynamics. If the infectiousness profile has a substantial pre-peak delay (as for influenza, COVID-19, and many respiratory pathogens), the renewal-equation model gives more accurate early-dynamics predictions. A practical test: compute T_g from the kernel and compare it with $1/(\gamma_R + \nu_m)$. Near threshold the growth rate scales as $1/T_g$, so a 10 percent difference in T_g means about a 10 percent error in the compartmental growth rate, and renewal-equation corrections are worth computing. For the influenza profile here, T_g is 3 days against 5, a 40 percent difference; for an exponential profile the two coincide.

15.8 Summary and Exercises

This chapter developed the continuous-structure framework that generalizes the compartmental approach.

1. Age-structured population dynamics are governed by the von Foerster equation (15.1), with the boundary condition encoding fertility and the stable age distribution given by the Lotka equation (15.4).

2. The age-structured SIR model (15.5) stratifies the disease compartments by chronological age, with the force of infection coupling all ages through an age-mixing kernel $C(a, a')$.

3. The time-since-infection PDE (15.7) tracks how the population of infectious individuals evolves with their time-since-infection variable, allowing variable infectiousness and arbitrary exit-rate distributions.

4. The Kermack-McKendrick integral equation (15.12) is the renewal equation for the new-infection rate, with the kernel $K(\tau) = A(\tau)F(\tau)$ encoding both the infectiousness profile and the survival distribution. It is the most general framework, with the standard SIR ODE recoverable as a special case.

5. The basic reproductive number is $\mathcal{R}_0 = \int_0^\infty K(\tau)d\tau$, the integral of the kernel. The threshold $\mathcal{R}_0 > 1$ for invasion follows from the Laplace-transform analysis of the characteristic equation.

6. The mean generation time T_g is a separate quantity from $\mathcal{R}_0$ that determines the outbreak's operational timescale. Different infectiousness profiles can have the same $\mathcal{R}_0$ but different T_g, with significant implications for outbreak-response timing.

7. Numerical methods include discretization in time-since-infection (yielding a high-dimensional ODE system) or direct solution of the renewal equation by quadrature.

8. A worked example with a peaked infectiousness profile shows that the compartmental approximation distorts the mean generation time by a factor of 5/3 for influenza-like infectiousness, while preserving $\mathcal{R}_0$.

This chapter completes the models of Part IV (Generalizations and Heterogeneous Populations). Chapter 16 closes the Part with the pitfall checklist; Part V then applies the full toolkit to specific diseases of major epidemiological importance.

Framework use in the renewal-equation setting. The Kermack-McKendrick renewal equation (15.12) is the framework's most general setting. The infected-viewpoint fitting structure carries over with no modification: in the time-since-infection formulation, the new-infection rate is

$$J(t) \;=\; \int_0^\infty A(\tau)\, i(t,\tau)\, d\tau,$$

and the total infectious population at time t is

$$I(t) \;=\; \int_0^\infty i(t,\tau)\, d\tau.$$

The ratio $J(t)/I(t)$ is a ratio of observable quantities with no explicit dependence on the susceptible population size. The infected-viewpoint estimator

$$\hat{\alpha}(t) \;=\; \frac{J(t)}{I(t)}$$

therefore retains its interpretation and its structural-immunity property in the renewal-equation setting, just as in the compartmental setting of Chapter 7. The quantity $\hat{\alpha}(t)$ in the renewal-equation setting is a time-dependent weighted average of the infectiousness profile $A(\tau)$, weighted by the time-since-infection density $i(t,\tau)$; the fitted value is the population-level effective force from infection at time t.

The sensitivity-analysis paper of Hyman [2026d] develops the full sensitivity argument in the compartmental setting; its generalization to the renewal-equation setting follows directly. For the variable-infectiousness case analyzed in Section 15.7, the fitted $\hat{\alpha}(t)$ absorbs the unknown detailed shape of the infectiousness profile $A(\tau)$ without requiring a committed value for S. This is the practical payoff of the renewal-equation formulation in outbreak response: when the infectiousness profile is uncertain (as for a novel pathogen), the infected-viewpoint fit continues to give S^*-independent estimates of the effective force from infection, which in turn drive the forecast of future incidence.

A reader focused on the compartmental models of Chapters 4 through 10 can take the renewal-equation analysis here as an assurance that the framework's central claims extend to the more general setting. The sensitivity-analysis paper of Hyman [2026d] develops the mathematics in the compartmental setting; this chapter gives the structural argument in the renewal-equation setting. For practical outbreak-response work, the compartmental formulations suffice in most cases, with the renewal-equation framework reserved for pathogens where the shape of the infectiousness profile is empirically known (HIV, hepatitis B, some respiratory pathogens with well-characterized shedding patterns).

Pitfall check

This chapter has met the count-to-density transition at boundary and source terms, the place where Pitfall: "Dimensional Consistency" and Pitfall: "Density vs. Frequency Dependence" are most often broken, because the variables look identical on the page (Section 15.2). It relaxed Pitfall: "The Independence of τ": a time-since-infection exit rate $\gamma(\tau)$ includes the constant γ_R, the exponential infectious period, as one case (Section 15.3). And it is the home of Pitfall: "Generation-Time-Distribution Misspecification". The renewal expression for early growth depends on the generation interval only through its mean; the compartmental $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$ is exact only for an exponential infectious period, so using it in an Erlang or near-deterministic model brings back the assumption the model set aside (Section 15.5). Before extending a model in age, space or time since infection, check the units of every boundary term, and ask which residence-time distribution each formula assumes.

Exercises

Five section-level exercises appear above, following Section 15.2, Section 15.3, Section 15.4, Section 15.5, and Section 15.6. The exercises below integrate across the chapter and have been ordered so that derivations and computational labs alternate (this chapter is derivation-heavy, so perfect alternation is not possible). Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[D]** *Stable age distribution and the Lotka equation.* Substitute the separable ansatz $n(t,a) = e^{\sigma_d t}w(a)$ into the von Foerster equation (15.1) to obtain the first-order ODE $w'(a) + (\sigma_d + \mu(a))w(a) = 0$ for w. Solve to recover the stable-age formula (15.3). Then impose the birth boundary condition (15.2) on the separable solution to derive the Lotka equation (15.4), identifying the step at which the unknown growth rate σ_d becomes determined.

2. **[D]** For a population with constant fertility $b(a) = b_0$ for $20 \le a \le 45$ (and zero elsewhere) and constant mortality $\mu(a) = \mu_0$, derive the equation that determines σ_d from b_0 and μ_0. For $b_0 = 0.05$/year and $\mu_0 = 0.012$/year, is the population growing or shrinking?

3. **[L]** Solve the renewal equation $J(t) = \int_0^\infty K(\tau)J(t-\tau)d\tau$ numerically for the influenza-like kernel of Section 15.7, with one index case infected at $t = 0$. Plot $J(t)$ over the first 30 days and compare with the compartmental SIR trajectory at the same $\mathcal{R}_0$.

4. **[D]** Compute $\mathcal{R}_0$ via the Kermack-McKendrick formula (15.13) for the following two infectiousness profiles, both with $\gamma_R = 0.1$/day and exponential exit:

   (a) Constant: $A(\tau) = A_0$.

   (b) Sharp: $A(\tau) = A_0\delta(\tau - 5)$ (all transmission at day 5).

   Show that they have the same $\mathcal{R}_0$ if A_0 is chosen appropriately, but different mean generation times.

5. **[D]** For the variable-infectiousness example of Section 15.7, compute the mean generation time T_g and verify that it differs from the compartmental approximation by a factor of 5/3.

6. **[L]** *Three-kernel comparison.* Define three infectiousness kernels with identical $\mathcal{R}_0 = 2.5$ but different shapes: (a) exponentially declining $A(\tau) = A_0e^{-\tau/4}$; (b) gamma-shaped (e.g., shape 3, scale 2 days); (c) constant over $[0, 8]$ days, which shares the mean of (a). Solve the renewal equation for each over 60 days and compute the mean generation time T_g, the doubling time, and the day by which cumulative infections from one index case reach 100. Tabulate the three quantities for the three kernels. This exercise quantifies the practical impact of the chapter's central observation that $\mathcal{R}_0$ alone does not determine outbreak speed.

7. **[D]** Discretize the age-structured SIR model (15.5) into 4 age groups (0-19, 20-39, 40-59, 60+) with appropriate age-mixing matrix $C(a, a')$. Show that this reduces to the multi-group framework of Chapter 11, with aging between groups as an added flow.

8. **[D]** For an age-structured population with childhood vaccination at age 1 (probability f of immunization), set up the modified SIR PDE with vaccination as a condition at age 1, a jump in s and r across $a = 1$. Discuss how to compute $\mathcal{R}_0$ in this setting. *(Challenge: this exercise requires graduate-level familiarity with boundary-value PDE problems.)*

Chapter 16

Common Pitfalls and How to Avoid Them

This chapter gathers the book's pitfalls in one place. Each earlier chapter teaches its method and, where an error is tempting, names the pitfall in one sentence and points here. The entries below give the rest: the tempting move, what goes wrong, the fix, and where in the book the pitfall is worked out.

The entries follow the catalogue's four families: regime, formulation, estimation and interpretation. Within a family they run in the order a modeler meets them, from writing the equations to reporting the result. Use the checklist in Table 16.1 before trusting a model, and the decision flow in Figure 16.1 for the order in which the main checks come. Section 16.6 takes one constructed model through both.

16.1 The Checklist and the Decision Flow

Table 16.1 asks one question per pitfall. A "no" or an "I don't know" sends you to the entry on the page given. Figure 16.1 puts six of the checks in the order a modeling project meets them.

16.2 Regime Pitfalls

16.2.1 The Stochastic-to-Deterministic Regime Transition

A deterministic compartmental model treats the population as a continuum, but transmission is stochastic while the number of infectious individuals is small.

The tempting move. Simulate or fit the deterministic model over the whole course of an outbreak, from the first case to the last.

What goes wrong. The deterministic model is reliable only inside a regime that recent work places at roughly three to six percent cumulative infection at the early end and roughly seventy to seventy-five percent at the late end. Outside that interior the deterministic trajectory is not reliable.

The fix. Restrict a fit to the valid window, or use stochastic-fitting machinery. Simulate outside the window with stochastic methods.

Where in the book. This chapter is its home. Previewed in the Preface, applied throughout Chapter 7 when fitting begins, and revisited wherever a modeling framework rests on a deterministic-

Table 16.1: The checklist: one question per pitfall, grouped by family, with the page of its entry.

Before you trust the model, ask	**Pitfall**	**Page**
Regime		
Is the infectious count large enough for a deterministic model over the whole window you use?	The Stochastic-to-Deterministic Regime Transition	347
Have you stated the closed-population and inflow assumptions, and do they hold over your horizon?	Short-Term Simulation Assumptions	350
Formulation		
Does every term in each rate equation carry the units of its derivative?	Dimensional Consistency	350
Is any dimensional quantity raised to a non-integer power?	Non-Integer Powers of Dimensional Quantities	350
Have you chosen density or frequency dependence, and said which?	Density vs. Frequency Dependence	351
Is transmission written in the frame you will fit?	Force of Infection versus Force from Infection	351
Are mean residence times the parameters, and the rates derived?	The Independence of τ	352
Do the contacts group A gives group B equal the contacts B receives from A?	Heterogeneous-Mixing Balance Conditions	352
If contact rates vary, does the model do more than use their mean?	Contact-Rate Averaging	353
Could behavior change the contact or transmission rates within your window?	Constant-Transmission Assumption Ignoring Behavioral Feedback	353
Can the trajectory leave the initial state you chose?	Natural Initial Conditions	354
Is the linear-looking form the true linearization in your regime?	Linearization Fidelity	354
Estimation		
Have you corrected for delays, weekday cycles, holidays and the recent edge?	Data Preprocessing	354
Are you fitting what the data report, at the resolution they report it?	Correct Data-Fitting Practice	355
Is each probability from the literature the marginal your model needs?	Conditional versus Marginal Probabilities	355
In a ratio estimator, are the two counts detected at the same rate?	Differential Ascertainment in a Ratio Estimator	355
Are the residuals weighted by their uncertainty?	Weighting Residuals by Their Uncertainty	356
When residuals fail a test, have you asked what that says about the mechanism?	Reading Residuals in a Mechanistic Model	356
Which parameter combinations can the data actually pin down?	Identifiability	357
Do the estimators you compare solve the same problem on the same data?	Estimator Honesty	357
Does the infectious-period shape match the measured generation-time distribution?	Generation-Time-Distribution Misspecification	358
Does the output you test answer your question, with τ, not γ, as the parameter?	Sensitivity Analysis Discipline	358
Can the parameters you vary one at a time really change independently?	Independence of Parameters of Interest	358
Is each parameter's effect monotone over the sampled range before you trust PRCC?	Hidden Monotonicity Assumption	359
Interpretation		
Are you reporting $\mathcal{R}_0$ at the disease-free state or $\mathcal{R}_e(t)$ at the current one?	$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$	359
Is the question invasion or long-run burden?	Invasion versus Equilibrium	360
Does anything besides susceptibles enter the population from outside?	NGM Applicability	360

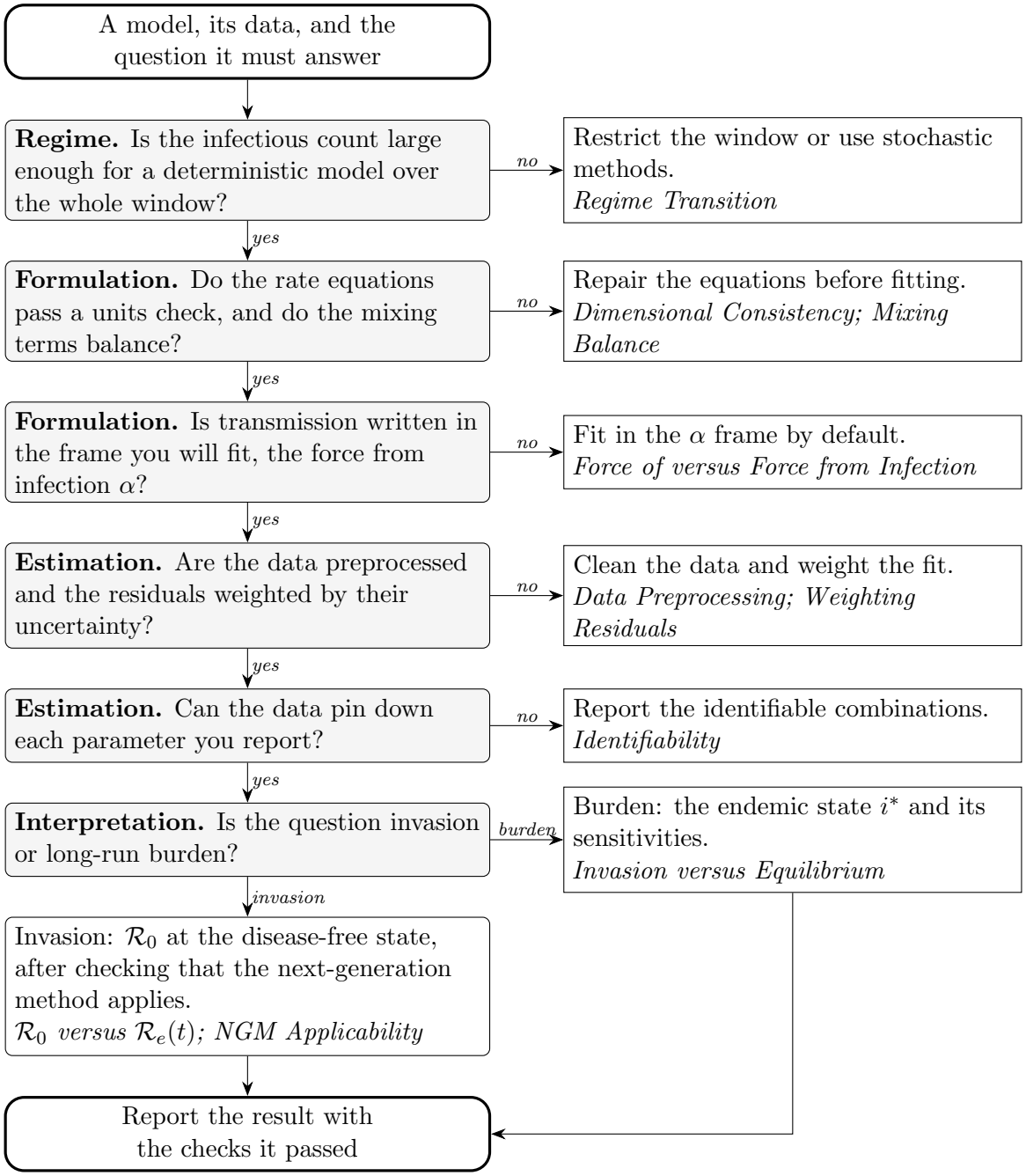


Figure 16.1: Decision flow for checking a model before its results are reported. A "no" leads to a repair and the pitfall that names it; after the repair, return to the same question. The flow shows six checks in order; Table 16.1 holds all of them.

regime assumption. Related: Density vs. Frequency Dependence, since the two are coupled in small populations.

16.2.2 Short-Term Simulation Assumptions

Simulations of a single outbreak routinely assume a conserved total population and, when migration is modeled, inflow of susceptibles only.

The tempting move. Leave both assumptions unstated, since over a short horizon they are reasonable.

What goes wrong. Both break down for multi-year endemic models and for populations with substantial movement of infectious individuals.

The fix. Name both assumptions explicitly whenever a model is set up, and re-examine them when the horizon lengthens or infectious individuals move.

Where in the book. This chapter is its home. Previewed in the Preface; revisited in Chapter 11, in Chapter 10, where disease-induced mortality and demographic turnover require the assumptions to be stated, and in Chapter 13, where vector lifespans short relative to host disease dynamics force them to be re-examined. Related: Conditional versus Marginal Probabilities, since holding mixture weights fixed while an outbreak runs is one of these assumptions.

16.3 Formulation Pitfalls

16.3.1 Dimensional Consistency

Every term summed into a rate equation must carry the dimensions of the derivative it feeds.

The tempting move. Put a transmission probability, a pure number, where the equation needs a transmission rate per unit time.

What goes wrong. The error is invisible under one choice of time unit and fatal under another. A period typed in place of the rate it determines is the same violation.

The fix. Audit the dimensions of every term at the moment the equations are first written. One audit removes this whole class of error before it can reach a simulation or a fit, and the book makes it the opening act of every model definition.

Where in the book. Chapter 2, with exercises there and in Chapter 3; applied as a routine check throughout the rest of the book. Related: Non-Integer Powers of Dimensional Quantities, its specialization, and Weighting Residuals by Their Uncertainty, for the units of the weights.

16.3.2 Non-Integer Powers of Dimensional Quantities

Raising a dimensional quantity to a non-integer power makes the units of any coefficient multiplying it depend on that power.

The tempting move. Read the fitted coefficient a of a phenomenological growth model, $\dot{C} = a\,C^p\,g(C)$ with g dimensionless and $p \in [0, 1]$, as a growth rate, and compare it across fits.

What goes wrong. Balancing units gives $[a] = [C]^{1-p}\,[T]^{-1}$, so a is a rate only when $p = 1$. For every $p < 1$, values of a fitted under different exponents carry different units and cannot be compared, and expressing C in a unit κ times larger multiplies a by κ^{p-1}. A fractional derivative

D_t^α of order $\alpha \in (0,1)$ raises the same issue: it carries $[T]^{-\alpha}$, so integer-order rate constants left in place unbalance the equation.

The fix. Ask whether the dimension of a coefficient depends on a fitted exponent. If it does, nondimensionalize the base quantity first, scaling C by a reference count C_{ref} or time by a reference time. The trajectories and forecasts are unchanged, and the coefficients become dimensionally honest and comparable once the reference scale is fixed. Nothing here bears on how well these models forecast; the caution is against reading meaning into coefficients whose units depend on an exponent.

Where in the book. This entry is its only treatment for now; a short development is planned for Chapter 7. A specialization of Dimensional Consistency.

16.3.3 Density vs. Frequency Dependence

The per-capita contact rate c either grows with population size or is independent of it, and the choice is a modeling commitment.

The tempting move. Pick one form, βsi for density dependence or $\beta si/n$ for frequency dependence, by habit, and carry it into any population.

What goes wrong. The choice changes the dynamics under any intervention that changes n, and the fitted transmission rate when populations of different size are compared. Density dependence fits livestock diseases, vector-borne diseases tied to host density, and small clustered populations. Frequency dependence fits sexually transmitted infections and most respiratory diseases in large modern populations. Many systems sit between.

The fix. State the scaling and its basis. This book uses frequency dependence throughout, on the assumption that the population is large enough for contact rates to have saturated. A livestock herd, a school, or a refugee camp calls for revisiting that assumption together with the regime transition.

Where in the book. Chapter 1, Section 1.3.1, with the density-frequency transition figure. Related: The Stochastic-to-Deterministic Regime Transition, since the deterministic framework is most reliable in the population-size regime where frequency dependence is empirically supported.

16.3.4 Force of Infection versus Force from Infection

One transmission process, two reference frames. Per susceptible, infection *arrives* at the force of infection $\lambda = c_S \beta P_I$; per infectious individual, it *departs* at the force from infection $\alpha = c_I \beta P_S$; detailed balance ties the frames together through $\lambda s = \alpha i$.

The tempting move. Treat the two forces as interchangeable. The identity $\lambda s = \alpha i$ says both give the same population-level rate of new infections, so it seems not to matter which frame a model is written and fitted in.

What goes wrong. The equivalence holds for the dynamics, not for estimation. Early in an outbreak $s \gg i$, so $\alpha \gg \lambda$, and the two estimation problems differ. The susceptible-viewpoint estimator $\hat{\lambda} = j/s$ depends on an assumed value of s, which early in an outbreak is often the most uncertain quantity in the model, and it inherits any misspecification of s^* at unit sensitivity. The infected-viewpoint estimator $\hat{\alpha} = j/i$ uses the observed infectious population and never invokes s, so it is structurally immune to that misspecification.

The fix. Fit in the α frame by default. Keep the λ frame for the settings where it is natural: seroincidence studies, and endemic-equilibrium analysis where s is well characterized. The immunity

belongs to the force estimator alone. Converting either force to a reproductive number brings the susceptible fraction back, since $\hat{\alpha}/(\gamma_R + \nu_m)$ returns $\mathcal{R}_e$ rather than $\mathcal{R}_0$ unless the population is naive.

Where in the book. Chapter 2 derives λ (Section 2.2). Chapter 3 derives α and proves the equivalence (Proposition 3.1). Chapter 7 works the two estimators and the conversion to a reproductive number, and Chapter 9 analyzes the asymmetry with sensitivity indices. Related: Linearization Fidelity, since freezing α early in an outbreak passes that test and freezing λ fails it.

16.3.5 The Independence of τ

The time an individual spends in a compartment is a random variable; its mean τ_x is the independent parameter, and the rate $\gamma_x = 1/\tau_x$ is derived from it.

The tempting move. Treat the rates that appear in the equations, such as γ_R and ν_m, as the model's parameters.

What goes wrong. The duration, not the rate, is what an epidemiologist or clinician estimates from data, and the reciprocal $\gamma_x = 1/\tau_x$ is exact only for an exponential residence time, the implicit assumption of every constant-rate compartment. A sensitivity taken with respect to a rate carries the same information as one taken with respect to the duration but inverts the apparent direction of the effect, since $\Upsilon^q_\tau = -\Upsilon^q_\gamma$. Measured latent periods are often more sharply peaked than the exponential allows: for SARS-CoV-2 the distribution is roughly gamma-shaped with a coefficient of variation near 0.5, against 1 for the exponential, and the exponential assumption biases estimates of $\mathcal{R}_0$ [Hong et al., 2024].

The fix. Fit τ, report confidence intervals on τ, and compute γ only for the solver. Take sensitivities with respect to τ. Where the exponential shape is inadequate, replace the compartment by an Erlang chain of k exponential stages, each of mean τ_x/k: the total residence time is gamma-distributed with mean τ_x and coefficient of variation $1/\sqrt{k}$, and τ_x stays the independent parameter, with k a second parameter for the shape.

Where in the book. Chapter 4, Table 4.1 (page 47); the Erlang chain in Chapter 10 (page 234); the sign identity as equation (9.2) in Chapter 9. Related: Generation-Time-Distribution Misspecification, which concerns the shape of the distribution rather than which parameter names its mean, and Sensitivity Analysis Discipline.

16.3.6 Heterogeneous-Mixing Balance Conditions

When groups mix, the contacts group 1 directs at group 2 must equal the contacts group 2 receives from group 1: $C_{12}N_1^c = C_{21}N_2^c$, a conservation law for the contact process.

The tempting move. Take a mixing matrix as given, whether from contact-survey data, from intuition, or from an AI tool, or set the per-person rates C_{12} and C_{21} equal when the groups differ in size.

What goes wrong. Such matrices can violate balance by tens of percent. Contacts then vanish from one side of the bookkeeping or are counted twice, the force of infection and the force from infection cease to be equivalent, and the $\mathcal{R}_0$ computed from the next-generation matrix can be biased by tens of percent. The effect is largest when the two groups differ substantially in size, where the per-person rates differ most.

The fix. Compute $C_{12}N_1^c$ and $C_{21}N_2^c$ and check that they agree to within survey precision. If they do not, adjust one off-diagonal entry, treating the better-supported of the two estimates as authoritative.

Where in the book. Chapter 11, equation (11.2) (page 251), with the next-generation matrix in Section 11.4; generalized to two sexes in Chapter 12 and to vector and host in Chapter 13. Related: Contact-Rate Averaging.

16.3.7 Contact-Rate Averaging

When people differ in how many contacts they make, a homogeneous model built on the mean contact rate $\langle c \rangle$ understates transmission.

The tempting move. Average the contact rate over the population and use the average as though everyone shared it.

What goes wrong. A person who makes more contacts is more likely both to be infected and to infect others, so each person's contact rate counts twice. Under proportionate mixing the rate that governs invasion is the activity-weighted mean $\langle c^2 \rangle / \langle c \rangle = \langle c \rangle + (\langle c^2 \rangle - \langle c \rangle^2)/\langle c \rangle$, and an $\mathcal{R}_0$ computed from $\langle c \rangle$ is too small by the factor $\langle c^2 \rangle / \langle c \rangle^2$. The error matters most where a small, highly active group can sustain transmission that the average says should fade out.

The fix. Ask of every rate whether it stands for an average over people who differ in how often they make contacts. If it does, use the activity-weighted mean, or keep the groups apart.

Where in the book. Chapter 11, Section 11.8, where the averaged rate gives $\mathcal{R}_0 \approx 1.87$ and the two-group computation gives 2.35. Related: Heterogeneous-Mixing Balance Conditions.

16.3.8 Constant-Transmission Assumption Ignoring Behavioral Feedback

Models conventionally hold the transmission rate β and the contact rates fixed, but behavior responds to the perceived state of the epidemic.

The tempting move. Fit constant β and constant contact rates through a period in which cases, hospitalizations, and public attention are changing.

What goes wrong. As case counts rise, people reduce contacts and adopt protective behaviors, and as they fall the behaviors reverse. A constant-β model cannot follow this loop, so its fitted parameters absorb the behavioral response as though it belonged to the pathogen and the population, which biases both transmissibility and final epidemic size. Published comparisons find that models with behavioral feedback forecast better out of sample and fit the first COVID-19 wave better across several geographies [Gozzi et al., 2025, Hamilton et al., 2024]. The bias is largest when the behavioral response is strong and smallest when behavior is nearly constant, early before awareness or at endemic equilibrium.

The fix. Name the assumption. This book keeps β constant on the premise that the analysis window is short enough for behavioral change to be negligible, or that its effect can be absorbed into a time-varying $\beta(t)$. Behavioral-feedback models are a separate tradition that the book does not develop.

Where in the book. Acknowledged throughout by the constant-β convention; flagged in Chapter 5 as the validity caveat for its $\mathcal{R}_0$ and $\mathcal{R}_e(t)$ analysis, and in Chapter 17 as the assumption under which its intervention scenarios are evaluated. Related: Reading Residuals in a Mechanistic Model, for which behavioral feedback is one common candidate.

16.3.9 Natural Initial Conditions

A model fitted to an outbreak must start from a state that could occur in the setting that produced the data.

The tempting move. Start the simulation at the disease-free equilibrium $(1, 0, 0)$.

What goes wrong. The disease-free state is an exact equilibrium of the deterministic system, so the trajectory never leaves it for any value of $\mathcal{R}_0$, and the simulation returns $i(t) \equiv 0$: correct, and useless for fitting. A second decision hides inside the first. The pathogen's entry, the epidemic's take-off, and the first reported case can lie days or weeks apart, so a model with the right starting state and the wrong time origin is fitted to the wrong window.

The fix. Start the fit at a time when a small infectious population is observable, set i there to the observed value, never to zero, and set s and r to the best surveillance estimates at that moment. In an Erlang-chain model, spread the initial infecteds across the stages in their growth-phase proportions rather than lumping them into the first stage or spreading them evenly.

Where in the book. Chapter 6, equation (6.1) (page 91); revisited in Chapter 7, where the choice interacts with uncertainty in the susceptible population, and for Erlang chains in Section 10.3.

16.3.10 Linearization Fidelity

A formulation that looks linear is not automatically the linearization of the nonlinear system it came from.

The tempting move. Freeze whichever quantity is convenient at its initial value and call the result the early-outbreak linearization.

What goes wrong. Freezing P_S at $P_S(0) \approx 1$ gives $\dot{i} \approx (c_I \beta P_S(0) - (\gamma_R + \nu_m))\, i$, whose growth rate is the correct linearization about the disease-free equilibrium, because P_S is the slow variable and i the fast one (Proposition 3.2). Freezing the force of infection $\lambda = c_S \beta P_I$ at $\lambda(0) \approx 0$ instead gives $\dot{i} \approx -(\gamma_R + \nu_m) i$, in which i decays, because P_I grows with the epidemic. That replaces the nonlinearity with zero; it does not linearize it.

The fix. Freeze only a quantity that is genuinely constant in the regime of interest. Early in an outbreak that is $\alpha = c_I \beta P_S$, not λ, which is why the fitted α carries structural meaning.

Where in the book. Chapter 5 (page 62); the fitting advantage it explains in Chapter 7. Related: Force of Infection versus Force from Infection, and NGM Applicability, since the next-generation construction is itself a linearization at the disease-free equilibrium.

16.4 Estimation Pitfalls

16.4.1 Data Preprocessing

Surveillance data are the output of a reporting pipeline, not a clean record of the epidemic.

The tempting move. Fit the reported counts as they arrive.

What goes wrong. The pipeline imposes its own structure: a day-of-week cycle from testing and laboratory schedules, holiday gaps filled in days later, and recent counts left incomplete while reports trickle in for weeks. A least-squares fit takes that structure as signal. The weekly spike and trough enter the transmission parameters, a holiday gap reads as a drop in transmission, and the dip at the latest points pulls $\mathcal{R}_e(t)$ down just when a forecast is wanted.

The fix. Separate reporting artifacts from epidemic signal before fitting, and report the preprocessing assumptions alongside the estimates, since they are part of the modeling commitment. This step is often the most time-consuming part of an applied fit and the part that most affects its conclusions.

Where in the book. Chapter 7 (page 115), with a worked example in which corrected and uncorrected data give substantially different estimates. Related: Identifiability, Estimator Honesty, and Differential Ascertainment in a Ratio Estimator.

16.4.2 Correct Data-Fitting Practice

Fit what the data actually report, at the resolution at which they are reported.

The tempting move. Fit whatever quantity is easiest to model, at whatever resolution the model runs.

What goes wrong. The common errors are fitting cumulative incidence when the reporting unit is weekly new cases, fitting at finer than the reporting resolution and inheriting reporting artifacts as model parameters, treating reported incidence as though it were prevalence, and ignoring known reporting structure such as delays or changes of case definition.

The fix. Match the fitted quantity and its resolution to the reporting unit, and account for the known structure of the reporting process.

Where in the book. Threaded through Chapters 7 and 8 as their unifying discipline; applied in Chapter 17. Related: Data Preprocessing.

16.4.3 Conditional versus Marginal Probabilities

A probability reported in the literature is almost always conditional on something, while a model usually needs the marginal over the population it runs on.

The tempting move. Use the published conditional probability as the model's parameter.

What goes wrong. The two agree only when the conditioning event is certain, and the substitution is silent: the units match, the magnitude looks plausible, and nothing in the simulation complains. For rubella, the disease probability conditional on first-trimester infection, used in place of the per-infection marginal, overstates the vertical disease burden by roughly $1/w_1$. The same error appears elsewhere. A fatality ratio computed over detected cases is not an infection fatality ratio [World Health Organization, 2020]; a secondary attack rate measured from symptomatic index cases does not transfer to all index cases [Madewell et al., 2020]; and a vaccine efficacy defined per exposure answers a different question from one defined on population incidence [Halloran et al., 1997].

The fix. Write the marginal as the mixture it is, $p = \sum_k w_k\, p_k$, and state where the weights w_k came from, since they belong to the setting rather than to the pathogen.

Where in the book. Chapter 14, Section 14.2.3. Related: Short-Term Simulation Assumptions, since holding the weights fixed while an outbreak runs is one, and Estimator Honesty, which guards against reporting a conditional estimate as though it answered the unconditional question.

16.4.4 Differential Ascertainment in a Ratio Estimator

When a parameter is estimated as a ratio of two observed counts, each count passes through its own detection process.

The tempting move. Trust the ratio because under-ascertainment cancels.

What goes wrong. Equal under-ascertainment does cancel, which is why a ratio estimator can survive a reporting level that defeats a count-based one. Unequal under-ascertainment does not cancel: it biases the estimate toward the more completely observed side. Markers of congenital Zika infection are 42 percent sensitive for first-trimester exposure, against 85 and 80 percent for the second and third, and the first trimester carries the highest risk, so a naive ratio understates exactly the cases that matter [Ades et al., 2021].

The fix. Carry an ascertainment factor on each side of the ratio and state what is assumed about each. Remember that a single multiplier cannot separate a small true ratio from thorough case-finding in the denominator.

Where in the book. Chapter 14, Section 14.2.3, with the multiplier correction in Chapter 8, Section 8.5. Related: Identifiability, since ascertainment and the parameter can trade off against each other; Data Preprocessing, the reporting-pipeline side; and, as a distinct question, Estimator Honesty.

16.4.5 Weighting Residuals by Their Uncertainty

Epidemic counts span orders of magnitude within a single outbreak, and so do the uncertainties of individual observations.

The tempting move. Fit by unweighted least squares, the first choice most modelers reach for.

What goes wrong. Unweighted least squares lets the largest observations decide the fit, is roughly twice as noisy as it needs to be, and reports a confidence interval that hides both facts. Losses under different weightings carry different dimensions, so their values cannot be compared with one another or across a change of reporting units. A weighting chosen once and forgotten is not safe either: Poisson weights are badly calibrated on the overdispersed data that surveillance systems produce.

The fix. Weight each squared residual by the inverse variance of its observation, $w_k = 1/\sigma_k^2$, the only choice that makes the loss dimensionless. Poisson counts have σ^2 proportional to the mean, constant-coefficient-of-variation data have it proportional to the square of the mean, and overdispersed counts lie between. Compare models through the likelihood or held-out prediction, never through the raw loss, and test the weighting with the scale factor (8.7).

Where in the book. Chapter 7; operationalized in Chapter 8, where the scale factor also flags data smoother than counting noise permits. Related: Estimator Honesty, for the interval failure, and Dimensional Consistency, for the units of the weights.

16.4.6 Reading Residuals in a Mechanistic Model

In a compartmental model, misfit lives in the structure of the model rather than in its error term.

The tempting move. Answer a failed residual test with the standard remedies: generalized least squares, a transformation, robust standard errors.

What goes wrong. Those remedies were built for models whose misfit lives in the noise, so no reweighting answers a test that failed because a mechanism is missing. Two readings are easy to get backwards. A failed test is strong evidence that something is missing and weak evidence about what, since rival mechanisms routinely whiten the same residuals. A passed test is weaker than it looks: only the part of a misspecification that varies across the fitting window reaches the residuals,

because a nearly constant discrepancy is absorbed by the fitted parameters, and overdispersion sets a noise floor that more data does not lower.

The fix. Read a failed test as grounds to reject the model, not to select its replacement, and read a passed test against the power that the weighting's scale factor predicts.

Where in the book. Chapter 8, Section 8.4. Depends on Weighting Residuals by Their Uncertainty, whose scale factor both gates the diagnostic and predicts its power; related to Identifiability, for why rival mechanisms can be indistinguishable, and Constant-Transmission Assumption Ignoring Behavioral Feedback, for one common candidate.

16.4.7 Identifiability

Fitting an epidemic model to data raises the question of which combinations of parameters the data can pin down and which they cannot. The data may be uninformative about individual parameters even when they are informative about combinations.

The tempting move. Minimize a loss over every parameter at once, then report each fitted value, to as many digits as the optimizer returns, as an estimate of that parameter.

What goes wrong. Early-outbreak data from the exponential-growth phase carry essentially two pieces of information: the growth rate and the starting time. A six-parameter inversion against that content cannot return six unique estimates. A value of β reported to six decimal places is then not an estimate of the per-contact transmission probability. It is an artifact of the path the optimizer took through a non-identifiable manifold. The diagnostics rarely give a clean verdict. More often the Hessian eigenvalues span many orders of magnitude: a few stiff directions that the data constrain, and a long tail of sloppy directions along which the loss barely changes and the optimizer drifts.

The fix. Report what the data identify, not what the optimizer returned. For early-outbreak data, that is the two or three dimensionless combinations the Buckingham Pi reduction makes explicit. Test before reporting: run multistart optimization and track where the parameters go, compute profile likelihoods, and monitor the condition number of the Hessian at the fitted minimum.

Where in the book. Chapter 7 develops the problem and works an example (Sections 7.3 and 7.3.2). Chapter 8 makes the diagnostics operational (Section 8.3), and an AI lab in the companion repository applies them.

16.4.8 Estimator Honesty

A comparison between estimators means something only when they solve the same inverse problem, at the same resolution, against the same data.

The tempting move. Make a comparison come out cleanly by converting one estimator through the other's assumptions, or by redefining the comparator to match the preferred one.

What goes wrong. Converting $\hat{\lambda}$ to $\mathcal{R}_0$ with the observed prevalence and the assumed population cancels exactly, so the two viewpoints agree and the comparison tests nothing; converting inconsistently manufactures a discrepancy at will. Redefining $\hat{\lambda}$ to use J_{obs}/s^*, with s^* chosen to match the data, builds in exactly the information the original procedure was supposed to test, and the redefined estimators agree on every dataset. The reverse error is to report one corruption factor as though it were a property of an estimator, when its size depends on the fitting window, the seed parameterization, and the loss function.

The fix. State the inverse problem each estimator solves before comparing them, and check that neither uses information its procedure would not have. Proposition 7.2 is stated this way, in terms of what each procedure can identify from the data it actually has.

Where in the book. Chapter 7, equation (7.19) (page 146) and Section 7.6.4, as the basis for the structural-immunity comparison; revisited in Chapter 8, where the same self-cancellation can enter identifiability diagnostics. Related: Force of Infection versus Force from Infection.

16.4.9 Generation-Time-Distribution Misspecification

Transfer out of a compartment at a constant rate γ_R assumes an exponentially distributed infectious period, and every reproductive-number estimate built on the model inherits that shape.

The tempting move. Keep the exponential shape because its mean is right.

What goes wrong. Measured generation intervals are rarely exponential: for most well-studied pathogens the distribution is peaked, with mean and variance free to differ. Estimates of $\mathcal{R}_0$ extracted from identical incidence data move by tens of percent as the assumed shape moves, and Hong et al. [2024] document the size of the bias for COVID-19. The pitfall lies in the shape of the distribution, not in which parameter names its mean.

The fix. Replace the rate constant with an infectiousness profile $A(\tau)$ over time since infection, which gives $\mathcal{R}_0 = \int_0^\infty K(\tau)\, d\tau$ for the empirical kernel $K(\tau)$.

Where in the book. Chapter 15, through the renewal-equation framework; Chapter 17, Section 17.4, where the SIR reading of a latent-period outbreak's growth rate gives 71 percent of $\mathcal{R}_0$. Related: The Independence of τ, which settles which parameter names the mean.

16.4.10 Sensitivity Analysis Discipline

The local sensitivity index $\Upsilon_p^q = (\partial q/\partial p)\,(p/q)$ yields useful conclusions only after two choices are made.

The tempting move. Report "the sensitivity of the model" to a parameter.

What goes wrong. The same parameter can matter very differently for different outputs: τ_R has unit elasticity for $\mathcal{R}_0$ but elasticity above 2 for i^* in the equal-contact-rate baseline, because invasion and burden are governed by structurally different parameter sets. A sensitivity taken with respect to a derived rate rather than a duration is not technically wrong, but it inverts the apparent direction of the effect, since $\Upsilon_{\gamma_R}^q = -\Upsilon_{\tau_R}^q$, and adds a translation step before the result can be acted on.

The fix. Choose the output the question asks about: $\mathcal{R}_0$ for emerging-pathogen surveillance, i^* for endemic burden, the time to peak or the peak prevalence for outbreak response. Take sensitivities with respect to the independent durations τ_R and τ_m, as this book does throughout.

Where in the book. Chapter 9 (page 190); applied in the case studies of Chapters 17 to 19. Related: The Independence of τ, Invasion versus Equilibrium, and Independence of Parameters of Interest.

16.4.11 Independence of Parameters of Interest

One-at-a-time sensitivity analysis assumes that the parameters can be changed independently of one another.

The tempting move. Vary each parameter of interest alone, holding the rest fixed, and read the indices side by side.

What goes wrong. When parameters are correlated, structurally, empirically, or by constraint, the indices misrepresent the response of the output to a joint change. In the SIR model, c_I and β enter the disease-free $\mathcal{R}_0$ only as the product $c_I\beta$: each has index $+1$, but the operationally meaningful index is that of the product, $\Upsilon^{\mathcal{R}_0}_{c_I\beta} = 1$, not the one-at-a-time sum, 2. For a dynamic output such as the integrated burden j, the same parameters separate through the contact-weighted denominator $C_{\text{total}} = c_S s + c_I i + c_R r$, so structural identifiability is a property of the output-parameter pair, not of the model alone.

The fix. Ask first whether the parameters can be varied independently in the system being modeled. Diagnose the correlation structure; reparameterize a structurally correlated pair as one composite parameter; and where empirical correlation can be quantified, report the total-derivative index $\Upsilon^q_{p_\ell}\big|_{\text{tot}} = \Upsilon^q_{p_\ell}\big|_{\text{std}} + \sum_{j\neq\ell} \Upsilon^q_{p_j} c_{\ell j}$, where $c_{\ell j}$ is the induced proportional change in p_j per unit proportional change in p_ℓ.

Where in the book. Chapter 9; the product structure first appears in Chapter 5, where the closed-form $\mathcal{R}_0$ is derived. Related: Hidden Monotonicity Assumption, whose variance-based remedy also addresses correlated parameters.

16.4.12 Hidden Monotonicity Assumption

The partial rank correlation coefficient (PRCC), the most common global sensitivity statistic in mathematical epidemiology, assumes through its rank construction that each parameter's effect on the output is monotonic.

The tempting move. Report a PRCC ranking from a Latin hypercube sample without looking at the shape of each response.

What goes wrong. When a parameter's effect rises and then falls across the sampled range, its PRCC collapses toward zero even if the parameter is highly influential, and the ranking misses it. The check for monotonicity is rarely shown in published applications. A small nonzero PRCC can also arise from sampling noise alone.

The fix. Plot the output against each parameter over the sample before reporting any ranking. Where the relationship is plainly non-monotonic, use a variance-based method such as eFAST [Marino et al., 2008], which needs no monotonicity and whose first-order and total indices $\mathcal{S}_\ell$ and $\mathcal{S}_{T_\ell}$ expose interactions through their gap. Include a dummy parameter in the design and report only PRCC magnitudes significantly larger than the dummy's.

Where in the book. Chapter 9, Sections 9.6.1 and 9.6.3. Related: Independence of Parameters of Interest.

16.5 Interpretation Pitfalls

16.5.1 $\mathcal{R}_0$ versus $\mathcal{R}_e(t)$

The basic reproductive number $\mathcal{R}_0$ is defined at the disease-free equilibrium and describes invasion; the effective reproductive number $\mathcal{R}_e(t) = \mathcal{R}_0\, P_S(t)$ describes transmission at the current state.

The tempting move. Report a reproductive number fitted from mid-outbreak data as $\mathcal{R}_0$, or read the threshold $\mathcal{R}_0 > 1$ as saying whether an outbreak already in progress is growing.

What goes wrong. The two coincide only at $t = 0$ in a naive population. An estimate fitted partway through an outbreak almost always estimates $\mathcal{R}_e$ at the data-time s/n, so reporting it as $\mathcal{R}_0$ inflates it by the factor $1/P_S$. The threshold theorem is a statement about invasion from a fully naive population; once susceptibles are depleted, whether the outbreak grows or declines is governed by $\mathcal{R}_e(t)$.

The fix. Answer public-health questions with $\mathcal{R}_e(t)$: whether the wave has crested, whether interventions can be relaxed, whether the epidemic is accelerating. Reserve $\mathcal{R}_0$ for invasion.

Where in the book. Chapter 5 (page 65); fitted estimates reframed in Chapter 7; applied throughout Chapter 17. Related: Invasion versus Equilibrium and NGM Applicability.

16.5.2 Invasion versus Equilibrium

Two distinct dynamical regimes earn distinct mathematical treatment.

The tempting move. Report a reproductive number as if it predicted endemic burden, or an endemic-burden estimate as if it predicted invasion threat. Both numbers come from the same model, so either one can seem to speak for the epidemic.

What goes wrong. The two regimes answer different questions with different machinery. The invasion regime asks whether a pathogen will establish itself in a never-infected population. Linear stability at the disease-free equilibrium settles it, and the parameters that matter are those in $\mathcal{R}_0$. The equilibrium regime asks for the long-run burden once the infection is established. Endemic-equilibrium analysis settles it, its time scale depends on demographic turnover, and the parameters that matter are those in i^*. The two parameter sets differ, so a parameter with high influence on $\mathcal{R}_0$ may have low influence on i^*, and the reverse.

The fix. Name the question before choosing the output. Analyze invasion through $\mathcal{R}_0$ and burden through i^*, and compute the sensitivities of the output that answers the question asked.

Where in the book. Chapter 5 makes the distinction formal (Sections 5.7 and 5.7.2), and Chapter 9 gives its quantitative version (Section 9.7). Related: NGM Applicability and Sensitivity Analysis Discipline.

16.5.3 NGM Applicability

The next-generation matrix method gives $\mathcal{R}_0$ as the spectral radius $\rho(\boldsymbol{F}\,\boldsymbol{V}^{-1})$ only inside its domain of validity, the five assumptions (A1) to (A5) [van den Driessche and Watmough, 2002].

The tempting move. Compute $\rho(\boldsymbol{F}\,\boldsymbol{V}^{-1})$ for any model and report it as $\mathcal{R}_0$.

What goes wrong. The assumption most often violated in applied work is (A4): with no infection present, nothing enters the infected compartments. A model with infected arrivals, such as refugees from a neighboring epidemic or infected migrants in cross-border surveillance, lies outside the domain, and its spectral radius is not $\mathcal{R}_0$. Arrivals into the recovered compartment, such as returning vaccinated travelers, do not break (A4); they move the disease-free state, and the construction must be evaluated there.

The fix. Check the assumptions before applying the formula. This book partitions arrivals as π_S, π_I, π_R, summing to one, and takes $\pi_S = 1$ unless stated otherwise, which keeps (A4) satisfied. A model with infected arrivals needs the construction itself revisited, not the formula rerun.

Where in the book. The in-migration form in Chapter 4; Chapter 5, Section 5.3.1, with a worked counterexample (page 68). Related: Linearization Fidelity, since the construction is a

linearization at the disease-free equilibrium; $\mathcal{R}_0$ versus $\mathcal{R}_e(t)$; and Invasion versus Equilibrium, since the method addresses the invasion threshold only and endemic-state interpretation needs the full-system Jacobian.

16.6 A Worked Audit

A checklist earns its place when it catches errors in a model that looks finished. This section takes one such model through Figure 16.1, check by check. The model is constructed. Every fault in it was put there on purpose, and every number below comes from a committed generator, so a reader can rerun the audit, change the faults and watch the checks respond.

The model and the report. Return to the town of ten thousand residents and its new respiratory infection. An analyst has written the book's SIR model with births and deaths for the town. Transmission is written per infectious resident, with force from infection $\alpha = c\beta s/n$. Residents leave the infectious compartment at the recovery rate γ_R and leave the population at the rate $\nu_m = 1/\tau_m$, with $\tau_m = 7300$ days. A contact survey gives $c = 10$ contacts a day, and clinical records give a mean infectious period of $\tau_R = 5$ days. The clinic reports every new case each day. The analyst fitted β to the first three weeks of counts and gave the health officer three findings. The first is $\beta = 0.519860$ and $\mathcal{R}_0 = 1.04$, so the outbreak will barely grow. The second is a sensitivity ranking: c, β and τ_R each have elasticity 1 for $\mathcal{R}_0$, and τ_m has elasticity 0.0007. The third, built on that ranking, is a forecast of long-run burden that leaves τ_m out. The town's true value is $\beta = 0.04$, so its basic reproductive number is $\mathcal{R}_0 = 2.00$. The analyst does not know this; the audit must find it.

Check 1: regime. The analyst fitted days 1 to 21, starting from the first reported case. By day 21 the town had 120 cumulative cases, 1.2 percent of its residents. Cumulative cases pass 3 percent on day 26 and 6 percent on day 30, so the whole fitting window lies below the early end of the deterministic regime (Regime Transition). The repair is to fit only inside the regime. Taking the low end of each band, that window runs from day 26, when cumulative cases pass 3 percent, to day 55, when they pass 70 percent.

Check 2: units and mixing. The rate equation for i fails the units check. The analyst typed the mean infectious period where the recovery rate belongs. The model therefore uses $\gamma_R = 5$, a number in days, and the term $\gamma_R i$ has units of person-days rather than persons per day. The rate is $\gamma_R = 1/\tau_R = 0.2$ per day (Dimensional Consistency; The Independence of τ). The fit did not reveal the error. Early counts fix the growth rate $c\beta - \gamma_R - \nu_m$, and the fit raised β until that rate matched. With $\gamma_R = 5$ per day the fitted β is 0.520; with 0.2 per day it is 0.038. The weighted sums of squares of the two fits differ by 0.38 over 21 days of counts, far too little to tell them apart. The units error moved the reported $\mathcal{R}_0$ from 1.90 to 1.04 and left the fit unchanged.

The mixing check passes, trivially, since the model has one group. It becomes live when the schools question splits the town into children and adults (Chapter 11).

Checks 3 and 4: frame and data. Both pass. Transmission is written in the α frame, the frame the fit uses (Force of Infection versus Force from Infection). The fit uses daily counts, each weighted by its Poisson variance (Weighting Residuals). The counts need no preprocessing only

Table 16.2: The worked audit at a glance. Every number comes from the committed generator.

Check	The report	What the check found	Pitfall
1 Regime	fits days 1 to 21	1.2 percent cumulative at day 21; 3 percent only on day 26	Regime Transition
2 Units, mixing	$\gamma_R = 5$	β 0.520 against 0.038, $\mathcal{R}_0$ 1.04 against 1.90, fits differ by 0.38; one group, so mixing passes	Dimensional Consistency; Independence of τ
3 Frame	α frame	passes	none
4 Data	Poisson-weighted daily counts	passes	none
5 Identifiability	τ_R fitted	runs to the 60-day bound; $\mathcal{R}_0$ from 1.18 to 4.60 over days 1 to 21, 1.98 to 2.17 over days 26 to 55	Identifiability
6 Interpretation	$\mathcal{R}_0$ ranking used for burden	τ_m elasticity 0.0007 for $\mathcal{R}_0$, -1.00 for i^*; $I^* = 3.4$	Invasion versus Equilibrium; Regime Transition

because the town was built to report every infection; real counts would. A pass is a finding too. It tells the officer which parts of the report need no more work.

Check 5: identifiability. Suppose the analyst, unsure of the clinical records, also fits τ_R. Over days 1 to 21 the fit runs to the upper bound placed on τ_R, 60 days, and returns $\mathcal{R}_0 = 11.7$. A fit that ends on a bound is a warning, and a profile shows why (Identifiability). Fix τ_R at each value from 1 to 20 days in steps of a quarter day, refit β and the initial count, and record the weighted sum of squares. Over the analyst's window every value fits within 0.04 of the best, so any $\mathcal{R}_0$ from 1.18 to 4.60 is consistent with the counts. Over the regime window, days 26 to 55, the same profile is sharp. A rise of 3.84 above the best fit marks a 95 percent interval, which runs from 4.75 to 6.00 days for τ_R and from 1.98 to 2.17 for $\mathcal{R}_0$. The repair of check 1 has repaired identifiability as well, because the regime window reaches past the exponential phase, where the counts carry more than a growth rate and a starting time. The report should still take $\tau_R = 5$ days from the clinical records, an independent source, and fit β over the regime window. That gives $\beta = 0.0403$ and $\mathcal{R}_0 = 2.02$, against the town's 2.00.

Check 6: invasion or burden. The report's last finding answers a burden question with an invasion tool. The elasticities of $\mathcal{R}_0$ say that τ_m hardly matters: 0.0007. The endemic state i^* says the opposite. Its elasticity to τ_m is -1.00, and its elasticity to τ_R is 2.00, twice that of $\mathcal{R}_0$ (Invasion versus Equilibrium). Leaving τ_m out of the forecast drops the parameter that sets the long-run burden. The endemic state also sends the audit back to check 1. At $I^* = 3.4$ infectious residents, the long-run question in a town this size concerns fade-out and reintroduction, a question for stochastic methods (Regime Transition).

The result. Table 16.2 gathers the six checks: four found faults and two passed. The corrected report tells the officer that $\mathcal{R}_0$ is about 2, not 1.04. The counts identify it once the window lies inside the deterministic regime and τ_R comes from the clinical records. Long-run burden turns on τ_m and τ_R, in a town too small for the deterministic endemic state to describe it. None of the four faults showed in the quality of the fit.

Part V

Recent Epidemics: Avoiding the Pitfalls in Applied Modeling

The degree of a belief is the extent to which we are prepared to act on it.

— Frank P. Ramsey

Three coupled differential equations cannot bear the weight of a forecast. The Lorenz system, distilled from the equations of atmospheric circulation, captures convective qualitative behavior — chaotic regimes, sensitive dependence, structural transitions — but no one would use it to predict tomorrow's weather. The same epistemic boundary applies to the three-equation SIR system. The case studies that follow apply the framework not to forecast outbreaks numerically, but to recognize the qualitative regimes of each pathogen and the common pitfalls that shaped the modeling choices. What we are prepared to act on, in Ramsey's sense, is the structural understanding the framework yields — not its quantitative output.

Chapter 17

Integrated Case Studies: Introduction and COVID-19

A forecast that disagreed with the outbreak was not a candidate for adjustment. It had been refuted.

The four preceding parts have developed the analytical framework: derivation of the SIR formulation from the infected viewpoint (Parts I-II), parameter fitting and sensitivity analysis (Part III), and extensions to heterogeneous populations, bipartite transmission cycles, and structured extensions (Part IV). Part V turns from apparatus to application. This chapter begins by motivating the case-study approach and introducing the common template, then applies the framework to COVID-19 as the first of three integrated case studies.

17.1 Why a Final Part on Case Studies?

The preceding chapters have developed a framework. The five-part development moves from the foundations of compartmental modeling (Part I), through the basic SIR model and its analysis (Part II), to the practical questions of fitting and sensitivity (Part III), to the extensions of the framework to richer biological structures (Part IV). Each chapter has been organized around a structural question: how do we derive the rate of new infections, how do we compute the basic reproductive number, how do we fit parameters when the susceptible population is uncertain, how do we partition parameters into invasion-controlling and burden-controlling groups, how do we extend the framework to heterogeneous populations or vector-borne pathogens.

Part V turns from structure to application. The three case studies in this part take three of the most-studied infectious diseases of the present era (COVID-19, dengue, and HIV) and apply the framework to each. The case studies do not aim to be authoritative epidemiological treatments of these diseases; specialist references such as Anderson and May [1991] and the disease-specific monographs provide that. The case studies aim instead to demonstrate how the framework works in practice on diseases whose biology actually makes use of its different components.

The case-study template. Each of the three disease chapters follows the same template. The first section sketches the epidemiological context: the pathogen, its transmission mode, its distinctive features, and the data typically available for fitting. The second section formulates the model, identifying which extensions of the basic SIR framework are required and presenting the resulting equations together with a compartmental block diagram. The third section addresses parameter

estimation: which parameters can be fit from the available data, which must be informed from outside the model, and what confidence can be assigned to each. The fourth section computes the basic reproductive number and conducts a sensitivity analysis. The fifth section evaluates a small set of intervention scenarios. The sixth section closes with the lessons that the case study illustrates about the framework. The exercises at the end of each chapter invite the reader to vary parameters, repeat the analyses with alternative model structures, or apply the framework to related pathogens.

What the case studies emphasize. The unifying methodological question across the four chapters is this: what can we estimate from the data in front of us, and how sensitive is that estimate to what we do not know? In each case study, the answer takes a specific form. For COVID-19, the asymptomatic compartment is invisible to incidence-based fitting, and the resulting identifiability problem shapes the entire analysis. For dengue, the vector population is rarely observed directly, and the model must be calibrated through human-side data alone. For HIV, the long natural history and the subgroup-structured contact network mean that fitting must integrate data from multiple sources at multiple time scales. In each case, the framework provides a principled answer to the identifiability question; the application yields conclusions that are both quantitative and honest about their uncertainty.

What the case studies are not. The case studies are not operational forecasts. We are not predicting the next variant or the next outbreak. We are using completed historical episodes (the early COVID-19 pandemic, the well-studied dengue dynamics in endemic settings, the HIV epidemic of the past four decades) as testbeds for the framework. Operational forecasting for active outbreaks combines the kind of mechanistic models developed in this book with statistical, ensemble, and machine-learning approaches that are outside the book's scope; Reich et al. [2019] surveys the state of operational infectious-disease forecasting.

The fourth chapter of Part V draws cross-cutting lessons from the three disease-specific case studies, identifying patterns in how the framework applies and where its limitations become binding. That chapter is the final substantive chapter of the book.

17.2 COVID-19: Epidemiological Context

SARS-CoV-2 emerged in late 2019 and became the central global health emergency of the early 2020s. It is a respiratory pathogen transmitted predominantly through respiratory droplets and aerosols, with a small contribution from fomite transmission. The basic biology that distinguishes it from earlier respiratory pandemics, and that drives the modeling choices in this chapter, is the combination of substantial pre-symptomatic and asymptomatic transmission, a strongly age-dependent severity profile, and rapid evolution of antigenically distinct variants over the first three years of the pandemic.

Asymptomatic and pre-symptomatic transmission. Estimates of the asymptomatic fraction of SARS-CoV-2 infections vary across studies, populations, and time periods, but the systematic review of Oran and Topol [2021] concludes that at least a third of infections proceed without symptoms severe enough to prompt testing or self-isolation. A separate phenomenon is pre-symptomatic transmission: even individuals who eventually develop symptoms shed virus and transmit infection in the day or two before symptom onset. Combined, pre-symptomatic and asymptomatic transmission account for roughly half of all transmission events in most settings, with the implication that

interventions targeting only symptomatic individuals (case isolation, symptom-based screening) cannot in principle suppress transmission below a substantial floor. This is the asymptomatic-reservoir problem we identified structurally in Chapter 10, Section 10.6; COVID-19 is its most consequential real-world instance.

Age structure of severity. Hospitalization and mortality from COVID-19 rise approximately exponentially with age. Population-averaged case fatality ratios for the wild-type variant ranged from approximately 0.5 percent to 2 percent depending on the age structure of the affected population and the adequacy of the medical response. Mortality in individuals over 80 years old can exceed 10 percent; mortality in children under 10 is below 0.01 percent. This age-dependence is biologically central but is not modeled explicitly in the case study below; an age-stratified extension is straightforward in the framework of Chapter 11 and is left as an exercise.

Variants and immune evasion. The Alpha, Delta, and Omicron variants of SARS-CoV-2 emerged at intervals of roughly six to twelve months during 2020–2022, each with distinguishable transmission and severity properties. From the framework's perspective, a new variant is captured by adjusting parameters: a higher per-contact transmission probability β (Alpha and Delta), a shorter latent period (Omicron), and varying degrees of escape from prior immunity. The structure of the model does not change; the parameter values do. The case study below treats the wild-type variant as representative; variant-specific re-fits would proceed identically. Detailed compartmental models that separately tracked detected versus undetected infections were central to real-time analysis during the pandemic's first wave [Giordano et al., 2020], and the waning-immunity literature of Huo et al. [2016] provides the template for extensions that incorporate imperfect cross-variant immunity.

Available data for fitting. Most national-level COVID-19 data streams during the first two years of the pandemic comprised: (i) confirmed case counts, with reporting delays and case-ascertainment biases that varied dramatically by jurisdiction; (ii) hospitalization counts, generally more reliable but lagged relative to incidence; (iii) all-cause and COVID-attributed death counts, the most reliable of the three but lagged by roughly three weeks from incidence; and (iv) seroprevalence surveys, which provide cross-sectional snapshots of cumulative incidence and are essential for grounding cumulative-attack-rate estimates. The case study below uses only the case-count data stream, illustrating both the power and the limitations of single-stream fitting.

The modeling choices implied. The framework provides most of the COVID-19-specific features with a natural representation. Asymptomatic transmission requires the SIAR structure of Chapter 10, Section 10.6. The latent period of three to five days warrants an exposed compartment, giving an SEIAR structure. Pre-symptomatic transmission is the exception: the latent compartment does not transmit, a limitation Section 17.7 returns to. Age structure, when relevant, layers the framework of Chapter 11 on top. Loss of immunity over months to years and immune escape by variants would warrant SIRS-style waning if the modeling time horizon were long enough; for the case study below, which addresses early-pandemic dynamics, we omit waning. The next section formalizes the resulting model.

Exercises for this section

1. [**C**] The COVID-19 chapter splits the infectious population into a symptomatic compartment i and an asymptomatic compartment A, each with its own contact rate. Show that, at the baseline, a single infectious class with the p-weighted mean contact rate gives the same $\mathcal{R}_0$. Then explain in plain English why the split still matters for the isolation scenario of Section 17.6.

17.3 Model Formulation

The case-study model is a six-compartment SEIAR structure that combines the latent compartment of Chapter 10, Section 10.3 with the asymptomatic-branching structure of Section 10.6. The compartmental layout is shown in Figure 17.1.

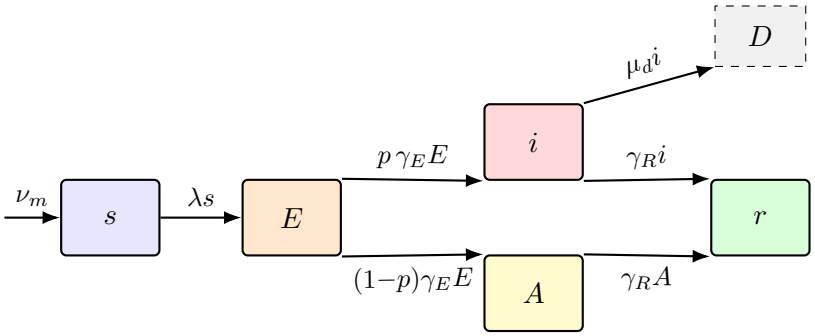


Figure 17.1: The SEIAR compartmental structure used for the COVID-19 case study. Susceptible individuals exposed to the pathogen at force of infection λ enter the exposed (latent) compartment E; on completion of the latent period at rate γ_E, a fraction p progress to the symptomatic compartment i and the remaining $(1-p)$ to the asymptomatic compartment A. Both transmit, with potentially distinct contact rates c_I and c_A. Symptomatic individuals can additionally exit to the dead sink D at rate μ_d (disease-induced mortality, dashed). Demographic outflows from each living compartment at rate ν_m are omitted for clarity.

The governing equations are

$$\dot{s} = \nu_m - \lambda s - \nu_m s, \tag{17.1a}$$
$$\dot{E} = \lambda s - \gamma_E E - \nu_m E, \tag{17.1b}$$
$$\dot{i} = p\,\gamma_E E - (\gamma_R + \mu_d + \nu_m) i, \tag{17.1c}$$
$$\dot{A} = (1-p)\,\gamma_E E - (\gamma_R + \nu_m) A, \tag{17.1d}$$
$$\dot{r} = \gamma_R (i + A) - \nu_m r, \tag{17.1e}$$

with the force of infection

$$\lambda \;=\; (c_I \beta_I i + c_A \beta_A A) \cdot \frac{P_S}{s} \;\approx\; \frac{c_I \beta_I i + c_A \beta_A A}{n}, \tag{17.2}$$

where the second form takes the standard $P_S \approx s/n$ approximation in the homogeneous-mixing limit and treats compartments-other-than-s as approximately uniform contributors to n. The probability p that a newly-infectious individual becomes symptomatic is taken as a parameter (best estimate

≈ 0.6 for SARS-CoV-2 wild type). Both infectious tracks recover at the same rate γ_R: the model assumes that asymptomatic infections stay infectious as long as symptomatic ones, the equal-rate case of Chapter 10, equation (10.14). Section 17.4 shows how much the headline share depends on it.

Notation note. Equation (17.2) defines λ, the force *of* infection: the per-susceptible rate that multiplies s in $\dot{s} = -\lambda s$. This is the natural object in a multi-source model, because the right-hand side aggregates contributions from both the symptomatic and asymptomatic compartments, and no single source population divides it. The book's primary symbol α is reserved throughout for the force *from* infection, the per-source rate satisfying $j = \alpha i$; the two are related by the identity $\lambda s = \alpha i$ of Chapter 3. Keeping them distinct matters here because this chapter uses both: the model is driven by λ, while the fitting estimator of Section 17.4 is $\hat{\alpha} = j/i$, a per-source quantity. Writing a single symbol for both would collapse exactly the distinction the book is about.

$$j(t) \;=\; \lambda\, s \;=\; (c_I \beta_I i + c_A \beta_A A) \cdot P_S,$$

consistent with the $j(t)$ defined in Chapter 3 equation (3.2). The fitting estimator $\hat{\alpha} = j/i$ used in Section 17.4 below builds on this $j(t)$ through the ratio of symptomatic incidence to symptomatic prevalence, and that section shows what the ratio measures. It is a per-source rate, distinct from the per-susceptible λ of equation (17.2).

Reproductive number. Linearizing the (E, i, A) subsystem at the disease-free equilibrium $(s, E, i, A) = (1, 0, 0, 0)$ and computing the next-generation matrix as in Chapter 5, Section 5.3, the basic reproductive number is

$$\mathcal{R}_0 \;=\; \frac{\gamma_E}{\gamma_E + \nu_m} \cdot \left[p\, \frac{c_I \beta_I}{\gamma_R + \mu_d + \nu_m} + (1 - p)\, \frac{c_A \beta_A}{\gamma_R + \nu_m} \right]. \tag{17.3}$$

The first factor is the probability that an exposed individual completes the latent period before migrating out of the population (close to unity for COVID-19 because $\nu_m \tau_E \approx 4 \cdot 3.4 \times 10^{-5} \approx 10^{-4}$). The bracket is the p-weighted contribution from each infectious track. The asymptomatic track's denominator does not include μ_d because asymptomatic individuals do not develop severe disease.

Connection to the framework. The two-track structure of (17.3) is the same structure analyzed in Chapter 10, Section 10.6, with the addition of the latent-period factor from the SEIR extension. The invasion-burden partition of Chapter 5, Section 5.7 extends naturally: the contact rates c_I, c_A and transmission probabilities β_I, β_A are invasion-facing; the recovery rate γ_R is mixed (as in the basic model); the symptomatic fraction p is invasion-facing in a non-trivial way (its sensitivity sign depends on the relative magnitudes of $c_I \beta_I$ and $c_A \beta_A$, as we examine in Section 17.5); and the demographic parameter ν_m is burden-facing but contributes negligibly at the time scales of the early pandemic. Proposition 10.2 in Chapter 10, Section 10.7 guarantees that the partition structure transfers from the basic model to this extension.

Exercises for this section

1. **[D]** Recompute $\mathcal{R}_0$ for the Delta variant, using $\beta_I = \beta_A = 0.10$ and $\tau_R = 5$ days (other parameters unchanged from the wild-type baseline). What is the new $\mathcal{R}_0$? What fraction now comes from the asymptomatic track?

2. **[D]** The baseline gives both infectious tracks the same period τ_R. Restore a distinct asymptomatic period $\tau_A = 1/\gamma_A$, as in Chapter 10, equation (10.13), and find the value of τ_A at which the asymptomatic track supplies exactly half of $\mathcal{R}_0$, and check it against the threshold quoted in Section 17.4. What does the answer imply for the claim that more than half of transmission proceeds without symptomatic warning?

17.4 Parameter Estimation and Fitting

The framework's distinctive contribution to parameter estimation is the infected-viewpoint estimator $\hat{\alpha} = j/i$ developed in Chapter 7, which fits the force-from-infection from the observable ratio of incidence to prevalence without requiring an estimate of the susceptible population. We illustrate that estimator here for the SARS-CoV-2 wild-type strain. When the estimator is applied to a region's record, the first weeks, with cumulative counts in the tens, lie in the stochastic regime and belong outside the deterministic fit (Pitfall: "The Stochastic-to-Deterministic Regime Transition"); Chapter 8, Section 8.7 gives the diagnostic.

Best-estimate baseline parameters. The following parameter values represent a consensus of early-pandemic estimates for the wild-type strain in a non-elderly adult population:

- Latent period: $\tau_E = 4$ days, so $\gamma_E = 0.25$ day^{-1}.
- Infectious period: $\tau_R = 7$ days, so $\gamma_R = 0.143$ day^{-1}.
- Symptomatic contact rate: $c_I = 5$ contacts day^{-1} (reflecting partial self-isolation by symptomatic individuals).
- Asymptomatic contact rate: $c_A = 10$ contacts day^{-1} (no behavioral reduction).
- Susceptible contact rate: $c_S = 10$ contacts day^{-1}, the survey-anchored adult value [Mossong et al., 2008, Prem et al., 2017].
- Per-contact transmission probability: $\beta_I = \beta_A = 0.05$ (taken equal in the absence of strong evidence for asymmetry; some studies suggest $\beta_A < \beta_I$ by a small factor).
- Symptomatic fraction: $p = 0.6$.
- Disease-induced mortality rate: $\mu_d \approx 7 \times 10^{-4}$ day^{-1}, corresponding to a CFR of approximately 0.5 percent in the symptomatic compartment (Pitfall: "Conditional versus Marginal Probabilities").
- Demographic turnover: $\tau_m = 80$ years, so $\nu_m \approx 3.4 \times 10^{-5}$ day^{-1} (negligible at pandemic timescales).

Computing the baseline $\mathcal{R}_0$. Substituting these values into (17.3),

$$\begin{aligned}
\mathcal{R}_0 &\approx 1.0 \cdot \left[0.6 \cdot \frac{5 \cdot 0.05}{0.1436} + 0.4 \cdot \frac{10 \cdot 0.05}{0.1429}\right] \\
&= 0.6 \cdot 1.741 + 0.4 \cdot 3.499 \\
&= 1.045 + 1.400 \\
&\approx 2.444.
\end{aligned}$$

The estimate $\mathcal{R}_0 \approx 2.44$ for wild-type SARS-CoV-2 is consistent with most early-pandemic estimates [Li et al., 2020, Riou and Althaus, 2020]. The asymptomatic compartment contributes $1.400/2.444 \approx 57$ percent of $\mathcal{R}_0$: more than half of transmission proceeds without symptomatic warning. That share rests on the model's assumption that both tracks stay infectious equally long (Section 17.3). If asymptomatic infectious periods are more than about a quarter shorter, with τ_A below about 5.2 days, the asymptomatic share falls below one half. A systematic review reports that direction, though not its size: most of its studies that compared the two found faster viral clearance in asymptomatic infection, and data on infectious virus in asymptomatic individuals remain scarce [Cevik et al., 2021].

Parameter-uncertainty interval on $\hat{\mathcal{R}}_0$. The point estimate $\mathcal{R}_0 \approx 2.44$ is computed from the consensus baseline parameters with no uncertainty propagated through. To quantify how far the stated parameter uncertainties carry through, we use the delta-method theorem 9.1 of Chapter 9: with the dominant uncertainties being τ_R ($\pm 20\%$, 95% CI), the symptomatic-track parameters c_I and β_I ($\pm 25\%$ each), the asymptomatic-track parameters c_A and β_A ($\pm 30\%$ each), and the symptomatic fraction p (± 0.10 around 0.6, equivalent to roughly $\pm 17\%$ relative), each half-width read as two standard deviations, and using the closed-form sensitivity indices computed in Section 17.5,

$$\begin{aligned} V^{\mathcal{R}_0} &\approx (\Upsilon^{\mathcal{R}_0}_{\tau_R})^2 V^{\tau_R} + (\Upsilon^{\mathcal{R}_0}_{c_I})^2 V^{c_I} + (\Upsilon^{\mathcal{R}_0}_{\beta_I})^2 V^{\beta_I} \\ &\quad + (\Upsilon^{\mathcal{R}_0}_{c_A})^2 V^{c_A} + (\Upsilon^{\mathcal{R}_0}_{\beta_A})^2 V^{\beta_A} + (\Upsilon^{\mathcal{R}_0}_{p})^2 V^{p} \\ &\approx (0.998)^2(0.10)^2 + (0.43)^2(0.125)^2 + (0.43)^2(0.125)^2 \\ &\quad + (0.57)^2(0.15)^2 + (0.57)^2(0.15)^2 + (-0.43)^2(0.085)^2 \\ &\approx 0.0100 + 0.0029 + 0.0029 + 0.0073 + 0.0073 + 0.0013 \\ &\approx 0.0317. \end{aligned}$$

The implied relative standard deviation is $\sqrt{0.0317} \approx 0.178$, giving a 95% parameter-uncertainty interval for $\mathcal{R}_0$ of approximately $\mathcal{R}_0 \cdot (1 \pm 1.96 \cdot 0.178) = \mathcal{R}_0 \cdot (1 \pm 0.349)$, or numerically $[1.59, 3.30]$ around the point 2.444. The dominant variance contributions are τ_R (31%), c_A and β_A (23% each), with c_I, β_I, p contributing the remaining $\sim 22\%$. This interval is consistent with the published range of early-pandemic $\mathcal{R}_0$ estimates [Li et al., 2020, Riou and Althaus, 2020], which span roughly $[1.4, 3.9]$ depending on assumed parameter values; the wider published range reflects parameter uncertainties beyond those tabulated here. A Monte Carlo propagation of the same ranges (Latin hypercube, $M = 200{,}000$, each half-width read as two standard deviations) gives $[1.66, 3.39]$, in good agreement with the delta-method approximation given here. The interval propagates the stated parameter uncertainties only: it assumes they are independent (Section 8.2 carries the correlated form), it takes no account of model-structure uncertainty, and because the inputs are plausible ranges rather than estimated standard errors it carries no coverage guarantee.

Fitting from incidence data. Suppose we observe daily case counts $j_{\text{obs}}(t)$ for the first thirty days of an outbreak in a previously naive population. The infected-viewpoint estimator is

$$\hat{\alpha}(t) \;=\; \frac{j_{\text{obs}}(t)}{i(t)}.$$

The estimator requires an estimate of the prevalence $i(t)$, which can be obtained either from independent surveys of symptomatic prevalence or by integrating the incidence over a presumed-known

infectious-period kernel. The estimator does not require an estimate of the susceptible population n, which would be needed by the susceptible-viewpoint estimator $\hat{\lambda} = j/s$.

For early-outbreak data where $s \approx n$, the two estimators produce nearly identical point estimates. The advantage of the infected-viewpoint estimator becomes apparent when the susceptible population is uncertain by a factor of two or three. Figure 1 in the Preface visualizes this advantage: a three-fold uncertainty in s produces a roughly three-fold uncertainty in the susceptible-viewpoint $\hat{\lambda}$, and whether it reaches $\mathcal{R}_0$ depends on the fit (Chapter 7), while the infected-viewpoint $\hat{\alpha}$ is nearly immune to the same uncertainty.

Identifiability of the asymptomatic compartment. A more difficult challenge for COVID-19 fitting is the invisibility of the asymptomatic compartment in case-count data. Symptomatic cases are reported; asymptomatic cases are not (except through serological surveys conducted weeks or months after infection). The case-count time series $j_{\text{obs}}(t)$ measures only the symptomatic incidence $p\gamma_E E$, not the total rate $\gamma_E E$ at which exposed individuals become infectious. Without independent information on p, the model is unidentifiable: any rescaling of p can be absorbed into the contact rates c_I and c_A to give an equivalent fit to the symptomatic case data.

It is tempting to conclude that a fit to symptomatic case counts sees only the symptomatic track, and so recovers only part of $\mathcal{R}_0$. The calculation says otherwise. During growth the symptomatic prevalence obeys $\dot{i} = \sigma i$, so the third equation of (17.1) gives $\hat{\alpha} = j_{\text{obs}}/i = \sigma + \gamma_R + \mu_d + \nu_m$ whatever the value of p: the ratio measures the growth rate. Two conditions keep it honest. First, the conversion from growth rate to $\mathcal{R}_0$ must carry the latent period. Read as in the SIR model, $\mathcal{R}_0 = \hat{\alpha}/(\gamma_R + \nu_m)$ (Chapter 7, equation (7.13)), the ratio gives 1.73 at the baseline, 71 percent of the true value, and the latent period accounts for the whole shortfall (Pitfall: "Generation-Time-Distribution Misspecification"). With the latent period included, $\mathcal{R}_0 \approx (1+\sigma\tau_E)(1+\sigma\tau_R)$ recovers $\mathcal{R}_0$ to about 0.1 percent. Because the two tracks share the infectious period τ_R (Section 17.3), this conversion needs neither s nor p. Second, the numerator and denominator must count the same people. Symptomatic onsets divided by a survey's total prevalence $i + A$ give a ratio smaller by about the factor p, and the same SIR reading then gives 1.04, about 42 percent of $\mathcal{R}_0$. Figure 17.2 shows the three readings in the chapter's model.

What case counts cannot reveal is how $\mathcal{R}_0$ divides between the tracks, since the rescaling of p described above leaves the growth rate unchanged. The split must come from outside the case counts: serological surveys for p, and contact-tracing studies that test asymptomatic close contacts for how much asymptomatic cases transmit. It is exactly what the isolation scenario of Section 17.6 turns on. The case study takes $p = 0.6$ as informed by the early-pandemic literature; a sensitivity to this assumption appears in Section 17.5.

AI Lab: Real-Time COVID-19 Fitting Workflow

Fitting a compartmental model to a fast-moving outbreak means working with reporting delays, shifting ascertainment, interventions that change parameters, and biology still being learned. The workflow is a *three-step audit-and-refine*.

Step 1: data-vintage inventory. Rank the sources by recency and reliability. Recent case counts are revised upward as late reports arrive; hospitalizations lag cases but depend less on testing; deaths lag most and are the most reliable. Ask the AI which series should be the fitting target, which a cross-check, and which a source on the symptomatic fraction p. A correct answer weighs recency (cases) against accuracy (deaths); fitting every series at once ignores the vintage problem.

Step 2: time-varying parameters. Interventions changed transmission during the case-study period, so a static fit averages across regimes. Ask the AI to date the regime changes from policy

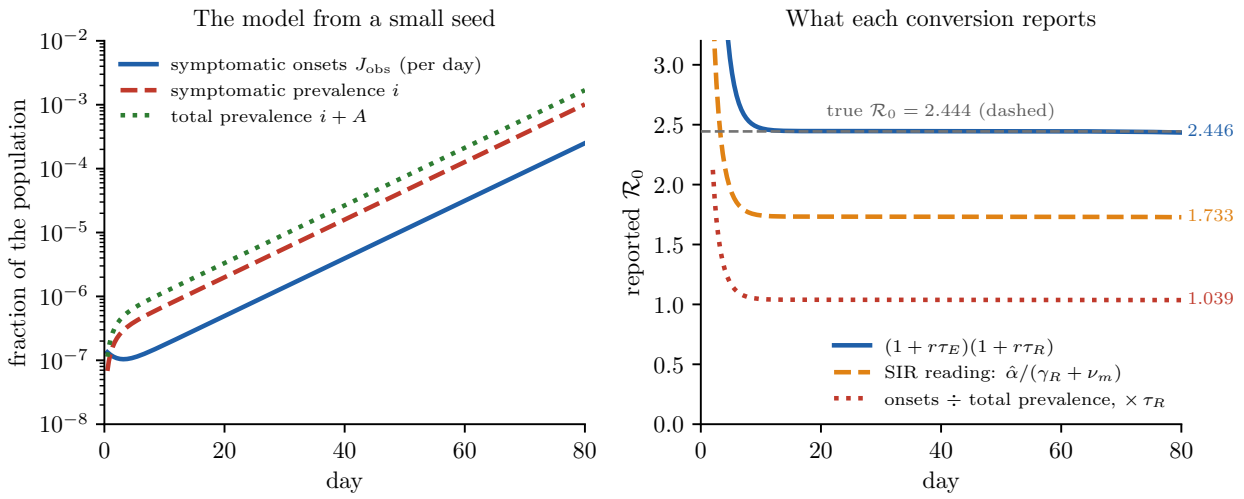


Figure 17.2: What the infected-viewpoint ratio recovers in the chapter's SEIAR model, at the baseline of Section 17.4, run from a small seed. Left: symptomatic onsets j_{obs}, symptomatic prevalence i and total prevalence $i + A$ grow at a common rate σ once the seed's transient has passed. Right: three readings of the case-count ratio against the true $\mathcal{R}_0 = 2.444$ (dashed). The conversion that carries the latent period recovers it. The SIR reading $\hat{\alpha}/(\gamma_R + \nu_m)$ gives 1.73, because it ignores the latent period. Symptomatic onsets divided by total prevalence give 1.04, because the numerator and denominator count different people. The relation $\hat{\alpha} = \sigma + \gamma_R + \mu_d + \nu_m$ holds whatever p is.

records and propose a piecewise parameter form. A correct answer names specific policy events and ties each to a parameter.

Step 3: forecast or nowcast. The case study nowcasts, which needs reporting delays and ascertainment shifts (Pitfall: "Correct Data-Fitting Practice"); forecasting needs assumptions about future behavior. Ask the AI which your task is; a correct answer decides and matches the method to it.

Disclose AI assistance, including AI-supplied regime dates and their policy sources (Appendix D, Section D.4).

17.5 Sensitivity Analysis

The sensitivity analysis uses the methodology of Chapter 9: relative sensitivity indices for $\mathcal{R}_0$, computed analytically from (17.3) where possible and supplemented by numerical Latin-hypercube-and-PRCC computation where the algebra is unwieldy.

Analytic sensitivities. For each parameter θ, the relative sensitivity is $\Upsilon_\theta^{\mathcal{R}_0} = (\partial \log \mathcal{R}_0)/(\partial \log \theta)$. Direct computation from (17.3) gives

$$\begin{aligned}
\Upsilon_{\beta_I}^{\mathcal{R}_0} &= \frac{p \cdot c_I \beta_I/(\gamma_R + \mu_d + \nu_m)}{\mathcal{R}_0} \approx 0.43, \\
\Upsilon_{\beta_A}^{\mathcal{R}_0} &= \frac{(1-p) \cdot c_A \beta_A/(\gamma_R + \nu_m)}{\mathcal{R}_0} \approx 0.57, \\
\Upsilon_{c_I}^{\mathcal{R}_0} &= \Upsilon_{\beta_I}^{\mathcal{R}_0} \approx 0.43, \\
\Upsilon_{c_A}^{\mathcal{R}_0} &= \Upsilon_{\beta_A}^{\mathcal{R}_0} \approx 0.57, \\
\Upsilon_{\tau_R}^{\mathcal{R}_0} &= \Upsilon_{\beta_I}^{\mathcal{R}_0} \frac{\gamma_R}{\gamma_R + \mu_d + \nu_m} + \Upsilon_{\beta_A}^{\mathcal{R}_0} \frac{\gamma_R}{\gamma_R + \nu_m} \approx 1.00, \\
\Upsilon_{\gamma_E}^{\mathcal{R}_0} &= \frac{\nu_m}{\gamma_E + \nu_m} \approx 1.4 \times 10^{-4}.
\end{aligned}$$

The sensitivity to p is more interesting. Differentiating (17.3) with respect to p and dividing by $\mathcal{R}_0$,

$$\Upsilon_p^{\mathcal{R}_0} = \frac{p}{\mathcal{R}_0} \left[\frac{c_I \beta_I}{\gamma_R + \mu_d + \nu_m} - \frac{c_A \beta_A}{\gamma_R + \nu_m} \right].$$

For our baseline parameters, the bracketed term is negative ($1.741 < 3.499$, with difference -1.758), so $\Upsilon_p^{\mathcal{R}_0}$ is negative: increasing the symptomatic fraction *decreases* $\mathcal{R}_0$, because shifting individuals from the high-contact asymptomatic track to the lower-contact symptomatic track reduces overall transmission. Numerically, with $p = 0.6$ and $\mathcal{R}_0 \approx 2.444$:

$$\Upsilon_p^{\mathcal{R}_0} = \frac{0.6}{2.444}(1.741 - 3.499) = 0.2455 \cdot (-1.758) \approx -0.432.$$

The magnitude $|\Upsilon_p^{\mathcal{R}_0}| \approx 0.43$ is comparable to that of c_I and β_I (the symptomatic-track parameters), reflecting the substantial influence of the symptomatic-fraction parameter on the asymptomatic-reservoir contribution to $\mathcal{R}_0$.

Tornado plot. Figure 17.3 ranks the parameters by the magnitude of their relative sensitivity. The dominant parameters are τ_R (sensitivity essentially unity) and the asymptomatic-track parameters c_A and β_A (sensitivity $\sim +0.57$ each). The index of τ_R is near one because the single τ_R stands for the infectious periods of both tracks; given separate periods, it would split about 0.43 to 0.57 between them. The symptomatic-track parameters c_I and β_I are smaller (sensitivity $\sim +0.43$). The latent-period parameter γ_E is negligible. The symptomatic fraction p has a substantial negative sensitivity of magnitude ~ 0.43 (comparable to the symptomatic-track parameters) reflecting the asymptomatic-reservoir effect with the influence one expects from a parameter that controls the split between two transmission tracks.

Implications for intervention design. The tornado plot translates directly into intervention recommendations. The largest invasion-facing sensitivities are to the asymptomatic-track parameters; interventions targeting only symptomatic individuals (case isolation, symptom-based screening) cannot in principle reduce these parameters and therefore cannot reduce the asymptomatic contribution to $\mathcal{R}_0$. Effective intervention strategies must include a component that reduces c_A (mass testing of asymptomatic individuals followed by isolation, or non-pharmaceutical reductions in contact such as remote work) or a component that reduces β_A uniformly (universal masking, ventilation, and similar measures that act on per-contact transmission regardless of symptom status). The intervention scenarios in Section 17.6 examine these strategies quantitatively.

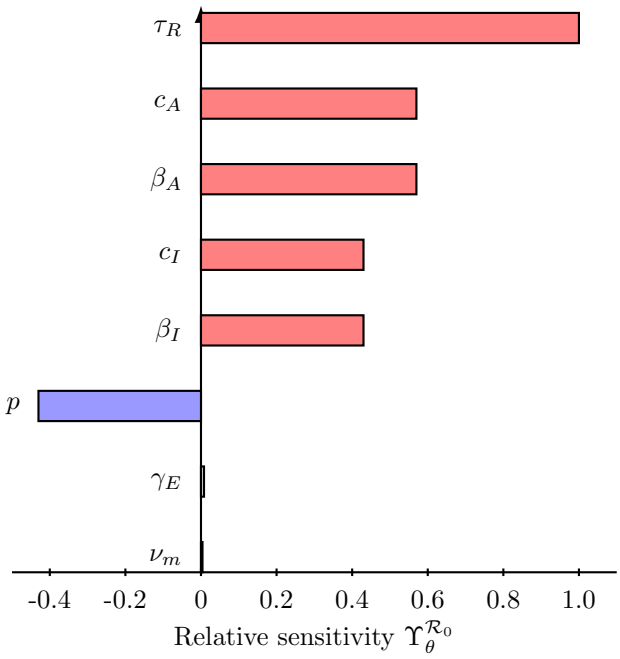


Figure 17.3: Tornado plot of relative sensitivity indices $\Upsilon_\theta^{\mathcal{R}_0} = \partial \log \mathcal{R}_0 / \partial \log \theta$ for the baseline COVID-19 parameters. Red bars indicate positive sensitivities (parameter increases drive $\mathcal{R}_0$ up); the blue bar for p indicates a negative sensitivity (increasing the symptomatic fraction reduces $\mathcal{R}_0$, since the asymptomatic track is more transmissive at baseline parameters). The infectious-period τ_R dominates with near-unit sensitivity. The asymptomatic-track parameters c_A and β_A contribute about a third more to $\mathcal{R}_0$ sensitivity than the symptomatic-track parameters, formalizing the asymptomatic-reservoir problem identified in Chapter 10, Section 10.6. The symptomatic-fraction parameter p has a substantial negative sensitivity (≈ -0.43), comparable in magnitude to the symptomatic-track parameters, reflecting the substantial influence of p on the asymptomatic-reservoir contribution. The latent-period rate γ_E and the demographic rate ν_m have negligible sensitivity at the time scales of the early pandemic.

Exercises for this section

1. **[L]** *LHS-PRCC analysis applied to the COVID-19 model.* Implement a Latin hypercube sampling and partial-rank-correlation-coefficient (LHS-PRCC) analysis of the SEIAR model (17.1) for both $\mathcal{R}_0$ and the peak symptomatic prevalence $\max_t I(t)/N^{\mathrm{c}}$. Use the methodology of Chapter 9, Section 9.6 and the implementation guidance of Appendix B, Section B.2. Take the baseline parameters of this section as the central values, vary each parameter except p independently within a factor of 2 above and below baseline (uniform on the multiplicative range, equivalent to log-uniform on the original scale), give p the range 0.6 ± 0.10 of the delta-method calculation above, and use $M = 5000$ samples. Report:

   (a) PRCC estimates with 95% confidence intervals for the eight parameters $\{\beta_I, \beta_A, c_I, c_A, \tau_R, \gamma_E, p, \nu_m\}$, separately for the two outputs $\mathcal{R}_0$ and peak symptomatic prevalence.

   (b) A side-by-side comparison with the closed-form local sensitivity indices given in this section: rank the parameters by absolute PRCC and by absolute local sensitivity, and identify any parameters whose ranking differs between the two methods.

   (c) The sign of the PRCC for p, the symptomatic fraction. The local analysis at baseline gave $\Upsilon_p^{\mathcal{R}_0} \approx -0.43$ (negative); verify whether the global analysis preserves this sign across the full parameter range, and identify any subregions of parameter space (if any) where the sign would flip. (Hint: consider the algebraic condition under which $c_I\beta_I/(\gamma_R + \mu_d + \nu_m) > c_A\beta_A/(\gamma_R + \nu_m)$, i.e. when the symptomatic-track contribution to $\mathcal{R}_0$ exceeds the asymptomatic-track contribution.)

   For (b), comment in a brief paragraph on whether the LHS-PRCC ranking would change the intervention recommendations of this section. Use the SEIAR model of the chapter's generator in the companion repository (`notebooks/ch17/ch17_figures.py`) as the forward model; the LHS sample design and PRCC computation follow the same recipe as the SIR implementation referenced in Chapter 9. Cite Saltelli et al. [2008] and McKay et al. [1979] as appropriate.

17.6 Intervention Scenarios

We evaluate three intervention strategies that span the asymptomatic-track and symptomatic-track parameter groups identified in Section 17.5 (Pitfall: "Sensitivity Analysis Discipline"). The methodology follows the model-based intervention-evaluation tradition exemplified for an earlier respiratory pandemic threat by Ferguson et al. [2005]: scenario simulation forward from a calibrated baseline with each intervention's parameter changes applied, compared against the no-intervention trajectory.

Scenario 1: universal masking (50 percent reduction in β_I and β_A). Suppose universal masking halves β_I and β_A. The factor is illustrative; the evidence that masks reduce per-contact transmission is reviewed by Howard et al. [2021]. The reproductive number (17.3) scales linearly in the bracket, so

$$\mathcal{R}_0^{\mathrm{mask}} \approx 0.5 \cdot 2.44 = 1.22.$$

Above the invasion threshold, but substantially reduced. Masking alone does not suppress the outbreak; combined with other measures, it can.

Scenario 2: aggressive symptomatic-case isolation (75 percent reduction in c_I). Reducing c_I from 5 to 1.25 contacts per day, the new bracket value is

$$0.6 \cdot \frac{1.25 \cdot 0.05}{0.143} + 0.4 \cdot \frac{10 \cdot 0.05}{0.143} \;=\; 0.262 + 1.399 \;=\; 1.661,$$

so $\mathcal{R}_0^{\text{isolate}} \approx 1.66$. The intervention substantially reduces the symptomatic-track contribution but cannot affect the asymptomatic-track contribution ($1.40/1.66 \approx 84$ percent of the post-intervention $\mathcal{R}_0$ comes from the asymptomatic track). This is the asymptomatic-reservoir problem in concrete numerical form.

Scenario 3: combined intervention. A combined strategy of universal masking ($\beta \to 0.025$), symptomatic case isolation ($c_I \to 1.25$), and asymptomatic detection-and-isolation reducing c_A to 5 (corresponding to mass testing identifying half of asymptomatic individuals and inducing equivalent contact reduction) gives

$$\mathcal{R}_0^{\text{combo}} \;=\; 0.6 \cdot \frac{1.25 \cdot 0.025}{0.143} + 0.4 \cdot \frac{5 \cdot 0.025}{0.143} \;=\; 0.131 + 0.350 \;=\; 0.481.$$

Below threshold; the combined strategy can suppress the outbreak. The conclusion that no single intervention is sufficient but a combination of complementary interventions can succeed is one of the central operational lessons of the pandemic, and it falls out of the framework directly through the additive structure of the two-track $\mathcal{R}_0$.

Why $\mathcal{R}_0$ is the right summary. The intervention scenarios are evaluated through their effect on $\mathcal{R}_0$, not through forward simulation of the full ODE system. Per Theorem 5.6 and Corollary 5.7, $\mathcal{R}_0$ is the right summary statistic for invasion-control interventions: an intervention that reduces $\mathcal{R}_0$ below unity prevents sustained transmission. In the unsuppressed case the final size depends on $\mathcal{R}_0$ alone (Section 5.5.4; the relation carries over to this model): about 89 percent of the population is eventually infected at the baseline, 67 percent under isolation and 34 percent under masking. Forward simulations would add quantitative detail to the burden estimates but would not change the qualitative conclusions drawn here.

Exercises for this section

1. [**I**] For the Omicron variant, use $\tau_E = 2$ days, $\tau_R = 5$ days, $\beta_I = \beta_A = 0.08$, and $p = 0.4$ (more asymptomatic). Compute $\mathcal{R}_0$ and identify which intervention scenario from Section 17.6, with its changes applied as factors (β halved, c_I quartered, c_A halved), would now be required for $\mathcal{R}_0 < 1$.

17.7 Lessons from the COVID-19 Case Study

The case study illustrates four general lessons about how the framework applies to a real pathogen.

First, the asymptomatic-reservoir problem of Chapter 10, Section 10.6 is the central practical constraint on COVID-19 control. The framework identifies it structurally as a property of the SIAR model class; the case study shows that for the wild-type SARS-CoV-2 it accounts for approximately 57 percent of $\mathcal{R}_0$, if asymptomatic infections stay infectious as long as symptomatic ones. The implication is that interventions targeting only symptomatic individuals cannot suppress transmission

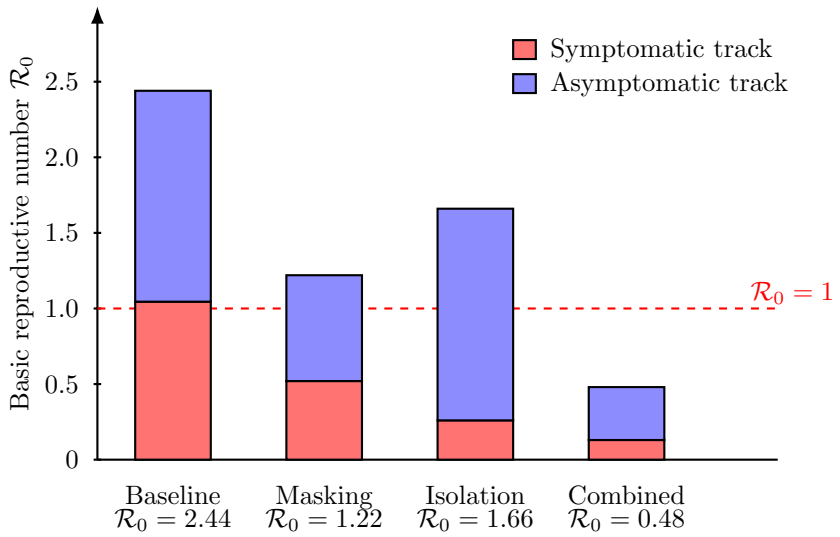


Figure 17.4: Intervention-scenario comparison for COVID-19. Each bar decomposes $\mathcal{R}_0$ into its symptomatic (red) and asymptomatic (blue) contributions per equation (17.3). The baseline $\mathcal{R}_0 = 2.44$ comprises 1.04 symptomatic + 1.40 asymptomatic (57% asymptomatic). Universal masking halves β and reduces both contributions proportionally, giving $\mathcal{R}_0 = 1.22$, near but above threshold. Aggressive symptomatic-case isolation reduces only the symptomatic-track contribution, leaving $\mathcal{R}_0 = 1.66$ with 84% of post-intervention transmission from the untouched asymptomatic reservoir. Only the combined strategy (masking + isolation + asymptomatic detection-and-reduction) drives $\mathcal{R}_0$ below the invasion threshold to 0.48. The figure visualizes the asymptomatic-reservoir lesson: interventions targeting only symptomatic individuals cannot control transmission by themselves, because the framework's two-track additive $\mathcal{R}_0$ structure leaves an uncontrolled second pathway.

below the level set by asymptomatic spread, and that effective control requires interventions that act on the asymptomatic track.

Second, the infected-viewpoint ratio $\hat{\alpha} = j/i$ recovers the growth rate from incidence data without an estimate of the susceptible population. With the latent period carried into the conversion, that growth rate gives $\mathcal{R}_0$ to about 0.1 percent. The pandemic also made the distinction between $\mathcal{R}_0$ and $\mathcal{R}_e(t)$ operational: agencies reported the effective number through the pandemic, written R_t, while $\mathcal{R}_0$ was a property of the wild-type strain in a fully susceptible population, and the two were often compared as if they measured the same thing (Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$").

Third, given the latent and infectious periods, case counts identify $\mathcal{R}_0$ but not how it divides between the tracks. The split, set by p and by $c_A\beta_A$ relative to $c_I\beta_I$, must be informed from outside the model (serological surveys, contact-tracing studies, random-sample testing), and it decides whether symptom-based interventions can work. This pattern (the model is identifiable from a multi-source data combination but not from a single source) is a recurring theme across the case studies in this part.

Fourth, the framework adapts to variants and to changing intervention regimes by adjusting parameters, not by changing structure. Alpha, Delta, and Omicron variants of SARS-CoV-2 are captured by re-fitted values of β, τ_E, and τ_R; the equations (17.1) and the reproductive number (17.3) remain unchanged. Operational forecasting that updates these parameters in real time as new data arrive is one direction of generalization; ensemble forecasting that combines the framework's predictions with statistical and machine-learning models is another. Both are outside the scope of this book. The same caution applies to the scenarios of Section 17.6: the declines observed after each wave's peak mixed depletion of the susceptible pool with behavioral response to perceived risk, and a constant-β model credits the whole decline to the intervention it was given (Pitfall: "Constant-Transmission Assumption Ignoring Behavioral Feedback"); Gozzi et al. [2025] quantify the gap for the first wave.

Limitations of the case study. The case study omits age structure, which is essential for predicting hospitalization and mortality and for evaluating age-targeted interventions (vaccine prioritization, school closure). An age-stratified extension layers the heterogeneous-population framework of Chapter 11 on top of the SEIAR structure used here, with age-specific contact matrices estimated from contact-survey studies and age-specific severity parameters from clinical data. The case study also omits spatial structure, which affects regional dynamics and intervention coordination but not single-population $\mathcal{R}_0$. The latent compartment is not infectious, so the model omits pre-symptomatic transmission. Isolation acts on all of the symptomatic track here, which overstates what isolation can achieve when part of that track transmits before symptoms appear. The case study uses parameter values representative of the wild-type strain in a non-elderly adult population; variant-specific re-fits would proceed identically and would yield different numerical conclusions but the same structural conclusions. Finally, the case study treats acute infection as the relevant clinical outcome; it does not develop a model of post-acute sequelae of SARS-CoV-2 infection (PASC, colloquially long COVID). PASC modeling is an active area of work, with Beams et al. [2024] showing through compartmental sensitivity analysis that some parameters (the duration and equilibrium prevalence of infection, the fraction of infections that develop PASC) dominate PASC trajectory uncertainty while others (the duration of immunity, secondary vaccine efficacy against PASC) contribute only modestly, an invasion-burden-style partition transposed to the chronic-sequela setting that fits naturally into this book's framework even though we do not develop it here.

Summary

The COVID-19 case study formulates a six-compartment SEIAR model that combines the latent-period and asymptomatic-branching extensions of the basic SIR framework. For the wild-type SARS-CoV-2 with consensus parameter values, the basic reproductive number is $\mathcal{R}_0 \approx 2.44$, of which approximately 57 percent comes from the asymptomatic track if both tracks stay infectious equally long. Incidence data pin this value through the growth rate, provided the conversion carries the latent period; they cannot reveal how much of it comes from the asymptomatic track. Sensitivity analysis identifies the asymptomatic-track parameters c_A and β_A as the dominant invasion-controlling levers (after the infectious period τ_R, which is biological rather than intervention-controlled). Intervention scenarios show that no single measure is sufficient to suppress transmission, but combinations targeting both tracks can drive $\mathcal{R}_0$ below unity. The asymptomatic-reservoir problem identified structurally in Chapter 10, Section 10.6 is the central practical constraint, and the framework's analysis of it transfers directly into operational intervention recommendations.

What this case study teaches structurally. The COVID-19 analysis illustrates a feature of the SIR framework that is generic, not specific to SARS-CoV-2: when a substantial fraction of infectious individuals is asymptomatic and therefore does not reduce its contact rate upon infection, the classical symptom-driven assumption $c_I \ll c_S$ fails for that subgroup. The invasion parameter c_A for the asymptomatic track remains close to the baseline c_S, and conventional symptom-based isolation interventions have no impact on that track. This structural observation (that transmission reduction requires reducing contact *with the infectious asymptomatic compartment specifically*, which demands testing-based rather than symptom-based identification) carries directly to other pathogens with substantial asymptomatic transmission: norovirus and many influenza variants. The pedagogical lesson is that symptom-based interventions have a structural ceiling set by the symptomatic track's share of $\mathcal{R}_0$ (43 percent here, although $p = 0.6$), and the framework makes this ceiling quantitative through the c_A-dependent contribution to $\mathcal{R}_0$.

Exercises

Five section-level exercises appear above, following Section 17.2, Section 17.3, Section 17.5, and Section 17.6. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

Notation. These exercises use the book's notation $\mathcal{R}_e(t)$ for the time-varying effective reproductive number (Chapter 5, Section 5.2). The COVID-19 nowcasting literature (EpiEstim, CDC and ECDC dashboards, RKI estimates) consistently writes this same quantity as R_t; when consulting those external references, read R_t as $\mathcal{R}_e(t)$.

1. **[C]** Reflecting on the chapter's structural Lessons, identify the single feature of the COVID-19 outbreak that most strongly motivated departures from the basic SIR model. Briefly state how the framework accommodates that feature without losing its central organizing structure.

2. **[D]** Stratify the model by two age groups (under 65, over 65) using the framework of Chapter 11. Choose plausible age-specific contact rates $c_S^{<65}$, $c_S^{\geq 65}$ and within-versus-between-group mixing parameters. Compute the age-stratified $\mathcal{R}_0$ via the next-generation matrix method.

3. **[I]** Suppose the symptomatic fraction p is in fact 0.4 (not 0.6), but case-count data are reported with a constant ascertainment rate $a < 1$ (only a fraction a of symptomatic cases

are reported). Show that, when the contact rates are fitted along with them, the parameters (p, a) are not jointly identifiable from case-count data alone. What independent data source would be needed to identify each?

4. **[D]** For the combined-intervention scenario in Section 17.6 that gives $\mathcal{R}_0 \approx 0.48$, compute the time to reduce the prevalence by half, assuming exponential decay at rate $-\sigma_2 = (\gamma_R + \nu_m)(1 - \mathcal{R}_0)$ (the SIR approximation). Then recompute it from the dominant eigenvalue of the linearized (E, i, A) system and explain the difference. Compare to historical data on time-to-suppression from regions that implemented combined interventions early in the pandemic.

5. **[L]** Implement the SEIAR model in your preferred scientific computing environment (Python, MATLAB, or R) and numerically verify the analytic $\mathcal{R}_0$ from (17.3) by computing the early-outbreak exponential growth rate and inverting it twice: through the SIR relation of Chapter 8, and through $(1 + \sigma\tau_E)(1 + \sigma\tau_R)$. Explain the gap between the two. The chapter's generator in the companion repository (`notebooks/ch17/ch17_figures.py`) is a reference implementation.

6. **[AI Lab]** *Real-time COVID-19 fitting workflow.* Choose a country or region with publicly available daily case, hospitalization, and death time series for a multi-month period during the pandemic (e.g., a US state from a CDC dashboard, a country from JHU CSSE). Execute the chapter's three-step workflow:

   (a) *Step 1: Data-vintage inventory.* Download the most recent two weeks of cases, hospitalizations, and deaths for the chosen region. Compare the values to the same data downloaded one week and four weeks earlier (most public dashboards retain revisions in their archives). Quantify the magnitude of revisions for each series. Prompt an AI tool: *"For a real-time COVID-19 fitting workflow using cases, hospitalizations, and deaths, which series should be the primary fitting target given the revision magnitudes I observed?"* Audit the response against your data; document agreement or disagreement.

   (b) *Step 2: Time-varying parameter elicitation.* Identify the three or four largest policy events for your region during the modeling period (lockdown, mask mandate, vaccine rollout milestones). Prompt an AI tool to recommend a piecewise-constant or piecewise-smooth parameter form for c_I that respects this policy structure, with effective dates and parameter-change magnitudes. Audit the AI's effective-date claims against public-health-department records; flag any AI hallucinations.

   (c) *Step 3: Forecast vs nowcast distinction.* Choose one specific modeling task (e.g., "estimate $\mathcal{R}_e(t)$ as of last week" or "predict hospital census two weeks ahead"). Prompt the AI tool to identify whether your task is nowcasting or forecasting and to recommend the methodology accordingly. Document whether the AI's recommendation distinguishes the two cases or conflates them.

   (d) *Implementation.* Implement the recommended workflow with the audit corrections from Steps 1–3. Compare the resulting $\mathcal{R}_e(t)$ estimates or forecast intervals against published estimates from public-health agencies (CDC, ECDC, etc.) for the same region and period. Document agreement, disagreement, and the source of any discrepancies.

   The exercise practices the full real-time fitting workflow with explicit AI-audit at each step. Disclosure: per Appendix D, Section D.4, document the AI system, all prompts, the AI-elicited regime-change time points, the policy-source-spot-check findings, and any hallucinations identified.

7. **[Verify the AI]** *Auditing an AI-generated real-time $\mathcal{R}_e(t)$ analysis.* Prompt an AI tool: *"I am tracking $\mathcal{R}_e(t)$ for a respiratory pathogen during an ongoing outbreak. I have daily case counts from the past 12 weeks, and I want to estimate $\mathcal{R}_e(t)$ over the most recent 4-week window. Recommend a workflow."* Save the response. Audit the recommended workflow against this chapter's three-error structure for real-time case-study fitting:

   (a) *Data-vintage audit.* Daily case counts at the time of analysis are systematically incomplete for the most recent 1–2 weeks because of reporting delays; deaths and hospitalizations have different (longer) lag structures. Does the AI's recommended workflow distinguish reporting-date from event-date data, or does it implicitly assume that recent low case counts reflect a true decline rather than incomplete reporting? The diagnostic question: *"If the most recent 7 days of data show case counts 30% below the prior 7-day average, what alternative explanations should I rule out before concluding that $\mathcal{R}_e(t)$ has fallen?"* A correctly-tuned response distinguishes (i) genuine transmission decline, (ii) right-truncation from reporting delay, (iii) weekday-vs-weekend effects, and (iv) ascertainment changes. An under-specified response treats the recent decline as a transmission signal without flagging the vintage caveats.

   (b) *Regime-change-detection audit.* The COVID-19 record is dominated by regime changes: NPI rollouts, variant emergence, vaccine campaigns, behavioral adaptation. Constant-parameter fits over a 12-week window will produce an averaged $\mathcal{R}_e(t)$ that may correspond to no actual point in time. Does the AI's recommended workflow include a regime-change detection step (changepoint analysis, rolling-window comparison, residual structure analysis), or does it propose a single-window fit? The diagnostic question: *"How would I detect, from the residual structure of an $\mathcal{R}_e(t)$ fit over the 12-week window, whether a regime change occurred mid-window? What does that detection imply for the reported $\mathcal{R}_e(t)$ value?"* A chapter-aware response identifies that residual autocorrelation at lags matching the suspected changepoint is the diagnostic, and that detected regime changes require the window to be split.

   (c) *Parameter-correlation audit.* In COVID-19 fitting workflows, the transmission-related parameters β and c_I are correlated with the ascertainment rate (the fraction of infections detected) and with the assumed infectious-period duration τ_R. Does the AI's recommended workflow report correlations among the fitted parameters, or does it report marginal CIs as if the parameters were independent? The diagnostic question: *"If I report $\hat{\mathcal{R}}_e(t)$ with a 95% CI based on the marginal posterior of $c_I\beta$ alone, what fraction of the true uncertainty does my CI represent when ascertainment rate and τ_R are also uncertain?"* A correctly-tuned response identifies that the marginal CI under-represents the joint uncertainty by a factor that depends on the correlation structure (often 1.5× to 3× tighter than the joint CI), and recommends reporting either a joint CI or a marginal CI with the conditioning explicit.

   The exercise practices the audit habit on the highest-impact failure modes in real-time COVID-19 $\mathcal{R}_e(t)$ analysis: data-vintage misinterpretation, regime-change blindness, and parameter-correlation under-reporting. Each contributed to widely-publicized $\mathcal{R}_e(t)$ estimates during the pandemic that were retrospectively shown to be misleading. Disclosure: per Appendix D, Section D.4, document the AI system, prompts, and audit findings.

8. **[Open Project]** *Re-analysis of a published COVID-19 $\mathcal{R}_0$ estimate.* A multi-week capstone project re-analyzing a published $\mathcal{R}_0$ or $\mathcal{R}_e(t)$ estimate from the COVID-19 literature using

the book's infected-viewpoint framework, identifying any methodological issues in the original analysis, and producing a corrected estimate with quantified uncertainty. The full workflow—six steps spanning parameter elicitation, mathematical analysis, numerical validation, AI-collaborative audit, disclosure, and final memo (≈15–25 hours over 3–4 weeks)—is specified in the companion repository at `capstone-projects/ch17-covid-reanalysis/`.

Chapter 18

Dengue

Constants are not always important. Dimensionless ratios usually are.

The COVID-19 case study of the previous chapter applied the framework to a respiratory pathogen with directly-transmitted dynamics and substantial asymptomatic transmission. This chapter applies the framework to a pathogen with a fundamentally different transmission mode: dengue is vector-borne, transmitted between humans by *Aedes* mosquitoes, with the dynamics of the vector population being as central to disease control as the dynamics of the human population. The Ross-Macdonald framework of Chapter 13 provides the natural starting point.

The case study follows the same template as Chapter 17: Section 18.1 sketches the epidemiological context, Section 18.2 formulates the model and computes the basic reproductive number, Section 18.3 addresses parameter estimation, Section 18.4 conducts the sensitivity analysis, Section 18.5 evaluates intervention scenarios, and Section 18.6 draws lessons. The framework's contributions to dengue analysis are sharper than to COVID-19 in one respect (the bipartite vector-host structure produces a square-root reproductive number whose sensitivity behavior differs systematically from direct-transmission models) and weaker in another (the multi-serotype structure of dengue's clinical importance is outside the single-serotype framework presented here).

Learning objectives. After reading this chapter, you will be able to:

- Apply the vector-host framework from Chapter 13 to an arboviral disease with bipartite human-mosquito dynamics.
- Compute dengue's $\mathcal{R}_0$ using the biologically transparent factored form (developed in Chapter 13, Section 13.4) and identify the intervention-target mapping for mosquito lifespan, vector competence, and host biting behavior.
- Evaluate interventions (source reduction, insecticide, Wolbachia replacement, behavior change) and predict their relative effectiveness through the sensitivity framework.
- Recognize why temperature-sensitive vector biology introduces structural forcing that homogeneous models cannot capture, and when this matters.

Prerequisites. Chapters 13 and 9; familiarity with the dengue transmission cycle.

18.1 Epidemiological Context

Dengue virus causes an estimated 100 to 400 million infections per year, with the wide range reflecting the gap between symptomatic clinical cases (which are reported) and total infections (which include a large asymptomatic and mild-symptomatic fraction) [Bhatt et al., 2013]. The geographic range covers most tropical and subtropical regions, with substantial concentration in Southeast Asia, the Indian subcontinent, the Caribbean, Central and South America, and parts of sub-Saharan Africa. Range expansion driven by climate change and by the global spread of *Aedes albopictus* has brought dengue into temperate regions of Europe, North America, and East Asia in the past two decades, and the trend is expected to continue.

The pathogen and its serotypes. Dengue virus exists as four antigenically distinct serotypes (DENV-1, DENV-2, DENV-3, DENV-4), each capable of independently sustaining transmission in suitable ecological conditions. Infection by one serotype confers lifelong immunity to that same serotype but only short-term cross-protection (months) against the other three. The clinical importance of this serotype structure cannot be overstated: most primary infections are asymptomatic or mild, while secondary infections by a heterologous serotype carry substantially elevated risk of severe disease (dengue hemorrhagic fever, dengue shock syndrome) through the antibody-dependent enhancement (ADE) mechanism [Guzman et al., 2010]. The fraction of secondary infections that progress to severe dengue is on the order of 1 to 5 percent, compared with under 0.5 percent of primary infections.

The vector and its biology. The principal vectors are *Aedes aegypti* (the global urban dengue vector, anthropophilic and predominantly day-biting) and *Aedes albopictus* (the Asian tiger mosquito, more catholic in host preference and active during a wider range of daylight hours). Both species breed in small artificial water containers in urban and peri-urban environments, which makes vector control a problem of municipal sanitation and household water management as much as of insecticide deployment. *Aedes* biting rates are approximately 0.5 per day per vector (one bite roughly every two days), and adult vectors live for approximately two weeks under typical urban conditions. The extrinsic incubation period for dengue in *Aedes* is approximately nine days, so the probability that an infected vector survives the EIP and becomes infectious is approximately $e^{-\mu_v \tau_v^E} \approx e^{-0.07 \cdot 9} \approx 0.53$ in the fixed-delay limit (Section 18.2 adopts 0.56). This survival probability is the rate-limiting biological factor in vector-side transmission and is the most temperature-sensitive parameter in the entire model. How rainfall drives mosquito density is modeled, for Rift Valley fever, by Chitnis et al. [2013], and comparative analyses of dengue and chikungunya dynamics in shared *Aedes* populations give a practical sense of how much of the disease-specific variation is attributable to vector biology versus pathogen parameters [Manore et al., 2014].

Transmission dynamics. A single vector takes blood meals from multiple human hosts during its infectious lifespan; a single human is bitten by multiple vectors during their infectious period. The result is a transmission graph that is bipartite (vector-to-human and human-to-vector, with no within-species transmission) but densely connected on each side. The Ross-Macdonald framework of Chapter 13 captures this structure with a small number of parameters and yields the reproductive number as a square root, which has direct implications for how interventions affect $\mathcal{R}_0$ relative to direct-transmission diseases.

Available data for fitting. Dengue surveillance is dominated by passive case reporting from clinical settings, with the systematic biases this implies: severely affected cases are over-represented, asymptomatic and mild cases are under-counted, and the symptomatic-case-to-true-infection ratio varies across settings, age groups, and serotypes [Murray et al., 2013]. Cross-sectional seroprevalence surveys provide cumulative-incidence anchors that are essential for calibrating long-term endemic-equilibrium estimates. Vector population estimates (the parameter $m = N_v/N_h$ in the Ross-Macdonald formulation) are typically obtained through ovitrap and adult-mosquito surveys; these are operationally expensive, subject to seasonal variation, and frequently the most uncertain quantity in the entire model. The framework's infected-viewpoint estimator $\hat{\alpha}_h = J_h/I_h$ exploits human-side incidence and prevalence data without requiring a committed estimate of N_v, which is one of the framework's distinctive practical contributions to vector-borne modeling (Chapter 13, Section 13.3, also discussed quantitatively in the sensitivity-analysis paper Hyman, 2026d).

Convention pointer. The subscripted forces in this chapter follow the vector-borne convention of Chapter 13: they are *per-target* rates, multiplying a susceptible population, and correspond to the force *of* infection λ in the book's primary notation. The per-source force from infection α of Chapter 3, which satisfies $J = \alpha I$, appears here only in the fitting estimators. The Notation and Conventions summary lists which chapters use which.

Modeling choices implied. The case study below uses the single-serotype Ross-Macdonald framework. Multi-serotype dynamics, antibody-dependent enhancement, and severe-disease prediction are central to operational dengue modeling but require a substantially larger compartment count (four-serotype models multiply the host compartments by four and add a 4×4 serotype-immunity matrix) and are outside the scope of a case study chapter; we discuss the multi-serotype extension in Section 18.6. Spatial heterogeneity (urban versus rural, household-clustering, climate-driven seasonality) is also outside the scope here. The single-serotype, well-mixed, single-vector-species model captures the core transmission biology and exposes the framework's distinctive contributions; richer models layer on top.

18.2 Model Formulation and Reproductive Number

The case-study model is the bipartite vector-host structure of Chapter 13. The compartmental layout is shown in Figure 18.1.

The governing equations are

$$\dot{S}_h = \nu_m N_h^c - \alpha_h S_h - \nu_m S_h, \tag{18.1a}$$
$$\dot{I}_h = \alpha_h S_h - \gamma_R I_h - \nu_m I_h, \tag{18.1b}$$
$$\dot{R}_h = \gamma_R I_h - \nu_m R_h, \tag{18.1c}$$
$$\dot{S}_v = \mu_v N_v^c - \alpha_v S_v - \mu_v S_v, \tag{18.1d}$$
$$\dot{E}_v = \alpha_v S_v - \gamma_v^E E_v - \mu_v E_v, \tag{18.1e}$$
$$\dot{I}_v = \gamma_v^E E_v - \mu_v I_v, \tag{18.1f}$$

with the cross-species forces

$$\alpha_h = a\, b_h \frac{I_v}{N_h}, \qquad \alpha_v = a\, b_v \frac{I_h}{N_h}. \tag{18.2}$$

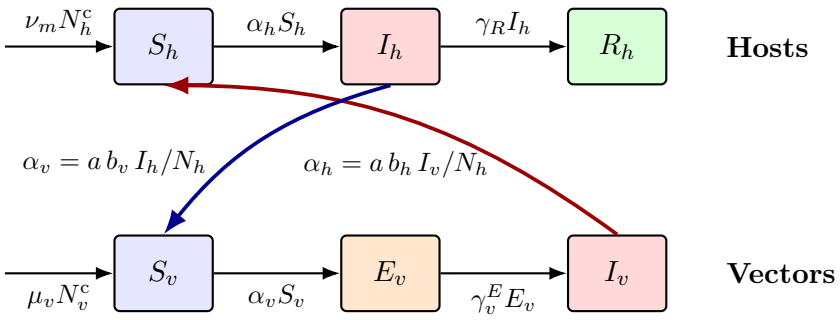


Figure 18.1: The bipartite vector-host structure used for the dengue case study. Hosts (humans) follow $S_h \to I_h \to R_h$ dynamics with permanent post-infection immunity (single-serotype assumption). The single-serotype, permanent-immunity assumption also fixes the susceptible pool over the window modelled: across multiple seasons, waning cross-immunity and serotype replacement make $S_h(t)$ genuinely time-varying, and a multi-season analysis must model that rather than inherit a constant from a single-outbreak fit. Vectors (*Aedes* mosquitoes) follow $S_v \to E_v \to I_v$ dynamics with no recovery; vector mortality at rate μ_v removes individuals from each vector compartment, and host turnover at rate ν_m from each host compartment (both omitted from the diagram for clarity). The exposed compartment E_v represents the extrinsic incubation period, during which a vector that has acquired infection is not yet infectious. Cross-species transmission (the curved arrows) couples the two subsystems through biting events at rate a per vector. The per-target force of infection on hosts, α_h, depends on the infectious vector population I_v; the per-target force on vectors, α_v, depends on the infectious host population I_h. Both forces are inversely proportional to N_h, the bite-dilution effect.

Effective vector-to-host transmission. The exposed-vector compartment E_v in equation (18.1) models the extrinsic incubation period (EIP) as a single Markovian compartment for ODE tractability. As Chapter 13 sets out, that choice sits at one end of a family: a single compartment assumes an exponentially distributed EIP with coefficient of variation one, while a deterministic delay assumes no dispersion at all, and the dengue data support neither. Pooled EIP observations imply a coefficient of variation near 0.5; we adopt the Erlang parameterization with $k = 3$ used there, whose coefficient of variation is $1/\sqrt{3} \approx 0.58$. The EIP-survival probability is

$$p_{\text{EIP}} \;=\; \left(1 + \frac{\mu_v \tau_v^E}{3}\right)^{-3}, \qquad b_h^{\text{eff}} \;=\; b_h \cdot p_{\text{EIP}}. \tag{18.3}$$

For dengue with $\mu_v = 0.07$ day^{-1} and $\tau_v^E = 9$ days, $\mu_v \tau_v^E \approx 0.63$, giving $p_{\text{EIP}} \approx 0.56$, consistent with the entomological literature [Watts et al., 1987]. We use $b_h^{\text{eff}} = b_h \cdot p_{\text{EIP}} \approx 0.226$ as the effective transmission probability in subsequent calculations.

> **On the EIP parameterization.** The two limiting forms bracket the value used here, and it is worth recording the spread because p_{EIP} enters $\mathcal{R}_0$ directly. At this chapter's baseline $\mu_v \tau_v^E \approx 0.63$, a deterministic delay gives $e^{-\mu_v \tau_v^E} \approx 0.53$, the single Markovian E_v compartment of (18.1) gives $1/(1+\mu_v \tau_v^E) \approx 0.61$, and the Erlang $k = 3$ form adopted above gives 0.56. Since $\mathcal{R}_0$ scales as $\sqrt{p_{\text{EIP}}}$ in the bipartite structure, the delay limit understates $\mathcal{R}_0$ by about three percent relative to the evidence-supported value and the single-compartment limit overstates it by about four percent. The E_v compartment is retained in (18.1) because it keeps the model an ODE system; where the closed-form $\mathcal{R}_0$ and the sensitivity calculations of Section 18.4 need a survival factor, they use the $k = 3$

value, and the discrepancy above is the price of the ODE convenience. The factor p_{EIP} in b_h^{eff} is not a second EIP factor beyond what E_v models; it is the survival probability emerging from the EIP-versus-mortality competition, evaluated in the parameterization the data support.

Reproductive number. The next-generation matrix calculation of Chapter 13, Section 13.4 gives the Ross-Macdonald reproductive number in the book's convention as the geometric mean of the host-to-vector and vector-to-host stage reproductive numbers (Chapter 11, Section 11.6):

$$\mathcal{R}_0 \;=\; a\sqrt{\frac{b_h^{\mathrm{eff}}\, b_v\, m}{\mu_v(\gamma_R+\nu_m)}}. \tag{18.4}$$

The square-root structure is the bipartite signature: a parameter that affects only one transmission direction enters $\mathcal{R}_0$ to the half power, not the first. Doubling the vector-to-host ratio m (twice as many mosquitoes per person) increases $\mathcal{R}_0$ by a factor of $\sqrt{2} \approx 1.41$, not by a factor of 2.

Connection to the framework. The Ross-Macdonald reproductive number (18.4) is the geometric mean of two stage reproductive numbers: $\mathcal{R}_0^{h\to v}$ counts the number of vectors a single infectious host infects, and $\mathcal{R}_0^{v\to h}$ counts the number of hosts a single infectious vector infects. The square root in the geometric-mean formulation reflects that an infection cycle requires both directions; neither direction alone produces invasion (Pitfall: "Heterogeneous-Mixing Balance Conditions"). The invasion-burden partition of Chapter 5, Section 5.7 extends to this bipartite model in modified form: the invasion-facing parameters are a, b_h, b_v, and m (each contributing to $\mathcal{R}_0$); the burden-facing parameters are ν_m (host demographic turnover) and the absolute population sizes; the mixed parameter is γ_R (in $\mathcal{R}_0$ via the host infectious period and in I_h^* via equilibrium balance). Proposition 10.2 from Chapter 10, Section 10.7 guarantees that the partition structure transfers; the square-root dampening modifies the magnitudes of the sensitivity indices but not the partition itself.

Exercises for this section

1. **[C]** The chapter argues that the bipartite Ross–Macdonald structure of Chapter 13 captures dengue's transmission dynamics with one important addition. Identify that addition and explain why it would be misleading to model dengue with only the basic two-compartment vector-host model.

18.3 Parameter Estimation and Fitting

The framework's distinctive contribution to vector-borne fitting is that the human-side estimator $\hat{\alpha}_h = J_h/I_h$ does not require an estimate of the susceptible vector population S_v. In operational dengue modeling, S_v is typically the most poorly known quantity in the entire system: vector populations vary on weekly time scales with rainfall and ovipositional habitat, are expensive to survey, and even systematic ovitrap programs produce density estimates with confidence intervals spanning factors of two or more [Barrera et al., 2014]. The infected-viewpoint estimator side-steps this uncertainty by fitting α_h from human-side incidence-and-prevalence data alone. The uncertainty is not made to vanish—S_v still influences the transmission the data record—but it no longer has to be quantified before an estimate can be produced, which is what matters operationally when the estimate is needed during an outbreak.

Best-estimate baseline parameters. The following parameter values represent typical urban-dengue conditions in a tropical setting:

- Vector biting rate: $a = 0.5$ day^{-1} (*Aedes* bites approximately every other day, more frequently than *Anopheles*).
- Vector-to-human transmission probability: $b_h = 0.4$ per infectious bite.
- Human-to-vector transmission probability: $b_v = 0.4$ per bite on infectious human (less asymmetric than malaria).
- Vector mortality: $\mu_v = 0.07$ day^{-1} ($\tau_v^m \approx 14$ days under typical urban conditions; *Aedes* is somewhat longer-lived than *Anopheles* in heated indoor microclimates).
- Extrinsic incubation period: $\tau_v^E = 9$ days ($\gamma_v^E \approx 0.11$ day^{-1}).
- EIP survival probability: $p_{\text{EIP}} = (1 + \mu_v \tau_v^E/3)^{-3} \approx 0.56$ (Erlang $k = 3$; see the note above).
- Effective vector-to-human transmission: $b_h^{\text{eff}} = b_h \cdot p_{\text{EIP}} \approx 0.226$.
- Human recovery rate: $\gamma_R = 1/7$ day^{-1} ≈ 0.143 day^{-1} ($\tau_R \approx 7$ days; dengue resolves more quickly than malaria).
- Vector-to-host ratio: $m = 2$ (highly variable; can be 5 to 10 in heavy infestation, less than 1 in well-managed urban areas).
- Host demographic turnover: $\nu_m \approx 3.4 \times 10^{-5}$ day^{-1} (negligible at outbreak time scales).

Computing the baseline $\mathcal{R}_0$. Substituting these values into (18.4),

$$\begin{aligned}
\mathcal{R}_0^2 &= \frac{a^2 b_h^{\text{eff}} b_v m}{\mu_v(\gamma_R + \nu_m)} \\
&\approx \frac{(0.5)^2 \cdot 0.226 \cdot 0.4 \cdot 2}{0.07 \cdot 0.143} \\
&= \frac{0.04516}{0.010002} \\
&\approx 4.515,
\end{aligned}$$

giving the per-generation reproductive number

$$\mathcal{R}_0 \approx 2.12.$$

This value is consistent with the published dengue-modeling literature, which typically reports per-generation $\mathcal{R}_0$ values in the range 2 to 4 depending on transmission setting [Ferguson et al., 2016]. Higher values (4 to 6 per generation) are reported for explosive urban outbreaks; lower values (1.5 to 2) for endemic transmission in well-managed urban areas with active vector control.

Parameter-uncertainty interval on $\hat{\mathcal{R}}_0$. The point estimate $\mathcal{R}_0 \approx 2.12$ is computed from a single committed parameter set. To quantify how far the stated parameter uncertainties carry through, we apply the delta-method theorem 9.1 of Chapter 9 with the dominant uncertainties being the vector-to-host ratio m ($\pm 50\%$, the most poorly-known dengue parameter), the biting rate a ($\pm 20\%$), the vector mortality μ_v ($\pm 25\%$), the host recovery rate γ_R ($\pm 20\%$), and the transmission probabilities b_h, b_v ($\pm 15\%$ each), each half-width read as two standard deviations; the EIP duration τ_v^E ($\pm 11\%$) adds a negligible term. Using the closed-form sensitivity indices computed in Section 18.4,

$$\begin{aligned} V^{\mathcal{R}_0} &\approx (\Upsilon_a^{\mathcal{R}_0})^2 V^a + (\Upsilon_{b_h^{\text{eff}}}^{\mathcal{R}_0})^2 V^{b_h^{\text{eff}}} + (\Upsilon_{b_v}^{\mathcal{R}_0})^2 V^{b_v} \\ &\quad + (\Upsilon_m^{\mathcal{R}_0})^2 V^m + (\Upsilon_{\mu_v}^{\mathcal{R}_0})^2 V^{\mu_v} + (\Upsilon_{\gamma_R}^{\mathcal{R}_0})^2 V^{\gamma_R} \\ &\approx (1.0)^2(0.10)^2 + (0.5)^2(0.075)^2 + (0.5)^2(0.075)^2 \\ &\quad + (0.5)^2(0.25)^2 + (-0.760)^2(0.125)^2 + (-0.5)^2(0.10)^2 \\ &\approx 0.0100 + 0.0014 + 0.0014 + 0.0156 + 0.0090 + 0.0025 \\ &\approx 0.0400. \end{aligned}$$

The implied relative standard deviation is $\sqrt{0.0400} \approx 0.200$, giving a 95% parameter-uncertainty interval for $\mathcal{R}_0$ of approximately $\mathcal{R}_0 \cdot (1 \pm 1.96 \cdot 0.200) = \mathcal{R}_0 \cdot (1 \pm 0.392)$, or numerically $[1.29, 2.96]$ around the point 2.12. The dominant variance contributions are m (39% of $V^{\mathcal{R}_0}$), a (25%), and μ_v (22%); these three together account for 87% of the uncertainty. The interval brackets the threshold $\mathcal{R}_0 = 1$ in its lower tail at slightly higher m-uncertainty assumptions: with m uncertain by a factor of 2 ($\pm 100\%$, read the same way, the upper end of dengue field-survey uncertainty per Barrera et al. 2014), the 95% CI extends to $[0.90, 3.35]$ and the threshold is no longer reliably above unity. This sensitivity to vector-density uncertainty is the operational reason vector surveillance matters for outbreak preparedness. The interval propagates the stated parameter uncertainties only: it assumes they are independent (Section 8.2 carries the correlated form), it takes no account of model-structure uncertainty, and because the inputs are plausible ranges rather than estimated standard errors it carries no coverage guarantee.

Fitting from human-side incidence. Using the infected-viewpoint estimator,

$$\hat{\alpha}_h(t) = \frac{J_h(t)}{I_h(t)},$$

where J_h is the daily incidence of human cases and I_h is the human prevalence. Two things must be said about what this ratio is. It is *computable* from human-side data alone: no vector-side quantity is required as an input, which is the operational advantage developed below. But it is not the mechanistic force from infection of the directly transmitted SIR model, because humans do not infect humans. New human infections are generated by infectious *vectors*, $J_h = a\, b_h^{\text{eff}} I_v (S_h/N_h)$, so $\hat{\alpha}_h$ measures the realized rate of new human infections *per current human case*, accumulated through the complete human–vector–human cycle. Two consequences follow. First, the quantity being estimated does depend on the vector population: at vector quasi-equilibrium $I_v = a b_v S_v I_h/(\mu_v N_h)$, so $\hat{\alpha}_h \approx a^2 b_h^{\text{eff}} b_v S_v S_h/(\mu_v N_h^2)$. What the human-side estimator avoids is the need to *know* S_v, not a dependence on it. Second, the vector stage carries a generation delay, so $\hat{\alpha}_h(t)$ reflects transmission seeded by humans infectious at earlier times; reading it as an instantaneous per-case rate requires vector dynamics fast relative to host dynamics, which the quasi-equilibrium assumption above makes explicit. To extract individual transmission parameters from $\hat{\alpha}_h$ requires further information: the biting rate a is determined from entomological landing-rate studies, the

vector-to-human transmission probability b_h^{eff} from controlled-feeding experiments or per-bite case-attack-rate studies, and the vector-to-host ratio m from ovitrap surveys. Each of these auxiliary parameters has its own uncertainty, but each is independently constrained by data sources that do not require fitting to the case time series.

Identifiability for vector-borne models. The key identifiability observation for vector-borne models, distinct from the COVID-19 case study, is that the host-side and vector-side parameters appear in $\mathcal{R}_0$ symmetrically (in a square-root). A doubling of b_h^{eff} and a halving of b_v produce the same $\mathcal{R}_0$. From human-side incidence data alone, b_h^{eff} and b_v are not separately identifiable: any rescaling preserving their product is equivalent. The framework's response is to constrain one of the two from independent entomological data (typically b_v, which is more easily measured by feeding mosquitoes on viremic patients and tracking vector-side infection rates) and back-compute the other from the host-side fit. The m parameter is similarly tied to b_v via the bipartite product $b_v m$ and requires independent constraint.

Exercises for this section

1. **[L]** *End-to-end dengue parameter-estimation workflow.* You are given a synthetic 12-week dengue outbreak time series consisting of weekly human-case incidence $J_h(t_k)$ and weekly human prevalence $I_h(t_k)$ for a city of $N_h^c = 200{,}000$. (Generate the synthetic series yourself by simulating the model (18.1) forward from the baseline parameters of this section with $I_h(0)/N_h^c = 10^{-4}$ and adding multiplicative log-normal noise with $\sigma = 0.15$ to both observed series; the companion repository provides a reference data-generation script.) Execute the full infected-viewpoint estimation workflow:

   (a) *Estimate $\hat{\alpha}_h$.* For each week k, compute the empirical force-from-infection $\hat{\alpha}_h(t_k) = J_h(t_k)/I_h(t_k)$. Plot $\hat{\alpha}_h(t)$ over the 12 weeks and identify the early-time plateau where the susceptible-fraction depletion is small (so $\hat{\alpha}_h$ is approximately constant); use this plateau to obtain a single point estimate $\bar{\alpha}_h$ with a bootstrap 95% confidence interval over the plateau weeks.

   (b) *Recover b_h^{eff} given auxiliary entomological data.* Suppose independent ovitrap and feeding-experiment data give $a = 0.5 \pm 0.1$ day^{-1}, $b_v = 0.4 \pm 0.05$, $\mu_v = 0.07 \pm 0.015$ day^{-1}, $\tau_v^E = 9 \pm 1$ days, and $m = 2 \pm 0.5$ (each with the indicated 1-σ uncertainty). From $\bar{\alpha}_h$ and the entomological inputs, back-compute b_h^{eff} from the quasi-equilibrium relation of Section 18.3, $\bar{\alpha}_h \approx a^2 b_h^{\text{eff}} b_v m/\mu_v$ (with $S_h \approx N_h$ and $S_v \approx N_v$). **Caution on QSS validity for dengue:** the rigorous justification of this reduction requires vector dynamics much faster than host dynamics ($\mu_v \gg \gamma_R$), which holds for malaria ($\mu_v/\gamma_R \sim 5$–10 with $\gamma_R \sim 0.015$ day^{-1}) but *not* for dengue at the baseline parameters of this section ($\mu_v = 0.07$ day^{-1} vs $\gamma_R = 0.143$ day^{-1}, so $\gamma_R \approx 2\mu_v$). For dengue the QSS reduction is an order-of-magnitude estimate; results should be cross-checked against full bipartite simulation (see the companion repository's Chapter 18 notebook). The fitting workflow below uses the QSS estimator as a starting point and reports the discrepancy from full-simulation values. Propagate the entomological uncertainties through to a 95% CI on b_h^{eff} via Monte Carlo (sample 10,000 entomological-input vectors, compute b_h^{eff} for each, and report the 2.5th and 97.5th percentiles).

   (c) *Compute $\mathcal{R}_0$ with full uncertainty.* Combine the b_h^{eff} distribution from (b) with the entomological inputs to evaluate $\mathcal{R}_0$ from (18.4). Report the 95% CI on $\mathcal{R}_0$ and decompose

the contributions to total $\mathcal{R}_0$ uncertainty by parameter (compute the variance contribution from each input). Identify the dominant uncertainty source.

(d) *Compare with naïve S_v-dependent fitting.* Repeat the estimation under the susceptible-viewpoint formulation $\lambda = c_S \beta P_I$, which would require a fit for the unobserved $S_v(t)$. Without performing the full optimization, explain what additional data would be needed and what the resulting confidence interval on $\mathcal{R}_0$ would be expected to look like compared to the infected-viewpoint workflow above. (Hint: see the framework's discussion of S_v uncertainty as the dominant operational obstacle.)

(e) *Validate the QSS approximation.* Because the dengue baseline violates the $\mu_v \gg \gamma_R$ premise of quasi-steady-state reduction (see the caution in step (b) above), the QSS estimate of b_h^{eff} should not be used as a final answer without comparison to a full bipartite simulation. Run the companion repository's Chapter 18 reference notebook to fit b_h^{eff} from the same synthetic data using the full bipartite ODE system (18.1) (no quasi-steady-state assumption). Report the full-simulation point estimate, the percent discrepancy from the QSS estimate of step (b), and identify whether the QSS estimate is biased high or low. State, in one sentence, the operational rule the comparison establishes: when can the QSS estimate be used as the final answer, and when must full-simulation fitting be carried out?

The exercise illustrates how the infected-viewpoint estimator decouples the well-measured host-side incidence from the poorly-measured vector population, with the entomological auxiliary data entering only through the secondary computation of individual transmission parameters rather than as a fitting target.

18.4 Sensitivity Analysis

The Ross-Macdonald formula (18.4) has logarithmic structure that makes sensitivity computation immediate: $\log \mathcal{R}_0 = \log a + \frac{1}{2}(\log b_h^{\text{eff}} + \log b_v + \log m - \log \mu_v - \log(\gamma_R + \nu_m))$, so the relative sensitivities are exactly the logarithmic-derivative coefficients (Pitfall: "Sensitivity Analysis Discipline"). An equivalent reading follows from the biologically transparent factoring of Chapter 13, Section 13.4 (equation (13.14)): each of the six factors (latent-period survival, daily transmission rate and infectious residence time, in each direction) contributes $\pm 1/2$ to $\Upsilon_p^{\mathcal{R}_0}$, and the biting rate a gets unit sensitivity because it enters both directional transmission rates.

Analytic sensitivities.

$$
\begin{aligned}
\Upsilon_{a}^{\mathcal{R}_0} &= 1 \quad \text{(unique unit-sensitivity parameter)},\\
\Upsilon_{b_h^{\text{eff}}}^{\mathcal{R}_0} &= +\tfrac{1}{2},\\
\Upsilon_{b_v}^{\mathcal{R}_0} &= +\tfrac{1}{2},\\
\Upsilon_{m}^{\mathcal{R}_0} &= +\tfrac{1}{2},\\
\Upsilon_{\mu_v}^{\mathcal{R}_0} &= -\tfrac{1}{2} \quad \text{(direct effect through denominator)},\\
\Upsilon_{\gamma_R}^{\mathcal{R}_0} &= -\tfrac{1}{2} \quad \text{(through host infectious period)},\\
\Upsilon_{\eta_I}^{\mathcal{R}_0} &= +\tfrac{1}{2} \quad \text{(compartment-specific biting exposure, via (13.15))}.
\end{aligned}
$$

The vector mortality μ_v has an additional indirect effect through b_h^{eff} via (18.3): increasing μ_v reduces p_{EIP} and therefore reduces b_h^{eff}. Applying the chain rule with the Erlang $k = 3$ parameterization $p_{\text{EIP}} = (1 + \mu_v \tau_v^E/3)^{-3}$ adopted in Section 18.2, the indirect sensitivity is

$$\Upsilon_{\mu_v}^{p_{\text{EIP}}} = \frac{\partial \log p_{\text{EIP}}}{\partial \log \mu_v} = -\frac{\mu_v \tau_v^E}{1 + \mu_v \tau_v^E/3} \approx -0.52$$

at the baseline, and the total relative sensitivity to μ_v is

$$\Upsilon_{\mu_v}^{\mathcal{R}_0,\,\text{total}} = -\tfrac{1}{2} + \tfrac{1}{2}\,\Upsilon_{\mu_v}^{b_h^{\text{eff}}} = -\tfrac{1}{2} + \tfrac{1}{2} \cdot (-0.52) = -\tfrac{1}{2} - 0.26 \approx -0.760.$$

The two effects reinforce: increasing vector mortality reduces $\mathcal{R}_0$ both directly (through the denominator) and indirectly (the reduction in b_h^{eff} has positive $+1/2$ sensitivity in $\mathcal{R}_0$, so a decrease in b_h^{eff} further decreases $\mathcal{R}_0$). Same-signed effects add constructively, giving net sensitivity approximately -0.760 in our parameter regime, substantially more negative than the direct denominator effect alone.

Tornado plot. Figure 18.2 ranks the parameters by absolute sensitivity. The biting rate a dominates when m is held fixed, with sensitivity exactly unity. The four parameters b_h^{eff}, b_v, m, and γ_R each have magnitude 0.5 from the bipartite square-root structure. The vector mortality μ_v has the larger magnitude 0.760 because its direct effect through the denominator is reinforced by an indirect effect through the EIP-survival reduction.

Implications for intervention design. The biting-rate sensitivity at unity makes biting reduction (bednets, repellents, indoor air-conditioning that keeps mosquitoes outside, environmental management to disrupt mosquito-human contact) the lever with the largest proportional effect on $\mathcal{R}_0$ per proportional reduction achieved. Whether it is also the most *cost-effective* lever is a separate question this analysis does not answer: an elasticity says how much $\mathcal{R}_0$ moves for a given fractional change in a parameter, not what that fractional change costs to deliver or how much of it is attainable in practice. Ranking interventions by cost-effectiveness requires programme costs and achievable coverage alongside the sensitivity indices. A 30 percent reduction in a produces a 30 percent reduction in $\mathcal{R}_0$. The vector-population reduction interventions (source reduction, larvicide treatment) work through m at half-sensitivity: a 50 percent reduction in m produces only a $\sqrt{0.5} - 1 \approx -29$ percent change in $\mathcal{R}_0$. Adult-vector control through insecticides works through μ_v at the larger net sensitivity of approximately -0.760: doubling μ_v produces approximately $1 - 2^{-0.760} \approx 41$ percent reduction in $\mathcal{R}_0$ by log-linear extrapolation of the sensitivity, and the full nonlinear computation gives 44 percent (Scenario 2 in Section 18.5). The square-root dampening through most parameters means that single interventions rarely achieve the threshold $\mathcal{R}_0 < 1$; combinations are typically necessary, as the next section quantifies.

Exercises for this section

1. **[C]** The dengue chapter exhibits a striking reinforcement: increasing vector mortality μ_v has a direct (negative) effect through the $\mathcal{R}_0$ denominator and an indirect (also negative) effect through reduced EIP-survival probability p_{EIP}, which reduces b_h^{eff} in the $\mathcal{R}_0$ numerator. Explain in plain English why both effects exist and have the same sign, and why this matters for the design of insecticide-based interventions.

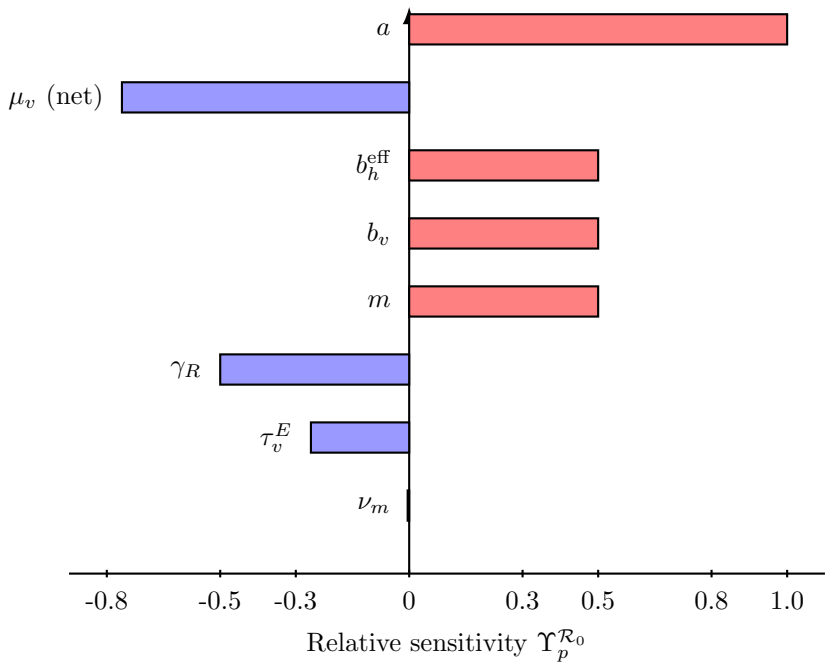


Figure 18.2: Tornado plot of relative sensitivity indices $\Upsilon_p^{\mathcal{R}_0} = \partial \log \mathcal{R}_0 / \partial \log p$ for the baseline dengue parameters. Red bars indicate positive sensitivities (parameter increases drive $\mathcal{R}_0$ up); blue bars indicate negative sensitivities (parameter increases reduce $\mathcal{R}_0$). The biting rate a has unique unit sensitivity, reflecting its appearance in both transmission directions of the bipartite cycle. The four parameters b_h^{eff}, b_v, m, and γ_R all have magnitude 0.5 (square-root dampening from the bipartite structure). The EIP duration τ_v^E enters only through p_{EIP}, with index -0.26. The vector mortality μ_v has the larger net magnitude 0.760 (computed under the Erlang $k = 3$ EIP convention adopted in Section 18.2: direct denominator effect $-1/2$ plus indirect EIP-mediated effect ≈ -0.26) because its direct denominator effect is reinforced by an indirect effect: increasing μ_v reduces the EIP-survival probability p_{EIP}, which reduces b_h^{eff}, which (with $+1/2$ sensitivity in $\mathcal{R}_0$) further reduces $\mathcal{R}_0$ in the same direction.

18.5 Intervention Scenarios

We evaluate three vector-control interventions, each targeting a different parameter group identified in Section 18.4. Labelling each scenario by which group of factors in the factoring (13.14) of Chapter 13 it modifies makes the structural effect on $\mathcal{R}_0$ transparent: source reduction acts on the "vectors per host" component of $\alpha_h^{(0)}$, insecticide acts on τ_{iv} (vector lifespan) and indirectly on P_v (EIP survival), and Wolbachia or behavior-change interventions reduce $\alpha_v^{(0)}$ or $\alpha_h^{(0)}$ through vector competence or biting-rate reduction. The square-root dampening of (13.14) means any single-direction intervention achieves at most a $\sqrt{q}$-fold reduction in $\mathcal{R}_0$ when the targeted factor is reduced to a fraction q of its baseline value.

Scenario 1: source reduction (50 percent reduction in vector population m). Larvicide treatment of breeding sites and removal of standing water (the recurring municipal-sanitation campaigns common in dengue-endemic cities) reduces m from 2 to 1. The $\mathcal{R}_0^2$ scales linearly in m, so

$$\mathcal{R}_0^{2,\,\text{source}} \;=\; \frac{0.25 \cdot 0.226 \cdot 0.4 \cdot 1}{0.07 \cdot 0.143} \;\approx\; 2.26,$$

and the per-generation reproductive number drops to $\mathcal{R}_0^{\text{source}} \approx 1.50$. Substantial reduction (a 29 percent decrease) but still above the invasion threshold. The square-root dampening means that even halving the vector population is not sufficient to suppress transmission; combination with other interventions is required.

Scenario 2: adult-vector control (insecticide doubling μ_v). Indoor and outdoor insecticide application doubles μ_v from 0.07 to 0.14 day^{-1}. The direct effect halves the denominator factor in (18.4); the indirect effect through the EIP reduces p_{EIP} from 0.56 to $(1{+}1.26/3)^{-3} \approx 0.35$, cutting b_h^{eff} from 0.226 to 0.140. Both effects are captured in:

$$\begin{aligned}\mathcal{R}_0^{2,\,\text{insect}} \;&=\; \frac{0.25 \cdot 0.140 \cdot 0.4 \cdot 2}{0.14 \cdot 0.143} \\ &\approx\; \frac{0.0279}{0.0200} \;\approx\; 1.40,\end{aligned}$$

giving $\mathcal{R}_0^{\text{insect}} \approx 1.18$. Still above threshold; the combined direct-plus-indirect effect of doubling vector mortality cuts $\mathcal{R}_0$ by more than 40 percent but does not suppress transmission on its own.

Fixed or compensating emergence. The model replaces every mosquito that dies: its birth term $\mu_v N_v^{\text{c}}$ doubles when μ_v doubles, so Scenario 2 holds m fixed. Adult emergence is set by larval habitat rather than by adult mortality, and if it stays fixed the adult population falls as $1/\mu_v$. Doubling μ_v then also halves m: $\mathcal{R}_0^{\text{insect}} \approx 0.84$, below threshold on its own, and the net sensitivity to μ_v is about -1.26, larger in magnitude than that of a. Which reading applies depends on how strongly larval density regulation refills the adult population.

Scenario 3: combined intervention (source reduction + insecticide). Combining the two interventions, $m = 1$ and $\mu_v = 0.14$ (with $b_h^{\text{eff}} = 0.140$):

$$\mathcal{R}_0^{2,\,\text{combo}} \;=\; \frac{0.25 \cdot 0.140 \cdot 0.4 \cdot 1}{0.14 \cdot 0.143} \;\approx\; 0.70,$$

giving $\mathcal{R}_0^{\text{combo}} \approx 0.84$. Below the invasion threshold; the combined strategy can suppress transmission. As with the COVID-19 case study (Chapter 17, Section 17.6), no single intervention is sufficient; combinations targeting independent parameter groups are required.

The biting-rate alternative. The intervention scenarios above target vector parameters. An alternative target is the biting rate a itself, through bed-net programs, household screens, repellents, and behavioral changes that reduce vector-human contact. With $\Upsilon_a^{\mathcal{R}_0} = 1$, a 30 percent reduction in a produces a 30 percent reduction in $\mathcal{R}_0$. From the baseline $\mathcal{R}_0 \approx 2.12$, a 50 percent reduction in a gives $\mathcal{R}_0 \approx 1.06$ (just above threshold); a 60 percent reduction gives $\mathcal{R}_0 \approx 0.85$ (below threshold). Because a enters with unit sensitivity rather than half-sensitivity, biting-rate interventions move $\mathcal{R}_0$ furthest per percentage of parameter change, with m held fixed. The operational difficulty is that sustained reductions in a require continuous behavioral or infrastructural intervention, while a one-time insecticide application or source-reduction campaign can produce a multi-month effect.

Scenario 4: behavior change among infected humans (reducing η_I). A more targeted intervention couples the biting-rate concept above with the compartment-specific extension of Chapter 13, Section 13.5.3. Symptomatic individuals who reduce vector exposure during the viremic phase (by staying indoors, using bed nets after symptom onset, or seeking treatment) reduce the exposure weight η_I in (13.15). Because η_I enters $\mathcal{R}_0^2$ linearly rather than quadratically, the sensitivity is $\Upsilon_{\eta_I}^{\mathcal{R}_0} = 1/2$: a 30 percent reduction in infected-host biting exposure ($\eta_I \to 0.7$) gives

$$\mathcal{R}_0^{\eta_I = 0.7} \;\approx\; \mathcal{R}_0\sqrt{0.7} \;\approx\; 0.84\,\mathcal{R}_0,$$

a 16 percent reduction from baseline. A 50 percent reduction yields $\mathcal{R}_0\sqrt{0.5} \approx 0.71\,\mathcal{R}_0$, a 29 percent reduction. The half-sensitivity is smaller than the unit sensitivity of uniform biting-rate reduction, but the intervention has two operational advantages: it requires coverage only of the infected subpopulation rather than the whole community, and the reduction takes effect at the point where viremia is highest (days 3–7 after symptom onset for dengue). Xue et al. [2018] analyzes this mechanism in combination with Wolbachia release for arboviral control, showing that the behavior-change component pushes $\mathcal{R}_0$ below threshold at lower Wolbachia-coverage levels than Wolbachia alone would require. Combined with the source-reduction-plus-insecticide Scenario 3 above ($\mathcal{R}_0^{\text{combo}} \approx 0.84$), a modest 30 percent reduction in η_I drives the combined intervention to $\mathcal{R}_0 \approx 0.70$, a robust margin below threshold.

Exercises for this section

1. **[I]** For the combined-intervention scenario in Section 18.5 that gives $\mathcal{R}_0 \approx 0.84$, compute the intervention duration required to drive prevalence I_h to half its initial value, assuming exponential decay at the bipartite rate near threshold, $(1 - \mathcal{R}_0^2)(\gamma_R + \nu_m)\mu_v/(\gamma_R + \nu_m + \mu_v)$, which leaves out the EIP. Then compute the dominant eigenvalue of the model linearized about the disease-free state, with the EIP as three Erlang stages (Section 18.2), and compare the two durations.

18.6 Lessons from the Dengue Case Study

The case study illustrates five general lessons.

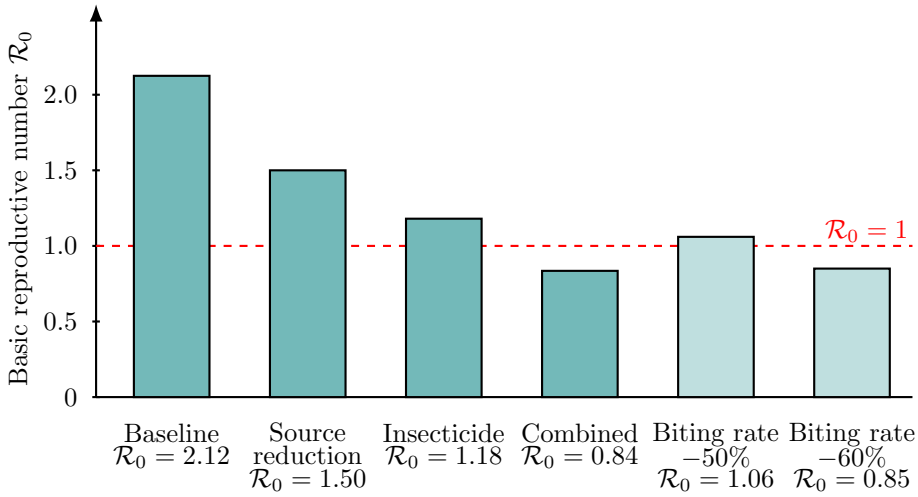


Figure 18.3: Intervention-scenario comparison for dengue. Three vector-targeted scenarios (source reduction halving m, insecticide doubling μ_v, and their combination) and two biting-rate interventions (50% and 60% reductions in a) are shown against the baseline $\mathcal{R}_0 \approx 2.12$. With emergence that replaces killed adults, neither single vector-parameter intervention brings $\mathcal{R}_0$ below the threshold: source reduction alone gives 1.50, insecticide alone gives 1.18. The combination gives 0.84. The biting-rate interventions (lighter bars) act on the unit-sensitivity parameter a: a 50% reduction leaves $\mathcal{R}_0$ just above the threshold (1.06), and a 60% reduction brings it to 0.85. Reducing m alone would take a cut of about 78% to reach the threshold. This is the bipartite square-root signature visualized: halving a (unit sensitivity) halves $\mathcal{R}_0$, but halving m (half-sensitivity) reduces $\mathcal{R}_0$ only by $1 - \sqrt{0.5} \approx 29\%$.

First, the bipartite vector-host structure produces a square-root reproductive number whose sensitivity behavior differs systematically from direct-transmission models. The biting rate a has unique unit sensitivity (because a couples both transmission directions), while all other invasion-facing parameters have ± 0.5 sensitivity. The implication is that biting-rate interventions have the largest proportional effect in vector-borne settings, while vector-population and per-bite transmission-probability interventions are subject to square-root dampening. This is a structural feature of bipartite transmission, not a feature specific to dengue.

Second, the framework's infected-viewpoint estimator $\hat{\alpha}_h = J_h/I_h$ resolves the most acute identifiability problem in operational dengue modeling: the susceptible vector population S_v is the most poorly known quantity in the system, and the human-side estimator does not require it as an input—though, as Section 18.3 makes explicit, the transmission it measures still depends on S_v through the vector stage. The sensitivity-analysis paper of Hyman [2026d] develops this advantage quantitatively for vector-borne models; the dengue case study illustrates it in an operational context. The remaining identifiability issues (the symmetric appearance of b_h^{eff} and b_v in the bipartite product) require auxiliary entomological data to resolve.

Third, the multi-serotype structure of dengue's clinical importance is outside the single-serotype framework presented here, but the framework extends naturally. A four-serotype model multiplies the host compartments by four (each individual is in one of the cross-product compartments determined by their immunity status with respect to each serotype) and adds a 4×4 serotype-cross-immunity matrix. The resulting system has 16 host compartments plus the vector subsystem, but the structural framework (bipartite transmission, NGM-derived $\mathcal{R}_0$, invasion-burden partition, infected-viewpoint fitting) applies identically. The four-serotype extension is the natural next step for any operational dengue analysis, and is implemented in several published modeling efforts [Nishiura, 2006, Ferguson et al., 2016]. Complementary to these multi-serotype efforts, outbreak-fitting studies grounded in single-serotype formulations [Chowell et al., 2004] provide the calibration anchors that any extended model must reproduce before being trusted for policy.

Fourth, the dengue case study highlights the framework's response to climate-driven parameter variation. Vector mortality μ_v, the extrinsic incubation period τ_v^E, and the vector-to-host ratio m all depend strongly on temperature and rainfall. Climate change projections that shift these parameters by 10 to 30 percent over decades produce $\mathcal{R}_0$ changes that the framework computes mechanically through the sensitivity indices of Section 18.4. Range expansion of *Aedes albopictus* into temperate regions, documented over the past two decades, can be evaluated with the same framework simply by computing $\mathcal{R}_0$ at the new local parameter values and identifying conditions under which $\mathcal{R}_0 > 1$ becomes possible.

Fifth, and relatedly, the steady-state case study calculations presented here use time-invariant parameters (Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$"): the compartmental ODE framework treats μ_v, τ_v^E, and m as constants throughout a simulation interval. For pathogens whose biological parameters vary substantially over the timescale of a single infection generation (influenza's sharp day-to-day infectiousness peak, or the time-varying extrinsic incubation period of dengue under seasonal temperature forcing), the renewal-equation framework of Chapter 15 captures the non-exponential infectious-period distribution that compartmental models approximate as exponential. For steady-state and long-horizon analyses, the compartmental approximation suffices. For seasonal forecasting, intra-outbreak climate variability, or fine-scale generation-time analysis, the renewal-equation formulation is the appropriate next tool.

Limitations of the case study. The case study omits the multi-serotype dynamics that drive dengue's clinical importance, restricts attention to a single vector species in a single ecological

setting, and uses a homogeneous-mixing approximation that masks the substantial spatial and household-level clustering observed in real dengue transmission. The parameter values are representative of typical urban tropical conditions and would differ in temperate range-edge settings, in heavily insecticide-treated environments, or in outbreak versus endemic regimes. The case study does not address dengue vaccines (the CYD-TDV controversy and second-generation candidates), which require a multi-serotype model to evaluate properly. Each of these limitations is a productive direction for extension within the framework rather than a problem with the framework itself.

Summary

The dengue case study formulates a bipartite vector-host Ross-Macdonald model for single-serotype dengue transmission. For typical urban tropical parameters, the per-generation basic reproductive number is $\mathcal{R}_0 \approx 2.12$, consistent with published dengue-modeling estimates. Sensitivity analysis identifies the biting rate a as the unique parameter with unit sensitivity, and identifies four parameters at ± 0.5 sensitivity reflecting the square-root dampening of bipartite reproductive numbers. Intervention scenarios show that source reduction alone or insecticide alone is insufficient to suppress transmission when emergence replaces killed adults, but combinations can drive $\mathcal{R}_0$ below unity. Biting-rate interventions have the largest proportional effect per percentage of parameter change; whether they are the cheapest is a question of cost and coverage, not of sensitivity. The infected-viewpoint estimator $\hat{\alpha}_h = J_h/I_h$ resolves the susceptible-vector-population identifiability problem that makes vector-borne fitting difficult. The single-serotype framework is the right scaffold; multi-serotype extension is the natural direction for operational dengue analysis.

What this case study teaches structurally. The dengue analysis illustrates the per-direction factoring $\mathcal{R}_0 = P \cdot \alpha \cdot \tau$ introduced in Chapter 13: each direction of the bipartite cycle contributes an EIP survival probability P, a transmission rate α, and an infectious residence time τ, and the bipartite $\mathcal{R}_0$ is the geometric mean of the two direction-products. The factoring assumes uniform exposure within each population; heterogeneity modifiers such as the biting-exposure weights η_I enter as multiplicative refinements inside the relevant directional rate rather than as additional factors of the same kind, so the three-factor reading survives but its factors are averages. This factoring is not a dengue-specific convenience. It applies to every bipartite transmission cycle in the book (sexual transmission in Chapter 12 and human-to-human-via-vector in every vector-borne disease). The structural lesson is that each factor corresponds to a distinct intervention class: P is targeted by shortening the vector's expected adult lifespan (insecticides, habitat control), α is targeted by reducing biting exposure (bed nets, repellents), and τ is targeted in the host direction by shortening the human infectious period (there is typically no analogue in the vector direction, since adult mosquitoes are infectious for life). Reading the factoring tells the outbreak responder where intervention effort can be directed with impact and where it cannot. The same logic applies to malaria, chikungunya, Zika, West Nile, and every other arbovirus that shares the bipartite transmission structure.

Exercises

Four section-level exercises appear above, following Section 18.2, Section 18.3, Section 18.4, and Section 18.5. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. [**D**] Recompute $\mathcal{R}_0$ for an outbreak-regime setting where m rises from 2 to 5 (high vector density). Compare to the baseline. What single-parameter change would produce an equivalent increase in $\mathcal{R}_0$?

2. [**D**] Suppose insecticide raises vector mortality by 50 percent (instead of doubling it). Recompute the direct, indirect, and net sensitivities to μ_v at this increased value, $1.5\mu_v$. Does the reinforcement between direct and indirect effects change qualitatively?

3. [**D**] Stratify the model into urban and rural populations using the two-group framework of Chapter 11. Choose plausible group-specific biting rates and vector-to-host ratios, and compute the stratified $\mathcal{R}_0$ via the multi-group NGM. Identify which subgroup acts as the transmission core.

4. [**I**] Extend the model to two serotypes (DENV-1 and DENV-2) with cross-protective immunity duration of 6 months. Write the equations for the host compartments (susceptible to both, monotypically immune to either, doubly immune) and identify which new parameters are required.

5. [**I**] Climate-projection sensitivity: suppose climate change increases ambient temperature by 2°C, with the result that μ_v decreases by 20 percent (longer-lived mosquitoes), τ_v^E decreases by 30 percent (faster pathogen development), and m increases by 50 percent (more breeding habitat). Compute the new $\mathcal{R}_0$ and identify which parameter change dominates the increase.

Chapter 19

HIV

A reusable routine is a trusted lemma in code.

The previous two case studies applied the framework to a directly-transmitted respiratory pathogen (COVID-19) and a vector-borne pathogen with bipartite human-mosquito dynamics (dengue). This chapter applies the framework to a third major transmission mode: a sexually-transmitted infection with no natural recovery, strong contact-rate heterogeneity across activity-level subgroups, and a long natural history that makes the standard SIR approximations break down in ways the earlier chapters have not addressed. HIV is the canonical example of an infection whose dynamics are dominated by core-group transmission within high-activity subnetworks; the homogeneous-mixing models of Parts II and III give wrong answers for HIV by underestimating $\mathcal{R}_0$ in the dominant transmission subgroup. The framework's response to this is the activity-stratified bipartite structure of Chapter 12, which we apply here.

The case study follows the template of Chapters 17 and 18, except that the sensitivity analysis comes before estimation: Section 19.1 sketches the epidemiological context, Section 19.2 formulates the model and computes the basic reproductive number, Section 19.3 conducts the sensitivity analysis, Section 19.4 addresses parameter estimation, Section 19.5 evaluates intervention scenarios, and Section 19.6 draws lessons. The framework's contributions to HIV analysis are sharpest in the treatment-as-prevention era: ART acts on the per-partnership transmission probability β, while behavioral interventions act on the partnership-formation rate c_H, and the framework lets us compare these levers directly through the sensitivity machinery of Chapter 9.

Notational reminder. Throughout this chapter a subscripted force such as α_M^H is per-target (per-susceptible) and multiplies a susceptible population, the reverse of Chapter 12, where α_M multiplies I_M; the one exception is the estimator $\hat{\alpha} = J/I$ of the per-source force from infection of Chapter 3, used in the fitting.

19.1 Epidemiological Context

HIV emerged as a global health emergency in the early 1980s and has been the most-studied infectious disease in mathematical epidemiology over the four decades since. About 40 million people (39.9 million) were living with HIV worldwide in 2023, with substantial geographic concentration in sub-Saharan Africa [UNAIDS, 2024]. The biology that distinguishes HIV from the pathogens of the previous two case studies, and that drives the modeling choices in this chapter, is the combination

of an essentially indefinite infectious period absent treatment, the dominance of sexual transmission in adults supplemented by parenteral and vertical routes, and the dramatic effect of antiretroviral therapy (ART) on both individual outcomes and population-level transmission.

Natural history without treatment. Untreated HIV infection follows a stereotyped progression: an acute phase of days to weeks with high viral load and substantial transmission risk, a chronic asymptomatic phase of years to a decade with moderate viral load and continuing transmission, and a late or AIDS phase of months to a few years with rising viral load and immunosuppressive complications. Without ART, median time from infection to AIDS-defining illness is approximately ten years, and median time from AIDS to death is approximately one to two years. The key implication for compartmental modeling is that the infectious period is essentially the entire residence time in the population: $\tau_R^* = 1/(\nu_m + \mu_d)$, the mean time to the first of death from other causes and disease-induced death. For untreated infection, τ_R^* is on the order of ten years, two to three orders of magnitude longer than the infectious periods for COVID-19 or dengue.

Transmission routes. Sexual transmission accounts for the large majority of new HIV infections globally, with parenteral transmission (shared needles in intravenous drug use, contaminated medical equipment) and vertical transmission (mother to child during pregnancy, delivery, or breastfeeding) as secondary routes. Per-contact transmission probabilities for sexual transmission are low: approximately 0.0008 for receptive vaginal exposure to an infected partner with detectable viral load, 0.0004 for insertive vaginal exposure, and substantially higher (about 0.014, 95% CI 0.010 to 0.019) for receptive anal intercourse [Patel et al., 2014]. Per-partnership transmission probabilities, which integrate over multiple contacts within a partnership, are accordingly higher: typically 0.025 to 0.05 per partnership for heterosexual contact, depending on contact frequency and the infected partner's viral load. We use per-partnership values throughout this case study.

Activity-level heterogeneity. Sexual contact networks are strongly heterogeneous in partnership-formation rate. Empirical studies consistently identify a small high-activity subpopulation (perhaps 10 to 20 percent of the population) with partnership rates an order of magnitude or more above the population average, and a larger low-activity population with rates well below the average [Anderson and May, 1992]. The high-activity subpopulation acts as a transmission core: HIV's $\mathcal{R}_0$ in the high-activity subgroup is above unity, while the homogeneous-population $\mathcal{R}_0$ averaged over both subgroups is below unity. Without the core-group structure, the population-average reproductive number would predict that HIV cannot sustain itself; with the core-group structure, the high-activity subgroup sustains transmission and serves as the engine that ultimately reaches the low-activity population. This is the canonical empirical signature of activity-level heterogeneity in modeling. The differential-infectivity formulation that made this subgroup structure analytically tractable was developed in Hyman et al. [1994] and extended to continuous activity-level distributions in Hyman et al. [1999]; the staged-progression companion to differential infectivity is treated by Martcheva and Thieme [2003].

The intervention landscape. Three intervention modalities have been deployed against HIV at population scale. *Behavioral interventions* reduce the partnership-formation rate c_H through education, partner reduction, and reduction in concurrent partnerships. *Barrier methods* (condom use) reduce the per-partnership transmission probability β by approximately 80 to 90 percent in adherent users. *Pharmaceutical interventions* include pre-exposure prophylaxis (PrEP) for uninfected high-risk individuals and antiretroviral therapy (ART) for infectious individuals. ART for

the infectious individual reduces their viral load to undetectable levels in adherent patients, which reduces per-partnership transmission to approximately zero. The treatment-as-prevention paradigm formalizes the intervention strategy of treating infectious individuals to prevent onward transmission, with the slogan "undetectable equals untransmissible" (U=U) capturing the per-individual implication [Eisinger et al., 2019]. ART also extends individual life expectancy to near normal in adherent patients, which from the model's perspective increases the residence time of infectious individuals in the population while suppressing their transmission potential.

Available data for fitting. HIV surveillance combines passive case reporting (with substantial under-ascertainment, especially in high-prevalence settings without universal testing access), HIV testing campaigns that produce point-prevalence estimates by age and sex, sentinel surveillance in antenatal clinics that monitor seroprevalence trends, and program-level data on ART coverage and viral suppression rates [UNAIDS, 2024, 2020]. The framework's infected-viewpoint estimator $\hat{\alpha} = J/I$ requires incidence and prevalence data; for HIV, prevalence is typically better measured than incidence (because chronic infections accumulate visibly), and the incidence-to-prevalence ratio gives a direct estimate of α that is structurally immune to uncertainty in the susceptible-population estimate. This advantage is operationally important in settings where the size of the high-activity subpopulation is itself uncertain.

19.2 Model Formulation and Reproductive Number

The case-study model uses an activity-stratified bipartite SI structure: bipartite because heterosexual sexual transmission is bipartite by construction, stratified because activity-level heterogeneity is the dominant feature shaping HIV dynamics, and SI rather than SIR because HIV has no natural recovery (the infectious period is the residence time in the population). Each sex contains two activity-level subgroups: high-activity (H) and low-activity (L), with the bulk of population in L and the bulk of transmission in H. The compartmental layout for the high-activity subgroup is shown in Figure 19.1.

The governing equations for the high-activity subgroup (with vertical transmission folded into an effective birth-rate term entering S_F^H; Pitfall: "Short-Term Simulation Assumptions") are

$$\dot{S}_M^H = \nu_b N_M^{H,c} - \alpha_M^H S_M^H - \nu_m S_M^H, \tag{19.1a}$$
$$\dot{I}_M^H = \alpha_M^H S_M^H - (\nu_m + \mu_d) I_M^H, \tag{19.1b}$$
$$\dot{S}_F^H = \nu_b N_F^{H,c}\big(1 - p_{\mathrm{MTCT}} I_F^H / N_F^H\big) - \alpha_F^H S_F^H - \nu_m S_F^H, \tag{19.1c}$$
$$\dot{I}_F^H = \nu_b N_F^{H,c}\, p_{\mathrm{MTCT}}\, I_F^H / N_F^H + \alpha_F^H S_F^H - (\nu_m + \mu_d) I_F^H, \tag{19.1d}$$

with the cross-sex forces of infection

$$\alpha_M^H \;=\; c_H\, \tilde{\beta}_{FM}\, \frac{I_F^H}{N_F^H}, \qquad \alpha_F^H \;=\; c_H\, \tilde{\beta}_{MF}\, \frac{I_M^H}{N_M^H}, \tag{19.2}$$

and the per-partnership transmission probabilities $\tilde{\beta}_{MF}$ (per partnership of an infected male with a susceptible female; the force on a susceptible female from an infected male source uses this M→F probability) and $\tilde{\beta}_{FM}$ (the reciprocal F→M; the force on a susceptible male from an infected female source uses this). The per-partnership transmission probabilities are the integrated effect of multiple per-contact events within a single partnership, with $\tilde{\beta} \approx 1 - (1 - \beta_{\text{per-contact}})^m$ for m contacts per partnership.

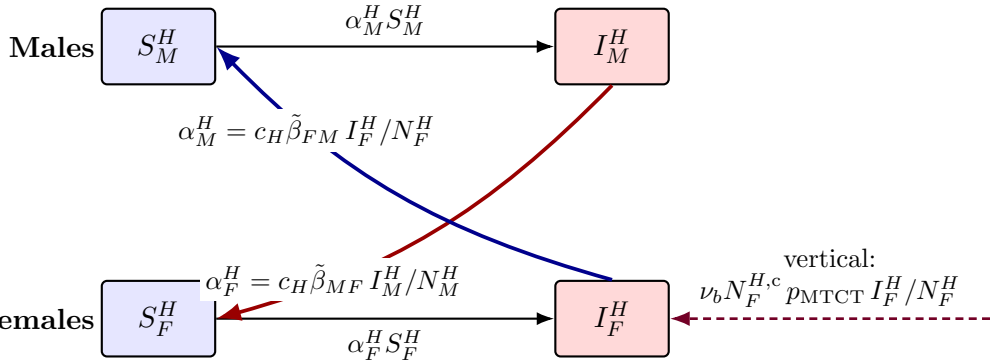


Figure 19.1: The activity-stratified bipartite SI structure used for the HIV case study, showing only the high-activity subgroup (H) of each sex. Male and female compartments couple through cross-sex transmission with sex-asymmetric per-partnership probabilities ($\tilde{\beta}_{MF}$, $\tilde{\beta}_{FM}$) and shared partnership-formation rate c_H. Vertical transmission (dashed purple) enters I_F^H directly as infected births, at rate $\nu_b N_F^{H,c} p_{\mathrm{MTCT}} I_F^H / N_F^H$. There is no natural recovery; the infectious period is set by mortality alone (rate $\nu_m + \mu_d$, omitted from the diagram for clarity, removing individuals from each I compartment). Births into S_M^H and S_F^H at rate ν_b, and their outflow at rate ν_m, are omitted as well. The low-activity subgroup (L, not shown) has the same structure but with much smaller c_L, and weak coupling to the H subgroup through inter-activity-level partnerships not modeled here.

Notation note. The forces α_M^H and α_F^H in (19.2) are per-susceptible rates (target-viewpoint): $\alpha_M^H S_M^H$ in the $\dot{I}_M^H$ equation is the rate of new male infections, and similarly for females. This is the natural convention in bipartite models because the source-target distinction matters at the equation level. The book's primary single-compartment convention (Chapter 3, Section 3.2) reserves α for the per-source-infectious rate $c_I \beta P_S$, where source and target coincide; the bipartite extension here generalizes that convention to a per-target rate. The multi-population conventions subsection of the front-matter notation summary (p. 484) collects the per-source/per-target distinction and the bridge identity for readers who arrive at this chapter from a non-sequential reading path. Population-level new-infection rates

$$J_M(t) \;=\; \alpha_M^H S_M^H, \qquad J_F(t) \;=\; \alpha_F^H S_F^H,$$

are convention-independent and consistent with the $J(t)$ defined in Chapter 3 equation (3.2). The fitting estimators $\hat{\alpha}_H^g = J_g / I_g$ used in Section 19.4 below build on these $J_g(t)$ via per-sex incidence-and-prevalence ratios; for the bipartite case, the per-sex estimator gives the contribution of cross-sex infections to that sex's compartment, and the bipartite $\mathcal{R}_0$ combines them via the 2×2 NGM as in equation (19.4) below.

Compartment-specific extension. The uniform partnership-formation rate c_H in (19.2) is the classical approximation. The SIR framework of Chapter 4 admits compartment-specific rates c_H^S, c_H^I within the high-activity subgroup (and parallel c_L^S, c_L^I for low-activity), with the uniform case $c_H^S = c_H^I = c_H$ used as the analytical baseline throughout this chapter. Extended, the force equations become

$$\alpha_M^H \;=\; c_H^I \, \tilde{\beta}_{FM} \, \frac{I_F^H}{N_F^H}, \qquad \alpha_F^H \;=\; c_H^I \, \tilde{\beta}_{MF} \, \frac{I_M^H}{N_M^H}, \tag{19.3}$$

where the rate-determining factor is the *infected* host's partnership formation, c_H^I, because the force-from-infection originates with the infectious individual. Setting $c_H^S = c_H^I$ recovers (19.2) exactly.

The distinction matters because known-HIV-positive individuals on ART and in care programs often modify partnership patterns (disclosure, serosorting, and condom-negotiated sex), giving $c_H^I < c_H^S$ in populations with high diagnosis and treatment coverage. This compartment-specific reduction is the c_H analogue of the treatment-as-prevention effect that ART exerts on $\tilde{\beta}$; intervention implications are developed in Section 19.5.

Reproductive number from the bipartite NGM. The next-generation matrix calculation at the disease-free equilibrium for the heterosexual bipartite system gives, in the high-activity subgroup,

$$\mathcal{R}_0^{\text{horiz}} = \frac{c_H \sqrt{\tilde{\beta}_{MF}\, \tilde{\beta}_{FM}}}{\nu_m + \mu_d}. \tag{19.4}$$

The square-root structure is the bipartite signature, the same as for the dengue Ross-Macdonald formula in Chapter 18. The shared partnership-formation rate c_H is unique because it appears in both transmission directions (Pitfall: "Heterogeneous-Mixing Balance Conditions"); it has unit sensitivity in $\mathcal{R}_0$. The asymmetric per-partnership transmission probabilities each have half-sensitivity.

The vertical contribution adds a separate term:

$$\mathcal{R}_0^{\text{vert}} = \frac{\nu_b\, p_{\text{MTCT}}}{\nu_m + \mu_d}, \tag{19.5}$$

counting the expected number of newborn infections per maternal infection over the infectious period. How the two contributions combine is taken up at the end of this section.

Connection to the framework. The structure of $\mathcal{R}_0^{\text{horiz}}$ in (19.4) mirrors the dengue Ross-Macdonald formula (18.4) in its square-root dependence on the bipartite per-direction transmission probabilities. The invasion–burden partition of Chapter 5, Section 5.7 (Pitfall: "Invasion versus Equilibrium") extends to this bipartite model, subject to the non-cancellation hypothesis of Proposition 10.2; that hypothesis holds here, because in (19.1) the new-infection terms involve c_H, $\tilde{\beta}_{MF}$, $\tilde{\beta}_{FM}$, ν_b and p_{MTCT}, and the transition terms only $\nu_m + \mu_d$, so no parameter scales both. Under it: the partnership-formation rate c_H and the per-partnership transmission probabilities $\tilde{\beta}_{MF}, \tilde{\beta}_{FM}$ are invasion-facing; the population sizes $N_M^{H,c}, N_F^{H,c}$ are burden-facing in the equilibrium prevalence but cancel from $\mathcal{R}_0^{\text{horiz}}$, since they enter the two cross-sex transmission terms as reciprocal ratios; the mortality rate $\nu_m + \mu_d$ is mixed in the same way that γ_R was mixed in the basic SIR model. Proposition 10.2 from Chapter 10, Section 10.7 gives the conditions under which the partition transfers, and they are met here; the bipartite structure modifies the magnitudes of the sensitivity indices through square-root dampening but preserves the qualitative partition.

Approximation caveat: the additive total. The total reproductive number is reported here as the sum of the horizontal and vertical contributions. That sum is an approximation, not an identity: the exact threshold quantity is the spectral radius of the full next-generation matrix, and for a matrix combining a two-sex transmission cycle with a vertical self-loop the spectral radius is strictly smaller than the sum of the cycle and loop contributions. With the vertical route contributing about 8 percent of the horizontal one, the additive figure overstates $\rho(\boldsymbol{K})$ by about 4 percent. The gap is small, but it decides a conclusion drawn below: the parameter-uncertainty interval sits close to the threshold, and the additive figure places its lower bound on the wrong side. This chapter therefore reports $\mathcal{R}_0 = \rho(\boldsymbol{K})$ for the parameter values of Section 19.3.

Note on the structure of $\mathcal{R}_0$ for HIV. The decomposition of $\mathcal{R}_0$ into horizontal and vertical contributions means that the HIV model is *not* a simple cyclic multistage model in the sense of Chapter 11, Section 11.6. The horizontal contribution $\mathcal{R}_0^{\text{horiz}}$ is itself bipartite (male and female activity-stratified stages) and admits the stage/cyclic/geometric-mean interpretation locally: within the horizontal sector alone, $\mathcal{R}_0^{\text{horiz}}$ is the geometric mean of male-to-female and female-to-male stage reproductive numbers. But the total $\mathcal{R}_0$, which also incorporates the vertical route as an additive branch, does not reduce to the root of a single cycle product; Remark 11.4 in Chapter 11 addresses this directly. The book's threshold quantity remains $\mathcal{R}_0 = \rho(\boldsymbol{K})$, computed from the full next-generation matrix that encompasses both transmission routes, and $\mathcal{R}_0 > 1$ if and only if the full infection system can invade the disease-free equilibrium.

Verify the AI

The per-sex $\mathcal{R}_0$ factoring trap. Asked for $\mathcal{R}_0$ in a bipartite sexually transmitted infection model, AI assistants routinely lose the square root that alternation between the sexes requires. Three errors recur: the sum $\mathcal{R}_0^{M\to F} + \mathcal{R}_0^{F\to M}$; the product $c_H\tilde{\beta}_{MF}\tilde{\beta}_{FM}/(\nu_m+\mu_d)$ without the root; and the root of the sum, $\sqrt{\tilde{\beta}_{MF}+\tilde{\beta}_{FM}}$.

Ask an AI assistant: *"With per-partnership transmission probabilities $\tilde{\beta}_{MF}$ and $\tilde{\beta}_{FM}$, partnership-formation rate c_H and infectious-period rate $\nu_m+\mu_d$, what is $\mathcal{R}_0$?"* A correct answer matches (19.4), $\mathcal{R}_0 = c_H\sqrt{\tilde{\beta}_{MF}\tilde{\beta}_{FM}}/(\nu_m+\mu_d)$, the spectral radius of the 2×2 next-generation matrix; each $\tilde{\beta}$ enters with sensitivity $+1/2$.

Structural verification. Multiply $\tilde{\beta}_{MF}$ by 4 with everything else fixed. The correct $\mathcal{R}_0$ doubles; the formula without the root quadruples it, and the other two change it by amounts that depend on the sizes of the two probabilities.

Exercises for this section

1. [**D**] Compute $\mathcal{R}_0$ for a model in which every individual has the average partnership-formation rate $\bar{c} = 1.4$ partnerships per year. Show that $\mathcal{R}_0 < 1$ despite the activity-stratified $\mathcal{R}_0^{\text{horiz, H}} \approx 1.58$ being above threshold. Then compute $\mathcal{R}_0$ under random mixing of the two activity classes, using $\langle c^2\rangle/\langle c\rangle$. What do the three values tell us about which model class is appropriate for HIV?

19.3 Sensitivity Analysis

Best-estimate baseline parameters. The following parameter values represent typical untreated heterosexual HIV transmission in a high-activity subpopulation. The values are illustrative, not specific to any one geographic setting.

- Population shares: high-activity subgroup is 20 percent of population by sex; low-activity is 80 percent.
- High-activity partnership formation rate: $c_H = 5$ partnerships per year, approximately 0.0137 day^{-1}.
- Low-activity partnership formation rate: $c_L = 0.5$ partnerships per year (an order of magnitude below high-activity), approximately 1.37×10^{-3} day^{-1}.
- Per-partnership transmission probabilities: $\tilde{\beta}_{MF} \approx 0.04$ (male-to-female per partnership), $\tilde{\beta}_{FM} \approx 0.025$ (female-to-male per partnership).
- Mortality rate (background plus disease-induced for untreated infection): $\nu_m+\mu_d \approx 0.10$ year^{-1}, approximately 2.74×10^{-4} day^{-1}, corresponding to a residence time of approximately 10 years from infection to death.
- Per-female birth rate: $\nu_b \approx 0.04$ year^{-1}, approximately 1.10×10^{-4} day^{-1}.
- Mother-to-child transmission probability without ART: $p_{\text{MTCT}} \approx 0.30$.

Computing the high-activity-subgroup $\mathcal{R}_0$. Substituting these values into (19.4),

$$\begin{aligned}\mathcal{R}_0^{\text{horiz}} &= \frac{c_H\sqrt{\tilde\beta_{MF}\,\tilde\beta_{FM}}}{\nu_m+\mu_d}\\ &\approx \frac{0.0137\cdot\sqrt{0.04\cdot 0.025}}{2.74\times10^{-4}}\\ &= \frac{0.0137\cdot 0.0316}{2.74\times10^{-4}}\\ &= \frac{4.33\times10^{-4}}{2.74\times10^{-4}}\\ &\approx 1.58.\end{aligned}$$

The horizontal $\mathcal{R}_0$ in the high-activity subgroup is approximately 1.58, just above the invasion threshold and consistent with the historical pattern of HIV concentrated transmission in core-group networks [Anderson and May, 1992, 1991]. The vertical contribution computed from (19.5),

$$\mathcal{R}_0^{\text{vert}} = \frac{1.10\times10^{-4}\cdot 0.30}{2.74\times10^{-4}} \approx 0.12,$$

adds about 8 percent to the horizontal value. The additive total is $1.58+0.12=1.70$. For this two-sex cycle with a vertical self-loop, the exact threshold quantity is

$$\mathcal{R}_0 = \rho(\boldsymbol{K}) = \tfrac{1}{2}\Big(\mathcal{R}_0^{\text{vert}}+\sqrt{(\mathcal{R}_0^{\text{vert}})^2+4\,(\mathcal{R}_0^{\text{horiz}})^2}\Big) \approx 1.64.$$

The bipartite reproductive-number formula (19.4) has the same logarithmic structure as the dengue formula. The relative sensitivities are immediate (Pitfall: "Sensitivity Analysis Discipline").

Analytic sensitivities.

$$\begin{aligned}\Upsilon_{c_H}^{\mathcal{R}_0^{\text{horiz}}} &= +1 \quad \text{(unique unit-sensitivity parameter)},\\ \Upsilon_{\tilde\beta_{MF}}^{\mathcal{R}_0^{\text{horiz}}} &= +\tfrac{1}{2},\\ \Upsilon_{\tilde\beta_{FM}}^{\mathcal{R}_0^{\text{horiz}}} &= +\tfrac{1}{2},\\ \Upsilon_{\nu_m+\mu_d}^{\mathcal{R}_0^{\text{horiz}}} &= -1,\\ \Upsilon_{\nu_b}^{\mathcal{R}_0^{\text{vert}}} &= +1,\\ \Upsilon_{p_{\text{MTCT}}}^{\mathcal{R}_0^{\text{vert}}} &= +1,\\ \Upsilon_{\nu_m+\mu_d}^{\mathcal{R}_0^{\text{vert}}} &= -1.\end{aligned}$$

The total reproductive number $\mathcal{R}_0=\rho(\boldsymbol{K})$ has sensitivities that are weighted averages of the horizontal and vertical ones, with weights equal to the elasticities of $\rho(\boldsymbol{K})$ with respect to $\mathcal{R}_0^{\text{horiz}}$ and $\mathcal{R}_0^{\text{vert}}$. Since $\mathcal{R}_0^{\text{horiz}}\approx 1.58$ dominates $\mathcal{R}_0^{\text{vert}}\approx 0.12$, the horizontal sensitivities transfer with weight approximately 0.96 and the vertical ones with weight approximately 0.04.

The mortality term $\nu_m+\mu_d$ has unit sensitivity (negative) in both $\mathcal{R}_0^{\text{horiz}}$ and $\mathcal{R}_0^{\text{vert}}$. This is the analogue of the SIR result that the infectious period τ_R has full sensitivity in the basic reproductive number; for HIV the infectious period is set by the mortality rate alone. ART's effect on mortality (extending the residence time of infectious individuals) increases $\mathcal{R}_0^{\text{horiz}}$, but ART's much larger effect on per-partnership transmission ($\tilde\beta\to 0$ in adherent patients) reduces $\mathcal{R}_0^{\text{horiz}}$ by an order of magnitude, with the net being a strong reduction. The interaction is captured cleanly in the framework.

Tornado plot. Figure 19.2 ranks the parameters by absolute sensitivity in the total reproductive number $\mathcal{R}_0 = \rho(\boldsymbol{K})$.

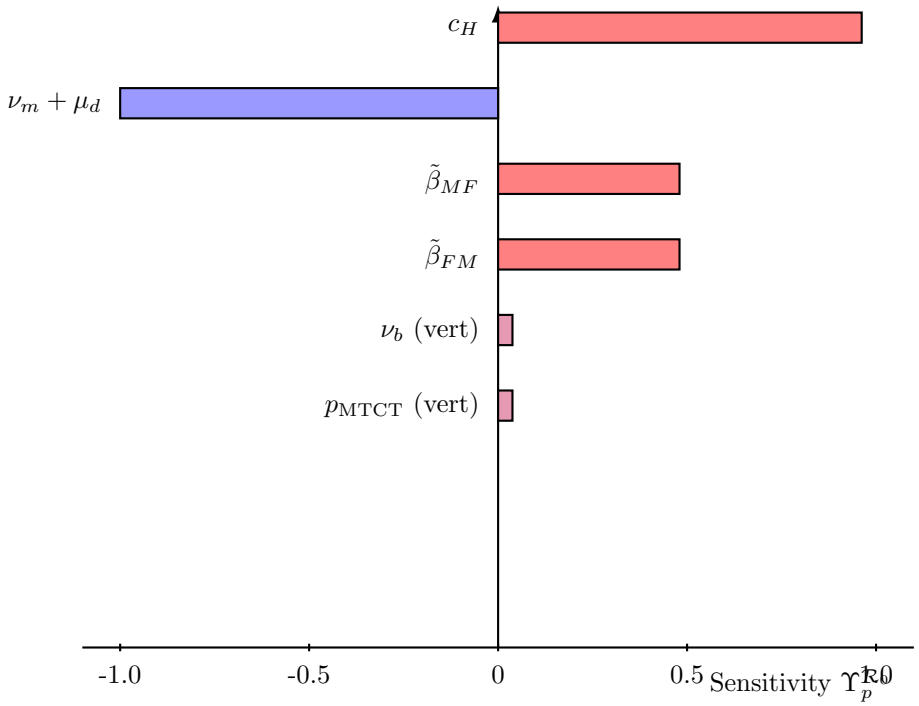


Figure 19.2: Tornado plot of relative sensitivity indices for the total HIV reproductive number $\mathcal{R}_0 = \rho(\boldsymbol{K})$ at the baseline high-activity-subgroup parameters. The mortality rate $\nu_m + \mu_d$ has unit negative sensitivity (extending the infectious period would proportionally increase $\mathcal{R}_0$, were mortality the only adjustable factor); this is the residence-time effect that makes HIV a long-natural-history pathogen. The partnership-formation rate c_H has near-unit positive sensitivity, weighted by the elasticity of $\mathcal{R}_0$ with respect to $\mathcal{R}_0^{\mathrm{horiz}}$. The per-partnership transmission probabilities have half-sensitivity weighted by the same factor (square-root dampening from the bipartite structure). The vertical-contribution parameters ν_b and p_{MTCT} have full unit sensitivity in $\mathcal{R}_0^{\mathrm{vert}}$ but contribute weakly to total $\mathcal{R}_0$ because the vertical contribution is only about 8 percent of the horizontal contribution at baseline.

Implications for intervention design. The unit sensitivity of c_H and the negative unit sensitivity of $\nu_m + \mu_d$ frame the strategic landscape. Behavioral interventions that reduce c_H act on the highest-sensitivity invasion parameter; reductions are difficult to sustain at population scale but produce the largest per-percent change in $\mathcal{R}_0$. Per-partnership transmission interventions (condom use, ART for treatment-as-prevention) act through $\tilde{\beta}_{MF}$ and $\tilde{\beta}_{FM}$ at half-sensitivity, but these interventions are typically highly effective per partnership (90 percent or more reduction with adherence), so the absolute effect on $\mathcal{R}_0$ can be larger than for behavioral interventions. The intervention scenarios in Section 19.5 compare these mechanisms quantitatively. Vertical interventions (PMTCT) act on a small contribution to total $\mathcal{R}_0$ but are essential for maternal-child clinical outcomes; they do not substitute for horizontal-transmission control if the goal is population-level elimination.

19.4 Parameter Estimation and Fitting

The framework's distinctive contribution to HIV fitting is twofold: the infected-viewpoint estimator $\hat{\alpha} = J/I$ resolves the activity-stratification identifiability problem (the size of the high-activity subpopulation $N_M^{H,c} + N_F^{H,c}$ is uncertain in operational settings, but the per-individual force-from-infection estimator does not depend on it), and the activity-stratified bipartite NGM allows us to compute the dominant transmission group's $\mathcal{R}_0$ separately from the population-average $\mathcal{R}_0$ that homogeneous-mixing models would compute.

Fitting from prevalence and incidence data. HIV's chronic nature makes prevalence the most reliably measured population-level quantity (cross-sectional testing surveys, antenatal sentinel surveillance, household surveys), with incidence harder to measure directly but inferable from longitudinal cohort studies and from the prevalence-trajectory derivative. The infected-viewpoint estimator $\hat{\alpha} = J/I$ uses both: from incidence-and-prevalence pairs in the high-activity subgroup, the per-individual force-from-infection is recovered without requiring an estimate of the high-activity subpopulation size $N_M^{H,c} + N_F^{H,c}$, which is operationally uncertain.

The remaining parameters of (19.4) are obtained from independent sources: the partnership-formation rate c_H from sexual-behavior surveys; per-partnership transmission probabilities from controlled-exposure cohort studies and viral-load data; mortality rates from cohort studies of untreated infection or from settings where ART coverage was historically low. Each independent input has its own uncertainty, but each is constrained by data sources that do not require fitting to incidence time series.

Parameter-uncertainty interval on $\hat{\mathcal{R}}_0$. The point estimate $\mathcal{R}_0 \approx 1.64$ for the high-activity subgroup is computed from a single committed parameter set. To quantify how far the stated parameter uncertainties carry through, we apply the delta-method theorem 9.1 of Chapter 9. The bipartite structure halves the per-direction transmission-probability sensitivities (each $\tilde{\beta}$ enters $\mathcal{R}_0^{\text{horiz}}$ with sensitivity $+1/2$), while c_H enters with unit sensitivity and the mortality combination $\nu_m + \mu_d$ enters with unit-magnitude negative sensitivity. The dominant uncertainties for the high-activity subgroup follow. Each is an approximate 95 percent range, so one relative standard deviation is half the stated value:

- Partnership-formation rate $c_H \approx 5$ partnerships/year: $\pm 30\%$ (reflecting between-study and between-setting variability in sexual-behavior surveys),
- Per-partnership transmission probabilities $\tilde{\beta}_{MF}, \tilde{\beta}_{FM}$: $\pm 40\%$ each (reflecting clinical and biological heterogeneity),
- Mortality combination $\nu_m + \mu_d$: $\pm 15\%$ (well-constrained for pre-ART populations from cohort survival studies),
- Vertical-transmission probability $p_{\text{MTCT}} \approx 0.30$ (without ART): $\pm 20\%$,
- Birth rate ν_b: $\pm 10\%$ (well-constrained from demographic data).

The horizontal-component variance is, using $\Upsilon_{c_H}^{\mathcal{R}_0^{\text{horiz}}} = +1$, $\Upsilon_{\tilde{\beta}_{MF}}^{\mathcal{R}_0^{\text{horiz}}} = \Upsilon_{\tilde{\beta}_{FM}}^{\mathcal{R}_0^{\text{horiz}}} = +1/2$, $\Upsilon_{\nu_m+\mu_d}^{\mathcal{R}_0^{\text{horiz}}} = -1$:

$$V^{\mathcal{R}_0^{\text{horiz}}} \approx (1)^2(0.15)^2 + (1/2)^2(0.20)^2 \cdot 2 + (-1)^2(0.075)^2 \approx 0.0225 + 0.0200 + 0.0056 \approx 0.0481.$$

The vertical-component variance, using $\Upsilon^{\mathcal{R}_0^{\text{vert}}}_{\nu_b} = +1$, $\Upsilon^{\mathcal{R}_0^{\text{vert}}}_{p_{\text{MTCT}}} = +1$, $\Upsilon^{\mathcal{R}_0^{\text{vert}}}_{\nu_m+\mu_d} = -1$, is

$$V^{\mathcal{R}_0^{\text{vert}}} \;\approx\; (0.05)^2 + (0.10)^2 + (0.075)^2 \;\approx\; 0.018.$$

Since $\mathcal{R}_0^{\text{horiz}}$ and $\mathcal{R}_0^{\text{vert}}$ are largely independent in their uncertainty inputs (excepting the shared $\nu_m + \mu_d$, which contributes a small correlation correction), and the total $\mathcal{R}_0 = \rho(\boldsymbol{K})$ weights them by the elasticities of Section 19.3, its variance is

$$V^{\mathcal{R}_0} \;\approx\; (0.96)^2 \cdot 0.0481 + (0.04)^2 \cdot 0.018 \;\approx\; 0.0444.$$

The implied relative standard deviation is $\sqrt{0.0444} \approx 0.211$, giving a 95% parameter-uncertainty interval of approximately $1.64 \cdot (1 \pm 0.414)$, or numerically $[0.96, 2.32]$. The dominant variance contributions are c_H and the $\tilde{\beta}$ pair (jointly $\sim 88\%$ of $V^{\mathcal{R}_0^{\text{horiz}}}$); vertical-transmission parameters contribute negligibly (well under 1 percent of $V^{\mathcal{R}_0}$). The interval's lower bound falls just below the threshold $\mathcal{R}_0 = 1$: the stated uncertainties cannot rule out a core group at or under threshold. That is a sharper form of the historical observation that HIV in core groups is only modestly above threshold, and of the case that interventions targeting c_H or $\tilde{\beta}$ can drive it below. ART expansion is the prototype intervention: in adherent patients, $\tilde{\beta}_{MF}, \tilde{\beta}_{FM} \to 0$ within months, dropping $\mathcal{R}_0^{\text{horiz}}$ by orders of magnitude. The interval propagates the stated parameter uncertainties only: it assumes they are independent (Section 8.2 carries the correlated form), it takes no account of model-structure uncertainty, and because the inputs are plausible ranges rather than estimated standard errors it carries no coverage guarantee.

The low-activity contrast. For the low-activity subgroup, $c_L = 0.5$ partnerships per year reduces the horizontal $\mathcal{R}_0$ by a factor of 10 relative to the high-activity subgroup:

$$\mathcal{R}_0^{\text{horiz, L}} \;\approx\; \frac{c_L \sqrt{\tilde{\beta}_{MF}\, \tilde{\beta}_{FM}}}{\nu_m + \mu_d} \;\approx\; 0.16.$$

Well below threshold. The full multi-group model captures this structure clearly: the high-activity subgroup sustains transmission with $\mathcal{R}_0 \approx 1.58$, while the low-activity subgroup alone could not sustain HIV. A model that gives everyone the population-average contact rate, $\bar{c} = 0.2 \cdot 5 + 0.8 \cdot 0.5 = 1.4$ partnerships per year, would compute $\mathcal{R}_0 \approx 0.44$, well below threshold, and would predict (incorrectly) that HIV cannot sustain itself in this population. The error lies in the averaging, not in the mixing. Under random (proportionate) mixing of the same two activity classes, partners are drawn in proportion to their activity, so the effective contact rate is $\langle c^2 \rangle / \langle c \rangle = 5.2/1.4 \approx 3.71$ and $\mathcal{R}_0 \approx 1.17$, above threshold. Replacing a distribution of contact rates by its mean is the canonical failure for activity-stratified pathogens.

Sanity check at the endemic equilibrium. For a chronic disease that has reached an approximate endemic equilibrium in a population (a regime that high-prevalence settings approach over decades), the effective reproductive number satisfies $\mathcal{R}_e^* = \mathcal{R}_0 \cdot S^*/N^* \approx 1$ at equilibrium. This identity follows directly from the equilibrium condition: at steady state, each infectious individual must replace itself on average. Operationally, the identity gives a one-line consistency check on a fitted model. Given an independent estimate of equilibrium susceptible fraction S^*/N^* from a recent serosurvey (typically 1 minus the cumulative seroprevalence in the at-risk subpopulation), the model-implied $\mathcal{R}_0$ should satisfy $\mathcal{R}_0 \approx N^*/S^*$. A fitted $\mathcal{R}_0$ that violates this relation by more than a factor of 1.5 in either direction signals that the model is not at equilibrium with the data,

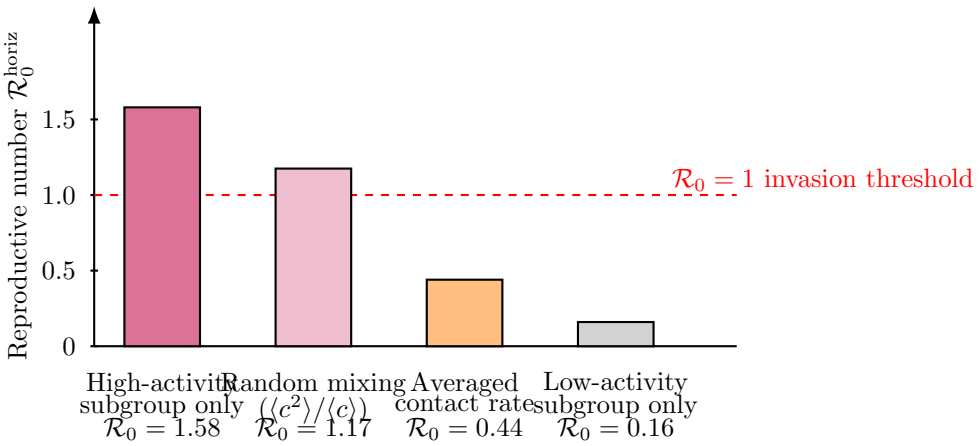


Figure 19.3: Contact-rate averaging for HIV. The high-activity subgroup alone sustains $\mathcal{R}_0 = 1.58$, above the invasion threshold; the low-activity subgroup alone has $\mathcal{R}_0 = 0.16$. Random (proportionate) mixing of the two classes gives $\mathcal{R}_0 \approx 1.17$, still above threshold, because the effective contact rate is the activity-weighted $\langle c^2\rangle/\langle c\rangle = 3.71/\text{yr}$. Replacing every individual's rate by the mean $\bar{c} = 0.2(5) + 0.8(0.5) = 1.4/\text{yr}$ gives $\mathcal{R}_0 = 0.44$, below threshold, and predicts incorrectly that the pathogen cannot invade. Averaging destroys the invasion-relevant structure: $\mathcal{R}_0$ follows $\langle c^2\rangle/\langle c\rangle$ under random mixing and approaches the high-activity value as mixing becomes fully assortative, never the mean.

that one of the input data products is biased, or that an unmodelled mechanism (treatment scale-up, behavior change, in-migration) is shifting the equilibrium. The check parallels the analogous endemic-equilibrium sanity check developed for the basic SIR model in Chapter 7, Section 7.8.

Balanced initial conditions for the multigroup model. A practical question arises whenever the activity-stratified SIR system is simulated forward from a chosen starting date: how should (S_j, I_j, R_j) be apportioned among the activity groups $j \in \{H, L\}$ and between sexes? Arbitrary initial apportionment injects a transient into the early dynamics that has nothing to do with the pathogen and contaminates fits to early-time data. The resolution is to choose *balanced* (or *natural*) initial conditions: the compartmental split across activity groups is set so that demographic flows and cross-group contact rates are in instantaneous equilibrium at $t = 0$, modulo the small epidemiological perturbation introduced by $I_j(0) > 0$. With balanced ICs the transient at the start of the simulation reflects epidemiological dynamics alone rather than a mismatch between the chosen split and the flow structure. Hyman et al. [2001] gives the first systematic development of this balancing procedure for multigroup HIV models and discusses its sensitivity properties; the methodology extends directly to the bipartite activity-stratified structure used throughout this chapter.

Identifiability of activity stratification. The most distinctive identifiability question for HIV models is the size of the high-activity subgroup (Pitfall: "Identifiability"). Sexual-behavior surveys consistently identify a long-tailed distribution of partnership-formation rates, but the cutoff between "high" and "low" activity is to some extent arbitrary, and the inferred subgroup sizes vary across studies and settings. The framework's response is to recognize that the high-activity-subgroup's transmission dynamics are largely insensitive to the subgroup size: $\mathcal{R}_0^{\text{horiz}}$ in (19.4) depends on the partnership-formation rate c_H and the per-partnership transmission probability, but not on the absolute size of the subgroup. Population-level burden estimates do depend on subgroup size, but

invasion-control conclusions (whether $\mathcal{R}_0 > 1$, what intervention strength is needed to drive $\mathcal{R}_0 < 1$) are subgroup-size-independent.

Exercises for this section

1. **[L]** *HIV parameter estimation with activity-stratified data.* You have access to two synthetic data products from a hypothetical HIV surveillance system covering a population of $N^{\mathrm{c}} =$ 1,000,000 adults aged 15–49: (i) annual seroprevalence by activity stratum and sex from cross-sectional surveys for years $t = 0, 1, \ldots, 20$ (giving $\hat{I}_j^g(t)/N_j^g$ for $j \in \{H, L\}$, $g \in \{M, F\}$); (ii) annual seroincidence in each stratum from longitudinal cohort follow-up (giving $\hat{J}_j^g(t)$). The activity-stratum cutoff is set at 2 partnerships per year by survey convention, with the high-activity stratum representing $\hat{\pi}_H = 18\%$ of each sex (uncertainty range 12% to 25%). Generate the synthetic data yourself by simulating the activity-stratified bipartite model of this chapter from the baseline parameters with the balanced initial conditions of Hyman et al. [2001], coupling the strata by random mixing, partners drawn in proportion to $c_k N_k$; add multiplicative log-normal noise with $\sigma = 0.10$. Execute the following stratified-data fitting workflow:

   (a) *Stratum-specific per-case transmission estimates.* For each year t and each (j, g) combination, compute $\hat{\alpha}_j^g(t) = \hat{J}_j^g(t)/\hat{I}_j^g(t)$. Note what this ratio is: in a bipartite model the new infections in stratum (j, g) are generated by infectious partners of the *opposite* sex, so $\hat{\alpha}_j^g$ is the realized rate of new infections per current case in that stratum, not a per-contact rate of the stratum's own members. It remains the right quantity to compare across strata, because the comparison is the point. Show numerically that, once the initial transient has passed, $\hat{\alpha}_H^g$ and $\hat{\alpha}_L^g$ converge to a common value, the growth rate plus $\nu_m + \mu_d$, as in Section 17.4; the stratification shows in the stratum prevalences, which do differ by about the factor c_H/c_L.

   (b) *Fit c_H from the high-activity stratum.* Using only the high-activity stratum's $\hat{\alpha}_H^g$ time series and the per-partnership transmission probabilities $\tilde{\beta}_{MF}$ and $\tilde{\beta}_{FM}$ from independent cohort studies (assume known with ±25% uncertainty each), recover an estimate $\hat{c}_H$ from the early-time plateau of $\hat{\alpha}_H^g$, using $\hat{\alpha}_H = c_H\sqrt{\tilde{\beta}_{MF}\tilde{\beta}_{FM}}$, which holds when the high-activity stratum mixes only within itself. Report a 95% CI via bootstrap. Compute the implied $\hat{\mathcal{R}}_0^{\mathrm{horiz},H}$ from (19.4), propagating the $\tilde{\beta}$ uncertainties.

   (c) *Robustness to subgroup size.* Repeat (b) under three alternative assumptions about the activity-stratum size: $\hat{\pi}_H = 12\%$, 18%, and 25%. Report the resulting $\hat{c}_H$ and $\hat{\mathcal{R}}_0^{\mathrm{horiz},H}$ in each case. Verify the framework's claim that $\hat{\mathcal{R}}_0^{\mathrm{horiz},H}$ is approximately invariant to $\hat{\pi}_H$ across this range, and quantify the deviation in percent. Identify which intermediate quantity in the fit *does* depend on $\hat{\pi}_H$, and explain why this dependence does not propagate to the invasion-relevant $\mathcal{R}_0^{\mathrm{horiz},H}$.

   (d) *Homogeneous-mixing comparison.* As a control, perform a homogeneous-mixing fit to the population-aggregated incidence $\hat{J}(t) = \sum_{j,g} \hat{J}_j^g(t)$ and prevalence $\hat{I}(t) = \sum_{j,g} \hat{I}_j^g(t)$, recovering a single population-averaged $\hat{\tilde{c}}$ and the implied $\hat{\mathcal{R}}_0^{\mathrm{homog}}$. Compare $\hat{\mathcal{R}}_0^{\mathrm{homog}}$ to $\hat{\mathcal{R}}_0^{\mathrm{horiz},H}$ and to the threshold value $\mathcal{R}_0 = 1$. Discuss which fit gives the operationally correct invasion conclusion (i.e., predicts that the pathogen will sustain transmission, as the synthetic data were generated to do). The exercise reproduces the canonical activity-stratification failure mode visualized in Figure 19.3.

The companion repository contains reference data-generation and fitting scripts; only the activity-stratification handling needs to be implemented from scratch.

19.5 Intervention Scenarios

We evaluate four intervention scenarios spanning the per-partnership transmission, partnership-formation, and vertical-transmission parameter groups identified in Section 19.3.

Scenario 1: treatment-as-prevention (90 percent reduction in per-partnership transmission). Antiretroviral therapy with high adherence reduces per-partnership transmission by approximately 90 percent for both transmission directions ($\tilde{\beta}_{MF} \to 0.004$, $\tilde{\beta}_{FM} \to 0.0025$). The horizontal $\mathcal{R}_0$ scales by $\sqrt{0.1 \cdot 0.1} = 0.1$:

$$\mathcal{R}_0^{\text{horiz, ART}} \;=\; 1.58 \cdot 0.1 \;=\; 0.158.$$

Combined with PMTCT prophylaxis (Scenario 4 below) reducing $\mathcal{R}_0^{\text{vert}}$ to approximately 0.008, the total is $\mathcal{R}_0 = \rho(\boldsymbol{K}) \approx 0.16$. Far below the invasion threshold; elimination is achievable in any subpopulation with high ART coverage. This is the analytical basis for the treatment-as-prevention paradigm.

Scenario 2: condom use (50 percent reduction in per-partnership transmission). Halving $\tilde{\beta}_{MF}$ and $\tilde{\beta}_{FM}$ through consistent condom use:

$$\mathcal{R}_0^{\text{horiz, condom}} \;=\; 1.58 \cdot \sqrt{0.5 \cdot 0.5} \;=\; 1.58 \cdot 0.5 \;=\; 0.79.$$

Below the invasion threshold; horizontal transmission alone cannot sustain HIV in the high-activity subgroup. With the small vertical contribution (~ 0.12), the total $\mathcal{R}_0 = \rho(\boldsymbol{K}) \approx 0.85$, still below threshold. The combined-condom-and-PMTCT scenario achieves elimination.

Scenario 3: behavioral intervention (50 percent reduction in c_H). Halving partnership-formation rate through behavioral intervention:

$$\mathcal{R}_0^{\text{horiz, behav}} \;=\; 1.58 \cdot 0.5 \;=\; 0.79.$$

The same effective reduction in horizontal $\mathcal{R}_0$ as Scenario 2. The unit sensitivity of c_H versus the half-sensitivity of $\tilde{\beta}$ means that behavioral and per-partnership-transmission interventions are equally effective per percentage of parameter change at the per-percent level. The operational difference is that a 50 percent reduction in c_H is harder to sustain at population scale than a 50 percent reduction in $\tilde{\beta}$ through condom use, while a 90 percent reduction in $\tilde{\beta}$ through ART exceeds what behavioral intervention alone can plausibly achieve.

Scenario 3b: targeted behavioral intervention on known positives (50 percent reduction in c_H^I). Scenario 3 applied a uniform 50 percent reduction to the single c_H, implicitly halving partnership formation for susceptibles and infecteds alike. Operationally, behavioral interventions in HIV programs are typically *targeted* at diagnosed-positive individuals (post-diagnosis counseling, partner-notification services, disclosure and serosorting support), reducing c_H^I specifically while

leaving c_H^S approximately unchanged. Under the compartment-specific forces (19.3), halving c_H^I yields

$$\mathcal{R}_0^{\text{horiz, targeted}} \;=\; 1.58 \cdot 0.5 \;=\; 0.79,$$

the same horizontal-$\mathcal{R}_0$ reduction as uniform Scenario 3 because the force-from-infection carries only the infected host's rate. This parallels the vector-borne case in Chapter 13, Section 13.5.3: interventions on infected hosts alone suffice when only infecteds emit infection. Two operational qualifications: the intervention reaches only *known* positives, so the effective reduction scales with diagnosis coverage (at 80 percent diagnosis, a 50 percent reduction in c_H^I among the diagnosed produces approximately 40 percent reduction in the population-averaged c_H^I, giving $\mathcal{R}_0^{\text{horiz}} \approx 0.95$); and the distinction between c_H^S and c_H^I is slow-acting in HIV because chronic infection means the diagnosed-to-treated-to-behaviorally-changed transition occurs over months rather than days. The framework makes the nominal-to-effective coverage conversion transparent through the compartment-specific rate.

Scenario 4: PMTCT alone (approximately 93 percent reduction in p_{MTCT}). Antiretroviral prophylaxis during pregnancy reduces p_{MTCT} from 0.30 to approximately 0.02 (a reduction of about 93 percent), dropping $\mathcal{R}_0^{\text{vert}}$ from 0.12 to approximately 0.008:

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}) \;\approx\; 1.59.$$

PMTCT alone does not change horizontal transmission and does not bring $\mathcal{R}_0$ below threshold. The intervention is essential for maternal-child clinical outcomes (preventing tens of thousands of pediatric infections globally each year) but does not substitute for horizontal-transmission control. The framework distinguishes the maternal-child clinical case from the population-elimination case cleanly through the decomposition of $\mathcal{R}_0$ into horizontal and vertical contributions.

The intervention complementarity. The four scenarios collectively show that ART-based treatment-as-prevention is the highest-impact single intervention (because the per-partnership transmission probability is the per-individual lever that ART acts on directly), behavioral intervention and condom use are comparable middle-impact strategies, and PMTCT addresses a separate clinical objective. Combinations are typically deployed in operational settings, with ART for known infectious individuals as the foundation, condom use and behavioral intervention as supplements, and PMTCT layered on for prevention of vertical transmission. The framework's contribution is to compare these levers on the same scale through their effect on $\mathcal{R}_0$ rather than through case counts in arbitrary scenarios.

Treatment-as-prevention: coverage-response curve. Figure 19.4 plots the total $\mathcal{R}_0 = \rho(\boldsymbol{K}(c))$ as a function of ART coverage $c \in [0, 1]$, under the assumption that ART achieves 96 percent suppression of horizontal transmission (the U=U benchmark) and 90 percent suppression of vertical transmission when both parents in infected partnerships receive ART. The dominant horizontal contribution scales approximately linearly with $(1 - 0.96\,c)$ because uniform ART coverage reduces the effective per-partnership transmission in both directions proportionally; the vertical contribution scales with $(1 - 0.90\,c)$. The invasion threshold $\mathcal{R}_0 = 1$ is crossed at approximately 41 percent coverage, a non-trivial but achievable target that has been approached or surpassed in several high-prevalence settings with mature national programs. The curve makes the intervention-effectiveness argument graphically: modest coverage suppresses $\mathcal{R}_0$ below threshold, and high coverage drives $\mathcal{R}_0$ near zero, consistent with the framework's prediction that $\tilde{\beta} \to 0$ reduces the bipartite $\mathcal{R}_0^{\text{horiz}}$ toward zero.

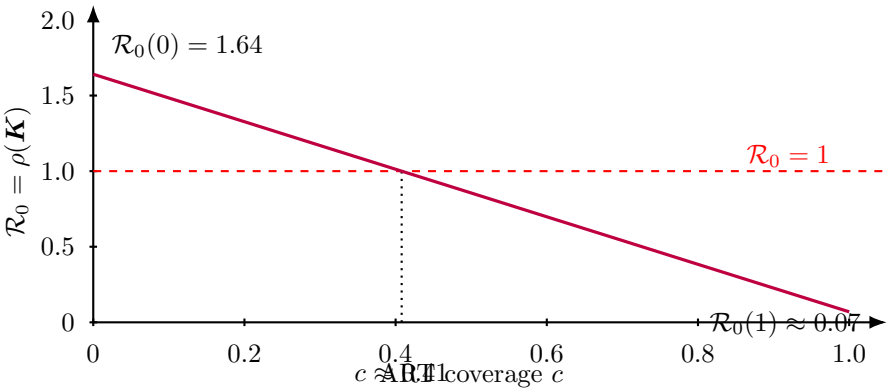


Figure 19.4: Treatment-as-prevention coverage-response curve for HIV in the high-activity subgroup. The threshold quantity $\mathcal{R}_0 = \rho(\boldsymbol{K}(c))$ is plotted against ART coverage $c \in [0, 1]$, with the horizontal cycle scaled by $(1 - 0.96c)$ (96 percent per-partnership-transmission suppression under U=U) and the vertical loop by $(1 - 0.90c)$. The invasion threshold is crossed at $c \approx 0.41$. At full coverage, residual $\mathcal{R}_0 \approx 0.07$ reflects the 4 percent non-suppression rate in adherent ART users; near-complete elimination is achievable but not absolute. The curve is nearly linear because both contributions scale approximately linearly with $(1 - c)$ under uniform coverage in a bipartite system where both partners are drawn from the same coverage population. This is the analytical structure underlying the empirical observation that national ART programs achieving 40–50 percent coverage can drive local incidence below replacement rate, while full elimination requires coverage approaching 90 percent given realistic adherence and transmission heterogeneity.

Exercises for this section

1. [C] Treatment-as-prevention (ART) extends an infectious individual's lifetime while suppressing transmission probability per contact. Explain qualitatively why these two effects pull $\mathcal{R}_0$ in opposite directions, and why the suppression effect typically dominates.

19.6 Lessons from the HIV Case Study

The case study illustrates four general lessons about how the framework applies to a chronic, sexually-transmitted, activity-stratified pathogen.

First, the activity-stratified bipartite structure recovers the high-activity subgroup as the transmission core, a mechanism that a model with an averaged contact rate cannot capture. With the contact rate averaged, $\mathcal{R}_0$ falls below threshold and the model predicts (incorrectly) that HIV cannot sustain itself; with the stratification, the high-activity subgroup sustains transmission and the dynamics are correctly captured. This is the canonical empirical signature of activity-level heterogeneity, formalized by the framework as a multi-group NGM computation.

Second, treatment-as-prevention is the highest-impact intervention in the framework's accounting, dropping $\mathcal{R}_0$ from approximately 1.6 to approximately 0.2 in subpopulations with high ART coverage. The framework recovers this conclusion from the basic per-partnership-transmission formula without invoking complex pharmacological modeling. The U=U slogan ("undetectable equals untransmissible") has its mathematical anchor in the framework's prediction that $\tilde{\beta} \to 0$ produces a near-zero contribution to $\mathcal{R}_0^{\text{horiz}}$.

Third, the decomposition of $\mathcal{R}_0$ into horizontal and vertical contributions separates the population-elimination question from the maternal-child clinical question. The vertical contribution is small at the population level (about 8 percent of horizontal at baseline) but PMTCT is essential for clinical outcomes; the framework distinguishes these without conflating them. Operational programs typically deploy both horizontal-transmission interventions (ART for the infected, condom use, behavioral) and PMTCT in parallel; the framework provides the analytical scaffold for combining them.

Fourth, the framework's response to HIV's long natural history is the residence-time perspective: the infectious period is set by mortality alone in the absence of natural recovery, $\tau_R^* \approx 1/(\nu_m + \mu_d)$. ART changes μ_d (extending residence time) at the same time as it changes $\tilde{\beta}$ (suppressing transmission); the net effect on $\mathcal{R}_0$ is captured cleanly through the framework but would be obscured in models that conflate "infectious period" and "residence time." For pathogens with even longer natural histories (HSV, HPV, HBV chronic carriage), the same residence-time formulation applies.

Limitations of the case study. The case study omits the time-since-infection variable-infectiousness structure (acute, chronic, AIDS phases) of Chapter 15, which would refine the per-partnership transmission probability calculation and is operationally important for understanding acute-phase transmission dynamics (Pitfall: "Conditional versus Marginal Probabilities"). The case study uses per-partnership rather than per-contact transmission probabilities, which simplifies but does not fully capture the within-partnership dynamics that determine whether a partnership transmits. The case study restricts attention to heterosexual transmission in a single high-activity subgroup; MSM populations and parenteral transmission (intravenous drug use) follow related but distinct framework specializations that build on the bipartite structure presented here. Geographic heterogeneity (concentrated epidemics in sub-Saharan Africa versus generalized epidemics elsewhere) is not addressed.

Exercises for this section

1. **[C]** The chapter shows that with an averaged contact rate, HIV's $\mathcal{R}_0 < 1$, but under activity-stratified mixing the high-activity subpopulation has $\mathcal{R}_0^{\text{horiz, H}} > 1$, and random mixing of the two classes also gives $\mathcal{R}_0 > 1$. Explain in plain English why this discrepancy is not a contradiction, and what it implies about which model class is appropriate for HIV epidemiology.

Summary

The HIV case study formulates an activity-stratified bipartite SI model focused on the high-activity subgroup where transmission is concentrated. For typical untreated heterosexual parameters, the horizontal $\mathcal{R}_0 \approx 1.58$ in the high-activity subgroup (above threshold) and approximately 0.16 in the low-activity subgroup (below threshold). Vertical transmission raises the high-activity $\mathcal{R}_0 = \rho(\boldsymbol{K})$ from 1.58 to 1.64, about 4 percent. Sensitivity analysis identifies the partnership-formation rate c_H as the unique unit-sensitivity invasion parameter (because it appears in both transmission directions of the bipartite cycle) and the mortality rate $\nu_m + \mu_d$ as the unit-sensitivity-but-negative parameter (the residence-time effect specific to chronic infections). Intervention scenarios show that treatment-as-prevention (90 percent reduction in per-partnership transmission through ART) is the highest-impact single intervention, with condom use and behavioral interventions producing comparable middle-magnitude effects. PMTCT addresses maternal-child clinical outcomes but does

not substitute for horizontal-transmission control. The framework's contribution is to compare these intervention levers on the same scale through their effect on $\mathcal{R}_0$.

What this case study teaches structurally. The HIV analysis illustrates activity-stratified core-group dominance: when a population contains a high-activity subgroup whose partnership-formation rate is large enough that the subgroup's internal $\mathcal{R}_0$ exceeds unity, and the between-group mixing is weak (as is typical for heterosexual transmission of HIV), the population-level transmission dynamics are set essentially by the core group alone. The population average $\bar{c}$, which is the quantity reported in national sexual-behavior surveys, is not the right scale on which to compute $\mathcal{R}_0$: using $\bar{c}$ gives a sub-threshold answer and predicts that HIV should not persist, which is empirically wrong. The stratified calculation gives a supra-threshold answer in the core group, and the core group then reseeds incidence into the low-activity subgroup at a rate controlled by between-group mixing. The pedagogical lesson transfers beyond HIV to any pathogen with substantial activity-level heterogeneity: gonorrhea, syphilis, and hepatitis C share this structure, and every intervention-effectiveness analysis must identify the core group first. The framework's tool for this identification is the bipartite NGM of Chapter 12 applied at the stratification that captures the relevant heterogeneity.

Pitfall check

This case study has applied six catalogue entries to one bipartite model. The partnership-formation rate c_H enters both transmission directions, so balance ties the two sides together (Pitfall: "Heterogeneous-Mixing Balance Conditions"), and vertical transmission is folded into an effective birth term (Pitfall: "Short-Term Simulation Assumptions"); both appear in Section 19.2, as does the extension of the invasion–burden partition to this model under its non-cancellation hypothesis (Pitfall: "Invasion versus Equilibrium"). The size of the high-activity subgroup is the identifiability question that matters most (Pitfall: "Identifiability"; Section 19.4), and the logarithmic form of $\mathcal{R}_0$ makes its relative sensitivities immediate (Pitfall: "Sensitivity Analysis Discipline"; Section 19.3). The study's stated limitation, a per-partnership transmission probability that ignores how infectiousness varies with time since infection, is where Pitfall: "Conditional versus Marginal Probabilities" enters (Section 19.6). Before fitting a model of this kind, ask which rates are shared across directions, and how well the data fix the size of the most active group.

Exercises

Four section-level exercises appear above, following Section 19.2, Section 19.4, Section 19.5, and Section 19.6. The exercises below integrate across the chapter. Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. **[I]** Recompute $\mathcal{R}_0^{\text{horiz}}$ assuming ART coverage is partial: 50 percent of infectious individuals on adherent ART. Use a weighted-average $\tilde{\beta}_{\text{eff}} = 0.5\tilde{\beta} + 0.5 \cdot 0.1\tilde{\beta} = 0.55\tilde{\beta}$ for both transmission directions. What ART coverage threshold drives $\mathcal{R}_0^{\text{horiz}} < 1$?

2. **[D]** Extend the model to include the MSM (men-who-have-sex-with-men) subpopulation as a third group with within-group transmission. Use the three-group framework of Chapter 12, Section 12.5. Identify the key bridge-population parameter and discuss why it is consequential for global HIV dynamics.

3. **[I]** Suppose that the residence time of an infectious individual in the population doubles due to widespread ART (so $\nu_m + \mu_d$ halves). Recompute $\mathcal{R}_0^{\text{horiz}}$ under (a) no other intervention, and (b) ART that simultaneously reduces $\tilde{\beta}$ by 90 percent. The interaction between residence-time extension and transmission suppression is the central feature of treatment-as-prevention; explain qualitatively why the (b) case still drives $\mathcal{R}_0$ below threshold.

4. **[I]** For PrEP (pre-exposure prophylaxis for uninfected high-risk individuals), construct a modified force-from-infection that accounts for PrEP-protected susceptibles: $\alpha_M^H \to (1 - f \cdot e_{\text{PrEP}})\alpha_M^H$, and likewise α_F^H, where f is the PrEP coverage and $e_{\text{PrEP}} \approx 0.9$ is per-individual efficacy. Compute the PrEP coverage threshold required to bring $\mathcal{R}_0^{\text{horiz}}$ below 1.

5. **[L]** Implement the bipartite high-activity model in your preferred scientific computing environment (Python, MATLAB, or R) and simulate forward from a small introduction of infected females (a deliberately naïve single-sex initial condition that suffices for the early-phase growth-rate verification asked for here, where the few-week transient from initial-condition imbalance is irrelevant; for fitting to multi-year longitudinal data, use the balanced initial conditions of Hyman et al. [2001] as in the parameter-estimation lab above, otherwise the demographic-equilibration transient, which can extend for one to two decades in HIV-time before settling, will contaminate the early-time fit). Compute the early-phase exponential growth rate, invert through the procedure of Chapter 7, Section 7.4 to estimate $\hat{\alpha}$, and verify it reproduces the analytic $\mathcal{R}_0^{\text{horiz}}$ from (19.4). The companion repository provides reference Python notebooks (Colab-executable) for this exercise.

6. **[I]** *Reasoning visibility.* HIV requires activity-stratified modeling because the high-activity subgroup sustains $\mathcal{R}_0 > 1$ while the population average $\mathcal{R}_0$ is sub-threshold (Figure 19.3). Influenza, by contrast, is conventionally modeled with homogeneous mixing despite real heterogeneity in contact rates (children have more contacts than adults; healthcare workers more than the general population). Articulate, in three paragraphs, the reasoning that justifies this asymmetric treatment:

   (1) State the structural test: under what condition on the contact-rate distribution does activity stratification matter for $\mathcal{R}_0$ even after correctly averaging? Reference the canonical formula relating the variance of the contact-rate distribution to the mean (for proportional mixing, $\mathcal{R}_0$ depends on $\langle c^2\rangle/\langle c\rangle$, not $\langle c\rangle$). State the magnitude of $\langle c^2\rangle/\langle c\rangle^2$ at which stratification becomes essential.

   (2) Apply the test to HIV and influenza. For HIV, the partnership-formation-rate distribution is order-of-magnitude long-tailed: a small subpopulation has rates $10\times$ the mean, so $\langle c^2\rangle/\langle c\rangle^2$ is large enough that the high-activity subgroup dominates $\mathcal{R}_0$. For influenza, contact rates vary across age and occupation but the variance is bounded: the highest-contact subgroup typically has rates at most 2–3$\times$ the mean, so $\langle c^2\rangle/\langle c\rangle^2$ is modest and the homogeneous-mixing $\mathcal{R}_0$ is approximately correct. Cite specific empirical values where you can.

   (3) State the observable that would tell you, for a new pathogen, whether stratification is required: what data product (sexual-behavior survey, contact-pattern study, work-and-school contact diary, age-stratified seroprevalence) would let you compute $\langle c^2\rangle/\langle c\rangle^2$ and decide? Identify the threshold value at which you would escalate from a homogeneous to a stratified model.

End with a one-sentence summary of the rule: when contact-rate variance dominates contact-rate mean, stratify; when contact-rate mean dominates contact-rate variance, average. The exercise tests whether you can move from the canonical examples (HIV stratified, influenza averaged) to a general principle that you can apply to a new pathogen.

7. [**Open Project**] *A combination-prevention portfolio for one HIV epidemic.* A multi-week capstone project fitting the activity-stratified model to published surveillance data for a single setting, then comparing ART, PrEP and behavioral coverage as a budget-constrained portfolio rather than singly.

Chapter 20

Cross-Pathogen Synthesis: Lessons from COVID, Dengue, and HIV

Each pathogen looks unique until you ask which common pitfalls apply to all of them.

Even simple epidemic models can mislead when basic pitfalls are overlooked. The three case studies of the preceding chapters made this concrete in three settings of strikingly different biology: COVID-19 (directly transmitted, respiratory, with substantial asymptomatic spread), dengue (vector-borne, with bipartite human-mosquito dynamics and temperature-sensitive vector biology), and HIV (sexually transmitted, with activity-stratified core-group dynamics and a natural history spanning a decade). Each case study showed the same pattern: a careful modeler who recognized the relevant common pitfalls from the catalog produced quantitative analyses that held up; a careless approach that overlooked one or more common pitfalls produced precise-looking but unreliable claims. This closing chapter draws the cross-cutting lessons together: which common pitfalls recurred in operational form, which framework machinery served them well, and which questions still require extensions beyond the deterministic compartmental scope.

This chapter is not another case study. It reads the three side by side. Section 20.1 compares the three pathogens, their model structures, and their principal results. Sections 20.2 to 20.5 take the pitfalls that recurred across the cases one at a time, and each compares the analytical tool that addressed its pitfall in all three diseases: the invasion-burden partition (Pitfall: "Invasion versus Equilibrium"), bipartite square-root dampening (Pitfall: "Heterogeneous-Mixing Balance Conditions"), reservoir heterogeneity (Pitfall: "Heterogeneous-Mixing Balance Conditions" and Pitfall: "Identifiability"), and infected-viewpoint fitting (Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Linearization Fidelity"). Sections 20.6 and 20.7 set out what the framework captures and where its limits lie, with a rule for when each extension is needed. Section 20.8 draws the broader lessons, and Section 20.9 closes with five open problems.

20.1 Comparative Summary Across the Three Diseases

The three case studies span the taxonomic categories of directly transmitted, vector-borne, and sexually transmitted pathogens. The principal numerical results are collected in Figure 20.1.

Observations from the comparative figure. The baseline $\mathcal{R}_0$ values cluster in a surprisingly narrow range despite the dramatic differences in pathogen biology: 2.12 for dengue, 1.64 for HIV

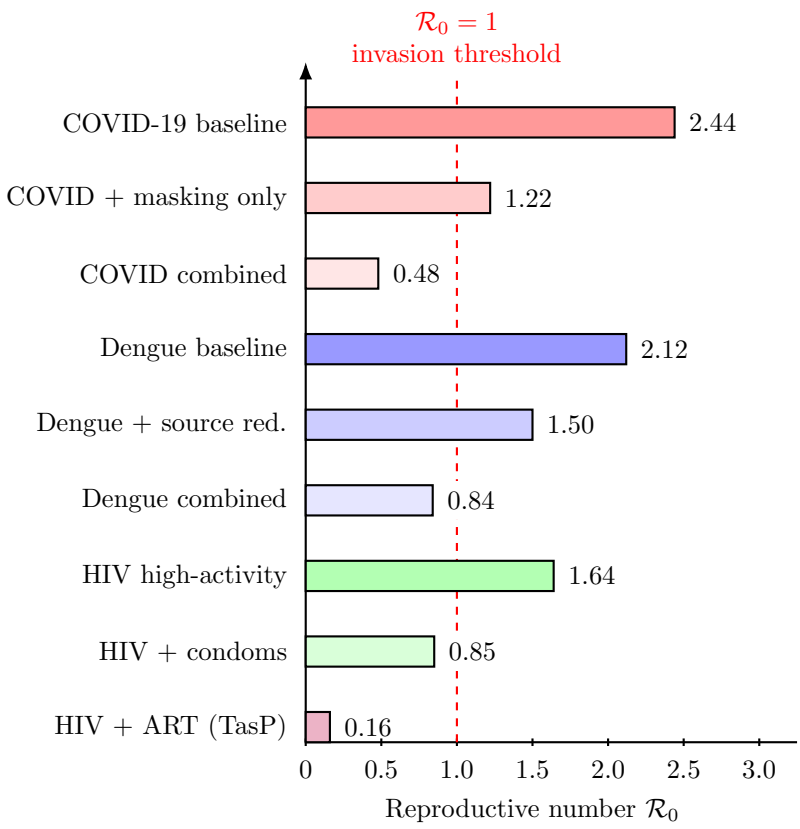


Figure 20.1: Comparative summary of $\mathcal{R}_0$ across the three case studies, showing baseline and post-intervention values. Red dashed line marks the invasion threshold $\mathcal{R}_0 = 1$. Baseline $\mathcal{R}_0$ values cluster around 1.6 to 2.4 for all three pathogens, with COVID-19 highest (respiratory transmission in a fully susceptible population) and HIV lowest (long natural history and low per-contact transmission compensating partially for high partnership-formation rates in the high-activity subgroup). Single interventions bring $\mathcal{R}_0$ close to threshold for COVID-19 (masking) and dengue (source reduction) but not below it; for HIV, condoms alone and ART alone bring it below. Combined interventions drive $\mathcal{R}_0$ below threshold for COVID-19 and dengue. The HIV treatment-as-prevention scenario achieves the largest relative reduction (10×) because ART acts directly on per-partnership transmission with near-total effectiveness. Per the convention of Chapter 5, Section 5.2, the post-intervention values shown here are technically the controlled reproductive number $\mathcal{R}_c$ for each scenario, the value of $\mathcal{R}_e$ at the post-intervention pre-depletion state. The figure uses the $\mathcal{R}_0$ notation throughout for visual consistency with the baseline-versus-post-intervention comparison; the interpretation is that each post-intervention bar is the $\mathcal{R}_0$ of the modified system (equivalently, the $\mathcal{R}_c$ of the original system under the indicated intervention regime). Numerical values are taken from the case-study chapters: COVID from Chapter 17, Section 17.6, dengue from Chapter 18, Section 18.5, HIV from Chapter 19, Section 19.5.

in the high-activity subgroup, and 2.44 for COVID-19. This narrowness is not a coincidence of the parameter choices but reflects a deeper pattern: pathogens that sustain endemic or outbreak transmission in human populations tend to have $\mathcal{R}_0$ in the range 1 to 5, with most between 1.5 and 3. The pathogens at the extremes (measles at $\mathcal{R}_0 \approx 15$; HIV in low-activity populations at $\mathcal{R}_0 \approx 0.2$) are the exceptions; pandemic influenza, near 1.5, sits at the low edge of the common range. The framework does not explain this empirical regularity, but it does accommodate it: the range of $\mathcal{R}_0$ values the framework handles analytically is unrestricted.

More consequentially, the figure shows how far one intervention can go. For COVID-19 and dengue, where $\mathcal{R}_0$ exceeds 2, no single measure in the case studies brings $\mathcal{R}_0$ below threshold (for dengue, when emergence replaces killed adults; Section 18.5), and only combinations do. HIV, whose core-group $\mathcal{R}_0$ is about 1.6, falls below threshold with condoms alone (0.85) and far below it with ART (0.16). A reader who takes away one operational message should take this one: the higher the baseline $\mathcal{R}_0$, the less a single intervention can do, and for pathogens with $\mathcal{R}_0$ above 2 combination strategies are the operational norm.

20.2 The Invasion-Burden Partition Across Diseases

The structural separation of invasion parameters from burden parameters, developed formally in Chapter 5, Section 5.7 as Theorem 5.6 and generalized in Chapter 10, Section 10.7 as Proposition 10.2, is the book's central structural contribution. The three case studies provide testbeds for the partition and demonstrate how it manifests under different transmission modes.

COVID-19: additive two-track structure. The COVID-19 $\mathcal{R}_0$ formula (17.3) decomposes into a symptomatic track ($p\,c_I\beta_I/(\gamma_R+\mu_d+\nu_m)$) and an asymptomatic track ($(1-p)\,c_A\beta_A/(\gamma_R+\nu_m)$). The invasion-facing parameters are the four transmission-related quantities $c_I, c_A, \beta_I, \beta_A$, each appearing linearly in $\mathcal{R}_0$ with sensitivity approximately proportional to its track's fractional contribution. The recovery rate γ_R is the mixed parameter: it appears in both $\mathcal{R}_0$ (at unit negative sensitivity) and in the endemic burden expression (where it determines residence time in the infectious compartment). The demographic rate ν_m is burden-facing at COVID's timescales but contributes only weakly because $\nu_m\tau_R \ll 1$.

The symptomatic fraction p is an unusual parameter: it is invasion-facing, but its sensitivity sign depends on the relative magnitudes of the two tracks. For the baseline parameters, where $c_A\beta_A > c_I\beta_I$, increasing p reduces $\mathcal{R}_0$ (shifting individuals to the lower-transmission track). This sign-dependence is specific to two-track models and is the reason the partition in COVID-19 requires more careful case-by-case analysis than in the basic SIR model.

Dengue: bipartite multiplicative structure. The dengue Ross-Macdonald $\mathcal{R}_0$ (18.4) has all invasion-facing parameters (the biting rate a, the per-bite transmission probabilities b_h and b_v, the vector-to-host ratio m) entering multiplicatively under the square root. The biting rate a is unique in appearing twice (once in each transmission direction), giving it unit sensitivity in the per-generation $\mathcal{R}_0$. The other invasion parameters each contribute at half-sensitivity. The vector mortality μ_v is invasion-facing twice over: directly, through the vector's infectious lifespan in the denominator, and indirectly, through the extrinsic-incubation survival probability in b_h^{eff}. The two effects are same-signed and reinforce, yielding the net sensitivity of approximately -0.760 with m held fixed, and about -1.26 if emergence stays fixed so that m falls as well (Chapter 18, Sections 18.4 and 18.5).

HIV: horizontal and vertical structure. The HIV $\mathcal{R}_0$ combines a horizontal bipartite contribution and a vertical mother-to-child contribution through the next-generation matrix, $\mathcal{R}_0 = \rho(\boldsymbol{K}) = \frac{1}{2}\left(\mathcal{R}_0^{\text{vert}} + \sqrt{(\mathcal{R}_0^{\text{vert}})^2 + 4(\mathcal{R}_0^{\text{horiz}})^2}\right)$ (Chapter 19); the sum $\mathcal{R}_0^{\text{horiz}} + \mathcal{R}_0^{\text{vert}}$ overstates it. The horizontal contribution follows the bipartite pattern of dengue (square-root dampening, unit-sensitivity for the shared partnership-formation rate c_H). The vertical contribution has full unit sensitivity in both its numerator parameters (ν_b and p_{MTCT}), and it enters the total with a weight of about 0.04. The mortality rate $\nu_m + \mu_d$ is the mixed parameter for HIV, appearing in both the horizontal and vertical denominators. Its sensitivity is nearly unity (negative) because HIV has no natural recovery.

Common structural pattern across all three pathogens. Across the three pathogens, the invasion-burden partition takes three different specific forms but preserves the same structural feature: the parameters that appear in $\mathcal{R}_0$ are the levers for prevention (invasion control), and the parameters that set the equilibrium residence time or the steady-state distribution are the levers for burden management. For acute pathogens (COVID-19, dengue), the two sets are nearly disjoint. For chronic pathogens (HIV), the mortality rate $\nu_m + \mu_d$ appears in both $\mathcal{R}_0$ and burden, and interventions that change μ_d (ART in the HIV case) affect both. The partition is not violated by this overlap; it is refined: the book's theorem accommodates the mixed category explicitly, in the τ_R and τ_m rows of Theorem 5.6 where both $\Upsilon_p^{\mathcal{R}_0}$ and $\Upsilon_p^{i^*}$ are non-zero.

20.3 Bipartite Square-Root Dampening

Both the dengue case study (Chapter 18) and the HIV case study (Chapter 19) yield reproductive-number formulas with square-root structure: $\mathcal{R}_0 \propto \sqrt{\tilde{\beta}_{MF}\,\tilde{\beta}_{FM}}$ for heterosexual HIV and $\mathcal{R}_0 \propto \sqrt{b_h\, b_v\, m}$ for vector-borne dengue. The COVID-19 case study, by contrast, has no square-root structure because direct transmission is not bipartite. The square-root dampening is the bipartite signature.

Operational implication: single-parameter interventions are less effective. For a directly transmitted pathogen, halving a single invasion parameter halves $\mathcal{R}_0$ (at unit sensitivity). For a bipartite pathogen, halving a single invasion parameter reduces $\mathcal{R}_0$ by only $1 - \sqrt{0.5} \approx 29\%$. The three case studies illustrate this quantitatively: the COVID-19 masking scenario (halving β) gave $\mathcal{R}_0 \approx 1.22$ (factor of 2 reduction from baseline 2.44), while the dengue source-reduction scenario (halving m) gave $\mathcal{R}_0 \approx 1.50$ (only 29% reduction from baseline 2.12). The difference is not due to the different intervention targets but to the bipartite structure of the dengue transmission formula.

The exception: unit-sensitivity parameters. Within the bipartite structure, one parameter typically has unit sensitivity: the shared parameter that appears in both transmission directions. For dengue, this is the biting rate a; for HIV, this is the partnership-formation rate c_H. Interventions targeting these parameters are as effective per-percent as direct-transmission interventions. The implication is that biting-rate reduction (bednets, repellents) moves dengue's $\mathcal{R}_0$ twice as far per percent as vector-population reduction with m held fixed; for HIV, reducing partnership formation moves $\mathcal{R}_0$ as far per percent as condom use, which acts on both transmission probabilities at once. Which is cheaper is a question of cost and coverage, not of sensitivity.

Why bipartite models require combinations. The square-root dampening means that a single halving of any one-sided invasion parameter produces less than a halving of $\mathcal{R}_0$, so a single intervention that achieves a 50 percent reduction in a target parameter is rarely sufficient to drive $\mathcal{R}_0$ below threshold if the baseline is above 2. Combination interventions, acting on multiple parameters simultaneously, are the operational norm. The case studies show this: dengue required combined source reduction and insecticide to achieve $\mathcal{R}_0 \approx 0.84$, while for HIV, whose baseline is lower, ART alone reaches $\mathcal{R}_0 \approx 0.16$ and condoms alone 0.85. The framework's analytical machinery (the NGM computation, the sensitivity indices) supports the combination-design process by identifying which parameter combinations maximize the cumulative $\mathcal{R}_0$ reduction per unit of intervention effort.

Exercises for this section

1. [**C**] The cross-cutting chapter argues that square-root dampening recurs in dengue and HIV but not in COVID-19. Explain in plain English what the unit-sensitivity parameter is in each of the two bipartite case studies, why the same dampening appears in two biologically different settings, and why it is absent from COVID-19. What feature of the next-generation matrix decides?

20.4 Reservoir Structures and Heterogeneity

The COVID-19 case study and the HIV case study share a structural feature that is not present in the dengue case study but that organizes intervention strategy profoundly when it applies: a subset of the population, distinguished by some biological or behavioral characteristic, sustains transmission at levels that the aggregate population cannot. In COVID-19, this is the asymptomatic compartment whose $c_A\beta_A$ product exceeds the symptomatic $c_I\beta_I$ at baseline parameters. In HIV, this is the high-activity subgroup whose c_H is three and a half times the population average $\bar{c}$ and ten times the low-activity rate.

The mathematical signature. In both cases, the mathematical signature is the same: next-generation-matrix analysis of a subsystem whose effective $\mathcal{R}_0$ exceeds unity, even when the population-averaged $\mathcal{R}_0$ does not. For COVID-19, the asymptomatic-only subsystem (the A-compartment dynamics in isolation from the i-compartment) gives $\mathcal{R}_0^{A\text{-only}} \approx 1.4$, above the invasion threshold. For HIV, the high-activity-subgroup-only subsystem gives $\mathcal{R}_0^{H\text{-only}} \approx 1.58$, above threshold. The population-averaged versions, taking weighted averages of contact rates across asymptomatic/symptomatic for COVID or high-activity/low-activity for HIV, would give $\mathcal{R}_0 \approx 2.44$ for COVID (above threshold because both tracks contribute) but $\mathcal{R}_0 \approx 0.44$ for HIV (below threshold, missing the core-group dynamics).

The HIV example is particularly instructive because it exposes the failure mode of homogeneous-mixing models: a homogeneous-mixing model with partnership rate $\bar{c} = 0.2 \cdot 5 + 0.8 \cdot 0.5 = 1.4$ partnerships per year would predict that HIV cannot sustain itself in this population, an empirically falsified prediction. The activity-stratified framework recovers the correct conclusion.

Operational consequence: the targeting paradox. A reservoir subpopulation that sustains $\mathcal{R}_0 > 1$ requires intervention specifically targeting that subpopulation. Interventions that target the general population, or that target visible cases (symptomatic patients in COVID, reported partnerships in HIV), leave the reservoir untouched. The COVID-19 case study's three intervention

scenarios illustrated this: case isolation alone reduces the symptomatic contribution but cannot affect the asymptomatic contribution, leaving $\mathcal{R}_0 \approx 1.66$ even with aggressive symptomatic-case isolation. Only universal interventions (universal masking, mass asymptomatic testing) can reach the reservoir.

For HIV, the analogous universal intervention is universal HIV testing with linkage to ART: reaching the high-activity subgroup requires testing and treating throughout the population, not only among those who present clinically. The U=U paradigm, operationalized through test-and-treat programs in high-prevalence settings, has this reservoir-targeting logic at its core.

The absence in dengue. The dengue case study does not exhibit reservoir structure in the same sense. The vector population is itself the "reservoir" in a trivial sense (infection persists in the vector pool), but the bipartite structure is fully characterized by the Ross-Macdonald formula; there is no subpopulation analogous to asymptomatic-COVID or high-activity-HIV that sustains $\mathcal{R}_0 > 1$ independently. Spatial and household heterogeneities do create within-population reservoirs in operational dengue settings (hot-spot clusters around specific breeding sites), but these are spatial features not captured in the homogeneous-mixing framework of the case study. Multi-serotype dynamics introduce a different kind of reservoir (past-infectious individuals susceptible to secondary infection by a different serotype), which is also outside the single-serotype framework.

The structural lesson. Reservoir structure emerges whenever transmission heterogeneity creates a subpopulation whose characteristic parameters are qualitatively different from the population average. The framework handles this case through the multi-group NGM extension (Chapter 11). Any population-level intervention strategy for a pathogen with reservoir structure must either target the reservoir specifically or deploy universal interventions that reach the reservoir by saturation. Targeting the reservoir requires identifying it, which itself requires data streams (asymptomatic surveillance for COVID, sexual-behavior surveys and partnership tracing for HIV) that are operationally difficult to obtain. Compartment-specific contact rates (c_S, c_I, c_R in the basic SIR setting and their multi-group, bipartite, and time-since-infection generalizations) are a framework-level feature, not a case-study special case: every transmission structure in this book admits compartment-specific rates, with the uniform case recovered as the analytical baseline and the compartment-specific case expressing behavior change, isolation, and treatment effects (Chapters 4, 11, 12, 13, 15, 19).

20.5 The Infected-Viewpoint Fitting Advantage

The framework's infected-viewpoint estimator $\hat{\alpha} = j/i$, introduced in Chapter 7, Section 7.4.3 and developed theoretically in the sensitivity-analysis paper [Hyman, 2026d], is the book's principal contribution to parameter estimation practice. The three case studies apply it to different data contexts and demonstrate its operational advantages.

COVID-19: early-pandemic uncertainty about s. In the early pandemic, the size of the susceptible population was effectively the entire population (no pre-existing immunity), but the effective-contact population through which transmission propagated was uncertain, depending on social-distancing behavior, lockdown stringency, and testing-reporting biases. The infected-viewpoint estimator $\hat{\alpha} = j/i$ is structurally immune to this uncertainty because s does not appear. The framework's COVID-19 demonstration computes $\mathcal{R}_0 \approx 2.44$ from consensus baseline parameters, independent of any assumed value for s.

Dengue: the susceptible-vector problem. Dengue's fitting challenge is asymmetric: the susceptible-human population is measurable through standard serological surveys, but the susceptible-vector population s_v is operationally the most uncertain quantity in the entire transmission system. Ovitrap surveys produce density estimates with confidence intervals spanning factors of two or more; weekly variation with rainfall and oviposition is substantial. The susceptible-viewpoint estimator $\hat{\lambda}_v = j_v/s_v$ on the vector side is nearly useless operationally. The infected-viewpoint estimator $\hat{\alpha}_h = j_h/i_h$ on the human side does not depend on s_v at all, and the forward-simulation through the bipartite model then determines what s_v must have been to produce the observed human-side dynamics. The framework's dengue analysis uses human-side data exclusively for fitting, with vector-side parameters taken from independent entomological sources.

Convention pointer. The subscripted forces in this chapter follow the vector-borne convention of Chapter 13: they are *per-target* rates, multiplying a susceptible population, and correspond to the force *of* infection λ in the book's primary notation. The per-source force from infection α of Chapter 3, which satisfies $j = \alpha i$, appears here only in the fitting estimators. The Notation and Conventions summary lists which chapters use which.

HIV: the activity-stratification problem. HIV's fitting challenge is different again: the high-activity subpopulation is defined by behavioral characteristics (partnership-formation rate above a threshold), but the exact cutoff and the exact size of the subgroup vary across studies and settings. The susceptible-viewpoint estimator would require knowing the absolute size of the high-activity subgroup, which is uncertain. The infected-viewpoint estimator $\hat{\alpha} = j/i$ within the high-activity subgroup produces a per-individual force-from-infection estimate that is independent of subgroup size; the framework then computes $\mathcal{R}_0^{\text{horiz, H}}$ from the per-individual α and the per-partnership parameters.

The common advantage. Across all three pathogens, the infected-viewpoint estimator sidesteps a different specific quantity whose uncertainty enters the susceptible-viewpoint estimate: the effective susceptible population for COVID-19, the susceptible-vector population for dengue, and the high-activity-subgroup size for HIV. Whether that uncertainty reaches $\mathcal{R}_0$ depends on the fit (Chapter 7). The framework's structural immunity to susceptible-population uncertainty is operational in all three case studies, though the specific quantity being side-stepped varies. This generality is the book's signature contribution to epidemic modeling practice, and the three case studies demonstrate it in contexts chosen precisely because they expose the limitations of the susceptible-viewpoint approach.

The source-normalized form of the principle. The three case studies make clear that the formula $\hat{\alpha} = j/i$ generalizes *only when j and i are measured on a consistent population.* For the basic single-compartment SIR model the consistent population is the only infectious compartment i, and there is no ambiguity. For multi-compartment, bipartite, and stratified extensions, the principle requires sharper articulation:

- *COVID-19 (multi-track)*: The natural fit quantity is $\hat{\alpha}_{\text{sym}} = j_{\text{sym}}/i_{\text{sym}}$. It measures the growth rate, so with the latent period carried into the conversion it recovers $\mathcal{R}_0$. What it cannot reveal is how $\mathcal{R}_0$ divides between the symptomatic and asymptomatic tracks (Chapter 17).

- *Dengue (bipartite vector-borne)*: The natural fit quantity is the human-side $\hat{\alpha}_h = j_h/i_h$, where j_h and i_h are the human-side incidence and prevalence. The vector-side $\hat{\alpha}_v = j_v/i_v$ is operationally inaccessible because vector incidence and prevalence are not measured. The framework's choice to fit human-side data exclusively is a consequence of this asymmetric data availability, not a structural property of the estimator.

- *HIV (bipartite sexually-transmitted, activity-stratified)*: The natural fit quantity is the per-sex per-activity-stratum estimator $\hat{\alpha}_M^H = j_M^H/i_M^H$ for the high-activity male subgroup, and similarly $\hat{\alpha}_F^H = j_F^H/i_F^H$ for high-activity females. Each gives the realized rate of new infections per current case in its stratum (Chapter 19, Section 19.4).

The unifying principle is that j and i must be measured on the same population, and that which population it is determines what the ratio means. Two cases arise, and conflating them is the error this section is guarding against.

When infection passes directly within one population, the source and target coincide, and $\hat{\alpha} = j/i$ is a genuine *per-source* rate: the denominator holds the infectious individuals who generate the numerator's new infections. This is the Chapter 7 formula.

When infection passes through an intermediate stage (a vector, or a partner stratum), the source and target populations differ, and no ratio of same-population quantities can be a per-source rate. The estimator is then a *per-target-population* rate: $\hat{\alpha}_h = j_h/i_h$ measures new human infections per current human case, accumulated through the complete human–vector–human cycle rather than generated by i_h directly (Chapter 18, Section 18.3). The source-side factor has not been eliminated; it has been absorbed into the quantity being estimated. For dengue it enters at the vector stage, so that at vector quasi-equilibrium $\hat{\alpha}_h$ carries the vector parameters and s_v within it; for HIV it enters through the partner stratum, so the stratum-specific $\hat{\alpha}_M^H$ carries the cross-stratum mixing. In both cases the numerator–denominator pairing must be stated explicitly, compartment by compartment, to avoid the source–target conflation flagged in Chapter 11, Section 11.3 and Chapter 12, Section 12.2.

The fitting advantage survives both cases, but it is worth stating precisely what survives. As *computed*, j/i requires no susceptible-population quantity in either the direct or the bipartite case: that is what makes it estimable from routinely collected data, and it is the whole of the operational benefit. What the bipartite case does not deliver is independence of the *estimand*. A per-target-population $\hat{\alpha}_h$ does depend on the source-side susceptibles it has absorbed, as the dengue algebra makes explicit. The distinction is between not needing to *know* a quantity and the quantity not *mattering*, and only the first is claimed here. The asymmetry between the two viewpoints is preserved throughout, since the susceptible-viewpoint estimator requires the susceptible population as an input before any fitting can begin.

Exercises for this section

1. [**C**] Identify the way the framework's structural-immunity advantage produced the largest practical estimation discrepancy across the three case studies. Your answer should reference the chapter's discussion of which side of the inverse problem (denominator or numerator) is the most uncertain in each case.

20.6 The Framework's Scope: What It Captures Cleanly

Across the three case studies, the framework captures cleanly five classes of phenomenon. Figure 20.2 summarizes the sensitivity comparison across the three diseases.

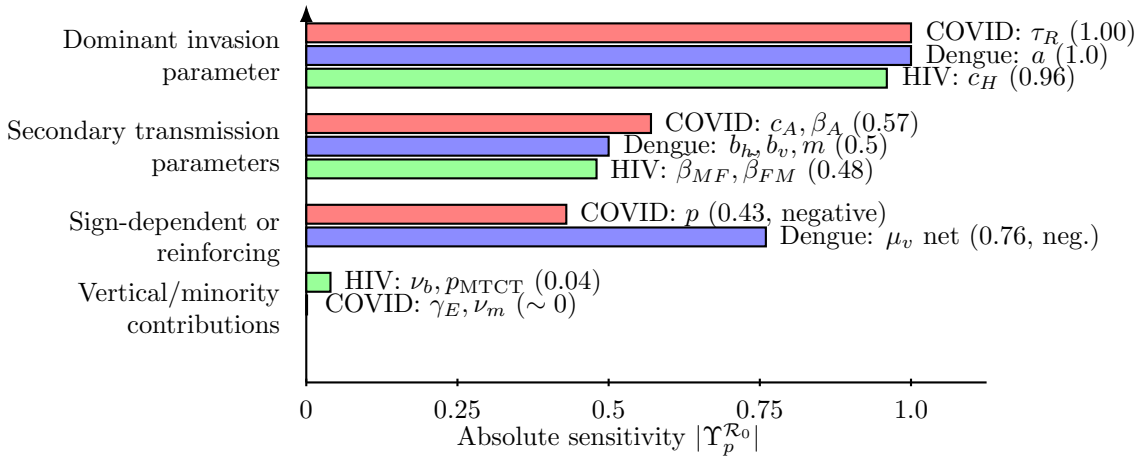


Figure 20.2: Sensitivity-index comparison across the three disease case studies, organized by role in the reproductive-number formula. Row 1: dominant invasion parameter for each disease. COVID-19's infectious period τ_R has near-unit sensitivity because it scales both tracks; HIV's partnership-formation rate c_H, because it appears in both transmission directions. Dengue's biting rate a has exact unit sensitivity as the shared bipartite parameter. Row 2: secondary transmission parameters. All three diseases show the signature of their transmission mode: COVID-19 with c_A and β_A at 0.57 reflecting the asymptomatic-track dominance, dengue and HIV with parameters at 0.5 reflecting bipartite square-root dampening. Row 3: parameters with sign-dependent or reinforcing effects. COVID-19's symptomatic fraction p has sign-dependent sensitivity depending on which track dominates transmission (magnitude 0.43 at the baseline, with the sign reflecting asymptomatic-track-dominant transmission); dengue's vector mortality μ_v has reinforcing direct and indirect effects through the EIP-survival probability, giving the larger net magnitude 0.760. Row 4: HIV's small vertical contribution and COVID-19's negligible latent-period and demographic terms illustrate that the framework preserves the hierarchy of effects across different disease biology.

Threshold structure. The framework captures the threshold structure of $\mathcal{R}_0$ relative to 1 as the invasion criterion, with the NGM computation providing a systematic way to compute $\mathcal{R}_0$ for any compartmental extension. All three case studies use this machinery, and the conclusions (baseline above threshold, combined interventions below threshold) depend essentially on it. Threshold analysis is the book's most-used analytical tool.

Sensitivity-based intervention ranking. The tornado plots in each case study rank parameters by absolute relative sensitivity, which translates directly into a ranking of interventions by expected effect per percent of parameter change. The framework makes these rankings quantitative and comparable across diseases, as Figure 20.2 shows. The practical advantage is that a modeler can compare intervention options on a common scale without re-deriving the sensitivity formulas for each pathogen. The comprehensive global-sensitivity methodology that complements the local-sensitivity rankings used in the case studies is surveyed in Saltelli et al. [2008], while Raue et al.

[2009] treat the companion problem of structural identifiability, i.e. whether the ranked parameters can in principle be recovered from the available data.

Structural transmission-mode differences. The framework's handling of direct (COVID), vector-borne (dengue), and sexual (HIV) transmission uses the same analytical machinery but with different compartmental structures. The structural differences between these modes, especially the bipartite square-root dampening discussed in Section 20.3, emerge from the framework's systematic application rather than requiring ad hoc pathogen-specific analysis.

Heterogeneity and reservoir structures. As discussed in Section 20.4, the framework handles activity-level and compartment-level heterogeneity through the multi-group NGM, and correctly identifies reservoir subpopulations that sustain transmission when the population average would predict fadeout.

Fitting structural immunity. The infected-viewpoint estimator $\hat{\alpha} = j/i$ provides s^*-independent parameter estimation across all three case studies, side-stepping different specific forms of susceptible-population uncertainty that enter the susceptible-viewpoint estimate; whether they reach $\mathcal{R}_0$ depends on the fit. This is discussed in Section 20.5.

Exercises for this section

1. **[I]** *Apply the framework to a pathogen of your choice.* Select a pathogen *not* treated in the three case-study chapters: candidates include measles, tuberculosis, hepatitis B, pertussis, gonorrhea, Chagas disease, leishmaniasis, Lyme disease, schistosomiasis, influenza, RSV, or seasonal coronaviruses. Working in the same synthesis style as the COVID, dengue, and HIV case studies, produce a complete framework application. The deliverable is a structured 4–6 page memo answering the following six questions in order:

   (a) *Transmission mode and structural choice.* What is the dominant transmission mode (direct respiratory, vector-borne, sexual, vertical, environmental, or some combination), and what compartmental structure does the framework recommend? Justify the choice by reference to the appropriate generalization chapter (Chapter 10–15). State which assumptions of the baseline SIR model (Chapter 4, Section 4.1) are violated and how the chosen extension addresses each violation.

   (b) *Model formulation.* Write out the full system of ODEs (or other equations as appropriate) for your chosen model, including initial and boundary conditions. Define every parameter symbol used; provide best-estimate values from the literature with brief citations or reasonable bracketed guesses with justifications. State which parameters you treat as independent and which as dependent (per Chapter 4, Section 4.3).

   (c) *Derive $\mathcal{R}_0$ via the next-generation matrix.* Identify the infected compartments, write out $\mathcal{F}$ and $\mathcal{V}$, compute $\boldsymbol{K} = \mathcal{F}\mathcal{V}^{-1}$, and report the spectral radius. Compare to any closed-form $\mathcal{R}_0$ expression available in the literature, and reconcile any apparent differences (often these arise from parameterization conventions). Cite Diekmann et al. [1990] and van den Driessche and Watmough [2002] as appropriate.

   (d) *Sensitivity analysis.* Compute the local sensitivity index $\Upsilon_p^{\mathcal{R}_0}$ for each independent parameter at your baseline. Produce a tornado plot in the same style as Figure 20.2 and

identify the dominant invasion parameter. State which row of Figure 20.2 (dominant-invasion, secondary-transmission, sign-dependent, vertical/minority) your dominant parameter falls into, and discuss whether the sensitivity profile resembles one of the three case-study patterns (COVID, dengue, HIV) or represents a fourth pattern.

(e) *Intervention analysis.* Identify two interventions: one targeting the dominant invasion parameter from (d) and one targeting a secondary parameter. For each, compute the parameter reduction needed to drive $\mathcal{R}_0$ below 1, using the linear approximation $\Delta\mathcal{R}_0/\mathcal{R}_0 \approx \Upsilon_p^{\mathcal{R}_0} \cdot \Delta p/p$. Discuss whether the intervention is operationally feasible and what auxiliary data would be needed to plan it.

(f) *Limits of the framework.* Identify at least one phenomenon central to your chosen pathogen that the framework does *not* capture cleanly, requiring extension beyond what is in the book. Examples might include explicit spatial transmission, multi-strain immune cross-reactivity, age-dependent severity, host genetic variation, or environmental persistence. Reference Section 20.7 on framework limits and indicate what additional methodology (specific paper or research direction, briefly cited) would be needed.

The exercise integrates content from the entire book and is designed to be completed over several days with literature consultation. AI tools may be used for literature retrieval and consistency checking, following the verification checklists of Appendix D. The final synthesis, especially the model-formulation and sensitivity-comparison reasoning, should be in the student's own words and arguments. Students must independently verify six things, whatever tool produced them: equations, numerical values, citations, model assumptions, sensitivity calculations, and interpretations. Submission to a course instructor would naturally accompany a 5–10 minute oral presentation. Successful execution of this exercise is the operational test of having internalized the framework: a researcher who can complete it for a novel pathogen has demonstrated the key skill the book aims to develop.

20.7 The Framework's Limits: What Requires Extensions

The framework's scope is broad but not universal. Five categories of phenomenon require extensions or replacements.

Variable infectiousness within an infection. The compartmental framework assumes that all individuals in the i compartment have the same per-capita transmission rate. For pathogens with substantial variation in infectiousness over the course of infection (HIV's acute-versus-chronic-versus-AIDS phases, influenza's sharp early peak, SARS-CoV-2's pre-symptomatic shedding), this is an approximation. The time-since-infection framework of Chapter 15 captures the variation explicitly through the infectiousness profile $A(\tau)$, and the renewal-equation formulation recovers the basic reproductive number as the integral of $A(\tau)F(\tau)$ over τ. For the case studies in this book, the compartmental approximation was sufficient; for research-grade modeling of specific pathogens, the time-since-infection extension is often necessary. *When to use it:* use a renewal or time-since-infection model when infectiousness varies substantially with time since infection.

Detailed age structure. The case studies used simple or absent age structure. For COVID-19, age structure is essential for predicting hospitalization and mortality, and for evaluating age-targeted interventions (vaccine prioritization, school closure). For HIV, age structure is essential for cohort

analysis and for evaluating interventions across age-stratified sexual networks. For dengue, age structure matters for serotype-cumulative-exposure dynamics. The age-structured PDE framework of Chapter 15, or the discrete age-group extension of Chapter 11, provides the natural extension. For operational research, age-stratification is typically the first extension applied beyond the basic compartmental framework. *When to use it:* use an age-structured model when age-specific susceptibility, severity, or contact patterns drive the question.

Multi-strain dynamics. Dengue's four-serotype structure and SARS-CoV-2's variant evolution both require multi-strain extensions with cross-immunity matrices. The framework does not present these extensions explicitly; the research literature develops them under the names of "multi-strain SIR" and "serotype-structured models" [Nishiura, 2006, Ferguson et al., 2016]. The mathematical apparatus (NGM computation on expanded compartment spaces, cross-immunity parameters) is a direct extension of the framework, but the notational complexity grows substantially. *When to use it:* use a multi-strain model when cross-immunity or competition between variants changes transmission.

Stochastic effects in small populations. The framework is deterministic throughout. For small populations, rare-event dynamics, and extinction probabilities, stochastic models (continuous-time Markov chains, branching processes, individual-based models) are required. The framework's deterministic predictions are the large-population limit of these stochastic models; the framework is silent on stochastic effects that are essential for understanding invasion dynamics in the early-outbreak regime and elimination dynamics in the end-game regime of pathogens approaching eradication. *When to use it:* use a stochastic model when extinction probabilities or small-number effects matter.

Network structure beyond contact-rate averaging. The framework uses contact rates (per-capita average rate of contact), not contact networks (specific partnership graphs). For pathogens where network structure matters (HIV in sparse sexual networks; hospital-acquired infections along clinical-contact networks), the framework's homogeneous-mixing assumption is an approximation that fails in specific ways. Network-structured models, including percolation models and adaptive-network models, are the research-literature response. The framework's contact-rate approach is the appropriate level of detail for most epidemiological modeling but is not the right level for questions specifically about network topology. *When to use it:* use a network model when partnership structure or local connectivity is central to the question.

The pattern. The five extension categories share a pattern: each handles a specific form of heterogeneity or stochasticity that the homogeneous-mixing compartmental framework averages over. The framework's scope is precisely the set of questions that can be addressed by averaging over these heterogeneities and stochasticities; its limits are the set of questions where the specific form of the heterogeneity matters. The case studies in this book fall within the framework's scope; research-grade analyses of specific operational questions typically require one or more of the extensions.

Exercises for this section

1. **[I]** Consider the framework's extension to the case of waning immunity (SIRS structure), introduced in Chapter 10, Section 10.2. Which of the three case studies in this part would

most benefit from this extension? Compute the qualitative effect on the intervention scenarios if immunity wanes on a 6-month time scale.

20.8 Closing Reflection

The three case studies, taken together, demonstrate the operational value of the common pitfalls introduced in the Preface and Chapter 16. Each case study is, in part, an extended worked example of the common pitfalls at work: the COVID-19 case study (Chapter 17) made visible the consequences of conflating $\mathcal{R}_0$ with $\mathcal{R}_e$ during a fast-moving outbreak Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$" and the importance of fitting practice that respects the reporting structure of incidence data Pitfall: "Correct Data-Fitting Practice"; the dengue case study (Chapter 18) developed the heterogeneous-mixing balance condition Pitfall: "Heterogeneous-Mixing Balance Conditions" in the bipartite vector-host setting and showed how the resulting square-root dampening modifies $\mathcal{R}_0$; the HIV case study (Chapter 19) made operational the identifiability constraint Pitfall: "Identifiability" under the multi-decade endemic dynamics of a chronic disease and the corresponding sanity check at the endemic equilibrium Pitfall: "Invasion versus Equilibrium". Across all three, the common pitfalls served as the navigational vocabulary: a reader who internalized them from Chapter 16 could anticipate, before each case study began, which common pitfalls would carry the operational weight. This is the practical test of whether the catalog framing pays for itself, and the case studies suggest it does.

Behind that operational success is a body of framework machinery that the book has developed to serve specific common pitfalls. The next-generation matrix construction of $\mathcal{R}_0$ in Chapter 5 is the analytical apparatus for Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$"; the invasion-burden partition (Theorem 5.6 and Proposition 10.2) is the analytical apparatus for Pitfall: "Invasion versus Equilibrium"; the infected-viewpoint formulation developed across Chapters 3–7 is the analytical apparatus for Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Linearization Fidelity"; the local-and-global sensitivity machinery of Chapter 9 is the analytical apparatus for Pitfall: "Sensitivity Analysis Discipline"; the renewal-equation framework of Chapter 15 relaxes the exponential-generation-time assumption underlying the constant-rate compartmental treatment. The framework machinery is the means; the common pitfalls are the ends. Both matter, and they are not interchangeable: a reader who learns only the machinery without the common pitfalls risks producing technically correct analyses that overlook the questions the framework cannot itself answer.

Two contributions of analytical machinery worth underscoring. Two pieces of analytical machinery developed in the book are worth underscoring in the closing for the role they play in serving the common pitfalls discussed here.

First, the invasion-burden partition, developed formally as Theorem 5.6 and generalized as Proposition 10.2, is the apparatus that gives Pitfall: "Invasion versus Equilibrium" its operational teeth. The partition separates the parameters that control invasion ($\mathcal{R}_0$ above or below unity) from the parameters that control endemic burden (steady-state prevalence). It applies universally across the case studies, with pathogen-specific manifestation, and yields Corollary 5.7's policy-asymmetry result: interventions that target invasion parameters have multiplicative effects on both $\mathcal{R}_0$ and burden, while interventions that target burden-only parameters affect burden but not invasion. The asymmetry has operational implications for intervention prioritization: when the goal is elimination, invasion-facing interventions dominate; when the goal is managing chronic disease, burden-facing in-

terventions compete. The partition is, in this sense, the analytical reason a modeler who internalizes Pitfall: "Invasion versus Equilibrium" can produce defensible intervention rankings.

Second, the infected-viewpoint fitting estimator $\hat{\alpha} = j/i$ is the analytical apparatus that gives Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Linearization Fidelity" their operational teeth. The susceptible-viewpoint alternative $\hat{\lambda} = j/s$ has been the default in most of the modeling literature. The infected-viewpoint estimator is not strictly new (variants have appeared in the literature going back to the 1990s), but the book's presentation establishes it as the appropriate default for modeling work under susceptible-population uncertainty, and its sensitivity-index foundation is developed independently in [Hyman, 2026d]. The three case studies demonstrate this across three different forms of susceptible-population uncertainty, and the conclusion is the same in all three: when the chosen formulation matters for the common pitfall being addressed, $\hat{\alpha} = j/i$ is the right default for fitting from incidence data, and $\hat{\lambda} = j/s$ is reserved for the special cases where s is better-characterized than i.

What the book is not. The book is not a comprehensive survey of infectious-disease epidemiology; it is a conceptual framework with worked examples. The case studies are not operational research papers; they are demonstrations that the framework handles the pathogens presented. Readers seeking pathogen-specific analyses for policy or forecasting purposes should consult the specialist literature for each pathogen; the framework provides the analytical scaffold on which such analyses build.

What comes next for the reader. A reader who has worked through this book to the closing chapter has internalized the common pitfalls and is equipped, more importantly, to recognize new variants of them in unfamiliar settings. Specifically, such a reader can (i) recognize when a deterministic compartmental model is the right tool for the question at hand and when it is not (Pitfall: "The Stochastic-to-Deterministic Regime Transition" and Pitfall: "Short-Term Simulation Assumptions"); (ii) formulate a compartmental model whose terms are dimensionally consistent and whose force-of-infection structure matches the available data (Pitfall: "Dimensional Consistency", Pitfall: "Force of Infection versus Force from Infection", Pitfall: "Linearization Fidelity"); (iii) initialize the model from physically realizable starting conditions and check the heterogeneous-mixing balance constraint when groups are involved (Pitfall: "Natural Initial Conditions" and Pitfall: "Heterogeneous-Mixing Balance Conditions"); (iv) fit parameters to available data with attention to identifiability and to the structure of the data-generating reporting process (Pitfall: "Correct Data-Fitting Practice", Pitfall: "Identifiability", and the generation-time pitfalls); (v) conduct sensitivity analysis that ranks interventions by their potential impact without overstating precision Pitfall: "Sensitivity Analysis Discipline"; and (vi) report reproductive-number quantities in a way that distinguishes invasion from current state and from endemic dynamics (Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$" and Pitfall: "Invasion versus Equilibrium"). These six capabilities together constitute the working modeler's toolkit, organized by the common pitfalls that the book has spent its previous chapters developing.

The mathematical foundations of the framework (differential equations, next-generation matrices, compartmental dynamics) are drawn from the broader literature and are not original to this book. What is original is the framing: the common pitfalls as the organizing spine, the per-chapter common-pitfall tags as the navigational aid, the case studies as worked examples of the common pitfalls in operational settings, and the analytical machinery as the apparatus that makes each common pitfall tractable. The framing is the book's contribution.

20.9 Open Problems

The framework opens several research directions that this book does not pursue. We name five that seem most worth a reader's attention, as dissertation projects or as follow-on papers. Each is given in three parts: the open question, why it matters, and the part of the framework it extends. The methods each would need are left to the references.

1. A rigorous comparative structural-immunity theorem for $\hat{\alpha} = j/i$ versus $\hat{\lambda} = j/s$. *Question.* For a specified joint distribution of the observables (j, i, s) and a specified loss, does the infected-viewpoint estimator strictly dominate the susceptible-viewpoint estimator, and by how much? *Why it matters.* Hyman [2026d] and Chapter 5 establish the sensitivity asymmetry $|\Upsilon_s^{\alpha}| = 1$ versus $|\Upsilon_s^{\lambda s}| = s/i$ only as a point estimate. Sharp constants would let a modeler predict, before fitting, how much the choice of estimator matters for a given outbreak. *Extends* the infected-viewpoint fitting of Section 20.5, Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Linearization Fidelity".

2. The sensitivity asymmetry under time-varying parameters. *Question.* When $\alpha(t)$ varies in time and must itself be fitted as a function, is the functional estimator $\widehat{\alpha(\cdot)}$ still structurally immune to uncertainty in $s(t)$? *Why it matters.* Theorem 5.6 assumes constant transmission parameters, but real transmission rates change with behavior, season, and control. Conrad et al. [2016] illustrates the point for the 2014–2015 West African Ebola epidemic, where behavior change was the dominant driver of epidemic arrest. The answer decides whether the fitting advantage carries over to operational forecasting. *Extends* Pitfall: "Linearization Fidelity" and Pitfall: "Sensitivity Analysis Discipline".

3. Optimal intervention allocation under the invasion-burden partition. *Question.* How should a fixed intervention budget be divided between invasion-facing and burden-facing interventions? *Why it matters.* Corollary 5.7 shows that invasion-facing interventions affect both $\mathcal{R}_0$ and endemic burden, while burden-facing interventions affect burden alone. When the objective is elimination, invasion-facing allocations dominate; when it is managing endemic disease, the two compete. An optimization in the control framework of Lenhart and Workman [2007] would turn the partition into rules of thumb for prioritizing interventions. *Extends* the invasion-burden partition of Section 20.2 and Pitfall: "Invasion versus Equilibrium".

4. Infected-viewpoint analogues in stochastic and network settings. *Question.* Does the infected-viewpoint fitting advantage persist when transmission is a continuous-time Markov chain, and does the invasion-burden partition have a clean statement on a network? *Why it matters.* The book is deterministic (Section 20.6). Branching-process theory addresses invasion on networks, but endemic prevalence on a network is structurally different from the mean-field case. Answers would carry the framework to the model classes surveyed by Allen [2010] and Britton and Pardoux [2019]. *Extends* Pitfall: "The Stochastic-to-Deterministic Regime Transition".

5. Practical rule-of-thumb calculators for fitting-error budgets. *Question.* For a data source with a specified measurement error (Poisson counts, a binomial prevalence survey, serology with imperfect sensitivity and specificity), what relative error does each estimator, $\hat{\alpha}$ or $\hat{\lambda}$, place on $\mathcal{R}_0$? *Why it matters.* A table of these errors for realistic cases would let a modeler choose between

data sources at the design stage. The calculations are standard; the work is in covering enough cases to be useful. *Extends* Pitfall: "Correct Data-Fitting Practice", Pitfall: "Identifiability", and Pitfall: "Sensitivity Analysis Discipline".

The five differ in scale. The first two are theoretical results, the third suits a dissertation, the fourth is a research program, and the fifth is sustained methodological work. All five build on the framework rather than replace it.

The three case studies of Chapters 17–19 make operational the common pitfalls introduced in the Preface and developed in Chapter 16. The pathogens differ in transmission mode (direct respiratory, vector-borne bipartite, sexually-transmitted bipartite with activity stratification), in natural history (acute, acute, chronic), and in the available data for fitting; the common pitfalls apply across all three with pathogen-specific manifestation. The cross-cutting lessons are organized by the common pitfall each one serves: the invasion-burden partition applies universally with pathogen-specific manifestation Pitfall: "Invasion versus Equilibrium"; bipartite transmission produces square-root dampening that requires combined interventions to be evaluated against the heterogeneous-mixing balance constraint Pitfall: "Heterogeneous-Mixing Balance Conditions"; reservoir subpopulations sustain transmission even when population averages do not, and identifiability constraints determine when the reservoir structure can in fact be inferred from the available data (Pitfall: "Heterogeneous-Mixing Balance Conditions" and Pitfall: "Identifiability"); the infected-viewpoint formulation sidesteps different specific forms of susceptible-population uncertainty across the three pathogens, with the choice of formulation determined by linearization-fidelity arguments (Pitfall: "Force of Infection versus Force from Infection" and Pitfall: "Linearization Fidelity"); the framework captures threshold dynamics ($\mathcal{R}_0$ versus $\mathcal{R}_e$, Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$"), sensitivity-based intervention ranking Pitfall: "Sensitivity Analysis Discipline", structural transmission-mode differences, heterogeneity, and the structural-immunity-based fitting advantage; and the framework requires extensions for variable infectiousness (a generalization of the time-in-compartment pitfalls), detailed age structure, multi-strain dynamics, stochastic effects Pitfall: "The Stochastic-to-Deterministic Regime Transition", and network topology. These extensions build on the framework rather than replacing it. The framework's scope is the set of questions that can be addressed by averaging over intra-compartment heterogeneity and stochasticity; its limits are the questions where specific heterogeneity or stochasticity matters, which is itself the operational form of Pitfall: "The Stochastic-to-Deterministic Regime Transition", returning the book to where it began.

The single sentence the book aims to leave the reader with is the one that opened it: even simple epidemic models can mislead when basic pitfalls are overlooked. The common pitfalls developed across the preceding chapters, illustrated in the case studies just summarized, and revisited here in operational form, are the practical equipment for recognizing the patterns of mistake before they propagate from a model into a published claim or a public-health recommendation. A reader who carries that vocabulary forward into their own work has taken what the book set out to teach.

Exercises

Four section-level exercises appear above, following Section 20.3, Section 20.5, Section 20.6, and Section 20.7. The exercises below integrate across the chapter and have been ordered to interleave derivation and interpretation exercises (this synthesis chapter is interpretation-heavy, so perfect alternation is not possible). Exercises are tagged by type: ***C*** *= Concept check,* ***D*** *= Derivation,* ***I*** *= Interpretation / modeling judgment,* ***L*** *= Computational lab.*

1. [**I**] For each of the three case studies, identify the three parameters with highest absolute sensitivity. Compute the maximum $\mathcal{R}_0$ reduction achievable by a 25 percent reduction in each parameter. Rank the pathogens by the intervention efficiency (maximum $\mathcal{R}_0$ reduction per unit of parameter change).

2. [**D**] Compare the square-root dampening in dengue and HIV. For each, compute the sensitivity of $\mathcal{R}_0$ to the shared unit-sensitivity parameter (a for dengue, c_H for HIV) and to a single half-sensitivity parameter of your choice. Quantify the intervention-effort trade-off.

3. [**I**] The invasion-burden partition predicts that intervention on invasion-facing parameters reduces both $\mathcal{R}_0$ and endemic burden, while intervention on burden-only parameters affects burden alone. For the HIV case study, show explicitly that ART acts on both categories (reducing $\tilde{\beta}$, an invasion parameter, and simultaneously extending residence time through reduced μ_d), and compute the net effect on $\mathcal{R}_0$ at 50 percent ART coverage.

4. [**D**] For the COVID-19 case study, the symptomatic fraction p has sign-dependent sensitivity. Derive the condition on $c_I, \beta_I, c_A, \beta_A$ under which $\Upsilon_p^{\mathcal{R}_0}$ changes sign, and identify the corresponding pathogen or intervention regime.

5. [**I**] The framework's fitting structural-immunity advantage depends on observing i accurately. For a pathogen where prevalence i is harder to measure than susceptible population s (for example, an endemic disease with low clinical attack rate and good serological surveillance), which viewpoint estimator is preferred? Identify the data-sources context in which the susceptible-viewpoint estimator becomes preferable.

6. [**I**] The reservoir structures identified in Section 20.4 arose for COVID (asymptomatic) and HIV (high-activity subgroup) but not for dengue. Propose a modification of the dengue model in which a reservoir structure emerges (hint: consider spatial heterogeneity or multi-serotype immunity). What parameter or subpopulation would act as the reservoir, and what intervention would target it?

7. [**I**] *(Research-level exercise.)* Propose a fourth case study from a disease not covered in Chapters 17–19 (candidates include Ebola, tuberculosis, pandemic influenza, measles, or gonorrhea). Outline the case study using the consistent template: epidemiological context, model formulation, parameter estimation, sensitivity analysis, interventions, lessons. Identify which chapters of the book the case study draws on and which extensions of the framework the disease biology requires.

Appendix A

Mathematical Preliminaries

This appendix provides a concise reference for the mathematical techniques used throughout the book. The treatment is intended as a refresher for readers familiar with the material at the level of a standard upper-undergraduate sequence in calculus, ordinary differential equations, and linear algebra. Readers encountering a topic for the first time should consult a dedicated textbook; the treatment here is too brief for initial learning.

Section A.1 reviews ordinary differential equations and the techniques used to analyze them. Section A.2 covers the linear-algebra tools needed for the next-generation matrix method. Section A.3 addresses dimensional analysis and the Buckingham Pi theorem. Section A.4 treats the integral and renewal equations of Chapter 15. Section A.5 reviews the stochastic-process concepts that underlie the derivation of the force from infection (Chapter 3).

A.1 Ordinary Differential Equations

The compartmental epidemic models throughout Chapters 4 through 20 are systems of ordinary differential equations (ODEs). This section reviews the essential ODE techniques used in their analysis.

A.1.1 First-Order Linear ODEs

A first-order linear ODE has the form

$$\dot{y}(t) + p(t)y(t) \;=\; q(t),$$

with general solution

$$y(t) \;=\; e^{-\int_0^t p(t')dt'}\left[y(0) + \int_0^t e^{\int_0^u p(t')dt'} q(u)\,du\right].$$

For constant coefficients $p(t) = p$ and $q(t) = q$, the solution simplifies to

$$y(t) \;=\; \frac{q}{p} + \left(y(0) - \frac{q}{p}\right)e^{-pt}.$$

The system approaches the equilibrium $y^* = q/p$ exponentially with time constant $1/p$.

This is the structure of, for example, the susceptible compartment in the SIR model when the infection flow αi is treated as approximately constant over the interval of interest. In the infected viewpoint the loss from s is the flow αi, not a per-susceptible rate acting on s, so

$$\dot{s} \;=\; \nu_m - \alpha i - \nu_m s$$

has the form above with $p = \nu_m$ and $q = \nu_m - \alpha i$. Writing the loss instead as $(\alpha + \nu_m)s$ would silently substitute the susceptible-viewpoint form λs for the infected-viewpoint flow αi; the two coincide only when $\alpha i = \lambda s$, which is the identity discussed in Chapter 3, not a licence to move α onto s.

A.1.2 Systems of ODEs and the Jacobian

A system of n ODEs is written

$$\dot{\mathbf{x}}(t) \;=\; \mathbf{f}(\mathbf{x}(t)),$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector and $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^n$ is the right-hand-side function (assumed smooth). An equilibrium $\mathbf{x}^*$ satisfies $\mathbf{f}(\mathbf{x}^*) = 0$.

The Jacobian matrix at the equilibrium is

$$\mathbf{J}(\mathbf{x}^*) \;=\; \left(\frac{\partial f_\ell}{\partial x_j} \right)_{\ell,j=1,\ldots,n} \Bigg|_{\mathbf{x}=\mathbf{x}^*},$$

an $n \times n$ matrix. The linearized dynamics near the equilibrium are

$$\dot{\boldsymbol{\xi}} \;=\; \mathbf{J}(\mathbf{x}^*)\boldsymbol{\xi}, \qquad \boldsymbol{\xi}(t) = \mathbf{x}(t) - \mathbf{x}^*,$$

with general solution $\boldsymbol{\xi}(t) = e^{\mathbf{J}t}\boldsymbol{\xi}(0)$.

A.1.3 Linear Stability via Eigenvalues

The eigenvalues of $\mathbf{J}(\mathbf{x}^*)$ determine the local stability of the equilibrium:

- If all eigenvalues have negative real part: the equilibrium is asymptotically stable (small perturbations decay).
- If any eigenvalue has positive real part: the equilibrium is unstable (small perturbations grow).
- If the largest real part is exactly zero: the linear analysis is inconclusive; nonlinear terms determine stability.

This is the framework used in Chapter 5, Section 5.4 to determine when the disease-free equilibrium is stable (corresponding to $\mathcal{R}_0 < 1$) and when the endemic equilibrium is stable (corresponding to $\mathcal{R}_0 > 1$).

A.1.4 Routh-Hurwitz Criterion

For higher-dimensional systems, computing eigenvalues directly may be difficult, but stability conditions can be derived from the coefficients of the characteristic polynomial. The characteristic polynomial of $\mathbf{J}$ is

$$p(\lambda) \;=\; \det(\mathbf{J} - \lambda\mathbf{I}) \;=\; \lambda^n + a_1\lambda^{n-1} + \ldots + a_n.$$

The Routh-Hurwitz criterion states that all eigenvalues of $\mathbf{J}$ have negative real part if and only if all coefficients a_ℓ are positive AND a certain sequence of determinants formed from the a_ℓ are all positive. For the $n = 2$ case, the criterion simplifies to:

$$\operatorname{tr}(\mathbf{J}) < 0 \text{ AND } \det(\mathbf{J}) > 0.$$

For higher n, the criterion is more elaborate; we cite Murray [2002] for the general statement.

A.1.5 Equilibria, Periodic Orbits, and Bifurcations

The behavior of nonlinear ODE systems can be more complex than approach to a single equilibrium:

- *Multiple equilibria*: a system may have two or more stable equilibria, with the basin of attraction depending on initial conditions. The SIR model has two equilibria (DFE and endemic) for $\mathcal{R}_0 > 1$, with the disease-free unstable and endemic stable.
- *Periodic orbits (limit cycles)*: stable closed trajectories. The basic SIR model does not exhibit limit cycles, but extensions such as the SIRS with strong waning immunity can. Seasonal forcing introduces periodic orbits in many epidemic models.
- *Bifurcations*: qualitative changes in dynamics as a parameter varies. The threshold $\mathcal{R}_0 = 1$ is a transcritical bifurcation: as $\mathcal{R}_0$ crosses 1, the disease-free equilibrium loses stability and the endemic equilibrium emerges from it.

A.2 Linear Algebra for the Next-Generation Matrix

The next-generation matrix method (Chapter 5, Section 5.3, applied throughout Chapters 11-13) uses several linear-algebra concepts. This section reviews them.

A.2.1 Matrix Notation

A matrix $\mathbf{A}$ is an $m \times n$ array of numbers $A_{\ell j}$. We use bold uppercase letters for matrices, lowercase bold for vectors. Matrix multiplication is defined for compatible dimensions: if $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{C} = \mathbf{AB}$ is $m \times p$ with $C_{\ell j} = \sum_k A_{\ell k} B_{kj}$.

The transpose $\mathbf{A}^T$ has entries $(A^T)_{\ell j} = A_{j\ell}$. The identity matrix $\mathbf{I}$ has $i_{\ell\ell} = 1$ and $i_{\ell j} = 0$ for $\ell \neq j$.

A.2.2 Eigenvalues and Eigenvectors

For a square matrix $\mathbf{A}$, an eigenvalue λ and corresponding eigenvector $\mathbf{v}$ satisfy

$$\mathbf{A}\mathbf{v} \;=\; \lambda \mathbf{v}, \qquad \mathbf{v} \neq 0.$$

The eigenvalues are roots of the characteristic polynomial $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$, an n-th-degree polynomial that has exactly n roots (counting multiplicities) in the complex numbers.

For the 2×2 case, the eigenvalues are

$$\lambda \;=\; \frac{\operatorname{tr}(\mathbf{A}) \pm \sqrt{\operatorname{tr}(\mathbf{A})^2 - 4 \det(\mathbf{A})}}{2},$$

with $\operatorname{tr}(\mathbf{A}) = A_{11} + A_{22}$ and $\det(\mathbf{A}) = A_{11}A_{22} - A_{12}A_{21}$.

A.2.3 Spectral Radius

The spectral radius of a square matrix $\mathbf{A}$ is

$$\rho(\mathbf{A}) \;=\; \max_{\ell} |\lambda_\ell|,$$

the largest absolute value of the eigenvalues. For non-negative matrices (all entries ≥ 0) of relevance to epidemic modeling, the Perron-Frobenius theorem guarantees that the spectral radius is itself a non-negative real eigenvalue with a corresponding non-negative eigenvector. This justifies the interpretation of $\mathcal{R}_0 = \rho(\boldsymbol{K})$ in the next-generation matrix method.

A.2.4 Matrix Inverse

If $\mathbf{A}$ is square and $\det(\mathbf{A}) \neq 0$, the inverse $\mathbf{A}^{-1}$ exists and satisfies $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. For the 2×2 case:

$$\mathbf{A}^{-1} \;=\; \frac{1}{\det(\mathbf{A})} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}.$$

A.2.5 The Next-Generation Matrix Construction

The NGM method constructs $\mathcal{R}_0$ as follows. Let the infected compartments be $\mathbf{x} = (x_1, \ldots, x_k)$. At the disease-free equilibrium, the dynamics linearize as

$$\dot{\mathbf{x}} \;=\; (\boldsymbol{F} - \boldsymbol{V})\mathbf{x},$$

where:

- $\boldsymbol{F}$: the $k \times k$ matrix of new-infection rates. $F_{\ell j}$ is the rate at which an individual in compartment j produces new infections in compartment ℓ.
- $\boldsymbol{V}$: the $k \times k$ matrix of all other transitions. $V_{\ell j}$ is the rate at which an individual in compartment j leaves through means other than new infection (recovery, mortality, etc.), with diagonal positive and off-diagonal negative for transitions *into* compartment ℓ from compartment j.

The next-generation matrix is

$$\boldsymbol{K} \;=\; \boldsymbol{F}\boldsymbol{V}^{-1}.$$

The basic reproductive number is the spectral radius:

$$\mathcal{R}_0 \;=\; \rho(\boldsymbol{K}).$$

This construction is the basis for the $\mathcal{R}_0$ derivations of all the heterogeneous-population chapters.

A.3 Dimensional Analysis and the Buckingham Pi Theorem

The choice of dimensionless parameters for fitting and sensitivity analysis (Chapter 7, Section 7.3, Chapter 9, Section 9.3) rests on dimensional-analysis principles.

A.3.1 Dimensions and Units

A dimension is a kind of physical quantity: time, population, etc. Units are specific scales for measuring dimensions: seconds, days, years for time; individuals, thousands of individuals for population.

Each parameter in the SIR model has dimensions:

- Population size S, I, R, N^c: dimension [population].
- Contact rate c_S, c_I, c_R: dimension [time^{-1}] (contacts per individual per unit time).
- Per-contact transmission probability β: dimensionless (a probability).
- Recovery rate γ_R, migration rate ν_m: dimension [time^{-1}].
- Recovery and migration times τ_R, τ_m: dimension [time].
- Force from infection $\alpha = c_I \beta P_S$: dimension [time^{-1}] (since c_I has [time^{-1}] and βP_S are dimensionless).
- Reproductive number $\mathcal{R}_0 = c_I \beta / (\gamma_R + \nu_m)$: dimensionless.

A.3.2 The Buckingham Pi Theorem

The Buckingham Pi theorem states: if a physical relationship involves n variables and k independent dimensions, then the relationship can be expressed in terms of $n - k$ dimensionless parameters.

For the SIR model, the variables are $c_S, c_I, c_R, \beta, \gamma_R, \nu_m, S(0), N^c$ (8 parameters) plus the state variables s, i, r (3 more). The independent dimensions are time and population (2). The Pi theorem predicts $n - k$ independent dimensionless parameters; for the parameters alone (8), this gives 6 dimensionless combinations.

The most natural dimensionless combinations for the SIR model are:

- $\mathcal{R}_0 = c_I \beta / (\gamma_R + \nu_m)$ (most important, threshold parameter).
- $F_0 = S(0)/N^c$ (initial susceptible fraction).
- $D = \nu_m / \gamma_R$ (timescale ratio).
- $c_S/c_I, c_R/c_I$ (contact-rate ratios; both equal 5 at the book's baseline, and near 1 only when symptomatic behavior is unchanged).

The sixth group can be taken as β itself, which is already dimensionless. It is listed for completeness rather than for fitting: β enters the SIR dynamics only through the product $c_I \beta$ in the force from infection, so it is not separately identifiable from epidemic data alone (Chapter 7, Section 7.3).

A.3.3 Why Use Dimensionless Parameters

Dimensionless parameters are the fundamental quantities of the model. Fitting them is more robust than fitting the dimensional parameters because:

- They are unaffected by the choice of time and population units.
- They have direct epidemiological interpretation ($\mathcal{R}_0$ is the invasion threshold; F_0 is the susceptible fraction).
- Their sensitivity indices are natural dimensionless elasticities (Chapter 9).

A.4 Integral and Renewal Equations

The Kermack-McKendrick formulation of Chapter 15 uses integral equations rather than differential equations. This section reviews the relevant techniques.

A.4.1 Convolution Integrals

The convolution of two functions f, g on $[0, \infty)$ is

$$(f * g)(t) \;=\; \int_0^t f(\tau) g(t-\tau)\, d\tau \;=\; \int_0^t f(t-\tau) g(\tau)\, d\tau,$$

the second equality by substitution $u = t - \tau$. Convolution is commutative, associative, and distributes over addition.

The Kermack-McKendrick equation for the new-infection rate j is

$$j(t) \;=\; \int_0^\infty K(\tau) j(t-\tau)\, d\tau \;=\; (K * j)(t),$$

expressing j as a convolution of itself with the kernel K.

A.4.2 The Laplace Transform

The Laplace transform of a function $f : [0, \infty) \to \mathbb{R}$ is

$$\widehat{f}(\sigma) \;=\; \int_0^\infty f(t) e^{-\sigma t}\, dt,$$

defined for σ values where the integral converges. The transform converts convolutions into ordinary products: $\widehat{(f * g)} = \widehat{f} \cdot \widehat{g}$.

For the renewal equation $j = K * j$, taking the Laplace transform gives $\widehat{j}(\sigma) = \widehat{K}(\sigma)\widehat{j}(\sigma)$, so $\widehat{j}(\sigma)(1 - \widehat{K}(\sigma)) = 0$. Non-trivial solutions require $\widehat{K}(\sigma) = 1$, the characteristic equation.

A.4.3 Solving Linear Renewal Equations

For a non-homogeneous renewal equation $j(t) = h(t) + \int_0^t K(\tau) j(t-\tau)\, d\tau$ with given forcing h, the Laplace transform gives

$$\widehat{j}(\sigma) \;=\; \frac{\widehat{h}(\sigma)}{1 - \widehat{K}(\sigma)}.$$

Inverting the Laplace transform recovers $j(t)$. The inverse-transform calculation is often technical; for this book's purposes, the threshold analysis at $\sigma = 0$ (giving $\mathcal{R}_0 = \widehat{K}(0)$) is the essential application.

A.5 Probability Background for Epidemic Models

The derivation of the force from infection (Chapter 3) and the per-contact transmission probability (Chapter 1, Section 1.4.1) use elementary stochastic-process concepts.

A.5.1 Exponential and Poisson Distributions

A random variable T has an exponential distribution with rate λ if its probability density is

$$f_T(t) \;=\; \lambda e^{-\lambda t}, \qquad t \geq 0.$$

The mean is $1/\lambda$ and the variance is $1/\lambda^2$. The exponential distribution has the memoryless property: $\Pr(T > u + t \mid T > u) = \Pr(T > t)$ for all $u, t \geq 0$. This property is the reason exponentially-distributed waiting times are convenient in compartmental modeling: an individual's expected residence time is independent of how long they have already been in the compartment.

A Poisson process with rate λ is a counting process where events occur at exponential intervals. The number of events in a time interval of length t is Poisson distributed with mean λt.

A.5.2 Thinning of Poisson Processes

If a Poisson process with rate λ is "thinned" by independently retaining each event with probability p (and discarding with probability $1 - p$), the result is a Poisson process with rate λp.

This is the technique used in Chapter 2 to derive the force of infection: contacts arrive as a Poisson process at rate c_S; only a fraction P_I are with infectious individuals; only a fraction β of those transmit. The resulting infection rate is

$$\lambda \;=\; c_S \cdot P_I \cdot \beta,$$

a Poisson process with rate λ.

A.5.3 Per-Contact Transmission and the Exponential Formula

For a contact of duration τ_c during which the pathogen has rate ρ of transmission per unit time, the transmission probability is

$$\beta \;=\; \Pr(\text{transmission during the contact}) \;=\; 1 - e^{-\rho\tau_c}.$$

This is the cumulative-distribution-function evaluation of an exponential distribution: the transmission time is exponential with rate ρ, and β is the probability the transmission time falls within the contact duration.

A.5.4 Branching Processes

A branching process tracks the descendants of an initial individual. Each individual produces a random number of "offspring" with a fixed distribution; the process either grows or dies out.

For epidemic modeling, the early-outbreak dynamics can be approximated as a branching process in which each infectious individual generates secondary infections according to some distribution (often Poisson or negative-binomial). The expected offspring count is $\mathcal{R}_0$. The branching process grows on average if $\mathcal{R}_0 > 1$ and dies out almost surely if $\mathcal{R}_0 < 1$. This recovers the threshold structure of the deterministic compartmental model and explains it from a stochastic perspective.

The branching-process framework is the foundation of more refined stochastic epidemic models that capture the finite-population effects and extinction probabilities not represented in the deterministic ODE framework.

A.6 Summary

This appendix has reviewed the main mathematical techniques used in the book:

1. ODEs and their analysis via Jacobian linearization, eigenvalue stability, and Routh-Hurwitz criteria.
2. Linear algebra for the next-generation matrix: eigenvalues, spectral radius, matrix inversion, and the Perron-Frobenius theorem.
3. Dimensional analysis and the Buckingham Pi theorem for choosing dimensionless parameters.
4. Integral equations, convolution, and Laplace transforms for the Kermack-McKendrick formulation.
5. Stochastic-process concepts (exponential distribution, Poisson processes, thinning, branching) underlying the force-from-infection derivation.

Readers needing a more thorough treatment of any topic should consult the standard references: Murray [2002] for ODE methods in mathematical biology; Horn and Johnson [2012] for matrix analysis; Barenblatt [2003] for dimensional analysis; Feller [1968] for probability and stochastic processes.

Appendix B

Numerical Methods: Guide to the Code Repository

A program without tests is a proof without lemmas.

The numerical methods used in this book (forward integration of compartmental models, computation of basic and effective reproductive numbers via the next-generation matrix, fitting of incidence data, sensitivity-index computation, Latin Hypercube Sampling with partial rank correlation coefficients (LHS-PRCC), and renewal-equation solvers) are provided as Python code in the book's companion GitHub repository, added chapter by chapter as the code base grows. Rather than reproducing code listings in print, this appendix provides:

1. A guide to locating each method's implementation in the repository (Section B.1),
2. The substantive methodological development of LHS-PRCC (Section B.2), which is mathematical rather than implementation content and belongs in the printed book,
3. Notes on software-engineering practices for research-quality reproducibility (Section B.3).

The choice to host code on GitHub rather than in print reflects two practical realities. First, code in print is read once and rarely re-typed; code on GitHub is downloaded and run. Second, printed code is fixed at publication; code on GitHub can be updated as Python libraries evolve, as bugs are found, and as community contributions accumulate. The book is the place for methodology that is stable across decades; the repository is the place for implementations that should evolve with their tools.

B.1 Guide to the GitHub Code Repository

The book's companion repository is hosted at the URL announced on the inside front cover and on the book's web page. Within the repository, code is organized by chapter so that a reader working through Chapter n can locate the corresponding implementation directly.

B.1.1 Repository structure

`notebooks/` Jupyter notebooks, organized into per-chapter subdirectories, that reproduce the book's figures and case studies and run end-to-end in Google Colab or a local Jupyter environment. For example, `notebooks/ch06/` holds the Chapter 5 analysis figures (bifurcation, phase

portrait, and the invasion–burden partition); `notebooks/ch07/` the Chapter 6 simulation figures; `notebooks/ch08/` the Chapter 7 fitting case studies; and `notebooks/ch10/` the Chapter 9 sensitivity analysis, including the LHS-PRCC implementation whose methodology is developed in Section B.2 below.

`src/` Shared Python modules imported by several notebooks, such as the SIR integrator built on `scipy.integrate.solve_ivp` with the standard tolerances (10^{-9} absolute, 10^{-8} relative).

`data/` Empirical datasets for case studies that use them, with provenance documentation. The stochastic case-study series in the book are instead generated within their notebooks from fixed random seeds, so they reproduce exactly without external files.

`requirements.txt` Pinned dependency versions (NumPy, SciPy, Matplotlib) for reproducibility of the book's numerical results.

`README.md` Repository-level documentation: setup, how to open a notebook in Colab, and pointers from each chapter to its code.

Application case studies (Chapters 17–19) and the code for later chapters are added to `notebooks/` as those chapters are developed.

B.1.2 Setup and conventions

The code uses Python 3.10 or later, with type hints where useful. The standard scientific Python stack is sufficient: NumPy for array operations and linear algebra, SciPy for ODE integration (`scipy.integrate.solve_ivp`) and statistical routines, and Matplotlib for plotting. Parameters are passed as Python dictionaries rather than as positional arguments, both for readability and to match the book's notation conventions (every parameter has a name; the dictionary key is the name).

For reproducibility, all stochastic methods set random seeds explicitly. The convention is `numpy.random.seed(42)` at the top of any notebook or script that uses random sampling.

B.1.3 Mapping book chapters to repository modules

The following table summarizes which repository module supports each chapter's numerical content:

Chapter	Repository location
Chapter 5 ($\mathcal{R}_0$, NGM, phase portrait)	`notebooks/ch06/`
Chapter 6 (Simulation)	`notebooks/ch07/`
Chapter 7 (Fitting)	`notebooks/ch08/`
Chapter 9 (Sensitivity, LHS-PRCC)	`notebooks/ch10/`

The `labs/` directory contains pedagogical notebooks that work through specific exercises end-to-end. Each lab notebook references the relevant chapter and exercise number, and includes setup, computation, and discussion in a form suitable for use in a classroom or for self-study.

B.2 Latin Hypercube Sampling and PRCC: Methodology

Chapter 9, Section 9.6 applies the LHS-PRCC method to the SIR model. This section develops the underlying methodology in full. The implementation appears in the Chapter 9 sensitivity notebook (`notebooks/ch10/`) of the companion repository.

B.2.1 Latin Hypercube Sampling

A Latin hypercube sample is a stratified sampling procedure for the joint parameter space that ensures good coverage of each marginal distribution even with modest sample sizes. For k parameters and M samples, the procedure is:

1. For each parameter p_j (with $j = 1, \ldots, k$), specify a distribution representing the modeler's uncertainty. Common choices are uniform distributions on $[p_j^{\min}, p_j^{\max}]$ and lognormal distributions centered at a baseline value.

2. For each parameter p_j, partition the cumulative distribution function into M equiprobable strata. A uniform on $[a, b]$ has strata at $[a+(\ell-1)(b-a)/M, a+\ell(b-a)/M]$ for $\ell = 1, \ldots, M$.

3. For each parameter, draw one sample from each stratum, producing M values $\{p_j^{(1)}, \ldots, p_j^{(M)}\}$.

4. Shuffle the samples independently for each parameter and pair the ℓth shuffled sample of each parameter to form the ℓth LHS sample $(p_1^{(\ell)}, \ldots, p_k^{(\ell)})$.

The result is M joint samples, each of which pairs one value from each of the k parameters. The samples are constructed so that the marginal distribution of each parameter is represented by exactly one point in each stratum, giving dense coverage even for small M.

Sample size. Typical choices for sensitivity studies are $M \in \{100, 500, 1000, 5000\}$ samples. Rule of thumb: M should be at least 10 times the number of parameters for reliable rankings, and 100 times for precise quantification. For the six-parameter SIR model, $M = 1000$ to 5000 is appropriate.

Why stratified rather than simple random. Simple random sampling (drawing each parameter independently from its distribution) gives good expected behavior but can produce individual samples that cluster in corners of the parameter space or miss important regions. Stratified sampling ensures uniform coverage of the marginal distributions: each stratum of each parameter is hit exactly once. For the same M, LHS gives smaller sampling variance than simple random sampling.

B.2.2 Running the Model on the LHS Design

For each of the M LHS samples, integrate the SIR model and compute the output of interest. This produces M values of the output:

$$\{q_1, q_2, \ldots, q_M\}, \quad q_\ell = q(p_1^{(\ell)}, \ldots, p_k^{(\ell)}).$$

The outputs and their corresponding parameter inputs form a dataset of M rows, each with $k+1$ columns (the k parameters and the output).

The computational cost is M full integrations, one per LHS sample. For the SIR model with $M = 1000$, this runs in seconds on a laptop; for more complex models it can run in minutes.

B.2.3 Partial Rank Correlation Coefficients

With the dataset in hand, the question is: which parameters contribute most to the variation in the output across the samples? The partial rank correlation coefficient (PRCC) answers this question.

Rank transformation. Convert each column (each parameter and the output) to ranks: the smallest value becomes rank 1, the next smallest becomes rank 2, and so on. Ranks transform arbitrary distributions to uniform distributions on the integers $\{1, \ldots, M\}$ and are robust to the scale of the original parameters.

Partial correlation. The correlation between a parameter p_j and the output q measures how strongly they move together across the samples. The *partial* correlation controls for the influence of the other parameters: it is the correlation between p_j and q after the linear influence of $p_1, \ldots, p_{j-1}, p_{j+1}, \ldots, p_k$ has been removed from both.

Formally, fit a linear regression of q on $(p_1, \ldots, p_{j-1}, p_{j+1}, \ldots, p_k)$ and take the residuals $\tilde{q}$; fit a linear regression of p_j on the same other parameters and take the residuals $\tilde{p}_j$; the partial correlation is the ordinary correlation between $\tilde{q}$ and $\tilde{p}_j$.

Putting it together. The PRCC combines the rank transformation with the partial correlation: it is the partial correlation between the ranks. The PRCC is bounded in $[-1, 1]$ and measures the strength and direction of the monotonic association between an input and the output after removing the linear influence of the other inputs. It is a partial *correlation* on ranks, not an elasticity: it carries no units of proportional change and does not estimate $\partial \log q / \partial \log p_j$. The normalized sensitivity indices of Chapter 9 are elasticities; PRCC is a complementary rank-based measure of association, and the two answer different questions. $\mathrm{PRCC}(p_j, q) = 1$ means that the rank of q is perfectly (and positively) explained by the rank of p_j after controlling for the others; $\mathrm{PRCC}(p_j, q) = 0$ means that p_j and q are uncorrelated after controlling for the others.

Significance testing. Under the null hypothesis that p_j and q are uncorrelated after controlling for the others, the PRCC is approximately normally distributed with mean 0 and standard deviation $1/\sqrt{M-k}$. For $M = 1000$ and $k = 6$, the standard deviation is about 0.032. A PRCC with absolute value above 0.1 is clearly non-zero; values below 0.05 are within sampling noise.

Implementation note. The Python implementation of LHS sampling and PRCC computation is in the Chapter 9 sensitivity notebook (`notebooks/ch10/`) of the companion repository. The implementation operates on parameter dictionaries for readability and produces both the PRCC values and bootstrap confidence intervals for each parameter.

B.3 Notes on Software Engineering for Reproducibility

For research-quality reproducibility, several practices beyond running the example code are recommended:

- **Pin dependency versions** in a `requirements.txt` or environment file. The companion repository's `requirements.txt` pins the versions of NumPy, SciPy, and Matplotlib that produced the numerical results in the book; substantially different versions may produce numerically different results due to library updates.
- **Use Jupyter notebooks for exploratory analysis** and standalone Python modules for reusable functions. Notebooks are well-suited to the iterative process of trying different parameter values, generating diagnostic plots, and writing narrative around results. Standalone modules are well-suited to the parts of the analysis that other notebooks or scripts will reuse.

- **Set random seeds explicitly** for any stochastic computation. Without an explicit seed, two runs of the same notebook can produce different numerical results, which is the antithesis of reproducibility. The convention used throughout the companion repository is `numpy.random.seed(42)` at the top of any notebook or script that uses random sampling.

- **Save intermediate results** (parameter samples, model trajectories, fit residuals) to disk for later analysis without re-running expensive computations. The companion repository uses `numpy.savez` for numerical arrays and `pandas.DataFrame.to_csv` for tabular results.

- **Use version control (Git)** for both code and accompanying data, with informative commit messages. The companion repository is hosted on GitHub for exactly this purpose.

- **Document the units of all parameters**, especially in code where unit ambiguity is most easily lost. The companion repository's parameter dictionaries include unit annotations as comments.

For collaborative or distributed projects, also consider containerization (e.g., Docker) for fully reproducible computational environments. Containerization is beyond the scope of this book's reproducibility recommendations but is the standard approach in the broader scientific computing community for ensuring that computational results can be reproduced years after the original work.

Appendix C

Glossary

This appendix provides definitions of the epidemiological and modeling terms used throughout the book, followed by a table of parameter symbols. Definitions are written in the book's own notation and reference the chapter sections where each term is developed. The terms (Section C.1) are collected in a single alphabetical list; the parameter symbols (Section C.2) are kept separate for quick reference against the equations.

For terms beyond the scope of this glossary, the reader is referred to specialized epidemiological references such as the National Collaborating Centre for Infectious Diseases (NCCID) glossary and the standard Modern Epidemiology textbook of Rothman et al. [2008].

C.1 Glossary of Terms

Antiretroviral therapy (ART). Pharmaceutical treatment for HIV that suppresses viral replication and substantially reduces per-contact transmission. The basis of treatment-as-prevention strategies. See Chapter 19.

Ascertainment; under-ascertainment. The fraction $\rho \in (0, 1]$ of true infections that are detected and reported. Under-ascertainment ($\rho < 1$) means the observed incidence systematically undercounts the truth, $J_{\text{obs}} = \rho\, J_{\text{true}}$. Because ρ enters the infected-viewpoint estimator $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$ as a shared factor, it cancels there. See Chapters 7 and 8.

Asymptomatic infection. An infection in which the host carries and may transmit the pathogen but does not develop characteristic symptoms. Asymptomatic individuals typically maintain normal contact patterns and may contribute substantially to transmission. Modeled in Chapter 10, Section 10.6 via an explicit asymptomatic compartment A.

Attack rate. The cumulative fraction of a population that becomes infected during an outbreak. For an SIR model with full epidemic resolution, the final attack rate satisfies $1 - s(\infty)/n = r(\infty)/n$, derivable from the SIR conservation law (Chapter 5).

Autocorrelation function (ACF). The correlation of a time series with lagged copies of itself, as a function of the lag. Applied to fitting residuals, nonzero autocorrelation at short lags signals unmodeled temporal structure—a misspecified model or correlated noise—rather than the independent errors that fitting assumes. See Chapter 8.

Backfilling. The upward revision of recently reported counts as delayed reports arrive, so the most recent days of a surveillance series are systematically incomplete when first observed.

A principal cause of the recent-edge bias introduced by fitting to raw real-time data. See Chapter 7; compare nowcasting.

Backward bifurcation. A qualitative feature of some epidemiological models in which a stable endemic equilibrium exists for a range of $\mathcal{R}_0$ values below 1, coexisting with the stable disease-free equilibrium. Contrasts with forward bifurcation, in which the endemic equilibrium exists only for $\mathcal{R}_0 > 1$. Arises from positive-feedback mechanisms at intermediate prevalence: imperfect vaccination, treatment saturation, differential susceptibility, or vector-host saturation. Produces a hysteresis phenomenon under interventions: reducing $\mathcal{R}_0$ below 1 is necessary but not sufficient for elimination. See Chapter 10, Section 10.1.

Basic reproductive number ($\mathcal{R}_0$). The expected number of secondary infections generated by a single infectious individual introduced into an entirely susceptible population. The threshold parameter for invasion: $\mathcal{R}_0 > 1$ implies an outbreak can sustain itself; $\mathcal{R}_0 < 1$ implies it dies out. Derived for the SIR model in Chapter 5 via two methods (heuristic individual-tracking and the next-generation matrix).

Bed nets, insecticide-treated (ITNs). Bed nets impregnated with insecticide. Reduce both biting rate a (mechanical barrier) and vector population (kills mosquitoes that attempt to bite). The most-sensitive single intervention for malaria control, given that a has unit sensitivity in $\mathcal{R}_0$ (Chapter 13, Section 13.5).

Behavioral feedback. Endogenous adjustment of transmission-relevant behavior (contact rates, use of protective measures, demand for non-pharmaceutical interventions) in response to the perceived state of an outbreak. Compartmental models with constant β and constant per-capita contact rates do not represent this feedback; published comparisons document systematic bias in fitted parameters when behavioral feedback is empirically strong. Named as a common pitfall in the catalog (Pitfall: "Constant-Transmission Assumption Ignoring Behavioral Feedback"); the book uses constant β throughout with the assumption made explicit. See Chapter 5 (validity-domain caveat) and Chapter 17 (intervention-scenario implications).

Bifurcation. A qualitative change in the behavior of a dynamical system as a parameter varies. The transition from disease-free to endemic equilibrium at $\mathcal{R}_0 = 1$ is a transcritical bifurcation. See Appendix A, Section A.1.

Bipartite transmission. A transmission mode in which the pathogen circulates between two distinct groups (male/female for sexually-transmitted infections, vector/host for vector-borne diseases) with little or no within-group transmission. Produces a distinctive square-root structure in the reproductive-number formula. Treated in Chapters 12 and 13.

Bootstrap. A resampling method for quantifying estimator uncertainty: refit the model to many synthetic datasets generated from the data (or from the fitted model) and use the spread of the resulting estimates as an empirical sampling distribution. Used to build confidence intervals for fitted parameters such as $\mathcal{R}_0$. See Chapter 8.

Branching process. A stochastic model of the descendants of an initial individual, where each individual produces a random number of offspring. In epidemic modeling, captures the early-outbreak dynamics where the susceptible pool is essentially infinite. The expected offspring count is $\mathcal{R}_0$. See Chapter 3, Section 3.3.

Buckingham Pi theorem. The theorem of dimensional analysis stating that a physical relationship involving n variables and k independent dimensions can be expressed in terms of $n - k$ dimensionless parameters. Applied in Chapter 7, Section 7.3 to identify the natural dimensionless parameters of the SIR model.

Burden. The endemic level of infection a disease sustains once established, measured for example by the equilibrium prevalence i^*. Governed by a different set of parameters than epidemic invasion (see invasion–burden distinction). Analyzed for the SIR model in Chapter 5.

Carrier. An individual who carries a pathogen and may transmit it but does not show clinical symptoms. May be persistent (chronic carriers, as in hepatitis B) or transient (incubating cases). Carrier states are modeled either as variants of the asymptomatic compartment or as long-duration infectious compartments.

Case isolation. Removal of identified infectious cases from contact with the general population. Reduces c_I and hence $\mathcal{R}_0$. Particularly effective for symptomatic infections; less so for diseases with substantial asymptomatic transmission.

Case-fatality ratio (CFR). The fraction of clinically apparent infections that result in death. For the SIR model with disease mortality (Chapter 10, Section 10.5), CFR $= \mu_d/(\mu_d + \gamma_R + \nu_m)$, where μ_d is the disease-induced mortality rate.

Catchment population. The epidemiologically relevant population actually at risk in an outbreak—the people who mix with the index cases—as distinct from an administrative or census population. It sets the susceptible denominator s in the susceptible-viewpoint estimator $\hat{\lambda} = J_{\text{obs}}/s$ and is often ill-defined; the infected viewpoint sidesteps it. See Chapter 7.

Chronic sequelae. Persistent post-acute health consequences of an infection that depend on the acute-infection compartmental model only indirectly through the cumulative or final-size attack rate. Long COVID / post-acute sequelae of SARS-CoV-2 infection (PASC) is the contemporary canonical example. The sensitivity-analysis framework of Chapter 9 applies to chronic-sequelae outputs without modification: only the operationally-relevant output changes. The book does not develop chronic-sequelae compartmental extensions; see Chapter 9, Section 9.7 for the structural analog of the invasion-burden partition transposed to the chronic-sequela setting.

Combination intervention. The simultaneous use of multiple intervention strategies that act on different parameters. Combined effect is multiplicative on $\mathcal{R}_0$. Often required to bring $\mathcal{R}_0$ below 1 when no single intervention is sufficient (Chapter 20).

Compartmental model. A class of dynamical-system models in which the population is partitioned into discrete compartments (susceptible, exposed, infectious, recovered, etc.), with flows between compartments governed by ODEs or stochastic transitions. The SIR model is the canonical example. Foundational formulation in Chapter 1; extensions surveyed in Chapter 10. Developed systematically beginning in Chapter 1, Section 1.1.

Concurrent partnership. A sexual relationship that overlaps temporally with another, in contrast to serial monogamy where partnerships are sequential. Concurrent partnerships substantially elevate the per-partnership transmission rate of sexually-transmitted infections by allowing simultaneous spread to multiple partners. Discussed in Chapter 12, Section 12.1.

Condom use. Behavioral intervention reducing per-contact transmission probability β for sexually-transmitted infections. Combined with other measures, can substantially reduce $\mathcal{R}_0$ (Chapter 12, Section 12.6).

Confidence interval. A range constructed from data that would contain the true value of a parameter in a specified proportion (e.g., 95%) of repeated experiments. It expresses uncertainty in an estimated quantity, in contrast to a prediction interval, which bounds a future observation. See Chapter 8.

Contact reduction. Behavioral intervention reducing the contact rate c_I (or c_S). The dominant intervention category for direct-transmission diseases, including school closures, work-from-home, and physical distancing.

Contact tracing. The systematic identification of individuals who have had contact with a confirmed case, with subsequent testing, isolation, or treatment of those contacts. An intervention modality that effectively reduces $\mathcal{R}_0$ by reducing the secondary-infection generation per case.

Controlled reproductive number ($\mathcal{R}_c$). The expected number of secondary infections per infectious individual under a specified intervention regime (vaccination, contact reduction, treatment, isolation), but with the population otherwise still naive. Mathematically the special case of $\mathcal{R}_e(t)$ at the post-intervention state before the susceptible pool has been substantially depleted by the epidemic itself. Useful for ranking interventions: the question "does intervention X bring $\mathcal{R}_c$ below 1" is the threshold question for whether intervention X alone can prevent an outbreak. Treated in Chapters 7 (estimation from data) and 9 (intervention ranking).

Cyclic reproductive number ($\mathcal{C}_0^{(\ell)}$). In a simple cyclic n-stage transmission model, the product of stage reproductive numbers around one complete cycle returning to stage ℓ. Counts the expected number of same-stage infections generated after one full cycle. Corresponds to what earlier literature sometimes called the "per-cycle reproductive number." See Chapter 11, Section 11.6.

Data preprocessing. The steps that turn raw surveillance counts into an incidence series suitable for fitting—removing day-of-week artifacts, correcting reporting delays and backfilling, and handling holiday gaps and missing values—producing the estimated incidence J_{est} from the observed J_{obs}. A recurring source of small, systematic error. See Chapter 7.

Deconvolution. Recovering an unobserved signal from a blurred or delayed version of it—here, recovering true incidence J_{true} from reported counts smeared by a reporting-delay kernel (the Richardson–Lucy algorithm is one stable approach). Central to nowcasting the recent edge of an outbreak. See Chapter 7.

Delta method. A first-order technique for propagating parameter uncertainty into a derived quantity: approximate the quantity by its Taylor expansion about the parameter estimates, so its variance is the gradient contracted with the parameter covariance. Cheap and local, in contrast to variance-based global methods. Stated as a theorem in Chapter 9.

Density-dependent transmission. An incidence form in which the contact rate rises with population density, so the force of infection scales with the *number* of infectious individuals rather than their fraction. Contrast frequency-dependent transmission. See Chapter 2.

Dimensional consistency. The requirement that quantities added, subtracted, or equated share the same dimensions. In this book's normalized/count convention, a normalized quantity is never equated with a count without the explicit N^{c} bridge; the check applies to model equations, generated code, and fitted observation models alike. See Chapter 7.

Disease-free equilibrium (DFE). The equilibrium state of a model in which no infection is present: $s = 1, i = r = 0$ for the SIR model. Stable if $\mathcal{R}_0 < 1$, unstable if $\mathcal{R}_0 > 1$. See Chapter 5.

eFAST (extended Fourier amplitude sensitivity test). A variance-based global sensitivity method that assigns each input a distinct oscillation frequency across the parameter space and reads first-order and total sensitivity indices from the Fourier spectrum of the model output. An alternative to Monte-Carlo Sobol' estimation. See Chapter 9.

Effective reproductive number ($\mathcal{R}_e$). The expected number of secondary infections per infectious individual at a given point in time, accounting for the depletion of the susceptible pool, the current behavioral or intervention state, and any acquired immunity. The operationally-primary reproductive-number quantity in this book: where transmission *during* an epidemic is the question (whether $i(t)$ is currently growing, when interventions become effective, when an endemic equilibrium is approached), the analysis is conducted in $\mathcal{R}_e(t)$. The basic reproductive number $\mathcal{R}_0$ is recovered as the special case $\mathcal{R}_e(0)$ in a fully naive population. For the SIR model, $\mathcal{R}_e(t) = \mathcal{R}_0 \cdot s(t)/n(t) = \mathcal{R}_0 \cdot P_S(t)$. Discussed in Chapter 5, Section 5.2.

Eigenvalue. A scalar λ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for some non-zero vector $\mathbf{v}$. Eigenvalues of the Jacobian determine local stability of equilibria. See Appendix A, Section A.2.

Endemic equilibrium. A steady state of an epidemic model in which infection persists at a stable prevalence rather than dying out or invading. For the SIR model, exists and is stable when $\mathcal{R}_0 > 1$. Computed in Chapter 5, Section 5.5.

Epidemic threshold. The critical value of $\mathcal{R}_0$ separating outbreak (sustained transmission) from extinction (decay to zero). Under the standard next-generation framework the threshold is $\mathcal{R}_0 = 1$: the disease-free equilibrium is locally stable for $\mathcal{R}_0 < 1$ and unstable for $\mathcal{R}_0 > 1$. This is a *local invasion* threshold, not a universal elimination criterion—in models exhibiting backward bifurcation an endemic state can persist for $\mathcal{R}_0 < 1$, so reducing $\mathcal{R}_0$ below one need not eliminate the infection. Derived for the SIR model in Chapter 5 and generalized via the next-generation matrix in Section 5.3. Derived for the basic SIR model in Chapter 5, Section 5.1.

Equilibrium. A steady state of a dynamical system, where all time derivatives vanish. The SIR model has two equilibria: disease-free (always) and endemic (when $\mathcal{R}_0 > 1$).

Erlang chain. A modeling construction that replaces a single compartment of mean residence time τ by a chain of k identical sub-compartments each with mean residence time τ/k. The total residence time is then gamma-distributed with shape parameter k, recovering an exponential at $k = 1$ and approaching a deterministic delay as $k \to \infty$. Used to relax the implicit exponential-residence-time assumption of basic ODE compartmental models without leaving the ODE framework. Also known as the linear-chain trick. Developed in Chapter 10, Section 10.3 in the context of the SEIR model's latent compartment.

Estimator. A rule for computing a parameter value from data. The book contrasts two estimators of transmission: the susceptible-viewpoint $\hat{\lambda} = J_{\text{obs}}/s$, which needs the susceptible denominator, and the infected-viewpoint $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$, which does not. See Chapters 3 and 7.

Event date and report date. The date an epidemiological event occurred (infection or symptom onset) versus the date it was recorded in surveillance. Indexing incidence by report date rather than event date shifts and distorts the inferred trajectory; the model's incidence is naturally an event-date quantity. See Chapter 7.

Exponential-growth phase. The early stage of an outbreak, while the susceptible pool is nearly undepleted, in which incidence grows approximately exponentially at a rate set by $\mathcal{R}_0$ and the generation time. The regime in which growth-rate estimates of $\mathcal{R}_0$ apply. See Chapter 5.

Exposed. In an SEIR model, an individual who has been infected but is not yet infectious. The compound description "exposed and infected" (or, equivalently, "infected but not yet infectious") is more precise than "exposed" alone, since the latter is sometimes interpreted in everyday usage to mean "possibly exposed but not yet known to be infected." Denoted E in the compartmental framework. The duration of the exposed state is the latent period; see *Latent period* and *Incubation period* for related but distinct concepts.

Extrinsic incubation period (EIP). For vector-borne diseases, the time interval between a vector acquiring the pathogen and the vector becoming infectious to subsequent hosts. The EIP is determined by within-vector pathogen development and is comparable to the vector lifespan for diseases like malaria, leading to the survival probability $p_{\text{EIP}} = e^{-\mu_v \tau_v^E}$. See Chapter 13, Section 13.2.

Final size relation. The implicit equation, due to Kermack and McKendrick, relating the total fraction of the population ultimately infected in an epidemic to $\mathcal{R}_0$, independent of the detailed time course. See Chapter 5.

Force from infection. The per-capita rate at which an infectious individual generates new infections. From the infected viewpoint, $\alpha = c_I \beta P_S$. Equivalent to the force of infection in producing the same population-level new-infection rate ($\alpha i = \lambda s$). The book's primary force formulation; developed in Chapter 3.

Force of infection. The per-capita rate at which a susceptible individual acquires infection. From the susceptible viewpoint, $\lambda = c_S \beta P_I$ for the SIR model. Developed in Chapter 2.

Frequency-dependent transmission. An incidence form in which each individual has a roughly fixed contact rate independent of population size, so the force of infection scales with the *fraction* infectious. The standard choice for directly transmitted and sexually transmitted infections. Contrast density-dependent transmission. See Chapter 2.

Generation time. The interval between an individual's infection and the infections that individual generates. The mean generation time T_g relates the doubling time of an outbreak to $\mathcal{R}_0$ via $\sigma \approx (\mathcal{R}_0 - 1)/T_g$. See Chapter 15, Section 15.5.

Geometric-mean reproductive number. For a simple cyclic n-stage model, the nth root of the cyclic reproductive number, equivalently the geometric mean of the n stage reproductive numbers. Coincides with $\mathcal{R}_0 = \rho(\boldsymbol{K})$ in the simple-cyclic case. Corresponds to what earlier

literature sometimes called the "per-generation reproductive number" in bipartite settings. For non-cyclic multigroup models (e.g., the HIV model of Chapter 19, where $\mathcal{R}_0 = \mathcal{R}_0^{\text{horiz}} + \mathcal{R}_0^{\text{vert}}$ is additive across transmission routes), $\mathcal{R}_0 = \rho(\boldsymbol{K})$ remains the threshold quantity but does not reduce to a single cycle root. See Chapter 11, Section 11.6.

Gillespie algorithm. An exact stochastic simulation method that generates individual event times (infection, recovery) for a continuous-time Markov model, reproducing the demographic stochasticity that deterministic ODEs omit. Used to study extinction and variability in small populations. See Chapter 6.

Global sensitivity analysis. Apportioning the variability of a model output over the full plausible ranges of its inputs, varied simultaneously, rather than through small perturbations about a single baseline. Captures interactions and nonlinearity that local sensitivity misses; methods include Sobol' indices, eFAST, and PRCC with Latin hypercube sampling. See Chapter 9.

Herd immunity. The level of population immunity (through prior infection or vaccination) sufficient to drive the effective reproductive number below 1. For a homogeneous population, the threshold is $1 - 1/\mathcal{R}_0$. For heterogeneous populations, the threshold for optimally targeted immunization is lower (Chapter 11, Section 11.7).

Heterogeneous population. A population partitioned into subgroups with different epidemiological parameters (contact rates, susceptibility, infectious period). Treated in Chapter 11 for two groups and extended in Chapters 12 and 13.

Homogeneous mixing. The assumption that every individual is equally likely to contact every other (a "well-mixed" population), so a single force of infection applies to all susceptibles. The baseline assumption of unstructured compartmental models, relaxed by mixing matrices and structured models. See Chapter 4, and Chapter 11 for its relaxation.

Identifiability. The property that the parameters of a model can be uniquely determined from the available data. Some parameter combinations are identifiable from incidence data alone; others require additional data streams or independent estimates. Treated in Chapter 7, Section 7.3.

Incidence. The rate of new infections per unit time, typically measured as cases per day or cases per week. In our notation, the *normalized* model incidence is $j(t) = \alpha i(t)$ (a fraction of N^{c} per unit time); its count form is $J = j\,N^{\text{c}}$. Defined and used throughout Chapter 7.

Incubation period. The interval between exposure to the pathogen and the appearance of symptoms. Distinct from the latent period (interval between exposure and onset of infectiousness). Incubation periods are clinically observable; latent periods are not directly measurable.

Index case. The first identified case of an outbreak. In modeling, the index case represents the initial infectious individual whose secondary infections constitute $\mathcal{R}_0$ at the moment of introduction.

Indoor residual spraying (IRS). Application of insecticide to indoor surfaces, killing vectors that rest indoors. Reduces vector population and increases vector mortality μ_v.

Infectious. An individual who is infected and capable of transmitting the pathogen, the central population in the compartmental framework's transmission dynamics. Denoted i in the compartmental framework. The compound description "infected and infectious" is sometimes used

in this book to emphasize that i refers specifically to those individuals who are both currently carrying infection and currently able to transmit, distinguishing them from individuals who are infected but not yet infectious (the E compartment of SEIR models, sometimes labelled "exposed"). See *infected viewpoint* for the methodological convention that takes i rather than s as the primary state variable for transmission analysis.

Infectious period. The duration during which an infectious individual is capable of transmitting the pathogen. In the SIR model, this is the time spent in the i compartment, with mean $\tau_R^* = 1/(\gamma_R + \nu_m)$. See Chapter 4 for the derivation and Chapter 15 for extensions with variable infectiousness.

Invasion–burden distinction. The observation, central to this book, that the parameters controlling whether a disease can *invade* (through $\mathcal{R}_0$: the contact and transmission parameters c_I, β) differ structurally from those controlling the endemic *burden* it sustains (through i^*: the residence-time parameters τ_R, τ_m). A parameter that dominates invasion may be nearly irrelevant to burden, and vice versa. Developed for the SIR model in Chapter 5.

Inverse problem. Inferring model parameters (or unobserved inputs) from observed outputs—the direction fitting runs—as opposed to the *forward* problem of computing outputs from known parameters. Inverse problems are frequently ill-posed. See Chapter 7.

Jacobian. The matrix of partial derivatives of the right-hand side of an ODE system, evaluated at a specific state. Determines the local linearization of the dynamics. See Appendix A, Section A.1.

Jensen's inequality. For a convex function ϕ, $\mathbb{E}[\phi(X)] \geq \phi(\mathbb{E}[X])$ (reversed for concave ϕ). It explains why averaging inputs and then applying a nonlinear model differs from applying the model and then averaging—a source of systematic bias when fitting or aggregating. See Chapter 8.

Kermack-McKendrick equation. The integral equation $j(t) = \int_0^\infty K(\tau) j(t-\tau) d\tau$ governing the new-infection rate in time-since-infection-structured models. The kernel $K(\tau) = A(\tau)F(\tau)$ encodes infectiousness and survival. The most general formulation of the SIR model. See Chapter 15, Section 15.4.

Larvicide. Chemical or biological agent killing immature vectors at breeding sites. Reduces the vector-to-host ratio m before adult vectors are produced.

Latent period. The interval between exposure and onset of infectiousness. Modeled in Chapter 10, Section 10.3 via the addition of an exposed compartment E. Distinct from the incubation period (time to symptoms), though for many pathogens the two are similar.

Latin hypercube sampling (LHS). A stratified sampling technique for the joint parameter space, ensuring uniform coverage of each marginal distribution. Used for global sensitivity analysis. See Chapter 9, Section 9.6 and Appendix B, Section B.2.

Least squares. Fitting by minimizing the sum of squared residuals between model and data. Its implicit likelihood is Gaussian with constant variance; when counts are Poisson (variance growing with the mean), least squares is less efficient than, and can bias differently from, maximum likelihood. See Chapter 7.

Ljung–Box test. A statistical test for whether a time series—for example, fitting residuals—shows autocorrelation up to a given lag, jointly across lags. A formal check of the independent-residuals assumption. See Chapter 8.

Local sensitivity analysis. Quantifying how a model output responds to small perturbations of each input about a fixed baseline, typically through partial derivatives or normalized sensitivity indices. Exact and cheap but blind to interactions and to behavior far from the baseline; contrast global sensitivity analysis. See Chapter 9.

Long COVID; PASC (post-acute sequelae of SARS-CoV-2 infection). Persistent symptoms following acute SARS-CoV-2 infection, lasting from weeks to indefinitely. PASC is the canonical contemporary example of *chronic sequelae* (q.v.): a post-acute outcome whose trajectory depends on the acute-infection compartmental model only indirectly. PASC modeling is an active research area. The book does not develop PASC compartmental extensions; sensitivity-analysis applications to PASC are surveyed in Chapter 9, Section 9.7 and the scope limitation is noted in Chapter 17.

Loss function. The scalar objective minimized in fitting, measuring the discrepancy between model output and data—the sum of squared residuals for least squares, or the negative log-likelihood for likelihood-based fitting. Its shape near the optimum governs parameter identifiability. See Chapter 7.

Mass-action transmission. The classical incidence law in which the rate of new infections is proportional to the product of susceptible and infectious counts, βsi. In this book's terms it is the density-dependent form, since the force of infection scales with the number of infectious individuals rather than their fraction. Contrast frequency-dependent transmission. See Density-dependent transmission and Chapter 2.

Maximum likelihood estimation (MLE). Choosing parameter values that maximize the probability the assumed observation model assigns to the data (equivalently, minimize the negative log-likelihood). The standard principled fitting method; with a Poisson observation model it accounts for the mean-dependent variance of case counts. See Chapters 7 and 8.

Missing data. Observations absent from a surveillance series (holidays, outages, suppressed small counts). Options include dropping the affected days, imputing them, or masking them in a likelihood-based fit; the right choice depends on whether values are missing at random. See Chapter 7.

Mixing matrix. A matrix encoding contact rates between subgroups in a heterogeneous population. The (j, k) entry C_{jk} is the rate of contacts of a group-j individual with group-k individuals, subject to the reciprocity constraint $C_{jk}N_j^c = C_{kj}N_k^c$. See Chapter 11, Section 11.2.

Mortality, disease-induced. Rate of death caused by the disease, distinct from background mortality. Denoted μ_d and added to the infectious-compartment outflow in Chapter 10, Section 10.5.

Mother-to-child transmission (MTCT). Transmission from an infected mother to her offspring during pregnancy, delivery, or breastfeeding. Per-pregnancy transmission probability p_{MTCT}. Modeled in Chapter 14.

Neonatal vaccination. Vaccination at birth, particularly for hepatitis B, to prevent vertical transmission outcomes (chronic carriage). See Chapter 14, Section 14.5.

Next-generation matrix (NGM). The matrix $\boldsymbol{K} = \boldsymbol{F}\boldsymbol{V}^{-1}$ whose spectral radius equals $\mathcal{R}_0$ for compartmental models with multiple infectious compartments. Introduced in Chapter 5, Section 5.3 and applied in Chapters 11 through 13.

Normalized sensitivity index. The dimensionless quantity $\Upsilon_p^q = (p/q)(\partial q/\partial p)$, measuring the percentage change in output q per percentage change in parameter p. See Chapter 9, Section 9.2.

Nowcasting. Estimating the true, complete incidence at the most recent, still-incomplete edge of a surveillance series by correcting for reporting delay and backfilling—a deconvolution problem. Distinct from forecasting the future. See Chapter 7.

Numerical integration. Approximating the solution of the model's differential equations by stepping forward in time, for example with a Runge–Kutta method. Step size and solver choice control accuracy; stiff systems require implicit solvers. See Chapter 6 (solver details in the Numerics appendix).

Outbreak. An epidemic in a defined population during a defined time interval. Distinguished from "epidemic" (often used for larger or longer events) and "pandemic" (geographically widespread epidemic).

Overdispersion. Count data whose variance exceeds its mean, violating the Poisson assumption of equal mean and variance. Common in case reports because of clustering and reporting heterogeneity; accommodated by a negative-binomial or quasi-Poisson observation model. See Chapter 8.

Partial rank correlation coefficient (PRCC). A global sensitivity measure: the correlation between rank-transformed parameter samples and rank-transformed model output, controlling for the influence of other parameters. See Chapter 9, Section 9.6.

Partner notification. Identification and treatment of sexual contacts of an STI case. Reduces τ_R for the partner population by accelerating treatment. See Chapter 12, Section 12.6.

Periodogram. A plot of the power (squared amplitude) of a time series against frequency, obtained from its discrete Fourier transform, showing how the series' variance is distributed across frequencies. Used to detect periodic reporting artifacts: a weekly reporting cycle appears as a peak at frequency $1/7$ day^{-1}. See Chapter 7.

Perron-Frobenius theorem. A theorem in linear algebra guaranteeing that a non-negative matrix has its spectral radius as a non-negative real eigenvalue with corresponding non-negative eigenvector. Justifies the interpretation of $\mathcal{R}_0 = \rho(\boldsymbol{K})$. See Appendix A, Section A.2.

Poisson observation model. The assumption that each reported count is a Poisson draw whose mean is the model's expected count, $J_{\text{obs}}(t_k) \sim \text{Poisson}\big(\rho\,\alpha\,i(t_k)\,N^{\text{c}}\,\Delta t\big)$. It makes the variance grow with the mean and underlies the likelihood used in fitting. See Chapter 8.

Practical non-identifiability. The situation in which parameters are structurally identifiable in principle but cannot be pinned down from the available, noisy data because the loss function is nearly flat along some direction in parameter space. Diagnosed with the profile likelihood; contrast structural identifiability. See Chapter 8.

Prediction interval. A range expected to contain a future or unobserved data point with a specified probability, combining parameter uncertainty with observation noise. Wider than a confidence interval for the underlying mean, because it also accounts for the scatter of individual observations. See Chapter 8.

Pre-exposure prophylaxis (PrEP). Antiretroviral drugs taken by uninfected individuals at high risk for HIV exposure, reducing per-contact transmission probability.

Prevalence. The number of currently infectious individuals at a given time, the count $I(t)$ (normalized fraction $i(t)$) in our notation. Distinct from incidence, the rate of new infections. Prevalence and incidence are related by $j = \alpha i$ when α is constant; Chapter 7 develops the fitting relationship in detail. Distinction developed in Chapter 4.

Prevention of mother-to-child transmission (PMTCT). Interventions to reduce vertical transmission, including ART during pregnancy (HIV) and neonatal vaccination (hepatitis B). See Chapter 14, Section 14.5.

Profile likelihood. The likelihood as a function of one parameter after re-optimizing all others at each of its values, used to obtain confidence intervals and to diagnose practical non-identifiability—a flat profile signals a poorly constrained parameter. See Chapter 8.

Quarantine. Restriction of movement of individuals exposed but not yet known to be infected, with the goal of identifying any who develop infection before they can transmit. Modeled in Chapter 10, Section 10.4 via a quarantine compartment Q.

Recovered. An individual who has cleared the infection and acquired (in the standard SIR framework) lasting immunity. Denoted r in the compartmental framework. The compound description "recovered and resistant" is sometimes used in this book to emphasize that r refers to individuals who are both no longer infectious and protected against reinfection. Models in which recovery does not confer immunity are described as SIS rather than SIR.

Recovery. The transition from the infectious to the recovered compartment, occurring at rate γ_R. May represent natural clearance, treatment-mediated cure, or progression to a non-infectious state.

Regime of validity. The range of conditions—parameter values, time horizon, and population regime—over which a model's assumptions hold and its predictions can be trusted. A model that is accurate within its regime of validity can fail badly outside it; stating the regime explicitly is one of the book's recurring cautions. See Section 16.2.1.

Renewal equation. An integral equation in which the unknown function appears under the integral as a convolution with a kernel. The Kermack-McKendrick equation is a renewal equation; see also the demographic Lotka equation in Chapter 15, Section 15.2.

Reporting delay. The lag between when a case occurs and when it appears in surveillance, described by a delay distribution. Convolving true incidence with this distribution blurs and shifts the observed series, especially at the recent edge; removing it is a deconvolution. See Chapter 7.

Reservoir. A subpopulation, species, or environment that maintains a pathogen and can serve as a source for transmission to other populations. The asymptomatic reservoir for COVID-19 (Chapter 17) and the high-activity-subgroup reservoir for HIV (Chapter 19) are examples within humans.

Residence time. The time an individual spends in a compartment before leaving it. Under the standard constant-rate assumption it is exponentially distributed with mean the inverse of the exit rate (e.g., recovery time $\tau_R = 1/\gamma_R$); Erlang chains produce more realistic, less-dispersed residence-time distributions. See Chapters 4 and 10.

Ring vaccination. A targeted-immunization strategy in which contacts (and contacts of contacts) of an identified case are vaccinated. Effective at containing localized outbreaks for diseases with targeted vaccines (smallpox, Ebola).

Ross–Macdonald model. The classical model of mosquito-borne transmission, coupling infection dynamics in hosts and vectors, from which vectorial capacity and the vector-borne $\mathcal{R}_0$ are derived. See Chapter 13.

Sensitivity analysis. The systematic study of how a model's outputs depend on its parameters. May be local (partial derivatives at a single baseline) or global (sampling broadly across parameter space). Developed in Chapter 9.

Sensitivity asymmetry. The book's result that the two viewpoints' transmission estimators respond oppositely to uncertainty in the susceptible count: the infected-viewpoint estimator has zero sensitivity to it, $\Upsilon_s^{\hat{\alpha}} = 0$, while the susceptible-viewpoint estimator is fully sensitive, $\Upsilon_s^{\hat{\lambda}} = -1$. This is the precise sense in which the infected viewpoint is robust to a poorly known susceptible population. See Chapter 9.

Seroprevalence. The fraction of the population that has antibodies against a pathogen, indicating prior infection. An estimator for $r(t)/n$ in the SIR framework, useful for fitting endemic-stage parameters and validating model assumptions.

SIR model. The book's core model: an SIR epidemic written from the infected viewpoint, with transmission driven by the force from infection α acting on the infectious population rather than the force of infection λ acting on susceptibles. Dynamically equivalent to the standard SIR, but with parameters and estimators tied to observable infected quantities. The subscript names the viewpoint, not a property of the disease: it does not abbreviate "with immunity". Derived in Chapter 4. Plain SIR is used where a statement holds for both formulations, or where the classical model of the literature is meant. See also SIR$_S$ *model*.

SIR$_S$ model. The same three-compartment model written from the susceptible viewpoint, with new infections generated by the force of infection $\lambda = c_S \beta P_I$ acting on susceptibles. Equivalent to SIR under the detailed-balance identity $\lambda s = \alpha i$, and not interchangeable with it in fitting: see Chapter 7.

Sobol' indices. Variance-based global sensitivity indices that partition the variance of a model output among its inputs. The *first-order* index is the fraction of output variance explained by an input alone; the *total* index also includes that input's interactions with all others. Estimated by Monte-Carlo sampling. See Chapter 9.

Spectral radius. The largest absolute value of the eigenvalues of a square matrix, $\rho(\mathbf{A}) = \max_\ell |\lambda_\ell|$. For non-negative matrices, the spectral radius equals the largest real eigenvalue. The basis for the NGM definition of $\mathcal{R}_0$.

Stage reproductive number ($\mathcal{R}_0^{\ell \to j}$). In a multistage transmission model, the expected number of secondary infections in stage j produced by a single infectious individual in stage ℓ when the stage-j population is fully susceptible. For bipartite sexual and vector-borne models, the stage reproductive numbers correspond to the two transmission directions ($M \to F$, $F \to M$, or $h \to v$, $v \to h$). See Chapter 11, Section 11.6.

Susceptible. An individual not currently infected and not immune. Denoted s in the compartmental framework. Susceptible individuals can become infected upon contact with infectious individuals.

Threshold, invasion. The condition $\mathcal{R}_0 > 1$ for sustained transmission. Below the threshold, the outbreak dies out; above, it grows. Proved for the SIR model in Chapter 5 and extended to structured populations via the next-generation matrix in Section 5.3.

Tornado plot. A horizontal-bar visualization of sensitivity indices, ordered by magnitude with the largest at the top. Standard tool for displaying local sensitivity results. See Chapter 9, Section 9.5.

Transmission dose. The per-contact quantity of pathogen transfer that sets the probability a susceptible–infectious contact results in infection, linking contact-level biology to the population-level transmission factor β. See Chapter 2.

Treatment as prevention. The strategy of treating infectious individuals to reduce their transmissibility, thereby reducing population-level $\mathcal{R}_0$. Most prominent for HIV with antiretroviral therapy. See Chapter 19.

Vaccination. Pre-exposure immunization that converts susceptible individuals to recovered (immune) individuals. Reduces effective $\mathcal{R}_0$ proportionally to the immunized fraction; achieves elimination at the herd-immunity threshold.

Variance decomposition. Expressing the variance of a model output as a sum of contributions from individual inputs and their interactions—the basis of variance-based global sensitivity indices such as Sobol' and eFAST. See Chapter 9.

Vector control. The collection of interventions targeting the vector population, including larvicides, adulticides, environmental management of breeding sites, and personal protection (bed nets, repellents, screens). The principal intervention category for vector-borne diseases (Chapter 13, Section 13.5).

Vectorial capacity. For vector-borne diseases, the daily rate of secondary host infections that a single host case would generate, $V = a^2 b_h b_v m e^{-\mu_v \tau_v^E} / \mu_v$. Related to $\mathcal{R}_0$ via $\mathcal{R}_0^2 = V/(\gamma_R + \nu_m)$. See Chapter 13, Section 13.4.

Vertical transmission. Transmission from parent to offspring, typically from mother to child. See MTCT. Modeled in Chapter 14.

Viewpoint (infected and susceptible). The book's organizing contrast between describing transmission from the susceptible side—the force of infection λ acting on s, which requires the susceptible count—and from the infected side—the force from infection α acting on i, which requires only infected quantities. The infected viewpoint yields estimators that avoid the hard-to-define susceptible denominator. Introduced in Chapters 2 and 3.

von Foerster equation. The first-order PDE governing age-structured population dynamics: $\partial_t n + \partial_a n = -\mu(a)n$. Underlies all age-structured epidemic models. See Chapter 15, Section 15.2.

Well-posed and ill-posed problems. A problem is *well-posed* (Hadamard) if a solution exists, is unique, and depends continuously on the data; it is *ill-posed* if any of these fails. Inverse problems such as deconvolution and parameter fitting are often ill-posed, so small data errors can produce large solution errors unless the problem is regularized. See Chapter 7.

C.2 Parameter Symbols

α**.** Force from infection, the per-capita rate at which an infectious individual generates new infections. $\alpha = c_I \beta P_S$ for the SIR model.

α_h, α_v**.** Host- and vector-side forces from infection in vector-borne models. They are the coupling between the host and vector compartmental dynamics, and the biting rate a enters both because biting is the rate-limiting step in each direction (Chapter 13).

α_M, α_F**.** Forces from infection generated by infectious males and females in heterosexual transmission models, $\alpha_M = c_{MF}\beta_{MF}\, s_F/n_F$ and $\alpha_F = c_{FM}\beta_{FM}\, s_M/n_M$ (Chapters 12 and 19).

a**.** Vector biting rate, in bites per vector per day. Used in vector-borne models (Chapter 13).

β**.** Per-contact transmission probability. The probability that a contact between an infectious and a susceptible individual transmits the pathogen. Dimensionless.

β_A, β_I**.** Per-contact transmission probabilities for asymptomatic and symptomatic infectious individuals. Paired with the contact rates c_A, c_I, they combine into a total force from infection when infectiousness differs by symptom status (Chapters 10 and 17).

β_{MF}, β_{FM}**.** Per-contact transmission probabilities for the two directions of heterosexual transmission, male-to-female and female-to-male. They are generally distinct; for HIV they differ roughly twofold because of anatomical asymmetry. Paired with the contact rates c_{MF}, c_{FM} (Chapters 12 and 19).

b_h, b_v**.** Per-bite transmission probabilities in vector-borne models: b_h is vector-to-host, b_v is host-to-vector.

c_S, c_I, c_R**.** Contact rates of susceptible, infectious, and recovered individuals, in contacts per individual per day. May differ across compartments to model behavioral changes (e.g., self-isolation when symptomatic).

c_{MF}, c_{FM}. Cross-sex contact rates in heterosexual STI models (Chapter 12).

C_{total}. Total contact rate of the population, $C_{\text{total}} = c_S s + c_I i + c_R r$, in contacts per unit time. It is the denominator of the contact-weighted fractions P_S, P_I, P_R (Chapter 4).

χ_I. Proportional reduction in contact rate during symptomatic periods, applied as $c \to (1 - \chi_I)\, c$. Since $\mathcal{R}_0 \to (1 - \chi_I)\, \mathcal{R}_0$, the reproductive number has unit sensitivity to χ_I (Chapter 12).

D. The dimensionless ratio $D = \nu_m/\gamma_R$ comparing demographic and epidemiological timescales. Used in fitting (Chapter 7).

Δt. A short time interval, with units of time: the discretization step in numerical integration, the interval in the rate-to-probability correspondence (the probability of one event in Δt is $r\,\Delta t + O(\Delta t^2)$), and the observation interval in the Poisson observation model (Chapters 8 and 15).

ϵ. A small dimensionless quantity: a relative-error tolerance when bounding an approximation, or a small multiplicative perturbation used to inject noise in uncertainty exercises. Its precise meaning is local to the passage where it appears (Chapters 7 and 9).

η_S, η_I, η_R. Relative biting-exposure weights for susceptible, infectious, and recovered hosts, normalized so that $\eta_S = 1$. They make the host classes differentially available to vectors (Chapter 13).

F_0. Initial susceptible fraction, $F_0 = S(0)/N^{\text{c}}$. A dimensionless parameter for fitting.

i_0. Initial infectious fraction, $i_0 = I(0)/N^{\text{c}}$. A dimensionless parameter for fitting (the initial index-case fraction; profiled in Chapter 8).

$\mathcal{F}, \mathcal{V}$. The new-infection vector field and the transition vector field in the NGM construction. These are vector-valued functions of the state $\mathbf{x}$: $\mathcal{F}(\mathbf{x})$ gives the rate of new infections, $\mathcal{V}(\mathbf{x})$ the rate of all other transitions. The ODE for the infected compartments is $\dot{\mathbf{x}} = \mathcal{F}(\mathbf{x}) - \mathcal{V}(\mathbf{x})$. Their Jacobians at the disease-free equilibrium are the matrices $\boldsymbol{F}, \boldsymbol{V}$ (below).

$\boldsymbol{F}, \boldsymbol{V}$. The Jacobians of the vector fields $\mathcal{F}, \mathcal{V}$ evaluated at the disease-free equilibrium. These are the matrices that appear in the NGM construction: $\boldsymbol{K} = \boldsymbol{F}\boldsymbol{V}^{-1}$. The basic reproductive number is $\mathcal{R}_0 = \rho(\boldsymbol{K}) = \rho(\boldsymbol{F}\boldsymbol{V}^{-1})$.

γ_R. Recovery rate, with units day^{-1}. Inverse of mean infectious duration: $\tau_R = 1/\gamma_R$.

γ_E. Rate of progression from exposed to infectious in SEIR models. $\tau_E = 1/\gamma_E$.

γ_Q. Rate of identification and quarantine of infectious individuals in SEIQR models.

γ_A. Recovery rate of asymptomatic infectious individuals, typically differing from the symptomatic recovery rate γ_R. The corresponding residence time is $\tau_A = 1/\gamma_A$ (Chapter 10).

γ_v^E. Rate at which exposed vectors become infectious, $\gamma_v^E = 1/\tau_v^E$: the reciprocal of the extrinsic incubation period (Chapter 13).

i^*. Endemic prevalence: the equilibrium value of i when $\mathcal{R}_0 > 1$.

j. *Model* incidence rate, $j = \alpha i$ = rate of new infections per unit time, *normalized* (a fraction of N^{c} per unit time); its count form is $J = j\,N^{\mathrm{c}}$ (units $\mathtt{P}\,\mathtt{T}^{-1}$). A flow, dimensionally distinct from the compartment sizes s, i, r, which are stocks. The book distinguishes the normalized model incidence j from the count-valued *true* (latent) incidence J_{true} it represents (through $j\,N^{\mathrm{c}}$), the *observed* reported incidence $J_{\mathrm{obs}}(t_k)$ (a noisy, ascertainment-scaled sample of J_{true}), and the *estimated* incidence $J_{\mathrm{est}}(t_k)$ obtained by preprocessing J_{obs} (the fitting target); see the Notation and Conventions summary.

$\boldsymbol{K}$. The next-generation matrix, $\boldsymbol{K} = \boldsymbol{F}\boldsymbol{V}^{-1}$.

$K(\tau)$. The kernel of the Kermack-McKendrick renewal equation, $K(\tau) = A(\tau)F(\tau)$.

λ. Force of infection (susceptible viewpoint), $\lambda = c_S \beta P_I$.

Λ. Recruitment rate into the population in models with demography, as in $\dot{N}^{\mathrm{c}}(t) = \Lambda(t) - \nu_m(t)\,N^{\mathrm{c}}(t)$ (Chapter 4). In the sensitivity analysis, $\Lambda_{jk} = \Sigma_{jk}/(p_j^* p_k^*)$ denotes the parameter relative-variance matrix (Chapter 9).

m. Vector-to-host ratio, $m = n_v/n_h$. Used in vector-borne models.

μ_d. Disease-induced mortality rate, used in models with explicit disease deaths.

ν_m. Migration rate (background outflow rate from the modeled population), with units day^{-1}. $\tau_m = 1/\nu_m$. See Appendix F (Notation and Conventions) for the Greek-letter rate convention placing ν_m in the demographic family.

μ_v. Vector mortality rate.

$\mu(a)$. Age-specific per-capita mortality rate in age-structured (PDE) models, with units day^{-1} (Chapter 15).

N^{c}. Carrying capacity of the population: the reference total population at which births balance deaths in the absence of disease mortality; the normalization scale: every fraction in the book is measured in units of N^{c}.

n^*. Endemic-equilibrium total population as a fraction of the carrying capacity, $n^* = s^* + i^* + r^*$. Equal to 1 in the basic model; below one when disease mortality is present.

S, I, R, N. The compartment and total variables as counts of individuals (uppercase), used where a specific population size matters (fitting to data, case studies, the derivations before the normalization, and the multi-population models); recovered from the fractions through the carrying capacity, as in $I = i\,N^{\mathrm{c}}$.

ν_b. Per-capita birth rate. Used in vertical-transmission models (Chapter 14).

ω. Rate of immunity waning in SIRS models. $\tau_{\mathrm{imm}} = 1/\omega$.

P_S, P_I, P_R. Contact-weighted fractions: $P_S = c_S s/(c_S s + c_I i + c_R r)$, etc. Sum to 1.

p. Symptomatic-fraction parameter in SIAR models: probability that an infectious individual becomes symptomatic.

p_{EIP}. Probability of vector survival through the extrinsic incubation period, $p_{\mathrm{EIP}} = e^{-\mu_v \tau_v^E}$.

p_{MTCT}. Per-pregnancy mother-to-child transmission probability.

Φ_M, Φ_F. Fluxes of new male and new female infections in heterosexual models, $\Phi_M = \alpha_F i_F$ and $\Phi_F = \alpha_M i_M$. The cross-coupling—male infections driven by infectious females, and conversely—is the hallmark of bipartite dynamics (Chapter 12).

$\mathcal{R}_0$. Basic reproductive number. The threshold parameter for invasion; equivalently $\mathcal{R}_e(0)$ in a fully naive population.

$\mathcal{R}_e$. Effective reproductive number at a given time. The operationally-primary reproductive-number quantity in this book; $\mathcal{R}_e(t) = \mathcal{R}_0 P_S(t)$ for homogeneous mixing.

$\mathcal{R}_c$. Controlled reproductive number: $\mathcal{R}_e$ under a specified intervention regime before substantial depletion of the susceptible pool.

ρ. Per-unit-time transmission rate during a single contact, used in the derivation of the per-contact transmission probability $\beta = 1 - e^{-\rho\tau_c}$. See Chapter 1, Section 1.4.1.

s, i, r. Compartments as fractions of the carrying capacity: susceptible, infectious, recovered (lowercase); the uppercase forms are counts (Appendix F).

Υ_p^q. Normalized sensitivity index of output q to parameter p.

σ. Generic exponential rate. In particular, σ_2 is the largest eigenvalue of the Jacobian at the disease-free equilibrium, governing early-outbreak growth.

σ_2. Early exponential-growth rate of an outbreak, $\sigma_2 = (\gamma_R + \nu_m)(\mathcal{R}_0 - 1)$: the positive eigenvalue of the linearization about the disease-free equilibrium, with units day^{-1} (Chapter 5).

σ_v, σ_h. In the Chitnis–Hyman–Cushing malaria model, σ_v is the number of blood meals a mosquito seeks per day and σ_h the number of bites a host tolerates per day; together they set the saturating biting rate (Chapter 13).

τ_c. Contact duration, used in the contact-duration treatment of β.

τ_E. Mean latent period, $\tau_E = 1/\gamma_E$.

τ_g, T_g. Mean generation time. See Chapter 15.

τ_R. Mean infectious duration, $\tau_R = 1/\gamma_R$.

τ_R^*. Effective infectious time, $\tau_R^* = 1/(\gamma_R + \nu_m)$. Includes both recovery and migration.

τ_m. Mean residence time in the modeled population, $\tau_m = 1/\nu_m$.

τ_v^E. Mean extrinsic incubation period for vector-borne pathogens.

τ_v^m. Mean vector lifespan, $\tau_v^m = 1/\mu_v$.

θ. The vector of parameters to be estimated, $\theta = (\theta_1, \ldots, \theta_p)$. The model prediction is written $j(t;\theta)$ when the parameter dependence matters (Chapters 7 and 9).

C.3 Closing Note

This glossary reflects the specific usage adopted in this book. Some terms have multiple definitions in the broader epidemiological literature; we have selected the definition most consistent with our compartmental and bipartite framework. For comprehensive epidemiological glossaries, we refer the reader to the National Collaborating Centre for Infectious Diseases (NCCID) glossary and to standard textbooks such as Rothman et al. [2008].

Appendix D

Verification Checklists for AI-Assisted Epidemic Modeling

A calculation without a check is only a rumor.

This appendix provides three short verification checklists to use when reviewing AI-assisted mathematical, fitting, or sensitivity-analysis work, plus a disclosure-template specification for assignments that involve AI assistance. The checklists are calibrated to the failure modes that arise specifically when AI tools are applied to mathematical epidemiology. For a fuller treatment of AI-aware learning (a comprehensive prompt-pattern library, a failure-mode catalog with worked examples across multiple disciplines, full worked disclosures, and study-routine guidance), see the companion Student Guide [Hyman, 2026c]. Instructors structuring AI-aware course assessments should additionally consult the companion Instructor Guide [Hyman, 2026b], which covers assignment design, the three assessment models (Oral-Defense, Evidence-of-Thinking, Hybrid), rubrics, and TA calibration.

The operational stance behind everything in this appendix is captured in a short mantra: *verify, verify, verify, then trust.* AI tools can produce derivations, code, fits, and analyses that look right on first read. The checklists in this appendix are the discipline that turns "looks right" into "is right." Apply them every time, not occasionally. The same posture applies to every *Verify the AI* box throughout the chapters of this book: each box is, in effect, an instance of the mantra at work, a place where the verification step is performed before the AI's output is accepted into the analysis.

The checklists share a common structure: each lists six to seven items roughly ordered by the frequency with which the item catches a real error. A reader applying the checklist to AI-generated work should expect the early items to fire often and the later items to fire less often but to indicate more serious problems when they do.

Each item names, at its right margin, which of the book's six verification categories it serves. Whatever tool produced the work, the reader verifies equations, numerical values, citations, model assumptions, sensitivity calculations, and interpretations (Chapter 20). Citations have no item of their own: confirm that every cited source exists and reports what is attributed to it.

D.1 Verification checklist: derivations

Use this checklist when reviewing any AI-assisted mathematical derivation.

1. *Every assumption used in the derivation appears in the assumption list.* AI-generated derivations frequently invoke unstated assumptions: homogeneous mixing, mass-action transmission, instantaneous recovery, exponential lifetime distributions. List the assumptions explicitly before reading the derivation, then check that every step uses only listed assumptions. *model assumptions*

2. *Every equation has units on both sides and the units match.* Write the units of every parameter and variable above its first appearance, then verify each equation. Unit mismatch is the single most common AI-generated error. *equations*

3. *Every variable used in step k has been defined or computed in steps 1 through $k-1$.* A linear scan with a definition tracker catches mid-derivation symbol introductions and rescinded definitions. *equations*

4. *The derivation reduces to a known special case under appropriate parameter limits.* An SIR derivation with distinct contact rates should reduce to the classical SIR formulas in the equal-contact-rate limit $c_S = c_I = c_R$. If the derivation fails the reduction, the general expression is wrong. *equations*

5. *Any claimed manipulation that is not algebraically obvious is verified in at least one numerical instance.* A single numerical substitution at baseline parameter values typically catches a wrong manipulation. *numerical values*

6. *Conclusions invoking a theorem state which theorem and verify the theorem's hypotheses for the specific setting.* "By the standard result . . . " is a flag; the standard result must be named and its hypotheses checked. *model assumptions*

7. *The notation is consistent with this book's conventions throughout, or any deviation is explicitly flagged and translated.* *equations*

A derivation that fails any of these checks is, at minimum, incompletely justified; the reader should treat it as a draft to be repaired rather than as a result. A derivation that passes all seven is not necessarily correct, but it is at least internally consistent and can be reasoned about further. The companion repository's failure-mode catalog provides epidemiology-specific worked examples of derivations that fail each of the seven checks; the broader subject-general treatment is in Hyman [2026c], in the section "Dangerous uses of AI" and "How to verify AI-assisted work."

D.2 Verification checklist: fitting

Use this checklist when reviewing AI-assisted parameter-fitting work. Fitting is the workflow where AI assistance is most often used in practice and where AI-generated errors propagate most rapidly into reported results.

1. *The objective function is the one the user intended.* Least-squares versus maximum-likelihood, weighted versus unweighted, log-scale versus linear-scale residuals: each pair of choices implies a different statistical model. AI-generated fitting code frequently picks one default without making the choice explicit. *model assumptions*

2. *The data being fit are the data the user intended.* Incidence versus prevalence versus cumulative case counts; same time alignment across sources; consistent units; same population denominator across the time series. *numerical values*

3. *Identifiability has been considered.* Parameters not identifiable from the available data should be either fixed externally (with the source documented) or explicitly flagged as non-identifiable. *model assumptions*

4. *The fitted parameter values lie within physically plausible ranges and agree with independently constrained values where applicable.* *numerical values*

5. *The fit residuals are inspected, not only the summary statistic.* Random-looking residuals are evidence that the model captures the systematic structure; visible trends indicate the model misses systematic structure that should be addressed before drawing inference. *interpretations*

6. *Sensitivity to the initial conditions and to the assumed susceptible population $s(0)$ is checked,* especially when the susceptible-viewpoint estimator $\hat{\lambda}$ is used. *sensitivity calculations*

7. *The code reproducing the fit is available, with random seeds set, solver tolerances explicit, and library versions documented.* *numerical values*

The infected-viewpoint estimator $\hat{\alpha} = j/i$ is structurally immune to $s(0)$ uncertainty (Chapter 7); the susceptible-viewpoint estimator $\hat{\lambda} = j/s$ is not. Item 6 is calibrated to this distinction. The companion repository's failure-mode catalog includes the COVID-19 case where an AI-generated workflow silently used $\hat{\lambda}$ with $s(0) = n$ and produced an $\mathcal{R}_0$ value depending on the assumed geographic scope.

D.3 Verification checklist: sensitivity analysis

Use this checklist when reviewing AI-assisted sensitivity-analysis work. The most consequential failure mode in this domain is normalized-vs-unnormalized index confusion, which produces inverted intervention rankings.

1. *The sensitivity quantity computed is named explicitly* and matches what the analysis intended to report. Raw partial $\partial y/\partial x$, normalized index $\Upsilon_x^y = (\partial y/\partial x)(x/y)$, partial rank correlation coefficient (PRCC), Sobol' index, and elasticity all answer different questions. *sensitivity calculations*

2. *The sampling distribution for global methods is specified for each parameter,* with units and ranges. Different priors produce different sensitivity rankings; the choice should be justified rather than left implicit. *sensitivity calculations*

3. *The reference value at which local sensitivities are evaluated is named* (typically the baseline parameter set). Local indices computed at non-baseline points can disagree substantially in nonlinear settings. *sensitivity calculations*

4. *The output quantity is defined precisely* and computable from the model state. "Peak prevalence" is ambiguous between $\max_t i(t)$ and $\max_t(i(t) + A(t))$ in models with asymptomatic compartments. *sensitivity calculations*

5. *Sign conventions are consistent.* Reversed sign conventions invert intervention-ranking conclusions. *sensitivity calculations*

6. *Any reduction of the sensitivity to a single "most important parameter" is checked against the invasion–burden partition* of Chapter 5, Section 5.7. The invasion-controlling parameters and the burden-controlling parameters are nearly disjoint; a ranking that mixes the two without acknowledging the partition has missed the structural point. *interpretations*

7. *Monte Carlo or LHS sample size is large enough that the reported indices have stabilized;* convergence is documented. A PRCC table reported from $M = 100$ samples is not a converged result. *numerical values*

The companion repository's failure-mode catalog includes worked examples of each failure mode, including the case where an AI-generated analysis compares the unnormalized partials (at the book's baseline, $\partial\mathcal{R}_0/\partial\beta \approx 48$, large because β is small, against $\partial\mathcal{R}_0/\partial c_I \approx 0.48$, smaller because c_I is large) and concludes that β is more influential. The conclusion is wrong because the units differ; the normalized indices give $\Upsilon_{\beta}^{\mathcal{R}_0} = \Upsilon_{c_I}^{\mathcal{R}_0} = 1$.

D.4 Disclosure-template specification

For each substantial use of an AI system in submitted work, state: (1) the question or task you posed to the AI; (2) the response you received, in summary; (3) which parts of the response you accepted, in your own words; (4) which parts you rejected, with a brief reason; and (5) which parts you verified independently and how. Disclosure is required for any AI-assisted derivation, code, fit, or numerical result; it is not required for routine notation lookup or syntax help. Disclosure templates and worked examples are in Hyman [2026c], in the section "Disclosure" and the templates in Appendices A–C. Instructors structuring AI-aware course assessments will find assignment-design frameworks, rubrics, and the three assessment models (Oral-Defense, Evidence-of-Thinking, Hybrid) in Hyman [2026b]; departments and curriculum committees adopting institution-wide policies will additionally find course-classification worksheets, syllabus templates, and an academic-integrity workflow in Hyman [2026a].

Appendix E

The Poisson Process and Thinning

Random points, sub-selected at random, are random in the same way.

The stochastic derivations of the force of infection (Chapter 2, Section 2.2) and the force from infection (Chapter 3, Section 3.2) rest on two facts about the *Poisson process*: what it is, and what happens when its events are randomly sub-selected (*thinned*). This appendix states both at an intuitive level, with no measure theory, so that those derivations can be read without a prior probability course. Readers who have seen Poisson processes may skip to Section E.3.

E.1 What a Poisson Process Is

A *Poisson process with rate* r is a model for events that occur one at a time, at random instants, at a constant average rate r (events per unit time), with the timing of each event carrying no memory of the others. Contacts made by an individual over a day, calls arriving at a switchboard, and clicks of a Geiger counter are the standard examples.

Three properties characterize it, and any one implies the others:

- *Constant rate.* In a short interval of length Δt, the probability of exactly one event is approximately $r\,\Delta t$, and the probability of two or more is negligible (of order $(\Delta t)^2$).

- *Independent increments.* The number of events in non-overlapping time intervals are independent. What happened in the last hour tells you nothing about the next.

- *Counts and gaps.* The number of events in an interval of length t follows a Poisson distribution with mean rt, and the waiting time between consecutive events follows an exponential distribution with mean $1/r$.

The last property is the one used in the main text: if a susceptible individual acquires infectious contacts as a Poisson process of rate λ, then the time until the first such contact is exponential with mean $1/\lambda$, and the per-unit-time hazard of infection is exactly λ. This is what licenses writing the deterministic loss term $-\lambda s$.

E.2 The Thinning Property

Thinning is the operation of keeping each event of a Poisson process independently with some fixed probability p and discarding it with probability $1 - p$. For example, a person's contacts form a process; keeping only those that are with an infectious individual is a thinning.

> **Thinning property.** If events arrive as a Poisson process of rate r, and each event is independently retained with probability p, then the retained events form a Poisson process of rate rp (and the discarded events form an independent Poisson process of rate $r(1-p)$).

The result is intuitive once stated. In a short interval Δt the chance of an event is $r\,\Delta t$; the chance that the event occurs *and* is retained is $r\,\Delta t \cdot p = (rp)\,\Delta t$. The retained events therefore arrive at constant rate rp, and because the original events were independent across disjoint intervals and each retention decision is made independently, the retained events inherit the independent-increments property. By the characterization of Section E.1, that makes them a Poisson process of rate rp. Thinning a Poisson process by a constant probability simply rescales the rate; it does not change the kind of process.

Thinning composes: applying two independent retention steps with probabilities p_1 and p_2 leaves a Poisson process of rate $r\,p_1 p_2$. The order does not matter, and the combined retention probability is the product. This composition is exactly the move the force-of-infection derivation makes twice.

E.3 Application to the Force of Infection

Fix a single susceptible individual. The individual makes contacts as a Poisson process of rate c_S (contacts per unit time). Two independent retention steps reduce this contact process to the process of *infection-producing* events:

1. Keep a contact only if it is with an infectious individual. In a well-mixed population this happens with probability P_I (the contact-weighted infectious fraction). After this thinning, infectious contacts arrive as a Poisson process of rate $c_S P_I$.

2. Keep an infectious contact only if transmission actually occurs, which happens with the per-contact transmission probability β. After this second thinning, infections arrive as a Poisson process of rate $c_S P_I \beta$.

By the composition rule, the surviving events (actual infections of this susceptible) form a Poisson process of rate

$$\lambda \;=\; c_S\,\beta\,P_I,$$

which is precisely the force of infection derived in Section 2.2. The same two-step thinning, applied to the contacts of a single *infectious* individual (retention probabilities P_S and β), yields the force from infection $\alpha = c_I \beta P_S$ of Section 3.2. In both cases the conceptually non-trivial step (reducing the raw contact process to the effective infection process) is just thinning, and the rate of the thinned process is the product of the original rate and the retention probabilities.

> *The contact process is Poisson; infection is what survives two rounds of thinning. The force of infection is the rate of the survivor.*

Appendix F

Notation and Conventions

Notation, notation, notation.

This section summarizes the notational conventions used throughout the book. A full glossary with chapter references appears in Appendix C.

A note on notation

The book's notational conventions are summarized on the pages following the table of contents and developed in full in Appendix C. Four conventions are worth noting at the outset. Subscripts on contact rates, contact probabilities, and compartment-specific parameters are uppercase ($c_S, c_I, c_R, P_S, P_I, P_R$) throughout, even when the classical literature uses lowercase. The reproductive numbers are denoted with a calligraphic letter: $\mathcal{R}_0$ for the basic reproductive number, $\mathcal{R}_e(t)$ for the time-varying effective reproductive number, and $\mathcal{R}_c$ for the controlled reproductive number that describes a state under specified intervention; the calligraphic letter is reserved for reproductive numbers, so r_0 denotes the recovered fraction at time zero, not a reproductive number. The population variables distinguish fractions from counts by case: lowercase s, i, r and the running total n are fractions of the carrying capacity N^{c}, the endemic-equilibrium total is written n^*, and uppercase S, I, R, N are counts of individuals, recovered through the carrying capacity, as in $I = i\,N^{\mathrm{c}}$. Compartments described by the language "infected" are distinguished from compartments described by the language "infectious" wherever the distinction matters: an individual in the E (exposed) compartment of an SEIR model is infected but not yet infectious, and an individual in the i compartment is both. The glossary entries for *Infectious*, *Recovered*, and *Exposed* formalize the distinction.

Typographic conventions

- **Compartment names** are uppercase roman when they label a compartment or a model (the S compartment, an SEIR model); as state variables they are lowercase fractions ($s(t) = 0.3$) or uppercase counts ($S(t)$).
- **Subscripts** on contact rates, contact probabilities, and compartment-specific rates are uppercase: $c_S, c_I, c_R, P_S, P_I, P_R$.
- **Reproductive numbers** are denoted with the calligraphic letter $\mathcal{R}$. Without the calligraphy, r and R are the recovered fraction and the recovered count.

- **Time dependence** is suppressed when clear from context. We write s rather than $s(t)$ except where the time argument is essential to the statement. The same applies to quantities built from the state. The contact-weighted fractions P_S, P_I, P_R and the forces α and λ are nonlinear functions of the state (s, i, r), while the contact-weighted total C_{total} is linear in it; all of them therefore vary in time. We write them without arguments unless a particular dependence is being emphasized, in which case the relevant argument is restored, as in $P_S(t)$ or $\lambda(t, s, i, r)$. Values at discrete observation times carry the time argument in parentheses, as in $j(t_k)$ at sample times t_k; a discrete index appears only in a summation or within the sample time t_k itself, never as a bare subscript such as j_k or j_t.
- **Vectors and matrices.** Column vectors are bold roman: $\mathbf{x}$ for the state vector, $\mathbf{p}$ for a parameter vector. Matrices (including those that get analyzed for eigenvalues, spectral radii, or norms) are also bold roman: $\boldsymbol{K}$ for the next-generation matrix, $\boldsymbol{J}$ for the Jacobian, $\boldsymbol{F}$ and $\boldsymbol{V}$ for the next-generation Jacobians at the DFE. The spectral radius is $\rho(\boldsymbol{K})$. The calligraphic forms $\mathcal{F}$ and $\mathcal{V}$ are reserved for the next-generation *vector fields*, the functions of the state that appear in the linearization decomposition $\dot{\mathbf{x}} = \mathcal{F}(\mathbf{x}) - \mathcal{V}(\mathbf{x})$. The Jacobians $\boldsymbol{F}, \boldsymbol{V}$ are obtained from these vector fields by differentiation at the disease-free equilibrium; the NGM is $\boldsymbol{K} = \boldsymbol{F}\,\boldsymbol{V}^{-1}$. The two-tier convention (calligraphic for vector fields, bold roman for matrices) matches the standard usage in the next-generation-matrix literature [van den Driessche and Watmough, 2002] and keeps the distinction between an object that takes the state as argument and an object that has been linearized at a fixed point.

Core state variables (stocks)

s, i, r	fraction	susceptible, infectious (i.e., infected and infectious), and recovered (i.e., recovered and resistant) compartments as fractions of the carrying capacity N^{c} (Section 4.5.3)
n	fraction	current total: $n = s + i + r$
N^{c}	individuals	carrying capacity: the reference population scale, at which births balance deaths in the absence of disease mortality. Every fraction in the book is measured against it
n^*	fraction	endemic-equilibrium total, $n^* = s^* + i^* + r^*$. Equal to 1 in the basic model and below one when disease mortality is present
S, I, R, N	individuals	the same quantities as counts, used where a specific population size matters; recovered from the fractions through the carrying capacity, as in $I = i\,N^{\text{c}}$

s, i, r, n, N^{c}, and n^* are *stocks*: extensive state variables. The case of the letter says which kind of number it is. Lowercase letters are fractions of N^{c}; uppercase letters S, I, R, N are counts of individuals. The carrying capacity is itself a count, the scale the fractions are measured against, so the bridge $I = i\,N^{\text{c}}$ says that a fraction times a count gives a count. Quantities *derived* from the state inherit the distinction. The equilibrium values s^*, i^*, r^*, n^* are fractions, because the equilibria are derived in the fraction variables (Chapter 5), and they stay lowercase wherever they are quoted. Initial conditions appear in both forms according to what is meant: $S(0)$ when a specific susceptible count is in play, as in the fitting workflow of Chapter 7, and $s(0)$ for the fraction. Ratios built from a count and a population scale, such as $F_0 = S(0)/N^{\text{c}}$, are dimensionless. A quantity built from a rate of change of the state (above all the incidence defined below) is a *flow*; it carries the same

distinction, so the incidence $j(t)$ (a fraction of N^{c} per unit time) has the count form $J(t) = j(t)\,N^{\mathrm{c}}$ in individuals per unit time. The per-capita transition rates (α, λ, γ_x, ν_x) carry dimensions of inverse time. These three dimensional classes (stock, flow, and per-capita rate) are never added to or equated with one another, and a fraction is never equated with a count without the carrying capacity as the bridge. The book applies that check throughout.

Where the book uses counts, and the letters with two roles

- **Counts before the normalization.** Chapters 1 to 3 and Chapter 4 up to Section 4.5.3 derive the model in counts, so their state letters are uppercase. From that section on the book works in fractions.
- **Counts in the multi-population models.** Chapters 11, 12, 13, 18, and 19 also work in counts. Every equation there carries its own group capacity N_k^{c}, and the group fractions appear as ratios such as S_k/N_k^{c}.
- **Densities.** In the structured models of Chapter 15, a lowercase letter with a structuring argument is a density, not a fraction: $s(t, a)$ counts susceptible individuals per unit age, and integrating it over age returns the count $S(t)$.
- **The time level n.** In the numerical schemes of Chapter 15, n also indexes time levels. There it appears only in subscripts and superscripts (t_n, u_j^n), never in base position, where n is the total fraction.
- **The letter j.** In base position j is the incidence fraction. As an index, j appears only in subscripts and group labels.
- **The exponential rate σ.** The rate in $e^{\sigma t}$ (a growth or decay rate, an eigenvalue of a linearization, or the transform variable whose roots are those rates) is written σ throughout. A standard deviation and a biting rate keep their own σ.
- **The stage count L.** In the cycle identity of Chapter 11, L is the number of stages in a transmission cycle.

Incidence (a flow)

$j(t)$	fraction / time	*model* incidence: the normalized population-level rate of new infections predicted by the model, $j = \alpha i$. Its count form is $J(t) = j(t)\,N^{\mathrm{c}}$. A *flow*, not a compartment: it has the dimensions of a rate, unlike the stocks s, i, r.
$J_{\mathrm{true}}(t)$	individuals / time	the *true* (latent) infection incidence, in counts; the estimand that the model's $J = j\,N^{\mathrm{c}}$ represents.
$J_{\mathrm{obs}}(t_k)$	individuals / time	the *observed* reported incidence (reported cases): a noisy, ascertainment-scaled, sampled observation of J_{true}.
$J_{\mathrm{est}}(t_k)$	individuals / time	the *estimated* incidence after preprocessing J_{obs} (removing day-of-week, reporting-delay, and backfill artifacts); the target the model is fit to (as $j\,N^{\mathrm{c}}$), with $J_{\mathrm{est}} \approx J_{\mathrm{true}}$.

The observed incidence J_{obs} is the true incidence scaled by an *ascertainment fraction* $\rho \in (0, 1]$ (the reported proportion of true infections) and blurred by reporting delay and observation noise; the

same obs/true/est convention applies to prevalence ($I_{\text{obs}}, I_{\text{true}}, I_{\text{est}}$, where I_{est} is prevalence reconstructed from incidence). The model works in normalized units and meets the count data through the bridge $J = j\,N^{\text{c}}$; the infected-viewpoint estimator $\hat{\alpha} = J_{\text{obs}}/I_{\text{obs}}$ (Chapter 7) is a ratio of counts, in which both the shared ρ and the population size N^{c} cancel. The symbol p without a subscript denotes the symptomatic fraction in the SIAR models of Chapters 10 and 17; subscripted forms such as p_{EIP} and p_{MTCT} are unrelated probabilities named by their subscripts. (The symbol ρ is overloaded and the collisions are worth naming. Besides the ascertainment fraction used throughout Chapters 7 and 8, ρ denotes the spectral radius $\rho(\mathbf{K})$ of a next-generation matrix, a correlation coefficient in the identifiability diagnostics, and in some contact-structure discussions a contact-level transmission rate. The ascertainment and contact-rate uses are the two that can appear near one another, and both are scalars, so context alone is not always enough: in the fitting chapters ρ is the ascertainment fraction unless explicitly stated otherwise.)

Contact structure

c_S, c_I, c_R	contact rates for susceptible, infectious, recovered individuals
β	per-contact transmission probability
ρ, τ_c	contact-level transmission rate and contact duration
C_{total}	total contact rate: $C_{\text{total}} = c_S s + c_I i + c_R r$
P_S, P_I, P_R	contact-weighted fractions: $P_X = c_X X / C_{\text{total}}$

Forces of and from infection

λ	force of infection (susceptible viewpoint): $\lambda = c_S \beta P_I$
α	force from infection (infected viewpoint, primary in book): $\alpha = c_I \beta P_S$
$\hat{\lambda}$	susceptible-viewpoint estimator: $\hat{\lambda} = j/s$
$\hat{\alpha}$	infected-viewpoint estimator: $\hat{\alpha} = j/i$

The algebraic identity $\lambda s = \alpha i$ holds by construction; Chapter 3 Proposition 3.1 gives the formal proof.

Multi-population conventions: per-source α versus per-target α

In single-compartment models (Chapters 2 through 9), the force from infection α is the *per-source-infectious* rate: αi is the rate of new infections, so $\alpha = c_I \beta P_S$ is the rate at which one infectious individual transmits. The book takes this as the primary convention because the source-and-target collapse to a single compartment in homogeneous-mixing models, and the per-source rate inherits the structural-immunity property developed in Chapter 7.

In two-population models with a structural source–target distinction (the vector–host coupling of Chapter 13 and the bipartite male–female coupling of Chapter 19) α appears with subscripts (α_h, α_v in vector–host models; α_M^H, α_F^H and parallel low-activity subscripts in HIV models) and is used in the *per-target-susceptible* convention: $\alpha_h s_h$ is the rate of new host infections and $\alpha_v s_v$ is the rate of new vector infections, so $\alpha_h = a b_h i_v / n_h$ and $\alpha_v = a b_v i_h / n_h$ are per-target rates. The convention switch is intentional. It aligns with the vector-borne and bipartite-mixing literatures, which write the Ross–Macdonald and NGM equations from the target side because the source and target are physically distinct populations.

To bridge the two conventions, both Chapter 13 and Chapter 19 provide per-source estimators $\hat{\alpha}_h^{\text{inf}} = j_h/i_v$, $\hat{\alpha}_v^{\text{inf}} = j_v/i_h$ (and per-sex analogues in HIV) that recover the primary-convention force-from-infection rate from each direction. The population-level new-infection rates $j_h = \alpha_h s_h$ and $j_v = \alpha_v s_v$ are convention-independent and are consistent with the $j(t)$ defined in Chapter 3 equation (3.2).

A reader arriving at Chapter 13 or Chapter 19 from a non-sequential reading path should expect the subscripted α_h, α_v, α_M^H, α_F^H to denote per-target rates and refer back to this subsection. One chapter departs from this: the bipartite model of Chapter 12 writes α_M and α_F as *per-source* rates, paired with i_M and i_F rather than with a susceptible population. That chapter flags the difference where the forces are introduced, but the similarity of α_M to α_M^H is close enough to be worth naming here as well for the bridge identity between the per-target and per-source forms.

Time constants and rates

τ_R, $\gamma_R = 1/\tau_R$	infectious period and recovery rate
τ_m, $\nu_m = 1/\tau_m$	mean residence time and migration rate
$\tau_R^* = 1/(\gamma_R + \nu_m)$	adjusted infectious time

The book treats the time spent in a compartment (τ_x) as the primary parameter and the corresponding rate ($\gamma_x = 1/\tau_x$) as the derived quantity. This direction reflects a methodological priority: the time in a compartment is what a clinician or epidemiologist actually measures, while the rate is what the ODE consumes. Throughout the book, when both are reported, the time constant is the parameter being estimated and the rate is computed from it.

Model names: SIR and SIR$_S$

Model names are set upright (SIR, SIS, SEIR), because an italic *SIR* would read as the product of three counts. The book writes SIR for the three-compartment model and, by default, means the formulation it uses throughout: the model written from the infected viewpoint, driven by the force from infection $\alpha = c_I \beta P_S$. Where the other formulation is meant, the subscript names it: SIR$_S$ is the same three-compartment model written from the susceptible viewpoint, driven by the force of infection $\lambda = c_S \beta P_I$. The subscript records *which force the model is written with*, not a property of the disease; immunity is what the letter r already encodes, and it is what distinguishes SIR from SIS.

The two formulations are algebraically equivalent. They are linked by the detailed-balance identity $\lambda s = \alpha i$ (Chapter 2), and given the same parameters and initial conditions they produce the same trajectories to numerical round-off (Chapter 6).

Equivalence is not interchangeability, and the distinction is the reason the subscript exists. The two formulations have different dimensional structures, different identifiability properties from incidence data, and different roles in dynamic analysis. Chapter 7 shows the consequence: fitting early-outbreak data, the infected-viewpoint estimator $\hat{\alpha} = j/i$ is structurally immune to uncertainty in the assumed susceptible fraction, while the susceptible-viewpoint estimator $\hat{\lambda} = j/s^*$ inherits that uncertainty at unit sensitivity. Choosing a viewpoint is a modeling decision with measurable consequences, which is why this book names the choice whenever it is not the default.

Greek-letter rate convention

Three Greek letters carry distinct meaning for rates throughout the book. Distinguishing among them is itself part of the modeling discipline the book teaches: the same numerical value can mean physically different things depending on which Greek letter wears it.

Letter	Family	Examples and scope
γ	Disease-progression rates	γ_R (recovery), γ_E (latent-to-infectious progression). One rate per disease state per outflow.
ν	Demographic rates	ν_m (migration in/out of the modeled population), ν_b (per-capita birth rate).
μ	Mortality rates	μ_d (host disease-induced mortality, additional to background turnover, used in models that track disease deaths separately), μ_v (vector mortality, all causes; vectors do not have separate disease-induced mortality in the standard models), μ_h (natural host mortality, used when natural death is tracked separately from migration).

The convention preserves alignment with the established mathematical-epidemiology literature (Brauer and Castillo-Chavez, Keeling and Rohani, and the vector-borne tradition descended from Ross and Macdonald), where μ denotes mortality and ν or the pair b/d denotes natural birth and death.

The mortality family of subscripts. The natural-death contribution to host-compartment outflow is, in the book's standard simulation model, absorbed into ν_m as part of net demographic turnover. The symbol μ_d appears only when the model needs to track host disease-induced mortality *in addition to* background demographic turnover: pathogens with non-negligible case-fatality ratios (Ebola, severe COVID-19, untreated HIV). When μ_d appears, the infectious-compartment outflow has the form $(\gamma_R+\nu_m+\mu_d)i$, and the case-fatality ratio is $\mu_d/(\gamma_R+\nu_m+\mu_d)$. The symbol μ_v is reserved for vector mortality, which is total mortality from all causes (vectors do not have separately-tracked disease-induced mortality in the standard vector-borne models, so μ_v captures the full vector death rate that limits the vector lifespan). The symbol μ_h is reserved for natural host mortality when a model needs to track it separately from migration, for example, in long-horizon endemic-disease simulations where the natural lifespan of the host population becomes substantial relative to disease dynamics.

Single- vs. two-subscript form for γ

In Chapters 1–5 we use the simpler single-subscript form $\gamma_x = 1/\tau_x$ for the total exit rate from a compartment with a single successor. This is the standard SIR-style notation: γ_R is the rate at which infectious individuals recover, period.

Beginning with the multi-exit compartments of Chapter 17, where the infectious compartment can branch into recovery, hospitalization, or death, we generalize to the two-subscript form

$$\gamma_{j\to k} \;=\; \frac{p_{j\to k}}{\tau_j},$$

the branching rate from state j to state k, where $p_{j\to k}$ is the branching probability satisfying $\sum_k p_{j\to k} = 1$. The total exit rate from j is $\gamma_j = \sum_k \gamma_{j\to k} = 1/\tau_j$, so the single-subscript form is recovered as the special case $p_{j\to k} = 1$.

Independent vs. dependent parameters

The book maintains a sharp distinction between the parameters that are independently specified and those that are derived from them. The distinction matters for sensitivity analysis (we take partial derivatives with respect to the independent parameters), for fitting (we estimate the independent parameters from data), and for interpretation (the independent parameters are the quantities a clinician or epidemiologist actually measures).

Class	Symbols	Why this class
Independent	τ_x, $p_{j\to k}$	Each is a quantity directly estimated from data: τ_x is the mean residence time in compartment x (clinical or demographic measurement); $p_{j\to k}$ is the branching probability from compartment j to compartment k (proportion of disease outcomes). Sensitivity analyses are performed with respect to these.
Dependent	γ_x, $\gamma_{j\to k}$	Derived from the independent parameters: $\gamma_x = 1/\tau_x$ (single-successor case); $\gamma_{j\to k} = p_{j\to k}/\tau_j$ (multi-successor case). Computed for use inside the ODE solver and in the next-generation-matrix construction.

The classification reflects Pitfall: "The Independence of τ" from the Catalogue of Common Pitfalls: the time spent in a compartment is the independent parameter, and the rate is derived. The reciprocal relationship $\gamma_x = 1/\tau_x$ is exact only under the exponential-residence-time assumption that the deterministic ODE compartmental model implicitly makes; Chapter 10 relaxes the exponential assumption via Erlang chains, where the independent parameter remains τ_x but a shape parameter k is added.

Directional subscripts: the arrow convention

Several quantities in the book are intrinsically directional: they describe a flow, transition, or transmission *from* one state or population *to* another. For these, we adopt the explicit arrow notation $j \to k$ to remove any ambiguity about which subscript names the source and which names the target. The convention is uniform across the book.

Symbol	Meaning
$p_{j\to k}$	Branching probability that an individual leaving compartment j goes to compartment k.
$\gamma_{j\to k}$	Rate at which individuals flow from compartment j to compartment k; equals $p_{j\to k}/\tau_j$.
$\beta_{j\to k}$	Per-contact transmission probability from a j-typed (e.g., infectious) individual to a k-typed (e.g., susceptible) contact, when this directional structure is needed (heterogeneous-mixing settings, vector–host models).
$c_{j\to k}$	Contact rate at which individuals in group j encounter individuals in group k (heterogeneous-mixing settings).
$\mathcal{R}_0^{j\to k}$	Type-reproductive number: expected number of secondary infections in population k produced by one infected individual in population j, when both populations are fully susceptible (heterogeneous-mixing and bipartite settings). For example, $\mathcal{R}_0^{v\to h}$ and $\mathcal{R}_0^{h\to v}$ in vector–host models, with $\mathcal{R}_0 = \sqrt{\mathcal{R}_0^{v\to h}\cdot\mathcal{R}_0^{h\to v}}$ as the geometric mean (Ross–Macdonald factoring; Chapter 13).

The arrow form is preserved in body prose throughout the book. We do not drop the arrow after a first introduction. The 4-character savings of writing p_{jk} for $p_{j\to k}$ is small, and the cost of dropping the arrow is real: a reader who arrives at Chapter 13 via the Reading Guide's case-study path may not have seen the $\mathcal{R}_0^{v\to h}$ symbol elsewhere, and the arrow form is unambiguous on first encounter without requiring the reader to remember a convention from an earlier chapter.

Reproductive numbers

$\mathcal{R}_0$	basic reproductive number: $\mathcal{R}_0 = \rho(\mathbf{K})$, the NGM spectral radius; equivalently $\mathcal{R}_e(0)$ in a fully naive population
$\mathcal{R}_e(t)$	effective reproductive number at time t: the operationally-primary quantity in this book (often denoted R_t in the epidemiological and nowcasting literature); for a homogeneously-mixing population, $\mathcal{R}_e(t) = \mathcal{R}_0\, s(t)/n(t) = \mathcal{R}_0 P_S(t)$
$\mathcal{R}_c$	controlled reproductive number: $\mathcal{R}_e$ under a specified intervention regime (vaccination, contact reduction, treatment) before substantial depletion of the susceptible pool
$\mathcal{R}_0^{\ell\to j}$	stage reproductive number: secondary infections in stage j per infected in stage ℓ
$\mathcal{C}_0^{(\ell)}$	cyclic reproductive number based at stage ℓ: product of stage reproductive numbers around one cycle
$\mathcal{C}_e^{(\ell)}(t)$	effective cyclic reproductive number at time t

For simple cyclic multistage models with n stages, $\mathcal{R}_0 = (\mathcal{C}_0^{(\ell)})^{1/N}$ is the geometric mean of the stage reproductive numbers. The three-level framework is developed in Chapter 11, Section 11.6.

Equilibria

DFE	disease-free equilibrium: $(s, i, r) = (1, 0, 0)$
EE	endemic equilibrium: (s^*, i^*, r^*) with $i^* > 0$
F_0	initial susceptible fraction: $F_0 = S(0)/N^{\mathrm{c}}$

Sensitivity analysis

Υ_p^Y	normalized sensitivity of output Y to parameter p
LHS	Latin hypercube sampling (Chapter 9, Appendix B)
PRCC	partial rank correlation coefficient

Units

The book uses a consistent unit notation: `C` for contacts, `P` for persons, `T` for time. Common derived units: contact rates c_X are $\mathtt{CP}^{-1}\mathtt{T}^{-1}$; forces α, λ are $\mathtt{T}^{-1}$; reproductive numbers are dimensionless.

For the complete glossary including historical context, alternate notations used in the classical literature, and chapter-by-chapter references, see Appendix C.

Appendix G

Notes for Instructors

What the book covers

The material is organized in five parts that develop the pitfalls in depth.

The book opens with an outbreak and the questions a health officer would ask of it. Chapter 1 builds from there the rate of new infections that every later model needs, counted first from the people who pass the infection on and then checked against the standard textbook form. **Part I (Foundations)** introduces the epidemiological compartments, the contact structure of a population, and the two ways the rate of new infections can be derived. The dimensional-consistency and force-of-vs-from pitfalls are introduced here.

Part II (The SIR Model) develops the three-compartment susceptible-infectious-recovered model in depth. The book chooses to derive most of its analytical results from the infected-viewpoint formulation (with α as primary) because that formulation supports the linearization-fidelity argument of Pitfall: "Linearization Fidelity" and clarifies the identifiability properties of Pitfall: "Identifiability". The susceptible-viewpoint formulation is treated carefully in Chapter 2 so the connection to the standard literature is preserved. Pitfall: "Natural Initial Conditions", Pitfall: "$\mathcal{R}_0$ versus $\mathcal{R}_e(t)$", and Pitfall: "Invasion versus Equilibrium" are introduced in this part.

Part III (Using the Model) addresses the practical questions that arise when the model must be fit to real data and analyzed for sensitivity. The data-fitting, identifiability, and sensitivity-discipline pitfalls are developed in depth here.

Part IV (Extensions) extends the framework to richer classes of models: loss of immunity, latent periods, disease-induced mortality, asymptomatic infection, heterogeneous populations (two-group mixing, sexual transmission, vector-host systems), vertical transmission, and continuous-structure formulations. Pitfall: "Heterogeneous-Mixing Balance Conditions" is developed in Chapter 11; the generation-time-distribution treatment is developed in Chapter 15 (the catalog handles this under Pitfall: "The Independence of τ" for the time-in-compartment principle and Pitfall: "Correct Data-Fitting Practice" for the data-resolution implications).

The pitfall chapter, placed between Part IV and Part V, reviews the book through its errors: one entry per pitfall, grouped by modeling stage (formulate, analyze, fit, extend), with a one-page checklist and a decision-flow figure.

Part V (Applications) closes with case studies that synthesize the apparatus: COVID-19, dengue, HIV, and a cross-cutting chapter that identifies patterns across diseases. The case studies revisit each pitfall in operational context.

Five appendices provide reference material. Appendix A (*Mathematical Preliminaries*) is a concise refresher on the ordinary-differential-equation, linear-algebra, and probability techniques the

book uses. Appendix B (*Numerical Methods*) documents the algorithms behind the book's simulations, reproductive-number computations, fits, and sensitivity analyses, and serves as a guide to the companion code repository that implements them. Appendix C is a glossary of the epidemiological and modeling terms used throughout, written in the book's own notation with chapter references. Appendix D, *Learning with AI*, develops the verification practice that AI-assisted study and modeling require—the same stance the book takes toward any computational result. Appendix E (*The Poisson Process and Thinning*) supplies the probability background underlying the stochastic derivations of the force of and from infection in Part I.

How this book compares with related texts

The models in this book descend from Kermack and McKendrick [1927], who introduced the three-compartment SIR system and proved its threshold theorem and final-size relation. They built on the mass-action principle of Hamer [1906] and on the transmission threshold that Ross [1911] derived for malaria. The standard textbook presentation descends from Anderson and May [1991], which made the susceptible viewpoint, with the force of infection λ as its central quantity, the canonical formulation. The texts discussed below follow the same convention. The infected viewpoint is not new. It is the natural object in branching-process approximations, in the next-generation matrix approach to $\mathcal{R}_0$ [Diekmann et al., 1990, van den Driessche and Watmough, 2002], and in most agent-based simulations. What the teaching literature lacks is a systematic development of the infected-viewpoint formulation for the cases where it offers an analytical advantage, together with a clear statement of when the susceptible viewpoint is the right place to start. The book aims to fill that gap. It introduces the infected viewpoint first, as the better default, and its opening chapter shows that the standard form gives the same rate.

The mathematical-epidemiology literature is well served by several standard references, and the present book draws on all of them.

Anderson and May [1991] remains a canonical reference for the biology of infectious disease alongside the mathematics, and is a natural starting point for readers who want a broad epidemiological survey. Keeling and Rohani [2008] is the most widely adopted textbook in mathematical epidemiology; it covers a much broader class of models than the present book (stochastic, network, metapopulation) and serves as an excellent first reading. Martcheva [2015] develops the classical compartmental framework carefully, with a particularly clear treatment of equilibrium analysis, bifurcation theory, and the next-generation matrix method. Brauer et al. [2019] covers similar ground with additional emphasis on epidemic control. Ma and Li [2009] is the standard reference from the Chinese mathematical-epidemiology school, curating contributions on qualitative analysis, pulse vaccination, and age-structured models, with application chapters on HIV/AIDS, SARS, and tuberculosis that complement the case studies of the present book. Diekmann and Heesterbeek [2000] develops reproductive-number theory with the rigor that readers need who want the mathematical foundations behind $\mathcal{R}_0$. Allen [2010] and Britton and Pardoux [2019] cover stochastic epidemic models, which the present book omits but which are essential for modeling early-outbreak dynamics and extinction probabilities.

The distinctive emphasis of the present book is the common-pitfalls-driven organization. The standard texts are organized around model classes (deterministic, stochastic, network) or around mathematical techniques (next-generation matrix, bifurcation analysis, equilibrium stability). The present book is organized around the recurring patterns of mistake that a modeler must learn to recognize and avoid. Each pitfall is flagged by name where the error is tempting, through the parenthetical `(Pitfall: ‘‘Name’’)` cross-reference convention, so that a reader can always trace

a specific lesson back to its entry in the pitfall chapter. The result is a book that is best used as a second course or as a practical reference: the basic compartmental machinery is developed for completeness, but the operational value lies in the pitfalls themselves and in the worked examples that show how each manifests in the case studies of Part V.

These choices reflect the book's intended audience: outbreak-response modelers, graduate students who will go on to do modeling work of their own, and instructors who want a one-semester text that emphasizes practical judgment alongside mathematical technique. The standard texts above are better starting points for a first course in mathematical epidemiology, for a broad survey of model classes, or for readers who need stochastic or network-based formulations.

Using this book as a course text

A one-semester course in epidemic modeling can use this book as its primary text. A recommended fifteen-week syllabus:

- Weeks 1-2: the Preface and Chapter 1 (the common-pitfalls preview and the foundations).
- Weeks 3-4: Chapters 2 and 3 (the two force derivations); the instructor may choose to do both or only the infected-viewpoint Chapter 3 on a first pass.
- Weeks 5-7: Chapter 4 (derivation), Chapter 5 (analysis), and Chapter 6 (simulations).
- Weeks 8-10: Chapter 7 (fitting) and Chapter 9 (sensitivity), with the supplementary Chapter 8 added if the class is project-oriented; this block develops the data-fitting, identifiability, and sensitivity-discipline pitfalls in depth and includes the central numerical comparison of fitted formulations.
- Weeks 11-12: Chapter 10 (model variants) and one or two heterogeneous-population chapters from Chapters 11, 12, 13, or 14.
- Weeks 13-14: Two case studies from Part V (Chapters 17 onward).
- Week 15: Cross-cutting synthesis (Chapter 20) and course review.

Chapter 15 (PDE and renewal-equation formulations) is more technically demanding and is usually omitted from a first-pass course. Instructors who include it should plan an extra week for the chapter and for the connection between the renewal-equation framework and the generation-time pitfalls (the catalog treats the time-in-compartment principle as Pitfall: "The Independence of τ" and the data-resolution implications under Pitfall: "Correct Data-Fitting Practice"). Chapter 8 is recommended for project-oriented courses; it can be taught alongside Chapter 7 or as the entry into the case-study chapters.

Chapters that carry the **[Open Project]** marker close with a capstone note. The fully specified projects—objectives, prerequisites, data guidance, starter code, rubric, and deliverable in the companion repository—are listed in Table G.1. The projects come in two tiers. A *chapter project* asks the student to demonstrate that a technique works; a *term project* asks them to analyze a disease and defend a conclusion about it. The hour ranges overlap, deliberately: tier is set by what the student must defend, not by how long it takes.

Projects can be done individually or in small teams. The six-step workflow gives a natural division of labor—elicitation, analysis, and validation parallelize well—while the AI-collaboration audit does not: in a team, the disclosure memo must account for each member's AI use separately.

Teams spread the load and sharpen internal review; solo work keeps a single, clean audit trail. The instructor decides.

Exercise conventions

Every chapter from Chapter 1 onward closes with two exercise blocks: section-level retrieval exercises distributed through the chapter's sections, and an end-of-chapter integrative block. Each exercise is tagged by type:

[C] Concept check. Short-answer conceptual questions, typically a few sentences or a single calculation, designed to verify the reader has understood the chapter's key ideas. These should be accessible immediately after reading the chapter.

[D] Derivation. Mathematical exercises that require carrying out a derivation, computation, or algebraic manipulation in the chapter's technical framework. Most exercises in the model-development chapters (Chapters 2–5) fall into this tier.

[I] Interpretation or modeling judgment. Exercises that require applying the chapter's framework to a new setting, formulating a modeling judgment, or interpreting a result in biological or operational terms. Often suitable as short project questions.

[L] Computational lab. Exercises that require writing and running code, usually to simulate a model, fit it to data, or conduct a sensitivity analysis. These require more time than classroom problems and may be assigned as homework or course projects. Starter notebooks for many **[L]** exercises are available in the companion repository (see Appendix B).

An instructor can build a problem set at a given level by selecting from the appropriate tiers; readers studying alone can work through **[C]** and **[D]** problems as consolidation and attempt **[I]** and **[L]** problems selectively.

A small subset of exercises carries one of three additional tags that mark exercises designed specifically around AI interaction. A **[Verify the AI]** exercise elicits AI output on a specific topic and asks the reader to identify where the output is wrong, ambiguous, or conflated, then produce a corrected version. An **[AI Lab]** exercise invites AI assistance in structuring a computational or modeling workflow, with independent verification of every mathematical and numerical choice. An **[Open Project]** exercise is a larger assignment where AI use is expected, accompanied by a short disclosure (what was asked, what was accepted, what was rejected, what was verified) in the spirit of the disclosure template in Appendix D. Exercises without one of these three tags are intended for unaided engagement.

The book's pedagogical stance on AI tool use is straightforward: AI systems can deepen understanding when used to clarify notation, generate alternative explanations, suggest counterexamples, prototype computations, identify edge cases, or pose self-test questions. They fail predictably on the conflation of force-of-infection with force-from-infection Pitfall: "Force of Infection versus Force from Infection", on dimensionally inconsistent ODE terms Pitfall: "Dimensional Consistency", on heterogeneous-setting reproductive-number formulas Pitfall: "Heterogeneous-Mixing Balance Conditions", on hallucinated citations, and on code that runs but implements wrong mathematics. The verification checklists in Appendix D and the in-chapter *Verify the AI* boxes provide concrete diagnostics for these failure modes. Readers and instructors who want a fuller treatment of AI-aware learning should consult the companion three-document package: the Student Guide

[Hyman, 2026c] for prompt-pattern libraries, worked failure-mode examples, disclosure templates with worked examples, and cognitive-science-grounded study habits; the Instructor Guide [Hyman, 2026b] for assignment design, the three assessment models (Oral-Defense, Evidence-of-Thinking, Hybrid), rubrics, and TA calibration; and the Institutional Guide [Hyman, 2026a] for departments and curriculum committees adopting institution-wide policies. All three guides are freely available, subject-general, and share a common vocabulary (the C/A/R framework, the AI Learning Loop, and the Red/Yellow/Green/Blue assignment categories).

Common-Pitfalls-by-chapter map

The common pitfalls introduced in the Preface and developed in Chapter 16 are revisited throughout the book. The table below maps each common pitfall to the chapter where it receives its principal development and to the chapters where it is revisited. A reader who wants to study a specific common pitfall in depth can use this map to navigate directly to the relevant chapters.

A note on software and reproducibility

The computational work in the book is meant to be reproducible rather than merely described, and the companion repository is organized to make that checkable. Each deposited figure is paired with the script that generates it, the random seed it uses, and a recorded hash of both, so a reader can regenerate the figure and confirm byte-for-byte that they have reproduced what was printed. Deposit is proceeding chapter by chapter and is furthest along for the analysis, fitting, and sensitivity chapters, where the numerical claims are load-bearing; the schematic figures are TikZ and reproduce with the manuscript itself. Where a computation is not yet deposited, the text gives the parameters and the method in enough detail to be re-implemented. The companion repository, described in Appendix B, supplies the numerical methods as Python notebooks that run in Google Colab with no local installation, so a reader can run, modify, and extend the code from any browser. The notebooks are heavily commented for readers new to scientific programming, and every numerical figure in the book was produced by a fixed-seed script, deposited in the repository as the chapter-by-chapter rollout described above reaches it.

The Companion Repository

The book is accompanied by a companion repository hosted on GitHub at `https://github.com/machyman/hyman2026avoiding`. The repository lets the reader reproduce the book's numerical figures, work through the computational lab exercises, and adapt the code to their own research applications. This short overview describes its contents, organization, and intended use; fuller documentation accompanies the code itself.

Language and platform

The methods developed in the book are language-agnostic at the algorithmic level: the simulations, fits, and sensitivity analyses of Chapters 6 through 9, and the case studies of Part V, translate directly into whichever scientific-computing environment the reader prefers. The companion repository provides reference implementations in *Python*, delivered as Jupyter notebooks that run in

Google Colab without any local installation. A reader following along on a tablet or a shared computer can open a notebook, run each cell, change parameters, and see the effect immediately; the same notebooks run in any local Jupyter environment with NumPy, SciPy, and Matplotlib installed (versions pinned in the repository's `requirements.txt`).

How the notebooks are written

The notebooks are written for readers who know the mathematics from the book but may have limited programming experience. Each notebook includes a header describing the chapter it accompanies and the parameter set it uses; inline comments that map the mathematical notation of the book to the variable names in the code; and, where useful, suggested parameter changes the reader can make to build intuition about the model's behavior. Explanations are not repeated across notebooks: a reader who works through one in detail has enough context to read the others.

Repository structure

Code is organized by chapter, so a reader working through Chapter n can find the corresponding code directly. The `notebooks/` directory holds the figure and case-study notebooks in per-chapter subdirectories; a small `src/` directory holds shared model code (such as the SIR integrator) that several notebooks import; and `requirements.txt` pins the dependency versions. The stochastic case-study series are generated within their notebooks from fixed random seeds, so they reproduce exactly without any external data files; empirical datasets, where a future case study uses them, are placed in a `data/` directory with provenance documentation.

Computational labs ([L]-tagged exercises)

The computational lab exercises marked with the **[L]** tag (the tier-label scheme is introduced at the start of Chapter 1's exercises) are supported in the repository by starter notebooks for the core labs, with the data-loading and parameter-definition cells already populated so the reader can concentrate on the algorithmic content. Reference-solution notebooks are provided for these core labs, structured so that a reader who attempts the exercise first can compare against the solution rather than read it as a walkthrough.

Citing the repository

A DOI is minted for the repository at publication. Readers who use the code in their own work should cite both the book and the repository; the recommended citation is included in the repository's `CITATION.cff` and README.

Versioning

The repository is versioned in tandem with the book, so the code corresponding to a given printing is available and identifiable. Errata corrections to the code are released as patch updates, and substantive additions accompany new printings or editions. Additional reference implementations and worked extensions may be added over time as readers identify common needs.

Table G.1: Capstone projects at a glance. Chapter projects demonstrate that a technique works; term projects analyze a disease and defend a conclusion about it. Full specifications, data guidance, and rubrics are in the companion repository.

Ch.	Tier	Project and deliverable	Hours
3	chapter	*Force of infection for a structurally richer contact model* — a derived force-of-infection expression for one richer contact structure, validated numerically against the chapter's baseline	20–30
4	chapter	*Audit an AI's exposition of the* SIR *model* — a corrected exposition, with an audit log of every defect found	6–10
6	term	*Endemic-equilibrium analysis for a chosen pathogen* — an endemic-equilibrium analysis for one pathogen, with the invasion–burden distinction argued explicitly	13–20
7	chapter	*Full* SIR *implementation, verification, and debugging project* — a working SIR simulator with a verification suite that provably catches seeded defects	15–25
8	chapter	*Full AI-collaborative fitting project* — a fitted SIR-type model with uncertainty quantified and identifiability addressed	15–25
10	term	*Full sensitivity-driven intervention recommendation* — an intervention recommendation driven by sensitivity analysis, with uncertainty in the ranking quantified	15–25
12	term	*Full two-group* SIR *fitting and intervention recommendation* — a fitted two-group model with a heterogeneity-aware intervention recommendation	20–30
14	term	*Full vector-host implementation and fitting project* — a fitted vector-host model with sensitivity-weighted intervention recommendations	20–30
17	term	*Re-analysis of a published COVID-19 $\mathcal{R}_0$ estimate* — a corrected $\mathcal{R}_0$ or $\mathcal{R}_e(t)$ estimate with quantified uncertainty, and a documented critique of the original	15–25

#	Common Pitfall	Principal	Revisited
Regime of validity			
1	Stochastic-to-deterministic regime transition	Preface	Chapter 7, 8
2	Short-term simulation assumptions	Preface	Chapter 11, 17
Building the model correctly			
3	Dimensional consistency	Chapter 2	throughout
4	Force *of* infection vs. force *from* infection	Chapter 3	Chapter 4–11
5	Linearization fidelity	Chapter 5	Chapter 7
6	Natural initial conditions	Chapter 6	Chapter 7, 8
7	Heterogeneous-mixing balance conditions	Chapter 11	Chapter 12, 13
Estimating from data			
8	Correct data-fitting practice	Chapter 7	Chapter 8, 17
9	Identifiability	Chapter 7	Chapter 5, 20
10	Generation-time-distribution misspecification	Chapter 15	Chapter 8, 17
11	Correct sensitivity analysis	Chapter 9	Chapter 8; throughout Part IV
Interpreting reproductive numbers			
12	Basic vs. effective reproductive number	Chapter 5	Chapter 7, 15, 17, 19
13	Equilibrium vs. invasion conflation	Chapter 5	Chapter 11, 13, 19

Table G.2: Map from the common pitfalls to the chapter where each is developed in depth. "Principal" is the chapter where the common pitfall receives its formal treatment; "Revisited" lists chapters where the pitfall is named and reinforced in operational context.

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Bibliography

Sam Abbott, Joel Hellewell, Robin N. Thompson, Katharine Sherratt, Hamish P. Gibbs, Nikos I. Bosse, James D. Munday, Sophie Meakin, Emma L. Doughty, June Young Chun, Yung-Wai Desmond Chan, Flavio Finger, Paul Campbell, Akira Endo, Carl A. B. Pearson, Amy Gimma, Tim Russell, Stefan Flasche, Adam J. Kucharski, Rosalind M. Eggo, and Sebastian Funk. Estimating the time-varying reproduction number of SARS-CoV-2 using national and subnational case counts. *Wellcome Open Research*, 5:112, 2020. doi: 10.12688/wellcomeopenres.16006.2.

A. E. Ades, Antoni Soriano-Arandes, Ana Alarcon, Francesco Bonfante, Claire Thorne, Catherine S. Peckham, and Carlo Giaquinto. Vertical transmission of Zika virus and its outcomes: a Bayesian synthesis of prospective studies. *The Lancet Infectious Diseases*, 21(4):537–545, 2021. doi: 10.1016/S1473-3099(20)30432-1.

Philipp Agyeman, A Duppenthaler, U Heininger, and C Aebi. Influenza-associated myositis in children. *Infection*, 32(4):199–203, 2004.

Linda J. S. Allen. *An Introduction to Stochastic Processes with Applications to Biology*. Chapman and Hall/CRC, Boca Raton, FL, 2 edition, 2010. ISBN 9781439818824.

Roy M Anderson and Robert M May. Vaccination and herd immunity to infectious diseases. *Nature*, 318(6044):323–329, 1985. doi: 10.1038/318323a0.

Roy M Anderson and Robert M May. *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press, Oxford, UK, 1991. ISBN 978-0198545996. Same content as the 1992 Oxford reprint; both years appear in the literature.

Roy M Anderson and Robert M May. *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press, 1992. ISBN 978-0198540403.

Joan L Aron and Robert M May. The population dynamics of malaria. In *The Population Dynamics of Infectious Diseases: Theory and Applications*, pages 139–179. Springer, 1982.

Leon Arriola and James M Hyman. Sensitivity analysis for uncertainty quantification in mathematical models. In *Mathematical and Statistical Estimation Approaches in Epidemiology*, pages 195–247. Springer, 2009. doi: 10.1007/978-90-481-2313-1_10.

Leon M Arriola and James M Hyman. Being sensitive to uncertainty. *Computing in Science & Engineering*, 9(2):10–20, 2007.

Nicolas Bacaër and Souad Guernaoui. The epidemic threshold of vector-borne diseases with seasonality: The case of cutaneous leishmaniasis in Chichaoua, Morocco. *Journal of Mathematical Biology*, 53(3):421–436, 2006. doi: 10.1007/s00285-006-0015-0.

G. I. Barenblatt. *Scaling*. Cambridge Texts in Applied Mathematics. Cambridge University Press, Cambridge, UK, 2003. ISBN 9780521533942.

Roberto Barrera, Manuel Amador, Veronica Acevedo, Belkis Caban, Gilberto Felix, and Andrew J. Mackay. Use of the CDC autocidal gravid ovitrap to control and prevent outbreaks of *Aedes aegypti* (Diptera: Culicidae). *Journal of Medical Entomology*, 51(1):145–154, 2014. doi: 10.1603/ME13096.

M. S. Bartlett. *An Introduction to Stochastic Processes*. Cambridge University Press, Cambridge, 1955.

Alexander B. Beams, David J. D. Earn, and Caroline Colijn. Uncertainty in COVID-19 transmission could undermine our ability to predict long COVID. *Journal of the Royal Society Interface*, 21 (221):20240438, 2024. doi: 10.1098/rsif.2024.0438.

Michael Begon, Malcolm Bennett, Roger G. Bowers, Nigel P. French, Sarah M. Hazel, and Joanne Turner. A clarification of transmission terms in host-microparasite models: numbers, densities and areas. *Epidemiology and Infection*, 129(1):147–153, 2002. doi: 10.1017/S0950268802007148.

Samir Bhatt, Peter W. Gething, Oliver J. Brady, Jane P. Messina, Andrew W. Farlow, Catherine L. Moyes, John M. Drake, John S. Brownstein, Anne G. Hoen, Osman Sankoh, et al. The global distribution and burden of dengue. *Nature*, 496(7446):504–507, 2013. doi: 10.1038/nature12060.

Matthew Biggerstaff, Simon Cauchemez, Carrie Reed, Manoj Gambhir, and Lyn Finelli. Estimates of the reproduction number for seasonal, pandemic, and zoonotic influenza: a systematic review of the literature. *BMC Infectious Diseases*, 14:480, 2014. doi: 10.1186/1471-2334-14-480.

Sally M. Blower and Hadi Dowlatabadi. Sensitivity and uncertainty analysis of complex models of disease transmission: an HIV model, as an example. *International Statistical Review*, 62(2): 229–243, 1994. doi: 10.2307/1403510.

Fred Brauer, Carlos Castillo-Chavez, and Zhilan Feng. *Mathematical Models in Epidemiology*, volume 69 of *Texts in Applied Mathematics*. Springer, New York, 2019. doi: 10.1007/978-1-4939-9828-9.

Tom Britton and Etienne Pardoux. *Stochastic Epidemic Models with Inference*, volume 2255 of *Lecture Notes in Mathematics*. Springer, Cham, Switzerland, 2019. doi: 10.1007/978-3-030-30900-8.

Stavros Busenberg and Carlos Castillo-Chavez. A general solution of the problem of mixing of subpopulations and its application to risk- and age-structured epidemic models for the spread of AIDS. *Mathematical Medicine and Biology: A Journal of the IMA*, 8(1):1–29, 1991.

Carlos Castillo-Chavez, Herbert W Hethcote, Viggo Andreasen, Simon A Levin, and Wei M Liu. Epidemiological models with age structure, proportionate mixing, and cross-immunity. *Journal of Mathematical Biology*, 27(3):233–258, 1989.

Muge Cevik, Matthew Tate, Ollie Lloyd, Alberto Enrico Maraolo, Jenna Schafers, and Antonia Ho. SARS-CoV-2, SARS-CoV, and MERS-CoV viral load dynamics, duration of viral shedding, and infectiousness: a systematic review and meta-analysis. *The Lancet Microbe*, 2(1):e13–e22, 2021. doi: 10.1016/S2666-5247(20)30172-5.

Youngjoon Cha, Mimmo Iannelli, and Fabio A Milner. Existence and uniqueness of endemic states for the age-structured S–I–R epidemic model. *Mathematical Biosciences*, 150(2):177–190, 1998.

Miranda Chan and Michael A. Johansson. The incubation periods of dengue viruses. *PLoS ONE*, 7(11):e50972, 2012. doi: 10.1371/journal.pone.0050972.

Wilbur H Chen, Franklin R Toapanta, Kari Ann Shirey, Lei Zhang, Angeliki Giannelou, Carly Page, Matthew B Frieman, Stefanie N Vogel, and Alan S Cross. Potential role for alternatively activated macrophages in the secondary bacterial infection during recovery from influenza. *Immunology Letters*, 141(2):227–234, 2012. doi: 10.1016/j.imlet.2011.10.009.

Nakul Chitnis, Jim M Cushing, and JM Hyman. Bifurcation analysis of a mathematical model for malaria transmission. *SIAM Journal on Applied Mathematics*, 67(1):24–45, 2006. doi: 10.1137/050638941.

Nakul Chitnis, James M Hyman, and Jim M Cushing. Determining important parameters in the spread of malaria through the sensitivity analysis of a mathematical model. *Bulletin of Mathematical Biology*, 70(5):1272–1296, 2008. doi: 10.1007/s11538-008-9299-0.

Nakul Chitnis, James M Hyman, and Carrie A Manore. Modelling vertical transmission in vector-borne diseases with applications to Rift Valley fever. *Journal of Biological Dynamics*, 7(1):11–40, 2013. doi: 10.1080/17513758.2012.733427.

Gerardo Chowell, Carlos Castillo-Chavez, Paul W Fenimore, Christopher M Kribs-Zaleta, Leon Arriola, and James M Hyman. Model parameters and outbreak control for SARS. *Emerging Infectious Diseases*, 10(7):1258–1263, 2004. doi: 10.3201/eid1007.030647.

Rebecca C. Christofferson, Christopher N. Mores, and Helen J. Wearing. Bridging the gap between experimental data and model parameterization for chikungunya virus transmission predictions. *The Journal of Infectious Diseases*, 214(suppl. 5):S466–S470, 2016. doi: 10.1093/infdis/jiw283.

Vittoria Colizza and Alessandro Vespignani. Epidemic modeling in metapopulation systems with heterogeneous coupling pattern: theory and simulations. In *Mathematical and Statistical Estimation Approaches in Epidemiology*, pages 221–244. Springer, Dordrecht, 2009.

Jessica R Conrad, Ling Xue, Jeremy Dewar, and James M Hyman. Modeling the impact of behavior change on the spread of Ebola. In Gerardo Chowell and James M Hyman, editors, *Mathematical and Statistical Modeling for Emerging and Re-emerging Infectious Diseases*, pages 5–23. Springer International Publishing, Cham, Switzerland, 2016. ISBN 9783319404134. doi: 10.1007/978-3-319-40413-4_2.

Jessica R. Conrad, Paul W. Fenimore, Kelly R. Moran, and Marisa C. Eisenberg. Verifying infectious disease scenario planning for geographically diverse populations. *Epidemics*, 52:100833, 2025. doi: 10.1016/j.epidem.2025.100833.

Anne Cori, Neil M Ferguson, Christophe Fraser, and Simon Cauchemez. A new framework and software to estimate time-varying reproduction numbers during epidemics. *American Journal of Epidemiology*, 178(9):1505–1512, 2013. doi: 10.1093/aje/kwt133.

Gaston De Serres, Isabelle Rouleau, Marie-Eve Hamelin, Caroline Quach, Danuta Skowronski, Louis Flamand, Nicole Boulianne, Yan Li, Julie Carbonneau, Anne-Marie Bourgault, Michel Couillard, Hugues Charest, and Guy Boivin. Contagious period for pandemic (H1N1) 2009. *Emerging Infectious Diseases*, 16(5):783–788, 2010. doi: 10.3201/eid1605.091894.

Sara Y Del Valle, James M Hyman, Herbert W Hethcote, and Stephen G Eubank. Mixing patterns between age groups in social networks. *Social Networks*, 29(4):539–554, 2007. doi: 10.1016/j.socnet.2007.04.005.

Méryl Delrieu, Jean-Philippe Martinet, Olivia O'Connor, Elvina Viennet, Christophe Menkes, Valérie Burtet-Sarramegna, Francesca D. Frentiu, and Myrielle Dupont-Rouzeyrol. Temperature and transmission of chikungunya, dengue, and Zika viruses: A systematic review of experimental studies on *Aedes aegypti* and *Aedes albopictus*. *Current Research in Parasitology & Vector-Borne Diseases*, 4:100139, 2023. doi: 10.1016/j.crpvbd.2023.100139.

Odo Diekmann and Johan Andre Peter Heesterbeek. *Mathematical Epidemiology of Infectious Diseases: Model Building, Analysis and Interpretation*, volume 5 of *Wiley Series in Mathematical and Computational Biology*. John Wiley & Sons, 2000. ISBN 9780471492412.

Odo Diekmann, J. A. P. Heesterbeek, and J. A. J. Metz. On the definition and the computation of the basic reproduction ratio R_0 in models for infectious diseases in heterogeneous populations. *Journal of Mathematical Biology*, 28(4):365–382, 1990. doi: 10.1007/BF00178324.

Klaus Dietz and Dieter Schenzle. Proportionate mixing models for age-dependent infection transmission. *Journal of Mathematical Biology*, 22(1):117–120, 1985.

Leah Edelstein-Keshet. *Mathematical Models in Biology*, volume 46 of *Classics in Applied Mathematics*. SIAM, Philadelphia, 2005. Original publication 1988, McGraw-Hill.

Robert W. Eisinger, Carl W. Dieffenbach, and Anthony S. Fauci. HIV viral load and transmissibility of HIV infection: Undetectable equals untransmittable. *JAMA*, 321(5):451–452, 2019. doi: 10.1001/jama.2018.21167.

William Feller. *An Introduction to Probability Theory and Its Applications*, volume I. Wiley, New York, 3 edition, 1968.

Neil M. Ferguson, Derek A. T. Cummings, Simon Cauchemez, Christophe Fraser, Steven Riley, Aronrag Meeyai, Sopon Iamsirithaworn, and Donald S. Burke. Strategies for containing an emerging influenza pandemic in Southeast Asia. *Nature*, 437(7056):209–214, 2005. doi: 10.1038/nature04017.

Neil M. Ferguson, Isabel Rodríguez-Barraquer, Ilaria Dorigatti, Luis Mier-y Teran-Romero, Daniel J. Laydon, and Derek A. T. Cummings. Benefits and risks of the Sanofi-Pasteur dengue vaccine: Modeling optimal deployment. *Science*, 353(6303):1033–1036, 2016. doi: 10.1126/science.aaf9590.

Christophe Fraser, Christl A. Donnelly, Simon Cauchemez, William P. Hanage, Maria D. Van Kerkhove, T. Déirdre Hollingsworth, Jamie Griffin, Rebecca F. Baggaley, Helen E. Jenkins, Emily J. Lyons, Thibaut Jombart, Wes R. Hinsley, Nicholas C. Grassly, François Balloux, Azra C. Ghani, Neil M. Ferguson, Andrew Rambaut, Oliver G. Pybus, Hugo Lopez-Gatell, Celia M. Alpuche-Aranda, Ietza Bojorquez Chapela, Ethel Palacios Zavala, Dulce Ma Espejo Guevara, Francesco Checchi, Erika Garcia, Stephane Hugonnet, Cathy Roth, and WHO Rapid Pandemic Assessment Collaboration. Pandemic potential of a strain of influenza A (H1N1): Early findings. *Science*, 324(5934):1557–1561, 2009. doi: 10.1126/science.1176062.

Daisuke Furushima, Shoko Kawano, Yuko Ohno, and Masayuki Kakehashi. Estimation of the basic reproduction number of novel influenza A (H1N1) pdm09 in elementary schools using the SIR model. *The Open Public Health Journal*, 11:64–72, 2017. doi: 10.2174/1874434601711010064.

Christian Geneus, Rhea Kataria, Zhuolin Qu, Asma Azizi, and James M. Hyman. Quantifying uncertainty in stochastic network epidemic models. *arXiv preprint*, 2025. Preprint; under review.

Giulia Giordano, Franco Blanchini, Raffaele Bruno, Patrizio Colaneri, Alessandro Di Filippo, Angela Di Matteo, and Marta Colaneri. Modelling the COVID-19 epidemic and implementation of population-wide interventions in Italy. *Nature Medicine*, 26(6):855–860, 2020. doi: 10.1038/s41591-020-0883-7.

Edward Goldstein, Jonathan Dushoff, Junling Ma, Joshua B. Plotkin, David J. D. Earn, and Marc Lipsitch. Reconstructing influenza incidence by deconvolution of daily mortality time series. *Proceedings of the National Academy of Sciences*, 106(51):21825–21829, 2009. doi: 10.1073/pnas.0902958106.

Nicolò Gozzi, Nicola Perra, and Alessandro Vespignani. Comparative evaluation of behavioral epidemic models using COVID-19 data. *Proceedings of the National Academy of Sciences*, 122 (24):e2421993122, 2025. doi: 10.1073/pnas.2421993122.

William D. Green, Neil M. Ferguson, and Anne Cori. Inferring the reproduction number using the renewal equation in heterogeneous epidemics. *Journal of the Royal Society Interface*, 19(192): 20210429, 2022. doi: 10.1098/rsif.2021.0429.

AB Gumel, Connell C McCluskey, and Pauline van den Driessche. Mathematical study of a staged-progression HIV model with imperfect vaccine. *Bulletin of Mathematical Biology*, 68(8):2105–2128, 2006. doi: 10.1007/s11538-006-9095-7.

Maria G. Guzman, Scott B. Halstead, Harvey Artsob, Philippe Buchy, Jeremy Farrar, Duane J. Gubler, Elizabeth Hunsperger, Axel Kroeger, Harold S. Margolis, Eric Martínez, et al. Dengue: a continuing global threat. *Nature Reviews Microbiology*, 8(12 Suppl):S7–S16, 2010. doi: 10.1038/nrmicro2460.

K. P. Hadeler and P. van den Driessche. Backward bifurcation in epidemic control. *Mathematical Biosciences*, 146(1):15–35, 1997. doi: 10.1016/S0025-5564(97)00027-8.

M. Elizabeth Halloran, Claudio J. Struchiner, and Ira M. Longini. Study designs for evaluating different efficacy and effectiveness aspects of vaccines. *American Journal of Epidemiology*, 146 (10):789–803, 1997.

William H. Hamer. The Milroy lectures on epidemic disease in England: the evidence of variability and of persistency of type. *The Lancet*, 167(4307):733–739, 1906. Often cited together with Hamer's 1928 *Epidemiology Old and New*.

Adam Hamilton, Fardad Haghpanah, Alex Tulchinsky, Nikoloz Kipshidze, Sandra Poleon, George Lin, Carla Tilchin, Anahita Khojandi, Madhav Marathe, and Eili Klein. Incorporating endogenous human behavior in models of COVID-19 transmission: a systematic scoping review. *Dialogues in Health*, 4:100179, 2024. doi: 10.1016/j.dialog.2024.100179.

James A. Hay, Pierre Nouvellet, Christl A. Donnelly, and Steven Riley. Potential inconsistencies in Zika surveillance data and our understanding of risk during pregnancy. *PLOS Neglected Tropical Diseases*, 12(12):e0006991, 2018. doi: 10.1371/journal.pntd.0006991.

Jon C. Helton and Freddie J. Davis. Latin hypercube sampling and the propagation of uncertainty in analyses of complex systems. *Reliability Engineering and System Safety*, 81(1):23–69, 2003. doi: 10.1016/S0951-8320(03)00058-9.

Herbert W Hethcote. An age-structured model for pertussis transmission. *Mathematical Biosciences*, 145(2):89–136, 1997.

Herbert W Hethcote. The mathematics of infectious diseases. *SIAM Review*, 42(4):599–653, 2000. doi: 10.1137/S0036144500371907.

Herbert W. Hethcote and James A. Yorke. Gonorrhea transmission dynamics and control. *Lecture Notes in Biomathematics*, 56, 1984.

Hyukpyo Hong, Hyeontae Park, and Jae Kyoung Kim. Overcoming bias in estimating epidemiological parameters with realistic history-dependent disease spread dynamics. *Nature Communications*, 15:8631, 2024. doi: 10.1038/s41467-024-53095-7.

Frank Hoppensteadt. *Mathematical Theories of Populations: Demographics, Genetics, and Epidemics*, volume 20. SIAM, 1975.

Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, UK, 2 edition, 2012. ISBN 9780521548236.

Jeremy Howard, Austin Huang, Zhiyuan Li, Zeynep Tufekci, Vladimir Zdimal, Helene-Mari van der Westhuizen, Arne von Delft, Amy Price, Lex Fridman, Lei-Han Tang, et al. An evidence review of face masks against COVID-19. *Proceedings of the National Academy of Sciences*, 118(4): e2014564118, 2021. doi: 10.1073/pnas.2014564118.

Ying-Hen Hsieh, David N. Fisman, and Jianhong Wu. On epidemic modeling in real time: An application to the 2009 novel A (H1N1) influenza outbreak in Canada. *BMC Research Notes*, 3: 283, 2010. doi: 10.1186/1756-0500-3-283.

Jing Huang and Jeffrey S. Morris. Infectious disease modeling. *Annual Review of Statistics and Its Application*, 12:19–44, 2025. doi: 10.1146/annurev-statistics-112723-034351.

Hai-Feng Huo, Rui Chen, and Xun-Yang Wang. Modelling and stability of HIV/AIDS epidemic model with treatment. *Applied Mathematical Modelling*, 40(13–14):6550–6559, 2016. doi: 10.1016/j.apm.2016.01.054.

James M. Hyman. *A University Framework for AI-Assisted Learning: Institutional Guide for Departments, Teaching Centers, and University Leaders*. Manuscript in preparation; freely available from the authors, 1 edition, 2026a. Freely available; CC-BY-4.0. Part of a three-document package with the Student Guide and Instructor Guide.

James M. Hyman. *Teaching and Assessing with AI: An Instructor Guide for AI-Assisted Learning*. Manuscript in preparation; freely available from the authors, 1 edition, 2026b. Freely available; CC-BY-4.0. Part of a three-document package with the Student Guide and Institutional Guide.

James M. Hyman. *Learning with AI: A Student Guide to Thinking, Verification, and Honest Work*. Manuscript in preparation; freely available from the authors, 1 edition, 2026c. Freely available; CC-BY-4.0. Part of a three-document package with the Instructor Guide and Institutional Guide.

James M. Hyman. Local and global sensitivity analysis of SIR models. Preprint in preparation, 2026d.

James M. Hyman and Leon M. Arriola. Foundations of sensitivity analysis: From local sensitivity to global uncertainty. Manuscript in progress, 2026. URL https://github.com/machyman/hyman2026foundations.

James M Hyman and E Ann Stanley. The effect of social mixing patterns on the spread of AIDS. In Carlos Castillo-Chavez, Simon A. Levin, and Christine A. Shoemaker, editors, *Mathematical Approaches to Problems in Resource Management and Epidemiology*, volume 81 of *Lecture Notes in Biomathematics*, pages 190–219. Springer, Berlin, Heidelberg, 1989.

James M Hyman, Jia Li, and E Ann Stanley. Threshold conditions for the spread of the HIV infection in age-structured populations of homosexual men. *Journal of Theoretical Biology*, 166 (1):9–31, 1994.

James M Hyman, Jia Li, and E Ann Stanley. The differential infectivity and staged progression models for the transmission of HIV. *Mathematical Biosciences*, 155(2):77–109, 1999. doi: 10.1016/S0025-5564(98)10057-3.

James M Hyman, Jia Li, and E Ann Stanley. The initialization and sensitivity of multigroup models for the transmission of HIV. *Journal of Theoretical Biology*, 208(2):227–249, 2001.

James M. Hyman, Zhuolin Qu, and Ling Xue. Companion repository: Foundations and Pitfalls in Epidemic Modeling. github.com/machyman/hyman2026avoiding, 2026. Python notebooks for all computational labs.

Mimmo Iannelli. *Mathematical Theory of Age-Structured Population Dynamics*, volume 7 of *Applied Mathematics Monographs*. Giardini editori e stampatori, Pisa, 1995. ISBN 978-8842702504.

Hisashi Inaba. Threshold and stability results for an age-structured epidemic model. *Journal of Mathematical Biology*, 28(4):411–434, 1990.

Michael A. Johansson, Luis Mier-y Teran-Romero, Jennita Reefhuis, Suzanne M. Gilboa, and Susan L. Hills. Zika and the risk of microcephaly. *New England Journal of Medicine*, 375(1):1–4, 2016. doi: 10.1056/NEJMp1605367.

Matt J. Keeling and Ken T. D. Eames. Networks and epidemic models. *Journal of the Royal Society Interface*, 2(4):295–307, 2005. doi: 10.1098/rsif.2005.0051.

Matt J. Keeling and Pejman Rohani. *Modeling Infectious Diseases in Humans and Animals*. Princeton University Press, Princeton, NJ, 2008. ISBN 9780691116174.

William Ogilvy Kermack and Anderson G McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A, Containing papers of a mathematical and physical character*, 115(772):700–721, 1927. doi: 10.1098/rspa.1927.0118.

István Z. Kiss, Joel C. Miller, and Péter L. Simon. *Mathematics of Epidemics on Networks: From Exact to Approximate Models*, volume 46 of *Interdisciplinary Applied Mathematics*. Springer, Cham, Switzerland, 2017. doi: 10.1007/978-3-319-50806-1.

Andrei Korobeinikov. Lyapunov functions and global properties for SEIR and SEIS epidemic models. *Mathematical Medicine and Biology*, 21(2):75–83, 2004. doi: 10.1093/imammb/21.2.75.

Christopher M Kribs-Zaleta and Maia Martcheva. Vaccination strategies and backward bifurcation in an age-since-infection structured model. *Mathematical Biosciences*, 177:317–332, 2002. doi: 10.1016/S0025-5564(01)00099-2.

Ana Lajmanovich and James A Yorke. A deterministic model for gonorrhea in a nonhomogeneous population. *Mathematical Biosciences*, 28(3-4):221–236, 1976.

Suzanne Lenhart. Next generation matrix method. Lecture notes, University of Tennessee, Knoxville; June 24, 2020. Provided to the authors by S. Lenhart in 2026. Slide 10 lists the five NGM-applicability assumptions; slide 17 presents a *Clostridium difficile* hospital-shelter model as a worked counterexample where arrivals into the diseased class violate the in-migration assumption, 2020.

Suzanne Lenhart and John T. Workman. *Optimal Control Applied to Biological Models*. Chapman and Hall/CRC, Boca Raton, FL, 2007. ISBN 9781584886402.

Michael Y. Li. *An Introduction to Mathematical Modeling of Infectious Diseases*. Mathematics of Planet Earth. Springer, Cham, 2018. ISBN 978-3-319-72121-7. doi: 10.1007/978-3-319-72122-4.

Qun Li, Xuhua Guan, Peng Wu, Xiaoye Wang, Lei Zhou, Yeqing Tong, Ruiqi Ren, Kathy S.M. Leung, Eric H.Y. Lau, Jessica Y. Wong, et al. Early transmission dynamics in Wuhan, China, of novel coronavirus–infected pneumonia. *New England Journal of Medicine*, 382(13):1199–1207, 2020. doi: 10.1056/NEJMoa2001316.

Zhien Ma and Jia Li, editors. *Dynamical Modeling and Analysis of Epidemics*. World Scientific, Singapore, 2009. ISBN 978-981-279-749-0.

Zachary J. Madewell, Yang Yang, Ira M. Longini, M. Elizabeth Halloran, and Natalie E. Dean. Household transmission of SARS-CoV-2: A systematic review and meta-analysis. *JAMA Network Open*, 3(12):e2031756, 2020. doi: 10.1001/jamanetworkopen.2020.31756.

Carrie A Manore, Kyle S Hickmann, Sen Xu, Helen J Wearing, and James M Hyman. Comparing dengue and chikungunya emergence and endemic transmission in *A. aegypti* and *A. albopictus*. *Journal of Theoretical Biology*, 356:174–191, 2014. doi: 10.1016/j.jtbi.2014.04.033.

Carrie A Manore, Kyle S Hickmann, James M Hyman, Ivo M Foppa, Justin K Davis, Dawn M Wesson, and Christopher N Mores. A network-patch methodology for adapting agent-based models for directly transmitted disease to mosquito-borne disease. *Journal of Biological Dynamics*, 9(1):52–72, 2015. doi: 10.1080/17513758.2015.1005698.

Simeone Marino, Ian B. Hogue, Christian J. Ray, and Denise E. Kirschner. A methodology for performing global uncertainty and sensitivity analysis in systems biology. *Journal of Theoretical Biology*, 254(1):178–196, 2008. doi: 10.1016/j.jtbi.2008.04.011.

Maia Martcheva. *An Introduction to Mathematical Epidemiology*, volume 61. Springer, 2015. ISBN 978-1489976116.

Maia Martcheva and Xue-Zhi Li. Linking immunological and epidemiological dynamics of HIV: the case of super-infection. *Journal of Biological Dynamics*, 7(1):161–182, 2013. doi: 10.1080/17513758.2013.820358.

Maia Martcheva and Horst R Thieme. Progression age enhanced backward bifurcation in an epidemic model with super-infection. *Journal of Mathematical Biology*, 46(5):385–424, 2003. doi: 10.1007/s00285-002-0181-7.

Hamish McCallum, Nigel Barlow, and Jim Hone. How should pathogen transmission be modelled? *Trends in Ecology and Evolution*, 16(6):295–300, 2001. doi: 10.1016/S0169-5347(01)02144-9.

Michael D McKay, Richard J Beckman, and William J Conover. Comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, 21(2):239–245, 1979. doi: 10.1080/00401706.1979.10489755.

Elizabeth Miller, John E. Cradock-Watson, and Thomas M. Pollock. Consequences of confirmed maternal rubella at successive stages of pregnancy. *The Lancet*, 320(8302):781–784, 1982. doi: 10.1016/S0140-6736(82)92677-0. Cited in some indexes as volume 2, issue 8302.

Joël Mossong, Niel Hens, Mark Jit, Philippe Beutels, Kari Auranen, Rafael Mikolajczyk, Marco Massari, Stefania Salmaso, Gianpaolo Scalia Tomba, Jacco Wallinga, Janneke Heijne, Małgorzata Sadkowska-Todys, Magdalena Rosińska, and W. John Edmunds. Social contacts and mixing patterns relevant to the spread of infectious diseases. *PLoS Medicine*, 5(3):e74, 2008. doi: 10.1371/journal.pmed.0050074.

James D. Murray. *Mathematical Biology I: An Introduction*, volume 17 of *Interdisciplinary Applied Mathematics*. Springer, New York, 3 edition, 2002.

N. E. A. Murray, M. B. Quam, and Annelies Wilder-Smith. Epidemiology of dengue: past, present and future prospects. *Clinical Epidemiology*, 5:299–309, 2013. doi: 10.2147/CLEP.S34440.

Hiroshi Nishiura. Mathematical and statistical analyses of the spread of dengue. *Dengue Bulletin*, 30:51–67, 2006.

Daniel P. Oran and Eric J. Topol. The proportion of SARS-CoV-2 infections that are asymptomatic: A systematic review. *Annals of Internal Medicine*, 174(5):655–662, 2021. doi: 10.7326/M20-6976.

Xiaoxi Pang, Yang Han, Elise Tressier, Nurin Abdul Aziz, Lorenzo Pellis, Thomas House, and Ian Hall. Time-varying reproduction number estimation: fusing compartmental models with generalized additive models. *Journal of the Royal Society Interface*, 22(222):20240518, 2025. doi: 10.1098/rsif.2024.0518.

Pragna Patel, Craig B. Borkowf, John T. Brooks, Arielle Lasry, Amy Lansky, and Jonathan Mermin. Estimating per-act HIV transmission risk: a systematic review. *AIDS*, 28(10):1509–1519, 2014. doi: 10.1097/QAD.0000000000000298.

Kiesha Prem, Alex R. Cook, and Mark Jit. Projecting social contact matrices in 152 countries using contact surveys and demographic data. *PLoS Computational Biology*, 13(9):e1005697, 2017. doi: 10.1371/journal.pcbi.1005697.

Andreas Raue, Clemens Kreutz, Thomas Maiwald, Julie Bachmann, Marcel Schilling, Ursula Klingmüller, and Jens Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, 2009. doi: 10.1093/bioinformatics/btp358.

Carrie Reed, Frederick J. Angulo, David L. Swerdlow, Marc Lipsitch, Martin I. Meltzer, Daniel Jernigan, and Lyn Finelli. Estimates of the prevalence of pandemic (H1N1) 2009, United States, April–July 2009. *Emerging Infectious Diseases*, 15(12):2004–2007, 2009. doi: 10.3201/eid1512.091413.

Nicholas G. Reich, Craig J. McGowan, Teresa K. Yamana, Abhinav Tushar, Evan L. Ray, Dave Osthus, Sasikiran Kandula, Logan C. Brooks, Willow Crawford-Crudell, Graham C. Gibson, et al. Accuracy of real-time multi-model ensemble forecasts for seasonal influenza in the U.S. *PLOS Computational Biology*, 15(11):e1007486, 2019. doi: 10.1371/journal.pcbi.1007486.

Steven Riley. Large-scale spatial-transmission models of infectious disease. *Science*, 316(5829):1298–1301, 2007. doi: 10.1126/science.1134695.

Julien Riou and Christian L. Althaus. Pattern of early human-to-human transmission of Wuhan 2019 novel coronavirus (2019-nCoV), December 2019 to January 2020. *Eurosurveillance*, 25(4):2000058, 2020. doi: 10.2807/1560-7917.ES.2020.25.4.2000058.

Michael A. Robert, Rebecca C. Christofferson, Paula D. Weber, and Helen J. Wearing. Temperature impacts on dengue emergence in the United States: Investigating the role of seasonality and climate change. *Epidemics*, 28:100344, 2019. doi: 10.1016/j.epidem.2019.05.003.

Ronald Ross. *The Prevention of Malaria*. John Murray, London, 2 edition, 1911.

Kenneth J. Rothman, Sander Greenland, and Timothy L. Lash. *Modern Epidemiology*. Lippincott Williams & Wilkins, Philadelphia, PA, 3 edition, 2008. ISBN 978-0781755641.

Andrea Saltelli, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola. *Global Sensitivity Analysis: the Primer*. John Wiley & Sons, 2008. ISBN 9780470059975.

Omar Saucedo, Amanda Laubmeier, Tingting Tang, Benjamin Levy, Lale Asik, Tim Pollington, and Olivia Prosper Feldman. Comparative analysis of practical identifiability methods for an SEIR model. *AIMS Mathematics*, 9(9):24722–24761, 2024. doi: 10.3934/math.20241204.

Elissa J. Schwartz, Boseung Choi, and Grzegorz A. Rempala. Estimating epidemic parameters: Application to H1N1 pandemic data. *Mathematical Biosciences*, 270(Part B):198–203, 2015a. doi: 10.1016/j.mbs.2015.01.006.

Elissa J. Schwartz, M. Morgan, and S. Lapin. Pandemic 2009 H1N1 influenza in two settings in a small community: The workplace and the university campus. *Epidemiology and Infection*, 143(8):1606–1609, 2015b. doi: 10.1017/S0950268814002684.

David L Smith, F Ellis McKenzie, Robert W Snow, and Simon I Hay. Revisiting the basic reproductive number for malaria and its implications for malaria control. *PLoS Biology*, 5(3):e42, 2007. doi: 10.1371/journal.pbio.0050042.

Ilya M Sobol. Sensitivity estimates for nonlinear mathematical models. *Mathematical Modelling and Computational Experiments*, 1(4):407–414, 1993.

R N Thompson, J E Stockwin, R D van Gaalen, J A Polonsky, Z N Kamvar, P A Demarsh, E Dahlqwist, S Li, E Miguel, T Jombart, J Lessler, S Cauchemez, and A Cori. Improved inference of time-varying reproduction numbers during infectious disease outbreaks. *Epidemics*, 29:100356, 2019. doi: 10.1016/j.epidem.2019.100356.

Nils Benjamin Tjaden, Stephanie Margarete Thomas, Dominik Fischer, and Carl Beierkuhnlein. Extrinsic incubation period of dengue: Knowledge, backlog, and applications of temperature dependence. *PLoS Neglected Tropical Diseases*, 7(6):e2207, 2013. doi: 10.1371/journal.pntd.0002207.

UNAIDS. Global HIV and AIDS statistics - 2020 fact sheet, 2020. URL https://www.unaids.org/en/resources/fact-sheet.

UNAIDS. The urgency of now: AIDS at a crossroads. Global AIDS update, Joint United Nations Programme on HIV/AIDS (UNAIDS), Geneva, Switzerland, 2024. URL https://www.unaids.org/en/resources/documents/2024/global-aids-update-2024.

Naveen K. Vaidya, M. Morgan, T. Jones, L. Miller, S. Lapin, and Elissa J. Schwartz. Modelling the epidemic spread of an H1N1 influenza outbreak in a rural university town. *Epidemiology and Infection*, 143(8):1610–1620, 2015. doi: 10.1017/S0950268814002568.

P. van den Driessche and James Watmough. A simple SIS epidemic model with a backward bifurcation. *Journal of Mathematical Biology*, 40(6):525–540, 2000. doi: 10.1007/s002850000032.

P. van den Driessche and James Watmough. Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Mathematical Biosciences*, 180(1–2):29–48, 2002. doi: 10.1016/S0025-5564(02)00108-6.

Alejandro F. Villaverde, Antonio Barreiro, and Antonis Papachristodoulou. Structural identifiability of dynamic systems biology models. *PLOS Computational Biology*, 12(10):e1005153, 2016. doi: 10.1371/journal.pcbi.1005153.

H von Foerster. Some remarks on changing populations. In Frank Stohlman, editor, *The Kinetics of Cellular Proliferation*, pages 382–407. Grune & Stratton, New York, 1959.

Jacco Wallinga and Marc Lipsitch. How generation intervals shape the relationship between growth rates and reproductive numbers. *Proceedings of the Royal Society B*, 274(1609):599–604, 2007. doi: 10.1098/rspb.2006.3754.

Wendi Wang and Xiao-Qiang Zhao. Threshold dynamics for compartmental epidemic models in periodic environments. *Journal of Dynamics and Differential Equations*, 20(3):699–717, 2008. doi: 10.1007/s10884-008-9111-8.

Douglas M. Watts, Donald S. Burke, Bruce A. Harrison, Robert E. Whitmire, and Ananda Nisalak. Effect of temperature on the vector efficiency of *Aedes aegypti* for dengue 2 virus. *The American Journal of Tropical Medicine and Hygiene*, 36(1):143–152, 1987. doi: 10.4269/ajtmh.1987.36.143.

GF Webb and CJ Browne. A model of the Ebola epidemics in West Africa incorporating age of infection. *Journal of Biological Dynamics*, 10(1):18–30, 2016. doi: 10.1080/17513758.2015.1105985.

Lander Willem, Frederik Verelst, Joke Bilcke, Niel Hens, and Philippe Beutels. Lessons from a decade of individual-based models for infectious disease transmission: a systematic review (2006–2015). *BMC Infectious Diseases*, 17:612, 2017. doi: 10.1186/s12879-017-2699-8.

Olivia C. Winokur, Bradley J. Main, Jay Nicholson, and Christopher M. Barker. Impact of temperature on the extrinsic incubation period of Zika virus in *Aedes aegypti*. *PLoS Neglected Tropical Diseases*, 14(3):e0008047, 2020. doi: 10.1371/journal.pntd.0008047.

World Health Organization. Estimating mortality from COVID-19, 2020. URL https://www.who.int/news-room/commentaries/detail/estimating-mortality-from-covid-19.

Ling Xue, Xin Fang, and James M Hyman. Comparing the effectiveness of different strains of *Wolbachia* for controlling chikungunya, dengue fever, and Zika. *PLOS Neglected Tropical Diseases*, 12(7):e0006666, 2018. doi: 10.1371/journal.pntd.0006666.

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